1. On the Formation of the form System of the Ternary Biquadratic form

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(Extract from the author's dissertation.)

Work by Gordan, Maisano, and Pascal[1] deals with the form system of the ternary biquadratic form. Gordan sets up the complete form system, consisting of 54 formations, for the special automorphic form

$$f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1,$$

on the basis of principles similar to those which he had given for form systems in the binary domain.

In Maisano's work, for the general biquadratic form, the forms up to and including the fifth order[2] are set up, together with some invariants, covariants, and contravariants of higher order, according to the method used by Gordan in volume I of the *Mathematische Annalen* for the ternary cubic form. Pascal, using Maisano's results, is chiefly concerned with the question of the decomposition of the biquadratic form into factors[3].

The aim of my investigations is to set up the form system for the general ternary biquadratic form; to begin with, only the main foundations are given, and a so-called "relatively complete system"[4] is established.

The basic idea for forming systems of ternary forms is the same as in the binary domain. Starting from a first relatively complete system – the system of forms taken over from the binary case – one passes, according to a definite law, to systems with ever higher modulus, until either the system of a modulus – taking the modulus as ground form – becomes finite and known, or until a modulus can be reduced to forms having invariants as factors. By transvection of the relatively complete system with the system of the modulus, in the first case one obtains the absolutely complete system, while in the second case the relatively complete and absolutely complete systems coincide. By Hilbert's general proof the finiteness of form systems, this procedure must necessarily come to an end.

In our case, the modulus (abc) of the first relatively complete system can be led back to the moduli

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2 \quad \text{and} \quad \nu = (abu)^4;$$

from this the relatively complete system modulo ν is readily obtained. As the sequence of moduli we now choose the forms

$$\nu; \qquad \nu^{(\nu)} = (\nu\nu_1 x)^4 = s_x^4,$$
$$\nu^{(s)} = (ss'u)^4 \ldots,$$

and adjoin, as further moduli, two quadratic forms occurring in the formation of the relatively complete system modulo s, namely u_σ^2 and t_x^2. One can then show that the modulus following s, namely $(ss'u)^4$, is reducible to the modulus (ϱ, t). Since, however, the simultaneous system of two quadratic forms is finite and known, the termination described above has thereby been reached. As the relatively complete system modulo (ϱ, t) one obtains 331 formations. —

[1] P. Gordan, "Über the volle Formensystem of the ternären biquadratischen form" $f = x_1^3x_2 + x_2^3x_3 + x_3^3x_1$ (Math. Annalen vol. XVII, pp. 217–233, 1880); G. Maisano, 1. "Sistemi completi dei primi cinque gradi della forma ternaria biquadratica e degl' invarianti, covarianti e contravarianti di sesto grado" (Giorn. di Battaglini XIX). 2. "Sui covarianti indipendenti di 6° grado nei coefficienti della forma biquadratica ternaria" (Rend. Circ. Mat. di Palermo I, 1887); E. Pascal, "Contributo alla teoria della forma ternaria biquadratica e delle sue varie decomposizioni in fattori" (Memoria premiata dalla R. Accademia delle scienze fisiche e matematiche di Napoli, 1905).

[2] By "order" the dimension in the coefficients is to be understood, and by "degree" the dimension in the variables.

[3] In his second short note Maisano attempts to prove a linear dependence among the three covariants of order 6 and degree 6. In contrast to this not quite complete proof, Pascal believes he has proved the linear independence of the three covariants of order 6, as well as that of the three invariants of order 9. That, however, a linear relation does in fact exist between the three covariants on the one hand and the three invariants on the other, emerged in the course of my computations; in Pascal's notation the explicit formulas are

$$0 = 20\Omega_1 + 6\Omega_2 - 3\Omega_3 + 4fC_1 - 12fC_2 - 4A \cdot \Delta,$$

$$0 = 4A^3 - 15AB + 30C - 90D + 9E.$$

[4] For the terminology, see Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie*, vol. II, p. 227.

I add a few details concerning the method used.

The fundamental process for producing forms, the folding process, can be defined in the ternary domain as follows:

If in a symbolic product

$$s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$$

one replaces the pairs of factors

	$s_x t_x$	$u_\sigma u_\tau$	$s_x u_\sigma$ or $s_x u_\tau$	$t_x u_\tau$ or $t_x u_\sigma$
respectively by	(stu)	$(\sigma\tau x)$	s_σ or s_τ	t_τ or t_σ
folding	I	II	III	IV,

then the forms so arising have been obtained from the original form by folding[5].

Here one may still always put $s_\sigma = 0;\ t_\tau = 0$. For the relation among the individual foldings, the following theorem holds:

The foldings I and II are fundamental foldings, from which, independently of the order of composition, the foldings III and IV can be composed. In other words: to form all forms arising from a given expression by folding, one need only apply foldings I and II[6].

By a "form sequence" we mean an initial form together with all forms arising from it by folding into itself. By the theorem, a form sequence $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$ can be arranged in a rectangular scheme. In proceeding, one obtains:

1. by moving one column to the right, all forms arising by folding I from the adjacent forms;

2. by moving one row downward, all forms arising by folding II from the forms standing above, and hence all forms arising by folding.

Under these assumptions, the theorems on reducers can be stated in their most general form, under the definition

"A reducer is a reducible form sequence,"

as follows:

If the initial form of a form sequence is reducible because one of its terms has arisen by folding with a reducer, and if the final form of the form sequence has arisen from this same term by folding, then the whole form sequence is reducible.

Further reduction methods to be mentioned are:

1. the so-called "double reduction," that is, the reduction of a form in two ways in order to obtain a relation between the higher forms;

2. "folding with decomposed forms," which partly has the character of reduction by reducers and partly that of "double reduction."

[5] Gordan, Math. Annalen, vol. XVII, p. 219.

[6] The theorem has an exception in the case where n and μ, respectively m and ν, vanish simultaneously; the "form sequence" then reduces to the diagonal term.

2. On the Formation of the form System of the Ternary Biquadratic form

Journal for the reine and angewandte Mathematik 134 (1908), pp. 23–90 and two tables

Introduction.

Work by Gordan, Maisano, and Pascal[7,8] deals with the form system of the ternary biquadratic form. Gordan sets up the complete form system, consisting of 54 formations, of the special automorphic form

$$f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1$$

on the basis of principles similar to those that he had given for form systems in the binary domain.

In Maisano's work, for the general biquadratic form, the forms up to and including the fifth order[9] are set up, together with some invariants, covariants, and contravariants of higher order, according to the method used by Gordan in volume I of the *Mathematische Annalen* for the ternary cubic form. Pascal, using Maisano's results, is chiefly concerned with the question of the decomposition of the biquadratic form into factors[10].

The aim of the following investigations is to set up the form system for the general ternary biquadratic form; in this work, namely, only the principal foundations are given, and a so-called "relatively complete system"[11] *is established.*

The work is closely connected with Gordan's work; however, the principles that were stated there only quite generally first had to be worked out in detail. With the aid of a theorem on the connection among the foldings and by introducing the "form sequence" (§ 1), the reduction theorems are sharply formulated and completed (§ 3), while the recursive establishment of special series expansions (§ 2) gives the computational means for actually carrying out the reductions.

The basic idea for forming systems of ternary forms is the same as in the binary domain. Starting from a first relatively complete system – the system of forms taken over from the binary case – one passes, according to a definite law, to systems with ever higher modulus, until either the system of a modulus – the modulus taken as ground form – becomes finite and known, or until a modulus can be reduced to forms having invariants as factors. By transvection of the relatively complete system with the system of the modulus, in the first case the absolutely complete system arises, while in the second case the relatively complete and absolutely complete systems coincide. By Hilbert's general proof the finiteness of form systems, this procedure must necessarily come to an end.

In our case the modulus (abc) of the first relatively complete system (§ 4) is reduced to the moduli

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2 \quad \text{and} \quad \nu = (abu)^4$$

(§ 5), and then the relatively complete system modulo ν is found (§ 6). These results are already given in Gordan's work for the general biquadratic form; our manner of deriving them, however, is more easily transferred to forms of higher degree.

As the sequence of moduli we now choose the forms (§ 7)

$$\nu, \qquad \nu(\nu) = (\nu\nu_1 x)^4 = s_x^4, \qquad \nu(s) = (ss'u)^4, \ldots .$$

[7] A short extract from the Introduction and Chapter I appeared in the *Sitzungsberichte of the physikalisch-medizinischen Sozietät Erlangen 1907*, pp. 176–179.

[8] P. Gordan, on the complete form system of the ternary biquadratic form

$$f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1.$$

(*Math. Annalen*, vol. XVII (1880), pp. 217–233.) G. Maisano, 1) *Sistemi completi dei primi cinque gradi della forma ternaria biquadratica e degl' invarianti, covarianti e contravarianti di sesto grado.* (Giorn. di Battaglini XIX.) 2) *Sui covarianti indipendenti di* 6° *grado nei coefficienti della forma biquadratica ternaria.* (Rend. Circ. Mat. di Palermo I, 1887.) E. Pascal, *Contributo alla teoria della forma ternaria biquadratica e delle sue varie decomposizioni in fattori.* (Memoria premiata dalla R. Accademia delle scienze fisiche e matematiche di Napoli. 1905.)

[9] By "order" the dimension in the coefficients is to be understood, and by "degree" the dimension in the variables.

[10] In his second short note, Maisano attempts to prove a linear dependence of the three covariants of order 6 and degree 6. In contrast to this not quite complete proof, Pascal believes he has proved the linear independence of the three covariants of order 6, and likewise that of the three invariants of order 9. That, however, a linear relation does in fact exist between the three covariants on the one hand and the three invariants on the other is shown by formulas (13), § 11, and (22), § 17 note, of the present work, where these relations are given explicitly. According to a communication from Pascal, the contradiction is explained by a numerical error in the expression for his contravariant p (called ϱ here), p. 46 of his paper.

[11] For the terminology see § 3a.

In forming the relatively complete system modulo s, two quadratic forms occurring in the system, u_ϱ^2 and t_x^2, are adjoined as moduli (§§ 9 and 10), and accordingly the relatively complete system modulo (s, ϱ, t) is formed. It is then shown (§ 17) that the modulus $(ss'u)^4$ is reducible to the moduli ϱ and t. The next higher system thereby passes over into a relatively complete system modulo (ϱ, t). Since, however, the simultaneous system of two quadratic forms is finite and known, and in the general case consists of 20 formations[12], the termination described above is reached with the establishment of the relatively complete system modulo (ϱ, t). The transvection of this system with the system of (ϱ, t) in order to form the absolutely complete system is reserved.

As the relatively complete system modulo (ϱ, t) one finds 331 formations, which in the appended table are ordered according to their degree in the variables x and u.

Finally we give an overview of the symbols introduced:

$$
\begin{aligned}
f &= a_x^4, \\
\theta &= \theta_x^4 u_\vartheta^2 = (abu)^2 a_x^2 b_x^2, \\
K &= K_x^6 u_k^3 = (a\theta u) u_\vartheta^2 a_x^3 \theta_x^3, \\
j &= u_j^6 = (a\theta u)^4 u_\vartheta^2, \\
\Delta &= \Delta_x^6 = a_\vartheta^2 a_x^2 \theta_x^4 = (abc)^2 a_x^2 b_x^2 c_x^2, \\
N &= N_x^8 u_\eta = (a\Delta u) a_x^3 \Delta_x^5, \\
\nu &= u_\nu^4 = (abu)^4, \\
H &= H_x^2 u_\eta^4 = (\nu\nu_1 x)^2 u_\nu^2 u_{\nu_1}^2, \\
L &= L_x^3 u_l^6 = (\nu\eta x) H_x^2 u_\nu^3 u_\eta^3, \\
g &= g_x^6 = (\nu\eta x)^4 H_x^2, \\
\sigma &= u_\sigma^6 = H_\nu^2 u_\nu^2 u_\eta^4, \\
s &= s_x^4 = (\nu\nu_1 x)^4, \\
Z &= Z_x^4 u_\zeta^2 = (ss'u)^2 s_x^2 s_x'^2, \\
\varrho &= u_\varrho^2 = \theta_\nu^4 u_\vartheta^2, \\
t &= t_x^2 = a_\eta^4 H_x^2, \\
i &= a_\nu^4, \\
J &= s_\nu^4.
\end{aligned}
$$

Chapter I. General Theorems on Ternary Forms.

§ 1. The folding process. form sequences.

The fundamental process for producing forms is the folding process, which in the ternary domain can be defined as follows. Let there be given a symbolic product

$$s_x^m t_x^n u_\sigma^\mu u_\tau^\nu.$$

If one replaces the pairs of factors

$$s_x t_x \qquad u_\sigma u_\tau \qquad s_x u_\sigma \text{ or } s_x u_\tau \qquad t_x u_\tau \text{ or } t_x u_\sigma$$

respectively by

$$(stu) \qquad (\sigma\tau x) \qquad s_\sigma \text{ or } s_\tau \qquad t_\tau \text{ or } t_\sigma,$$

folding

$$\text{I} \qquad \text{II} \qquad \text{III} \qquad \text{IV},$$

then the resulting forms have arisen from the original one by folding[13].

[12] Cf. Clebsch–Lindemann, *Vorlesungen über Geometrie.* (Vol. I, p. 288 ff.) System of two cogredient forms. Gordan, "Über Büschel of Kegelschnitten." (*Math. Ann.* vol. XIX, p. 530.) System of two contragredient forms.

[13] Gordan, *Math. Annalen* vol. XVII, p. 219.

Here the expression $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$ can be regarded either as an actual product of two forms $S \cdot T$, or as a single form with several rows of symbols:

$$s_x^m t_x^n u_\sigma^\mu u_\tau^\nu = A_x^{m+n} u_x^{\mu+\nu}.$$

In the first case we speak of the folding of the form S with T, in the second case of the "folding of the form A in itself." For brevity we always assume in what follows, as indeed is the case for all forms later to be considered, that

$$s_\sigma^\lambda = 0, \qquad t_\tau^\lambda = 0 \tag{a}$$

for any λ and x distinct from 0, and in conjunction with any other foldings. This says that the form $s_x^m t_y^n u_\sigma^\mu u_\tau^\nu$ is a so-called "normal form," that is, it vanishes under application of the Ω-process, both with respect to the variables x and u and with respect to the variables y and v.

Thus condition (a) represents no restriction, but can always be achieved[14].

For the connection among the individual foldings the following theorem holds:

Theorem I. Foldings I and II are fundamental foldings from which, independently of the order of composition, foldings III and IV can be composed. In other words: in order to form all forms arising by folding from a given expression, one has to apply only foldings I and II.

Proof: By the identity theorem, respectively the product theorem for matrices, taking (a) into account, one obtains

$$\begin{aligned} (\widehat{st}\sigma x) &= -t_\sigma, \qquad & (\widehat{\sigma\tau} su) &= -s_\tau, \\ (\widehat{st}\tau x) &= s_\tau, \qquad & (\widehat{\sigma\tau} tu) &= t_\sigma, \end{aligned}$$

$$(stu)(\sigma\tau x) = s_\tau + t_\sigma - u_x\, s_\tau t_\sigma.$$

Thus by composing foldings I and II one obtains foldings III and IV, and this holds both when applying folding II to folding I, i.e. to the factors $(stu)u_\sigma u_\tau$, and when applying folding I to folding II, to the factors $(\sigma\tau x)s_x t_x$.

By a *form sequence*[15] we mean an initial form together with all forms that have arisen from it by folding in itself, and we denote the form sequence by the initial form. A "higher form" is to mean here any form with more folding in itself, while among the equally entitled foldings s_τ and t_σ one has to be normalized as higher. According to the law of formation of the form sequence it follows that:

1) Each form of the form sequence is linear in the symbols of the initial form.

2) The form sequence is uniquely determined for initial forms with two rows each of contragredient symbols; for initial forms with more rows of symbols it is to be uniquely normalized by distinguishing special foldings among those connected by the identity theorem.

By Theorem I, a form sequence $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$ $(m > n;\ \mu > \nu)$ can be arranged according to the following rectangular scheme:

$$\begin{array}{c|c|c|c} s_x^m t_x^n u_\sigma^\mu u_\tau^\nu & (stu) & (stu)^2 & \cdots \\ (\sigma\tau x) & t_\sigma,\ s_\tau & t_\sigma(stu),\ s_\tau(stu) & \cdots \\ (\sigma\tau x)^2 & t_\sigma(\sigma\tau x),\ s_\tau(\sigma\tau x) & t_\sigma^2,\ t_\sigma s_\tau,\ s_\tau^2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \\ (\sigma\tau x)^\nu & t_\sigma(\sigma\tau x)^{\nu-1},\ s_\tau(\sigma\tau x)^{\nu-1} & t_\sigma^2(\sigma\tau x)^{\nu-2},\ t_\sigma s_\tau(\sigma\tau x)^{\nu-2},\ s_\tau^2(\sigma\tau x)^{\nu-2} & \cdots \\ \vdots & t_\sigma(\sigma\tau x)^\nu & t_\sigma^2(\sigma\tau x)^{\nu-1},\ t_\sigma s_\tau(\sigma\tau x)^{\nu-1} & \cdots \end{array}$$

and the correspondingly continued lower part[16]

$$\begin{array}{c|c|c} (stu)^n & \cdots & \cdots \\ t_\sigma(stu)^{n-1},\ s_\tau(stu)^{n-1} & s_\tau(stu)^n & \cdots \\ t_\sigma^2(stu)^{n-2},\ t_\sigma s_\tau(stu)^{n-2},\ s_\tau^2(stu)^{n-2} & t_\sigma s_\tau(stu)^{n-1},\ s_\tau^2(stu)^{n-1} & s_\tau^2(stu)^n. \end{array}$$

Here one obtains, when one moves

1) one column to the right, all forms arising by folding I from the adjacent forms,

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[14] Cf. Gordan, *Math. Annalen* vol. V, p. 104.

[15] Cf. Clebsch, "Über eine Fundamentalaufgabe of the Invariantentheorie" (Abh. of the Gött. Ges. d. Wiss. vol. XVII): § 17 and end of § 18. The "form sequence" differs from the "properly reduced equivalent system" introduced there by the principle of ordering according to higher forms.

[16] Here, and similarly in what follows, the lower part marked with an asterisk is to be set at the correspondingly marked place at the upper right of the scheme.

2) one row downward, all forms arising by folding II from the forms above,

and hence all forms arising by folding.

An arbitrary form of the total form sequence, $t_\sigma^\kappa s_\tau^\lambda (stu)^\mu$ or $t_\sigma^\kappa s_\tau^\lambda (\sigma\tau x)^\nu$, forms the beginning of a new form sequence, consisting of all those forms of the rectangular scheme with $t_\sigma^\kappa s_\tau^\lambda (stu)^\mu$ or $t_\sigma^\kappa s_\tau^\lambda (\sigma\tau x)^\nu$ as left upper corner, which have the factor $t_\sigma^\kappa s_\tau^\lambda$.

Supplementary remark. Theorem I has an exception in the case where the numbers n and μ (respectively m and ν) become equal to zero, so that foldings I, II, and IV no longer exist, while folding III (respectively IV) still does. In this case we designate as the form sequence the diagonal member of the scheme:

$$\begin{array}{ccccccc} s_x^m u_\tau^\nu & & & & & & t_x^n u_\sigma^\mu \\ & s_\tau & & & t_\sigma & & \\ & s_\tau^2 & & & t_\sigma^2 & & \\ & & \ddots & & & & \\ & & & s_\tau^\nu & & t_\sigma^\mu . & \end{array}$$

In accordance with the absence of foldings I and II, the series expansion according to polars of the form sequence in this case always reduces to the initial member; however, the theorems on reducers remain valid.

§ 2. Series expansions according to polars of the form sequence.

For forms with n variables two kinds of series expansions have been set up explicitly[17]:

1) for forms with one row each of contragredient variables, $s_x^m u_\tau^\nu$, a series progressing by powers of u_x, whose coefficients are "normal forms";

2) for forms with two rows of cogredient variables, $s_x^m t_x^n$, an expansion according to polars of the elementary covariants (polars of the form sequence $s_x^m t_x^n$), which in the ternary case reads as follows[18]:

$$s_x^m t_x^{n-\lambda} t_y^\lambda = \sum_\varrho \frac{\binom{m}{\varrho}\binom{\lambda}{\varrho}}{\binom{m+n-\varrho+1}{\varrho}} \left[(st\widehat{xy})^\varrho s_x^{m-\varrho} t_x^{n-\varrho}\right]_{y^{\lambda-\varrho}}. \tag{I}$$

We combine the two expansions by setting up, for forms with two rows each of contragredient variables,

$$s_x^m t_x^{n-\lambda} t_y^\lambda u_\sigma^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa,$$

expansions according to powers of u_x, whose coefficients are composite polars of the form sequence $s_x^m t_x^n u_\sigma^\mu u_\tau^\nu$, taken with respect to the variables x and u.

It is enough to set up the series for two contragredient rows of symbols (respectively variables), since, by combining each two rows of symbols (variables) into a single new row, forms with more rows of symbols (variables) can be reduced to this case.

In order to ensure that all forms of the form sequence contain only two rows of symbols, respectively variables, we specialize:

$$\text{a) } \sigma = \widehat{st}, \quad \text{a') } y = \widehat{uv},$$
$$\text{b) } s = \widehat{\sigma\tau}, \quad \text{b') } v = \widehat{xy},$$

and carry out the expansion for the specialization a), a').

By direct transfer of expansion I one obtains composite polars for forms $(stu)^\mu s_x^m t_x^{n-\lambda} t_y^\lambda$, whereas for forms $(stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa s_x^m t_x^{n-\lambda} t_y^\lambda$, instead of composite polars, terms of such a kind arise whose evaluation makes necessary the introduction of ever new auxiliary variables that are to be set equal only at the end, thereby complicating the calculation unnecessarily.

We therefore take an indirect route, by representing the composite polars of all forms of the form sequence, that is, expressions

$$\left[s_\tau^\varrho (stu)^{\mu-\varrho+r}\right]_{y^\lambda}^{v^\kappa},$$

[17]Cf. Gordan, "Über Kombinanten," §§ 2 and 5 (*Math. Ann.* vol. V).

[18]Cf. Study, *Methoden to the Theorie of the ternären Formen* (Teubner 1889) II. § 7. In distinction to the derivation there, ours gives the method for the rapid computational determination of the numerical coefficients $C_{pr\varrho}$. The Clebsch–Study expansion, upon specialization a), a'), would read

$$(stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa s_x^m t_x^{n-\lambda} (tuv)^\lambda = \sum_{p,r,\varrho} C_{pr\varrho} \left[s_\tau^\varrho (stu)^{\mu+r-\varrho}\right]_{\widehat{uv}^{\lambda-r+\varrho}\, v^{\nu-\varrho},\, u^p,\, (vw=\widehat{xw})},$$

and therefore does not permit the simplification that occurs already in the course of our calculation, once one knows that all terms with the factor u_x are to be omitted.

by the initial members of these polars, and by inversion of the resulting recursive system of equations obtaining the desired expansion according to polars.

By the transfer principle, for the λ-th polar of a form $s_x^m t_x^n : [s_x^m t_x^n]_{y^\lambda}$ $(\lambda \leq n)$ the following expansion holds (the case $\lambda > n$ is reduced to the first by the expansion of $[s_y^m t_y^n]_{x^{m+n-\lambda}}$)[19]:

$$[s_x^m t_x^n]_{y^\lambda} = \sum_r (-1)^r \frac{\binom{m}{r}\binom{\lambda}{r}}{\binom{m+n}{r}} (st\widehat{xy})^r s_x^{m-r} t_x^{n-\lambda} t_y^{\lambda-r},$$

and correspondingly for the κ-th polar of $u_\sigma^\mu u_\tau^\nu : [u_\sigma^\mu u_\tau^\nu]_{v^\kappa}$

$$[u_\sigma^\mu u_\tau^\nu]_{v^\kappa} = \sum_\varrho (-1)^\varrho \frac{\binom{\mu}{\varrho}\binom{\kappa}{\varrho}}{\binom{\mu+\nu}{\varrho}} (\sigma\tau\widehat{uv})^\varrho u_\sigma^{\mu-\varrho} u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho}.$$

By multiplication it follows, with introduction of the specialization a) and a'),

$$[s_x^m t_x^n (stu)^\mu u_\tau^\nu]^{v^\kappa}_{\widehat{uv}^\lambda} = \sum_{r,\varrho} c_{r\varrho}\, (st\widehat{uv})^r (st\widehat{uv})^\varrho s_x^{m-r} t_x^{n-\lambda} (tuv)^{\lambda-r} (stu)^{\mu-\varrho} u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho},$$

and from this, by expansion according to powers of u_x with the first member distinguished ($\sum'_{r,\varrho}$ means that the member $r=0$, $\varrho=0$ is to be omitted) and taking account of (a) on p. 26,

$$\begin{aligned}
&s_x^m t_x^{n-\lambda}(tuv)^\lambda (stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa \\
&\quad= [s_x^m t_x^n (stu)^\mu u_\tau^\nu]^{v^\kappa}_{\widehat{uv}^\lambda} \\
&\qquad - \sideset{}{'}\sum_{r,\varrho} c_{r\varrho}\, s_\tau^\varrho (stu)^{\mu-\varrho+r} u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho} s_x^{m-r} t_x^{n-\lambda} (tuv)^{\lambda-r+\varrho} v_x^r \\
&\qquad + u_x \sum_\varrho \sum_{r=1}^{\lambda} c_{r\varrho}\, s_\tau^\varrho (stu)^{\mu-\varrho+r-1} (stv) u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho} s_x^{m-r} t_x^{n-\lambda} (tuv)^{\lambda-r+\varrho} v_x^{r-1} \\
&\qquad \vdots \\
&\qquad - (-1)^\lambda u_x^\lambda \sum_\varrho c_{\lambda\varrho}\, s_\tau^\varrho (stu)^{\mu-\varrho} (stv)^\lambda u_\tau^{\nu-\kappa} v_\tau^{\kappa-\varrho} s_x^{m-\lambda} t_x^{n-\lambda} (tuv)^\varrho,
\end{aligned} \tag{II}$$

where

$$c_{r\varrho} = (-1)^{r+\varrho} \cdot \frac{\binom{m}{r}\binom{\lambda}{r}}{\binom{m+n}{r}} \cdot \frac{\binom{\mu}{\varrho}\binom{\kappa}{\varrho}}{\binom{\mu+\nu}{\varrho}} \qquad \begin{matrix}\text{for } \lambda \leq n, \\ \kappa \leq \nu.\end{matrix}$$

and correspondingly for the remaining combinations of the numbers $\lambda, n; \kappa, \nu$. Thus the expression $s_x^m t_x^{n-\lambda}(tuv)^\lambda (stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa$ is reduced to a composite polar, and, by applying the identity

$$(stv)u_\tau = (stu)v_\tau - s_\tau(tuv),$$

to a sum of analogously formed expressions corresponding to higher forms of the form sequence.

By iteration one arrives at the expansion

$$s_x^m t_x^{n-\lambda}(tuv)^\lambda (stu)^\mu u_\tau^{\nu-\kappa} v_\tau^\kappa = \sum_{p,r,\varrho} u_x^p\, C_{pr\varrho} \cdot [s_\tau^\varrho (stu)^{\mu-\varrho+r}]_{\widehat{uv}^{\lambda-r+\varrho} v^{\kappa-\varrho+p},\, v_x^{r-p}}. \tag{III}$$

The numerical coefficients $C_{pr\varrho}$ are computed uniquely and recursively from the coefficients $c_{r\varrho}$, $c'_{r\varrho}$ of the system of equations arising by iteration of (II) (cf. the example on p. 42). Analogous expansions arise for the other specializations.

§ 3. Reduction theorems.

The known theorems on the reduction of forms and form systems shall now be brought into connection with §§ 1 and 2, for which some definitions taken over from the theory of binary forms are necessary.

a) We call a system of forms a *relatively complete system* modulo a prescribed sequence of forms if it has the property that all formations arising by folding an arbitrary product of these forms can be expressed as

[19] Gordan, *The Resultante binärer Formen* (Chap. I, § 5). Rend. Circ. Mat. di Palermo XXII. 1906.

integral rational functions of the forms of the system, up to an additive term consisting of expressions that have arisen by folding the system forms with the system of the modulus[20].

b) We call a form *reducible* when it can be expressed by forms having invariants as factors or by "higher forms," that is, by higher forms of the total form sequence to which the form belongs, or by forms containing the symbols of the modulus in higher order.

The theorems on reducers[21] can now be stated in their most general form as follows:

Definition: *A reducer is a reducible form sequence.*

Theorem II. If the initial form of a form sequence is reducible because one of its members has arisen by folding with a reducer (has a reducer as factor), and if the final form of the form sequence has arisen from exactly this member by folding, then the total form sequence is reducible[22].

Proof: The form sequence arises from the initial member by folding in itself, hence by higher folding with the reducer on the one hand, and by folding the reducer in itself on the other hand. In both cases, by § 2, all forms of the form sequence can be represented as polars (transvections) over the form sequence of the reducer.

The inference from the initial form to the total form sequence no longer holds for the remaining reduction methods. These are:

1) so-called "double reduction."

By this we mean the reduction, in two different ways, of an expression that has two mutually independent reducers as factors, whereby relations arise between the higher forms to which one has reduced.

By systematic application of "double reduction" one must, on the one hand, obtain all relations[23]; on the other hand, if "double reduction" applied to different expressions supplies no new relations, one has both a criterion for the independence of the higher forms and a check on the calculation.

2) *Folding with decomposable forms.*

By "decomposable forms" are to be understood – with a slight generalization of the usual concept – forms that can be expressed by products of forms of lower degree and by "higher" forms. Products of forms pass into products under one folding; likewise products of nonlinear forms under twofold folding in the specializations a') and b') (§ 2), according to the product theorem:

$$a_y b_y a_x b_x = \frac{1}{2} a_y^2 b_x^2 + \frac{1}{2} b_y^2 a_x^2 - \frac{1}{2} (ab\widehat{yx})^2.$$

First polars (transvections) and certain second polars of decomposable forms are therefore again decomposable forms. It follows that:

A member of a onefold (respectively twofold) transvection over a decomposable form, arising by onefold (respectively twofold) folding with a decomposable form, itself becomes a decomposable form:

a) if the next-higher forms in the form sequence of the decomposable form decompose or become reducible, or also if, under onefold folding, the specializations $y = \widehat{uv}$ or $v = \widehat{xy}$ occur;

b) if the remaining onefold foldings lead to higher forms (cf. § 12, B. H_k and $H_k(k\eta x)$).

Such a member of a transvection can also decompose:

c) if the remaining onefold foldings lead to lower forms that decompose independently of the folding with the decomposable form, that is, that can be expressed by products and by the form to be reduced, although in each case one must first verify by calculation that the numerical coefficient of the member concerned does not vanish identically. This is nothing other than "double reduction" of the lower form (cf. § 23, C. $H_q(\theta H u)(\theta s u)$).

Decomposable forms arise by all onefold foldings with functional determinants[24], and moreover by certain higher foldings. One has

$$(st\widehat{yx})s_y = \frac{1}{2}\, t\, s_y^2 - \frac{1}{2}\, s\, t_y^2 + \frac{1}{2}(st\widehat{yx})^2,$$

[20]Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie.* Vol. II. p. 227.

[21]Gordan, *Math. Annalen* vol. XVII, p. 222.

[22]The simplification that results from Theorem II can be seen in the following example. For the special form $f = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1$, $a_y^2 a_x^2 u_y^2$ is a reducer. It follows by Theorem II that the form sequence

$$\theta_\nu (\vartheta \nu x) \theta_x^3 u_\vartheta u_\nu^2 = a_\nu^2 (abu) - a_\nu b_\nu (aba),$$

is reduced, since $\theta_\nu^4 = a_\nu^2 b_\nu^2 (abu)^2$ has arisen from the member $a_\nu^2 (abu)$ by folding. In Gordan, loc. cit., p. 231, by contrast, the reduction of the forms

$$\theta_\nu (\vartheta \nu x), \quad \theta_\nu (\vartheta \nu x)^2, \quad \theta_\nu^2, \quad \theta_\nu^2 (\vartheta \nu x), \quad \theta_\nu^2 (\vartheta \nu x)^2$$

is carried out individually.

[23]Thus, for example, the following were foand by "double reduction": the reduction of the form a_η^4 (§ 10), the only form included superfluously in Maisano's system; and also the relations mentioned in the note to the Introduction between the three covariants and the three invariants.

[24]cf. Gordan, loc. cit. § 3.

$$(st\widehat{yx})^2 s_y^2 = \frac{1}{3}\, t\, s_y^3 - \frac{1}{3}\, s\, t_y^3 + s_y (st\widehat{yx})^2 - \frac{1}{3}(st\widehat{yx})^3.$$

Analogous formulas hold for the dualistic forms

$$(\sigma\tau\widehat{\nu u})v_\sigma \quad \text{and} \quad (\sigma\tau\widehat{\nu u})v_\sigma^2.$$

From the theorems of this section the following rule follows for setting up all irreducible forms that arise from the folding of S with T:

Beginning with the lowest foldings, proceed along any previously fixed path (for example row by row, or symmetrically with respect to the diagonal member) in the formation of the form sequence ST until one reaches a form that has a reducer as factor. The form sequence defined by this form is to be omitted as soon as the final form satisfies the condition stated in Theorem II. After all reducible form sequences have been eliminated, all remaining reducible forms, and likewise all decomposable forms, are to be removed by applying double reduction. Places in the total form sequence that are covered twice by independently reducible form sequences give rise to the reduction of higher forms (cf. § 11, D).

Theorem III. *The forms taken over from the binary domain, that is, those forms which arise from the system forms of the corresponding binary form by bordering according to Clebsch's transfer principle, form a relatively complete system modulo* (abc).

Proof. To a binary relation asserting that the forms $Q'_1, \ldots, Q'_n$ form a complete system of a binary ground form,

$$Q' - F(Q') = 0 = \sum_\lambda \{(ab)c_x + (bc)a_x + (ca)b_x\}\varphi^*_\lambda,$$

there corresponds, by bordering, the ternary relation

$$Q - F(Q_i) = \sum_\lambda u_x \cdot (abc)\varphi_\lambda.$$

This is precisely the defining equation for a relatively complete system modulo (abc). For the biquadratic form, the system of the forms taken over consists of the following forms:

$$f = a_x^4, \qquad \theta = \theta_x^4 u_\vartheta^2 = (abu)^2 a_x^2 b_x^2, \qquad K = K_x^6 u_k^3 = (a\theta u) u_\vartheta^2 a_x^3 \theta_x^3,$$
$$j = u_j^6 = (a\theta u)^4 u_\vartheta^2, \qquad \nu = u_\nu^4 = (abu)^4,$$

among which the first four form a relatively complete system modulo $((abc), \nu)$.

§ 5. Reduction of the modulus (abc) to the moduli Δ and ν.

The modulus (abc), given only by a bracket factor, is to be reduced, in agreement with the definition (§ 3, a), to forms of the system. In other words, the form sequence (abc) is to be normalized uniquely (cf. § 1).

$$a_s^2 a_x u_x = \frac{1}{3} a_s^3 u_x = \frac{1}{3} S \cdot u_x. \tag{2}$$

$$\left.\begin{aligned} (a\Delta u)^2 &= a_s - \frac{S}{6} u_x^2, \\ (a\Delta u)^3 &= 0. \end{aligned}\right\} \tag{3}$$

$$\left.\begin{aligned} \theta(s) &= (ss, x)^2 = 2a_t + \frac{1}{3} S \cdot \theta - \frac{1}{3} T u_x^2, \\ \theta(t) &= (tt, x)^2 = -\frac{1}{3} S \cdot a_t + \frac{2}{3} T \cdot a_s + \frac{1}{18} S^2 \cdot \theta - \frac{1}{18} u_x^2 \cdot S \cdot T. \end{aligned}\right\} \tag{4}$$

Sequence of moduli: (abc); Δ; s; t (the system of the modulus t consists of the single form t).[25]

It will be shown that the forms

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2, \qquad \nu = (abu)^4$$

suffice as moduli, that is, that the forms

$$\Delta = (abc)^2 a_x^2 b_x^2 c_x^2, \qquad a_\nu = (abc)(bcu)^3 a_x^3,$$
$$a_\nu^2 = (abc)^2 (bcu)^2 u_x^2, \qquad i = a_\nu^4 = (abc)^4$$

define the form sequence (abc). In the form sequence a_ν, the form a_ν^3 is missing according to the identity

$$a_\nu^3 = (abc)^3 a_x (bcu) = \frac{1}{3} u_x \cdot (abc)^4 = \frac{1}{3} i \cdot u_x.$$

For reducing a modulus to a higher one, according to the definition, we have to reduce the most general foldings, or all special foldings coming into consideration, with the modulus to foldings with the higher modulus.

In the present case the most general foldings arise from the expressions

$$(abc) a_x^3 b_y^3 c_z^3, \qquad (abc) a_x^2 b_y^2 c_z^2 a_z b_x c_x$$

(taken over from the cubic forms),

$$(abc) a_x b_y c_z a_x^2 b_x^2 c_x^2$$

[25] Gordan–Kerschensteiner, loc. cit., p. 134.

(taken over from the quadratic forms), and from the polarisation of these expressions.

For the calculation of these expressions by the polars of Δ and of the form sequence a_ν, respectively of their initial terms, we take the indirect route, as in § 2. We expand the polar terms

$$(abc)^2a_x^2b_y^2c_z^2, \qquad a_\nu(\nu xy)(\nu yz)(\nu zx)a_xa_ya_z = (abc)(bc\widehat{x}u)(bc\widehat{y}z)(bc\widehat{z}x)a_xa_ya_z \quad \text{etc.}$$

as linear aggregates of expressions with the symbolic factor (abc) and obtain, by inverting the system of equations, the desired relations, as well as a relation between the polars taken according to different combinations of the variables. The identity theorem and the product theorem for determinants are used in the evaluation.

The system of equations is, for $(abc)a_x^3b_y^3c_z^3$:

$$1)\quad \sum_3 a_\nu(\nu yz)^3a_x^3 = 6(abc)a_x^3b_y^3c_z^3 - 6\sum_3(abc)a_x^3b_y^2b_zc_z^2c_y^*,$$

$$\begin{aligned}2)\quad 3(abc)^2a_x^2b_y^2c_z^2\cdot(xyz) - \frac{1}{2}\sum_3 a_\nu^2(\nu yz)^2\nu_x^2(xyz) &= 2\sum_3(abc)a_x^3b_y^2b_zc_z^2c_y\\ &\quad - 4\sum_3(abc)a_xa_ya_zb_y^2b_zc_z^2c_x,\end{aligned}$$

$$3)\quad a_\nu(\nu xy)(\nu yz)(\nu zx)a_xa_ya_z = 2\sum_3(abc)a_xa_ya_zb_y^2b_zc_z^2c_x,$$

$$4)\quad \sum_3 a_\nu(\nu yz)^3u_x^3 - \frac{3}{2}\sum_3 a_\nu^2(\nu yz)^2\nu_x^2(xyz) + \frac{1}{6}i\cdot(xyz)^3 = 6\sum_3(abc)a_xa_ya_zb_y^2b_zc_z^2c_x.$$

It follows for

$$(abc)a_x^3b_y^3c_z^3,$$

and by analogous calculation for

$$(abc)a_x^2b_y^2c_z^2a_zb_xc_x, \qquad (abc)a_xb_yc_za_z^2b_x^2c_x^2:$$

$$\begin{aligned}(abc)a_x^3b_y^3c_z^3 &= \frac{3}{2}(abc)^2a_x^2b_y^2c_z^2(xyz) + \frac{3}{2}a_\nu(\nu xy)(\nu yz)(\nu zx)a_xa_ya_z\\ &\quad - \frac{1}{12}i\cdot(xyz)^3,\\ \text{(IV.)}\quad (abc)a_x^2b_y^2c_z^2a_zb_xc_x &= \frac{2}{3}(abc)^2a_x^2b_y^2c_z^2c_x\cdot(xyz)\\ &\quad + \frac{1}{3}a_\nu(\nu xy)(\nu yz)(\nu zx)a_x^3 - \frac{1}{6}a_\nu^2(\nu xy)(\nu zx)a_x^2\cdot(xyz),\\ (abc)a_xb_yc_za_z^2b_x^2c_x^2 &= \frac{1}{6}(abc)^2a_x^2b_z^2c_z^2\cdot(xyz).\end{aligned}$$

We have therefore obtained a relatively complete system modulo (Δ, ν), consisting of the four forms f, θ, K, j. It follows from this that all transvections of f over θ not taken over from the binary domain, as well as the transvection $(a\theta u)^2$ reducible in the binary case to $(ab)^4$, can be expressed by the symbols Δ and ν.[26]

We obtain the reduction formulas fundamental for what follows by specializing expansion IV, or more shortly by direct calculation (interchanging the symbols), for $a_\vartheta(a\theta u)^2$, $a_\vartheta^2((a\theta u)^2$, $(a\theta u)^2$, according to the following ansatz:

$$\begin{aligned}a_\vartheta(a\theta u)^2 &= (abc)(bcu)(abu)^2 - \frac{1}{3}(abc)(bcu)\{(bcu)a_x - u_x(abc)\}^2,\\ a_\vartheta^2(a\theta u)^2 &= (abc)^2(abu)^2 - \frac{1}{3}(abc)^2\{(bcu)a_x - u_x(abc)\}^2,\end{aligned}$$

for $((a\theta u)^2)^*$:

$$\begin{aligned}(abu)a_y^3b_x^3 &= \frac{3}{2}u_\vartheta(\vartheta yx)\theta_y^2 + \frac{1}{4}(\nu yx)^3,\\ ((abu)(acu))b_x^3 &= \frac{3}{2}u_\vartheta(\vartheta\widehat{a}ux)(\theta au)^2 + \frac{1}{4}(\nu\widehat{a}ux)^3.\end{aligned}$$

---

[26] $\sum_3$ refers to the sum of three terms obtained from the initial term by cyclic interchange of the variables. The equations also hold individually for each term of the sum.

$$\left.\begin{aligned}
(a\theta u)^2 &= \frac{1}{6}\nu\cdot f - \frac{2}{3}u_x\cdot a_\nu + \frac{2}{3}u_x^2\cdot a_\nu^2 - \frac{i}{18}u_x^4,\\
a_\vartheta &= \frac{1}{3}u_x\cdot\Delta,\\
a_\vartheta(a\theta u) &= 0,\\
a_\vartheta^2 &= \Delta,\\
(a\theta u)^3 &= 0,\\
a_\vartheta(a\theta u)^2 &= -\frac{5}{6}a_\nu + \frac{7}{6}u_x\cdot a_\nu^2 - \frac{i}{9}u_x^3,\\
a_\vartheta^2(a\theta u) &= 0,\\
(a\theta u)^4 &= j,\\
a_\vartheta(a\theta u)^3 &= 0,\\
a_\vartheta^2(a\theta u)^2 &= \frac{2}{3}a_\nu^2 - \frac{i}{9}u_x^2.
\end{aligned}\right\} \tag{1}$$

We add the formula, likewise obtained from the reduction of the modulus (abc):

$$a_\nu^3 a_x u_\nu = \frac{1}{3}i\cdot u_x. \tag{2}$$

From formulas (1.) and (2.) and from the formulas formed analogously for the ground form ν, all later reduction formulas are derived by polarisation, except for the relations for the decomposed forms. From formulas (1.) we see:

The form sequence leads:

$$\begin{array}{rcl}
a_\vartheta(a\theta u)^2 & \text{to} & \text{symbols } \nu \text{ alone,}\\
a_\vartheta & \text{to} & \text{symbols } \Delta \text{ and } \nu,\\
(a\theta u)^2 & \text{to} & \text{symbols } j, \Delta \text{ and } \nu,\\
(a\theta u)^2 \text{ under folding I} & \text{to} & \text{symbols } j \text{ and } \nu,\\
(a\theta u)^2 \text{ under folding II} & \text{to} & \text{symbols } \Delta \text{ and } \nu.
\end{array}$$

We can therefore replace:

1. modulo (Δ,ν): foldings with j are replaced by corresponding foldings with $(a\theta u)^2$ or $(a\theta u)^3$;
2. modulo (ν): foldings with Δ are replaced by corresponding foldings with a_ϑ or $a_\vartheta(a\theta u)$, or also by special foldings with $(a\theta u)^2$ (which lead only to a single folding I).

In particular:

$$(a\theta u)^2\theta_y^2\theta_x^2 = \frac{1}{3}(jyx)^2 \pmod{\nu}, \qquad (a\theta u)^2 u_y\theta_y = -\frac{1}{6}(jyx)^2 \pmod{\nu},$$

$$(a\theta u)^2\nu_y^2 = \frac{1}{3}(\Delta u\nu)^2 \pmod{\nu}, \qquad (a\theta u)(a\theta\nu)u_\vartheta\nu_\vartheta = -\frac{1}{6}(\Delta u\nu)^2 \pmod{\nu},$$

$$(a\theta u)^2 u_\nu^2\vartheta_\nu^2 = \frac{1}{3}(\Delta u\nu)^2\Delta_\nu \pmod{\nu}.$$

§ 6. Reduction of the modulus (Δ,ν) to the modulus (ν).

The reduction formulas of the preceding paragraph give us the means to set up the relatively complete system modulo ν directly; we shall see that to the system of the transferred forms one has only to add the forms Δ and $(a\Delta u) = N$ (analogously as for the cubic form, and indeed valid in general).

For the reduction of the modulus Δ we consider all special foldings coming into consideration, that is, the foldings of the relatively complete system modulo (Δ,ν) with the system of Δ, and begin with the forms lowest in the coefficients, the foldings of f with Δ.

For the reduction of the forms $(a\Delta u)^2$, $(a\Delta u)^3$, $(a\Delta u)^4$, we replace the symbols Δ by lower forms of the form sequence according to § 5; that is, we develop the expressions

$$a_\vartheta b_\vartheta(b\theta u)^2, \qquad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^2, \qquad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^3$$

1) by polars of the form sequence a_ϑ (symbols Δ and ν),

2) by polars of the form sequence $b_\vartheta(b\theta u)^2$ (symbols ν),

and obtain the reduction formulas (calculation below):

$$\begin{aligned} &\text{a)} \quad (a\Delta u)^2 = \frac{1}{2}(\vartheta\nu x)^2 - \frac{2}{5}f\cdot a_\nu^2 - \frac{4}{5}u_x\cdot\theta_\nu(\vartheta\nu x)^2 \\ &\qquad\qquad + u_x^2\left\{\frac{1}{6}i\cdot f - \frac{1}{40}S\right\}, \\ &\text{b)} \quad (a\Delta u)^3 = -\frac{3}{10}\theta_\nu(\vartheta\nu x), \\ &\text{c)} \quad (a\Delta u)^4 = \frac{21}{10}\theta_\nu^2 - \frac{3}{10}\Pi - \frac{6}{5}u_x\cdot\theta_\nu^3 + \frac{1}{10}u_x^2\cdot\theta. \end{aligned} \tag{3}$$

(The same results are also reached by the double reduction of the expressions $a_\vartheta^2(b\theta u)^2$, $a_\vartheta^2(b\theta u)^3$, $a_\vartheta^2((a\theta u)^2(b\theta u)^2$ and $a_\vartheta^2((a\theta u)(b\theta u))^3$ after elimination of a_ϑ^2.)

From formulas (3.) it follows that $(a\Delta u)^2$ is a reducer with respect to the modulus ν. Hence:

1) The reduction of the system of Δ: the form sequence $(\Delta\Delta' u)^2 = a_\vartheta^2((a\Delta u)^2$ has the reducer $(a\Delta u)^2$ as a factor.

2) The reduction of the system arising by folding with the modulus Δ:

The form sequence $(\theta\Delta u)^2$ has the reducer $(a\Delta u)^2$ as a factor.

For the forms $(\theta\Delta u)$ and Δ_ϑ one obtains:

$$(a\Delta u)^2(abu) = (\theta\Delta u) - u_x\cdot\Delta_\vartheta(\theta\Delta u),$$

$$(a\Delta u)(a\Delta b) = -\frac{1}{2}\Delta_\vartheta + \frac{1}{2}u_x\cdot\Delta_\vartheta^2.$$

From the reducer $(\theta\Delta u)$, however, follows the reduction of the system arising by folding with the modulus Δ.

The relatively complete system modulo ν therefore consists of the six forms:

$$f,\ \theta,\ K,\ j,\ \Delta,\ N.$$

As an example of the sequence expansion in § 2 we give the derivation of formula (3.)a in full; in later calculations the final formula III will be set down directly. (Polars of vanishing forms have already been omitted during the calculation; the first line gives the values inserted according to Expansion II, and the second line gives the final values found recursively, beginning with the last equation of the system.) It follows, according to Expansion II:

$$\begin{aligned} &1) \quad m=3, \quad n=4, \quad \lambda=2, \quad \mu=0, \quad \nu=\chi=1, \\ a_\vartheta b_\vartheta(b\theta u)^2 &= [a_\vartheta]_{b^2}b + \frac{6}{7}\{a_\vartheta b_\vartheta(a\theta u)(b\theta u) - u_x a_\vartheta b_\vartheta(a\theta b)(b\theta u)\} \\ &\quad - \frac{1}{7}\{a_\vartheta(a\theta u)^2 b_\vartheta - 2u_x a_\vartheta(a\theta u)(a\theta b)b_\vartheta + u_x^2 a_\vartheta(a\theta b)^2 b_\vartheta\} \\ &= \frac{2}{3}(a\Delta u)^2 + \frac{1}{5}[a_\vartheta(a\theta u)^2]b + \frac{1}{30}f[a_\vartheta^2(a\theta u)^2] - \frac{2}{5}u_x[a_\vartheta(a\theta u)^2]b^2 \\ &\quad - \frac{1}{15}u_x[a_\vartheta^2(a\theta u)^2]b + \frac{1}{5}u_x^2[a_\vartheta(a\theta u)^2]b^3 + \frac{1}{30}u_x^2[a_\vartheta^2(a\theta u)^2]b^2; \end{aligned}$$

$$\begin{aligned} &2) \quad m=2, \quad n=3, \quad \lambda=1, \quad \mu=1, \quad \nu=\chi=1, \\ a_\vartheta(a\theta u)b_\vartheta(b\theta u) &= \frac{2}{5}\{a_\vartheta(a\theta u)^2 b_\vartheta - u_x a_\vartheta(a\theta u)(a\theta b)b_\vartheta\} + \frac{1}{2}a_\vartheta^2(b\theta u)^2 \\ &\quad - \frac{1}{5}\{a_\vartheta^2(b\theta u)(a\theta u) - u_x a_\vartheta^2(a\theta b)(b\theta u)\} \\ &= \frac{1}{2}(a\Delta u)^2 + \frac{2}{5}[a_\vartheta(a\theta u)^2]b + \frac{1}{15}[a_\vartheta^2(a\theta u)^2]f \\ &\quad - \frac{2}{5}u_x[a_\vartheta(a\theta u)^2]b^2 - \frac{1}{10}u_x[a_\vartheta^2(a\theta u)^2]b + \frac{1}{30}u_x^2[a_\vartheta^2(a\theta u)^2]b^2; \end{aligned}$$

$$3)\quad m=2,\quad n=3,\quad \lambda=1,\quad \mu=1,\quad \nu=1,\quad \chi=2,$$

$$\begin{aligned}a_\vartheta(a\theta b)b_\vartheta(b\theta u) &= \frac{2}{5}\{a_\vartheta(a\theta b)(a\theta u)b_\vartheta - u_x a_\vartheta(a\theta b)^2 b_\vartheta\}\\ &= \frac{2}{5}[a_\vartheta(a\theta u)^2]b^2 + \frac{1}{30}[a_\vartheta^2(a\theta u)^2]b - \frac{2}{5}u_x[a_\vartheta(a\theta u)^2]b^3 - \frac{1}{30}u_x[a_\vartheta^2(a\theta u)^2]b^2;\end{aligned}$$

$$4)\quad m=1,\quad n=2,\quad \lambda=0,\quad \mu=2,\quad \nu=\chi=1,$$

$$\begin{aligned}a_\vartheta(a\theta u)^2 b_\vartheta &= [a_\vartheta(a\theta u)^2]b + \frac{2}{3}a_\vartheta^2(a\theta u)(b\theta u)\\ &= [a_\vartheta(a\theta u)^2]b + \frac{1}{6}[a_\vartheta^2(a\theta u)^2]f - \frac{1}{6}u_x[a_\vartheta^2(a\theta u)^2]b;\end{aligned}$$

$$5)\quad m=1,\quad n=2,\quad \lambda=0,\quad \mu=2,\quad \nu=1,\quad \chi=2,$$

$$\begin{aligned}a_\vartheta(a\theta u)(a\theta b)b_\vartheta &= [a_\vartheta(a\theta u)^2]b^2 + \frac{1}{3}a_\vartheta^2(a\theta b)(b\theta u)\\ &= [a_\vartheta(a\theta u)^2]b^2 + \frac{1}{12}[a_\vartheta^2(a\theta u)^2]b - \frac{1}{12}u_x[a_\vartheta^2(a\theta u)^2]b^2;\end{aligned}$$

$$6)\quad m=1,\quad n=2,\quad \lambda=0,\quad \mu=2,\quad \nu=1,\quad \chi=3,$$

$$a_\vartheta(a\theta b)^2 b_\vartheta = [a_\vartheta(a\theta u)^2]b^3;$$

$$7)\quad m=2,\quad n=4,\quad \lambda=2,\quad \mu=\nu=\chi=0,\quad \text{(according to Expansion I)},$$

$$a_\vartheta^2(b\theta u)^2 = [a_\vartheta^2]_{ub^2} + \frac{1}{10}\{[a_\vartheta^2(a\theta u)^2]f - 2u_x[a_\vartheta^2(a\theta u)^2]b + u_x^2[a_\vartheta^2(a\theta u)^2]b^2\};$$

$$8)\quad m=1,\quad n=3,\quad \lambda=1,\quad \mu=1,\quad \nu=\chi=0,\quad \text{(Expansion I)},$$

$$a_\vartheta^2(a\theta u)(b\theta u) = \frac{1}{4}[a_\vartheta^2(a\theta u)^2]f - \frac{1}{4}u_x[a_\vartheta^2(a\theta u)^2]b;$$

$$9)\quad m=1,\quad n=3,\quad \lambda=1,\quad \mu=\chi=1,\quad \nu=0,\quad \text{(Expansion I)},$$

$$a_\vartheta^2(a\theta b)(b\theta u) = \frac{1}{4}[a_\vartheta^2(a\theta u)^2]b - \frac{1}{4}u_x[a_\vartheta^2(a\theta u)^2]b^2.$$

From 1) and 4) it follows:

$$\begin{aligned}(a\Delta u)^2 &= \frac{6}{5}[a_\vartheta(a\theta u)^2]b + \frac{1}{5}f[a_\vartheta^2(a\theta u)^2] + \frac{3}{5}u_x[a_\vartheta(a\theta u)^2]b^2 - \frac{3}{20}u_x[a_\vartheta^2(a\theta u)^2]b\\ &\quad - \frac{3}{10}u_x^2[a_\vartheta(a\theta u)^2]b^3 - \frac{1}{20}u_x^2[a_\vartheta^2(a\theta u)^2]b^2,\\ (a\Delta u)^2 &= -a_\nu b_\nu + \frac{3}{5}fa_\nu^2 + \frac{4}{5}u_x a_\nu^2 b_\nu - \frac{3}{20}u_x^2 a_\nu^2 b_\nu^2 - \frac{1}{12}u_x^2 if.\end{aligned}$$

(For conversion into the symbols θ cf. § 8 and § 9.) Formula (3.)a can be computed still more briefly by direct transfer of Expansion I, introducing the auxiliary variable $w = x\widehat{u}b$ whereas already for formulas (3.)b and (3.)c the path taken here becomes the simpler one; for (3.)b one knows in advance, by § 9, that all terms with factor u_x are to be omitted. For the calculation of (3.)b and (3.)c one obtains:

$$\begin{aligned}&(3.)\text{b}\quad 1)\quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^2 = \frac{1}{2}(a\Delta u)^3 + \frac{4}{5}[a_\vartheta(a\theta u)^2]_{ub}b + \frac{7}{360}[a_\vartheta^2(a\theta u)^2]_{ub^2},\\ &\qquad\quad\ \ 2)\quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^2 = [a_\vartheta(a\theta u)^2]_{ub}b + \frac{13}{36}[a_\vartheta^2(a\theta u)^2]_{ub^2};\end{aligned}$$

$$\begin{aligned}&(3.)\text{c}\quad 1)\quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^3 = \frac{1}{2}(a\Delta u)^4 + \frac{6}{5}[a_\vartheta(a\theta u)^2]_{ub^2}b + \frac{53}{120}[a_\vartheta^2(a\theta u)^2]_{ub^3}\\ &\qquad\qquad - \frac{6}{5}u_x[a_\vartheta(a\theta u)^2]_{ub^3}b^2 - \frac{3}{5}u_x[a_\vartheta^2(a\theta u)^2]_{ub^2}b + \frac{19}{120}u_x^2[a_\vartheta^2(a\theta u)^2]_{ub^3}b^2,\\ &\qquad\quad\ \ 2)\quad a_\vartheta(a\theta u)b_\vartheta(b\theta u)^3 = \frac{3}{4}[a_\vartheta^2(a\theta u)^2]_{ub^2}.\end{aligned}$$

Supplementary remark: the transvectants of f over j that are reducible according to § 5 are computed analogously. One obtains

$$\begin{aligned}
g_\nu &= -\theta_\nu + u_x\left(\frac{9}{2}\theta_\nu^2 - \frac{1}{2}H\right) - 3u_x^2\theta_\nu^3 + \frac{1}{2}u_x^3\theta, \\
g_\nu^2 &= \frac{21}{10}\theta_\nu^2 - \frac{3}{10}H - 2u_x\theta_\nu^3 + \frac{1}{2}u_x^2\theta, \\
g_\nu^3 &= -\frac{7}{10}\theta_\nu^3 + \frac{1}{2}u_x^2\theta, \qquad g_\nu^4 = \frac{3}{5}\theta.
\end{aligned} \tag{4}$$

$$g_\nu^3 = -\frac{7}{10}\theta_\nu^3 + \frac{1}{2}u_x^2\theta, \qquad g_\nu^4 = \frac{3}{5}\theta. \tag{4}$$

From (3.) and (4.) it follows, by transfer from the binary domain, that

$$(\theta\theta' u)^2 = \frac{1}{3}\, j \cdot f \quad (\mathrm{mod}\ \nu).$$[27]

The remaining forms of the form sequence $(\theta\theta' u)^2$ and the form θ'_ν are reducible to symbols ν. We can therefore replace:

mod ν : foldings with $f \cdot j$ by the corresponding foldings with $(\theta\theta' u)^2$.

Furthermore,

$$(\vartheta\vartheta' x)^2 = \frac{4}{3} f \cdot \Delta, \qquad \theta_{g\nu}(\vartheta\vartheta' x) = -N.$$

Chapter III. The relatively complete system modulo (s, ϱ, t).

§ 7. Overview of the formation of the system.

In order to pass from the relatively complete system modulo ν to the relatively complete system with the next higher modulus, one has, by Definition § 3, to form the system of ν with respect to this higher modulus and to fold it with the forms of the relatively complete system modulo ν. But $\nu = u_\nu^4$, as a contravariant of the fourth degree, is dual to the form f. If therefore we regard ν as the ground form, we obtain a relatively complete system of six forms modulo $\nu(\nu) = (\nu\nu_1 x)^4 = s_x^4$.

Thus, for the formation of the relatively complete system modulo s (System III), we have to transvect

System I: $f,\ \theta,\ K,\ j,\ \Delta,\ N$ over

System II: $\nu,\ H = (\nu\nu_1 x)^2 u_\nu^2 u_{\nu_1}^2,\ L = (\nu\eta x) H_x^2 u_\nu^3 u_\eta^3,$
$g = (\nu\eta x)^4 H_x^2,\ \sigma = H_\eta^2 u_\nu^2 u_\eta^4,\ (\nu\sigma x).$

It will be shown (formula (13.)) that the form g is reducible, and hence that System II consists of only five forms.

In the course of the calculation the quadratic forms

$$\varrho = u_\varrho^2 = \theta_\nu^4 u_\vartheta^2, \qquad t = t_x^2 = a_\eta^4 H_x^2$$

are adjoined as moduli, so that one obtains a relatively complete system modulo (s, ϱ, t); and the modulus (ϱ, t) is to be regarded as a higher modulus than the modulus s.

The arrangement of the individual forms is to follow the order in the coefficients. We normalize the folding

$$\theta_\nu(K_\nu, \theta_\nu, \ldots)$$

as higher than the folding

$$H_y(H_y, L_y, \ldots).$$

Consequently, by § 1 and § 3b, the order of the individual forms of the same order is uniquely determined.

The formation of the forms of the same order is carried out in tabular form:

I. Indication of which forms of Systems I and II are to be folded with each other in order to obtain the prescribed order in the coefficients.

[27] Gordan–Kerschensteiner, loc. cit., p. 181.

II. Listing of the irreducible forms according to the scheme of the form sequence.

III. Reduction of the reducible or decomposable form sequences or forms, that is, of those places where the total form sequence breaks off, thereby proving that all irreducible constructions have been exhausted by those indicated.

IV. Consequences: 1) indication of newly arising reducers; 2) indication of those forms which do not enter, in products with forms of the same system, into folding with forms of the other system.

It will be shown that only the following occur:

1) foldings of one form of System I with one form of System II;

2) foldings of powers of θ, respectively H, or of two forms of one system with one form of the second system.

We shall obtain the following scheme for folding:

System I		folded with	System II	
1. order	f		2. order	ν
2. ,,	θ		4. ,,	H
3. ,,	K, j, Δ		6. ,,	L, σ
4. ,,	$N, \theta^2, f \cdot j$		8. ,,	$(\nu\sigma x), H^2$
5. ,,	$\theta \cdot K$		10. ,,	$H \cdot L$
6. ,,	$\theta^3, \Delta \cdot j$		12. ,,	H^3.

For checking the calculation, besides the "double reduction", the two special forms serve:

$$\text{I.}\quad \begin{aligned} f &= \sum_3 x_1^3 x_2, & u_\nu^2 &= \tfrac{i}{6} u_x^2, & s &= \tfrac{i}{3} f, \\ a_\eta^2 &= -\tfrac{1}{9} i\theta, & H_\nu^2 &= \tfrac{i}{6}\theta, & \sigma &= -\tfrac{i}{6} j, \\ g &= -\tfrac{2}{3} i\Delta, & \varrho &= 0, & t &= 0, \end{aligned}$$

$$\text{II.}\quad \begin{aligned} f &= \sum_3 x_1^4, & s &= \tfrac{4}{3} i f, & a_\eta^2 &= \tfrac{2}{9} i\theta, \\ \Delta_\nu^2 &= \tfrac{1}{15} i\theta, & \sigma &= \tfrac{4}{3} i j, & g &= \tfrac{4}{3} i\Delta, \\ \varrho &= 0, & t &= 0. \end{aligned}$$

Supplementary remark: for the ground form ν, formulas (1.), (2.), (3.), (4.) correspond to the following:

$$\begin{array}{r|c|c} (\nu\eta x)^2 = A & H_\nu(\nu\eta x) = 0 & H_\nu^2 = \sigma \\ (\nu\eta x)^3 = 0 & H_\nu(\nu\eta x)^2 = B & H_\nu^2(\nu\eta x) = 0 \\ (\nu\eta x)^4 = g & H_\nu(\nu\eta x)^3 = 0 & H_\nu^2(\nu\eta x)^2 = C, \end{array} \tag{5}$$

$$A = \frac{1}{6} s\nu - \frac{2}{3} u_x s_\nu + \frac{2}{3} u_x^2 s_\nu^2 - \frac{J}{18} u_x^4, \quad B = -\frac{5}{6} s_\nu + \frac{7}{6} u_x s_\nu^2 - \frac{J}{9} u_x^3, \quad C = \frac{2}{3} s_\nu^2 - \frac{J}{9} u_x^2.$$

$$s_\nu^3 u_x s_\nu = \frac{1}{3} u_x s_\nu^4 = \frac{1}{3} J u_x. \tag{6}$$

$$\begin{aligned} &\text{a)} \quad (\nu\sigma x)^2 = \frac{1}{2}(Hsu)^2 - \frac{2}{5}\nu \cdot s_\nu^2 - \frac{4}{5} u_x\, s_\eta (Hsu)^2 + u_x^2 \left\{ \frac{1}{6} J \cdot \nu - \frac{1}{40}(ss'u)^4 \right\}, \\ &\text{b)} \quad (\nu\sigma x)^3 = -\frac{3}{10} s_\eta (Hsu), \\ &\text{c)} \quad (\nu\sigma x)^4 = \frac{21}{10} s_\eta^2 - \frac{3}{10} Z - \frac{6}{5} u_x s_\eta^3 + \frac{1}{10} u_x^2 s_\eta^4. \end{aligned} \tag{7}$$

$$\begin{aligned} &\text{a)} \quad g_\nu = -s_\eta + u_x \left(\frac{9}{2} s_\eta^2 - \frac{1}{2} Z \right) - 3 u_x^2 s_\eta^3 + \frac{1}{2} u_x^3 s_\eta^4, \\ &\text{b)} \quad g_\nu^2 = \frac{21}{10} s_\eta^2 - \frac{3}{10} Z - 2 u_x s_\eta^3 + \frac{1}{2} u_x^2 s_\eta^4, \\ &\text{c)} \quad g_\nu^3 = -\frac{7}{10} s_\eta^3 + \frac{1}{2} u_x s_\eta^4, \\ &\text{d)} \quad g_\nu^4 = \frac{3}{5} s_\eta^4. \end{aligned} \tag{8}$$

§ 8. Forms of the third order (System III).

Folding of f with ν.
Irreducible forms:

$$\begin{array}{cccc} a_\nu & \cdot & \cdot & \cdot \\ \cdot & a_\nu^2 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & a_\nu^4 = i. \end{array}$$

Reduction:

$$a_\nu^3 = \frac{1}{3} i u_x \qquad \text{(formula (2.))}.$$

Consequences: 1) a_ν^3 is a reducer. 2) f enters into folding with ν only in products with contragredient forms (j). The same holds for ν with respect to f.

§ 9. Forms of the fourth order (System III).

Folding of θ with ν.
Irreducible forms:

$$\begin{array}{cccc} (\theta\nu x) & \theta_\nu & \cdot & \cdot \\ (\theta\nu x)^2 & \theta_\nu(\theta\nu x) & \theta_\nu^2 & \cdot \\ \cdot & \theta_\nu(\theta\nu x)^2 & \cdot & \theta_\nu^3. \end{array}$$

reductions: From the reducer a_ν^3, by double reduction of the expressions $a_\nu^3(abu)$, $a_\nu^3 b_\nu$, $a_\nu^3 b_\nu(abu)$ according to the ansatz:

$$\begin{aligned} \theta_\nu^2(\vartheta\nu x) &= a_\nu^2(\widehat{a}b\nu x)(abu) - \frac{1}{3}(\nu\nu_1 x)^3 = a_\nu b_\nu(\widehat{a}b\nu x)(abu) + \frac{1}{6}(\nu\nu_1 x)^3 \\ &= a_\nu^3(abu) - a_\nu^2 b_\nu(abu) = 2a_\nu^2 b_\nu(abu) = \frac{2}{3}a_\nu^3(abu), \\ \theta_\nu^2(\vartheta\nu x)^2 &= a_\nu^2(\widehat{a}b\nu x)^2 - \frac{1}{3}(\nu\nu_1 x)^4 = a_\nu b_\nu(\widehat{a}b\nu x)^2 + \frac{1}{6}(\nu\nu_1 x)^4 \\ &= if - 2a_\nu^3 b_\nu + a_\nu^2 b_\nu^2 - \frac{1}{3}s = 2a_\nu^3 b_\nu - 2a_\nu^2 b_\nu^2 + \frac{1}{6}s = \frac{2}{3}if - \frac{2}{3}a_\nu^3 b_\nu - \frac{1}{6}s, \\ \theta_\nu^3(\vartheta\nu x) &= a_\nu^2 b_\nu(\widehat{a}b\nu x)(abu) = a_\nu^3 b_\nu(abu). \end{aligned}$$

The formulas obtained are:

$$\begin{array}{l|l|l} \theta_\nu^2(\theta\nu x) = 0 & \theta_\nu^3(\theta\nu x) = 0 & \theta_\nu^4 u_x^2 = u_\varrho^2 = \varrho \quad \text{(adjoined modulus, § 7)}, \\ \theta_\nu^2(\theta\nu x)^2 = \frac{4}{9}if - \frac{1}{6}s & & \end{array} \tag{9}$$

Consequences: 1) $\theta_\nu^2(\theta\nu x)^2$ is a reducer. 2) ν enters into folding with θ only in products with contragredient forms (g).

§ 10. Forms of the fifth order (System III).

$$\text{Folding of } \begin{cases} K, j, \Delta \text{ with } \nu, \\ f \text{ with } H. \end{cases}$$

Irreducible forms:

$$\text{A)} \quad \begin{array}{cccc} \cdot & K_\nu & \cdot & \cdot \\ (k\nu x)^2 & \cdot & K_\nu^2 & \cdot \\ (k\nu x)^3 & K_\nu(k\nu x)^2 & \cdot & \cdot \\ \cdot & K_\nu(k\nu x)^3 & \cdot & \cdot \end{array} \qquad \text{B)} \quad \begin{array}{c} (j\nu x) \\ (j\nu x)^2 \\ (j\nu x)^3 \\ \cdot \end{array} \qquad \text{C)} \quad \begin{array}{c} \Delta_\nu \\ \cdot \\ \cdot \\ \cdot \end{array}$$

$$\text{D)} \quad \begin{array}{cccc} (aHu) & (aHu)^2 & \cdot & \cdot \\ a_\eta & a_\eta(aHu) & a_\eta(aHu)^2 & \cdot \\ \cdot & a_\eta^2 & \cdot & \cdot \end{array}$$

reductions:

A) form sequence K_ν^3: reducer a_ν^3.
form $K_\nu^2(k\nu x)^2$: reducer $\theta_\nu^2(\vartheta\nu x)^2$.

$$(k\nu x), \quad K_\nu(k\nu x), \quad K_\nu^2(k\nu x)$$

are decomposable forms according to § 3.
B) form

$$\begin{aligned}(j\nu x)^4 = (\widehat{a}\theta\nu x)^4 &= a_\nu^4\theta + \theta_\nu^4 f - 4a_\nu^3\theta_\nu - 4a_\nu\{a_\nu\theta_x - (\widehat{a}\theta\nu x)\}^3 \\ &+ 6a_\nu^2\{a_\nu\theta_x - (\widehat{a}\theta\nu x)\}^2 = \varrho f + \frac{5}{3}i\theta - 6a_\nu^2(\widehat{a}\theta\nu x)^2 + 4a_\nu(\widehat{a}\theta\nu x)^3,\end{aligned}$$

and therefore

$$(j\nu x)^4 = \varrho f + \frac{5}{3}i\theta \ \mathrm{mod}(\Delta, \nu), \qquad \text{(cf. § 5 end).}$$

C) form Δ_ν^4: reducer θ_ν^4.
forms Δ_ν^2 and Δ_ν^3: reducer a_ν^3 or θ_ν^4 by replacing the form Δ by lower forms of the form sequence (§ 5 end), i.e. by double reduction of the expressions

$$a_\vartheta a_\nu^3, \qquad a_\vartheta a_\nu^3\theta_\nu, \qquad a_\vartheta\theta_\nu^4.$$

The double calculation of Δ_ν^3 gives rise to the reduction of the higher form a_η^3.
Reduction formulas:

$$\begin{aligned}&\text{a)} \quad \Delta_\nu^2 = \frac{3}{5}a_\eta^2 - \frac{3}{10}(asu)^2 + \frac{1}{3}i\theta + \frac{1}{5}u_x^2(2a_\varrho^2 - t),\\ &\text{b)} \quad \Delta_\nu^3 = -\frac{3}{10}a_\varrho + \frac{3}{5}u_x\left(\frac{13}{10}a_\varrho^2 - \frac{2}{5}t\right),\\ &\text{c)} \quad \Delta_\nu^4 = a_\varrho^2 - \frac{2}{5}t.\end{aligned} \tag{10}$$

Derivation of the formulas:

$$\begin{aligned}\text{a)} \quad a_\vartheta a_\nu^3 = \frac{1}{3}i\theta &= [a_\vartheta]_{\nu^3} + \frac{6}{7}[a_\vartheta^2]_{\nu^2} + \frac{6}{5}[a_\vartheta(a\theta u)^2]_{\nu^2}\, ux^2 + \frac{11}{30}[a_\vartheta^2(a\theta u)^2]_\nu\, \nu x^2\\ &- \frac{6}{7}u_x[a_\vartheta^2]_{\nu^3} - \frac{1}{6}u_x[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2,\\ \frac{1}{3}i\theta &= \Delta_\nu^2 - \frac{2}{3}u_x\Delta_\nu^3 - a_\nu a_{\nu_1}(\nu\nu_1 x)^2 + \frac{2}{5}a_\nu^2(\nu\nu_1 x)^2 + \frac{1}{5}u_x^2 a_\nu^2 a_{\nu_1}(\nu\nu_1 x)^2;\end{aligned}$$

$$\begin{aligned}\text{b)} \quad a_\vartheta a_\nu^3\theta_\nu = 0 &= [a_\vartheta]_{\nu^4} + \frac{9}{14}[a_\vartheta^2]_{\nu^3} + \frac{3}{5}[a_\vartheta(a\theta u)^2]_{\nu^2}\, \nu x^2 + \frac{17}{120}[a_\vartheta^2(a\theta u)^2]_\nu\, \nu x^2\\ &- \frac{9}{14}u_x[a_\vartheta^2]_{\nu^4} - \frac{1}{24}u_x[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2,\\ 0 &= \frac{5}{6}\Delta_\nu^3 - \frac{1}{2}u_x\Delta_\nu^4 - \frac{1}{4}a_\eta^3 + \frac{1}{20}u_x a_\eta^4;\end{aligned}$$

$$\begin{aligned}\text{b')} \quad a_\vartheta\theta_\nu^4 = a_\varrho &= a_\vartheta\theta_\nu\{a_\nu - (a\theta\nu x)\}^3\\ &= -\frac{3}{2}[a_\vartheta]_{\nu^3} + \frac{3}{5}[a_\vartheta(a\theta u)^2]_{\nu^2}\, \nu x^2 - \frac{37}{120}[a_\vartheta^2(a\theta u)^2]_\nu\, \nu x^2\\ &+ \frac{3}{2}u_x[a_\vartheta^2]_{\nu^4} + \frac{49}{120}u_x[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2,\\ a_\varrho &= -\frac{3}{2}\Delta_\nu^3 + \frac{3}{2}u_x\Delta_\nu^4 - \frac{11}{20}a_\eta^3 + \frac{7}{20}u_x a_\eta^4;\end{aligned}$$

$$\begin{aligned}\text{c)} \quad a_\vartheta^2\theta_\nu^4 = a_\varrho^2 &= [a_\vartheta^2]_{\nu^4} + \frac{3}{5}[a_\vartheta^2(a\theta u)^2]_{\nu^2}\, \nu x^2,\\ a_\varrho^2 &= \Delta_\nu^4 + \frac{2}{5}a_\eta^4.\end{aligned}$$

D) reduction of a_η^3 by double reduction of Δ_ν^3 (C).
The remaining reductions are obtained by a computation dualistically opposite to that of § 9.

Reduction formulas:

$$
\left.\begin{aligned}
a_\eta^2(aHu) &= -\frac{1}{6}(asu)^3, \\
a_\eta^3 &= -a_\varrho + \frac{3}{5}u_x a_\varrho^2 + \frac{1}{5}u_x t,
\end{aligned}\right|
\begin{aligned}
a_\eta^2(aHu)^2 &= \frac{4}{9}i\nu - \frac{1}{6}(asu)^4, \\
a_\eta^3(aHu) &= 0, \\
a_\eta^4 &= t \quad \text{(adjoined as a modulus)}.
\end{aligned}
\tag{11}
$$

Consequences:

1) $(j\nu x)^4$, Δ_ν^2, $a_\eta^2(aHu)$ are reducers.

2) ν enters only in products with contragredient forms in foldings with K, j, Δ; the same holds for f in relation to H, and for j in relation to ν, since $\theta_\nu(\theta \cdot j, \nu, x^3)$ becomes reducible according to § 11 B.

§ 11. Forms of the sixth order (System III).

$$
\text{Folding of} \left\{ \begin{array}{l} \theta^2, f \cdot j, N \text{ with } \nu, \\ \theta \text{ with } H. \end{array} \right.
$$

Irreducible forms:

$$
\text{A) } (\vartheta^2\nu x)^3, \quad \theta_\nu(\vartheta^2\nu x)^3 \quad \text{B) } a_\nu(j\nu x), \quad a_\nu^2(j\nu x) \quad \text{C) } N_\nu
$$

$$
\text{D)} \quad
\begin{array}{c|c|c|c}
 & (\theta H u) & (\theta H u)^2 & \cdot \\
(\vartheta\eta x) & \theta_\eta & H_\vartheta(\theta H u),\ \theta_\eta(\theta H u) & \theta_\eta(\theta H u)^2 \\
(\vartheta\eta x)^2 & H_\vartheta(\vartheta\eta x),\ \theta_\eta(\vartheta\eta x) & \theta_\eta^2 & \theta_\eta^2(\theta H u) \\
\cdot & \theta_\eta(\vartheta\eta x)^2 & \cdot & \cdot
\end{array}
$$

reductions:

A) $(\vartheta^2\nu x)^4$ decomposes according to formula (3.)a:

$$
(a\Delta\vartheta x)^4 = \frac{1}{2}(\vartheta^2\nu x)^4 - \frac{3}{5}f \cdot a_\nu^2(\nu\vartheta x)^2 = \frac{1}{3}\Delta^2 + f\Delta_\vartheta^2.
$$

B) $a_\nu(j\nu x)^2$ decomposes according to formula (9.)a, indem wir according to § 6 end $f \cdot j$ by $(\theta\theta' u)^2$ ersetzen, and the reducibility of θ'_ϑ take account of:

$$
\frac{1}{3}a_\nu(j\nu x)^2 = (\theta\theta'\nu x)^2\theta_\nu = \theta_\nu^3\theta - \theta_\nu^2\theta'_\nu = \theta_\nu^3\theta + \theta_\nu^2(\widehat{j}\nu\theta' u) = \theta_\nu^3\theta - \frac{2}{3}\theta_\nu^3\theta.
$$

forms $a_\nu(j\nu x)^3$ and $a_\nu^2(j\nu x)^2$: reducer a_j and $(j\nu x)^4$:

$$
a_\nu(j\nu x)^3 = a_j(j\nu x)^3 - (j\nu x)^3(j\nu\widehat{a}u),
$$

$$
a_\nu^2(j\nu x)^2 = a_j^2(j\nu x)^2 - 2a_j(j\nu x)^2(j\nu\widehat{a}u) + (j\nu x)^2(j\nu\widehat{a}u)^2.
$$

C) form sequence N_ν^2; reducer Δ_ν^2. $(n\nu x)$ and $N_\nu(n\nu x)$ decompose as a transvection over functional determinants according to § 3.

D) *Overview of the reductions:*

a) form sequence $\theta_\eta^2 H_\vartheta$: reducer $a_\eta^2(aHu)$. (Leads to forms with higher symbols.)

b) form sequence $\theta_\eta H_\vartheta$: reducer $a_\eta^2(aHu)$ by double reduction. (Leads to forms with higher symbols and to higher forms of the total form sequence, i.e. to the form sequence θ_η^2.) Instead of $\theta_\eta H_\vartheta$ we consider the form sequence

$$
\theta_\eta(\vartheta\eta x)(\theta H u) = \theta_\eta H_\vartheta + \theta_\eta^2,
$$

ersetzen in it the symbols θ by the lower form (abu), gelangen so to the double reduction of the form sequence

$$
a_\eta^2(aHu)(abu) = \theta_\eta H_\vartheta + \frac{3}{2}\theta_\eta^2 \mod s
$$

up to the terminal form

$$
a_\eta^3 b_\eta(bHu)(abu) = \theta_\eta^3 H_\vartheta + \frac{1}{2}\theta_\eta^4.
$$

c) form sequence θ_η^3: By double reduction of those forms which simultaneously belong to the form sequences $\theta_\eta H_\vartheta$ and $\theta_\eta^2 H_\vartheta$ - i.e. the forms of the form sequence $\theta_\eta^2 H_\vartheta$ - to forms with higher symbols on the one hand, and to forms with higher symbols and to the higher form sequence θ_η^3 on the other hand.

d) forms $\theta_\eta^2(\vartheta\eta x)$, $\theta_\eta^2(\vartheta\eta x)^2$, $\theta_\eta^3(\vartheta\eta x)$: By double reduction by means of the reducers a analogous to the computation of § 9.

e) forms $\theta_\eta^2(\theta Hu)^2$ and $\theta_\eta^2(\vartheta\eta x)^2$: By double reduction of the expressions $\Delta_\nu^2(a\Delta u)^4$ and $(a\Delta\nu x)^4$ by means of the reducers Δ_ν^2 and $(a\Delta u)^4$.

f) forms H_ϑ and H_ϑ^2: Decomposition of forms. Is obtained for H_ϑ^2 by double reduction of the expression $\Delta_\nu^2(a\Delta u)^2$ or also by direct calculation of the products $(a_\nu^2)^2$, $a_\nu^2 b_{\nu_1}$ by forms of the system. (The analogous calculation of $(a_\nu)^2$ gives a relation between decomposable forms.)

g) Reduction of higher forms by the fact that forms of the form sequence $\theta \cdot H$ admit two reduction possibilities (belong to two reducible form sequences), namely:

> g: by double reduction of $\theta_\eta^2(\vartheta\eta x)^2$; allows reduction d) and e) zu.
>
> $s_\vartheta^2(\theta su)$: by double reduction of $\theta_\eta^3(\vartheta\eta x)$; allows reduction c) and d) zu.
>
> $s_\vartheta^2(\theta su)^2$: by double reduction of $\theta_\eta^2 H_\vartheta^2$; allows reduction a) and has the reducers $\theta_\nu^2(\vartheta\nu x)^2$ as a factor.
>
> s_ν^2: by double reduction of $\theta_\eta^3 H_\vartheta$; allows reduction a) and c) zu.

The proof of the independence of the remaining forms is obtained by double reduction of the form sequence $\Delta_\nu^2(a\Delta u)^2 = \theta_\eta^2$.

Execution of the reductions:

The existence of reductions a), b), c), d) is clear a priori, since according to the stated law of formation the relations arising from double reduction are independent of one another. For reductions e), f), g), it must be shown by computation that no identities arise.

Proceeding symmetrically with the diagonal term in the form sequence $\theta \cdot H$ up to the form θ_η, and partly indicating the calculation, we also give the formulas obtained by reductions a), b), c), d), since they are used partly in reductions e), f), g), and partly later.

1) H_ϑ and H_ϑ^2 according to f). According to the ansatz[28]

$$\begin{aligned}
a_\nu b_{\nu_1}^2 &= \nu a_\nu b_\nu^2 - (aHu)(bHu)b_\eta \sim \nu a_\nu b_\nu^2 - f a_\eta (aHu)^2 - (abu)(aHu)b_\eta + (abu)^2 b_\eta, \\
a_\nu^2 b_{\nu_1}^2 &= b_\nu^2\{a_\nu u_\nu - (a\nu\nu_1 u)\}^2 \sim \nu\{a_\nu^2 b_\nu^2 - 2a_\nu b_\nu (aHu)(bHu) - \frac{1}{6}(asu)^2(bsu)^2\} \\
&\sim \nu a_\nu^2 b_\nu^2 - 2a^2(aHu)(abu) - \frac{1}{6}(asu)^2(bsu)^2 + (\vartheta\eta x)^2(\theta Hu)^2
\end{aligned}$$

after substitution of the value of $\theta_\eta H_\vartheta$, the formulas arise:

$$\begin{aligned}
&\text{a)}\quad H_\vartheta \sim 2a_\nu a_\varrho^2 + 2\nu b_\nu(\vartheta\eta x)^2 + 2f a_\eta (aHu)^2 - 2\theta_\eta, \\
&\text{b)}\quad H_\vartheta^2 \sim (a^2)^2 + 2\theta_\eta^2 + \frac{2}{3}(\theta su)^2 + \frac{1}{12}s\nu - \frac{1}{9}if\nu - \frac{1}{6}f(asu)^4.
\end{aligned} \tag{12}$$

2) $\theta_\eta H_\vartheta$ according to b):

$$\begin{aligned}
a_\eta^2(aHu)(abu) &\sim [a_\eta^2(aHu)]_{bu} + \frac{1}{2}f[a_\eta^2(aHu)^2] = a_\eta b_\eta(aHu)(abu) \\
&\quad + a_\eta(\widehat{ab}\eta x)(abu)(aHu) \sim \theta_\eta(\vartheta\eta x)(\theta Hu) + \frac{1}{2}\theta_\eta^2 + \frac{1}{4}(\nu\eta x)^2.
\end{aligned}$$

According to the formulas (5.) and (11.) follows:

$$\theta_\eta H_\vartheta \sim -\frac{3}{2}\theta_\eta^2 - \frac{1}{4}(\theta su)^2 - \frac{1}{12}s\nu + \frac{2}{9}if\nu.$$

3) $\theta_\eta H_\vartheta(\theta Hu)$ is computed according to b), analogously to $\theta_\eta H_\vartheta$:

$$\theta_\eta H_\vartheta(\theta Hu) \sim -\theta_\eta^2(\theta Hu) - \frac{1}{6}(\theta su)^3.$$

4) $\theta_\eta H_\vartheta(\vartheta\eta x)$ according to b); $\theta_\eta^2(\vartheta\eta x)$ according to d) (cf. § 9):

$$\theta_\eta H_\vartheta(\vartheta\eta x) \sim -\theta_\eta^2(\vartheta\eta x) - \frac{1}{6}s_\vartheta(\theta su) \sim -\frac{1}{3}(\vartheta\varrho x) - \frac{1}{6}s_\vartheta(\theta su),$$

[28] The sign $\sim$ in place of the equality sign means that products arising from u_x with higher forms, other than those arising by foldings III and IV, are omitted.

$$\theta_\eta^2(\vartheta\eta x) \sim \frac{2}{3}a_\eta^3(abu) \sim -\frac{1}{3}(\vartheta\varrho x).$$

5)

$\theta_\eta H_\vartheta^2$ according to b)	$\theta_\eta^2 H_\vartheta$ according to a)	$\theta_\eta^2 H_\vartheta$ according to b) and thereby θ_η^3 according to c)
from $a_\eta^2(aHb)(bHu)$	from $a_\eta^3(Hab)(abu)$	from $a_\eta^3(bHu)(abu)$

$$\theta_\eta H_\vartheta^2 \sim -\frac{3}{2}\theta_\eta^2 H_\vartheta - \frac{1}{4}s_\vartheta(\theta su)^2 + \frac{1}{6}s_\nu - \frac{4}{9}ia_\nu \sim -\theta_\varrho - \frac{1}{6}s_\nu - \frac{1}{9}ia_\nu,$$

$$\theta_\eta^2 H_\vartheta \sim \frac{2}{3}\theta_\varrho - \frac{1}{6}s_\vartheta(\theta su)^2 + \frac{2}{9}s_\nu - \frac{2}{9}ia_\nu,$$

$$\theta_\eta^3 H_\vartheta \sim \frac{2}{3}\theta_\varrho - \frac{2}{3}\theta_\eta^3 + \frac{5}{36}s_\nu, \qquad \theta_\eta^3 \sim \frac{1}{4}s_\vartheta(\theta su)^2 - \frac{1}{8}s_\nu + \frac{1}{3}ia_\nu.$$

6) $\theta_\eta^2(\theta Hu)^2$ according to e):

$$\Delta_\nu^2(a\Delta u)^2 = [(a\Delta u)^4]_{\nu^2} = -\frac{12}{5}\theta_\eta^2(\theta Hu)^2 - \frac{3}{10}\sigma - \frac{1}{20}(\theta su)^4 + \nu\varrho \quad \text{(according to formula (3.)c)},$$

$$\Delta_\nu^2(a\Delta u)^4 = [\Delta_\nu^2]_{aa}^4 = -\frac{3}{5}\theta_\eta^2(\theta Hu)^2 + \frac{1}{5}\sigma - \frac{3}{10}(\theta su)^4 + \frac{1}{3}ij \quad \text{(according to formula (10.)a)},$$

$$\theta_\eta^2(\theta Hu)^2 = -\frac{1}{6}\sigma + \frac{1}{12}(\theta su)^4 - \frac{1}{9}ij + \frac{1}{3}\nu\varrho.$$

7) $\theta_\eta^2(\vartheta\eta x)^2$ according to d) and e). From this reduction of the higher form g.
By a computation analogous to that in § 9, it follows:

$$\begin{aligned}\theta_\eta^2(\vartheta\eta x)^2 &= \frac{1}{2}if - \frac{1}{2}a_\eta^2 b_\eta - \frac{1}{12}g = \frac{2}{3}tf - \frac{2}{3}a_\eta^3 b_\eta - \frac{1}{6}g\\ &= -\frac{1}{6}g - \frac{1}{3}(\vartheta\varrho x)^2 + \frac{4}{15}fa_\varrho^2 + \frac{8}{15}ft \quad \text{(according to formula (11.))}.\end{aligned}$$

By the corresponding computation as above follows:

$$((a\Delta\nu x)^4) = [(a\Delta u)^4]_{\nu x^4} = \frac{21}{10}\theta_\eta^2(\vartheta\eta x)^2 + \frac{7}{10}s_\vartheta^2 - \frac{3}{10}g \quad \text{(according to formula (3.)c)},$$

$$\begin{aligned}(a\Delta\nu x)^4 &= -\frac{1}{3}i\Delta + 6[\Delta_\nu^2]a^2 - 4[\Delta_\nu^3]a + \Delta_\nu^4 f\\ &= -\frac{36}{5}\theta_\eta^2(\vartheta\eta x)^2 - \frac{9}{5}s_\vartheta^2 - \frac{3}{5}g - \frac{3}{5}(\vartheta\varrho x)^2\\ &\quad + \frac{5}{3}i\Delta + \frac{37}{25}fa_\varrho^2 + \frac{74}{25}ft \quad \text{(according to formula (10.))}.\end{aligned}$$

Thus

$$\begin{aligned}\frac{93}{10}\theta_\eta^2(\vartheta\eta x)^2 &= -\frac{3}{10}g - \frac{5}{2}s_\vartheta^2 - \frac{3}{5}(\vartheta\varrho x)^2\\ &\quad + \frac{5}{3}i\Delta + \frac{37}{25}fa_\varrho^2 + \frac{74}{25}ft,\end{aligned}$$

and by comparing the two values of $\theta_\eta^2(\vartheta\eta x)^2$ according to g):

$$0 = g - 2s_\vartheta^2 + 2(\vartheta\varrho x)^2 + \frac{4}{3}i\Delta - \frac{4}{5}fa_\varrho^2 - \frac{8}{5}ft. \tag{13}$$

8) $\theta_\eta^2 H_\vartheta(\theta Hu)$ according to a) and b), from this $\theta_\eta^3(\theta Hu)$ according to c) (forms with the Faktor u_x vanish):

$$\theta_\eta^2 H_\vartheta(\theta Hu) = -\frac{1}{6}s_\vartheta(\theta su)^3 - \frac{1}{3}(\nu\varrho x),$$

$$\theta_\eta^2 H_\vartheta(\theta Hu) = -\frac{1}{3}\theta_\eta^3(\theta Hu) - \frac{1}{3}(\nu\varrho x),$$

$$\theta_\eta^3(\theta Hu) = \frac{1}{2}s_\vartheta(\theta su)^3.$$

9) $\theta_\eta^2 H_\vartheta(\vartheta\eta x)$ according to a) and b), from this $\theta_\eta^3(\vartheta\eta x)$ according to c). $\theta_\eta^3(\vartheta\eta x)$ according to d), from this reduction of the higher form $s_\vartheta^2(\theta su)$ according to g) (forms with the Faktor u_x vanish):

$$\begin{aligned}
\theta_\eta^2 H_\vartheta(\vartheta\eta x) &= \frac{1}{3}(atu),\\
\theta_\eta^2 H_\vartheta(\vartheta\eta x) &= \frac{1}{3}(atu) + \frac{1}{3}\theta_\eta^3(\vartheta\eta x) - \frac{1}{6}s_\vartheta^2(\theta su),\\
\theta_\eta^3(\vartheta\eta x) &= \frac{1}{2}s_\vartheta^2(\theta su),\\
\theta_\eta^3(\vartheta\eta x) &= \frac{2}{5}\theta_\varrho(\vartheta\varrho x) + \frac{1}{5}(atu),\\
s_\vartheta^2(\theta su) &= \frac{4}{5}\theta_\varrho(\vartheta\varrho x) + \frac{2}{5}(atu).
\end{aligned} \tag{14}$$

10) $\theta_\eta^2 H_\vartheta^2$ according to a) and by reducer $\theta_\nu^2(\vartheta\nu x)^2$, $\theta_\eta^3 H_\vartheta$ according to a) and c), θ_η^4 according to c), from this reduction of the higher forms $s_\vartheta^2(\theta su)^2$ and s_ν^2 according to g):

$$\begin{aligned}
\text{According to a)}\quad \theta_\eta^2 H_\vartheta^2 &= [a_\eta^2(aHu)^2]b^2 + \frac{1}{3}[a_\eta^4]_{bu^2} - \frac{1}{3}H_\nu^2(\nu\eta x)^2\\
&= \frac{4}{9}ia_\nu^2 - \frac{1}{6}s_\vartheta^2(\theta su)^2 - \frac{5}{18}s_\nu^2 + \frac{1}{3}(atu)^2 + \frac{1}{54}Ju_x^2.
\end{aligned}$$

$$\begin{aligned}
\text{By } \theta_\nu^2(\vartheta\nu x)^2:\quad \theta_\eta^2 H_\vartheta^2 &= [\theta_\nu^2(\vartheta\nu x)^2]_{\nu_1} + \frac{1}{3}[\theta_\nu^4]_{\nu_1}\hat{\nu}_1 x^2 - \frac{1}{3}s_\vartheta^2(\theta su)^2\\
&= \frac{4}{9}ia_\nu^2 - \frac{1}{6}s_\nu^2 - \frac{1}{3}s_\vartheta^2(\theta su)^2 + \frac{1}{3}(\nu\varrho x)^2.
\end{aligned}$$

$$\begin{aligned}
\text{According to a)}\quad \theta_\eta^3 H_\vartheta &= -[a_\eta^2(aHu)]_{bu}b^2 - \frac{1}{2}[a_\eta^2(aHu)^2]b^2 - \frac{1}{3}[a_\eta^3]b + \frac{1}{12}[a_\eta^4]_{bu^2}\\
&\quad + \frac{1}{2}u_x[a_\eta^2(aHu)^2]b^3\\
&= -\frac{2}{9}ia_\nu^2 + \frac{1}{12}s_\nu^2 + \frac{4}{15}\theta_\varrho^2 - \frac{7}{90}(\nu\varrho x)^2 + \frac{1}{30}(atu)^2 + \frac{2}{27}i^2u_x^2 - \frac{1}{36}Ju_x^2.
\end{aligned}$$

$$\begin{aligned}
\text{According to c)}\quad \theta_\eta^3 H_\vartheta : a_\eta^3(Hab)(bHu) &= \frac{1}{2}(atu)^2 = \theta_\eta^2 H_\vartheta(\theta Hu)(\vartheta\eta x) + \frac{1}{4}H_\nu^2(\nu\eta x)^2\\
&\quad + \frac{1}{2}\theta_\eta^2 H_\vartheta^2 = \frac{3}{2}\theta_\eta^2 H_\vartheta^2 + \theta_\eta^3 H_\vartheta - u_x[a_\eta^2(aHu)^2]b^3\\
&\quad + \frac{1}{4}H_\nu^2(\nu\eta x)^2,
\end{aligned}$$

and therefore

$$\begin{aligned}
\theta_\eta^3 H_\vartheta = &-\frac{2}{3}ia_\nu^2 + \frac{5}{12}s_\nu^2 + \frac{1}{2}(\nu\varrho x)^2 - \frac{1}{2}(atu)^2\\
&+ \frac{4}{27}i^2u_x^2 - \frac{1}{12}Ju_x^2.
\end{aligned}$$

According to c)

$$\begin{aligned}
\theta_\eta^4 = &\frac{4}{9}ia_\nu^2 - \frac{1}{6}s_\nu^2 + \frac{16}{15}\theta_\varrho^2\\
&- \frac{14}{45}(\nu\varrho x)^2 + \frac{2}{15}(atu)^2.
\end{aligned}$$

According to g), from the two values for $\theta_\eta^2 H_\vartheta^2$:

$$\begin{aligned}
s_\vartheta^2(\theta su)^2 &= \frac{2}{3}s_\nu^2 - 2(atu)^2 + 2(\nu\varrho x)^2 - \frac{1}{9}Ju_x^2\\
&= \frac{8}{9}ia_\nu^2 + \frac{8}{15}\theta_\varrho^2 - \frac{14}{15}(atu)^2 + \frac{38}{45}(\nu\varrho x)^2 - \frac{4}{27}i^2u_x^2.
\end{aligned} \tag{15}$$

According to g), from the two values for $\theta_\eta^3 H_\vartheta$:

$$s_\nu^2 = \frac{4}{3}ia_\nu^2 + \frac{4}{5}\theta_\varrho^2 + \frac{8}{5}(atu)^2 - \frac{26}{15}(\nu\varrho x)^2 + \frac{1}{6}Ju_x^2 - \frac{2}{9}i^2u_x^2. \tag{16}$$

Formula (16.) gives the basis for the reduction of the modulus $(ss'u)^4$, and formula (13.) gives the basis for the reduction of the system of s. (§ 17.)

Consequences:

1) $a_\nu(j\nu x)^3$, $\theta_\eta H_\vartheta$, $\theta_\eta^2(\vartheta\eta x)$, $\theta_\eta^2(\theta H u)^2$ are reducers, $a_\nu(j\nu x)^2$, H_ϑ and H_ϑ^2 decomposable forms.

2) The form of the system II: $j(\nu) = g = (\nu\eta x)^4 H_x^2$ is reducible.

3) Since the only contragredient form g becomes reducible, ν does not enter at all in powers or in products with forms of the same system in folding with System I.

θ enters into foldings only as a contravariant in products or powers; correspondingly, H enters only as a covariant (cf. § 13 B).

§ 12. Forms of the seventh order.

$$\text{Folding of} \begin{cases} \theta\cdot K \text{ with } \nu, \\ K, j, \Delta \text{ with } H, \\ f \text{ with } L, \sigma. \end{cases}$$

Irreducible forms:

$$\text{B)}\quad \begin{array}{c|c|c|c|c} \cdot & \cdot & (KHu)^2 & \cdot & \cdot \\ \cdot & K_\eta & \cdot & K_\eta(KHu)^2 & \cdot \\ (k\eta x)^2 & \cdot & K_\eta^2 & \cdot & \cdot \\ (k\eta x)^3 & H_k(k\eta x)^2,\ K_\eta(k\eta x)^2 & \cdot & \cdot & \cdot \\ \cdot & K_\eta(k\eta x)^3 & \cdot & \cdot & \cdot \end{array}$$

$$\text{C)}\quad \begin{array}{cc} (j\eta x) & H_j \\ \cdot & H_j(j\eta x) \\ \cdot & H_j^2 \end{array} \qquad \text{D)}\quad \begin{array}{c} (\Delta H u) \\ \Delta_\eta \end{array}$$

$$\text{E)}\quad \begin{array}{cccc} a_i & (aLu)^2 & (aLu)^3 & \cdot \\ \cdot & \cdot & a_i(aLu)^2 & a_i(aLu)^3 \\ \cdot & a_i^2 & \cdot & \cdot \end{array}$$

reductions:

A) $(\vartheta\cdot k, \nu, x)^4$ decomposes as single Überschiebung over the decomposable form $(\vartheta^2\nu x)^4$.

B) form sequence $H_k K_\eta$: reducer $H_\vartheta\theta_\eta$.

form sequence $K_\eta^2(KHu)$: reducer $a_\eta^2(aHu)$.

form sequence $K_\eta^2(k\eta x)$: reducer $\theta_\eta^2(\vartheta\eta x)$.

The from this arising double reduction of the form sequence K_η^3 gives rise to the reduction of the higher form sequence $(j\eta x)^3$.

(KHu), $(k\eta x)$, $K_\eta(k\eta x)$ decomposable forms according to § 3.

$H_k(KHu)$, $K_\eta(KHu)$ decompose as arising by single folding with the decomposable form $K_\nu(k\nu x)$ and the functional determinant

$$K_\nu^2 = \theta_\nu^2(a\theta u) \equiv a_\nu^2(a\theta u) \mod \nu^2.$$

A relation between decomposable forms corresponds to the double representation of K_ν^2. One obtains:

$$H_k(KHu) = 2K_\nu(\nu\nu_1 k) = 2\theta_{\nu_1}(\nu\nu_1 a\theta) = 2\theta_\nu^2 a_\nu - a_\nu^2\theta_\nu + f\theta_\eta(\theta Hu)^2 - f\nu\theta_\vartheta^3 \mod \nu^3,$$

$$\begin{aligned} K_\eta(KHu) &= -K_\nu^2(\nu\nu_1 x) = -a_\nu^2(a\theta u)(\nu\nu_1 x) \\ &= a_\eta(aHu)^2\theta + a_\nu^2\theta_\nu = -\theta_\nu(\theta Hu)^2 f - \theta_\nu^2 a_\nu. \end{aligned}$$

Therefore

$$0 = a_\nu^2\theta_\nu + \theta_\nu^2 a_{\nu_1} + f\theta_\eta(\theta Hu)^2 + \theta a_\eta(aHu)^2 \mod \nu^3. \tag{17}$$

The forms H_k, $H_k(k\eta x)$, H_k^2, $H_k^2(k\eta x)$ decompose as arising by single folding with the decomposable forms H_ϑ and H_ϑ^2 (cf. § 3. 2), a) and b)).

Es is

$$H_k = H_\vartheta(a\theta u) \sim [H_\vartheta]_{ua} - fH_\vartheta(\theta Hu)$$

(specialization $y = uv$ of 2)a).

$$H_k(k\eta x) = H_\vartheta a_\eta - H_\vartheta\theta_\eta f, \qquad H_\vartheta a_\eta = \frac{5}{4}[H_\vartheta]a - \frac{1}{4}H_\vartheta a_\vartheta \quad (\S\ 3.2)b),$$

$$H_\vartheta^2(a\theta u) = [H_\vartheta^2]_{ua}, \qquad H_\vartheta^2 a_\eta = [H_\vartheta^2]a = H_k^2(k\eta x) + H_\vartheta^2\theta_\eta f.$$

C) form sequence $(j\eta x)^3$: Reducible by double reduction of the form sequence K_η^3 according to B); according to the ansatz:

$$0 = a_\eta^3(a\theta u) = \theta_\eta^3(a\theta u) = \theta_\eta + (a\theta\eta x)^3(a\theta u) - \theta_\eta^3(a\theta u) = \frac{1}{2}(j\eta x)^3 \mod(\nu^3, s, \varrho, t, i, J).$$

The analogous ansatz holds up to the terminal form of the form sequence:

$$0 = H_\vartheta^2(a\theta\eta x)a_\eta^3 = H_\vartheta^2(a\theta\eta x)\theta_\eta^3 - \frac{1}{2}H_\vartheta^2(j\eta x)^4 \mod(\nu^3, s, \varrho, t, i, J).$$

form $(j\eta x)^2$: decomposes 1) by twofold polarization of the relation for the decomposable form H_ϑ^2 ((12.) b);

2) by direct calculation of the products $a_\nu\theta_\nu^3$; thereby Decomposition of the higher form $(\Delta Hu)^2$. One obtains:

$$\left.\begin{aligned} &\text{a)}\quad \frac{1}{3}(\Delta Hu)^2 + \frac{1}{3}(j\eta x)^2 = \theta_\nu^2 a_{\nu_1}^2, \\ &\text{b)}\quad (j\eta x)^2 = \frac{4}{3}\theta_\nu^3 a_{\nu_1}, \end{aligned}\right\} \mod(\nu^3, s, \varrho, t, i, J). \tag{18}$$

according to the ansatz:

$$H_\vartheta^2(a\theta u)^2 - \frac{1}{3}(\Delta Hu)^2 = 2\theta_\eta^2(a\theta u)(aHu) + \theta_\nu^2 a_{\nu_1}^2 = (a\theta u)(aHu)\{a_\eta - (a\theta\eta x)^2\} + \theta_\nu^2 a_{\nu_1}^2,$$

$$\begin{aligned} a_{\nu_1}\theta_\nu^3 &= -(a\widehat{\nu}\nu_1 u)\theta_\nu^2(\theta\widehat{\nu}\nu_1 u) - \frac{1}{2}(u\nu\nu_1 u)\theta_\nu\theta_{\nu_1}(\vartheta\nu\nu_1 u) \\ &= -\frac{3}{2}\theta_\eta^2(\theta Hu)(aHu) = -\frac{3}{2}\theta_\eta^2(\theta Hu)(a\theta u) \\ &= -\frac{3}{2}\{a_\eta - (a\theta\eta x)^2\}\{(aHu) - (a\theta u)\}(a\theta u). \end{aligned}$$

D) form sequence $\Delta_\eta(\Delta Hu)$: reducer Δ_ν^2.

$(\Delta Hu)^2$ decomposable form according to C).

E) form sequence $a_i^2(aLu)$: reducer $a_\eta^2(aHu)$. forms (aLu) and $a_i(aLu)$: Decomposition as a transvectionen over functional determinants according to § 3.

F) form sequence a_σ^3: reducer a_ϱ^3.

forms a_σ^2 and a_σ^3: reducer a_ν^3 and a_η^3 by double reduction of the corresponding expressions

$$H_\nu a_\nu^3 \quad\text{and}\quad H_\nu a_\eta^3, \qquad H_\nu a_\nu^3 a_\nu \quad\text{and}\quad H_\nu a_\eta^4$$

(cf. § 10. C) reduction of Δ_ν^2 and Δ_ν^3).

From this reduction for the higher forms $a_\nu s_\nu(asu)^2$, $a_\nu^2 s_\nu(asu)^2$ and for forms in symbolsn $\varrho, t(a_\nu, a_\eta^2)$ taking account of (16.):

$$\left.\begin{aligned} a_\nu s_\nu(asu)^2 &= \frac{1}{3}i\theta_\nu^2 - \frac{1}{18}iH, \\ a_\nu^2 s_\nu(asu)^2 &= 0, \end{aligned}\right\} \mod(\varrho, t). \tag{19}$$

form a_σ: decomposes by polarization of (12.)b or, more briefly, by direct expansion of the products $a_\nu^2\theta_\nu^3$ according to the ansatz:

$$a_\nu^2\theta_{\nu_1}^3 = (a_\nu^2\theta_{\nu_1})a_{\nu_1} - (a\theta\nu_1 x)^2 = -2a_\eta(aHu)(a\theta H)(a\theta\eta x) + (a\theta\nu_1 x)^2 a_\nu^2\theta_{\nu_1}$$

and taking account of the reduction of $H_j(j\eta x)^2$ (C):

$$a_\sigma = 2a_\nu^2\theta_{\nu_1}^3 \mod(s, \varrho, t, i). \tag{20}$$

Consequences:

1) $(j\eta x)^3$, $\Delta_\eta(\Delta Hu)$, a_σ^2 are reducers, $(j\eta x)^2$, $(\Delta Hu)^2$, a_σ decomposable forms.

2) f enters into folding with L only in products with contragredient forms; therefore f, and likewise K and N, do not enter at all into folding with σ and consequently with $(\nu\sigma x)$. The form j enters only in products with contragredient forms; Δ does not enter at all in products in folding with H and L (this follows from $\Delta_\eta(\Delta Hu)$ and $(\Delta Hu)^2$). Likewise σ, and consequently $(\nu\sigma x)$, do not enter in products in folding with any form of System I. Thus, among products in System II, taking account of the consequences of § 11, there remain only powers of H and products of H with L.

§ 13. Forms of the eighth order (System III).

$$\text{Folding of} \begin{cases} \Delta \cdot j \text{ with } \nu, \\ \theta^2, f \cdot j, N \text{ with } H, \\ \theta \text{ with } L, \sigma. \end{cases}$$

Irreducible forms:

A) $\Delta_\nu (j\nu x)$ B) $\begin{matrix} (\vartheta^2 \eta x)^3 \\ (\vartheta^2 \eta x)^4 \end{matrix}$ $H_\vartheta (\vartheta^2 \eta x)^3, \; \theta_\eta (\vartheta^2 \eta x)^3,$

C) $(aHu)H_j, \quad a_\eta (aHu)H_j$ D) $H_n, \; N_\eta,$

$$\text{E)} \quad \begin{array}{c|c|c|c} \cdot & (\theta Lu)^2 & (\theta Lu)^3 & \cdot \\ \theta_i & \cdot & L_\vartheta (\theta Lu)^2, \; \theta_i (\theta Lu)^2 & \theta_i (\theta Lu)^3 \\ (\vartheta lx)^2 & \theta_i^2 & \cdot & \cdot \\ \cdot & \theta_i (\vartheta lx)^2 & \cdot & \cdot \end{array}$$

reductions:

A) $\Delta_\nu (j\nu x)^3$: reducer $a_\nu (j\nu x)^3$.

$\Delta_\nu (j\nu x)^2$ is reducible by the decomposable form $a_\nu (j\nu x)^2$ according to § 3. 2)b, taking account of the reducer $\theta_{\vartheta j}$. Indeed, according to § 11 B), one obtains:

$$\frac{3}{10} \Delta_\nu (j\nu x)^2 + \frac{3}{5} a_\nu (j\nu x) a_\vartheta (j\nu x) + \frac{1}{10} a_\nu (j\nu x)^2 = \frac{3}{5} \theta_\nu^3 \theta_{\vartheta j} + \frac{2}{5} \theta_\nu^3 \theta_{\vartheta j} \theta_{\vartheta j},$$

(reducer a_j and $\theta_{\vartheta j}$).

B) The forms $H_\vartheta (\vartheta' \eta x)^2$, $H_\vartheta^2 (\vartheta' \eta x)$, $H_\vartheta^2 (\vartheta' \eta x)^2$ decompose as arising by successive single folding with the decomposable forms H_ϑ and H_ϑ^2, respectively $H_\vartheta a_\eta$ and $H_\vartheta^2 a_\eta$ according to § 3. 2)b) and according to the relation:

$$H_\vartheta (\vartheta' \eta x)^2 = 2 H_\vartheta a_\eta^2 f - 2 H_\vartheta a_\eta b_\eta.$$

C) form sequence $a_\eta (j\eta x)^2$: reducers a_j and $(j\eta x)^3$.

form $a_\eta (j\eta x)$ decomposes: reducer a_j and single folding with the decomposable form $(j\eta x)^2$ according to § 3, 2)a (specialization $y = uv$).

form $a_\eta H_j$ decomposes: 1) by single folding with the decomposable form $a_\nu (j\nu x)^2$;

2) by a special twofold folding with the decomposable form $(j\eta x)^2$; from this one obtains a relation between decomposable forms.

From $a_\nu (j\nu x)^2$:

$$0 = \theta'_\nu \theta_\nu + 2 a_\eta H_j \mod (s, \varrho, t, i, J).$$

From $(j\eta x)^2$:

$$0 = \theta'_\nu \theta_\nu - \frac{3}{2} a_\eta H_j \mod (s, \varrho, t, i, J)$$

(products with contravariants are omitted), according to the ansatz:

$$0 = \frac{1}{2} a_\nu (abu)^2 \theta_\nu^3 + \frac{1}{2} u_x (ubu) \theta_{\nu_1}^3 (\theta bu) - \frac{3}{4} (\widehat{j} \eta bu)^2 + \frac{3}{4} H_j (\widehat{j} \eta bu) - \frac{1}{8} f H_j^2$$

and by interchanging the symbols in $a_\nu (ubu)(\theta bu) \theta_{\nu_1}^3$.

D) form sequence $N_\eta (NHu)$: reducer Δ_ν^2.

(NHu), $(n\eta x)$, $N_\eta (n\eta x)$, $H_\eta (NHu)$ decomposable forms (cf. § 12 B), $(NHu)^2$ decomposes by single folding with the form $(\Delta Hu)^2$.

E) form sequences $\theta_i L_\vartheta$, $\theta_i^2 (\theta Lu)^2$, $\theta_i^2 (\vartheta lx)$:

reducers: $\theta_\eta H_\vartheta$, $\theta_\eta^2 (\theta Hu)^2$, $\theta_\eta^2 (\vartheta \eta x)$.

forms (θLu), (ϑlx), L_ϑ, $L_\vartheta (\theta Lu)$, $\theta_i (\theta Lu)$, $L_\vartheta (\vartheta lx)$, $\theta_i (\vartheta lx)$, L_ϑ^2, $L_\vartheta^2 (\theta Lu)$, $\theta_i^2 (\theta Lu)$ decompose. (Dualistically opposite to the decomposable forms of § 12 B).

F) form sequence $\theta_\sigma (\vartheta \sigma x)$: reducer a_σ^2.

forms $(\vartheta \sigma x)$ and $(\vartheta \sigma x)^2$ decompose, as arising by single folding with the decomposable form a_σ; θ_σ, by twofold folding arising, decomposes, since the form sequence

$$a_{\nu_2}^2 (a\theta u) \theta_{\nu_1}^3 \equiv H_j^2 (j\eta x) \equiv 0 \mod u_x (s, \varrho, t, i, J):$$

$$\theta_\sigma = a_\sigma(abu)^2 = a_\nu^2(abu)^2\theta_\nu^3.$$

Consequences:

θ^3 or $\theta^2 K$ does not enter into folding with H; θ enters into folding with L only as a contravariant in powers or products, and not at all into folding with σ and $(\nu\sigma x)$.

N does not enter in products in folding with irgend einer form of System II enters, ebensowenig wie with powers or products of forms of System II. There remain an products in System I, taking account of the consequences of the last paragraphs, only products $f \cdot j$, $\Delta \cdot j$, powers of θ and products of $\theta \cdot K$.

§ 14. Forms of the ninth order (System III).

$$\text{Folding of} \begin{cases} \theta \cdot K \text{ with } H, \\ K, j, \Delta \text{ with } L, \\ j, \Delta \text{ with } \sigma, \\ f \text{ with } H^2. \end{cases}$$

Irreducible forms:

A) $(\vartheta \cdot k, \eta, x)^4,\ H_k(\vartheta \cdot k, \eta, x)^4$ B) $(KLu)^3,\ K_i(KLu)^3,\ K_i^2,\ (klx)^3,\ K_i(Klx)^3,$

C) $L_j,\ L_j^2$ D) $\Delta_i,$

E) $(j\sigma x)$ G) $(aH^2u)^3,\ (aH^2u)^4,\ a_\eta(aH^2u)^3.$

reductions:

A) the forms $H_\vartheta(k\eta x)^3$, $H_\vartheta^2(k\eta x)^2$, $H_\vartheta^2(k\eta x)^3$ decompose as arising by successive single folding with decomposable forms.

B) form sequences K_iL_k, $K_i^2(KLu)$, $K_i^2(klx)$:

reducers: $\theta_\eta H_\vartheta$, $a_\eta^2(aHu)$, $\theta_\eta^2(\vartheta\eta x)$.

forms L_i, $K_i(KLu)$, $K_i(klx)$, $(KLu)^2$, $(klx)^2$ decompose as a transvectionen over functional determinants (§ 3).

forms L_k, $L_k(KLu)$, $L_k(KLu)^2$, $L_k(klx)$, $L_k(klx)^2$, L_k^2, $L_k^2(KLu)$, $L_k^2(klx)$, L_k^3, $K_i(KLu)^2$, $K_i(klx)^2$ decompose as arising by single folding with the decomposable forms:

$$L_\vartheta,\ H_k(KHu),\ L_\vartheta(\vartheta lx),\ H_k^2,\ L_\vartheta^2,\ L_\vartheta^2(\theta Lu),\ K_\eta(KHu),\ \theta_i(\vartheta lx)$$

according to § 3. 2), a) and b).

C) form sequence $(jlx)^3$: reducer $(j\eta x)^3$.

forms $(jlx)^2$ and $L_j(jlx)$: arising by single folding with the decomposable form $(j\eta x)^2$.

D) form sequence $\Delta_i(\Delta Lu)$: reducer Δ_ν^2.

forms $(\Delta Lu)^2$, $(\Delta Lu)^3$: Single folding with the decomposable form $(\Delta Hu)^2$.

E) form sequence $(j\sigma x)^2$: reducer a_σ^2 by replacing j by lower forms of the form sequence (§ 5):

$$a_\sigma^2(a\theta u)^2 = \frac{1}{3}(j\sigma x)^2, \qquad a_\sigma^2(a\theta\sigma x)^2 = \frac{1}{3}(j\sigma x)^4.$$

F) form sequence Δ_σ^2: reducer a_σ^2.

form Δ_σ: reducer a_σ^2 by replacing Δ by lower forms of the form sequence:

$$a_\sigma a_\varrho^2 = \frac{2}{3}\Delta_\sigma \mod \nu.$$

Consequences: Only the form j enters in folding with σ and $(\nu\sigma x)$ enters.

N does not enter in folding with L enters, since the Δ_i correspondinglye form N_i decomposes.

§ 15. Forms of the tenth order (System III).

$$\text{Folding of} \begin{cases} \theta^2,\ f \cdot j \text{ with } L, \\ \theta \text{ with } H^2. \end{cases}$$

Irreducible forms:

A) $(\vartheta^2 lx)^4,\ \theta_i(\vartheta^2 lx)^4$ B) $(aLu)^2L_j,\ a_i(aLu)^2L_j,$

C) $(\theta H^2u)^3,\ (\theta H^2u)^4,\ H_\vartheta(\theta H^2u),\ \theta_\eta(\theta H^2u)^3.$

reductions:

A) and B): Decomposition of the remaining forms by single folding with decomposable forms.

C) Decomposition of the remaining forms according to § 13 B) by the dualistically opposite consideration.

Consequences:

$\theta^2 K$ does not enter into folding with L; correspondingly, $H^2 L$ does not enter into folding with K. The forms H^3 and $H^2 L$ do not enter into folding with θ. The folding scheme of § 7 is therefore proved.

§ 16. Forms of the 11th, 12th, 13th, and 15th orders (System III).

11th order.

$$\text{Folding of } \begin{cases} \theta \cdot K \text{ with } L, \\ K, j \text{ with } H^2, \\ j \text{ with } (\nu\sigma x), \\ f \text{ with } H \cdot L. \end{cases}$$

Irreducible forms:

$$\begin{array}{llll} \text{A)} & (\vartheta \cdot k, l, x)^5,\ \theta_i(\vartheta \cdot k, l, x)^5 & \text{B)} & (KH^2u)^4,\ K_\eta(KH^2u)^4, \\ \text{C)} & (\nu\sigma j), & \text{D)} & (a_\eta H \cdot L, u)^4. \end{array}$$

12th order.

$$\text{Folding of } \begin{cases} \theta^3 \text{ with } L, \\ \theta \text{ with } H \cdot L. \end{cases}$$

Irreducible forms:

$$\text{E) } (\vartheta^3 lx)^6, \qquad \text{F) } (\theta, H \cdot L, u)^4, \qquad L_\vartheta(\theta, H \cdot L, u^4).$$

13th order.

$$\text{Folding of } K \text{ with } H \cdot L.$$

Irreducible forms:

$$\text{G) } (K, H \cdot L, u)^5, \qquad K_\eta(K, H \cdot L, u)^5.$$

15th order.

$$\text{Folding of } K \text{ with } H^3.$$

Irreducible form:

$$\text{H) } (KH^3u)^6.$$

reductions:

The forms H_j^2, H_j^2 have reducer L_j^3, from the relation:

$$(\nu lx)L_x^3 = -\frac{1}{2}H^2 \mod(\sigma, s).$$

Decomposition or reducibility of the remaining forms follows from the considerations of the earlier paragraphs.

Consequences:

According to the folding scheme of § 7 and the consequences of §§ 8–15, the irreducible forms of System III (117 forms) are thereby exhausted.

Chapter IV. The relatively complete system mod (ϱ, t).

§ 17. Reduction of the modulus $(ss'u)^4$ and of the system of s.

For the formation of the next higher relatively complete system, one must, in analogy with § 7, form the system of s with respect to the next modulus

$$\nu(s) = (ss'u)^4$$

and fold it with the forms of the relatively complete system mod s. It will be shown that, after introduction of the modulus (ϱ, t) as the higher modulus,

A) the modulus $(ss'u)^4$ becomes reducible,
B) the system of s consists of the single form s.

A) The reduction of the modulus $(ss'u)^4$ follows at once from the reducer s_ν^2 found by formula (16.), § 11. From

$$s_\nu^2 = \frac{4}{3}ia_\nu^2 + \frac{4}{5}\theta_\varrho^2 + \frac{8}{5}(atu)^2 - \frac{26}{15}(\nu\varrho x)^2 + \frac{1}{6}J\cdot u_x^2 - \frac{2}{9}i^2\cdot u_x^2$$

there follows, by polarization from x to ν_1,

$$\begin{aligned}(ss'u)^4 &= -\frac{2}{3}J\cdot\nu + 6s_\nu^2 s_{\nu_1}^2\\ &= \frac{24}{5}\theta_\nu^2\theta_\varrho^2 + \frac{48}{5}a_\nu^2(atu)^2 - \frac{52}{5}H_\varrho^2 + \frac{4}{3}i\cdot(asu)^4 + \frac{1}{3}J\cdot\nu - \frac{4}{9}i^2\cdot\nu,\end{aligned} \tag{21}$$

and with this the reduction of the modulus $(ss'u)^4$.

B) The reduction of

$$Z = (ss'u)^2 = \theta(s)$$

and hence the reduction of the system of s is obtained by replacing the form Z by the lower form g_ν^2, according to formula (8.)b, § 7, taking into account the relation

$$s_\eta^2 = s_\nu^2(\nu\nu_1x)^2 - \frac{1}{3}Z.$$

The form g_ν^2 has, by formula (13.), § 11, the reducer s_ν^2 as a factor. By formula (8.) and (16.) one obtains

$$g_\nu^2 = -Z + \frac{14}{5}ia_\eta^2 + \frac{14}{15}i(asu)^2 \mod(\varrho, t),$$

and by formulas (13.), (14.), (15.), (16.),

$$\begin{aligned}g_\nu^2 &= 2[s_\vartheta^2]_{\nu^2} - \frac{4}{3}i\Delta_\nu^2 = 2s_\vartheta^2 s_\nu^2 - \frac{6}{5}[s_\vartheta^2(\theta su)^2]\widehat{\nu x}^{\,2} - \frac{4}{3}i\Delta_\nu^2\\ &= \frac{4}{5}i\cdot a_\eta^2 - \frac{2}{5}i\cdot(asu)^2 + \frac{1}{3}J\cdot\theta \mod(\varrho, t).\end{aligned}$$

Thus

$$Z = 2i\cdot a_\eta^2 + \frac{4}{3}i\cdot(asu)^2 - \frac{1}{3}J\cdot\theta \mod(\varrho, t). \tag{23}$$

The relatively complete system mod $((ss'u)^4, \varrho, t)$ therefore passes into a relatively complete system mod (ϱ, t), and from this, by transvection over the known system of two quadratic forms, the absolutely complete system arises. In order to form the relatively complete system mod (ϱ, t), since by formula (23.) the system of s reduces to the form s, we must transvect the relatively complete system mod (s, ϱ, t) over s and over powers of s. It will be shown that powers of s enter into folding only with System I.

The announced relation is

$$\begin{aligned}(ass')^4 &= \frac{24}{5}\theta_\nu^2\theta_\varrho^2 a_\nu^2 a_\varrho^2 + \frac{48}{5}a_\nu^2 b_\nu^2(tab)^2 - \frac{52}{5}H_\varrho^2 a_\nu^4 + \frac{5}{3}J\cdot i - \frac{4}{9}i^3,\\ (ass')^4 &= \frac{24}{5}a_\varrho^2 a_{\varrho'}^2 - \frac{12}{5}t_\varrho^2 + \frac{5}{3}J\cdot i - \frac{4}{9}i^3.\end{aligned} \tag{22}$$

[28] From formula (21.) there follows the relation, mentioned in the note to the introduction, between the three invariants of order 9, with use of formula (10.)c.

Among forms of the same order and the same degree, we define as "higher forms" those which have arisen from higher forms of the relatively complete system mod (s, ϱ, t) by folding with s, regarding the folding with s merely as a polarization of the original forms. Thus the order of the individual forms is uniquely determined (cf. § 1, form sequences 2). For instance,

$$(\theta_\eta(\theta Hu), s, u)^2 \text{ is higher than } s_\eta((\theta Hu)^2, s, u),$$

$$(\theta_\eta(\theta Hu)^2, s, u) \text{ is higher than } (\theta_\eta(\theta Hu), s, u)^2.$$

We divide the resulting system into the four subsystems

System s_I :	obtained by folding s and powers of s with System I;
System s_{II} :	obtained by folding s with System II;
System s_{III} :	obtained by folding s with the individual forms (without products) of System III and with products of one form each from Systems I and II;
System s_{IV} :	obtained by folding s with products of one form each from Systems I and III, II and III, III and III.

It should also be noted that from now on all formulas are to be understood mod (ϱ, t); from formula (27.) on, they are to be understood mod $(\varrho, t, i, J,$ higher forms), without this being expressly stated each time.

For the reduction we collect the formulas obtained earlier:

$$\begin{aligned}
s_\nu^2 &= \frac{4}{3} i a_\nu^2 + \frac{1}{6} J \cdot u_x^2 - \frac{2}{9} i^2 \cdot u_x^2, & s_\nu^3 &= \frac{1}{3} J \cdot u_x, \quad s_\nu^4 = J, \\
g &= 2 s_\vartheta^2 - \frac{4}{3} i \Delta, & s_\vartheta^2(\theta s u) &= 0, \\
s_\vartheta^2(\theta s u)^2 &= \frac{8}{9} i \cdot a_\nu^3 - \frac{4}{27} i^2 \cdot u_x^2, \\
Z &= 2i \cdot a_\eta^2 + \frac{4}{3} i (asu)^2 - \frac{1}{3} J \cdot \theta, \\
g_\nu &= -s_\eta + u_x \left\{ 2 i a_\eta^2 - \frac{2}{3} i (asu)^2 + \frac{2}{3} J \cdot \theta \right\}, \\
g_\nu^2 &= \frac{4}{5} i a_\eta^2 - \frac{2}{5} i (asu)^2 + \frac{1}{3} J \cdot \theta, & g_\nu^3 &= 0, \qquad g_\nu^4 = 0.
\end{aligned} \tag{24}$$

Consequences:

$$s_\nu^3, \quad s_\vartheta^2(\theta s u), \quad s_\vartheta^2 s_\nu, \quad s_\vartheta^2 \theta_\nu$$

are reducers.

§ 18. System s_I.

$$\text{Folding of} \quad \begin{cases} s \text{ with } f;\ \theta;\ K, j, \Delta;\ f \cdot j, N;\ \Delta \cdot j, \\ s^3 \text{ with } \theta, K. \end{cases}$$

Irreducible forms.

Order 5:

$$(asu), \qquad (asu)^2, \qquad (asu)^3, \qquad (asu)^4.$$

Order 6:

$$\begin{array}{cccc}
(\theta s u) & (\theta s u)^2 & (\theta s u)^3 & (\theta s u)^4 \\
s_\vartheta & s_\vartheta(\theta s u) & s_\vartheta(\theta s u)^2 & s_\vartheta(\theta s u)^3 \\
\cdot & s_\vartheta^2 & \cdot & \cdot
\end{array}$$

Order 7:

$$\text{A)} \begin{array}{cccc}
\cdot & (Ksu)^2 & (Ksu)^3 & (Ksu)^4 \\
s_k & s_k(Ksu) & s_k(Ksu)^2 & s_k(Ksu)^3 \\
\cdot & s_k^2 & \cdot & \cdot
\end{array} \qquad \begin{array}{l} \text{B) } s_j, \\ \text{C) } (\Delta s u),\ (\Delta s u)^2. \end{array}$$

Order 8:

$$\text{A)}\quad s_j(asu),\ s_j(asu)^2,\ s_j(asu)^3,$$
$$\text{B)}\quad (Nsu)^2,\quad s_n,\quad s_n(Nsu).$$

Order 10:

$$\text{A)}\ s_j(\Delta su),\qquad \text{B)}\ s_\vartheta(\theta s'u)^4.$$

Order 11:

$$(Ks^2u)^6,\qquad s_k(Ks^2u)^5,\qquad s_k(Ks^2u)^6.$$

Reductions:

The form sequences

$$s_\vartheta^2(\theta su),\quad s_k^2(Ksu),\quad s_j^2,\quad (\Delta su)^3,\quad (Nsu)^3$$

have the reducer $s_\vartheta^2(\theta su)$ as a factor; s_j^2 and $(\Delta su)^3$ are obtained by going back from j and Δ to lower forms of the form sequence:

$$\begin{aligned} s_\vartheta^2(\theta su)(a\theta u)^3 &= -\frac{1}{2}s_j^2, \\ s_\vartheta^2(\theta su)(a\theta u)^2 &= \frac{1}{3}(\Delta su)^3, \\ s_\vartheta^2(\theta su)^2(a\theta u)^2 &= \frac{1}{3}(\Delta su)^4+\frac{1}{3}s_j^2. \end{aligned}$$

Forms $s_\vartheta^2 s_{\vartheta'}$ and $s_\vartheta^2 s_{\vartheta'}^2$: reducers $s_\vartheta^2(\theta su)$ and $\theta_{\vartheta'}$:

$$\begin{aligned} s_\vartheta^2 s_{\vartheta'} &= s_\vartheta^2\theta_{\vartheta'} - s_\vartheta^2(\theta s\widehat{\vartheta}'x), \\ s_\vartheta^2 s_{\vartheta'}^2 &= s_\vartheta^2 s_{\vartheta'}\theta_{\vartheta'} - s_\vartheta^2 s_{\vartheta'}(\theta s\widehat{\vartheta}'x). \end{aligned}$$

Form sequence $s_j(\Delta su)^2$: reducers Δ_j and $(\Delta su)^3$.

Consequences:

s does not enter into powers in folding with f, j, Δ, N. The forms f, j, Δ, N enter into folding only in products with contragredient forms; however, in products with f those forms may still be quadratic in x, and in products with Δ they may still be linear in x. Thus the following products of forms are to be considered:

$$f\cdot(0,\lambda),\quad f\cdot(1,\lambda),\quad f\cdot(2,\lambda),\quad j\cdot(\mu,0),\quad N,\quad \Delta\cdot(0,\lambda),\quad \Delta\cdot(1,\lambda)\quad (\lambda>0),$$

where (μ,λ) denotes a form of degree μ in x and of degree λ in u.

Neither θ nor K enters into powers or products with arbitrary forms in folding with s or s^2. For products with the functional determinant K, $K\cdot\varphi_x$ ($\varphi\neq f$ or θ), the relation between decomposing forms is

$$K\cdot\varphi_x=(a\theta u)\varphi_x=f\cdot(\varphi\theta u)-\theta\cdot(\varphi au).$$

The above scheme for folding System s_I is thereby proved.

§ 19. System s_{II}.

Folding of s with ν; H; L; H^2.

Irreducible forms:

Order 6	Order 8	Order 10	Order 12
s_ν	$(Hsu),\ (Hsu)^2$	$(Lsu)^2,\ (Lsu)^3$	$(H^2su)^3$
.	s_η	s_l	.
.	$s_\nu^4=J$	.	.

Reductions:

Form sequences $s_\eta(Hsu)$ and $s_l(Lsu)$: reducer s_ν^2.

Form sequence s_σ: reducer s_ν^2 from H, $s_\nu^2=\dfrac{2}{3}s_\sigma$.

$(H^2su)^4$ decomposes by formula (7.)a (cf. § 11 A). Hence $(H\cdot L,s,u)^4$ also decomposes.

Consequences:

s^2 does not enter into folding with System II. The form ν enters only in products with contragredient forms. The form H, regarded as a covariant, enters into folding in products with forms $(1,\lambda)$, $(2,\lambda)$, $(3,\lambda)$, with arbitrary λ. The forms σ and $(\nu\sigma x)$ do not enter at all, and L, as a functional determinant, does not enter into products in folding; the full transvections over covariants of System III become reducible. The products of Systems I and II that remain are

$$f \cdot \nu, \qquad \Delta \cdot \nu.$$

§ 20. Forms of order 7 (System s_{III}).

Folding of s with $f \cdot \nu$, a_ν, a_ν^2.
Irreducible forms:

$$s_\nu(asu), \quad a_\nu(asu), \quad s_\nu(asu)^2, \quad a_\nu(asu)^2, \quad s_\nu(asu)^3, \quad a_\nu(asu)^3,$$
$$a_\nu s_\nu, \quad a_\nu s_\nu(asu), \quad a_\nu^2(asu), \quad a_\nu^2(asu)^2.$$

Reductions:

The evident reductions that arise by direct folding with the reducers s_ν^2 and $s_\eta(Hsu)$ will not be mentioned, nor will the decomposition of transvections with functional determinants.

For the forms $a_\nu s_\nu(asu)^2$ and $a_\nu^2 s_\nu(asu)^2$, see formula (19.), § 12; moreover,

$$a_\nu s_\nu(asu)^3 = a_\nu(a\widehat{\nu}_1\nu_2 u)^3(\nu_1\nu_2\nu) = 0.$$

Form sequence $a_\nu^2 s_\nu$: reducer $s_\vartheta^2(\theta su)$, obtained by going back from a_ν^2 to lower forms, i.e. by double reduction of the expressions

$$s_\vartheta^2(a\theta s)(a\theta u), \qquad s_\vartheta^2(a\theta s)(a\theta u)^2$$

according to the Ansatz

$$\begin{aligned}
s_\vartheta^2(a\theta s)(a\theta u) &= [(a\theta u)^2]s^3 + \frac{1}{5}[a_\vartheta(a\theta u)^2]s^2 + \frac{1}{60}[a_\vartheta^2(a\theta u)^2]s \\
&= \frac{1}{2}a_\nu^2 s_\nu - \frac{2}{3}a_\nu^2 s_\nu^2, \\
s_\vartheta^2(a\theta s)(a\theta u)^2 &= \frac{4}{5}[a_\vartheta(a\theta u)^2]_{us}s^2 + \frac{23}{180}[a_\vartheta^2(a\theta u)^2]_{us}s \\
&= -\frac{1}{2}a_\nu^2 s_\nu(asu).
\end{aligned}$$

Here $a_\nu^2 b_\nu(abu) = 0$.

The formulas obtained are

$$\begin{aligned}
a_\nu^2 s_\nu &= \frac{4}{9}i\, a_\nu^2 b_\nu, & a_\nu s_\nu(asu)^2 &= \frac{1}{3}i\theta_\nu^2 - \frac{1}{18}iH, \\
a_\nu s_\nu(asu)^3 &= 0, & a_\nu^2 s_\nu(asu) &= 0, \\
a_\nu^2 s_\nu(asu)^2 &= 0.
\end{aligned} \tag{25}$$

Consequences:

1) $a_\nu s_\nu(asu)^2$ and $a_\nu^2 s_\nu$ are reducers.

2) The values of the forms $(\Delta su)^3$, $(\Delta su)^4$, already recognized as reducible in § 19, are thereby obtained; likewise for the corresponding form sequence $(agu)^2$, which, when folded with ν, gives rise to double reduction through the reducer g_ν:

$$\begin{aligned}
&\text{a)} \quad (\Delta su)^3 = -\frac{6}{5}a_\nu^2(asu) + 3a_\nu s_\nu(asu) + 2i\theta_\nu(\theta\nu x), \\
&\text{b)} \quad (\Delta su)^4 = -\frac{2}{5}a_\nu^2(asu)^2 + \frac{4}{3}i\theta_\nu^2 + \frac{1}{9}iH.
\end{aligned} \tag{26}$$

From formula (24.) and (26.), mod (i, J) (cf. p. 71),

$$\begin{aligned}
&\text{a)} \quad (agu)^2 = \frac{2}{3}(\Delta su)^2 - \frac{4}{3}a_\nu s_\nu + \frac{3}{5}s \cdot a_\nu^2, \\
&\text{b)} \quad (agu)^3 = 2a_\nu s_\nu(asu) - a_\nu^2(asu)^2, \\
&\text{c)} \quad (agu)^4 = -a_\nu^2(asu)^2.
\end{aligned} \tag{27}$$

§ 21. Forms of 8th order (System s_{III}).

Folding of s with the forms of 4th order (System III). (§ 9.)

Irreducible forms:

$$\begin{array}{ll} (\vartheta\nu x)s_\nu, & ((\vartheta\nu x), s, u)^2, \\ ((\vartheta\nu x)^2, s, u), & ((\vartheta\nu x)^2, s, u)^2,\ \theta s_\nu, \\ (\vartheta\nu x)^2 s_\nu, & ((\vartheta\nu x)^2, s, u)s_\nu, \end{array}$$

$$\begin{array}{ll} ((\vartheta\nu x), s, u)^3,\ (\theta_\nu su)^2, & ((\vartheta\nu x), s, u)^4,\ (\theta_\nu su)^3, \\ ((\vartheta\nu x)^2, s, u)^3,\ (\theta_\nu(\vartheta\nu x), s, u)^2,\ (\theta_\nu^2 su), & ((\vartheta\nu x), s, u)^3,\ (\theta_\nu^2 su)^2,\ (\theta_\nu(\vartheta\nu x), s, u)^4. \end{array}$$

Reductions:

$((\vartheta\nu x), s, u)s_\nu$ decomposes as a transvection over functional determinants, by § 3.

Form sequence $((\vartheta\nu x), s, u)^2 s_\nu$: reducers s_ν^2 and $s_\vartheta^2\theta_\nu$:

$$(\widehat{\vartheta}\nu su)s_\nu(\theta su) = s_\nu s_\vartheta(\theta su) = -\frac{1}{2}(\widehat{\vartheta}\nu su)^2(\theta su),$$

$$\theta_\nu(\widehat{\vartheta}\nu su)s_\nu = \theta_\nu s_\nu s_\vartheta = -\frac{1}{2}\theta_\nu(\widehat{\vartheta}\nu su)^2.$$

Form sequence $\theta_\nu s_\nu(\theta su)$: reducer $a_\nu s_\nu(asu)^2$ by double reduction of the expressions

$$a_\nu s_\nu(asu)^2(abu) \quad \text{and} \quad a_\nu s_\nu(asu)^2(abu)(sbu);$$

$\theta_\nu s_\nu(\theta su)^3$ by double reduction of $(agu)^3(abu)(bgu)^3$.

Form sequence $\theta_\nu(\vartheta\nu x)s_\nu$: reducer $a_\nu^2 s_\nu$ from $\theta_\nu(\vartheta\nu x)s_\nu = a_\nu^2(abu)s_\nu$.

Form sequence $(\theta_\nu(\vartheta\nu x)^2, s, u)$: double reduction of the forms of the sequence $\theta_\nu(\widehat{\vartheta}\nu su)s_\nu$, which at the same time belong to the form sequences $((\vartheta\nu x)su)^2 s_\nu$ and $\theta_\nu(\vartheta\nu x)s_\nu$.

The form $(\theta_\nu(\vartheta\nu x), s, u)$ decomposes as a single transvection over the functional determinant $\theta_\nu(\vartheta\nu x) = a_\nu^2(abu)$.

The form $((\vartheta\nu x)^2, s, u)^4$: double reduction of the expression $(a\Delta u)^2(\Delta su)^4$, or also of $(\theta gu)^4$, from formulas (27.) and analogously to the reduction of $(j\eta x)^4$ (§ 12 C)).

Double reduction gives the formulas:

$$\begin{aligned} 0 &= \theta_\nu s_\nu(\theta su) - \frac{1}{6}(\widehat{\vartheta}\nu su)^2(\theta su) - \frac{1}{12}(Hsu), \\ 0 &= \theta_\nu s(\theta su)^2 - \frac{1}{4}(\widehat{\vartheta}\nu su)^2(\theta su)^2 - \frac{1}{24}(Hsu)^2, \\ 0 &= (\widehat{\vartheta}\nu su)^2(\theta su)^2 + 2\theta_\nu(\widehat{\vartheta}\nu su)(\theta su)^2 + 3\theta_\nu^2(\theta su)^2 - \frac{1}{3}H(su)^2, \\ 0 &= \theta_\nu s_\nu(\theta su)^3 + \frac{1}{2}\theta_\nu(\widehat{\vartheta}\nu su)(\theta su)^3, \\ 0 &= \theta_\nu s_\nu s_\vartheta - \frac{1}{4}s_\eta, \quad 0 = \theta_\nu s_\nu s_\vartheta(\theta su), \quad 0 = \theta_\nu s_\nu s_\vartheta(\theta su)^2. \end{aligned} \tag{28}$$

The last group corresponds to $((\theta_\nu(\vartheta\nu x)^2, s, u)^2, \ldots)$.

Consequences:

1) $\theta(\vartheta\nu x)s_\nu$, $(\widehat{\vartheta}\nu su)(\theta su)s_\nu$, $\theta_\nu(\widehat{\vartheta}\nu su)^2$, $(\widehat{\vartheta}\nu su)^2(\theta su)^2$ are reducers.

2) The forms of 4th order $(5, \lambda)$, $(6, \lambda)$ etc. do not enter into folding with s^2.

3) For the form sequence $(\theta gu)^2$, formulas (27.), with the reductions of this paragraph taken into account, give the following formulas:

$$\begin{aligned} &\text{a)} \quad (\theta gu)^2 = \frac{4}{3}\theta_\nu s_\nu + \frac{2}{3}s_\nu s_\vartheta - \frac{1}{3}s\theta_\nu^2 - \frac{1}{9}sH - \frac{1}{3}f\, a_\nu^2(asu)^2 + \frac{1}{3}a_\nu^2(asu)^2, \\ &\text{b)} \quad (\theta gu)^3 = -\frac{1}{3}(\widehat{\vartheta}\nu su)^2(\theta su) - \theta_\nu(\widehat{\vartheta}\nu su)(\theta su) - \theta_\nu^2(\theta su), \\ &\text{c)} \quad (\theta gu)^4 = 2\theta_\nu^2(\theta su)^2 - \frac{1}{3}(Hsu)^2, \\ &g\vartheta(\theta gu) = (\widehat{\vartheta}\nu su)(\vartheta\nu x)s_\nu - \theta_\nu(\vartheta\nu x)(\widehat{\vartheta}\nu su), \\ &g\vartheta(\theta gu)^2 = \frac{2}{3}s\theta_\nu^3, \quad g\vartheta(\theta gu)^3 = \frac{1}{3}\theta_\nu^3(\theta su), \quad g\vartheta(\theta gu)^4 = 0. \end{aligned} \tag{29}$$

§ 22. Forms of 9th order (System s_{III}).

Folding of s with the forms of 5th order (System III) and the product $\Delta \cdot \nu$ (§ 10).
Irreducible forms:

A) $(k\nu x)^2 s_\nu, ((k\nu x)^3, s, u), (k\nu x)^3 s_\nu, ((k\nu x)^2, s, u)^2, K_\nu s_\nu,$
$((k\nu x)^3, s, u)^2, ((k\nu x)^2, s, u) s_\nu, (K_\nu (k\nu x)^3, s, u),$
$(K_\nu su)^2,\ (K_\nu su)^3,\ (K_\nu su)^4, ((k\nu x)^2, s, u)^3, (K_\nu^2 su)^4,$

B) $(j\nu x) s_\nu, ((j\nu x)^2, s, u), (j\nu x)^3, s, u, ((j\nu x)^2, s, u)^2,$

C) $s_\nu(\Delta su), \quad (\Delta_\nu su),$

D) $(aHu)s_\eta, (a_\eta su), ((aHu), s, u)^2, ((aHu)^2, s, u),$
$(aHu)^2 s_\eta, (a_\eta su)^2, (a_\eta^2 su), ((aHu), s, u)^3, ((aHu)^2, s, u)^2,$
$((aHu), s, u)^4, (a_\eta su)^3, (a_\eta (aHu), s, u)^2, (a_\eta (aHu)^2, s, u), (a_\eta (aHu), s, u)^3.$

Reductions:

A) The reductions follow directly from the reductions of § 21 by single folding. The form sequence $K_\nu s_\nu (Ksu)^2$ has the reducers $a_\nu s_\nu (asu)^2$ and $\theta_\nu s_\nu (\theta su)^2$ as factors. Hence the higher forms $K_\nu^2 (Ksu)^2$, $K_\nu^2 (Ksu)^3$ decompose according to the Ansatz:

$$0 = a_\nu s_\nu (asu)^2 (a\theta u) = K_\nu s_\nu (Ksu)^2 - \theta_\nu s_\nu (\theta su)^2 (a\theta u).$$

B) Form sequence $(j\nu x)^2 s_\nu$: reducer $a_\nu^2 s_\nu$ from

$$(j\nu x)^2 s_\nu = 3 a_\nu^2 s_\nu (a\theta u)^2.$$

$((j\nu x)^3, s, u)^2$: reducer $a_\nu^2 s_\nu$ from

$$((j\nu x)^3, s, u)^2 = -6 a_\nu^2 s_\nu (a\theta u)^2 (\theta su).$$

C) Forms $\Delta_\nu (\Delta su)^2$ and $s_\nu (\Delta su)^2$: reducer g_ν from formula (27.)a.

Form $\Delta_\nu s_\nu$: 1) reducer $a_\nu^2 s_\nu$ from $a_g a_\nu^2 s_\nu = \frac{2}{3} \Delta_\nu s_\nu$; 2) decomposes by formula (27.)a through double reduction of the expression $(\Delta s \widehat{\nu} x)^2$.

Hence the higher form $a_\eta s_\eta$ decomposes.

D) Form sequence $(aHu)^2 s_\eta (asu)$: reducer $a_\nu s_\nu (asu)^2$ by double reduction of the form sequence $[a_\nu s_\nu (asu)^2]_{\nu_1 x^2}$; reduction of $(aHu)^2 s_\eta (asu)^2$ from $a_\nu s_\nu (asu)^2$ and $a_\nu s_\nu (asu)^3$. Hence the reduction of $(a_\eta, s, u)^4$.

Form sequence $a_\eta (aHu) s_\eta$: reducers $a_\nu^2 s_\nu$, $a_\eta^2 s_\eta$ by double reduction of the expression $(\Delta s \widehat{\nu} x)^3$, since $a_\eta^2 s_\eta$ does not arise by folding from the reducible term $a_\nu^2 s_\nu (\nu \nu_1 x)$ of the form $a_\eta (aHu) s_\eta$.

Form sequence $(a_\eta^2 su)^2$: reducer $a_\nu^2 s_\nu$ from

$$a_\nu^2 s_\nu s_{\nu_1} = -\frac{1}{2} (a_\eta^2 (Hsu))^2.$$

The form $(a_\eta (aHu), s, u)$ decomposes as a transvection over a functional determinant.

The reduction formulas obtained are:

$$\begin{aligned}
&\text{a)} \quad 0 = (aHu)^2 (asu) s_\eta - a_\eta (asu)(Hsu)^2 - a_\eta (aHu)(asu)(Hsu), \\
&\text{b)} \quad 0 = (aHu)^2 (asu)^2 s_\eta - a_\eta (aHu)(asu)^2 (Hsu), \\
&\text{c)} \quad 0 = a_\eta (asu)^2 (Hsu)^2, \\
&\qquad 0 = a_\eta s_\eta + \frac{1}{4} a_\nu^2 g - \frac{1}{4} s \cdot a_\eta^2, \\
&\qquad 0 = a_\eta (aHu) s_\eta - \frac{1}{2} a_\eta^2 (Hsu), \quad 0 = a_\eta^2 (Hsu)^2, \ldots, \quad 0 = a_\eta^2 s_\eta.
\end{aligned} \tag{30}$$

Consequences:

1) $(j\nu x)^2 s_\nu$, $\Delta_\nu s_\nu$, $\Delta_\nu (\Delta su)^2$, and consequently $N_\nu (Nsu)^2$, $(aHu)^2 (asu) s_\eta$, $a_\eta (aHu) s_\eta$, $a_\eta^2 (Hsu)^2$, are reducers; $a_\eta s_\eta$ is a decomposable form that passes into decomposable forms under all single and double foldings.

2) The forms of 5th order $(5, \lambda)$, $(6, \lambda)$ etc. do not enter into folding with s^2.

§ 23. Forms of the tenth order (System s_{III}).

Folding of s with the forms of the sixth order (System III) (§ 11).
Irreducible forms:

A) $(\vartheta^2\nu x)^3 s_\nu, ((\vartheta^2\nu x)^3, s, u), ((\vartheta^2\nu x)^3, s, u)^2, (\vartheta^2\nu x)^3 s_\nu s_\vartheta, \theta_\nu(\vartheta^2\nu x)^3 s_\vartheta,$
B) $a_\nu(j\nu x)s_\nu, (a_\nu(j\nu x), s, u)^2, (a_\nu(j\nu x), s, u)^3, (a_\nu(j\nu x), s, u)^4, (a_\nu^2(j\nu x), s, u)^2, (a_\nu^2(j\nu x), s, u)^3,$
C) $((\vartheta\eta x)^2, s, u), ((\vartheta\eta x), s, u)^2, (\theta Hu)s_\eta, (\theta_\eta su), ((\theta Hu), s, u)^2, ((\theta Hu)^2, s, u),$
$((\theta\eta x), s, u)^3, (\theta_\eta su)^2, (\theta_\eta^2 su), ((\theta Hu), s, u)^3, ((\theta Hu)^2, s, u)^2, (H_\vartheta(\theta Hu), s, u)^2,$
$(\theta_\eta(\theta Hu), s, u)^2, (\theta_\eta(\theta Hu)^2, s, u), ((\theta Hu), s, u)^4, (\theta_\eta su)^4, (\theta_\eta(\theta Hu), s, u)^3, (\theta_\eta^2(\theta Hu), s, u)^2.$

Reductions: A) The forms $((\vartheta^2\nu x)^3, s, u)^2$, $((\vartheta^2\nu x)^3, s, u)^3$ decompose:

$$(\vartheta\nu x)(\widehat{\vartheta\nu su})(\widehat{\vartheta\nu su}) = (\vartheta\nu x)s_\vartheta s_\nu - (\vartheta\nu x)s_\nu s_\vartheta = -2\theta(\vartheta\nu x)s_\nu s_\vartheta + (\vartheta\nu x)s_\vartheta^2.$$

The remaining forms either have reducers as factors or are single foldings with decomposed forms.

B) Reducer $a_\nu^2 s_\nu$ and $(\widehat{j\nu su})s_\nu$.

C) 1. The form sequence $(\theta Hu)^2 s_\eta$: a) reducer $(aHu)^2(asu)s_\eta$; b) double reduction of the form sequences $(\theta gu)^3 g_\nu$ and $g_\nu(agu)^3(bgu)^3(abu)$. Hence reduction of the higher forms $(\theta, s, u)^3$, $(H_\vartheta(\theta Hu), s, u)^3$ and $(a_\nu b_{\nu_1}^2, s, u)^4$ [the latter form replacing $(H_\vartheta su)^4$].

2. $(\vartheta\eta x, s, u)^4$: reducer $(a_\eta su)^4$.

3. $((\vartheta\eta x)^2, s, u)^3$: reducer $s_\vartheta^2 s_\nu$ from $(\widehat{\vartheta\eta su})^2(Hsu)$. The forms $(\vartheta\eta x)s_\eta$, $(\vartheta\eta x)^2 s_\eta$, $((\vartheta\eta x)^2, s, u)^2$, $\theta_\eta s_\eta$ decompose by folding with the decomposed form $a_\eta s_\eta$.

4. Form sequences $H_\vartheta(\theta Hu)s_\eta$, $\theta_\eta(\theta Hu)s_\eta$, $H_\vartheta(\vartheta\eta x)s_\eta$, $\theta_\eta(\vartheta\eta x)s_\eta$: reducers $a_\eta(aHu)s_\eta$ and $a_\eta^2 s_\eta$.

The form sequence $(\theta_\eta(\vartheta\eta x), s, u)^2$: reducer $a_\eta^2(Hsu)^2$ [except for $(\theta_\eta^2(\theta Hu), s, u)^2$, which arises from the term $a_\eta^2(\theta su)(Hsu)$ by folding].

5. $(H_\vartheta(\vartheta\eta x), s, u)$, $(H_\vartheta(\vartheta\eta x), s, u)^2$: double reduction of $a_\eta(aHu)s_\eta(abu)$ and $g_\vartheta(\theta gu)g_\nu$.

The form sequence $(H_\vartheta(\vartheta\eta x), s, u)^3$: reducer $((\vartheta\nu x)^2, s, u)^2 s_\nu$.

6. $(H_\vartheta(\theta Hu), s, u)$ decomposes by single folding with the decomposed form $(\theta_\nu(\vartheta\nu x), s, u)$ according to § 3. 2), c), that is, by double reduction of the lower decomposed form $(H_\vartheta su)^2$:[29]

$$\begin{aligned}
0 &= \theta_\nu(\vartheta\nu_1)(\theta su) + \frac{2}{3}\theta_\nu^2(\vartheta\nu_1 x)(\theta su) + \theta_\nu(\vartheta\nu x)(\theta su)s_{\nu_1} \\
&\equiv -\frac{1}{2}H_\vartheta(\theta Hu)(\theta su) + \theta_\eta(\theta su)(Hsu) + \frac{1}{2}H_\vartheta(\theta su)(Hsu) \\
&\equiv -\frac{1}{2}H_\vartheta(\theta Hu)(\theta su) + \theta_\eta(\theta su)(Hsu) + \frac{1}{2}[H_\vartheta]_{sx}^2 - \frac{1}{2}\frac{3}{5}H_\vartheta(\theta Hu)(\theta su) \\
&\equiv -\frac{3}{5}H_\vartheta(\theta Hu)(\theta su).
\end{aligned}$$

By double reduction, according to 1. and 5., the following formulae arise:

$$\begin{aligned}
0 &= \theta_\eta(\theta su)(Hsu)^2 + \frac{1}{4}H_\vartheta(\theta Hu)(\theta su)^2 \\
&\quad - \frac{3}{4}\theta_\eta(\theta Hu)(\theta su)(Hsu) - \frac{5}{4}\theta_\eta(\theta Hu)^2(\theta su)^3 \\
&\quad - (asu)^3 b_\eta(\theta Hu)^2 - a_\nu(asu)^3 b_\nu^2, \\
0 &= H_\vartheta(\theta Hu)(\theta su)^3 - 7\theta_\eta(\theta Hu)(\theta su)^2(Hsu), \\
0 &= a_\nu(asu)^2 b_{\nu_1}^2(bsu)^2 - \theta_\eta(\theta su)^2(Hsu)^2 \\
&\quad + 6\theta_\eta(\theta Hu)(\theta su)^2(Hsu), \\
0 &\equiv (H_\vartheta(\vartheta\eta x), s, u), \qquad 0 \equiv H_\vartheta(\widehat{\vartheta\eta su})(Hsu) - 4\theta_\eta^2(Hsu).
\end{aligned} \tag{31}$$

Consequences:

1. $(\theta Hu)^2 s_\eta$, $H_\vartheta(\theta Hu)s_\eta$, $\theta_\eta(\theta Hu)s_\eta$, $H_\vartheta(\vartheta\eta x)s_\eta$, $\theta_\eta(\vartheta\eta x)s_\eta$, $(H_\vartheta(\vartheta\eta x), s, u)$, $(\theta_\eta(\vartheta\eta x), s, u)^2$ are reducers.

2. s^2 does not enter into folding with the forms $(5, \lambda)$, etc., of order 6.

[29]The sign $\equiv$ means that products of forms of lower degree have been omitted.

§ 24. Forms of the 11th order (System s_{III}).

Folding of s with the forms of the 7th order (System III) (§ 12).
Irreducible forms:

A) $((k\eta x)^3, s, u)$, $(K_\eta(k\eta x)^3, s, u)$, $(K_\eta su)^2$, $((KHu)^2 su)^2$,
$((KHu)^2, s, u)^3$, $((KHu)^2, s, u)^4$, $(K_\eta(KHu)^2, s, u)^3$,

B) $((j\eta x), s, u)^2$, $(H_j su)$, $((j\eta x), s, u)^3$,

C) $(\Delta_\eta su)$,

D) $(aLu)^2 s_\eta$, $(a_\eta su)^2$, $((aLu)^2, s, u)^2$, $((aLu)^3, s, u)^2$, $((aLu)^3, s, u)^3$, $(a_l(aLu)^2, s, u)^2$.

Reductions:

A) Reduction or decomposition by single folding with the corresponding forms in the symbols $\theta - H$.

$(K_\eta(KHu)^2, s, u)$ and $(K_\eta(KHu)^2, s, u)^2$: decomposition according to § 3. 2), a), by folding with $K_\nu^2(Ksu)$, $K_\nu^2(Ksu)^2$.

$(K_\eta(KHu), s, u)^4 \equiv (a_\nu^2\theta_{\nu_1}, s, u)^4$: decomposition according to § 3. 2), b), from the decomposed form $K_\nu^2(Ksu)^3$.

B) Form sequence $(j\eta x)s_\eta$: reducer $(j\nu x)^2 s_\nu$.

Form $H_j(\widehat{j\eta su})(Hsu)$: reducer $(j\nu x)(\widehat{j\nu su})^2$.

Form $H_j(j\eta x)(Hsu)$: decomposes by reducing the decomposed form $((j\eta x)^2, s, u)^2$ by the reducer $(j\nu x)^2 s_\nu$ (formula (18.)a):

$$0 = -2(j\nu x)^2 s_\nu s_{\nu_1} = [(j\eta x)^2]_{su^2} + H_j(j\eta x)(Hsu).$$

C) Reducers $\Delta_\nu s_\nu$ and $\Delta_\nu(\Delta su)^2$.

D) Reduction and decomposition by single folding with the corresponding forms in the symbols $f - H$.

E) From (30.)c one obtains the reduction of the decomposed form sequence $(a_\sigma su)^2$:

$$0 = a_\eta(Hsu)^2(as\widehat{\nu x})^2 = a_\eta(Hsu)^2(a\widehat{\nu\eta}u)^2 + 2a_\eta(a\widehat{\nu\eta}u)(\widehat{\eta\sigma}u)(Hsu)^2 = \frac{1}{3}a_\sigma(asu)^2,$$

$$0 = a_\eta a_\nu(Hsu)^2(as\widehat{\nu x})(asu) = a_\eta(a\widehat{\nu\eta}u)^2(Hsu)^2(asu) + a_\eta(a\widehat{\nu\eta}u)(\widehat{\eta\sigma}u)(Hsu)^2(asu) = \frac{1}{3}a_\sigma(asu)^3.$$

Consequence: s^2 does not enter into folding with the forms of order 7.

§ 25. Forms of the 12th, 13th, 14th, and 15th orders (System s_{III}).

Folding of s with the forms of orders 8, 9, 10, 11 (System III) (§§ 13–16).
Irreducible forms:
12th order.

A) $((\vartheta^2\eta x)^4, s, u)$, $\theta_\eta(\vartheta^2\eta x)^3 s_\vartheta$,

B) $((aHu)H_j, s, u)^2$, $((aH\cdot u)H_j, s, u)^3$,

C) $(\theta_\nu su)^2$, $((\theta Lu)^2, s, u)^2$, $((\theta Lu)^3, s, u)$, $((\theta Lu)^2, s, u)^3$, $(\theta_l(\theta Lu)^2, s, u)^2$.

Reductions: The reductions arising by direct folding with reducers or decomposed forms are no longer to be mentioned.

B) Decomposition of $((aHu)H_j, s, u)$ and $(a_\eta(aHu)H_j, s, u)$ as transvections over functional determinants.

Decomposition of $(a_\eta(aHu)H_j, s, u)^2$: double reduction of the decomposed form $[\theta_\nu\theta_{\nu_1}^3]_{su^3}$ (§ 13. C):

$$\begin{aligned} 0 &\equiv [\theta_{\nu_1}^3\theta_\nu]_{su^3} \equiv -\frac{3}{4}\theta_\nu(\theta su)^2(\theta\theta' u)\theta_{\nu_1}^3 \\ &\equiv -\frac{9}{8}(\theta\theta'\eta su)(\theta\theta' u)(\theta Hu)(\theta' Hu)\theta'_\nu(\theta su) \equiv -\frac{3}{8}a_\eta(aHu)H_j(asu)^2. \end{aligned}$$

C) From formulae (31.) follows the reduction of the decomposed forms

$$[L_\vartheta(\theta Lu)]_{su^3} \quad \text{and} \quad [\theta_l(\theta Lu)]_{s^2u^3},$$

that is, of the products $[\theta_\nu^2 H]_{su^3}$ and $[a_\nu^2(bHu)^2]_{su^3}$:

$$\begin{aligned}
&0 \equiv \theta_\nu^2(\theta su)^2(Hsu), 0 && \equiv [L_\vartheta(\theta Lu)]_{su^4} - 7[\theta_l(\theta Lu)]_{su^4},\\
&0 \equiv a_\nu^2(asu)^2(bHu)^2(bsu) \\
&\quad + 6\theta_l(\theta Lu)^2(\theta su)(Lsu), \\
&0 \equiv \theta_\nu^3(\theta su)(Hsu)^2 \\
&\quad \text{from } 0 \equiv \theta_l^2(\theta Lu)(\theta su)(Lsu)^2 \\
&\quad (\text{reducer } a_\eta^2(Hsu)^2).
\end{aligned} \tag{32}$$

13th order.

$$\text{A)}\ ((KLu)^3, s, u)^2,\ ((KLu)^3, s, u)^3, \qquad \text{B)}\ (L_j su)^2, \qquad \text{C)}\ ((aH^2u)^3, s, u)^2.$$

14th order.

$$\text{A)}\ ((aLu)^2 L_j, s, u)^2, \qquad \text{B)}\ ((\theta H^2 u)^3, s, u)^2.$$

15th order.

$$((KH^2u)^4, s, u)^2.$$

Reductions: Decomposition of $(K_l(KLu)^3, s, u)^2$, $(H_\vartheta(\theta H^2u)^3, s, u)$, $((K, H \cdot L, u)^5, s, u)$ according to § 3. 2), c); that is, with simultaneous consideration of the decomposed forms $(K_l(KLu)^2, s, u)^3$, $(H_\vartheta(\theta H'u)^2, s, u)^2$, $((K, H \cdot L, u)^4, s, u)^2$.
Consequence: s^2 does not enter into folding with individual forms of System III.

§ 26. System s_{IV}.

1) *Folding of s with System I–III.*
Reductions: According to the reductions of the last paragraphs ($a_\nu^2 s_\nu(aHu)^2(asu)s_\eta$, etc.), the forms $(0, \lambda)$, $(1, \lambda)$, $(2, \lambda)$ do not permit foldings I and III simultaneously; hence, by § 18, the forms f, Δ, N do not enter in products in folding to form System s_{IV}.

The form j does not enter into folding, since, by the same reductions, the foldings contragredient to j,

$$(K_\nu(k\nu x)^3, s, u), \qquad ((\vartheta^2\eta x)^4, s, u) \quad \text{etc.},$$

correspond to the foldings cogredient to j:

$$K_\nu(k\nu x)^2 s_k, \qquad (\vartheta^2\eta x)^3 s_\vartheta \quad \text{etc.}$$

2) *Folding of s with System II–III*, that is, of s with $H \cdot (1, \lambda)$, $H \cdot (2, \lambda)$, $H \cdot (3, \lambda)$.
Irreducible forms:

$$\begin{aligned}
\text{A)}\quad & (a_\nu H, s, u)^4, ((aHu)^2H, s, u)^3, ((aHu)^2H, s, u)^4,\\
& ((aLu)^2H, s, u)^4, ((aLu)^3H, s, u)^3, ((aH^2u)^3H, s, u)^4,\\
\text{B)}\quad & (a_\nu^2 H, s, u)^3, (a_\eta(aHu)^2H, s, u)^3, (a_l(aLu)^2H, s, u)^4,\\
\text{C)}\quad & (\theta_\nu H, s, u)^4, ((\theta Hu)^2H, s, u)^3, ((\theta Hu)^2H, s, u)^4,\\
& ((\theta Lu)^2H, s, u)^4, ((\theta Lu)^3H, s, u)^3, ((\theta H^2u)^3H, s, u)^4,\\
\text{D)}\quad & (\theta_\eta(\theta Hu)^2H, s, u)^3, (\theta_l(\theta Lu)^2H, s, u)^4,\\
\text{E)}\quad & ((KLu)^3H, s, u)^4, ((KH^2u)^4H, s, u)^4,\\
\text{F)}\quad & ((j\nu x)^2H, s, u)^3, ((j\nu x)^2H, s, u)^4, ((j\eta x)H, s, u)^4,\\
& (H_jH, s, u)^3, (L_jH, s, u)^4.
\end{aligned}$$

Reductions: ν does not enter into folding, analogously to j.

B) and D). $(a_\nu^2 H, s, u)^4$ and $(\theta_\nu^2 H, s, u)^4$ decompose by formula (27.)c) and (29.)c), taking into account $(gHu)^2 = -\frac{1}{3}s\sigma$:

$$(a_\nu^2 H, s, u)^4 = \frac{1}{3}(asu)^4\sigma, \qquad (\theta_\nu^2 H, s, u)^4 = -\frac{1}{6}(\theta su)^4\sigma;$$

hence decomposition of $(a_\eta(aHu)H, s, u)^4$, $(\theta_\eta(\theta Hu)H, s, u)^4$.

$(\theta_\nu^2 H, s, u)^3$, $(\theta_\nu^3 H, s, u)^3$, and hence $(\theta_\eta^2(\theta Hu)H, s, u)^4$, decompose according to § 25, forms of order 12, C).

3) *Folding of s with System III–III*, that is, of s with forms of System III:

$$(3,\lambda)(3,\lambda), \quad (3,\lambda)(2,\lambda), \quad (3,\lambda)(1,\lambda), \quad (2,\lambda)(2,\lambda), \quad (2,\lambda)(1,\lambda), \quad (1,\lambda)(1,\lambda).$$

Irreducible forms:

$$\begin{aligned}
&\text{A)} \quad (a_\nu^2 a_\nu^2, s, u)^3, (a_\nu^2 a_\nu^2, s, u)^4, (a^2 a_\eta (aHu)^2, s, u)^3,\\
&\text{B)} \quad (a_\nu H_j, s, u)^4, ((aHu)^2 H_j, s, u)^3, ((aLu)^2 H_j, s, u)^4, ((aLu)^3 H_j, s, u)^2, ((aH^2u)^3 H_j, s, u)^3.
\end{aligned}$$

Reductions:

Products II–III, for the same order in the symbols ν and f, are to be considered higher than products III–III.

The reductions arise partly from relations between products (syzygies), partly by replacing the products by the corresponding decomposed forms and reducing the folding of these forms with s. Products containing contravariants are always omitted, since they pass into products.

A) f quadratic, symbols ν of arbitrary order. By § 11, formulae (12.), and by analogous computation, one has:

$$\begin{aligned}
&1) \quad 0 = a_\nu b_{\nu_1} + \frac{1}{2} f (aHu)^2 - \frac{1}{4}\theta H + \frac{1}{2} u_x H_\vartheta - \frac{1}{4} u_x^2 H_\vartheta^2,\\
&2) \quad 0 = a_\nu b_{\nu_1}^2 + f a_\eta (aHu)^2 - \theta_\eta - \frac{1}{2} H_\vartheta,\\
&3) \quad 0 = a_\nu^2 b_{\nu_1}^2 - H_\vartheta^2 + 2\theta_\eta^2.
\end{aligned}$$

It follows that

$$\begin{aligned}
4) \quad 0 &= a_\nu^2 b_{\nu_1}^2 (j\nu_1 x),\\
5) \quad L_\vartheta(\theta L u) &= -a_\nu b_\eta (bHu)^2 + \frac{3}{4} a_\nu^2 (bHu)^2 + \frac{3}{4}\theta_\nu^2 H,\\
6) \quad L_\vartheta(\theta L u) &= -6 a_\nu b_\eta (bHu)^2 + 2 a_\nu^2 (bHu)^2 - \frac{1}{2}\theta_\nu^2 H,\\
7) \quad 0 &= a_\nu (bHu)^2 + f(aLu)^3 + \theta_\nu H,\\
8) \quad 0 &= a_\nu (bLu)^3 + \frac{3}{4}(\theta H u)^2 H,\\
9) \quad 0 &= (aHu)^2 (bHu)^2 + 2(\theta H u)^2 H,\\
10) \quad 0 &= a_\eta (aHu)^2 b_\eta (bHu)^2,\\
11) \quad 0 &\equiv [a_\nu^2 b_\eta (bHu)]_{\widehat{su^4}}.
\end{aligned}$$

The reductions of

$$(H_\vartheta s u)^4, \qquad (L_\vartheta(\theta L u), s, u)^3, \qquad (L_\vartheta(\theta L u), s, u)^4$$

(formulae (31.) and (32.)) together with formulae 1) to 11) give the reduction of all foldings of s with products that are quadratic in the symbols f and arbitrary in ν.

B) Products of two of the symbols f, θ, j; ν in arbitrary order. By formulae (17.), (18.), (20.), and by direct computation, the relations are:

$$\begin{aligned}
&1) \quad 0 = a_\nu \theta_{\nu_1} + \frac{1}{4}\theta (aHu)^2 + \frac{1}{4} f (\theta H u)^2,\\
&2) \quad 0 = \theta_\nu \theta_{\nu_1} + \frac{1}{2}\theta(\theta H u)^2.
\end{aligned}$$

Therefore:

$$\begin{aligned}
&3) \quad 0 = 3 a_\nu (\theta H u)^2 + \theta (aLu)^3 + 2 f (\theta L u)^3,\\
&4) \quad 0 = 3\theta_\nu (aHu)^2 + f(\theta L u)^3 + 2\theta (aLu)^3,\\
&5) \quad 0 = a_\nu (\theta L u)^3, \qquad\qquad 0 = \theta_\nu (aLu)^3, \qquad 0 = (aHu)^2 (\theta H u)^2,\\
&6) \quad 0 = a_\nu \theta_\nu^2 + \theta_\nu a_\nu^2 + f\theta_\eta (\theta H u)^2 + \theta a_\eta (aHu)^2,\\
&7) \quad 0 = \theta_\nu \theta_\nu^2 + \theta\theta_\eta (\theta H u)^2 + \frac{1}{6} f H_j,
\end{aligned}$$

8) $0 = a_\nu (j\nu x)^2 + f H_j,$

9) $0 = (aHu)^2 (j\nu x)^2 - 2a_\nu H_j,$

10) $0 = \theta_\nu (j\nu x)^2,$ $\qquad 0 = \theta H_j,$

11) $0 = a_\nu \theta_\nu^3 - \frac{3}{4}(j\eta x)^2,$

12) $0 = a_\nu^2 \theta_\nu^2 - \frac{1}{3}(j\eta x)^2 - \frac{1}{3}(\Delta H u)^2,$

13) $0 = a_\nu^2 \theta_\nu^3 - \frac{1}{2} a_\sigma,$

14) $0 = \theta_\nu \theta_\nu^3,$ $\qquad 0 = a_\eta H_j.$

The formulae have been carried far enough for the products to become polarizable: that is, by folding only one new product arises, because a) the folding of the product into itself becomes reducible, and b) the remaining products arising by folding become reducible or lead to contravariants (cf. § 3. 2), a) and b)).

Formulae 1) to 5) give the reduction of all foldings of s with products arising from $a_\nu \theta_\nu$ by folding. Formulae 6) to 10) give the reduction of those products arising from $a_\nu \theta_\nu^2$ by folding. By § 24 A), taking formulae 6) to 10) into account, the foldings of s with products arising from $a_\nu^2 \theta_\nu$ decompose.

One has:

$$0 \equiv a_\nu^2 (asu)^2 \theta_{\nu_1} (\theta su)^2, \qquad 0 \equiv a_\nu^2 (asu) \theta_{\nu_1} (\theta su)^3 - K_\eta (KHu)^2 (Ksu)^3,$$
$$0 \equiv a_\nu^2 (asu)^2 \theta_{\nu_1} (\theta su), \qquad 0 \equiv a_\nu^2 (asu) \theta_{\nu_1} (\theta su)^2,$$
$$0 \equiv a_\nu^2 (asu) \theta_{\nu_1} (\theta su).$$

The reducers

$$((j\eta x)^2, s, u)^3 \text{ [from } (\widehat{\vartheta\eta su})^2 (Hsu),\ \S\ 23.\ C3],$$
$$(H_j (j\eta x), s, u)^2 \ [\S\ 24,\ B], \quad ((\Delta H u)^2, s, u)^2 \ [\S\ 24,\ C], \quad (a_\sigma su)^2 \ [\S\ 24,\ E]$$

together with formulae 11) to 14) give the reduction of all products arising from $a_\nu \theta_{\nu_1}^3$, $a_\nu^2 \theta_{\nu_1}^2$, $a_\nu^2 \theta_{\nu_1}^3$.

Our preceding computations can be summarized in the following *conclusion*:

The relatively complete system modulo (ϱ, t) consists of the following 331 forms and subsystems, which are also reproduced below in tabular order.

Table I (cf. p. 90)

$u \backslash x$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
0	i				f		Δ		$K_\nu(k\nu x)^3$		$K_\eta(k\eta x)^3$				$(\vartheta^3\eta x)^4$	$H_k(k\vartheta, \eta x)^4$		$\theta_l(\vartheta k, lx)^5$				$(\vartheta^2 lx)^6$
1		u_x				$\theta_\nu(\vartheta\nu x)^2$		$\theta_\eta(\vartheta\eta x)^2$	N	$(k\nu x)^3$	$\theta_\nu(\vartheta^2\nu x)^3$	$(k\eta x)^3$	$H_\vartheta(\vartheta^2\eta x)^3$ $\theta_\eta(\vartheta^2\eta x)^3$		$\theta_l(\vartheta^2 lx)^4$		$(\vartheta k, \eta, x)^4$		$(\vartheta k, l, x)^5$			
2			a_ν^2		θ a_η^2		$(\vartheta\nu x)^2$	$K_\nu(k\nu x)^2$	$(\vartheta\eta x)^2$	$H_k(k\eta x)^2$ $K_\eta(k\eta x)^2$		$(\vartheta^2\nu x)^3$ $K_l(klx)^3$		$(\vartheta^2\eta x)^3$		$(\vartheta^2 lx)^4$						
3		θ_ν^3		a_ν	$\theta_\nu(\vartheta\nu x)$	Δ_ν a_η	K $H_\vartheta(\vartheta\eta x)$ $\theta_\eta(\vartheta\eta x)$	Δ_η	$(k\nu x)^2$ $\theta_l(\vartheta lx)^2$		$(k\eta x)^2$		$(klx)^3$									
4	ν		H θ_ν^2	$(j\nu x)^3$ $a_\eta(aHu)$	θ_η^2	$(\vartheta\nu x)$ a_l^2		N_ν $(\vartheta\eta x)$		H_η N_η $(\vartheta lx)^2$												
5		$a_\eta(aHu)^2$	$\theta_\eta^2(\theta Hu)$	θ_ν	$\frac{K_\nu^2}{(aHu)}$	θ_η	K_η^2 (ΔHu) a_l		Δ_l													
6	j σ		$(j\nu x)^2$ $(aHu)^2$	L $a_\nu^2(j\nu x)$ $H_\vartheta(\theta Hu)$ $\theta_\eta(\theta Hu)$	$\frac{K_\nu}{\theta_l^2}$			K_η														
7		$\theta_\eta(\theta Hu)^2$	$H_j(j\eta x)$ $a_l(aLu)^2$		$a_\nu(j\nu x)$ (θHu)		$\Delta_\nu(j\nu x)$ θ_l	K_l^2														
8	H_j^2 $a_l(aLu)^3$	$(\nu\sigma x)$ $(j\nu x)$	$(\theta Hu)^2$	$K_\eta(KHu)^2$ $(j\eta x)$ $(aLu)^2$																		
9		H_j $(aLu)^3$	$a_\eta(aHu)H_j$ $L_\vartheta(\theta Lu)^2$ $\theta_l(\theta Lu)^2$		(KHu)																	
10	$\theta_l(\theta Lu)^3$	L_j^2 $(j\sigma x)$ $a_\eta(aH^2u)^3$		$H_j(aHu)$ $(\theta Lu)^2$																		
11		$(\theta Lu)^3$	$K_l(KLu)^3$ L_j $(aH^3u)^3$																			
12	$(aH^2u)^4$	$a_\eta(aLu)^2L_j$ $H_\vartheta(\theta H^2u)^3$ $\theta_\eta(\theta H^2u)^3$		$(KLu)^3$																		
13	$(\nu\sigma j)$		$(aLu)^2L_j$ $(\theta H^3u)^3$																			
14	$(\theta H^2u)^4$	$K_\eta(KH^2u)^4$ $(a_\eta H\cdot L, u)^4$																				
15	$L_9(\theta, H\cdot L, u)^4$		$(KH^2u)^4$																			
16	$(\theta, H\cdot L, u)^5$																					
17	$K_\eta(K, H\cdot L, u)^5$																					
18		$(K, H\cdot L, u)^5$																				
19																						
20																						
21	$(KH^3u)^6$																					

(The numbers in the uppermost horizontal row denote the degree in the x; the numbers in the first vertical row denote the degree in the u.)

Table II (cf. p. 90)

u \ x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	J				s		s_j^3					s_ν	$(k\nu x)^3 s_\nu$	$(\vartheta^3\nu x)^2 s_\vartheta$ $(\vartheta^2\nu x)^2 s_\vartheta$		$\theta_\eta(\vartheta^2\eta x)^2 s_\eta$	
1							(asu)	s_η	s_l^2 (Δsu)	$s_\eta(Nsu)$ $(\vartheta\nu x)^2 s, u$	$((k\nu x)^3, s, u)_s$ $(K_\nu(k\nu x)^3, s, u)$		$K_\eta(k\eta x)^3, s, u$		$(\vartheta^2\nu x)^2 s_\nu$		$(\vartheta^2\eta x)^4, s, u$
2					$(asu)^2$	$s_\theta(\theta su)$	$(\Delta su)^2/a_\nu s_\nu$	$(\vartheta\nu x)^2, s, u$ $(\theta_l(\vartheta\nu x)^2, s, u)$		$\theta_\eta(\vartheta^2\eta x)^2, s, u$		$(k\eta x)^2 s_\nu$ $((k\nu x)^2, s, u)$		$(k\eta x)^3, s, u$			
3			$(asu)^3$	$s_\vartheta(\theta su)^2$ s_ν	$a_\nu s_\nu(asu)$ $a_l^2(asu)$	s_η	(θsu) $a_l^2 su$		$(Nsu)^2$ $(\vartheta\eta x)s_\nu$ $((\vartheta\nu x)^2, s, u)$	$((k\nu x)^2, s, u)^2$	$(\vartheta^2\eta x)^2, s, u$						
4	$(asu)^4$	$s_\vartheta(\theta su)^3$	$a_\nu^2(asu)^2$	$(\theta_\nu^2 su)$	$(\theta su)^2$	$s_k(Ksu)^3$ $a_\nu(asu)$ $s_\nu(asu)$	$((\vartheta\nu x)^2, s, u)^2$	$s_\nu(\Delta su)$ (Δsu) $(aHu)s_\eta$ $(a_\nu su)$		$(\Delta_\eta su)$							
5			$(\theta su)^3$	$s_k(Ksu)^3, s_j$ $s_\vartheta(\theta s' u)^4$ $s_\nu(asu)^2$ $a_\nu(asu)^2$	(Hsu) $((\vartheta\nu x)^2, s, u)$ $(\theta_\nu(\vartheta\nu x)^2, s, u)$ $(\theta_\nu^2 su)$	$((j\nu x)^2, s, u)$ $(aHu)^2 s_\nu$ $(a_\eta su)^2$	$(Ksu)^2$ s_η $(\theta_\eta^2 su)$		$((k\nu x)^2, s, u)^2$ $K_\nu s_\nu$								
6	$(\theta su)^4$	$s_\nu(asu)^3$ $a_\nu(asu)^3$	$(Hsu)^2$ $(\theta_\nu(\vartheta\nu x), s, u)^3$ $(\theta_\nu^2 su)^2$	$a_\eta s_\nu^2$ $(a_\eta(aHu), s, u)^2$ $(a_\eta(aH^2u), s, u)$	$(Ksu)^3$	$s_\eta(asu)$ $((\vartheta\nu x), s, u)^3$ $(\theta_\eta su)$	$((k\nu x)^2, s, u)^3$	$s_\nu(\Delta su)$ $a_\nu(j\nu x)s_\nu$ $(\theta Hu)s_\eta$ $(\theta_\eta su)$									
7	$\theta_\nu(\vartheta\nu x), s, u)^4$	$(a_\eta(aHu), s, u)^3$	$(Ksu)^4$ $(\theta_\eta^2(\theta Hu), s, u)^2$ $(a_\eta^2 - a_l^2, s, u)^3$	$s_j(asu)^3$ $s_k(K^2u)^3$ $((j\nu x), s, u)^2$ $(\theta_\eta su)^2$	$((j\nu x), s, u)$ $((j\eta x), s, u)$ $(aHu), s, u)$ $(aH^2u), s, u)$	$((\vartheta\eta x), s, u)^3$	$(aLu)^3 s_l/(asu)^3$										
8	$(a^2 - a_l^2, s, u)^4$	$s_j(asu)^3$ $s_k(K^2u)^3$ $((j\nu x), s, u)^4$ $(\theta_\eta su)^3$	$((j\nu x)^2, s, u)^2$ $(aHu), s, u)^3$ $((aHu)^2, s, u)^3$	$(Lsu)^2$ $(a_l^2(j\nu x), s, u)^3$ $H_\vartheta(\theta Hu), s, u)^3$ $\theta_l(\theta Hu), s, u)^3$	$(asu)^3$	$(K, su)^3$		$(K_\vartheta su)^2$									
9	$K_j^2 su^4$ $((aHu), s, u)^4$	$(Lsu)^4$ $(a_l^2(j\nu x), s, u)^4$ $(\theta_\nu(\theta Hu), s, u)^2$	$(K_\nu^3 u)^6$ $(a_\eta(aLu)^2, s, u)^3$ $(a_l^2 H, s, u)^3$	$(K, su)^3$	$(a_\nu(j\nu x), s, u)^3$ $(\theta Hu), s, u$ $(\theta Hu)^2, s, u$		$(\theta_l su)^3$										
10			$(K, su)^4$ $(a_l^2 a_\eta(aHu)^2, s, u)^3$	$((j\eta x), s, u)^3$ (H_j, su) $((aLu)^2, s, u)^2$ $((aLu)^3, s, u)$													
11	$(a_\nu(j\nu x), s, u)^4$ $((\theta Hu), s, u)^4$	$(K_l(KHu)^2, s, u)^3$ $((j\eta x), s, u)^3$ $((aLu)^2, s, u)^4$ $(a_\eta H, s, u)^3$	$(H^2 su)^3$ $\theta_l(\theta Lu)^2, s, u)^3$		$((KHu)^2, s, u)^3$												
12		$(a_\eta(aHu)^2 H, s, u)^3$	$((KHu)^3, s, u)^3$	$((aHu)H_j, s, u)^2$ $((\theta Lu)^2, s, u)^3$													
13	$((KHu)^3, s, u)^4$	$((aHu)H_j, s, u)^3$ $((\theta Lu)^2, s, u)^4$ $(\nu \cdot H, s, u)^3$	$(L_j su)^2$ $((aH^3u)^3, s, u)^3$ $((j\eta x)H, s, u)^3$ $((aHu)^2 H, s, u)^3$														
14	$((j\nu x)^2 H, s, u)^4$ $((aHu)^2 H, s, u)^4$	$(\theta_\eta(\theta Hu)^2 H, s, u)^3$		$((KLu)^3, s, u)^2$													
15	$(a_l(aLu)^3 H, s, u)^4$	$((KLu)^3, s, u)^3$	$((aLu)^2 L_j, s, u)^2$ $((\theta Hu)^3 H, s, u)^3$ $((\theta Hu)^2 H, s, u)^2$														
16	$((\theta Hu)^2 H, s, u)^4$ $(\sigma \cdot H_j, s, u)^4$	$((j\eta x)H, s, u)^4$ $(H_j H, s, u)^3$ $((aLu)^2 H, s, u)^4$ $((aLu)^3 H, s, u)^3$															
17	$(\theta(\theta Lu)^2 H, s, u)^4$		$((KH^3u)^3, s, u)^3$														
18		$((\theta Lu)^2 H, s, u)^4$ $((\theta Lu)^3 H, s, u)^3$ $(aHu)^2 H, s, u)^4$															
19	$((aH^3u)^3 H, s, u)^4$ $(L_j H, s, u)^5$																
20		$((KLu)^3 H, s, u)^5$	$((aLu)^3 H, s, u)^5$														
21	$((\theta H^3u)^3 H, s, u)^4$ $((aLu)^2 H_j, s, u)^4$																
22																	
23	$((KH^3u)^4 H, s, u)^4$	$((aH^3u)^4 H, s, u)^3$															

(The numbers in the uppermost horizontal row denote the degree in the x; the numbers in the first vertical row denote the degree in the u.)

3. On the Invariant Theory of Forms in n Variables[30]

Jahresbericht der DMV 19 (1910), pp. 101–104

In the projective invariant theory of forms in n variables the main problem has been solved: the finiteness of the system of forms has been proved. A series of further questions, however, have not been treated, namely those concerned with methods for investigating the connection among forms. I would like briefly to communicate the results that I have obtained with respect to such questions, but for the sake of clarity I shall first sketch the questions in the ternary domain, where they are known.

I have first in mind the two fundamental theorems of the symbolic method: the theorem that all invariant formations can be represented symbolically, and the second theorem that all relations existing among invariants are obtained by successive application of a small number of symbolic identities; thus the symbolic method gives all relations without leaving the domain of invariants.[31] Built upon this are the processes for generating the forms, the symbolic folding process and the nonsymbolic differentiation processes, the theory of series expansions, and the like. In this connection I should also mention a theorem proved in my dissertation,[32] namely that all foldings can be generated by successive application of the two possible "cogredient" foldings; by this, for a symbolic product $s_x t_x u_\sigma u_\tau$, I mean the two foldings (stu) and $(\sigma\tau x)$. On this theorem one can build the theory of reducers and of the reduction of systems of forms, in analogy with the binary domain, as I showed in my dissertation.

When we now pass to the n-ary domain, even the first theorem of the symbolic method holds only in a conditional measure. It is well known that already for quaternary forms there occur, besides the point and plane coordinates x and u, line coordinates p_{ik}, the subdeterminants from two rows of point coordinates. Analogously, in the n-ary domain we have rows of variables p_ρ, whose individual elements are the subdeterminants from ρ rows x, and rows of symbols S^ρ, whose elements behave like subdeterminants from ρ rows s cogredient to the u. The symbolic representation by determinants is known to hold only when the symbol rows S^ρ are resolved into the rows s and these are distributed among different determinants; only such aggregates of determinants as depend on the S^ρ then have real meaning. But which aggregates of determinants accomplish this cannot be surveyed, and thereby all the other theorems mentioned in the ternary domain are lost.

I would now like to state a simple principle by which one obtains a symbolic representation explicit in the rows S^ρ, and thereby also recovers the remaining theorems. This is an application of the familiar theorem on corresponding matrices: "To each row of variables p_ρ there corresponds one-to-one another row $q_{n-\rho}$ whose individual elements are subdeterminants from $n-\rho$ rows u connected with the ρ rows x by equations $(u \mid x) = 0$." Correspondingly, to each symbol row S^ρ there is assigned another $S_{n-\rho}$, behaving as if it were composed from $n-\rho$ rows cogredient to the x.

The theorem also permits a second formulation, suited for applications: "To every matrix product $(v^{(1)} \cdots v^{(\rho)} \mid y^{(1)} \cdots y^{(\rho)})$,[33] where the rows of the second matrix are contragredient to those of the first, there is assigned a determinant; and conversely each determinant can be transformed into a matrix product with a prescribed number ρ of rows." Thus determinant and matrix product appear as fully equivalent, and the first theorem of the symbolic method assumes the form: "All invariant formations can be represented symbolically by matrix products." From two symbol rows S^σ, T^τ one can now, with the aid of rows of variables, very easily form a matrix product explicitly containing these symbol rows and hence representing an invariant formation of the form $(S^\sigma \mid p_\sigma)(T^\tau \mid p_\tau)$, namely the following:

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}), \qquad (\lambda = 1, 2, \ldots, \tau \text{ or } n-\sigma). \tag{1}$$

More important is the converse: that all invariant formations can be represented explicitly by matrix products, so that the above process is the extension of the binary and ternary folding process. To prove this converse it is necessary to transfer the second theorem of the symbolic method to the n-ary domain, namely to set up the totality of the independent indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable. These identities arise from the known ones containing only rows x and u,[34] by replacing the product theorem for determinants by a system of product theorems for matrices, and by contracting different

[30] Lecture delivered at the Salzburg meeting of natural scientists, 1909.

[31] Study: *Methoden zur Theorie der ternären Formen.* II. § 6. (Leipzig, Teubner, 1889.)

[32] *Über die Bildung des Formensystems der ternären biquadratischen Form.* § 1. (Crelle's Journal Bd. 134. 1908.)

[33] That is, the sum over the products of corresponding determinants, formed from the matrix of the ρ rows v and that of the ρ rows y.

[34] E. Pascal in *Memorie d. R. Acc. d. Lincei.* (4) 5 (1888), p. 375.

matrix products into a single one by means of precisely these product theorems. One obtains in this way two dualistic formulas, of which only one is given:

$$
\begin{aligned}
(S_1^{\rho_1} S_2^{\rho_2} \cdots S_k^{\rho_k} \mid x p_{\rho-1}) &= \sum_{i=1}^{k} \varepsilon \,(S_1^{\rho_1} \cdots S_{i-1}^{\rho_{i-1}} S_{i+1}^{\rho_{i+1}} \cdots S_k^{\rho_k} q_{n-\rho_i+1} \mid S_{n-\rho_i} x), \\
\rho = \sum_{i=1}^{k} \rho_i &< n, \qquad \varepsilon = \pm 1.
\end{aligned} \tag{2}
$$

The only specialization contained in these formulas, namely that a row x enters, cannot be avoided; for completely general symbol and variable rows we arrive at a "decomposition identity," an identity in which a row p_ρ is decomposed into the individual rows x. The simplest special case of the decomposition identity was already used by Clebsch in his "fundamental problem of invariant theory" to derive the elementary covariants in the theory of series expansions. With the aid of these identities, which mediate the transition from the non-explicit determinant expressions to the matrix product, one can in fact show that the desired explicit symbolic representation is obtained with the matrix representation.

For the folding process, however, a theorem entirely analogous to that in the ternary domain holds, and the reduction theorems can likewise be built upon it analogously. If in (1) one denotes the number λ as the defect of the folding, and calls foldings for which $\sigma = \tau$ cogredient, then all foldings can be generated by successive application of the cogredient foldings of defect 1. The proof of this theorem rests essentially on the fact that the most general forms can be replaced by "normal forms," that is, by forms that vanish identically under the application of all invariant differential operations. In the quaternary domain this latter fact was proved by Mertens;[35] our knowledge of the most general differential operations–which, as is well known, arise from the folding processes by replacing the symbol rows by differential symbols–allows us, using the decomposition identity, to transfer Mertens's proof almost verbatim to the n-ary domain.

[35] Über invariante Gebilde quaternärer Formen. Wiener Berichte. Bd. 98. 1889.

4. On the Invariant Theory of Forms in n Variables

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Introduction.

In the projective invariant theory of forms in n variables, the questions adjoining the main problem–the proof of finiteness of the system of forms–have scarcely been treated when one goes beyond the binary and ternary domains. These are the questions concerning the mutual dependence of the forms and the methods for controlling this connection among forms. These methods rest essentially–as has been completely proved in the binary and ternary domains–on two theorems designated by Study as the "fundamental theorems of the symbolic method," namely the theorem on the symbolic representability of the forms and the theorem on symbolic identities.[36] The latter theorem says that all relations existing among integral invariant formations are obtained by successive application of a small number of symbolic identities; thus the symbolic method, and only it, gives knowledge of the connection among forms without leaving the domain of invariants.

Gordan, in his Erlangen program,[37] already pointed to the reason for the difficulty in passing to n variables: it lies in the fact that the first theorem of the symbolic method is no longer valid in its general form. In the n-ary domain there occur, as is well known, variable rows of higher level p_ρ[38] and analogously symbol rows of higher level S^ρ; one arrives at these rows both when, as originally with Clebsch,[39] one starts from forms with arbitrarily many rows p_ρ, and when one starts, following F. Deruyts and A. Capelli,[40] from forms containing only arbitrarily many cogredient rows x. A symbolic representation that would contain the rows p_ρ, S^ρ explicitly[41] does not exist; it is necessary to resolve the p_ρ, S^ρ into the individual rows x and the s contragredient to them, and to distribute these among different determinants. It is indeed possible, by means of differential conditions (cf. formula (19.)), to verify that a given aggregate of determinants has the property of depending only on the rows p_ρ, S^ρ; but in this way one obtains no overview of the totality of invariant formations and of the processes for their generation.[42]

We shall now show in §§ 2 and 6 how, using the "theorem on corresponding matrices,"[43] the first fundamental theorem is recovered in its generality by replacing representation by determinants with representation by "matrix products." In § 2 it is shown that every matrix product explicitly containing the rows S^ρ, p_ρ is an invariant formation of the ground form; the converse given in § 6, that every invariant formation can be represented explicitly by matrix products, succeeds only through the transfer of the second fundamental theorem (§§ 3–5).

This theorem is known for forms that contain only variable rows x.[44] The general problem, to set up the totality of the indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable, was formulated by Study;[45] at the same time he sees in the number of identities that occur a measure of the difficulty of the methods in the n-ary domain. Since it is now possible to combine the totality of the identities in a single formula (36.), this difficulty is at least pressed down to a minimum. In applications, however, certain distinguished special cases of (36.) will often occur, such as (20.), (21.), characterized by the fact that they admit explicit representation without substitution of new rows; or formula (31.), designated as the "decomposition identity," since a row p_ρ appears to be decomposed into the rows y.

Upon the two fundamental theorems the further theory can now easily be built. The first theorem gives directly the processes for generating the forms, the symbolic folding process and the nonsymbolic differentiation processes,[46] while the decomposition identity eliminates, among these processes, all equivalent

[36] E. Study, *Methoden zur Theorie der ternären Formen* (Leipzig, Teubner, 1889). II. § 6. For the occurrence of these fundamental theorems also in the invariant theory of special transformation groups, cf. Study, *Das Apollonische Problem* (Math. Ann. Bd. 49 [1897]) and *Zur Differentialgeometrie der analytischen Kurven* (Transactions of the American Math. Society, Bd. 1 [1909]).

[37] P. Gordan, *Über das Formensystem binärer Formen.* S. 46. (Leipzig, Teubner, 1875).

[38] For the notation see § 1.

[39] A. Clebsch, über eine Fundamentalaufgabe der Invariantentheorie. (Abh. der Ges. d. Wiss. zu Göttingen, Bd. XVII, 1872).

[40] Cf. A. Capelli, *Lezioni sulla teoria delle forme algebriche* (Napoli 1902), § 22–24, and the literature cited there.

[41] More precise definition of "explicit representation" in § 2.

[42] A computationally very valuable development of the symbolism due to Study (communicated by F. Engel, Ber. d. K. Sächs. Ges. d. Wiss., math.-phys. Kl., 1900, p. 126, and for the quaternary domain by E. Study, *Geometrie der Dynamen*, p. 135) is also not an explicit representation in our sense. It would, for example, hardly lead to the nonsymbolic differentiation processes for generating the forms.

[43] Apart from Grassmann's *Ausdehnungslehre* of 1862, no. 112, first in A. Brill, über zwei Berührungsprobleme, § 2 (Math. Ann. Bd. 4. 1871).

[44] E. Pascal, Giorn. di mat. 26 (1888), pp. 33, 102; Rendic. d. Accad. d. Linc. (4) 4 (1888), p. 119; ib. Mem. (4) 5 (1888), p. 375.

[45] "Methoden . . . ", p. 204, note 17).

[46] In the paper "Invariante Gebilde von Nullsystemen" (Sitzungsber. d. Ak. d. Wiss. Wien, Bd. 97, Abt. IIa, 1888), Mertens arrived by another path, not directly generalizable, at the quaternary folding process. The alternating invariant of 6 rows S^2 occurring there differs from the one arising with us by a single application of substitution (11.) by an aggregate of even invariants. For a special case the quaternary folding process is also found

ones (§ 6). Knowledge of the differentiation processes further yields the transfer of the concept of "normal form" to the n-ary domain and, by means of a proof procedure given by Mertens[47] in the quaternary domain, the theorem that a system of invariant formations can be replaced by an equivalent system of normal forms (§ 7). Under this latter assumption I showed in my dissertation[48] that the ternary folding process can be generated by successive application of two fundamental foldings. On the basis of this theorem and the theory of series expansions, I then developed, by introducing the concept of the "form row," the principal reduction theorems for systems of forms. Entirely analogously, in the n-ary domain the folding process can be generated from $n-1$ fundamental foldings (§ 8), and the reduction theorems can be set up (§ 9).

Afterword. On the occasion of the Salzburg communication, Prof. E. Müller of Vienna drew my attention to the fact that the calculation with matrix products underlying this paper is, in principle, identical with the calculus of Grassmann's theory of extension.[49] Indeed, Grassmann's "combinatorial product" $[S^{\rho_1} S^{\rho_2} \cdots S^{\rho_i}]$ can be regarded as a matrix given by these rows (cf. § 2), specifically the "progressive product" as the matrix of the rows themselves, the "regressive" as the matrix of the assigned rows $S_{n-\rho_1}, S_{n-\rho_2}, \ldots, S_{n-\rho_i}$, while the matrix corresponding to the "mixed product" arises when several rows S are combined by substitution (9.) into new rows P, Q. Our "assigned row" corresponds to Grassmann's "complement," our "matrix product" to a "mixed product of level number 0."

Under this correspondence, the development given in the *Ausdehnungslehre* of 1862, no. 113, becomes identical with a formula dualistic to our formula (32.). In the special case $\rho = 1$, (32.) goes over into (17.), the dualistic formula into (16.). From these special cases of Grassmann's development, Müller[50] has recently derived the identities (20.), (21.) mentioned above.

In contrast to the Grassmann–Müller derivation, our derivation at the same time gives the proof of completeness of the identities, which appears to be the essential point for the questions considered here, and it proceeds from (20.), (21.) further to the most general indecomposable identities (36.), which do not seem to have been noticed before now.

§ 1. Notations and definitions. Summary of known results.

We denote, as usual, by the variable row x the row of elements $x_1, x_2, \ldots, x_n$, and analogously by the variable row p_ρ $(\rho = 1, 2, \ldots, n-1)$ the row of the $\binom{n}{\rho}$ elements $p_{i_1 i_2 \ldots i_\rho}$, where $p_{i_1 i_2 \ldots i_\rho}$ denotes the $\binom{n}{\rho}$ determinants formed from the matrix of the ρ rows $x^{(1)}, x^{(2)}, \ldots, x^{(\rho)}$,

$$\begin{vmatrix} x_{i_1}^{(1)} & x_{i_2}^{(1)} & \cdots & x_{i_\rho}^{(1)} \\ x_{i_1}^{(2)} & x_{i_2}^{(2)} & \cdots & x_{i_\rho}^{(2)} \\ \vdots & \vdots & & \vdots \\ x_{i_1}^{(\rho)} & x_{i_2}^{(\rho)} & \cdots & x_{i_\rho}^{(\rho)} \end{vmatrix}, \qquad (i_1 < i_2 < \cdots < i_\rho = 1, 2, \ldots, n).$$

According to the theorem on corresponding matrices,[51] to every row p_ρ there is assigned one-to-one a second row $q_{n-\rho}$ by the relation, valid for each element of the row,

$$p_{i_1 i_2 \ldots i_\rho} = (-1)^{\frac{\rho(\rho+1)}{2} + i_1 + i_2 + \cdots + i_\rho} q_{j_1 j_2 \ldots j_{n-\rho}}, \tag{1}$$

where the i and j are each in good order and together form a complete permutation of the numbers 1 through n. The row $q_{n-\rho}$ consists of the $\binom{n}{\rho}$ determinants of degree $n-\rho$, $q_{j_1 j_2 \ldots j_{n-\rho}}$, formed from the matrix of $n-\rho$ rows $u^{(1)}, u^{(2)}, \ldots, u^{(n-\rho)}$ contragredient to the x. These rows satisfy the $\rho(n-\rho)$ condition equations

$$(u^{(k)} \mid x^{(l)}) = \sum_{\alpha=1}^{n} u_\alpha^{(k)} x_\alpha^{(l)} = 0; \qquad (k = 1, 2, \ldots, n-\rho;\ l = 1, 2, \ldots, \rho). \tag{2}$$

in P. Gordan, *Das simultane System von zwei quadratischen quaternären Formen* (Math. Ann. Bd. 56. 1903).

[47] F. Mertens, Über invariante Gebilde quaternärer Formen. (Sitzber. d. Ak. d. Wiss. Wien. Bd. 98. Abt. IIa. 1889.)

[48] Über die Bildung des Formensystems der ternären biquadratischen Form. (Dieses Journal, Bd. 134. 1908.)

[49] Hermann Graßmanns gesammelte mathematische und physikalische Werke. Ersten Bandes zweiter Teil: Die Ausdehnungslehre von 1862. In Gemeinschaft mit H. Graßmann d. Jüngeren herausgegeben v. Fr. Engel (Leipzig, Teubner 1896).

[50] E. Müller, Beiträge zur Graßmannschen Ausdehnungslehre. 1. Mitteilung: Einige allgemeine Sätze. (Sitzungsber. d. Ak. d. Wiss. Wien; Bd. 118. Abt. IIa. Juli 1909), Satz 16 und 18.

[51] Cf. for example Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie*, Bd. I, p. 99.

We shall assume them normalized (with Clebsch[52]) so that the proportionality factor otherwise entering in (1.) is set equal to one.

According to Clebsch[53] (and likewise according to the later investigations of F. Deruyts and A. Capelli, cf. the Introduction), the most general forms can be reduced to those that contain at most one of each of the $n-1$ rows p_ρ, and hence can be represented symbolically as follows:

$$(S^1 \mid p_1)^{m_1}(S^2 \mid p_2)^{m_2}\cdots(S^{n-1} \mid p_{n-1})^{m_{n-1}}, \tag{3}$$

where, in analogy with (2.), we have set

$$(S^\rho \mid p_\rho) = \sum S^{i_1 i_2 \dots i_\rho} p_{i_1 i_2 \dots i_\rho}.$$

In consequence of the nonlinear identical relations existing between the elements of a row p_ρ, the symbol rows S^ρ can be normalized so that they satisfy the very same identities, hence so that they behave like the determinants of a ρ-rowed matrix whose rows $s^{(1)}, s^{(2)}, \dots, s^{(\rho)}$ are contragredient to the x.[54] Thus to each row S^ρ there can be assigned one-to-one another $S_{n-\rho}$ through the relation

$$S_{i_1 i_2 \dots i_{n-\rho}} = (-1)^{\frac{(n-\rho)(n-\rho+1)}{2}+i_1+i_2+\dots+i_{n-\rho}} S^{j_1 j_2 \dots j_\rho}, \tag{4}$$

where again the i and j are each in good order and together exactly exhaust the numbers 1 through n. The elements $S_{i_1 i_2 \dots i_{n-\rho}}$ of the row $S_{n-\rho}$ behave like the determinants formed from $(n-\rho)$ rows $\sigma^{(1)}, \sigma^{(2)}, \dots, \sigma^{(n-\rho)}$ cogredient to the x, and the rows s and σ are linked by the $\rho(n-\rho)$ conditions

$$(s^{(k)} \mid \sigma^{(l)}) = 0, \qquad (k = 1, 2, \dots, \rho;\ l = 1, 2, \dots, n-\rho). \tag{5}$$

In consequence of the relations (1.) and (4.), each factor of (3.) is capable of the equivalent fourfold symbolic representation

$$(S^\rho \mid p_\rho) = (S^\rho q_{n-\rho}) = (S_{n-\rho} p_\rho) = (-1)^{\rho(n-\rho)}(q_{n-\rho} \mid S_{n-\rho}), \tag{6}$$

where, as always in what follows, $(S^\rho q_{n-\rho})$ and $(S_{n-\rho} p_\rho)$ are abbreviations for the determinants $(s^{(1)} \cdots s^{(\rho)} u^{(1)} \cdots u^{(n-\rho)})$ and $(\sigma^{(1)} \cdots \sigma^{(n-\rho)} x^{(1)} \cdots x^{(\rho)})$; by Laplace expansion these prove to depend only on the rows $S^\rho, q_{n-\rho}$ and respectively $S_{n-\rho}, p_\rho$, as the above notation is meant to indicate. The expressions $(S^\rho \mid p_\rho)$ and $(q_{n-\rho} \mid S_{n-\rho})$ mean, according to the preceding discussion, the products of the matrices of the rows s and x, respectively of the rows u and σ, where by matrix product is meant the sum over the products of corresponding determinants, that is, the value of the determinant of the product matrix.

The characteristic number $\rho(n-\rho)$ belonging to a symbol row (respectively variable row) will be called, following Clebsch, the weight of the row;[55] the number ρ, which gives the number of rows s, the upper weight index; the number $n-\rho$, which gives the number of rows σ, the lower weight index. It follows from the relations (4.) that a symbol row can occur in one and the same relation once with upper and once with lower weight index; the same holds for the occurrence of the corresponding rows p_ρ and $q_{n-\rho}$. It remains to note that we generally denote variable rows of higher level whose associated matrix consists of rows x by p, those whose matrix consists of rows u by q; symbol rows cogredient to the x by small Greek letters $\sigma, \tau, \alpha, \beta$; symbol rows cogredient to the u by small Latin letters s, t, a, b; all other symbol rows by capital Latin letters with weight index: S^ρ, T^τ, A^ρ or $S_{n-\sigma}$, $T_{n-\tau}$, $A_{n-\rho}$. Corresponding to an upper or lower weight index, we also write the individual elements of a row with upper or lower indices, as for instance in (4.).

§ 2. Representation by matrix products.

In what follows we always understand by "matrix product" the sum over the products of corresponding determinants of two matrices with the same number of rows, where the rows of the second matrix are contragredient to those of the first. The rows of each matrix may be: 1. individually given; 2. combined, as in (6.), into a row S^ρ; 3. united into different rows $S^\rho, S^\sigma \dots$. The matrix product is to be denoted by the symbol $(\ \mid\ )$ used in (6.). The representation (6.) is a special application of a second formulation of the theorem on corresponding matrices: "To every ρ-rowed matrix product $(v^{(1)} v^{(2)} \cdots v^{(\rho)} \mid y^{(1)} \cdots y^{(\rho)})$ there is

[52]a. a. O. § 5.

[53]a. a. O. § 12.

[54]Clebsch, a. a. O. § 4.

[55]a. a. O. § 11.

assigned a determinant; and conversely to every determinant there can be assigned a matrix product with prescribed number ρ of rows ($\rho = 1, 2, \dots, n-1$)."[56] Indeed, one has only to combine the rows y into a row p_ρ (respectively the rows v into a row q_ρ) and to carry out the substitutions given by (1.) in order to pass from a given matrix product to the determinant and conversely. The value of the determinant is, however, not given by its individual elements, but first results from Laplace expansion. Since determinant and matrix product therefore prove to be equivalent, the theorem on the symbolic representability of the forms (the first fundamental theorem) is expressed also as follows:

Theorem I. "All invariant formations can be represented symbolically by matrix products, and conversely all matrix products are invariant formations."[57]

Representation by matrix products allows us to overcome the essential difficulty of the non-explicit representation caused by the determinant representation.[58] By "explicit representation" we mean a representation into which only the rows $S^\rho, T^\tau, \dots$ themselves enter, or rows R^ρ uniquely derivable from these, but in which the individual component rows $s, t, \dots$ no longer occur. We shall obtain in § 6, for all invariant formations that depend only on the symbol rows $S^\rho, T^\tau, \dots$, a representation explicit in these symbol rows, and thereby the transfer to the n-ary domain of the generating processes, the folding process and the differentiation processes. The possibility of explicit representation is evident from the fact that only the simplest matrix products can be assigned precisely to the determinants occurring in the determinant representation, so that all other matrix products are combinations of different determinants into a single expression. For if one resolves the symbol and variable rows into the individual rows s, u, x, then, according to the first fundamental theorem in its usual form, an aggregate of determinants must necessarily arise. The proof that, conversely, all determinant aggregates that represent invariant formations of prescribed ground forms can be combined into explicit matrix products follows by using the second fundamental theorem (§ 6).

In order to obtain from two symbol rows S^σ, T^τ, with the help of variable rows, an invariant formation of the ground form $(S^\sigma \mid p_\sigma)(T^\tau \mid p_\tau)$ that explicitly contains all rows (the folding process), we need only form a matrix product of the third kind according to the definition given at the beginning of the paragraph:

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}), \qquad (\lambda = 1, 2, \dots, \tau, n-\sigma). \tag{7}$$

Expanded, (7.) gives:

$$\begin{aligned}(v^{(1)} \cdots v^{(n-\sigma-\lambda)} s^{(1)} \cdots s^{(\sigma)} \,|\, \alpha^{(1)} \cdots \alpha^{(n-\tau)} y^{(1)} \cdots y^{(\tau-\lambda)}) \\ = \sum_{i,k,l} \varepsilon\, q_{k_1 \dots k_{n-\sigma-\lambda}} S^{k_{n-\sigma-\lambda+1} \dots k_{n-\lambda}} T_{l_1 \dots l_{n-\tau}} p^{l_{n-\tau+1} \dots l_{n-\lambda}},\end{aligned} \tag{8}$$

that is, an expression depending only on the rows S, T, q, p. Here $\varepsilon = \pm 1$[59] and the sum is to be understood in such a way that the indices of each row $S \dots$ are in good order, and that for each combination $i_1, i_2, \dots, i_{n-\lambda}$ the k as well as the l individually run through all then possible permutations of the numbers i. The number λ entering into (7.) will be called the defect of the folding in the case $\sigma > \tau$; the reason for this latter restriction appears in § 6.[60]

The determinants of the two matrices entering in (7.) can be regarded as elements of new rows $Q^{n-\lambda}, P_{n-\lambda}$ whose associated rows are Q_λ, P^λ. According to (8.), these rows Q, P can be expressed through the original S and q, respectively T and p. We indicate this by the equations

$$\begin{aligned} q_{n-\sigma-\lambda} S^\sigma &= Q^{n-\lambda} \sim Q_\lambda, &\quad \text{or also} \quad Q_\lambda &= [q_{n-\sigma-\lambda} S^\sigma]_\lambda, \\ T_{n-\tau} p_{\tau-\lambda} &= P_{n-\lambda} \sim P^\lambda, &\quad \text{or also} \quad P^\lambda &= [T_{n-\tau} p_{\tau-\lambda}]^\lambda. \end{aligned} \tag{9}$$

Taking (4.) into account, one then has, analogously to (6.), the equivalent fourfold symbolic representation for the matrix product (7.):

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}) = (q_{n-\sigma-\lambda} S^\sigma P^\lambda) = (Q_\lambda T_{n-\tau} p_{\tau-\lambda}) = (-1)^{\lambda(n-\lambda)} (P^\lambda \,|\, Q_\lambda). \tag{10}$$

A further substitution of new rows can be performed on (7.) by setting

$$(q_{n-\sigma-\lambda} S^\sigma \,|\, T_{n-\tau} p_{\tau-\lambda}) = (R^{\sigma+\lambda} \,|\, p_{\sigma+\lambda})(R^{\tau-\lambda} p_{\tau-\lambda}). \tag{11}$$

[56] For $\rho > n$ the matrix product is known to vanish; for $\rho = n$ it goes into the product of two determinants.

[57] Invariant formations of prescribed ground forms are, of course, only those aggregates of matrix products that depend on the symbol rows $S^\rho \dots$.

[58] Cf. the Introduction.

[59] This notation is to be retained throughout.

[60] The number λ always runs up to the smaller of the two last indicated numbers.

Here too (8.) shows that the products of the elements of the rows R can be expressed uniquely through the rows S and T. The substitutions (9.) and (11.) will be used especially when a further folding with other rows is performed with (7.).

The matrix representation enables us to state invariantly the quadratic relations between the elements of a row p_ρ (from which, as is known, all the others follow), hence, according to § 1, the relations linear in the coefficients of the ground form that the rows S^ρ satisfy. These are the identities

$$(q_{n-\rho-\lambda}S^\rho \mid S_{n-\rho}p_{\rho-\lambda}) = 0, \qquad (\lambda = 1, 2, \ldots, \rho, n-\rho), \tag{12}$$

where the p and q are to be regarded as arbitrary quantities. We shall see in § 5 that the identities with odd defect number λ are consequences of those with higher defect. The relations (12.) are a direct consequence of the relations (5.); for one has

$$(q_{n-\rho-\lambda}S^\rho \mid S_{n-\rho}p_{\rho-\lambda}) = (v^{(1)}\cdots v^{(n-\rho-\lambda)}s^{(1)}\cdots s^{(\rho)} \mid \sigma^{(1)}\cdots\sigma^{(n-\rho)}y^{(1)}\cdots y^{(\rho-\lambda)}),$$

and the application of the product theorem gives, in consequence of (5.), a vanishing $(n-\lambda)$-rowed determinant. The identities (12.) say that in order to find all types of foldings it suffices to fold the product of forms linear in each variable row; or also, by § 6, that the normalized form $(S^\rho \mid p_\rho)^m$ possesses no invariant formation linear in the coefficients.[61]

The conditions (12.) are also sufficient to characterize the rows p_ρ. For the property that the individual elements of p_ρ are determinants of a matrix is an invariant property, and therefore it must be expressible invariantly; but by Theorem VI, the conditions (12.) give the greatest possible invariant restriction for the p_ρ, namely that the linear form formed from the row p_ρ as coefficients possesses no invariant formations at all.

§ 3. The symbolic identities.

We now have to transfer the second fundamental theorem of the symbolic method, namely to set up the totality of the irreducible indecomposable identities in which all rows S^ρ, p_ρ are regarded as indecomposable. We also stipulate the following. Corresponding to the meaning of symbol and variable rows, those identities are to be regarded as composite that arise when one specializes the variable row p_ρ to $p_{\rho-1}y$ and applies one of the identities of the system to the resulting expressions; while, on the other hand, identities containing k symbol rows are to count as indecomposable even if they arise from identities with fewer symbol rows by the above process. The rows S^ρ, p_ρ need not necessarily enter the identities explicitly; for their interpretation as indecomposable quantities it suffices to show that the matrix products that occur depend only on these rows. We shall show, however, that in this case, partly by applying substitution (11.), an explicit representation can always be obtained. The general identities are obtained, by means of the principle of matrix representation, from the special identities given by Pascal, which contain only rows x and u and constitute a direct transfer of those given by Study for the ternary domain:[62]

$$\sum_{i=1}^{n}(-1)^{i+1}(a^{(i)} \mid x)(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(n)}u) = (-1)^{n+1}(u \mid x)(a^{(1)}\cdots a^{(n)}), \tag{13}$$

$$\sum \pm (a^{(1)} \mid x^{(1)})(a^{(2)} \mid x^{(2)})\cdots(a^{(n)} \mid x^{(n)}) = (a^{(1)}\cdots a^{(n)})(x^{(1)}\cdots x^{(n)}), \tag{14}$$

$$\sum_{i=1}^{n}(-1)^{i+1}(u \mid a^{(i)})(a^{(1)}\cdots a^{(i-1)}a^{(i+1)}\cdots a^{(n)}x) = (-1)^{n+1}(u \mid x)(a^{(1)}\cdots a^{(n)}). \tag{15}$$

Two further identities arise from (13.) and (15.) through the substitutions $(a^{(i)} \mid x) = (a^{(i)}q_{n-1})$ and respectively $(u \mid a^{(i)}) = (p_{n-1}a^{(i)})$, and so by our interpretation they are not to be regarded as essentially different. (14.) represents the product theorem for determinants; we shall show how (13.) and (14.) (and, dualistically, (15.) and (14.)) can be replaced by the system of suitably formed product theorems for matrices. The product theorem, formed once for ρ-rowed matrices and once for $(\rho-1)$-rowed matrices, reads

$$(a^{(1)}a^{(2)}\cdots a^{(\rho)} \mid xp_{\rho-1}) = (a^{(1)}a^{(2)}\cdots a^{(\rho)} \mid xy^{(2)}\cdots y^{(\rho)}) = \sum \pm (a^{(1)} \mid x)(a^{(2)} \mid y^{(2)})\cdots(a^{(\rho)} \mid y^{(\rho)}),$$

$$\sum \pm (a^{(2)} \mid y^{(2)})\cdots(a^{(\rho)} \mid y^{(\rho)}) = (a^{(2)}\cdots a^{(\rho)} \mid y^{(2)}\cdots y^{(\rho)}) = (a^{(2)}\cdots a^{(\rho)}p_{\rho-1}) = (a^{(2)}\cdots a^{(\rho)}q_{n-\rho+1});$$

[61] Cf. an addendum by Study in Grassmann's works, first volume, second part, p. 510 (condition that a quantity S^ρ is simple).

[62] For formulation and literature of the problem, cf. the Introduction.

and from this follows the system of product theorems, where $\rho = 2, 3, \ldots, n$:

$$\begin{aligned}(a^{(1)}a^{(2)} \cdots a^{(\rho)} \,|\, xp_{\rho-1}) &= \sum_{i=1}^{\rho} (-1)^{i+1} (a^{(i)} \,|\, x)(a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho)} \mid p_{\rho-1}) \\ &= \sum_{i=1}^{\rho} (-1)^{i+1} (a^{(i)} \,|\, x)(a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho)} q_{n-\rho+1}).\end{aligned} \tag{16}$$

For $\rho = n$, (16.) goes over into (13.); if we further specialize $p_{n-1} = y^{(2)} p_{n-2}$, $p_{n-2} = y^{(3)} p_{n-3}$, etc., and apply the identities (16.) to the expressions $(a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(n)} \mid y^{(2)} p_{n-2})$, $(a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(h-1)} a^{(h+1)} \cdots a^{(n)} \mid y^{(3)} p_{n-3})$, etc., then we pass from (13.) to (14.). Thus the identity (14.) is, by our definition, composite from the identities (16.); equivalently, the system (16.) is equivalent to the identities (13.) and (14.). The system (16.) satisfies one of the conditions required for the general identities: it contains the row $p_{\rho-1}$ as an indecomposable quantity and at the same time in explicit form. It is the only system satisfying this condition and only this condition. For we can form no further determinants or matrix products from the rows a, x, p; and between these no relations independent of (16.) can exist, since otherwise, by eliminating $(a^{(1)} \cdots a^{(\rho)} \mid xp_{\rho-1})$, there would arise a relation between ρ ($\rho < n$) expressions $(a^{(i)} \mid x)(a^{(1)} \cdots a^{(i-1)} a^{(i+1)} a^{(\rho)} q_{n-\rho+1})$, whereas by (13.) at least $n+1$ such expressions must enter into a relation. From analogous considerations one obtains the system equivalent to (15.) and (14.):

$$\begin{aligned}(uq_{\rho-1} \,|\, a^{(1)} \cdots a^{(\rho)}) &= \sum_{i=1}^{\rho} (-1)^{i+1} (u \,|\, a^{(i)})(q_{\rho-1} \mid a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho)}) \\ &= \sum_{i=1}^{\rho} (-1)^{i+1} (u \,|\, a^{(i)})(p_{n-\rho+1} a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho)}).\end{aligned} \tag{17}$$

In order to pass from (16.) to identities between arbitrary rows $A^{\rho_1}, B^{\rho_2}, \ldots, x, p$, we observe that these identities must become identical with (16.) as soon as we resolve $A^{\rho_1} = a^{(1)} a^{(2)} \cdots a^{(\rho_1)}$, $B^{\rho_2} = b^{(1)} b^{(2)} \cdots b^{(\rho_2)}$, etc., and that they can also be combined into final expressions only by the operations determined by (16.) as soon as the individual expressions are of the type entering into (16.). Thus we obtain, if we also set

$$\rho = \rho_1 + \rho_2 + \cdots + \rho_k \leq n, \tag{18}$$

$$\begin{aligned}&(a^{(1)} \cdots a^{(\rho_1)} b^{(1)} \cdots b^{(\rho_2)} \cdots k^{(1)} \cdots k^{(\rho_k)} \,|\, xp_{\rho-1}) = (A^{\rho_1} B^{\rho_2} \cdots K^{\rho_k} \,|\, xp_{\rho-1}) \\ &\quad = \sum_{i=1}^{\rho_1} (-1)^{i+1} (a^{(i)} \,|\, x)(a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho_1)} B^{\rho_2} \cdots K^{\rho_k} q_{n-\rho+1}) \\ &\quad\quad + (-1)^{\rho_1\rho_2} \sum_{i=1}^{\rho_2} (-1)^{i+1} (b^{(i)} \,|\, x)(b^{(1)} \cdots b^{(i-1)} b^{(i+1)} \cdots b^{(\rho_2)} A^{\rho_1} \cdots K^{\rho_k} q_{n-\rho+1}) \\ &\quad\quad + \cdots \\ &\quad\quad + (-1)^{(\rho_1+\cdots+\rho_{k-1})\rho_k} \sum_{i=1}^{\rho_k} (-1)^{i+1} (k^{(i)} \,|\, x)(k^{(1)} \cdots k^{(i-1)} k^{(i+1)} \cdots k^{(\rho_k)} A^{\rho_1} \cdots K^{\rho_{k-1}} q_{n-\rho+1}).\end{aligned}$$

Each individual sum (but not already a part of the sum) does in fact contain the rows $A^{\rho_1}, B^{\rho_2}, \ldots$ as indecomposable quantities; for it satisfies the necessary and sufficient conditions[63]

$$\left(a^{(i+1)} \middle| \frac{\partial}{\partial a^{(i)}} \right) = 0; \; \ldots \; ; \left(k^{(j+1)} \middle| \frac{\partial}{\partial k^{(j)}} \right) = 0. \quad (i = 1, 2, \ldots, \rho_1 - 1; \; j = 1, 2, \ldots, \rho_k - 1) \tag{19}$$

We show that it also contains all rows explicitly. For if we set $B^{\rho_2} \cdots K^{\rho_k} q_{n-\rho+1} = q_{n-\rho_1+1}$, then by (16.) and (10.) we obtain

$$\begin{aligned}\sum (-1)^{i+1} (a^{(i)} \,|\, x)(a^{(1)} \cdots a^{(i-1)} a^{(i+1)} \cdots a^{(\rho_1)} q_{n-\rho+1}) &= (A^{\rho_1} \,|\, xp_{\rho_1-1}) \\ &= (A_{n-\rho_1} x p_{\rho_1-1}) = (-1)^{(\rho_1+1)(n+1)} (q_{n-\rho_1+1} \,|\, A_{n-\rho_1} x)\end{aligned}$$

[63] Capelli, a. a. O. § 23, gives sufficient conditions: $\left(a^{(k)} \mid \frac{\partial}{\partial a^{(i)}}\right) = 0$, $(k > i, \; i = 1, 2, \ldots, \rho_1 - 1)$. These most simply imply the necessary and sufficient conditions by application of the Poisson bracket process after Wellstein, Von den Differentialgleichungen der projektiven Invarianten, Math. Ann. 67 (1909) § 3.

$$= (-1)^{(\rho_1+1)(n+1)}(B^{\rho_2} \cdots K^{\rho_k} q_{n-\rho+1} \,|\, A_{n-\rho_1} x).$$

If we compute the remaining sums analogously and subsequently set, since the rows are already characterized as distinct by the weight indices, $B^{\rho_2} = A^{\rho_2}$, $C^{\rho_3} = A^{\rho_3}, \ldots$, then we obtain the desired identities:

$$\begin{gathered}(A^{\rho_1} A^{\rho_2} \cdots A^{\rho_k} \,|\, x p_{\rho-1}) = \sum_{i=1}^{k} (-1)^{\delta_i} (A^{\rho_1} \cdots A^{\rho_{i-1}} A^{\rho_{i+1}} \cdots A^{\rho_k} q_{n-\rho+1} \,|\, A_{n-\rho_i} x), \\ \delta_i = (\rho_1 + \rho_2 + \cdots + \rho_{i-1})\rho_i + (\rho_i + 1)(n+1),\end{gathered} \tag{20}$$

and from (17.) the dualistic identities:

$$\begin{gathered}(u q_{\sigma-1} \,|\, A_{\sigma_1} A_{\sigma_2} \cdots A_{\sigma_k}) = \sum_{i=1}^{k} (-1)^{\varepsilon_i} (u A^{n-\sigma_i} \,|\, p_{n-\sigma+1} A_{\sigma_1} \cdots A_{\sigma_{i-1}} A_{\sigma_{i+1}} \cdots A_{\sigma_k}), \\ \varepsilon_i = (\sigma_1 + \sigma_2 + \cdots + \sigma_{i-1})\sigma_i + (\sigma_i + 1)(n + \sigma).\end{gathered} \tag{21}$$

Theorem II. With (20.) and (21.) the totality of the indecomposable identities between the rows A^{ρ_i}, x, p, respectively A_{σ_i}, u, q, is exhausted, and at the same time the totality of those indecomposable identities that admit explicit representation without carrying out substitution (11.).

The first part of the theorem follows from the considerations leading to (20.) and (21.); the second part from the impossibility of an identity between two symbol rows:

$$(q_\rho A^\sigma \,|\, B_{\rho+\sigma-\tau} p_\tau) = \varepsilon_1 (q_\rho B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_\tau) + \varepsilon_2 (q_\rho q_{n-\tau} \,|\, B_{\rho+\sigma-\tau} A_{n-\sigma}),$$

where $\rho \leq \tau \leq \sigma$ and none of the rows has weight $1 \cdot (n-1)$. (Other assumptions about ρ, τ, σ can be reduced to this one by interchanging the row designations.) This impossibility follows, for example, by expanding (8.) under the specialization $q_\rho = l M^{\rho-1}$, $q_{n-\tau} = l N^{n-\tau-1}$, if one sets $l_1 = 1$, $M^{2,3,\ldots,\rho} = 1$, $A^{\rho+1,\ldots,\rho+\sigma} = 1$, all other elements of the rows l, M, A equal to zero, while N and B are arbitrary. Since identities between arbitrary rows, by resolving a row p_τ into $y^{(1)} y^{(2)} \cdots y^{(\tau)}$, pass into identities composite from (20.), the second part of the assertion is proved as soon as (20.) can be generated from identities with only two symbol rows. In fact, from

$$(A^{\rho_1} A^{\rho_2} \,|\, x p_{\rho-1}) = \varepsilon (A^{\rho_1} q_{n-\rho+1} \,|\, A_{n-\rho_2} x) + (A^{\rho_1} \,|\, x p_{\rho_1-1}), \quad \text{where } p_{\rho_1-1} \sim A^{\rho_2} q_{n-\rho+1}, \tag{20a}$$

by specializing $A^{\rho_1} = B^{\sigma_1} B^{\sigma_2}$, that is, by

$$(B^{\sigma_1} B^{\sigma_2} \,|\, x p_{\rho_1-1}) = \varepsilon (B^{\sigma_1} q_{n-\rho_1+1} \,|\, B_{n-\sigma_2} x) + (B^{\sigma_1} \,|\, x p_{\sigma_1-1}), \quad \text{where } p_{\sigma_1-1} \sim B^{\sigma_2} q_{n-\rho_1+1},$$

one obtains an identity (20.) in three symbol rows; and by further specializations of $B^{\sigma_1} = C^{\tau_1} C^{\tau_2}$, etc., the general identities (20.). The nonexistence of the explicit representation of an identity between two symbol rows and two arbitrary variable rows thus implies the nonexistence in the case of arbitrarily many symbol rows, provided that among the variable rows no row x or u is contained.

§ 4. The decomposition identities.

We now consider the most general case of identities among arbitrary symbol and variable rows in whose final expressions, without performing substitution (11.), a row p_τ occurs resolved into its individual rows $y^{(1)} \cdots y^{(\tau)}$; we call these identities "decomposition identities." Following the remark at the end of the preceding paragraph, we first determine them for the case of only two symbol rows, that is, we expand the expression $(q_\rho A^\sigma \mid B_{\rho+\sigma-\tau} p_\tau)$. We set:

$$q_{\rho-\alpha}^{(i_1 \dots i_\alpha)} \sim p_{n-\rho}\, y^{(i_1)} \cdots y^{(i_\alpha)}, \qquad p_{\tau-\alpha}^{(i_1 \dots i_\alpha)} = y^{(i_{\alpha+1})} \cdots y^{(i_\tau)}, \qquad p_\tau = y^{(1)} \cdots y^{(\tau)}. \tag{22}$$

With respect to the numbers ρ, σ, τ four cases must be distinguished:

$$\begin{aligned}
&1)\quad \rho+\sigma \le n, \quad \tau \le \rho, \quad \tau \le \sigma;\\
&2)\quad \rho+\sigma \le n, \quad \tau \ge \rho, \quad \tau \le \sigma;\\
&3)\quad \rho+\sigma \le n, \quad \tau \le \rho, \quad \tau > \sigma;\\
&4)\quad \rho+\sigma \le n, \quad \tau \ge \rho, \quad \tau \ge \sigma.
\end{aligned}$$

We shall single out the first case, and then briefly characterize the others from it. By (20.), (10.) and (9.) we obtain, when we decompose the row $y^{(1)} y^{(2)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}$ into $y^{(1)}$ and $y^{(2)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}$:

$$\begin{aligned}
(q_\rho A^\sigma \mid y^{(1)} y^{(2)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}) &= (q_\rho^{(1)} A^\sigma \mid y^{(2)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau})\\
&\quad + (-1)^\rho (q_\rho [A_{n-\sigma} y^{(1)}]^{\sigma-1} \mid y^{(2)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}).
\end{aligned} \tag{23}$$

The last term in (23.) now has exactly the form of the original expression–only σ is replaced by $\sigma - 1$, and τ by $\tau - 1$–and therefore again admits equation (23.). Thus, since $\tau \le \sigma$, we can apply operation (23.) τ times and, taking account of (10.), obtain the relation:

$$\begin{aligned}
(q_\rho A^\sigma \mid p_\tau B_{\rho+\sigma-\tau}) &= (-1)^{\rho\sigma + (\sigma+\tau)(n+1)} (q_\rho B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_\tau)\\
&\quad + \sum_{i_1=1}^{\tau} (-1)^{\rho(i_1-1)} (q_{\rho-1}^{(i_1)} [A_{n-\sigma} y^{(1)} \cdots y^{(i_1-1)}]^{\sigma-i_1+1} \mid y^{(i_1+1)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau}).
\end{aligned} \tag{24}$$

Every term under the summation sign is again of the type of the original expression; ρ is replaced by $\rho - 1$, σ by $\sigma - i_1 + 1$, and τ by $\tau - i_1$, so that condition 1) is even more strongly satisfied. Since $\tau \le \rho$, we can therefore apply operation (24.) τ times and obtain the desired decomposition identity:

$$\begin{aligned}
(q_\rho A^\sigma \mid p_\tau B_{\rho+\sigma-\tau}) &= \varepsilon_{i_0} (q_\rho B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_\tau)\\
&\quad + \sum_{i_1=1}^{\tau} \varepsilon_{i_1} (q_{\rho-1}^{(i_1)} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_{\tau-1}^{(i_1)})\\
&\quad + \sum_{i_2=i_1+1}^{\tau} \varepsilon_{i_2} (q_{\rho-2}^{(i_1 i_2)} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_{\tau-2}^{(i_1 i_2)}) + \cdots\\
&\quad + \varepsilon_{i_\tau} (q_{\rho-\tau}^{(i_1 \dots i_\tau)} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma})\\
&= \sum_{\alpha=0}^{\tau} \sum_{i_\alpha = i_{\alpha-1}+1}^{\tau} \varepsilon_{i_\alpha} (q_{\rho-\alpha}^{(i_1 \dots i_\alpha)} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_{\tau-\alpha}^{(i_1 \dots i_\alpha)}).
\end{aligned} \tag{25}$$

For the sign factor ε one obtains from (24.):

$$\begin{aligned}
\varepsilon_{i_\alpha} = (-1)^{\delta_{i_\alpha}}, \qquad \delta_{i_\alpha} &= \sum_{\beta=1}^{\alpha} (\rho - \beta + 1)(i_{\beta-1} + i_\beta + 1)\\
&\quad + (\rho - \alpha)(\sigma + i_\alpha + \alpha) + (\sigma + \tau + \alpha)(n+1),
\end{aligned} \tag{26}$$

where $i_0 = 0$ is to be put. The summation sign in (25.) is to be understood as follows: to each combination $i_1 i_2 \dots i_{\alpha-1}$, where $i_1 < i_2 < \cdots < i_{\alpha-1}$, all values i_α for which $i_{\alpha-1} < i_\alpha \le \tau$ are adjoined. Taking (26.) into account, one easily verifies that the individual sums in (25.) satisfy the differential equations analogous to (19.):

$$\left(\frac{\partial}{\partial y^{(i)}} \middle| y^{(i+1)} \right) = 0, \qquad (i = 1, 2, \dots, \tau - 1), \tag{27}$$

and hence depend only on the row p_τ. Thus the decomposition identity is an indecomposable identity among the rows A, B, q, p; we shall return to its explicit representation in the next paragraph.

In case 2), operation (23.) can likewise be applied τ times, but operation (24.) stops after the ρ-th application. Thus in the final expression of (25.) the first summation sign must be replaced by $\sum_{\alpha=0}^{\rho}$. In cases 3) and 4), operation (23.) can be carried out only σ times; in place of (24.) one then has the relation:

$$\begin{aligned}(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) = (-1)^{\rho\sigma}&(q_\rho^{(\sigma+1\dots\tau)} B^{n-\rho+\tau-\sigma} \,|\, A_{n-\sigma} p_{\tau-\sigma}^{(\sigma+1\dots\tau)}) \\ &+ \sum_{i_1=1}^{\sigma} (-1)^{\rho(i_1-1)} (q_{\rho-1}^{(i_1)} [A_{n-\sigma} y^{(1)} \cdots y^{(i_1-1)}]^{\sigma-i_1+1} \,|\, y^{(i_1+1)} \cdots y^{(\tau)} B_{\rho+\sigma-\tau});\end{aligned} \tag{28}$$

repeating (28.) $\tau-\sigma$ times, where the summation sign becomes $\sum_{i_\beta=i_{\beta-1}+1}^{\sigma+\beta-1}$–as the $(\sigma-i_1+1)$-fold application of (23.) to the terms under the summation sign in (28.) shows–gives all terms of the individual sum in (25.) corresponding to the index $\alpha = \tau-\sigma$, with the corresponding sign. Then cases 3) and 4) pass over into cases 1) and 2), for the numbers corresponding to σ, τ become $\sigma' = \sigma - i_{\tau-\sigma} + \tau - \sigma$, $\tau' = \tau - i_{\tau-\sigma} = \sigma'$; and as the procedure continues, $\tau' < \sigma'$. Thus in the final expression of (25.) the first summation sign must be replaced by $\sum_{\alpha=\tau-\sigma}^{\tau,\rho}$.

Analogously, from (21.) one obtains the dual decomposition identity, in which a row q_ρ is resolved into the individual rows $v^{(1)}, \dots, v^{(\rho)}$. If we put

$$\dot{p}_{\tau-\beta}^{(i_1\dots i_\beta)} \sim v^{(i_1)} \cdots v^{(i_\beta)} q_{n-\tau}; \qquad \dot{q}_{\rho-\beta}^{(i_1\dots i_\beta)} = v^{(i_{\beta+1})} \cdots v^{(i_\rho)}; \qquad q_\rho = v^{(1)} \cdots v^{(\rho)}, \tag{29}$$

then

$$(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) = \sum_{\beta=\max(0,\tau-\sigma)}^{\min(\tau,\rho)} \sum_{i_\beta=i_{\beta-1}+1}^{\rho} \varepsilon (\dot{q}_{\rho-\beta}^{(i_1\dots i_\beta)} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} \dot{p}_{\tau-\beta}^{(i_1\dots i_\beta)}), \tag{30}$$

where the summation index β runs from the larger of the lower values to the smaller of the upper values. All terms of an individual sum in (25.) and (30.) are polars of a single form, produced by the polar processes

$$\left(\frac{\partial}{\partial p_{\tau-\alpha}} \middle| p_{\tau-\alpha}^{(i_1\dots i_\alpha)}\right), \qquad \left(q_{\rho-\alpha}^{(i_1\dots i_\alpha)} \middle| \frac{\partial}{\partial q_{\rho-\alpha}}\right).$$

To express this, we denote by Π an aggregate of such polar operations, multiplied by constants, and obtain, in place of (25.) and (30.), the relation

$$(q_\rho A^\sigma \,|\, p_\tau B_{\rho+\sigma-\tau}) = \sum_{\alpha=\max(0,\tau-\sigma)}^{\min(\tau,\rho)} \Pi (q_{\rho-\alpha} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_{\tau-\alpha}). \tag{31}$$

§ 5. The decomposition identities in explicit form.

If in (25), in case 4), we specialize τ to $\sigma+\rho$, we obtain

$$(q_\rho A^\sigma \,|\, y^{(1)} \cdots y^{(\rho+\sigma)}) = \sum_{1\le i_1<\dots<i_\rho\le\rho+\sigma} \varepsilon_{i_\rho} (q_\rho \,|\, y^{(i_1)} \cdots y^{(i_\rho)}) (A^\sigma \,|\, y^{(i_{\rho+1})} \cdots y^{(i_{\rho+\sigma})}), \tag{32}$$

where $\varepsilon_{i_\rho} = (-1)^{\delta_{i_\rho}}$ and $\delta_{i_\rho} = \rho(\rho+1)/2 + i_1 + \dots + i_\rho$. This is a more general form of the product theorem for matrices than (16.) and (17.), and follows by Laplace expansion of the determinant of the product matrix

$$\sum \pm (v^{(1)} y^{(1)}) \cdots (v^{(\rho)} y^{(\rho)}) (a^{(1)} y^{(\rho+1)}) \cdots (a^{(\sigma)} y^{(\rho+\sigma)}).$$

Thus the more general product theorem is composed out of the identities (20.) and (21.), respectively, and the selection of the special form (16.), (17.) is thereby explained.

Relation (32.) now gives the means for explicitly representing the decomposition identities. For this purpose we regard the forms occurring in (31.) as ground forms; that is, we perform substitution (11.), which changes nothing in the explicit representation in the rows A, B, since by (8.) the newly introduced rows R can be expressed through the rows A, B themselves and not through their component rows $a^{(1)} a^{(2)} \dots$, $b^{(1)} b^{(2)} \dots$. Thus let

$$(q_{\rho-\alpha} B^{n-\rho-\sigma+\tau} \,|\, A_{n-\sigma} p_{\tau-\alpha}) = (R_{\rho-\alpha} p_{n-\rho+\alpha} \,|\, R^{\tau-\alpha} p_{\tau-\alpha}). \tag{33}$$

Then, for the individual sum of (25.) corresponding to the index α, we get

$$\begin{aligned}&\sum \varepsilon_{i_\alpha}(R_{\rho-\alpha}p_{n-\rho}y^{(i_1)}\cdots y^{(i_\alpha)}\,|\,R^{\tau-\alpha}y^{(i_{\alpha+1})}\cdots y^{(i_\tau)})\\&\quad=\sum \varepsilon_{i_\alpha}([R_{\rho-\alpha}p_{n-\rho}]^{\alpha}y^{(i_1)}\cdots y^{(i_\alpha)}\,|\,R^{\tau-\alpha}y^{(i_{\alpha+1})}\cdots y^{(i_\tau)})\\&\quad=\varepsilon([R_{\rho-\alpha}p_{n-\rho}]^{\alpha}R^{\tau-\alpha}\,|\,p_\tau)=\varepsilon(R^{\tau-\alpha}q_{n-\tau}\,|\,R_{\rho-\alpha}p_{n-\rho}).\end{aligned}\tag{34}$$

Thus an explicit final expression is in fact obtained, and the symmetric form in which q_ρ and p_τ occur shows that (30.) leads to the same final expression, which is therefore independent of the original decomposition. There is a difference between (25.) and (30.) only as long as the rows y and v have independent meaning, so that the decomposition identity passes, by our definition, into a decomposable identity. To arrive at identities with more symbol rows, one need only, according to § 3 (end), resolve a row q_ρ into $q_{\rho_1}C^{\sigma_1}$, or p_τ into $p_{\tau_1}C_{\tau-\tau_1}$. The resulting expressions

$$(q_{\rho_1}C^{\sigma_1}\,|\,[R^{\tau-\alpha}q_{n-\tau}]_\lambda R_{\rho+\sigma_1-\alpha-\lambda}),\qquad ([R_{\rho-\alpha}p_{n-\rho}]^{\lambda}R^{\tau-\alpha}\,|\,p_{\tau_1}C_{\tau-\tau_1})\tag{35}$$

are exactly of the type that occurs for two symbol rows, and therefore again admit explicit representation after a substitution corresponding to (33.). Repetition therefore gives explicit representation for arbitrarily many rows. It may also be noted that the first and last sums in (25.) and (30.), respectively, give the explicit representation without applying (33), solely by formula (32), or else reduce altogether to a single term containing all rows explicitly. The relation obtained from (32.) by resolving q_ρ into $q_{\rho_1}C^{\sigma_1}$ and so forth can be regarded as the product theorem in its most general form.

In summary we shall show:

Theorem III: With the identities

$$(q_\rho A^\sigma\,|\,p_\tau B_{\rho+\sigma-\tau})=\sum_{\alpha=\max(0,\tau-\sigma)}^{\min(\tau,\rho)}\varepsilon(R^{\tau-\alpha}q_{n-\tau}\,|\,R_{\rho-\alpha}p_{n-\rho}),\tag{36}$$

where the rows R are determined by (33), together with those obtained from them by resolving q_ρ, p_τ into $q_{\rho_1}C^{\sigma_1}$, etc., the totality of indecomposable identities is exhausted.

Indeed, the resulting identities are the most general of their kind, since the individual rows are no longer subject to any restriction as to weight or number, as was still the case in (20.) and (21.). Nor do any further identities among these rows exist; for when a row p_τ is resolved into $y^{(1)}\cdots y^{(\tau)}$, the most general identities are composed out of (20.), hence by § 3 (end) out of (20a.); and the composition discussed following (16.) is precisely the one carried out in § 4. The transition to arbitrarily many rows is supplied, as the discussion of (20a.) shows, by applying always the same operation to the expressions that arise from resolving q_ρ into $q_{\rho_1}C^{\sigma_1}$, and so on. It also follows that the direct transition from (20.) instead of from (20a.) gives nothing new. This is easily verified by calculation, by specializing once a row q_ρ in (25.), and another time carrying out the transition from (20.) by the method of § 4. In both cases the same sums arise, which pass by application of the general product theorem (see above) into the identities obtained from (36.). The identities (20.), (21.) correspond to the special cases $\tau=1$, $\rho=1$; only two individual sums then occur, and hence, in accordance with an earlier remark, explicit representation without the substitution (33.), agreeing with what was found before.

At the end let us briefly mention two special cases of the decomposition identity. Put in (31): $\tau=\sigma$, $\rho=n-\sigma$, and denote the row p_τ by $p'_\sigma\sim q'_{n-\sigma}$. Then

$$(A^\sigma\,|\,p_\sigma)(B^\sigma\,|\,p'_\sigma)-(A^\sigma\,|\,p'_\sigma)(B^\sigma\,|\,p_\sigma)=\sum_{\alpha=1}^{\min(\sigma,n-\sigma)}\Pi(q_{n-\sigma-\alpha}B^\sigma\,|\,A_{n-\sigma}p_{\sigma-\alpha}).\tag{37}$$

This is Clebsch's formula[64] for expanding a form with two cogredient rows p_σ, p'_σ into elementary covariants. The elementary covariants belonging to a form $(A^\sigma\,|\,p_\sigma)^m(B^\sigma\,|\,p'_\sigma)^n$ are then the following:

$$\left\{\sum_{\alpha=1}^{\min(\sigma,n-\sigma)}\varepsilon(q_{n-\sigma-\alpha}B^\sigma\,|\,A_{n-\sigma}p_{\sigma-\alpha})\right\}^{\rho}(A^\sigma\,|\,p_\sigma)^{m-\rho}(B^\sigma\,|\,p'_\sigma)^{n-\rho},\qquad(\rho=0,1,\ldots,m,n).\tag{38}$$

[64]Clebsch, loc. cit., pp. 25–32. Clebsch proves that the individual terms on the right-hand sides depend only on the rows A, B; the explicit representation would be obtained by applying (32.) to his expressions Φ_h.

They arise, as we shall briefly say, from the ground form by "cogredient folding of defect ρ" (cf. § 2).

Second, put $\tau = \sigma - \lambda$, $\rho = n - \sigma - \lambda$, $B^\sigma = A^\sigma$. Then

$$\begin{aligned}(q_{n-\sigma-\lambda}A^\sigma \mid A_{n-\sigma}p_{\sigma-\lambda}) = (-1)^\lambda(q_{n-\sigma-\lambda}A^\sigma \mid A_{n-\sigma}p_{\sigma-\lambda}) \\ + (-1)^{(n-\sigma)(\sigma-\lambda)} \sum_{\alpha=1}^{\min(\sigma-\lambda,n-\sigma-\lambda)} \Pi(q_{n-\sigma-\lambda-\alpha}A^\sigma \mid A_{n-\sigma}p_{\sigma-\lambda}).\end{aligned} \tag{39}$$

Thus, as noted in § 2, the conditions (12.) characterizing the symbol rows for odd defect λ are consequences of the conditions of higher defect. For a linear form $(A^\sigma \mid p_\sigma) = (B^\sigma \mid p_\sigma)$, (39.) implies that its irreducible invariant formations that are quadratic in the coefficients reduce to ones of even defect; this agrees with the known fact that a linear form in the binary and ternary domains has no invariant formations.

It should be noted that the general identities were originally derived under the assumption that the individual rows satisfy conditions (12.). Since, however, these individual rows enter the identities only linearly, the latter also hold for the more general rows that do not satisfy conditions (12.).

§ 6. Explicit representation of the invariant formations. Folding and differentiation processes.

The results of the preceding paragraphs allow us to complete the theorem stated in § 2 on symbolic representability (the first fundamental theorem) by adding explicit representability; namely, we show:

Theorem IV: "All invariant formations of arbitrary ground forms can be represented explicitly by matrix products; that is, aside from variable rows p_ρ, only the rows $S^\sigma, \dots, T^\tau$ of the original forms, or the rows P, Q, R derived from them by substitution (9.) or (11.), enter the final expressions."

For the proof, suppose that an invariant formation depends, besides variable rows, on the rows $S^\sigma, \dots, T^\tau$; and suppose that these rows have been resolved sufficiently far into individual rows $S^\sigma = S^{\sigma_1} \cdots S^{\sigma_i}, \dots, T^\tau = T^{\tau_1} \cdots T^{\tau_k}$ to make representation by determinants possible. Let the rows be ordered in a sequence $S^\sigma, \dots, T^\tau$ that is arbitrary in itself but fixed once and for all. We pick out an aggregate of determinants that depends on the first total row $S^\sigma = S^{\sigma_1} \cdots S^{\sigma_i}$, not merely on individual component rows. This dependence, however, is by § 3–§ 5 a consequence of the general identities. If we now resolve a variable row p_ρ into the rows y, v, or one of the later rows T into the individual rows t, respectively τ, then the most general identities are composed out of (20.) and (21.). But these latter identities always lead to an expression containing the row $S^\sigma = S^{\sigma_1} \cdots S^{\sigma_i}$ explicitly, namely the left-hand side of (20.) and (21.); for by (19.) one verifies that only the complete sum on the right–not a part of it corresponding to a term of (20a.)–has the property of depending only on S^σ. As the procedure continues, all rows that have once entered explicitly remain explicit; application of the identities can at most require the combining of several rows into a single one, i.e. substitution (9.). If finally only a single row T^τ remains to be brought to explicit form, two cases are possible. There may still occur a free variable row, that is, one not connected with other rows by substitution (9.); resolving it into component rows y or v completes the procedure and gives, as in (31.), polars of one or more forms containing all rows explicitly. If no free variable row remains, then from the manner in which they arose we recognize, in the totality of the expressions that contain the individual rows t or τ but depend on the row T^τ, the terms of a decomposition identity; by § 5, however, these always admit explicit representation in all rows after applying substitution (11.). This proves the theorem in full generality. It should also be noted that when only two symbol rows occur, substitution (11.) never appears. For, since $\sigma, n-\sigma, \tau, n-\tau < n$, no variable rows can enter only when the two symbol rows are united in a single determinant and thus occur explicitly; but as soon as variable rows enter, (34.) and (33.) show that substitution (11.) becomes unnecessary.

It follows from what has just been said that, in order to generate all invariant formations, it suffices to unite the symbol rows, after adjoining variable rows, into all possible matrix products. The most general matrix product is now of the form

$$(P^{\mu_1} \cdots P^{\mu_i} \mid Q_{\nu_1} \cdots Q_{\nu_k}), \qquad \sum \mu = \sum \nu, \tag{40}$$

where the P and Q may be rows of the original forms, or may have arisen from the original rows by a sequence of substitutions (9.) or (11.), or finally may be variable rows. But the expressions (40.) can be generated by successively uniting two symbol rows at a time into a matrix product, with the help of variable rows; for from

$$(q_{n-\mu-\lambda}P^\mu \mid Q_{\nu_1}p_{n-\mu-\nu_1-\lambda}) = (R_\lambda Q_{\nu_1} \mid p_{n-\mu-\nu_1-\lambda})$$

one obtains, by uniting with the row Q_{ν_2}, an expression

$$([R_\lambda Q_{\nu_1}]^{n-\lambda-\nu_1} \mid Q_{\nu_2} p_{n-\mu-\nu_1-\nu_2-\lambda}) = (q_{n-\mu-\lambda} P^\mu \mid Q_{\nu_1} Q_{\nu_2} p_{n-\mu-\nu_1-\nu_2-\lambda}),$$

and hence, by continuing the procedure, an expression (40.). If P and Q are not rows of the original forms, then (9.) and (11.), in conjunction with what was just shown, imply that they too arose through a sequence of unions of two symbol rows at a time. Thus:

Theorem V: The process for generating all invariant formations, the folding process, consists in successively uniting two symbol rows at a time into a matrix product, with the help of variable rows.

From two symbol rows S^σ, T^τ, where $\sigma \geq \tau$, the following matrix products can be formed:

$$\begin{aligned} &(q_{n-\sigma-\lambda} S^\sigma \mid T_{n-\tau} p_{\tau-\lambda}), && (\lambda = 1, 2, \ldots, \min(\tau, n-\sigma)), && (41)\\ &(q_{n-\tau-\mu} T^\tau \mid S_{n-\sigma} p_{\sigma-\mu}), && (\mu = \sigma - \tau + \lambda), && (42)\\ &(q_{n-\tau-\nu} T^\tau \mid S_{n-\sigma} p_{\sigma-\nu}), && (\nu = 1, 2, \ldots, \sigma - \tau). && (43)\end{aligned}$$

From (31.), however, the following values are obtained for (42.) and (43.):

$$(q_{n-\sigma-\lambda} S^\sigma \mid T_{n-\tau} p_{\tau-\lambda}) + \sum_\alpha \Pi(q_{n-\sigma-\lambda-\alpha} S^\sigma \mid T_{n-\tau} p_{\tau-\lambda-\alpha}), \tag{42a}$$

$$\Pi(q_{n-\sigma} S^\sigma)(T_{n-\tau} p_\tau) + \sum_\beta \Pi(q_{n-\sigma-\beta} S^\sigma \mid T_{n-\tau} p_{\tau-\beta}). \tag{43a}$$

Thus the forms (42.) can be expressed linearly by (41.) and polars of (41.), and the forms (43.) by polars of the ground form and of the forms (41.). Likewise (31.) gives the linear dependence of the forms (41.) on (42.) and polars of (42.). Hence, according to Clebsch's definition,[65] the forms (42.) are equivalent to the forms (41.), while (43.) leads back to the ground form. The generating process is therefore given with equal right by (41.) or (42.); we shall define the process (41.), which is simpler with respect to the number λ, as the folding process. The number λ, the defect of the folding (cf. § 2), gives the number of rows missing from the determinant product, namely in that matrix product which, for the given folding, contains the maximal number of rows. Since λ runs only up to the smaller of the two values $\tau, n-\sigma$, where $\sigma \geq \tau$, we have

$$\lambda \leq \frac{n}{2},\ n \text{ even}; \qquad \lambda \leq \frac{n-1}{2},\ n \text{ odd}. \tag{44}$$

The reduction of (42.) and (43.) to (41.) remains valid even when, through further folding, the rows p, q have passed into symbol rows; for by (36.) one has

$$(A^{n-\tau-\varkappa} T^\tau \mid S_{n-\sigma} B_{\sigma-\varkappa}) = \sum_{\alpha=\max(0,\varkappa-\sigma+\tau)}^{\min(\tau,n-\sigma)} \varepsilon(R^{\tau-\alpha} B^{n-\sigma+\varkappa} \mid A_{\tau+\varkappa} R_{n-\sigma-\alpha}),$$

where the rows R have arisen from S and T according to (33.). Thus these forms containing symbol rows only are obtained by folding $(q_\tau A^{n-\tau-\varkappa} \mid B_{\sigma-\varkappa} p_{n-\sigma})$ or $(q_{\tau-\alpha} B^{n-\sigma+\varkappa} \mid A_{\tau+\varkappa} p_{n-\sigma-\alpha})$–according as $\sigma - \tau \gtreqless 2\varkappa$–with the ground form and the forms (41.).

From the symbolic folding process one obtains, in the usual way, the invariant differentiation processes if one merely replaces the symbol rows $S^\sigma, T_{n-\tau}$ by the rows of differentiation symbols $\frac{\partial}{\partial p_\sigma}, \frac{\partial}{\partial q_{n-\tau}}$; as is well known, the product $\frac{\partial}{\partial p_{i_1\ldots i_\sigma}} \frac{\partial}{\partial q_{j_1\ldots j_{n-\tau}}}$ then has the real meaning $\frac{\partial^2}{\partial p_{i_1\ldots i_\sigma} \partial q_{j_1\ldots j_{n-\tau}}}$. Since the rows $\frac{\partial}{\partial p_\sigma}$ and $\frac{\partial}{\partial q_{n-\tau}}$ are cogredient with the rows $S^\sigma, T_{n-\tau}$, they satisfy the same identities as these, in particular also the ones existing between (42.) and (42a.), and between (43.) and (43a.). In summary we obtain:

Theorem VI: All invariant formations of arbitrary ground forms arise from these by repeated application of the symbolic folding process (41.), or also of the invariant differentiation processes

$$\left(q_{n-\sigma-\lambda} \frac{\partial}{\partial p_\sigma} \middle| \frac{\partial}{\partial q_{n-\tau}} p_{\tau-\lambda}\right), \qquad (\sigma \geq \tau,\ \lambda = 1, 2, \ldots, \min(\tau, n-\sigma)). \tag{45}$$

It remains to note that in every folding (41.) the total weight of the form, that is, the sum of the weights of the individual variable rows (cf. § 1), decreases by the positive number $2(\lambda^2 + \lambda(\sigma - \tau))$. This implies,

[65] Clebsch, loc. cit., § 7.

independently of the general finiteness proof, the finiteness of the form row, that is, of the totality of invariant formations of a given ground form that are linear in the coefficients.[66]

It should further be noted that Theorem VI still holds for forms that do not satisfy conditions (12.). For each factor in (3), $(S^\rho \,|\, p_\rho)^{m_\rho}$, can be replaced by a product of m_ρ linear factors $(A^\rho \,|\, p_\rho)(B^\rho \,|\, p_\rho)\cdots(M^\rho \,|\, p_\rho)$; the invariant formations thereby pass into those of a system of linear forms, which need only afterwards be set equal to one another. For every single linear form, however, conditions (12.)–which, when differential symbols are introduced, contain only second-order differential quotients–are always satisfied.

§ 7. Expansion of the general forms according to normal forms.

In what follows (§ 8) we shall derive a principal property of the folding process (41.), namely its composability from fundamental foldings. For this we need the transfer to the n-ary domain of an important theorem given by Mertens[67] in the quaternary domain. This theorem says that any finite system of forms can be replaced by an equivalent system of normal forms. By a "normal form" we mean–generalizing the binary and ternary definition[68]–a form all of whose invariant formations linear in the coefficients vanish identically, with the exception of the ground form itself. Thus by Theorem VI (§ 6) a normal form φ satisfies the system of differential equations

$$\left(q_{n-\sigma-\lambda}\frac{\partial}{\partial p_\sigma}\,\middle|\,\frac{\partial}{\partial q_{n-\tau}}p_{\tau-\lambda}\right)\varphi = 0, \qquad (\sigma \ge \tau,\ \lambda = 1, 2, \ldots, \min(\tau, n-\sigma)), \tag{46}$$

where the rows $q_{n-\sigma-\lambda}, p_{\tau-\lambda}$ are to be regarded as arbitrary quantities. In the quaternary domain–taking (39.) into account for $\sigma = \tau = 2$–these differential equations agree completely with the ten differential equations given by Mertens without introduction of the arbitrary rows p, q (loc. cit., formula 28). Their knowledge, together with Theorem VI, allows us to transfer his proof almost word for word to n variables. The only thing to add is the verification, obtained by the decomposition identity (30.), that the constructed forms are normal forms; in the quaternary domain this verification still follows without further transformation. It is therefore enough to give a short exposition of the proof, and for simplicity in the case relevant below, namely that of a single ground form f and its form row (cf. § 6, end); for invariant formations of arbitrary degree in the coefficients of arbitrarily many ground forms, the proof remains the same.

The basic idea of the proof is to replace, by introducing the transformed coefficients, a form with $n-1$ rows p_ρ by forms with n rows ξ cogredient to the x, namely the rows of the transformation quantities. From these forms the sought normal forms are obtained directly by invariant differentiation processes. The expansion of the general forms according to normal forms is mediated by a series expansion for forms with ρ ($\rho < n$) cogredient rows ξ; invariant differentiation processes lead back to the forms in the original rows p_ρ.

Let $\xi^{(1)}, \ldots, \xi^{(n)}$ be the rows of transformation quantities, so that

$$\begin{aligned} x_i &= \xi_i^{(1)}x_1' + \xi_i^{(2)}x_2' + \cdots + \xi_i^{(n)}x_n', \\ p_{i_1\ldots i_\rho} &= \sum_j (\xi_{i_1}^{(j_1)}\cdots\xi_{i_\rho}^{(j_\rho)})p'_{j_1\ldots j_\rho}. \end{aligned} \tag{47}$$

The transformed coefficients of the forms $f, \varphi, \ldots$ will be denoted by $(f), (\varphi), \ldots$, and let

$$f = (S^1 \,|\, p_1)^{m_1}(S^2 \,|\, p_2)^{m_2}\cdots(S^{n-1} \,|\, p_{n-1})^{m_{n-1}}.$$

Then

$$\begin{aligned} (f) = {} & (S^1 \,|\, \xi^{(1)})^{m_{11}}\cdots(S^1 \,|\, \xi^{(n)})^{m_{1n}}(S^2 \,|\, \xi^{(1)}\xi^{(2)})^{m_{21}}\cdots(S^{n-1} \,|\, \xi^{(2)}\cdots\xi^{(n)})^{m_{n-1,n}} \\ & \cdot (s^{(1)} \,|\, \xi^{(1)})^{k_1}(s^{(2)} \,|\, \xi^{(2)})^{k_2}\cdots(s^{(n)} \,|\, \xi^{(n)})^{k_n}, \end{aligned} \tag{48}$$

where $m_{11} + \cdots + m_{1n} = m_1$, and so on. From each coefficient (f) for which

$$k_1 \ge k_2 \ge k_3 \ge \cdots \ge k_n, \tag{49}$$

the differentiation process exhausting exactly the rows ξ,

$$\left(\frac{\partial}{\partial\xi^{(1)}}\middle| p_1\right)^{k_1-k_2}\left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\middle| p_2\right)^{k_2-k_3}\cdots\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(n-1)}}\middle| p_{n-1}\right)^{k_{n-1}-k_n}\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(n)}}\right)^{k_n} \tag{50}$$

[66] Cf. Clebsch, loc. cit., § 11 and 12: finiteness of the system of elementary covariants.

[67] F. Mertens, *Über invariante Gebilde quaternärer Formen* (see the Introduction), cited as "Mertens."

[68] Study, loc. cit., p. 54.

gives the form

$$\varphi = \bigl(s^{(1)} \mid p_1\bigr)^{k_1-k_2}\bigl(s^{(1)}s^{(2)} \mid p_2\bigr)^{k_2-k_3} \cdots \bigl(s^{(1)}\cdots s^{(n-1)} \mid p_{n-1}\bigr)^{k_{n-1}-k_n}\bigl(s^{(1)}s^{(2)}\cdots s^{(n)}\bigr)^{k_n}. \tag{51}$$

The forms φ are the sought normal forms. Indeed, they have the following properties defining normal forms.

1. They are invariant formations of the ground form f that are linear in the coefficients. This follows from the fact that they arise from f by the invariant differentiation processes $\left(\frac{\partial}{\partial p_\rho}\Big|\xi^{(i_1)}\xi^{(i_2)}\cdots\xi^{(i_\rho)}\right)$ and (50.). Thus (by § 6, end) they can be expressed linearly by polars of the finite form row of f. These polars are obtained by regarding the variables p_ρ entering φ as coefficients of a form decomposed into linear forms with the variable rows p_ρ:

$$H = (q_{n-1} \mid p_{n-1})^{k_1-k_2}(q_{n-2} \mid p_{n-2})^{k_2-k_3}\cdots(q_1 \mid p_1)^{k_{n-1}-k_n}. \tag{52}$$

The simultaneous invariants of the form rows f and H give all relevant polars, since the latter must be of the same dimension in the individual rows p_ρ as φ itself.

2. They satisfy the differential equations (46.). If the differential process (45.) is applied to φ, then

$$\varphi' = \bigl(q_{n-\sigma-\lambda}s^{(1)}\cdots s^{(\sigma)} \mid [s^{(1)}\cdots s^{(\tau)}]_{n-\tau}p_{\tau-\lambda}\bigr), \qquad (\sigma > \tau).$$

From the decomposition identity (30.) it follows, if the rows $q_\sigma = s^{(1)}\cdots s^{(\sigma)}$, $p_{n-\tau} = [s^{(1)}\cdots s^{(\tau)}]_{n-\tau}$ formed out of the rows s are identified with the q_ρ, p_τ occurring there, that

$$\varphi' = \sum_{\beta=\sigma-\tau+\lambda}^{n-\tau,\sigma}\ \sum_{i_\beta}\varepsilon\,\bigl(q_{\sigma-\beta}^{(i_1\ldots i_\beta)}q_{n-\tau+\lambda} \,|\, p_{\sigma+\lambda}p_{n-\tau-\beta}^{(i_1\ldots i_\beta)}\bigr).$$

Here

$$p_{n-\tau-\beta}^{(i_1\ldots i_\beta)} \sim s^{(1)}\cdots s^{(\tau)}s^{(i_1)}\cdots s^{(i_\beta)},$$

where $i_1,\ldots,i_\beta$ denote any β distinct numbers chosen from the numbers 1 to σ. Since $\beta > \sigma-\tau$, it follows that among these numbers $i_1,\ldots,i_\beta$ there must be some between 1 and τ; hence $p_{n-\tau-\beta}^{(i_1\ldots i_\beta)}$ vanishes identically, and therefore every individual matrix product, and with it φ', vanishes.

For the equivalence of the system of normal forms with the form row f, it remains only to show that all forms of the form row can be expanded in polars of the normal forms in the sense fixed under 1. Let Π be an aggregate of polar operations

$$\left(\frac{\partial}{\partial\xi^{(i)}}\Big|\xi^{(k)}\right), \qquad (i \neq k;\ i = 1,2,\ldots,\rho;\ k = 1,2,\ldots,\rho), \tag{53}$$

and let Π^σ be an aggregate of the special operations (53.) for which $i=\sigma$ and $k<\sigma$. For a form F of the ρ rows $\xi^{(1)},\ldots,\xi^{(\rho)}$ that has degree k_ρ in the row $\xi^{(\rho)}$, the expansion[69] holds:

$$F = \sum_{\alpha=0}^{k_\rho}\Pi F_\alpha, \qquad (\rho \le n), \tag{54}$$

where F_α has degree α in the last row $\xi^{(\rho)}$ and arises from F by the invariant differential operations

$$\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(\rho)}}\Big|\xi^{(1)}\cdots\xi^{(\rho)}\right)^\alpha \tag{55}$$

and Π^ρ. If, in constructing F_α, one replaces process (55.) by

$$\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(\rho)}}\Big|p_\rho\right)^\alpha, \tag{56}$$

then a form $F_\alpha(p_\rho)$ with only $\rho-1$ rows $\xi^{(1)},\ldots,\xi^{(\rho-1)}$ is obtained, to which (54.) can again be applied. Continuing the procedure leads to forms $G(p_\rho,p_{\rho-1},\ldots,p_1)$. In order to obtain from these the expansion of F according to (54.), one replaces the individual rows p_ρ by $\xi^{(1)},\ldots,\xi^{(\rho)}$ and applies to the resulting forms

[69] F. Mertens, *Über eine Formel der Determinantentheorie.* Sitzungsberichte der Akademie der Wissenschaften, Vienna, Math.-Naturw. Kl., vol. 91, 2nd section (1885), formula 20).

the polar process Π (53.); this gives (cf. (48.)) a sum of transformed coefficients (G). For $\rho = n$, the process (55.) remains at the first step. If c denotes numerical constants and we put, for brevity,

$$(\xi^{(1)}\xi^{(2)}\cdots\xi^{(n)}) \sim \Delta; \qquad \left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\cdots\frac{\partial}{\partial\xi^{(n)}}\right) = \nabla, \tag{57}$$

then, in the case $\rho = n$,

$$F = \sum c\Delta^{\alpha}(G), \tag{58}$$

whereas for $\rho < n$ only $\alpha = 0$, and hence $\sum c(G)$, appears on the right.

Applying (58.) to a transformed coefficient (f) (48.), and using the commutativity of the polar processes Π^{σ} with all operations (56.) for which $\rho \geq \sigma$, as well as the fact that polars of transformed coefficients again give transformed coefficients, one obtains

$$G = \left(\frac{\partial}{\partial\xi^{(1)}}\Big|p_1\right)^{\beta_1}\left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\Big|p_2\right)^{\beta_2}\cdots\left(\frac{\partial}{\partial\xi^{(1)}}\cdots\frac{\partial}{\partial\xi^{(n-1)}}\Big|p_{n-1}\right)^{\beta_{n-1}}\nabla^{\beta_n}\Pi(f) = \sum c\varphi,$$

and from (58.) it follows that

$$(f) = \sum c\Delta^{\alpha}(\varphi) \qquad \text{(Mertens, formula 23)).} \tag{59}$$

To pass from (59.) to the expansion of f itself, we again regard the variables p_ρ as coefficients of a form H (52.) with the degrees $m_1, \ldots, m_{n-1}$ corresponding to f, and put $m = m_1 + \cdots + m_{n-1}$. Then, by the definition of invariants,

$$f \cdot \Delta^m = \sum (f)(H) = \sum c\Delta^{\alpha}(\varphi)(H),$$

and application of the process ∇^m that exactly exhausts the rows ξ gives

$$f = \sum c\Phi, \tag{60}$$

where the forms Φ are derived from φ and H by invariant differentiation processes with respect to the variables, and hence are simultaneous invariants of the form rows φ and H.[70] But since φ is a normal form, the form row φ reduces to the form φ itself; the form row H contains all possible invariant variable combinations. Thus the simultaneous invariants of φ and H are the polars of φ in the sense indicated under 1.; in other words, (60.) gives the expansion of f in polars of the normal forms. It remains to verify the same expansion for an arbitrary form θ of the form row f. Let Γ be the normal forms derived from (θ) by (50.), so that

$$(\theta) = \sum c\Delta^{\beta}(\Gamma). \tag{61}$$

The forms θ are characterized by the relation, valid for every transformed coefficient,

$$(\theta)\Delta^{\sigma} = \sum c(f) \qquad \text{(Mertens, formula 9)).}$$

But every form F of the n rows ξ which contains the factor Δ^{σ} also contains the second $\nabla^{\sigma}\Pi F$ (Mertens, formula 20)); hence

$$(\theta) = \sum c\nabla^{\sigma}(f), \qquad \text{therefore} \qquad \Gamma = \sum c\varphi,$$

and from (61.) we obtain the relation corresponding to (59.),

$$(\theta) = \sum c\Delta^{\beta}(\varphi),$$

which implies for θ the expansion (60.) in polars of the normal forms φ. Thus:

Theorem VII. Every form row f can be replaced by an equivalent system of normal forms.

[70] Instead of the direct conclusion, Mertens, § 7, introduces a series of further differential processes that essentially serve to form the form row H; this is done by our Theorem VI.

§ 8. Reduction of the folding process to fundamental foldings.

Following Theorem VII, we now carry out the reduction of the folding process to fundamental foldings mentioned at the beginning of § 7. We call (cf. § 5) every folding of two cogredient rows S^ρ, T^ρ a "cogredient folding", and specifically the cogredient folding of defect 1 of two rows S^ρ, T^ρ a "fundamental folding of type ρ". We then prove, completing Theorem VI:

Theorem VIII: The general folding process (41.) can be composed from the $(n-1)$ fundamental foldings of type ρ, where $\rho = 1, 2, \ldots, n-1$. In other words, to generate all invariant formations it suffices to apply successively the $n-1$ fundamental foldings.

In the binary domain the folding process (41.) coincides directly with the only possible fundamental folding; in the ternary domain I proved Theorem VIII in § 1 of my dissertation (cf. the Introduction). There, because of (44.), the general cogredient foldings are still identical with the fundamental foldings, and it is noteworthy that Theorem VIII for n variables gives the reduction to fundamental foldings, not merely the much weaker reduction to cogredient foldings. The proof is modeled in principle on the ternary proof: the most general foldings are constructed from the fundamental foldings, using the symbolic identities. We shall generate:

1. arbitrary foldings of defect 1 by cogredient foldings of defect 1, that is, by fundamental foldings (the analogue of the ternary case),
2. arbitrary foldings of defect λ by cogredient foldings of defect λ and arbitrary foldings of defect 1,
3. cogredient foldings of defect λ by cogredient foldings of defect $\lambda - 1$ and arbitrary foldings of defect 1.

As is essential, we assume by Theorem VII that the two forms S and T to be folded are normal forms. Thus, by (46.), the relations hold:

$$\begin{aligned} (q_{n-\sigma_1-\lambda}S^{\sigma_1} \,|\, S_{n-\sigma_2}p_{\sigma_2-\lambda}) = 0, &\quad (\sigma_1 \geq \sigma_2,\ \lambda = 1, 2, \ldots, \sigma_2, n-\sigma_1), \\ (q_{n-\tau_1-\lambda}T^{\tau_1} \,|\, T_{n-\tau_2}p_{\tau_2-\lambda}) = 0, &\quad (\tau_1 \geq \tau_2,\ \lambda = 1, 2, \ldots, \tau_2, n-\tau_1). \end{aligned} \tag{62}$$

It is further sufficient, by § 2 (end), to suppose that the forms S and T are linear in all variable rows; that is, the constructed forms belong to the form row

$$f = (S^1 \,|\, p_1)(S^2 \,|\, p_2) \cdots (S^{n-1} \,|\, p_{n-1})(T^1 \,|\, p_1)(T^2 \,|\, p_2) \cdots (T^{n-1} \,|\, p_{n-1}).$$

1) Generation of $(q_{n-\sigma-1}S^\sigma \,|\, T_{n-\tau}p_{\tau-1})$, where $\sigma > \tau$, from fundamental foldings of type $\tau + \alpha$, $\alpha = 0, 1, \ldots, \sigma - \tau$. From the fundamental folding of type τ

$$(q_{n-\tau-1}S^\tau \,|\, T_{n-\tau}p_{\tau-1})$$

we obtain, by the substitution $w \sim T_{n-\tau}p_{\tau-1}$, a form of the form row f with three rows of upper weight index (cf. § 1) $\tau + 1$:

$$(wS^\tau \,|\, p_{\tau+1})(S^{\tau+1} \,|\, p_{\tau+1})(T^{\tau+1} \,|\, p_{\tau+1}). \tag{63}$$

A fundamental folding of type $\tau + 1$ applied to (63.) gives

$$(q_{n-\tau-2}wS^\tau \,|\, S_{n-\tau-1}p_\tau);$$

from (31.), by the substitution $q_{n-\tau-2}w = q'_{n-\tau-1}$, this has the value

$$\varepsilon(q_{n-\tau-2}wS^{\tau+1} \,|\, S^\tau p_\tau) + \sum_{\alpha=1}^{\tau, n-\tau-1} \Pi(q'_{n-\tau-1-\alpha}S^{\tau+1} \,|\, S_{n-\tau}p_{\tau-\alpha}).$$

The expression under the summation sign vanishes identically by (62.); thus we have reached a form

$$(wS^{\tau+1} \,|\, p_{\tau+2})(S^{\tau+2} \,|\, p_{\tau+2})(T^{\tau+2} \,|\, p_{\tau+2}),$$

which is obtained from (63.) by changing τ into $\tau + 1$. Repeating the process $\sigma - \tau$ times therefore leads to the desired folding:

$$(wS^\sigma \,|\, p_{\sigma+1}) = \varepsilon(q_{n-\sigma-1}S^\sigma \,|\, T_{n-\tau}p_{\tau-1}).$$

We indicate the process just carried out by the formula

$$T^\tau S^\tau, \quad S^{\tau+1}, \quad S^{\tau+2}, \ldots, S^\sigma; \tag{64}$$

and see that only the normal-form property of S, not that of T, is used. A process dual to (64.) can be performed, using only the normal-form property of T, in the reverse order, namely:

$$S^\sigma T^\sigma, \quad T^{\sigma-1}, \quad T^{\sigma-2}, \ldots, T^\tau, \tag{65}$$

according to the relations:

$$\begin{aligned}
(q_{n-\sigma-1}S^\sigma \,|\, T_{1-\sigma}p_{\sigma-1}) &= \varepsilon(T_{n-\sigma}y \,|\, p_{\sigma-1}), \quad y \sim q_{n-\sigma-1}S^\sigma,\\
(q_{n-\sigma}T^{\sigma-1} \,|\, T_{n-\sigma}yp_{\sigma-2}) &= \varepsilon(T_{n-\sigma+1}y \,|\, p_{\sigma-2}),\\
&\vdots\\
(q_{n-\tau-1}T^\tau \,|\, T_{n-\tau-1}yp_{\tau-1}) &= \varepsilon(q_{n-\sigma-1}S^\sigma \,|\, T_{n-\tau}p_{\tau-1}).
\end{aligned}$$

We should also point out the connection with the decrease of total weight given in § 6 (end): for foldings of defect 1 this decrease has the value $2(1+(\sigma-\tau))$, and for fundamental foldings the value $2\cdot 1$. Therefore, if composition out of fundamental foldings is possible at all, exactly $\sigma-\tau+1$ such foldings are necessarily required, as was carried out above.

2) Generation of general foldings from cogredient and fundamental foldings. The weight decrease for a general folding (41) is $2(\lambda^2+\lambda(\sigma-\tau)) = 2[\lambda^2+(\sigma-\tau)(1+(\lambda-1))]$. This gives the possibility of decomposing it into a cogredient folding of defect λ (weight decrease $2\lambda^2$) and $\sigma-\tau$ foldings of defect 1 with difference of weight indices $\lambda-1$ (weight decrease $2\{1+(\lambda-1)\}$). In the case $\lambda=1$ this decomposition agrees with the one given under 1); we show that it can indeed be realized.

From the cogredient folding of defect λ

$$(q_{n-\tau-\lambda}S^\tau \,|\, T_{n-\tau}p_{\tau-\lambda})$$

we obtain, by the substitution $R^\lambda \sim T_{n-\tau}p_{\tau-\lambda}$, a form with the two rows of upper weight indices $\tau+\lambda$ and $\tau+1$:

$$(R^\lambda S^\tau \,|\, p_{\tau+\lambda})(S^{\tau+1} \,|\, p_{\tau+1}). \tag{66}$$

The row with the lower weight index $\tau+1$ belongs to a normal form; hence (66.) admits a folding of defect 1 composed out of λ fundamental foldings, in the order (65):

$$(R^\lambda S^\tau)S^{\tau+\lambda}, \quad S^{\tau+\lambda-1}, \quad S^{\tau+\lambda-2}, \ldots, S^{\tau+1}.$$

Taking account of (31.) and (62.), this gives

$$(q_{n-\tau-\lambda-1}R^\lambda S^\tau \,|\, S_{n-\tau-1}p_\tau) = \varepsilon(R^\lambda S^{\tau+1} \,|\, p_{\tau+\lambda+1}),$$

that is, a form obtained from (66.) by replacing τ by $\tau+1$, just as in 1) for (63.). Repeating the process $\sigma-\tau$ times leads to the desired folding:

$$(R^\lambda S^\sigma \,|\, p_{\sigma+\lambda}) = \varepsilon(q_{n-\sigma-\lambda}S^\sigma \,|\, T_{n-\tau}p_{\tau-\lambda}).$$

In the whole process only the normal-form property of S was used. Viewing T as a normal form gives the dual process, with complete reversal of the order, according to the formulas

$$\begin{aligned}
(q_{n-\sigma-\lambda}S^\sigma \,|\, T_{n-\sigma}p_{\sigma-\lambda}) &= \varepsilon(T_{n-\sigma}R_\lambda \,|\, p_{\sigma-\lambda}), \quad R_\lambda \sim q_{n-\sigma-\lambda}S^\sigma,\\
(q_{n-\sigma}T^{\sigma-1} \,|\, T_{n-\sigma}R_\lambda p_{\sigma-\lambda-1}) &= \varepsilon(T_{n-\sigma+1}R_\lambda \,|\, p_{\sigma-\lambda-1}),\\
&\vdots\\
(q_{n-\tau-1}T^\tau \,|\, T_{n-\tau-1}R_\lambda p_{\tau-\lambda}) &= \varepsilon(q_{n-\sigma-\lambda}S^\sigma \,|\, T_{n-\tau}p_{\tau-\lambda}).
\end{aligned}$$

3) Generation of cogredient foldings of defect λ from cogredient foldings of defect $\lambda-1$ and fundamental foldings. The weight decrease for cogredient foldings of defect λ is $2\lambda^2 = 2((\lambda-1)^2+2\lambda-1)$, giving the possibility of a decomposition into a cogredient folding of defect $\lambda-1$ and $2\lambda-1$ fundamental foldings.

This is most easily found by first putting $n = 2\lambda$. The only possible cogredient folding of defect λ is $(S^\lambda \,|\, T_\lambda) = (S^\lambda \,|\, T_{n/2})$; it contains one row u and one row x fewer than the folding of defect $\lambda - 1$ of the same rows, $(uS^\lambda \,|\, T_\lambda x)$. Conversely, we compose the folding of defect λ out of fundamental foldings which effect the disappearance of a row u and a row x (weight decrease $2(n-1) = 2(2\lambda - 1)$) and at the same time generate a new symbol row R^λ, together with a cogredient folding of defect $\lambda - 1$. The simplest folding that does this is the following one of defect 1:

$$(sT^\lambda \,|\, \sigma p_\lambda) = \varepsilon(S^{2\lambda-1} \,|\, R_{\lambda-1} p_\lambda) = \varepsilon(R^\lambda \,|\, p_\lambda),$$

where

$$R^{\lambda+1} = sT^\lambda, \quad s = S^1, \quad \sigma = S_1, \quad R^\lambda \sim \sigma R_{\lambda-1}.$$

By (65.) and (64.) it is composed out of the $2\lambda - 1$ fundamental foldings

$$T^\lambda S^\lambda, S^{\lambda-1}, \ldots, S^1; \qquad R^{\lambda+1} S^{\lambda+1}, S^{\lambda+2}, \ldots, S^{2\lambda-1}. \tag{67}$$

A folding of defect $\lambda - 1$, taking account of (21.) and (62.), gives

$$(uS^\lambda \,|\, \sigma R_{\lambda-1} x) = \varepsilon(R^{\lambda+1} \,|\, S_\lambda x) = \varepsilon(sT^\lambda \,|\, S^\lambda x) = \varepsilon(S^\lambda \,|\, T_\lambda).$$

The general case, for two rows S^ρ, T^ρ and arbitrary n, is obtained directly from the special one. In place of (67.) one has the following $2\lambda - 1$ fundamental foldings:

$$T^\rho S^\rho, S^{\rho-1}, \ldots, S^{\rho-\lambda+1}; \qquad R^{\rho+1} S^{\rho+1}, S^{\rho+2}, \ldots, S^{\rho+\lambda-1}. \tag{68}$$

The first half of the foldings gives the form

$$(q_{n-\rho-1} T^\rho \,|\, S_{n-\rho+\lambda-1} p_{\rho-\lambda}),$$

from which, by the substitutions

$$S_{n-\rho+\lambda-1} p_{\rho-\lambda} \sim w, \qquad T^\rho w = R^{\rho+1},$$

and application of the second half of the foldings, one obtains

$$(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \,|\, R_{n-\rho-1} p_\rho).$$

Substituting

$$q_{n-\rho-\lambda} S^{\rho+\lambda-1} \sim y,$$

one obtains, by a cogredient folding of defect $\lambda - 1$,

$$(q_{n-\rho-\lambda+1} S^\rho \,|\, y R_{n-\rho-1} p_{\rho-\lambda+1}). \tag{69}$$

We show that this is the desired folding of defect λ. Substitute

$$R_{n-\rho-1} p_{\rho-\lambda+1} = R_{n-\lambda}$$

and observe that resolving y gives

$$(q_{n-\rho-\lambda+1} R^\lambda \,|\, S_{n-\rho} y) = \varepsilon(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \,|\, S_{n-\rho} p'_{\rho-1}) = 0.$$

Thus by (20.) the value of (69.) is

$$\varepsilon(S^\rho R^\lambda \,|\, p_{\rho+\lambda-1} y) = \varepsilon(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \,|\, [S^\rho R^\lambda]_{n-\rho-\lambda} p_{\rho+\lambda-1}).$$

This folding is of type (43.), hence equivalent to

$$(q_{n-\rho-\lambda} S^\rho \,|\, R_{n-\rho-1} p_{\rho-\lambda+1}) = \varepsilon(wT^\rho \,|\, R_\lambda p_{\rho-\lambda+1}), \qquad R_\lambda \sim q_{n-\rho-\lambda} S^\rho.$$

Here R_λ is already of the desired final form; the expression corresponds to the form $(sT^\lambda \,|\, S_\lambda x)$ in the special case. Now, observing that resolving w and R_λ gives

$$(wR^{n-\lambda} \,|\, T_{n-\rho} p_{\rho-\lambda+1}) = \varepsilon(q'_{n-1} R^{n-\lambda} \,|\, S_{n-\rho+\lambda-1} p_{\rho-\lambda}) = \varepsilon(q''_{n-\sigma-1} S^\rho \,|\, S_{n-\rho+\lambda-1} p_{\rho-\lambda}) = 0,$$

we get by (21.) the value

$$(wq_{n-\rho+\lambda-1}\,|\,T_{n-\rho}R_\lambda) = \varepsilon(q_{n-\rho+\lambda-1}[T_{n-\rho}R_\lambda]^{n-\lambda}\,|\,S_{n-\rho+\lambda-1}p_{\rho-\lambda}) \quad \text{equivalent to} \quad (q_{n-\rho-\lambda}S^\rho\,|\,T_{n-\rho}p_{\rho-\lambda}).$$

In the whole process only the normal-form property of S, not that of T, was used. Thus the cogredient folding of defect $\lambda-1$ used to generate (69.) can be replaced by $2\lambda-3$ fundamental foldings and a cogredient folding of defect $\lambda-2$, which in turn admits a decomposition into fundamental foldings, and so on. Analogously, the generation is obtained by using the normal-form property of T and applying the dualistic process, that is, the fundamental foldings

$$S^\rho T^\rho, \quad T^{\rho-1},\dots,T^{\rho-\lambda+1}; \qquad R^{\rho-1}T^{\rho-1}, \quad T^{\rho-2},\dots,T^{\rho-\lambda+1},$$

and a folding dual to (69.). The folding

$$(q_{n-\rho-\lambda}T^\rho\,|\,S_{n-\rho}p_{\rho-\lambda}),$$

equivalent to the preceding one, is then obtained.

Adding the decomposition obtained under 2), we have indeed obtained a generation of the most general foldings from fundamental foldings. It remains to show that this generation is unique, apart from permutation of the order, i.e. that to each folding there corresponds one and only one number α_ρ of fundamental foldings of type ρ, where $\rho = 1,2,\dots,n-1$.

Let m_ρ be the degrees in the variable rows p_ρ of the initial form f, and l_ρ the degrees of the form produced by folding. Each fundamental folding of type ρ transforms the degrees $m_{\rho-1}, m_\rho, m_{\rho+1}$ respectively into $m_{\rho-1}+1, m_\rho-2, m_{\rho+1}+1$. Thus the α satisfy the system of equations

$$m_\rho - l_\rho = -\alpha_{\rho-1} + 2\alpha_\rho - \alpha_{\rho+1}, \qquad (\rho = 1,2,\dots,n-1), \tag{70}$$

where α_0 and $\alpha_n = 0$ are to be put, and whose determinant has value n.[71] The α are thereby uniquely determined, and indeed, by Theorem VIII, as positive integers; this gives conditions for the differences $m_\rho - l_\rho$.

The results of this paragraph hold by virtue of relations (62.); however, if the relations (62.) do not hold, the final expressions are by no means equivalent to the initial expressions. The definition of equivalence given in § 6 also admits the formulation:[72] "Two forms are equivalent if and only if they differ by forms of higher folding, i.e. by forms that exhibit a greater weight decrease." This greater weight decrease always occurs when the two rows $q_{n-\sigma-i}, p_{\tau-\lambda}$ entering the folding are really variable rows, as happens in (41.), (42.), in the foldings appearing in this paragraph as equivalent, and also in the equivalent forms of Theorem VII. But it need not occur when these rows are derived from symbol rows by substitution, as was the case for the forms of this paragraph which could be expressed in terms of one another by virtue of (62.). One easily sees that the vanishing forms even have, in part, a higher total weight.

§ 9. Form rows. Reduction theorems.

We now define the form row introduced in § 6 (end) somewhat more sharply as "the totality of invariant formations of a given initial form that are linear in the coefficients, i.e. the totality of forms that arise from the initial form by folding in itself, arranged according to higher forms."

To explain the higher forms, suppose by Theorem VII that the initial form is given as a product of two normal forms $S \cdot T$. By a higher form we mean forms with higher folding in itself, and among forms with the same folding in itself, specially normalized ones. More precisely: suppose the form A has arisen by α_ρ fundamental foldings of type ρ, and the form B by β_ρ such foldings. Then B is higher than A:

1. if, in the system of inequalities $\beta_\rho \geq \alpha_\rho$, $\rho = 1,2,\dots,n-1$, the inequality sign holds for at least one value of ρ;[73]

2. if the equality sign holds for all values ρ, but fewer symbol rows have been combined into a single matrix product. This normalization proves necessary because of the reductions in § 8. Accordingly, for example, $(S^{n-1}\,|\,T_{n-1})$ is higher than $(S^{n-1}\,|\,[S^1T^\rho]_{n-\rho-1}p_\rho)$;

[71] If Δ_n denotes the n-rowed determinant, then the recurrence formula holds: $\Delta_n = 2\Delta_{n-1} - \Delta_{n-2}$; that is, $\Delta_n - \Delta_{n-1} = \Delta_{n-1} - \Delta_{n-2} = \cdots = \Delta_2 - \Delta_1 = 1$, since $\Delta_2 = 3$, $\Delta_1 = 2$.

[72] Cf. the consideration at the end of the next paragraph.

[73] If in the system of inequalities the sign $>$ holds in part and the sign $<$ in part, then the forms are not comparable, since a form arising from another by folding always has a positive value system of fundamental foldings, hence in our case positive or vanishing values $\beta_\rho - \alpha_\rho$.

3. if the equality sign holds for all values ρ and the number of symbol rows combined into one matrix product is the same, then B is made higher than A by a normalization arbitrary in itself but fixed once and for all. This normalization is always possible because of the finite number of forms belonging to a value system α; it can be achieved, for instance, by privileging the one form S (larger number of symbol rows S, larger sum of the upper weight indices of the rows S, and so on).

It is useful to think of the form row as arranged as an $(n-1)$-dimensional manifold of lattice points, where the $n-1$ directions correspond to the $n-1$ fundamental foldings. To every form there then corresponds one and only one value system α, i.e. a positive integral lattice point, while the different forms belonging to a value system α are uniquely determined in their order by the above normalization. An arbitrary form of the form row gives the initial form of a new form row, contained as a part of the total form row; indeed it is contained as the part consisting of the forms obtained from the new initial form by proceeding in the $n-1$ positive directions.

By virtue of Theorem VIII and the general theory of series expansions, exactly the reduction theorems of the binary and ternary domains (cf. dissertation, § 3) hold for form rows. We shall briefly derive the principal theorem. We first state the definition:

"In a given system of forms, suppose a certain finite number of forms has been combined into a module. We call a form reducible if it can be expressed by forms that have invariants as factors, or by 'higher forms'; that is, by higher forms of the total form row to which the form belongs, or by forms that contain the symbols of the module in higher order. A reducible form row is called a reducer." Then:

Theorem IX: If the initial form of a form row is reducible by having arisen through folding with a reducer, then the entire form row arising from this initial form is reducible.

For the proof, observe that a form h arising by folding a form f with a second form g can, because of (45.), be regarded as a form f containing several cogredient variable rows p_ρ, p'_ρ. Such forms can, by the general theory of series expansion, be represented as polars of the elementary covariants of f, by applying the series expansion successively to each pair of cogredient variable rows p_ρ, p'_ρ. By (38.), the elementary covariants that occur are the forms generated by cogredient folding. Since, however, the order in which the series expansion is applied is still arbitrary, we may in particular choose an order corresponding to the generation of the general foldings from fundamental foldings. The resulting system of elementary covariants is then identical with the form row f; the forms that arise by cogredient folding of higher defect are arranged among those that arise by fundamental foldings. But one also obtains polars of the form row f for any order of the series expansion, to which a second system of elementary covariants may correspond. Since the form row exhausts the totality of invariant formations, every form of the second system can be expressed linearly by polars of the form row, and indeed by polars of those forms that show the same or a greater weight decrease. The latter follows because (cf. § 7) the polars can be regarded as simultaneous invariants of a form row H (52.), with the degree numbers corresponding to the form h, and of the form row f. Every folding of H in itself corresponds to a weight decrease which, after substitution (11.) is performed, is transferred to the symbol rows and hence to the forms of f.[74]

An arbitrary form of the form row h has arisen by folding h in itself; that is, on the one hand by a higher folding of g with f, and on the other by folding f or g in itself. In both cases the series expansion leads to polars of the form row f, just as was shown above for the initial form h. If in particular f is a reducer, then an expansion in polars of the reducer results; hence the form row h becomes reducible.

Applications of the theorem to the reduction of complete systems of forms follow analogously to the binary and ternary domains.

[74] This consideration also underlies the second formulation of the equivalence concept (§ 8, end). Further explanations of the different systems of elementary covariants in the ternary domain are found in Study, loc. cit., II, § 7, Theorems 7) and 8). By our Theorem VII the same results also hold for n variables.

5. Rational Function Fields

Jahresbericht der DMV 22 (1913), pp. 316–319

The following questions originally go back to conversations with E. Fischer. Some of the questions, moreover, were already raised and settled by E. Steinitz for the special case of one indeterminate — and indeed under more general assumptions on the coefficient field.[75]

By a "rational function field" I understand a field whose elements are rational functions of n indeterminates, with coefficients from a prescribed number field, which in particular may also comprise all complex numbers. Examples of such fields are the field of symmetric functions of n quantities, or more generally the totality of rational functions of n indeterminates that allow the permutations of a certain group (Lagrange's *Gattungsbereiche*); furthermore, the invariant field.

Between the first two fields just mentioned and the latter there is a characteristic difference: the former contain n algebraically independent functions, the elementary symmetric functions; whereas the number of algebraically independent invariants is always smaller than the number of indeterminates. It is therefore important that, in general, one can associate such fields of the second kind one-to-one with fields of the first kind, by replacing some of the indeterminates by numbers. In what follows I shall therefore restrict myself to fields of the first kind, i.e. to fields with n algebraically independent functions; by virtue of the association, the results then hold in general.

The questions group themselves around three basis concepts: rational basis, minimal basis, and integrality basis.

1. By a "rational basis" I understand a finite number of functions of the field such that every function of the field can be represented as a rational combination of this finite number, with coefficients from the prescribed coefficient field. By simple considerations — which in the case of one indeterminate are also found in E. Steinitz, loc. cit. — one proves the existence of a rational basis for every rational function field. For example, by Lagrange's theorem, the elementary symmetric functions and one function belonging to the group form a rational basis for the Lagrange *Gattungsbereiche* mentioned earlier.

2. Every rational basis must contain (for fields of the first kind) at least n functions; if it contains exactly n functions, which then must be algebraically independent, I call it a "minimal basis." The question of the existence of a minimal basis can be answered in part by theorems of Lüroth, Castelnuovo, and Enriques.[76] According to these, for fields of one indeterminate there always exists a minimal basis: the Lüroth function of the field, uniquely determined up to fractional linear transformation (cf. Steinitz § 24). For fields of two indeterminates, a minimal basis exists always, and in general only, if one permits an algebraic extension of the coefficient field, whereas for three and more indeterminates a minimal basis in general does not exist at all. No attempt has yet been made to characterize the special fields with minimal basis.

I have pursued further the question of the minimal basis for Lagrange's *Gattungsbereiche*. Here it acquires the following meaning: "If the Lagrange *Gattungsbereich* belonging to the group G possesses a minimal basis, then one can construct rationally, by a parametric representation, the totality of equations with prescribed group G; and indeed for every number field that contains the coefficient field of the minimal basis — hence in particular for any arbitrary number field if this coefficient field is the field of rational numbers."

The simplest example is provided by the symmetric group; here the elementary symmetric functions form a minimal basis with rational numerical coefficients. From this one obtains as parametric representation simply the equation with indeterminate coefficients. Hilbert's irreducibility theorem shows how one passes from this to arbitrarily many affectless equations over every number field.

Now the existence of the minimal basis for all groups occurring in equations of degrees 3 and 4 follows directly from the theorems of Lüroth and Castelnuovo; at the same time it further appears that this minimal basis has rational numerical coefficients. Thus, for every arbitrary number field, one can rationally construct the totality of equations of degrees 3 and 4 with prescribed group. For the cyclic and dihedral groups there exists a minimal basis with the field of the nth roots of unity as coefficient field; the same holds for a few further metacyclic groups. The question whether the minimal basis is at all characteristic for certain groups must remain undecided.

3. To arrive at the last basis concept, I consider the totality of polynomials contained in the field. By

[75] E. Steinitz: Algebraische Theorie der Körper, especially § 24. Crelle's Journal 137. 1910.

[76] J. Lüroth: Beweis eines Satzes über rationale Kurven. Math. Ann. 9. 1875. — G. Castelnuovo: Sulla razionalità delle involuzioni piane. Math. Ann. 44. 1893. — F. Enriques: Sopra una involuzione non razionale dello spazio. Rend. Acc. Linc. vol. 21. 21 Jan. 1912.

an "integrality basis" I understand a finite number of these polynomials such that every polynomial of the field can be represented as an integral rational combination of this finite number, with coefficients from the prescribed coefficient field. The question of the existence of an integrality basis — which, for example, contains as a special case the finiteness of the invariant system — was posed by Hilbert in his "Mathematical Problems"[77] in a somewhat different formulation, as the problem of relatively integral functions.

For the moment I can indicate only one class of fields for which an integrality basis exists. Namely, if the field contains a system of n polynomials such that the homogeneous resultant of the terms of highest dimension of these polynomials is different from zero, then an integrality basis exists; it is given by the coefficients of all u in the equation, irreducible in the field, for the linear form:

$$u_0 = u_1x_1 + u_2x_2 + \cdots + u_nx_n.$$[78]

If, in particular, the value of the resultant is equal to a unit of the coefficient field, then the integrality of the representation is also secured. An example of this class of fields is furnished by Lagrange's *Gattungsbereiche*, since the resultant of the elementary symmetric functions has the value ± 1. A further example (without integrality) is given by all fields that contain only one algebraically independent polynomial; here the integrality basis is identical with the Lüroth function of the smallest subfield containing all polynomials. Thus, as is well known, all integral rational projective invariants of a quadratic form in n variables are integral functions of the discriminant.

The condition just given is only sufficient, by no means necessary, for the existence of an integrality basis. One can easily construct fields in which the resultant of every n polynomials vanishes, and which nevertheless possess an integrality basis given by the coefficients of that irreducible equation. The question arises whether perhaps even in the most general case the coefficients of that equation furnish the integrality basis.

[77] D. Hilbert: Mathematische Probleme. Lecture, Paris 1900. Problem 14. Göttinger Nachr. 1900.

[78] This irreducible equation is found, in the case of one indeterminate, in E. Steinitz's proof of Lüroth's theorem.

6. Fields and Systems of Rational Functions

Math. Ann. 76 (1915), pp. 161–196

The present paper treats basis questions for arbitrary systems of rational and integral rational functions. The methods of field theory used here make the settlement of these questions for fields of rational functions — rational function fields[79] — appear as the essential matter, while the generalization of the results to arbitrary systems arises as a consequence.

Among basis questions for general systems, up to now only the existence of a module basis, guaranteed by Hilbert's theorem (Math. Ann. 36), has been known. In what follows, the question of rational representability is answered completely by the existence of a rational basis for every arbitrary system (§ 7); the rational basis of fields is already obtained in § 4. This existence of the rational basis permits us throughout to start from the abstractly defined field or system, and thereby to avoid difficulties that are caused only by the special choice of the rational basis and not by the system itself, such as the occurrence of special denominators or of fundamental points of the basis functions.

For fields the question of the minimal basis is also treated, i.e. of a rational basis consisting of algebraically independent functions (§ 6). The methods of field theory further lead to a distinguished rational basis, the involution basis[80] (§ 5), which becomes especially important for integrality domains consisting of polynomials (§ 8). In that setting it gives a representation with a fixed denominator (a power of a function determined by the integrality domain), as is known for example in the special case of the typical representation of invariants.

From rational representability we now draw, as far as possible, consequences for integral rational representability, i.e. for the question of finiteness in the narrower sense.

The finiteness theorems known up to now all rest on the fact that Hilbert's theorem on module bases guarantees finiteness for all systems in which a representation

$$F = A_1 f_1 + A_2 f_2 + \cdots + A_k f_k$$

entails a second representation

$$F = B_1 f_1 + B_2 f_2 + \cdots + B_k f_k,$$

where the B_i belong to the system and are of lower degree than F.

By contrast, the theorem on the rational basis and the methods of field theory permit one to infer finiteness under assumptions of an entirely different kind.

Thus, as a partial answer to a problem of Hilbert (Mathematical Problems 14), one obtains a class of relatively integral functions (relatively integral domains of the first kind) which are finite integrality domains and whose integrality basis is given by the involution basis mentioned above (§ 10). In analogy with Hilbert's proof of Kronecker's theorem on the fundamental system of algebraic integers (Complete invariant systems, § 2; Math. Ann. 42), it can further be shown that all regular systems of polynomials — i.e. all systems that can be mapped to ones without fundamental points — possess an integrality basis (§ 12), whereas nonregular systems without integrality basis can easily be specified. As a simple example of this fact one may mention the theorem: "In every arbitrary system of infinitely many polynomials in one indeterminate, there exists a finite number of these polynomials such that every polynomial of the system can be represented as an integral rational combination of this finite number." Further examples are found in § 13.

Finally, the last paragraphs (§§ 14 and 15) give a sharpening of the basis theorems with respect to integrality.

An essential tool of the investigation is the transfer principle of § 3, according to which it suffices to consider all basis questions raised here for those systems in which the number of indeterminates coincides with the number of algebraically independent functions of the system.

The impetus for the present work came from conversations with Mr. E. Fischer, in particular from a question of his concerning the minimal basis of Lagrange's *Gattungsbereiche*. The investigations relating to this, which led to the construction of equations with prescribed group (cf. the cited communication on "Rational Function Fields," part 2), are reserved for a further publication.

[79] Cf. a preliminary communication on the occasion of the Vienna meeting of natural scientists: Rationale Funktionenkörper, Jahresber. d. D. Math.-Ver. 22 (1913), where an overview is given of the questions and of the results concerning function fields.

[80] The name was chosen because, in its geometric interpretation, the rational function field represents the most general involution in a linear space.

The essential extension of the present paper over the original communication consists in the fact that the basis theorems stated there only for fields or special integrality domains are here transferred to arbitrary systems. This transfer succeeds in a simple way by means of the concept of the smallest containing field of an arbitrary system, as a generalization of the quotient field of an integrality domain (§ 7). I was led to form this concept by the fact that Mr. K. Hentzelt, after learning the rational basis of fields, was able to prove the existence of the rational basis for arbitrary systems by a detour through Hilbert's module basis. I was also led to the theorem on the integrality basis of regular systems by K. Hentzelt's observation that the arguments I had applied to integrality domains were valid for arbitrary systems subject to the same assumptions; likewise I owe to Mr. Hentzelt a few isolated smaller remarks.

Finally, let it be mentioned that in the case of one indeterminate the rational basis and minimal basis of fields are found in E. Steinitz's "Algebraische Theorie der Körper," § 24 (Crelle's Journal 137); there the coefficient field is taken to be a field in the most general sense. The question how far the theorems given here remain valid under these more general assumptions is not touched on in what follows; in any event the proofs require modification, since, for example, the tools from the theory of functional determinants fail for fields of characteristic p.

§ 1. Fields $\mathfrak{K}_{n\rho}$ and systems $\mathfrak{S}_{n\rho}$.

In what follows we are concerned with the investigation of systems of rational functions in n indeterminates, especially fields of rational functions.

By $f(x_1, \ldots, x_n)$, $g(x_1, \ldots, x_n)$, ..., or abbreviated $f(x)$, $g(x)$, ..., one should therefore always understand rational functions of $x_1, \ldots, x_n$; these are assumed to be in reduced representation — that is, numerator and denominator relatively prime. Integral rational functions (polynomials) will be denoted by capital letters, $F(x)$, $G(x)$, ... The coefficient field Ω is assumed to be some number field; in particular, it may also contain all complex numbers. The field Ω may also contain a finite number of parameters.

The quantities from Ω, hence all functions of degree zero, are always counted as belonging to the system.

With these stipulations, the field consisting of rational functions — the rational function field — can be defined abstractly.

Definition I: A system of rational functions is called a field if it satisfies the conditions:

1. *Together with $f(x)$, and for every quantity c from Ω, it also contains $c \cdot f(x)$.*

2. *Together with $f(x)$ and $g(x)$, it always also contains $f(x) + g(x)$, $f(x) \cdot g(x)$ and — for $g(x) \neq 0$ — the quotient $f(x) : g(x)$.*[81]

Special fields are those which arise by adjoining a finite number of rational functions

$$f_1(x),\ f_2(x), \ldots, f_k(x)$$

to Ω; they will be denoted by

$$\Omega(f_1(x), \ldots, f_k(x)) \quad \text{or} \quad \Omega(f_1, \ldots, f_k).$$

[82] Among these special fields we also mention the field arising by adjoining $x_1, \ldots, x_n$ to Ω,

$$\Omega(x_1, \ldots, x_n),$$

which is identical with the totality of rational functions of $x_1, \ldots, x_n$ with coefficients from Ω.

We also introduce (following E. Steinitz) the concept of an intermediate field:

> "A field $\mathfrak{M}$ lies between Ω_1 and Ω_2 if it contains Ω_1 and is contained in Ω_2;"

then Definition I also admits the following formulation:

The field of rational functions of n indeterminates is an intermediate field between Ω and $\Omega(x_1, \ldots, x_n)$ which effectively contains the n indeterminates in at least one function.

Beside the number of indeterminates, there arises for systems of rational functions a further characteristic number, the number of algebraically independent functions, or the algebraic rank, by the following definition.

[81] Here $f(x)$ and $g(x)$ may also be of degree zero, i.e. quantities from Ω.

[82] That these "special fields" already exhaust the totality of all fields follows in § 4.

Definition II: A system of rational functions has algebraic rank ρ if ρ functions of the system can be specified that are algebraically independent, but every $(\rho+1)$ functions of the system are algebraically dependent.[83]

Systems of (finitely or infinitely many) rational functions in n indeterminates and of algebraic rank ρ will be denoted by the double index

$$\mathfrak{S}_{n\rho}.$$

In particular, by

$$\mathfrak{K}_{n\rho}$$

one is to understand a field of n indeterminates and of algebraic rank ρ. The inequality holds:

$$1 \leq \rho \leq n.$$

We give a few more examples of fields $\mathfrak{K}_{nn}$ and $\mathfrak{K}_{n\rho}$:

1. Fields $\mathfrak{K}_{nn}$ are Lagrange's *Gattungsbereiche*, which can be defined as

> "the totality of rational (and integral rational) functions of $x_1, \ldots, x_n$ that allow the permutations of a permutation group G of $x_1, \ldots, x_n$."

They always contain the field of symmetric functions, hence also the n algebraically independent elementary symmetric functions.

2. Fields $\mathfrak{K}_{n\rho}$ are the projective invariant fields, which are formed from the totality of the rational (and integral rational) invariants of one or more ground forms.[84] Here $\rho < n$, since the number of algebraically independent invariants is always smaller than the number of coefficients.

The corresponding statement holds for the invariant fields with respect to subgroups of the projective group.

§ 2. Auxiliary theorem on algebraically dependent functions.

The auxiliary theorem of this paragraph will show in § 3 that the algebraic rank of a system is its properly characteristic number. Namely, the auxiliary theorem shows that by a successive numerical specialization of some indeterminates, such that ρ algebraically independent functions remain algebraically independent, an arbitrary function algebraically dependent on these ρ functions can neither vanish identically nor become indeterminate.

Let therefore the system

$$g_1(x),\ g_2(x), \ldots, g_\rho(x) \tag{1}$$

have algebraic rank ρ, where $\rho < n$, and let $h(x)$ be a rational function algebraically dependent on the functions (1). By known theorems, the rank of the functional matrix of (1) is equal to ρ. The indeterminates may therefore be named, for instance, so that

$$\frac{\partial(g_1, g_2, \ldots, g_\rho)}{\partial(x_1, x_2, \ldots, x_\rho)} = \frac{\Gamma(x)}{G(x)^{\rho+1}} \neq 0, \tag{2}$$

where $G(x)$, as usual, denotes the common denominator of the functions (1).

Now choose the number a so that the product $G(x) \cdot \Gamma(x)$ is not divisible by $(x_i - a)$, where i is understood to be one of the numbers $\rho+1, \rho+2, \ldots, n$. Thus, for $x_i = a$, the common denominator of the functions (1) cannot vanish, and the ρ functions

$$[g_1(x)]_{x_i=a}, \ldots, [g_\rho(x)]_{x_i=a} \tag{3}$$

are likewise algebraically independent. As we shall say, the algebraic rank of the system (1) is preserved under the substitution $x_i = a$.

[83] ρ functions $f_1(x), \ldots, f_\rho(x)$ are called algebraically independent if from every relation

$$F\big(f_1(x), \ldots, f_\rho(x)\big) = 0 \quad \text{identically in } x_1, \ldots, x_n$$

it follows that

$$F(\lambda_1, \ldots, \lambda_\rho) = 0 \quad \text{identically in } \lambda_1, \ldots, \lambda_\rho.$$

In the opposite case they are called algebraically dependent.

[84] It is expedient to consider only transformations of determinant 1; then every rational combination of arbitrarily many invariants again admits the transformation, and one does indeed obtain a field. The homogeneous, isobaric invariants taken by themselves do not form a field, but they do form a system $\mathfrak{S}_{n\rho}$.

Let the irreducible equation satisfied by $h(x)$ with respect to the functions (1) be given by

$$\begin{aligned} A_0(g_1,\ldots,g_\rho)\,h(x)^\tau + A_1(g_1,\ldots,g_\rho)\,h(x)^{\tau-1} + \cdots \\ + A_\tau(g_1,\ldots,g_\rho) = 0, \end{aligned} \tag{4}$$

where $A_0(g_1,\ldots,g_\rho)$ and $A_\tau(g_1,\ldots,g_\rho)$ do not vanish identically. In order to pass from this to an identity between polynomials, we set

$$g_i(x) = G_i(x) : G(x); \qquad h(x) = H_1(x) : H_2(x), \tag{5}$$

and note that — because of the assumed reduced representation of $h(x)$ — $H_1(x)$ and $H_2(x)$ are relatively prime. By virtue of (5), (4) becomes

$$\begin{aligned} &B_0(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_0}H_1(x)^\tau \\ &\quad + B_1(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_1}H_1(x)^{\tau-1}H_2(x) + \cdots \\ &\quad + B_\tau(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_\tau}H_2(x)^\tau = 0, \end{aligned} \tag{6}$$

where the B_k are homogeneous forms in G, G_i, and at least one of the nonnegative exponents σ_k must be zero.

Suppose now that $H_1(x)$ were divisible by $(x_i - a)$. Then, by (6),

$$B_\tau(G,G_1,\ldots,G_\rho)\,G(x)^{\sigma_\tau}H_2(x)^\tau$$

would also be divisible by $(x_i - a)$. This is excluded by assumption for $G(x)$; hence from the divisibility of B_τ there would also follow the divisibility of

$$A_\tau = B_\tau : G^\lambda,$$

and between the functions (3) there would exist the relation $A_\tau = 0$, contrary to the assumed algebraic independence. Finally $H_2(x)$, being relatively prime to $H_1(x)$, also cannot be divisible by $(x_i - a)$;[85] thus the assumption that $H_1(x)$ is divisible by $(x_i - a)$ leads to a contradiction. In the same way one shows that $H_2(x)$ also cannot be divisible by $(x_i - a)$. Hence the function $[h(x)]_{x_i=a} = h'(x)$ can neither vanish identically nor become indeterminate.

The above conclusion can now be applied again to the system (3) and the function $h'(x)$, again assumed to be in reduced representation, and continuation of the procedure leads finally to the following auxiliary theorem.

Auxiliary theorem: The two systems

$$g_1(x),\ g_2(x),\ldots,g_\rho(x),$$

$$g_1(x),\ g_2(x),\ldots,g_\rho(x),\ h(x)$$

shall each have algebraic rank ρ, where $\rho < n$, and suppose, for instance, that

$$\frac{\partial(g_1,\ldots,g_\rho)}{\partial(x_1,\ldots,x_\rho)} = \frac{\Gamma(x)}{G(x)^{\rho+1}} \neq 0.$$

The numbers

$$a_n,\ a_{n-1},\ldots,a_{\rho+1}$$

are to be chosen so that, under the substitution

$$x_n = a_n, \quad x_{n-1} = a_{n-1},\ldots, \quad x_{\rho+1} = a_{\rho+1},$$

the product $G(x)\cdot\Gamma(x)$ does not vanish — that is, the algebraic rank of the system (1) is preserved. In the successive substitutions

$$\begin{aligned} [h(x)]_{x_n=a_n} = h^{(n-1)}(x), \qquad [h^{(n-1)}(x)]_{x_{n-1}=a_{n-1}} = h^{(n-2)}(x),\ldots, \\ [h^{(\rho+1)}(x)]_{x_{\rho+1}=a_{\rho+1}} = h^*(x_1,x_2,\ldots,x_\rho), \end{aligned}$$

[85] This last conclusion, resting on the reduced representation, would no longer be possible for a substitution $x_i = a$, $x_k = b$ while the remaining assumptions are preserved. Then $H_1(x)$ and $H_2(x)$ could vanish simultaneously, and the quotient would become indeterminate under the substitution, equal to $\frac{0}{0}$; for example

$$h(x) = \frac{x_3(\alpha x_1 + \beta x_2) + x_4(\gamma x_1 + \delta x_2)}{x_3(\alpha' x_1 + \beta' x_2) + x_4(\gamma' x_1 + \delta' x_2)}$$

under the substitution $x_3 = 0$, $x_4 = 0$.

let the functions $h(x)$, $h^{(n-1)}(x)$, ..., $h^{(\rho+1)}(x)$ each be put into reduced representation. Under these assumptions, in the function $h^*(x_1, \ldots, x_\rho)$ neither numerator nor denominator can vanish identically, and the function also cannot become $\frac{0}{0}$.[86]

§ 3. One-to-one mapping of systems $\mathfrak{S}_{n\rho}$ onto systems $\mathfrak{S}^*_{\rho\rho}$.

From the auxiliary theorem we shall immediately obtain a mapping of the systems $\mathfrak{S}_{n\rho}$ onto systems $\mathfrak{S}^*_{\rho\rho}$, by replacing some of the indeterminates by numbers; the algebraic rank ρ thereby presents itself as the properly characteristic number.

Since $\mathfrak{S}_{n\rho}$ has algebraic rank ρ, there exist ρ functions from $\mathfrak{S}_{n\rho}$,

$$g_1(x), \ldots, g_\rho(x), \tag{1}$$

which are algebraically independent. Further, if $f(x)$ denotes an arbitrary function from $\mathfrak{S}_{n\rho}$, then the system

$$g_1(x), \ldots, g_\rho(x),\ f(x) \tag{2}$$

also has algebraic rank ρ, and the systems (1) and (2) satisfy the conditions of the auxiliary theorem.[87]

If, therefore, one determines the numbers

$$a_n,\ a_{n-1}, \ldots, a_{\rho+1}$$

according to the auxiliary theorem in such a way that the algebraic rank ρ of the system (1) is preserved by the substitution — which is possible in arbitrarily many ways — then in the function defined by

$$[f(x)]_{x_n=a_n} = f^{(n-1)}(x), \qquad [f^{(n-1)}(x)]_{x_{n-1}=a_{n-1}} = f^{(n-2)}(x); \ldots,$$
$$[f^{(\rho+1)}(x)]_{x_{\rho+1}=a_{\rho+1}} = f^*(x_1, \ldots, x_\rho),$$

namely

$$f^*(x_1, \ldots, x_\rho), \tag{3}$$

neither numerator nor denominator can vanish identically.

Now let $f(x)$ run through all (finitely or infinitely many) functions from $\mathfrak{S}_{n\rho}$. We consider the system consisting of the totality of all functions (3); it has the following properties:

1. It has algebraic rank ρ; for it contains the functions arising from (1),

$$g_1^*(x_1, \ldots, x_\rho), \ldots, g_\rho^*(x_1, \ldots, x_\rho),$$

which, because of the choice of $a_n, a_{n-1}, \ldots, a_{\rho+1}$, are algebraically independent. The system is therefore to be denoted by

$$\mathfrak{S}^*_{\rho\rho}.$$

[88]

2. The correspondence of the functions $f(x)$ and $f^*(x)$ is without exception one-to-one.

For if $f_1(x)$ and $f_2(x)$ belong to $\mathfrak{S}_{n\rho}$, then their difference

$$d(x) = f_1(x) - f_2(x)$$

is also algebraically dependent on the system (1); moreover

$$d^*(x) = f_1^*(x) - f_2^*(x).$$

Since $d(x)$, together with the system (1), satisfies the conditions of the auxiliary theorem, it follows from

$$d(x) \neq 0$$

[86] For example, in the function $h(x)$ of the preceding note, the successive substitution gives

$$[h(x)]_{x_4=0} = h^{(3)}(x) = \frac{\alpha x_1 + \beta x_2}{\alpha' x_1 + \beta' x_2}, \qquad h^*(x_1, x_2) = \frac{\alpha x_1 + \beta x_2}{\alpha' x_1 + \beta' x_2}.$$

[87] If necessary, the indeterminates are to be renumbered.

[88] Correspondingly, by $f^*(x)$, $g^*(x)$, ... one should throughout understand mapping functions, i.e. functions of $x_1, \ldots, x_\rho$.

necessarily that

$$d^*(x) \neq 0;$$

in other words, different functions from $\mathfrak{S}_{n\rho}$ correspond to different functions from $\mathfrak{S}^*_{\rho\rho}$.

3. From every relation between functions from $\mathfrak{S}^*_{\rho\rho}$,

$$F\big(f_1^*(x), f_2^*(x), \ldots, f_k^*(x)\big) = 0 \quad \text{identically in } x_1, \ldots, x_\rho,$$

it follows, for the uniquely corresponding functions from $\mathfrak{S}_{n\rho}$, that

$$F\big(f_1(x), f_2(x), \ldots, f_k(x)\big) = 0 \quad \text{identically in } x_1, \ldots, x_n.$$

For with $f_1(x), f_2(x), \ldots, f_k(x)$, every rational combination of these functions is also algebraically dependent on the system (1).

It further follows from

$$\begin{aligned} h(x) &= F\big(f_1(x), \ldots, f_k(x)\big) : \\ h^*(x) &= F\big(f_1^*(x), \ldots, f_k^*(x)\big). \end{aligned} \tag{4}$$

By the auxiliary theorem, therefore,

$$h(x) \neq 0 \quad \text{implies} \quad h^*(x) \neq 0;$$

q.e.d.

In summary, we obtain

Theorem I: To every system $\mathfrak{S}_{n\rho}$ of (finitely or infinitely many) rational functions there can be assigned — in arbitrarily many ways — a one-to-one system $\mathfrak{S}^*_{\rho\rho}$, in such a way that all relations existing between functions from $\mathfrak{S}^*_{\rho\rho}$ are also preserved in $\mathfrak{S}_{n\rho}$ and conversely. In particular, by (4), a field $\mathfrak{K}_{n\rho}$ again corresponds to a field $\mathfrak{K}^*_{\rho\rho}$.

From Theorem I we draw the consequence that simplifies all further proofs: *properties of systems $\mathfrak{S}_{n\rho}$ which are characterized by finitely or infinitely many relations between functions of the system hold for the systems $\mathfrak{S}_{n\rho}$ as soon as they hold for a mapping system $\mathfrak{S}^*_{\rho\rho}$.*

We call this consequence briefly the "transfer principle."

The simplest example of this mapping is furnished by the theory of algebraic invariants. In fact, by a linear transformation one can always numerically fix some coefficients of the ground form in such a way that all relations valid for the special form also remain valid in general.

§ 4. Rational basis of the fields $\mathfrak{K}_{n\rho}$.

In the following investigations, grouped around basis concepts, we shall consistently use the "transfer principle" of § 3. We first prove the existence of the basis for fields, but state the definitions generally:

Definition III: A finite number of functions from $\mathfrak{S}_{n\rho}$ is called a rational basis if every function from $\mathfrak{S}_{n\rho}$ can be represented as a rational combination of this finite number, with coefficients from Ω.

The rational basis must contain ρ algebraically independent functions, since the algebraic rank of a system derived from a rational basis is equal to the algebraic rank of the rational basis.

We first show that the existence proof for the rational basis admits the transfer principle. Namely, Definition III can be formulated as follows:

The functions from $\mathfrak{S}_{n\rho}$

$$g_1(x),\ g_2(x), \ldots, g_k(x)$$

form a rational basis if and only if the infinitely many relations

$$f(x) = \gamma(g_1(x), \ldots, g_k(x)) \tag{1}$$

are satisfied, where $f(x)$ runs through all functions from $\mathfrak{S}_{n\rho}$ and γ denotes a rational function of k arguments, varying with f.

Thus, if in the mapping system $\mathfrak{S}^*_{\rho\rho}$ a rational basis is given by

$$g^*_1(x),\ g^*_2(x), \ldots, g^*_k(x),$$

which entails the relations

$$f^*(x) = \gamma(g^*_1(x), \ldots, g^*_k(x)),$$

then by Theorem I the relations (1) are fulfilled for $\mathfrak{S}_{n\rho}$. Thus the *transfer principle for the rational basis* says:

From the existence of a rational basis for a mapping system $\mathfrak{S}^*_{\rho\rho}$ follows the rational basis for the original system $\mathfrak{S}_{n\rho}$.[89]

To prove the existence of the rational basis of all fields $\mathfrak{K}_{n\rho}$, we may therefore restrict ourselves to fields $\mathfrak{K}^*_{\rho\rho}$; here a direct generalization of an argument used by E. Steinitz leads to the goal.[90]

Let

$$g^*_1(x_1, \ldots, x_\rho), \ldots, g^*_\rho(x_1, \ldots, x_\rho) \tag{2}$$

be a system of ρ algebraically independent functions from $\mathfrak{K}^*_{\rho\rho}$. By eliminating $x_1, \ldots, x_\rho$ one obtains ρ relations:

$$\begin{aligned} &F_1(g^*_1(x), \ldots, g^*_\rho(x), x_1) = 0, \ldots, \\ &F_\rho(g^*_1(x), \ldots, g^*_\rho(x), x_\rho) = 0, \end{aligned} \tag{3}$$

where in each relation $F_i = 0$ the indeterminate x_i actually occurs; otherwise, contrary to the assumption, there would be an algebraic dependence among the functions (2). The relations (3) say that

$$x_1, x_2, \ldots, x_\rho$$

are algebraic elements with respect to

$$\Omega(g^*_1(x), \ldots, g^*_\rho(x)).$$

Hence the field arising by adjoining $x_1, \ldots, x_\rho$,

$$\Omega(g^*_1, \ldots, g^*_\rho, x_1, \ldots, x_\rho) = \Omega(x_1, \ldots, x_\rho),$$

is a finite algebraic extension of $\Omega(g^*_1, \ldots, g^*_\rho)$. By a known theorem,[91] $\mathfrak{K}^*_{\rho\rho}$, as an intermediate field between $\Omega(g^*_1, \ldots, g^*_\rho)$ and $\Omega(x_1, \ldots, x_\rho)$, is itself an algebraic extension of $\Omega(g^*_1, \ldots, g^*_\rho)$, while $\Omega(x_1, \ldots, x_\rho)$ is an

[89] The transfer principle is formulated correspondingly for every basis question treated here.

[90] E. Steinitz, Algebraische Theorie der Körper, § 24, Crelle's Journal 137 (1909).

[91] Compare, for example, Weber, Lehrbuch d. Algebra 1, § 144 (1st ed.) or § 55 (small edition).

algebraic extension of $\mathfrak{K}^*_{\rho\rho}$. Thus $\mathfrak{K}^*_{\rho\rho}$ arises from $\Omega(g_1^*, \ldots, g_\rho^*)$ by adjoining a finite number of elements from $\mathfrak{K}^*_{\rho\rho}$, or also by adjoining one suitably chosen function; i.e. one has

$$\mathfrak{K}^*_{\rho\rho} = \Omega(g_1^*, \ldots, g_\rho^*, g_{\rho+1}^*, \ldots, g_k^*) = \Omega(g_1^*, \ldots, g_\rho^*, g_0^*).$$

By the transfer principle we therefore have

Theorem II: Every field $\mathfrak{K}_{n\rho}$ possesses a rational basis of algebraic rank ρ; the rational basis can in particular be chosen so that it contains at most $\rho + 1$ functions.

Let an arbitrary system of ρ algebraically independent functions from $\mathfrak{K}_{n\rho}$ be given by $h_1(x), \ldots, h_\rho(x)$, and suppose it can be extended by the above method to a rational basis

$$h_1(x), \ldots, h_\rho(x); h_{\rho+1}(x), \ldots, h_\sigma(x).$$

By the defining property, relations exist:

$$\begin{aligned} h_i(x) &= \chi_i(g_1(x), \ldots, g_k(x)) \quad (i = 1, 2, \ldots, \sigma), \\ g_j(x) &= \psi_j(h_1(x), \ldots, h_\sigma(x)) \quad (j = 1, 2, \ldots, k), \end{aligned}$$

where χ_i and ψ_j denote rational functions of their arguments. Thus:

From any rational basis one obtains every further rational basis by a birational transformation; and:

If

$$h_1(x), \ldots, h_\rho(x)$$

is any system of ρ algebraically independent functions from $\mathfrak{K}_{n\rho}$, then $\mathfrak{K}_{n\rho}$ is a finite algebraic extension of

$$\Omega(h_1(x), \ldots, h_\rho(x)).$$

§ 5. Involution form and involution basis of the fields $\mathfrak{K}_{n\rho}$.

Whereas every rational basis constructed in § 4 had the disadvantage of depending on the arbitrarily chosen starting functions, we now show the existence of a *rational basis uniquely determined by the field.*

By § 4, $\Omega(x_1, \ldots, x_\rho)$ is an algebraic extension — say of degree i — of $\mathfrak{K}^*_{\rho\rho}$, and, because

$$\mathfrak{K}^*_{\rho\rho}(x_1, \ldots, x_\rho) = \Omega(x_1, \ldots, x_\rho),$$

arises by adjoining the elements $x_1, \ldots, x_\rho$, which are algebraic with respect to $\mathfrak{K}^*_{\rho\rho}$. We therefore consider the integral function $\Phi^*(z, u)$ irreducible with respect to $\mathfrak{K}^*_{\rho\rho}$, with the zero

$$z = u_1 x_1 + u_2 x_2 + \cdots + u_\rho x_\rho = u(x),$$

which is of ith degree in z. Further, as the product of the quantities conjugate to $[z - u(x)]$, $\Phi^*(z, u)$ is a homogeneous form of dimension i in $z, u_1, \ldots, u_\rho$; hence as coefficient of z^i it contains a function from $\mathfrak{K}^*_{\rho\rho}$ that is free of u. If we make this highest coefficient equal to 1 by division, then the irreducible function $\Phi^*(z, u)$ is uniquely determined. The homogeneous form so defined,

$$\begin{aligned} \Phi^*(z, u) = z^i + {\sum}' g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x_1, \ldots, x_\rho) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \\ (\alpha + \alpha_1 + \cdots + \alpha_\rho = i), \end{aligned} \tag{1}$$

is to be called the *involution form*[92] of $\mathfrak{K}^*_{\rho\rho}$; from (1) one obtains

$$\begin{aligned} \Phi(z, u) = z^i + {\sum}' g_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \\ (\alpha + \alpha_1 + \cdots + \alpha_\rho = i), \end{aligned} \tag{2}$$

as an *involution form* of $\mathfrak{K}_{n\rho}$.

The coefficients of the involution form $\Phi^*(z, u)$ will turn out to be a rational basis of $\mathfrak{K}^*_{\rho\rho}$; i.e. the relation

$$\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x_1, \ldots, x_\rho)) = \mathfrak{K}^*_{\rho\rho} \quad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i) \tag{3}$$

[92]The reason for the name will be clear at the end of the paragraph; ${\sum}'$ means that the summation is to extend over all terms except z^i.

holds. For the proof we observe that $\Phi^*(z,u)$ has coefficients from $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$ and, because of its irreducibility with respect to $\mathfrak{K}^*_{\rho\rho}$, is all the more irreducible with respect to this subfield. Hence $\Phi^*(z,u)$ gives the irreducible equation of $u(x)$ with respect to $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$. It follows that $\Omega(x_1,\ldots,x_\rho)$ is an algebraic extension — of degree i — of $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$. Since $\mathfrak{K}^*_{\rho\rho}$ lies between $\Omega(g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x))$ and $\Omega(x_1,\ldots,x_\rho)$, the identity of degrees is known to imply the identity of the fields, hence relation (3). By the transfer principle, the coefficients of $\Phi(z,u)$ correspondingly yield a rational basis of $\mathfrak{K}_{n\rho}$; i.e. we have

Theorem III: The coefficients of the involution form constitute a rational basis determined by the field, the involution basis; for fields $\mathfrak{K}^*_{\rho\rho}$ this basis is uniquely determined.[93]

We add an irrational interpretation of this fact, which establishes the connection with the geometric theory of involutions.

The factorization of $\Phi^*(z,u)$ into linear factors is given in an extension field by

$$\begin{aligned}\Phi^*(z,u) = (z - u_1x_1 - \cdots - u_\rho x_\rho)(z - u_1x_1^{(1)} - \cdots - u_\rho x_\rho^{(1)})\cdots \\ \cdot (z - u_1x_1^{(i-1)} - \cdots - u_\rho x_\rho^{(i-1)}).\end{aligned} \tag{4}$$

Comparison with (1) shows that the functions

$$g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x_1,\ldots,x_\rho) \quad (\alpha+\alpha_1+\cdots+\alpha_\rho = i)$$

are identical with the elementary symmetric functions of the rows of quantities

$$\begin{matrix} x_1 & x_2 & \cdots & x_\rho \\ x_1^{(1)} & x_2^{(1)} & \cdots & x_\rho^{(1)} \\ \vdots & \vdots & & \vdots \\ x_1^{(i-1)} & x_2^{(i-1)} & \cdots & x_\rho^{(i-1)}. \end{matrix} \tag{5}$$

Thus every function from $\mathfrak{K}^*_{\rho\rho}$ is a rational combination of these elementary functions, symmetric in the rows (5), and conversely every symmetric function of the rows (5) belongs to $\mathfrak{K}^*_{\rho\rho}$.

Thus, as

$$f^*(x) = \frac{F^*(x)}{H^*(x)}$$

runs through all functions from $\mathfrak{K}^*_{\rho\rho}$, there is a system of infinitely many relations:

$$\frac{F^*(x)}{H^*(x)} = \frac{F^*(x^{(1)})}{H^*(x^{(1)})} = \cdots = \frac{F^*(x^{(i-1)})}{H^*(x^{(i-1)})}, \tag{6}$$

or, summarized,

$$F^*(x^{(k)})H^*(x^{(l)}) - F^*(x^{(l)})H^*(x^{(k)}) = 0 \quad (k,l = 0,1,\ldots,i-1),$$

where neither $F^*(x^{(k)})$ nor $H^*(x^{(k)})$ vanishes identically, since they are conjugate to $F^*(x)$ and $H^*(x)$ and formally symmetric.

Conversely, for every row of quantities ξ satisfying the infinitely many relations

$$F^*(x)H^*(\xi) - H^*(x)F^*(\xi) = 0, \qquad H^*(\xi) \neq 0, \tag{7}$$

one obtains membership in the rows of quantities (5).

Indeed, if $G^*(x)$ denotes the common denominator of the coefficients of $\Phi^*(z,u)$, then by assumption (7) the function

$$G^*(\xi)\cdot\Phi^*(z,u;\xi),$$

which is also integral in ξ, does not vanish identically — that is, the coefficients of $\Phi^*(z,u;\xi)$ do not become indeterminate — and has the zero

$$z = u_1\xi_1 + u_2\xi_2 + \cdots + u_\rho\xi_\rho.$$

Because of (7), $\Phi^*(z,u;x)$ therefore also has the zero $z = u(\xi)$; i.e. ξ belongs to the rows of quantities (5). The corresponding assertion holds by (6) if ξ satisfies a system of equations with respect to a conjugate row of quantities:

$$F^*(x^{(k)})H^*(\xi) - H^*(x^{(k)})F^*(\xi) = 0; \qquad H^*(\xi) \neq 0.$$

[93]For fields $\mathfrak{K}_{n\rho}$ the involution form may depend on the mapping; uniqueness can be obtained by a mapping with indeterminate coefficients.

Thus the rows of quantities (5) are defined as the solutions ξ of the system of equations

$$F^*(x)H^*(\xi) - H^*(x)F^*(\xi) = 0; \qquad H^*(\xi) \neq 0,$$

where $F^*(x) : H^*(x)$ runs through all functions from $\mathfrak{K}^*_{\rho\rho}$; here the row x may also be replaced by any conjugate row.

Geometrically this says the following, interpreting each row x as a point: the solutions of (7) represent, in a ρ-dimensional linear space, ∞^ρ groups of i points each, in such a way that any point of the group determines the individual group uniquely. Such a system of point-groups is called an "*involution in a linear space*."[94]

Correspondingly, a field $\mathfrak{K}_{n\rho}$ characterizes an involution in a linear space consisting of curves, surfaces, and so on.

By the preceding, the degree i of $\Omega(x_1, \ldots, x_\rho)$ with respect to $\mathfrak{K}^*_{\rho\rho}$ is equal to the number of solution systems of (7) satisfying the side condition $H^*(\xi) \neq 0$ — which we may call the solution systems of $\mathfrak{K}^*_{\rho\rho}$. Hence the argument used to prove the existence of the involution basis also permits the following irrational formulation:

"If $\mathfrak{L}^*$ is a divisor of $\mathfrak{K}^*_{\rho\rho}$ and every solution system of the field $\mathfrak{L}^*$ is also a solution system of $\mathfrak{K}^*_{\rho\rho}$, then $\mathfrak{L}^*$ and $\mathfrak{K}^*_{\rho\rho}$ are identical."

The corresponding statement holds, by the transfer principle, for divisors $\mathfrak{L}$ of $\mathfrak{K}_{n\rho}$.

§ 6. Minimal basis of the fields $\mathfrak{K}_{n\rho}$.

This paragraph gives an overview of known theorems, but in an expanded formulation made possible by the transfer principle and the existence of the rational basis.

Definition IV: A rational basis of $\mathfrak{K}_{n\rho}$ is called a minimal basis if it contains only the minimum number ρ of functions, which must then be algebraically independent.

From theorems of Lüroth, Castelnuovo, and Enriques[95] one obtains the facts:

I. Every field $\mathfrak{K}_{n1}$ possesses a minimal basis, the Lüroth function of the field, uniquely determined up to linear fractional transformation.

II. For fields $\mathfrak{K}_{n2}$, a minimal basis exists always, and in general only, if one permits an algebraic extension of the coefficient field Ω.

III. For fields $\mathfrak{K}_{n\rho}$ ($\rho \geq 3$), in general no minimal basis exists. A characterization of the special fields $\mathfrak{K}_{n\rho}$ with minimal basis has not yet been attempted.

We briefly indicate the proof of Lüroth's theorem for fields $\mathfrak{K}^*_{11}$ given by E. Steinitz:[96]

Let $\Phi^*(z, u)$ be the involution form of $\mathfrak{K}^*_{11}$, and let $\psi^*(x)$ be any coefficient of $\Phi^*(z, u)$ that actually contains the indeterminate x. It turns out that the degree of $\Omega(x)$ with respect to $\Omega(\psi^*(x))$ is equal to the degree of $\Omega(x)$ with respect to $\mathfrak{K}^*_{11}$, whence the identity of the fields follows. Further, if $\Omega(\chi^*(x))$ is equal to $\Omega(\psi^*(x))$, then the mutual rational dependence of $\chi^*(x)$ and $\psi^*(x)$ implies the linear fractional dependence of the two functions.

By the transfer principle, Lüroth's theorem therefore also admits the formulation:

The minimal basis of the fields $\mathfrak{K}_{n1}$ consists of any function of the involution basis; and any two functions of the involution basis of $\mathfrak{K}_{n1}$ depend on one another by a linear fractional transformation.

G. Castelnuovo proves Fact II, by means of the geometric theory of involutions, for fields $\mathfrak{K}^*_{22}$ derived from a rational basis of three functions. Since the transfer principle also remains valid under finite algebraic extension of the coefficient field Ω, the existence of the rational basis makes the proof complete for all fields $\mathfrak{K}_{n2}$.

Regarding III, it should be noted that Enriques constructs a nonrational involution in three-dimensional space; that is, a field $\mathfrak{K}_{33}$ which has no minimal basis.

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[94] The excluded solutions of (7), for which $F^*(\xi) = 0$, $H^*(\xi) = 0$, and likewise the excluded solutions of the system of equations corresponding to the involution basis, are called fundamental points of the involution; this also includes those arising by homogenization, the points "lying at infinity."

[95] J. Lüroth: Beweis eines Satzes über rationale Kurven, Math. Ann. 9 (1875). — G. Castelnuovo: Sulla razionalità delle involuzioni piane, Math. Ann. 44 (1893). — F. Enriques: Sopra una involuzione non razionale dello spazio, Rend. Acc. Linc., Vol. 21, 21 Jan. 1912.

[96] Loc. cit., § 24.

§ 7. Arbitrary system $\mathfrak{S}_{n\rho}$, linear family $\mathfrak{L}_{n\rho}$, and integrality domain $\mathfrak{J}_{n\rho}$ of rational functions. Existence of the rational basis.

From the existence of the rational basis of the fields $\mathfrak{K}_{n\rho}$ we shall now obtain the rational basis for arbitrary systems of rational and integral rational functions. We arrange these systems in the order in which they successively arise from one another through the elementary operations of addition, multiplication, and division.

I. As always, let $\mathfrak{S}_{n\rho}$ denote an arbitrary system of rational (or integral rational) functions of n indeterminates and of algebraic rank ρ. The quantities from Ω are counted as belonging to the system.

II. The system $\mathfrak{L}_{n\rho}$ is called a *linear family* if it satisfies the conditions:

1. Together with $f(x)$, $\mathfrak{L}_{n\rho}$ always also contains $c \cdot f(x)$, where c is an arbitrary quantity from Ω;
2. Together with $f(x)$ and $g(x)$, $\mathfrak{L}_{n\rho}$ always also contains $f(x) + g(x)$.

III. The linear family $\mathfrak{J}_{n\rho}$ is called an *integrality domain* under the further condition:

3. Together with $f(x)$ and $g(x)$, $\mathfrak{J}_{n\rho}$ always also contains $f(x) \cdot g(x)$.

IV. The integrality domain $\mathfrak{K}_{n\rho}$ is called a *field* under the further condition:

4. Together with $f(x)$ and $g(x)$ — where $g(x) \neq 0$ — $\mathfrak{K}_{n\rho}$ always also contains the quotient $f(x) : g(x)$.

Conditions 1 through 4 are those stated in § 1 for the field.[97]

These domains are now to be derived successively from one another.

a) An arbitrary system $\mathfrak{S}_{n\rho}$ is enlarged to a linear family $\mathfrak{L}_{n\rho}$ by the requirement:

$\mathfrak{L}_{n\rho}$ contains, in addition to $\mathfrak{S}_{n\rho}$, all functions $h(x)$ and only those that can be represented linearly with coefficients from Ω by a finite number of functions from $\mathfrak{S}_{n\rho}$:

$$h(x) = c_1 f_1(x) + c_2 f_2(x) + \cdots + c_k f_k(x),$$

where $f_1(x), \ldots, f_k(x)$ run through all functions from $\mathfrak{S}_{n\rho}$, and $c_1, \ldots, c_k$ through all quantities from Ω.

For adding rational combinations of functions of the system preserves the algebraic rank of a system. $\mathfrak{L}_{n\rho}$ satisfies conditions 1 and 2, and is therefore a linear family. Moreover, every linear family containing $\mathfrak{S}_{n\rho}$ also contains $\mathfrak{L}_{n\rho}$ by 1 and 2; hence $\mathfrak{L}_{n\rho}$ is the smallest linear family containing $\mathfrak{S}_{n\rho}$.

b) Correspondingly, from a linear family $\mathfrak{L}_{n\rho}$ arises the smallest containing integrality domain $\mathfrak{J}_{n\rho}$ by the requirement:

$\mathfrak{J}_{n\rho}$ contains, in addition to $\mathfrak{L}_{n\rho}$, all functions $h(x)$ and only those that can be represented as an *integral rational combination* — with coefficients from Ω — of a finite number of functions from $\mathfrak{L}_{n\rho}$, $f_1(x), \ldots, f_k(x)$, where $f_1(x), \ldots, f_k(x)$ run through all functions from $\mathfrak{L}_{n\rho}$.

c) Finally, from an integrality domain $\mathfrak{J}_{n\rho}$ arises the smallest containing field $\mathfrak{K}_{n\rho}$ by the requirement: $\mathfrak{K}_{n\rho}$ contains, in addition to $\mathfrak{J}_{n\rho}$, all functions $h(x)$ and only those that can be represented as quotients of two functions from $\mathfrak{J}_{n\rho}$.

Combining the three extensions into one step gives:

Every system $\mathfrak{S}_{n\rho}$ can be enlarged to a smallest containing field $\mathfrak{K}_{n\rho}$ by the requirement:

$\mathfrak{K}_{n\rho}$ contains, in addition to $\mathfrak{S}_{n\rho}$, all functions $h(x)$ and only those that can be represented as rational combinations — with coefficients from Ω — of a finite number of functions from $\mathfrak{S}_{n\rho}$, $f_1(x), \ldots, f_k(x)$, where $f_1(x), \ldots, f_k(x)$ run through all functions from $\mathfrak{S}_{n\rho}$.

From the latter fact the existence of the rational basis for arbitrary domains follows immediately:

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{S}_{n\rho}$, and let a rational basis of $\mathfrak{K}_{n\rho}$ be given by

$$h_1(x),\ h_2(x), \ldots, h_\sigma(x). \tag{1}$$

By the definition of $\mathfrak{K}_{n\rho}$, relations exist:

$$\begin{aligned} h_1(x) &= r_1(g_1(x), \ldots, g_\lambda(x)); \ldots; \\ h_\sigma(x) &= r_\sigma(g_\mu(x), \ldots, g_\nu(x)), \end{aligned} \tag{2}$$

where $r_1, \ldots, r_\sigma$ are rational functions of their arguments, and

$$g_1(x), \ldots, g_\lambda(x), \ldots, g_\mu(x), \ldots, g_\nu(x) \tag{3}$$

all belong to $\mathfrak{S}_{n\rho}$. Because of the relations (2), (3), in addition to (1), also represents a rational basis of $\mathfrak{K}_{n\rho}$; because the functions (3) belong to $\mathfrak{S}_{n\rho}$, this is at the same time a rational basis of $\mathfrak{S}_{n\rho}$. We therefore have

[97] $f(x)$ and $g(x)$ may also be of degree zero, i.e. quantities from Ω.

Theorem IV: An arbitrary system $\mathfrak{S}_{n\rho}$ of rational (or integral rational) functions possesses a rational basis, consisting of a finite number of functions of the system, of algebraic rank ρ.

Now let $\mathfrak{L}_{n\rho}$ in particular be a linear family, and let

$$g_1(x), \dots, g_\rho(x),\ g_{\rho+1}(x), \dots, g_\nu(x) \tag{4}$$

be a rational basis of $\mathfrak{L}_{n\rho}$. Suppose, for instance, that

$$g_1(x), \dots, g_\rho(x)$$

are algebraically independent. If one then sets

$$g(x) = c_1 g_{\rho+1}(x) + c_2 g_{\rho+2}(x) + \cdots + c_{\nu-\rho} g_\nu(x),$$

then the c_i can be chosen as numbers from Ω in such a way that

$$g_1(x), \dots, g_\rho(x),\ g(x) \tag{5}$$

becomes a rational basis of the smallest containing field $\mathfrak{K}_{n\rho}$. But since $\mathfrak{L}_{n\rho}$ is a linear family, $g(x)$ belongs to $\mathfrak{L}_{n\rho}$, and (5) represents a rational basis of $\mathfrak{L}_{n\rho}$; that is,

Theorem V: The rational basis of every linear family $\mathfrak{L}_{n\rho}$ of rational (or integral rational) functions — hence in particular the rational basis of every integrality domain $\mathfrak{J}_{n\rho}$ and every field $\mathfrak{K}_{n\rho}$ — can be chosen in particular so that it contains at most $\rho + 1$ functions (and at least ρ).

The bound on the number of functions in the rational basis found in § 4 for fields therefore rests on the property of the field of being a linear family, whereas the restriction to ρ functions in the case of the existence of the minimal basis uses all assumptions of the field.

It remains to note that, by § 3 (4), all characteristic properties of the systems and smallest containing systems are preserved under the mapping.

§ 8. Involution basis of the integrality domains $\mathfrak{J}_{n\rho}$ consisting of polynomials.

In the following paragraphs we are concerned with systems $\mathfrak{S}_{n\rho}$ whose elements are integral rational functions (polynomials); first with integrality domains $\mathfrak{J}_{n\rho}$ consisting of polynomials. For these domains $\mathfrak{J}_{n\rho}$, the divisibility notions are first to be fixed.

Let $F(x), G(x)$ be two polynomials from $\mathfrak{J}_{n\rho}$, and let $L(x)$ be a common divisor of these polynomials, so that one has

$$F(x) = L(x) \cdot F_1(x), \qquad G(x) = L(x) \cdot G_1(x).$$

$L(x)$ is called a *common divisor in* $\mathfrak{J}_{n\rho}$ if and only if the three polynomials $L(x), F_1(x), G_1(x)$ belong to $\mathfrak{J}_{n\rho}$.

Two polynomials from $\mathfrak{J}_{n\rho}$ are called *relatively prime in* $\mathfrak{J}_{n\rho}$ if they possess no common divisor in $\mathfrak{J}_{n\rho}$.

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{J}_{n\rho}$. Then, from the definition, every function from $\mathfrak{K}_{n\rho}$ has a representation

$$f(x) = F_1(x) : F_2(x),$$

where $F_1(x), F_2(x)$ belong to $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$. Correspondingly, several functions from $\mathfrak{K}_{n\rho}$ can be brought to a common denominator in $\mathfrak{J}_{n\rho}$:

$$f_1(x) = \frac{F_1(x)}{F(x)}, \dots, f_k(x) = \frac{F_k(x)}{F(x)}.$$

$F(x)$ is called *a least common denominator in* $\mathfrak{J}_{n\rho}$ if and only if the polynomials from $\mathfrak{J}_{n\rho}$,

$$F(x), F_1(x), \dots, F_k(x),$$

are relatively prime in $\mathfrak{J}_{n\rho}$.

Such a representation can be possible in different ways, since in $\mathfrak{J}_{n\rho}$ the laws of unique factorization into irreducible functions do not hold in general.

We now investigate the significance of the involution basis of $\mathfrak{K}_{n\rho}$ for $\mathfrak{J}_{n\rho}$, by transferring the notions of involution form and involution basis to integrality domains. Let

$$\Phi(z, u) = z^i + \sum{}' g_{\alpha\alpha_1\dots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \dots + \alpha_\rho = i)$$

be an involution form of $\mathfrak{K}_{n\rho}$, and let $G(x)$ be a least common denominator in $\mathfrak{J}_{n\rho}$ of the coefficients of $\Phi(z, u)$. By multiplication with $G(x)$, $\Phi(z, u)$ passes into a form $\Psi(z, u)$ whose coefficients are polynomials from $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$:

$$G(x) \cdot \Phi(z, u) = \Psi(z, u) = G(x) z^i + \sum{}' G_{\alpha\alpha_1\dots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \dots + \alpha_\rho = i).$$

Every form $\Psi(z, u)$ arising from an involution form $\Phi(z, u)$ of $\mathfrak{K}_{n\rho}$ by multiplication with a polynomial from $\mathfrak{J}_{n\rho}$ in such a way that the coefficients of $\Psi(z, u)$ are polynomials from $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$, shall be called an *involution form* of $\mathfrak{J}_{n\rho}$; and the coefficients of $\Psi(z, u)$ shall be called a corresponding *involution basis.*

It is first clear from the meaning of the involution basis of $\mathfrak{K}_{n\rho}$ that every involution basis of $\mathfrak{J}_{n\rho}$ is at the same time a rational basis. For the further investigation we consider a mapping domain $\mathfrak{J}^*_{\rho\rho}$ for whose smallest containing field $\mathfrak{K}^*_{\rho\rho}$ the irreducible equation with zero

$$z = u_1 x_1 + \dots + u_\rho x_\rho$$

is given by $\Phi^*(z, u) = 0$.

According to § 5, the coefficients of

$$\Phi^*(z, u) = z^i + \sum{}' g^*_{\alpha\alpha_1\dots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \qquad (\alpha + \alpha_1 + \dots + \alpha_\rho = i)$$

represent the elementary symmetric functions of the rows of quantities conjugate with respect to $\mathfrak{K}^*_{\rho\rho}$,

$$x, \quad x^{(1)}, \dots, x^{(i-1)}.$$

Let $F^*(x)$ be any polynomial from $\mathfrak{K}^*_{\rho\rho}$. Since

$$F^*(x) = \frac{1}{i}\Big(F^*(x) + F^*(x^{(1)}) + \dots + F^*(x^{(i-1)})\Big),$$

it follows that $F^*(x)$ is an integral symmetric function of these rows of quantities; consequently it can be expressed as an integral rational combination of the elementary symmetric functions, that is, of the functions

$$g^*_{\alpha\alpha_1\ldots\alpha_\rho}(x).$$

If now

$$G^*(x)\cdot\Phi^*(z,u)=\Psi^*(z,u)=G^*(x)z^i+{\sum}'G^*_{\alpha\alpha_1\ldots\alpha_\rho}(x)z^\alpha u_1^{\alpha_1}\cdots u_\rho^{\alpha_\rho},\qquad(\alpha+\alpha_1+\cdots+\alpha_\rho=i)$$

is an involution form of $\mathfrak{J}^*_{\rho\rho}$, then every polynomial from $\mathfrak{K}^*_{\rho\rho}$, and in particular every polynomial from $\mathfrak{J}^*_{\rho\rho}$, is an integral rational combination of the quotients

$$G^*_{\alpha\alpha_1\ldots\alpha_\rho}(x):G^*(x).$$

Taking the transfer principle into account, this gives

Theorem VI: To every integrality domain of polynomials $\mathfrak{J}_{n\rho}$ there exists at least one distinguished rational basis, the involution basis, such that in the rational representation of all polynomials from $\mathfrak{J}_{n\rho}$ by the functions of the involution basis only a power of one fixed function (the highest coefficient of the corresponding involution form) occurs in the denominator.

§ 9. Relatively integral domains $\mathfrak{G}_{n\rho}$ consisting of polynomials.

In what follows we consider special integrality domains consisting of polynomials, which form an intermediate stage between integrality domains and fields.

An integrality domain of polynomials $\mathfrak{G}_{n\rho}$ is called a "relatively integral domain" under the condition:

4a) $\mathfrak{G}_{n\rho}$ contains, along with $F(x)$ and $G(x)$ — where $G(x)\neq 0$ — the quotient $F(x):G(x)$ always and only when this quotient is integral in the indeterminates, that is, is a polynomial.

First the identity of the relatively integral domains with the totality of the polynomials of a field $\mathfrak{K}_{n\sigma}$ $(\sigma\geq\rho)$ of rational functions is to be proved.

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{G}_{n\rho}$. For every function $f(x)$ from $\mathfrak{K}_{n\rho}$ there is a representation as quotient of two polynomials from $\mathfrak{G}_{n\rho}$:

$$f(x)=F_1(x):F_2(x),$$

where polynomials of degree zero are also allowed. As soon as $f(x)$ becomes a polynomial, it belongs to $\mathfrak{G}_{n\rho}$ by condition 4a); and since $\mathfrak{G}_{n\rho}$ can contain no further polynomials, *every relatively integral domain $\mathfrak{G}_{n\rho}$ is identical with the totality of the polynomials of the smallest containing field.*

Conversely, let $\mathfrak{K}_{n\sigma}$ be any field of rational functions (therefore not necessarily the smallest containing field of a given system), and let $\mathfrak{J}$ be the totality of the polynomials contained in $\mathfrak{K}_{n\sigma}$. $\mathfrak{J}$ satisfies the conditions 1, 2, 3 of an integrality domain and condition 4a), and so is a relatively integral domain; that is, *the totality of the polynomials contained in a field $\mathfrak{K}_{n\sigma}$ forms a relatively integral domain $\mathfrak{G}_{n\rho}$ $(\rho\leq\sigma)$.*[98]

We further show the identity of the relatively integral domains with Hilbert's "integrality domains of relatively integral functions."[99]

Let

$$G_1(x),\ldots,G_k(x)\tag{1}$$

be a rational basis of $\mathfrak{G}_{n\rho}$. By conditions 1, 2, 3, 4a, every rational combination of the polynomials (1) which is integral in the indeterminates, that is, becomes a polynomial, belongs to $\mathfrak{G}_{n\rho}$. Conversely, every polynomial from $\mathfrak{G}_{n\rho}$ can, by the notion of rational basis, be rationally expressed through the polynomials (1); hence $\mathfrak{G}_{n\rho}$ is *identical with the totality of those rational combinations of the polynomials (1) which are integral in the indeterminates, that is, are polynomials.* Thus we have Hilbert's definition starting from the special rational basis, whereas our definition uses only abstract properties of the domain.

For relatively integral domains, the definition of a common divisor given in § 8 for general integrality domains simplifies:

Let $F(x),G(x)$ be two polynomials from $\mathfrak{G}_{n\rho}$ and let $L(x)$ be a common divisor of these polynomials. Then $L(x)$ is called a *common divisor in $\mathfrak{G}_{n\rho}$* if and only if $L(x)$ belongs to $\mathfrak{G}_{n\rho}$.

[98] It may also be $\rho=0$; that is, the totality of the polynomials from $\mathfrak{K}_{n\sigma}$ consists of elements of the coefficient domain Ω.

[99] D. Hilbert: Mathematische Probleme. Lecture Paris 1900. Problem 14 (Göttinger Nachrichten 1900).

For then the membership of the quotients $F(x) : L(x)$ and $G(x) : L(x)$ follows from condition 4a).

Another essential property of relatively integral domains is that the notion of being "algebraically integral with respect to $\mathfrak{G}_{n\rho}$" can be defined by means of any equation exactly as is known for algebraic number fields or for the field $\Omega(x_1, \dots, x_n)$.

Definition V: A quantity ξ is called algebraically integral with respect to $\mathfrak{G}_{n\rho}$ — or also relatively algebraically integral — if it satisfies some equation whose highest coefficient is the unit and whose remaining polynomials are from $\mathfrak{G}_{n\rho}$.

Suppose, namely, that the relatively algebraically integral quantity ξ satisfies, for example, the equation

$$L(z) = z^\lambda + G_1(x)z^{\lambda-1} + \cdots + G_\lambda(x) = 0.$$

$L(z)$ is divisible by the integral function irreducible with respect to the smallest containing field $\mathfrak{K}_{n\rho}$,

$$M(z) = z^\mu + H_1(x)z^{\mu-1} + \cdots + H_\mu(x).$$

But from this divisibility it follows by the known extension of Gauss's theorem[100] that the rational functions $H_1(x), \dots, H_\mu(x)$ are polynomials from $\mathfrak{K}_{n\rho}$, and therefore belong to $\mathfrak{G}_{n\rho}$. Thus the definition of the relatively algebraically integral quantity ξ is obtained from the irreducible equation. In particular, every polynomial $F(x)$ from $\mathfrak{G}_{n\rho}$ is also algebraically integral with respect to $\mathfrak{G}_{n\rho}$, since it satisfies the equation $z - F(x) = 0$.

It should further be mentioned that, for an integrality domain $\mathfrak{J}_{n\rho}$ of polynomials, analogously to the smallest containing field, the smallest containing relatively integral domain $\mathfrak{G}_{n\rho}$ is also defined by the requirement:

$\mathfrak{G}_{n\rho}$ contains, along with $\mathfrak{J}_{n\rho}$, all polynomials $H(x)$, and only those, which can be represented as quotients of two polynomials from $\mathfrak{J}_{n\rho}$.

From this definition it follows further: From every involution form and involution basis of $\mathfrak{J}_{n\rho}$ a corresponding one of $\mathfrak{G}_{n\rho}$ arises by splitting off, at most, a common divisor in $\mathfrak{G}_{n\rho}$ of all polynomials of the basis.

For $\mathfrak{J}_{n\rho}$ and $\mathfrak{G}_{n\rho}$ possess the same smallest containing field $\mathfrak{K}_{n\rho}$; and every involution form of $\mathfrak{J}_{n\rho}$ arises from an involution form of $\mathfrak{K}_{n\rho}$ by multiplication with a polynomial from $\mathfrak{J}_{n\rho}$, and so has coefficients from $\mathfrak{G}_{n\rho}$, which however may have a common divisor in $\mathfrak{G}_{n\rho}$.[101]

Finally it should be noted that the *mapping domain* of a relatively integral domain $\mathfrak{G}_{n\rho}$, which by § 7 is again an integrality domain $\mathfrak{J}^*_{\rho\rho}$ of polynomials, need not itself be a relatively integral domain. For if $F(x)$ and $G(x)$ are two polynomials from $\mathfrak{G}_{n\rho}$ whose quotient is not a polynomial and consequently does not belong to $\mathfrak{G}_{n\rho}$, then the quotient $F^*(x) : G^*(x)$ also does not belong to $\mathfrak{J}^*_{\rho\rho}$. But this quotient, arising by specialization of some indeterminates, may very well be a polynomial, and for $\mathfrak{J}^*_{\rho\rho}$ condition 4a) is then not fulfilled.[102]

§ 10. Relatively integral domains of the first kind and their integrality basis.

The notion introduced in § 9 (Definition V), "relatively algebraically integral," makes it possible to give a class of relatively integral domains which are finite integrality domains, and thereby to answer positively, at least for this class, a question raised by D. Hilbert (Math. Problems, Problem 14). We give the following

Definition VI: A finite number of polynomials from $\mathfrak{G}_{n\rho}$ is called an integrality basis if every polynomial from $\mathfrak{G}_{n\rho}$ can be represented as an integral rational combination of this finite number, with coefficients from Ω. An integrality domain of polynomials which possesses an integrality basis is called a finite integrality domain.

The class of relatively integral domains to be considered, for which the involution basis will prove to be an integrality basis, we designate as relatively integral domains of the first kind according to the following

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[100] Compare, for instance, J. König: Einleitung in die allgemeine Theorie der algebraischen Größen (Leipzig, B. G. Teubner, 1903), Kap. IX, § 2, or Weber (small edition), § 20.

[101] For example, for the integrality domain $\mathfrak{J}_{22}$ consisting of all integral rational combinations of x_1^2 and x_1x_2, the following is obtained as an involution form:

$$\Psi(z, u) = x_1^2z^2 - x_1^4u_1^2 - 2x_1^3x_2u_1u_2 - x_1^2x_2^2u_2^2.$$

Since x_1^2 belongs to $\mathfrak{J}_{22}$, and therefore all the more to the smallest containing relatively integral domain $\mathfrak{G}_{22}$, and consequently is a common divisor in $\mathfrak{G}_{22}$, the following results as an involution form of $\mathfrak{G}_{22}$:

$$\Psi(z, u) = z^2 - x_1^2u_1^2 - 2x_1x_2u_1u_2 - x_2^2u_2^2.$$

[102] Let $\mathfrak{G}_{22}$ consist, for example, of all polynomials which are rational combinations of $F = x_1^2x_2$ and $G = x_1^2(1 + x_3)$. The admissible specialization $x_3 = 0$ gives $F^* : G^* = x_2$, whereas $F : G$ is not a polynomial. $\mathfrak{J}^*_{22}$ is therefore not a relatively integral domain.

Definition VII: A relatively integral domain $\mathfrak{G}_{n\rho}$ is called of the first kind if, as mapping domain, it possesses a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$ such that every arbitrary polynomial in $x_1, \ldots, x_\rho$ (with coefficients from Ω) is algebraically integral over $\mathfrak{G}^*_{\rho\rho}$.

According to this definition, the indeterminates $x_1, \ldots, x_\rho$, and consequently also the linear form $u(x) = u_1x_1 + \cdots + u_\rho x_\rho$, are algebraically integral over $\mathfrak{G}^*_{\rho\rho}$. The irreducible integral function with zero $z = u(x)$ — the involution form of the smallest containing field $\mathfrak{K}^*_{\rho\rho}$ — therefore has the unit as highest coefficient and, as its further coefficients, polynomials from $\mathfrak{G}^*_{\rho\rho}$; it is therefore at the same time an involution form of $\mathfrak{G}^*_{\rho\rho}$, and in $\mathfrak{G}^*_{\rho\rho}$ no further involution form can exist. By the transfer principle, therefore, $\mathfrak{G}_{n\rho}$ also possesses an involution form whose highest coefficient is the unit; and since, by Theorem VI, in the rational representation of all polynomials from $\mathfrak{G}_{n\rho}$ by the corresponding involution basis only this highest coefficient enters into the denominator, the involution basis becomes an integrality basis. Thus:

Theorem VII. Every relatively integral domain of the first kind is a finite integrality domain; the integrality basis is given by the involution basis obtained from the characterizing mapping domain. In particular, the integrality basis of relatively integral domains of the first kind $\mathfrak{G}_{\rho\rho}$ is given by the uniquely defined involution basis of $\mathfrak{G}_{\rho\rho}$.

We add a criterion for relatively integral domains of the first kind. Let $\mathfrak{G}_{n\rho}$ be of the first kind and let an integrality basis of $\mathfrak{G}_{n\rho}$ be given by

$$F_1(x), F_2(x), \ldots, F_k(x). \tag{1}$$

Then, by a Hilbert auxiliary theorem,[103] there exist ρ algebraically independent polynomials from $\mathfrak{G}_{n\rho}$,

$$G_1(x), G_2(x), \ldots, G_\rho(x), \tag{2}$$

such that the polynomials (1), and hence every polynomial from $\mathfrak{G}_{n\rho}$, are algebraically integral over the polynomials (2). The same holds for every polynomial of the mapping domain $\mathfrak{G}^*_{\rho\rho}$ with respect to the corresponding polynomials

$$G^*_1(x), G^*_2(x), \ldots, G^*_\rho(x). \tag{3}$$

By Definition VII, therefore, every arbitrary polynomial in $x_1, \ldots, x_\rho$ is also algebraically integral over the polynomials (3).

Conversely, suppose a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of an arbitrary relatively integral domain $\mathfrak{G}_{n\rho}$ contains ρ polynomials (3) over which every polynomial in $x_1, \ldots, x_\rho$ is algebraically integral. Then we show that $\mathfrak{J}^*_{\rho\rho}$ thereby becomes a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$, and further that $\mathfrak{G}^*_{\rho\rho}$, and consequently $\mathfrak{G}_{n\rho}$, are of the first kind.

Indeed, let $F^*(x)$ be a polynomial from the smallest field $\mathfrak{K}^*_{\rho\rho}$ containing $\mathfrak{J}^*_{\rho\rho}$. By assumption it is algebraically integral over the polynomials (3). Since the corresponding function from $\mathfrak{K}_{n\rho}$ is then algebraically integral over ρ polynomials from $\mathfrak{G}_{n\rho}$, it is a polynomial $F(x)$ and hence belongs to $\mathfrak{G}_{n\rho}$; therefore $F^*(x)$ belongs to $\mathfrak{J}^*_{\rho\rho}$. The mapping domain $\mathfrak{J}^*_{\rho\rho}$ therefore contains all polynomials of $\mathfrak{K}^*_{\rho\rho}$ and is a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$. For $\mathfrak{G}^*_{\rho\rho}$, and hence also for $\mathfrak{G}_{n\rho}$, it follows immediately from Definitions V and VII and from the assumption that they are of the first kind; that is:

The relatively integral domains of the first kind $\mathfrak{G}_{n\rho}$ are characterized by the fact that in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ there exist ρ polynomials over which every arbitrary polynomial in $x_1, \ldots, x_\rho$ is algebraically integral. From this assumption the mapping domain itself results as a relatively integral domain.[104]

§ 11. Auxiliary theorem on algebraically integral dependence.

We now give a criterion for every arbitrary polynomial in $x_1, \ldots, x_\rho$ to be algebraically integral over ρ polynomials in $x_1, \ldots, x_\rho$,

$$G^*_1(x), \ldots, G^*_\rho(x). \tag{1}$$

If the polynomials (1) are, in particular, homogeneous forms, then for every special system of values which makes (1) vanish, all algebraically integrally dependent forms vanish at the same time; hence in particular also $x_1, \ldots, x_\rho$.

[103] D. Hilbert: Über die vollen Invariantensysteme, Math. Ann. 42 (1893), § 1. The auxiliary theorem stated there only for homogeneous polynomials holds likewise for inhomogeneous ones.

[104] For example, in the domain $\mathfrak{G}_{22}$ mentioned in the closing remark to § 9, the admissible specialization $x_1 = 1$ gives the two polynomials $F^* = x_2$, $G^* = 1 + x_3$, over which every arbitrary polynomial in x_2, x_3 is algebraically integral. $\mathfrak{G}_{22}$ is therefore of the first kind, indeed with $F(x)$ and $G(x)$ as integrality basis.

The forms (1), therefore, cannot vanish for any system of values other than

$$x_1 = 0, \ldots, x_\rho = 0;$$

their resultant must be different from zero.

Conversely, this condition is also sufficient and admits an extension to inhomogeneous polynomials according to the following

Auxiliary theorem: A sufficient condition that every arbitrary polynomial in $x_1, \ldots, x_\rho$ be algebraically integral over ρ polynomials in $x_1, \ldots, x_\rho$

$$G_1^*(x), \ldots, G_\rho^*(x)$$

is the non-vanishing of the resultant, taken with respect to $x_1, \ldots, x_\rho$, of the terms of highest dimension

$$H_1^*(x), \ldots, H_\rho^*(x)$$

of the polynomials (1):

$$R_0 = \begin{bmatrix} H_1^* & \cdots & H_\rho^* \\ x_1 & \cdots & x_\rho \end{bmatrix} \neq 0. \tag{2}$$

For homogeneous forms (1) the condition (2) is necessary.[105]

For the proof we form the polynomials

$$\begin{gathered} \bar{G}_1^*(x) = G_1^*(x) - \lambda_1, \ldots, \bar{G}_\rho^*(x) = G_\rho^*(x) - \lambda_\rho, \\ \bar{\Phi}^*(x) = \Phi^*(x) - \mu, \end{gathered} \tag{3}$$

where $\lambda_1, \ldots, \lambda_\rho, \mu$ denote indeterminates and $\Phi^*(x)$ denotes an arbitrary polynomial in $x_1, \ldots, x_\rho$. Let the resultant of the polynomials (3) with respect to $x_1, \ldots, x_\rho, 1$[106] be given by

$$R(\lambda, \mu) = \begin{bmatrix} \bar{G}_1^* & \cdots & \bar{G}_\rho^* & \bar{\Phi}^* \\ x_1 & \cdots & x_\rho & 1 \end{bmatrix}; \tag{4}$$

it is an integral rational function of $\lambda_1, \ldots, \lambda_\rho, \mu$. According to F. Mertens, in the expansion of (4) by powers of μ, the highest coefficient can be explicitly specified;[107] one obtains

$$(-1)^\tau R(\lambda, \mu) = R_0^\sigma \mu^\tau + A_1(\lambda)\mu^{\tau-1} + \cdots + A_\tau(\lambda), \tag{5}$$

where $A_1(\lambda), \ldots, A_\tau(\lambda)$ denote integral integer-valued functions of $\lambda_1, \ldots, \lambda_\rho$ and the coefficients of the polynomials (3). This expansion first shows that under the assumption (2), $R_0 \neq 0$, also $R(\lambda, \mu)$ cannot vanish identically. The identity in $x_1, \ldots, x_\rho, \lambda_1, \ldots, \lambda_\rho, \mu$,

$$R(\lambda, \mu) \equiv 0 \pmod{\bar{G}_1^*(x), \ldots, \bar{G}_\rho^*(x), \bar{\Phi}^*(x)},$$

passes, by the substitution

$$\lambda_i = G_i^*(x), \qquad \mu = \Phi^*(x),$$

into

$$\begin{aligned} R(G_i^*(x), \Phi^*(x)) = R_0^\sigma \Phi^{*\tau}(x) + A_1(G_i^*(x))\Phi^{*\tau-1}(x) + \cdots \\ + A_\tau(G_i(x)) = 0, \end{aligned} \tag{6}$$

and this proves the auxiliary theorem, since R_0 is, by assumption, a number from Ω.

[105] For resultant theory, compare F. Mertens, Zur Theorie der Elimination (I. and II. Teil), Sitzungsber. d. k. Ak. d. Wiss. Wien, Math.-naturw. Kl. Abt. IIa, Bd. 108 (1899). Partly reproduced in J. König, Theorie d. algebr. Größen, Kap. VI.

[106] That is, the homogeneous resultant with respect to $x_1, \ldots, x_\rho, x_0$, if the polynomials (3) are homogenized by means of an additional indeterminate x_0 so that the degree in the x is preserved.

[107] A. a. O. § 3 (proof of Theorem I), and explicitly § 6 (determination of the totality of all terms of the resultant R which contain a given power product $a_{\lambda\lambda}^{p_\lambda} a_{\mu\mu}^{p_\mu} \cdots a_{\sigma\sigma}^{p_\sigma}$ as factor). In our case $p_\lambda = 0$, $p_\mu = 0, \ldots, p_\rho = \tau$, $a_{\sigma\sigma} = (-\mu)$ is set. Reproduced in König, Kap. VI, § 9.

§ 12. The integrality basis of regular systems $\mathfrak{S}_{n\rho}$ consisting of polynomials.

The auxiliary theorem of § 11, together with the conclusion of § 10, immediately yields the fact:

If there exist, in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of $\mathfrak{G}_{n\rho}$, ρ polynomials such that the homogeneous resultant of their terms of highest dimension does not vanish identically — $R_0 \neq 0$ —, then $\mathfrak{G}_{n\rho}$ is of the first kind and consequently is a finite integrality domain, with the involution basis as integrality basis.

The condition $R_0 \neq 0$ now permits, by means of the auxiliary theorem, a second existence proof for the integrality basis, essentially due to D. Hilbert;[108] this proof, to be sure, does not make the involution basis recognizable as such, but it holds uniformly for all systems $\mathfrak{S}_{n\rho}$ with the indicated property.

Indeed, let the ρ polynomials from $\mathfrak{J}^*_{\rho\rho}$ for which $R_0 \neq 0$, and which consequently are algebraically independent, be given by

$$G_1^*(x), \ldots, G_\rho^*(x). \tag{1}$$

By the auxiliary theorem, every arbitrary polynomial in $x_1, \ldots, x_\rho$, hence in particular every polynomial of the smallest field $\mathfrak{K}^*_{\rho\rho}$ containing $\mathfrak{J}^*_{\rho\rho}$, is algebraically integral over the polynomials (1). If the corresponding polynomials in $\mathfrak{S}_{n\rho}$ are therefore

$$G_1(x), \ldots, G_\rho(x), \tag{2}$$

then, by the transfer principle, every polynomial of the smallest field $\mathfrak{K}_{n\rho}$ containing $\mathfrak{S}_{n\rho}$, and hence in particular every polynomial from $\mathfrak{S}_{n\rho}$, is algebraically integral over the polynomials (2). Conversely, every function from $\mathfrak{K}_{n\rho}$ which is algebraically integral over the polynomials (2) is also a polynomial. Thus, if one regards $\mathfrak{K}_{n\rho}$ (according to § 4) as an algebraic field over $\Omega(G_1, \ldots, G_\rho)$, the totality of the polynomials from $\mathfrak{K}_{n\rho}$ — the relatively integral domain $\mathfrak{G}_{n\rho}$ — is identical with the algebraically integral quantities of this field. Let $G(x)$ be a polynomial from $\mathfrak{K}_{n\rho}$ which completes (2) to a rational basis, and let $D(x)$ be the discriminant of the irreducible equation of degree k which $G(x)$ satisfies with respect to $\Omega(G_1, \ldots, G_\rho)$. Then every polynomial from $\mathfrak{K}_{n\rho}$, and in particular every polynomial $F(x)$ from $\mathfrak{S}_{n\rho}$, as an algebraically integral quantity, admits the representation

$$F(x) = \frac{\Gamma_1(G_i)G(x)^{k-1} + \Gamma_2(G_i)G(x)^{k-2} + \cdots + \Gamma_k(G_i)}{D(x)}. \tag{3}$$

Let $F(x)$ now run through all polynomials from $\mathfrak{S}_{n\rho}$. Then the application of Hilbert's Theorem I on the module basis (Math. Ann. 36) to the system formed from the corresponding functions

$$v_1\Gamma_1(G_i) + v_2\Gamma_2(G_i) + \cdots + v_k\Gamma_k(G_i)$$

shows the existence of a finite number of polynomials from $\mathfrak{S}_{n\rho}$,

$$F_1(x), F_2(x), \ldots, F_\lambda(x), \tag{4}$$

such that every polynomial from $\mathfrak{S}_{n\rho}$ admits a representation

$$F(x) = A_1(G_i)F_1(x) + A_2(G_i)F_2(x) + \cdots + A_\lambda(G_i)F_\lambda(x), \tag{5}$$

where $A_1, \ldots, A_\lambda$ denote polynomials in the $G_i(x)$. The polynomials (2) and (4) therefore form an integrality basis of $\mathfrak{S}_{n\rho}$; that is:

Theorem VIII: If there exist, in a mapping system $\mathfrak{J}^*_{\rho\rho}$ of the system of polynomials $\mathfrak{S}_{n\rho}$, ρ polynomials such that the resultant of the terms of highest dimension does not vanish identically — $R_0 \neq 0$ —, then $\mathfrak{S}_{n\rho}$ possesses an integrality basis.

This theorem allows a slight generalization. It is enough that the ρ polynomials (1) for which $R_0 \neq 0$ belong to the integrality domain $\mathfrak{J}^*_{\rho\rho}$ which is the smallest containing domain of $\mathfrak{S}^*_{\rho\rho}$. In fact, the arguments leading to Theorem VIII show that the representation (5) still holds then. But by the definition, every polynomial $L(x)$ of the smallest containing integrality domain $\mathfrak{J}_{n\rho}$ can be represented integrally and rationally by a finite number — depending on $L(x)$ — of polynomials from $\mathfrak{S}_{n\rho}$; hence there exist σ polynomials from $\mathfrak{S}_{n\rho}$,

$$K_1(x), K_2(x), \ldots, K_\sigma(x), \tag{6}$$

[108] Über die vollen Invariantensysteme, § 2. In Hilbert's case, under the assumption of the otherwise obtained existence of the integrality basis of the invariant field, the point is only an interpretation of this basis through the Kronecker fundamental system of the algebraically integral quantities of a field.

such that the polynomials $G_i(x)$ become integral rational combinations of (6). Thus (6) and (4) give an integrality basis. Introducing the following terminology:

Definition VIII: A system $\mathfrak{S}_{n\rho}$ of polynomials is called regular if in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of the smallest containing integrality domain there exist ρ polynomials such that the resultant of their terms of highest dimension does not vanish identically — $R_0 \neq 0$ —

we therefore have:

Theorem IX: Every regular system $\mathfrak{S}_{n\rho}$ of polynomials possesses an integrality basis.[109]

We further give, analogously to § 5 for the involution, a geometric interpretation of regular systems. For this purpose we consider, as there in (7), the system of equations

$$L^*(x) - L^*(\xi) = 0, \tag{7}$$

where $L^*(x)$ runs through all polynomials from $\mathfrak{J}^*_{\rho\rho}$. Homogenizing by means of x_0, ξ_0, let (7) pass into the system of equations between homogeneous forms

$$\xi_0^m M^*(x) - x_0^m M^*(\xi) = 0, \tag{8}$$

where, if $H^*(x)$ denotes the terms of highest dimension of $L^*(x)$, the relation

$$H^*(x) = M^*(x) \pmod{x_0} \tag{9}$$

holds.

As *fundamental points* of $\mathfrak{J}^*_{\rho\rho}$ (compare the note to § 5), we designate the solutions of (8) that are independent of x — thus different from the rows conjugate to x. Fundamental points are therefore the systems of values different from $\xi_1 = 0, \dots, \xi_\rho = 0$ for which simultaneously

$$\xi_0^m = 0, \qquad M^*(\xi) = 0$$

holds, or, by (9),

$$\xi_0 = 0, \qquad H^*(\xi) = 0.$$

If now there exist ρ forms $H^*(\xi)$ for which $R_0 \neq 0$, hence with no common system of solutions, then $\mathfrak{J}^*_{\rho\rho}$ can possess no fundamental point. If, in particular, $\mathfrak{S}^*_{\rho\rho}$ consists of homogeneous forms, then for $R_0 \neq 0$ also $\mathfrak{S}^*_{\rho\rho}$ can possess no fundamental point, since such a point would at the same time be a fundamental point of $\mathfrak{J}^*_{\rho\rho}$. But the converse also holds. For this we consider the system $\mathfrak{J}(H^*)$ formed from the totality of the forms $H^*(x)$; this is again an integrality domain.

From Hilbert's Theorem I on the module basis and Hilbert's auxiliary theorem (compare the citation in § 10), there follows the existence of ρ forms from $\mathfrak{J}(H^*)$ whose vanishing has the vanishing of all forms from $\mathfrak{J}(H^*)$ as consequence. If now $\mathfrak{J}^*_{\rho\rho}$ possesses no fundamental point, these ρ forms cannot vanish simultaneously; their resultant $R_0 \neq 0$.

Thus:

The regular systems $\mathfrak{S}_{n\rho}$ of polynomials are characterized by the fact that a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of the smallest containing integrality domain possesses no fundamental point. For regular systems $\mathfrak{S}^*_{\rho\rho}$ of homogeneous forms, the system itself already possesses no fundamental point.[110]

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[109] That there are non-regular systems without integrality basis was shown by Hilbert with an example (Gött. Nachr. 1891, p. 233). The following is even simpler:

$$\mathfrak{S}_{22} = x_1, x_1x_2, x_1x_2^2, \dots, x_1x_2^\nu, \dots \quad \text{in inf.}$$

The existence of the integrality basis here would be equivalent to the existence of a positive integer ν such that the identities

$$x_1x_2^{\nu+\sigma} = \sum C \cdot x_1^{i_0}(x_1x_2)^{i_1}(x_1x_2^2)^{i_2}\cdots(x_1x_2^\nu)^{i_\nu} \quad (\sigma = 1, 2, \dots)$$

hold. Comparing the dimensions in x_1, x_2 already for $\sigma = 1$ leads to the two conditional equations for the exponents

$$1 = i_0 + i_1 + \cdots + i_\nu, \qquad \nu + 1 = i_1 + 2i_2 + \cdots + \nu i_\nu,$$

which plainly have no solution in non-negative integers. Thus $\mathfrak{S}_{22}$ possesses no integrality basis.

[110] The non-regular system $\mathfrak{S}_{22}$ given in the preceding note possesses the fundamental point

$$\xi_1 : \xi_2 : \xi_0 = 0 : 1 : 0.$$

§ 13. Examples of relatively integral domains of the first kind and of regular systems.

1. As a first example of a relatively integral domain of the first kind $\mathfrak{G}_{nn}$, we consider the *totality of the polynomials in $x_1, \ldots, x_n$ that admit a given permutation group G* — thus the totality of the polynomials of a Lagrange genus domain. Since $x_1, \ldots, x_n$, by virtue of the identity

$$x_k^n - \sigma_1(x) x_k^{n-1} + \sigma_2(x) x_k^{n-2} + \cdots \pm \sigma_n(x) = 0 \qquad (k = 1, 2, \ldots, n), \tag{1}$$

are algebraically integral over the elementary symmetric functions belonging to $\mathfrak{G}_{nn}$, and hence by Definition V are algebraically integral over $\mathfrak{G}_{nn}$, $\mathfrak{G}_{nn}$ is of the first kind. Moreover $\mathfrak{G}_{nn}$, as a domain of the first kind, since its elements are homogeneous forms and $n = \rho$, forms a regular system (by the conclusion of § 10). Indeed, because of (1), the resultant R_0 of the elementary symmetric functions cannot vanish identically; by the rules of calculation of resultant theory, R_0 is computed to be ± 1. The *involution form* of $\mathfrak{G}_{nn}$ is uniquely determined — since $\mathfrak{G}_{nn}$ is of the first kind and $n = \rho$ — as the involution form of the corresponding Lagrange genus domain. Since the quantities conjugate to $x_1, \ldots, x_n$ with respect to this domain arise through the permutations of the group G, this involution form is therefore given by

$$\begin{aligned} \Psi(z, u) &= \prod_i (z - u_1 x_{i_1} - u_2 x_{i_2} - \cdots - u_n x_{i_n}) \\ &= z^i + \sum{}' G_{\alpha\alpha_1\ldots\alpha_n}(x) z^\alpha u_1^{\alpha_1} \cdots u_n^{\alpha_n}, \end{aligned} \tag{2}$$

where the product is to be extended over all permutations of G. Formula (2) represents the "Galois form of the group G," and hence the following fact holds:

All polynomials in $x_1, \ldots, x_n$ which admit a permutation group G are integral rational combinations of the coefficients $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ of the Galois form of the group.

2. As an example of *regular systems*, consider the systems $\mathfrak{S}_{n1}$ of polynomials, that is, systems which contain only one algebraically independent polynomial.

Indeed, let $\mathfrak{S}^*_{11}$ be a mapping system of $\mathfrak{S}_{n1}$ and let

$$F^*(x) = a_0 x^k + a_1 x^{k-1} + \cdots + a_k \qquad (a_0 \neq 0)$$

be a polynomial from $\mathfrak{S}^*_{11}$. Then

$$R_0 = a_0 \neq 0. \tag{3}$$

Thus $\mathfrak{S}_{n1}$ is regular by Definition VIII, and consequently:

Every system $\mathfrak{S}_{n1}$ of polynomials, in particular every system of infinitely many polynomials in one indeterminate, possesses an integrality basis.[111]

3. We now specialize the systems $\mathfrak{S}_{n1}$ to relatively integral domains $\mathfrak{G}_{n1}$, which, as regular systems by § 12, are relatively integral domains of the first kind. Their integrality basis is therefore given by an involution basis, which is at the same time the involution basis of the smallest containing field $\mathfrak{K}_{n1}$. But by § 6, any function of the involution basis of $\mathfrak{K}_{n1}$ is at the same time a minimal basis — we called it the Lüroth function of the field — and every function of the involution basis depends on this Lüroth function by a fractional-linear transformation. In the present case the Lüroth function, since it belongs to $\mathfrak{G}_{n1}$, is a polynomial; every further polynomial of the involution basis therefore depends fractional-linearly, and by (3) algebraically integrally, hence integrally and linearly, on this Lüroth function $L(x)$, which thus becomes the integrality basis. The involution form of $\mathfrak{G}^*_{11}$, setting also $(u_1 = 1)$, is

$$\Psi^*(z) = L^*(z) - L^*(x),$$

from which $L(x)$ again appears as integrality basis. Thus:

The relatively integral domains $\mathfrak{G}_{n1}$ are of the first kind; their integrality basis consists of one polynomial, the Lüroth function of the smallest containing field.

4. As an example of a relatively integral domain $\mathfrak{G}_{n1}$, let us mention specifically the integral rational projective invariants of a quadratic form in s variables. In agreement with the preceding, it is well known that they can be represented as integral rational combinations of the determinant $|a_{ik}|$ of the form; and this determinant is at the same time the Lüroth function of the corresponding invariant field.

[111] The simplest possible non-regular systems without integrality basis are therefore systems $\mathfrak{S}_{22}$. That such systems actually exist is shown by Hilbert's example and by the example given in the note to § 12.

§ 14. Systems consisting of integer polynomials.

The basis theorems obtained are now to be sharpened with respect to integrality over the integers.

The coefficient domain Ω will therefore in what follows be assumed to be a (finite) algebraic number field; and let $[\Omega]$ denote the totality of the algebraic integers from Ω, which we shall briefly call whole numbers. By an integer polynomial is meant a polynomial with coefficients from $[\Omega]$; from now on $F(x), G(x), \ldots$ shall denote only integer polynomials.

In analogy with § 7, we consider the different systems made up of integer polynomials:

An arbitrary system of integer polynomials shall be denoted as an integer system $\mathfrak{S}_{n\rho}$.

The integer linear family $\mathfrak{L}_{n\rho}$ is an integer system which, along with $F_1(x)$ and $F_2(x)$, always contains $c_1F_1(x) + c_2F_2(x)$, where c_1, c_2 are arbitrary whole numbers from $[\Omega]$.

The integer integrality domain $\mathfrak{J}_{n\rho}$ is an integer linear family which, along with $F(x)$ and $G(x)$, always contains $F(x) \cdot G(x)$.

The integer relatively integral domain $\mathfrak{G}_{n\rho}$ is an integrality domain which, along with $F(x)$ and $G(x)$ — where $G(x) \neq 0$ — always and only contains the quotient $F(x) : G(x)$ when this quotient is an integer polynomial.

Correspondingly as in § 7, the smallest containing integer integrality domain $\mathfrak{J}_{n\rho}$ of an integer system $\mathfrak{S}_{n\rho}$ consists of all polynomials $H(x)$ which are integral rational, integer combinations of a finite number of polynomials from $\mathfrak{S}_{n\rho}$;

the smallest containing integer relatively integral domain $\mathfrak{G}_{n\rho}$ consists of all integer polynomials $H(x)$ which are rational combinations of a finite number of polynomials from $\mathfrak{S}_{n\rho}$.

We now pass to the transfer of the basis theorems. With respect to the rational basis, it is to be noted that the notions of field, smallest containing field, and rational basis, which rest only on "rationality," undergo no change. Furthermore, the mapping system $\mathfrak{S}^*_{\rho\rho}$ of an integer system $\mathfrak{S}_{n\rho}$ can in arbitrarily many ways again be chosen as an integer system. Since, also for the integer linear family, the arguments leading to the bound on the number of generators remain valid, one obtains:

Theorem V′: Every integer system $\mathfrak{S}_{n\rho}$ possesses a rational basis; the rational basis of an integer linear family $\mathfrak{L}_{n\rho}$ can in particular be chosen so that it contains at most $\rho + 1$ polynomials.

We now investigate the analogue of the involution basis of integrality domains of polynomials (§ 8). For this purpose the theorem on *symmetric functions of rows of quantities* must be extended with respect to *integrality over the integers* by the following

Auxiliary theorem. The integral integer symmetric functions of n rows of quantities $(yz \cdots t)$:

$$\begin{matrix} y_1 & z_1 & \cdots & t_1 \\ y_2 & z_2 & \cdots & t_2 \\ \vdots & \vdots & & \vdots \\ y_n & z_n & \cdots & t_n \end{matrix} \tag{1}$$

are integral integer functions of a finite number among them, namely of the elementary symmetric functions

$$\begin{gathered} \sigma_1(y), \sigma_2(y), \ldots, \sigma_n(y); \quad \sigma_1(z), \sigma_2(z), \ldots, \sigma_n(z); \quad \ldots; \\ \sigma_1(t), \ldots, \sigma_n(t); \end{gathered} \tag{2}$$

and of the functions

$$\chi_1(y, z, \ldots, t), \ldots, \chi_k(y, z, \ldots, t), \tag{3}$$

which are algebraically integral and integer over the functions (2).

Indeed, the n quantities $y_1, \ldots, y_n$ depend algebraically integrally and integrally over the integers on $\sigma_1(y), \ldots, \sigma_n(y)$; the corresponding assertion holds for $z_1, \ldots, z_n$ with respect to $\sigma_1(z), \ldots, \sigma_n(z)$, and so on. The field of symmetric functions of the rows of quantities (1) therefore forms an algebraic field over the field determined by the functions (2), in such a way that the algebraically integral and integer quantities of this field are identical with the integral integer symmetric functions of the rows of quantities (1). If one therefore takes (3) as Kronecker's fundamental system, the auxiliary theorem is proved.

Let now $\mathfrak{J}_{n\rho}$ be an integer integrality domain, $\mathfrak{J}^*_{\rho\rho}$ an integer mapping domain, and

$$\Psi^*(z, u) = G^*(x)z^i + \sum{}' G^*_{\alpha\alpha_1\ldots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}$$

an involution form of $\mathfrak{J}^*_{\rho\rho}$; here $G^*(x)$ may also have dimension zero, that is, be a number from $[\Omega]$. The elementary symmetric functions corresponding to the functions (2) are then given by certain ones among the quotients

$$G^*_{\alpha\alpha_1\ldots\alpha_\rho}(x) : G^*(x). \tag{4}$$

Since by the auxiliary theorem the functions (3) depend algebraically integrally and integrally over the integers on (2), the functions from $\mathfrak{K}^*_{\rho\rho}$ corresponding to (3) are given — by the extension of Gauss's theorem to polynomials with algebraically integral coefficients — by

$$K^*_\nu(x) : (G^*(x))^\sigma, \tag{5}$$

where $K^*_\nu(x)$ are integer polynomials from $\mathfrak{J}^*_{\rho\rho}$. Thus $K^*_\nu(x)$ belong to the domain if the case is that of relatively integral domains; in general they are quotients of polynomials from $\mathfrak{J}^*_{\rho\rho}$.

If now $F^*(x)$ is an arbitrary integer polynomial from $\mathfrak{J}^*_{\rho\rho}$, then one has

$$F^*(x) = \frac{1}{i}\Big(F^*(x) + F^*(x^{(1)}) + \cdots + F^*(x^{(i-1)})\Big),$$

and therefore $i \cdot F^*(x)$ becomes an integral integer symmetric function of the rows of quantities $x, x^{(1)}, \ldots, x^{(i-1)}$.

By the auxiliary theorem and the transfer principle, therefore:

Theorem VI′: For integer integrality domains $\mathfrak{J}_{n\rho}$, each involution form determines a distinguished rational basis in such a way that every polynomial $F(x)$ from $\mathfrak{J}_{n\rho}$ admits a representation

$$F(x) = \frac{\Gamma(H(x), H_1(x), \ldots, H_s(x))}{i \cdot H(x)^i},$$

where Γ denotes an integral integer function. For integer relatively integral domains $\mathfrak{G}_{n\rho}$, with integer relatively integral domain $\mathfrak{G}^*_{\rho\rho}$ as mapping domain, the denominator $H(x)$ is given by the highest coefficient of the corresponding involution form.

§ 15. Integer relatively integral domains of the first kind and regular systems.

For integer systems, the integer integrality basis takes the place of the integrality basis; that is, a finite number of integer polynomials of the system such that every integer polynomial of the system can be represented as an integral integer combination of this finite number.

The theorems on the integer integrality basis run essentially parallel to the earlier ones, and only the proofs require slight modifications.

First, Definition V (§ 9) transfers directly.

Definition V′: A quantity ξ is called algebraically integral and integer with respect to the integer relatively integral domain $\mathfrak{G}_{n\rho}$ if it satisfies some equation whose highest coefficient is a unit from $[\Omega]$ and whose remaining polynomials are from $\mathfrak{G}_{n\rho}$.

The extension of Gauss's theorem to polynomials with algebraically integral coefficients shows, as in § 9, that the definition at the irreducible equation can be obtained from this.

From Definition V′ one further obtains the transfer of Definition VII.

Definition VII′: An integer relatively integral domain $\mathfrak{G}_{n\rho}$ is called of the first kind if, as mapping domain, it possesses a relatively integral domain $\mathfrak{G}^*_{\rho\rho}$ such that every arbitrary integer polynomial in $x_1, \ldots, x_\rho$ is algebraically integral and integer over $\mathfrak{G}^*_{\rho\rho}$.

To show the finiteness of these domains of the first kind, we first note that, as in § 10, it follows from the definition that the highest coefficient of the involution form is a unit from $[\Omega]$.

By Theorem VI′ (§ 14), every polynomial $F(x)$ from $\mathfrak{G}_{n\rho}$ therefore admits a representation

$$F(x) = \frac{\Gamma(H_1(x), \ldots, H_\sigma(x))}{i}.$$

Applying here to the system formed from the integer polynomials $\Gamma(H_1, \ldots, H_\sigma)$ Hilbert's Theorem II on the integer module basis (Math. Ann. 36), originally stated for integral rational numerical coefficients, in its extension to algebraically integral polynomials,[112] the existence of the integer integrality basis follows; that is:

[112] If $w_1, \ldots, w_k$ denotes a fundamental system of the algebraic integers from Ω, and one sets

$$\Gamma(H_1, \ldots, H_\sigma) = \Gamma_1(H)w_1 + \cdots + \Gamma_k(H)w_k,$$

Theorem VII′: Every integer relatively integral domain of the first kind possesses an integer integrality basis.

We further transfer the definition of regular systems:

Definition VIII′: An integer system $\mathfrak{S}_{n\rho}$ is called integer-regular if, in a mapping domain $\mathfrak{J}^*_{\rho\rho}$ of the smallest containing integer integrality domain, there exist ρ polynomials such that the resultant of the terms of highest dimension is a unit from $[\Omega]$ — $R_0 = \varepsilon$.

In order to transfer the existence proof of the integrality basis, we note that the auxiliary theorem of § 11 admits the following sharper formulation, which follows from equations (5) and (6) in § 11 and from the fact that $A_1(\lambda), \ldots, A_\tau(\lambda)$ are integral integer functions of $\lambda_1, \ldots, \lambda_\rho$:

A sufficient condition that every arbitrary integer polynomial in $x_1, \ldots, x_\rho$ depend algebraically integrally and integrally over ρ integer polynomials is that the resultant of the terms of highest dimension of these polynomials be a unit from $[\Omega]$ — $R_0 = \varepsilon$.

From this auxiliary theorem, exactly as in § 12, for every polynomial $F(x)$ of an integer-regular system, representation (3) of § 12 follows, where now $\Gamma_1(G_i), \ldots, \Gamma_k(G_i)$ are integer polynomials in the G_i. Furthermore, as in § 12, the application of Hilbert's Theorem II (Math. Ann. 36) in its extension to algebraically integral polynomials, together with the definition of the smallest containing integer integrality domain, gives the existence of the integer integrality basis; that is:

Theorem IX′: Every integer-regular system possesses an integer integrality basis.

As an example of an integer relatively integral domain of the first kind, in analogy with example 1 in § 13, we refer to the totality $\mathfrak{G}_{nn}$ of integer polynomials of a Lagrange genus domain. That $\mathfrak{G}_{nn}$ is integer of the first kind follows from the identity

$$x_k^n - \sigma_1(x)x_k^{n-1} + \cdots \pm \sigma_n(x) = 0 \qquad (k = 1, 2, \ldots, n);$$

since the resultant of the elementary symmetric functions has the value ± 1, $\mathfrak{G}_{nn}$ is also an integer regular system.

A further example is given, by the auxiliary theorem of § 14, by the totality of the integral integer symmetric functions of n rows of quantities.

Erlangen, May 1914.

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then $\Gamma_1(H), \ldots, \Gamma_k(H)$ have integral rational numerical coefficients. Applying Theorem II to the system formed from the forms $v_1\Gamma_1(H) + \cdots + v_k\Gamma_k(H)$ directly shows the validity of the extension.

7. The finiteness theorem for the invariants of finite groups

Math. Ann. 77 (1916), pp. 89–92

In what follows an entirely elementary proof of finiteness for the invariants of finite groups is to be given — one based only on the theory of symmetric functions — and at the same time it gives an actual specification of the full system; whereas the usual proof, based on Hilbert's theorem on module bases (Ann. 36), is only an existence proof.[113]

Let the finite group $\mathfrak{H}$ consist of the h linear transformations (with non-vanishing determinant) $A_1, \ldots, A_h$, where A_k denotes the linear transformation

$$x_1^{(k)} = \sum_{\nu=1}^{n} a_{1\nu}^{(k)} x_\nu, \ldots, x_n^{(k)} = \sum_{\nu=1}^{n} a_{n\nu}^{(k)} x_\nu,$$

or, in abbreviated form, $(x^{(k)}) = A_k(x)$. The group $\mathfrak{H}$ therefore carries the row (x) with elements $x_1, \ldots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \ldots, x_n^{(k)}$. Since the identity must be contained among $A_1, \ldots, A_h$, the row (x) is also contained among the rows $(x^{(k)})$. By an integral rational (absolute) invariant of the group is meant an integral rational function of $x_1, \ldots, x_n$ which remains identically unchanged under the application of $A_1, \ldots, A_h$; for such an invariant $f(x)$ one therefore has

$$f(x) = f(x^{(1)}) = \cdots = f(x^{(h)}) = \frac{1}{h} \sum_{k=1}^{h} f(x^{(k)}). \tag{1}$$

1.

Formula (1) says that $f(x)$ is an integral rational symmetric function of the rows of quantities $(x^{(1)}), \ldots, (x^{(h)})$ — and indeed, since each summand $f(x^{(k)})$ contains only the one row $(x^{(k)})$, this is the simplest case, usually called the *one-form* case. By the known theorem on symmetric functions of rows of quantities,[114] $f(x)$ is therefore integrally and rationally representable by the elementary symmetric functions of these rows, that is, by the coefficients $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ of the "Galois resolvent"

$$\begin{aligned} \Phi(z, u) &= \prod_{k=1}^{h} (z + u_1 x_1^{(k)} + \cdots + u_n x_n^{(k)}) \\ &= z^h + \sum G_{\alpha\alpha_1\ldots\alpha_n}(x) z^\alpha u_1^{\alpha_1} \cdots u_n^{\alpha_n}, \end{aligned} \qquad \left(\begin{array}{c} \alpha + \alpha_1 + \cdots + \alpha_n = h \\ \alpha \neq h \end{array} \right),$$

where the $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ are invariants of degree $\alpha_1 + \cdots + \alpha_n$ in the x's. Thus it is proved:

The coefficients $G_{\alpha\alpha_1\ldots\alpha_n}(x)$ of the Galois resolvent form a full system of invariants of the group, in such a way that every invariant can be represented integrally and rationally by these finitely many invariants.

2.

Starting from (1), one may also make the following still more elementary consideration,[115] which at the same time leads to a second full system. Put

$$f(x) = a + b x_1^{\mu_1} \cdots x_n^{\mu_n} + \cdots + c x_1^{\nu_1} \cdots x_n^{\nu_n},$$

where $a, b, \ldots, c$ denote constants. Then by (1) one has

$$\begin{aligned} h \cdot f(x) &= h \cdot a + b \sum_{k=1}^{h} (x_1^{(k)})^{\mu_1} \cdots (x_n^{(k)})^{\mu_n} + \cdots + c \sum_{k=1}^{h} (x_1^{(k)})^{\nu_1} \cdots (x_n^{(k)})^{\nu_n} \\ &= h \cdot a + b \cdot J_{\mu_1\ldots\mu_n} + \cdots + c \cdot J_{\nu_1\ldots\nu_n}. \end{aligned}$$

Every invariant can therefore be composed integrally and linearly from the special invariants

$$J_{\mu_1\ldots\mu_n} = \sum_{k=1}^{h} (x_1^{(k)})^{\mu_1} \cdots (x_n^{(k)})^{\mu_n},$$

[113] Cf., for example, Weber, *Lehrbuch der Algebra* (2nd ed.), vol. 2, § 57.

[114] Cf. the note to 2.

[115] This amounts to a proof of the theorem on symmetric functions of rows of quantities in the above-mentioned one-form case.

and it is enough to prove the assertion for these special ones. But, apart from a numerical factor, $J_{\mu_1\dots\mu_n}$ is the coefficient of $u_1^{\mu_1}\cdots u_n^{\mu_n}$ in the expression

$$S_\mu = \sum_{k=1}^{h} (u_1 x_1^{(k)} + \cdots + u_n x_n^{(k)})^\mu,$$

where $\mu = \mu_1 + \cdots + \mu_n$, and this expression is the μ-th power sum of the h linear forms

$$\xi_1 = u_1 x_1^{(1)} + \cdots + u_n x_n^{(1)}, \dots, \xi_h = u_1 x_1^{(h)} + \cdots + u_n x_n^{(h)}.$$

It is known that the infinitely many power sums S_μ are integral rational combinations of

$$S_1, S_2, \dots, S_h,$$

whose coefficients are given by the invariants

$$J_{\mu_1\dots\mu_n} \qquad (\mu_1 + \cdots + \mu_n \leq h).$$

Thus a second full system has been obtained:

A full system of invariants of the group is given by all invariants $J_{\mu_1\dots\mu_n}$ for which $\mu_1 + \cdots + \mu_n \leq h$, where h denotes the order of the group.

The connection with 1. is established by observing that the power sums $S_1, \dots, S_h$ can be replaced by the elementary symmetric functions of $\xi_1, \dots, \xi_h$, which leads to the full system of the $G_{\alpha\alpha_1\dots\alpha_n}(x)$ given there. Both results show that all invariants can be expressed integrally and rationally by those whose degree in the x's does not exceed the order h of the group.

3.

Consequences for rational representation can be drawn from what has just been proved. Every rational absolute invariant can be represented — as is seen immediately by multiplying numerator and denominator by the conjugates of the denominator with respect to the group — as the quotient of two integral rational absolute invariants, not necessarily relatively prime. It follows that every rational absolute invariant can be expressed rationally by the coefficients $G_{\alpha\alpha_1\dots\alpha_n}(x)$ of the Galois resolvent $\Phi(z,u)$. This theorem can also be proved easily in various ways without passing through the finiteness theorem; it is already formulated in Weber II, § 58.[116]

4.

Finally, it should be mentioned that the results derived here are implicitly contained in my paper "Fields and systems of rational functions";[117] more precisely, the full system given in 1. is contained in Theorems VI and VII, and a proof of rational representation independent of this finiteness theorem is contained in Theorem III.[118] In the present special case, however, the course of the proof and the result simplify considerably in comparison with the general theory. I was led to apply the general investigations to the invariants of finite groups by conversations with E. Fischer, from whom, in substance, the proof given under 2 — in the general case the full system that appears there is not rationally definable — and the note to 3 also originate.

Erlangen, May 1915.

[116]The proof given there, however, is not sound. It is only shown that the function $\Psi(t)$ in formula (7) has invariants as coefficients, but not that these invariants are rationally expressible by the coefficients of $\Phi(t)$. This gap is avoided if, in representing the $x_i^{(k)}$ by ξ_k, one uses a known differentiation procedure instead of Lagrange's interpolation formula. This is the relation obtained by differentiating the identity $\Phi(-\xi_k, u) = 0$ with respect to all u:

$$\left[\frac{\partial\Phi}{\partial u_i} - x_i^{(k)}\frac{\partial\Phi}{\partial z}\right]_{z=-\xi_k} = 0.$$

Accordingly, for each rational function $\omega(x)$ the following formula must replace formulas (7) and (8) in Weber:

$$\omega(x_1^{(k)}, \dots, x_n^{(k)}) = \omega\left(\frac{\partial\Phi/\partial u_1}{\partial\Phi/\partial z}, \dots, \frac{\partial\Phi/\partial u_n}{\partial\Phi/\partial z}\right)\Big|_{z=-\xi_k},$$

which contains only coefficients of $\Phi(z,u)$, and whose summation over all k, for invariants $\omega(x)$, gives the desired representation.

[117]Math. Ann. 76, p. 161 (1915).

[118]For relative invariants, Theorems VIII and IX contain a proof of finiteness which is likewise based on a different foundation from the usual one, but is also only an existence proof; the reason is that the relative invariants do not form a field.

8. On integral rational representation of the invariants of a system of arbitrarily many ground forms

Math. Ann. 77 (1916), pp. 93–102

At the end of his contribution to the Schwarz Festschrift, “On the invariants of a system of arbitrarily many ground forms”, D. Hilbert states the conjecture that these invariants can be represented integrally and rationally by the finitely many invariants of the system (J, PJ), where J denotes the full system of invariants of the linearly independent ground forms, and the invariants PJ arise from J by polar processes.

I show below, in I, that this conjecture is in fact correct, and indeed as a simple consequence of the reduction theorem that every form in arbitrarily many rows of n variables each can be represented as a sum of polars of forms which contain only n fixed rows and which themselves are derived from the original form by polar processes. This reduction theorem is a special case, first formulated by F. Mertens, of the generalization, given by A. Capelli, F. Mertens, and J. Deruyts, of the Clebsch–Gordan series expansion to forms in arbitrarily many rows of n variables each; this generalization is proved by formal transformation of the differentiation processes.[119]

In II I also give a more conceptual proof of the reduction theorem, by treating the question as a problem concerning the equivalence of linear families of forms. This point of view, and also an essential part of the considerations used, I take from a paper “On algebraic module systems”[120] and from unpublished theorems of E. Fischer adjoining it, whose use he kindly permitted me.

Finally, in III, I show how the corresponding considerations also lead to the most general series expansions.

I.

Let θ be a form homogeneous in each of the $N > n$ rows

$$A_1, A_2, \ldots, A_N,$$

where in general the row A_k consists of the n elements

$$A_k^{(1)}, A_k^{(2)}, \ldots, A_k^{(n)}.$$

The coefficients of θ may be indeterminates or quantities of a given domain of rationality. Let P further denote an integral rational function — with rational integer coefficients — of the polar processes

$$P_{hk} = \left(A_h \frac{\partial}{\partial A_k}\right) = A_h^{(1)} \frac{\partial}{\partial A_k^{(1)}} + A_h^{(2)} \frac{\partial}{\partial A_k^{(2)}} + \cdots + A_h^{(n)} \frac{\partial}{\partial A_k^{(n)}},$$

where the product $P_{hk} P_{ij} \theta$ is to be understood as the application of P_{hk} to $P_{ij}\theta$.

Then the reduction theorem just mentioned is given by the identity

$$\theta = \sum PZ, \tag{1}$$

where the forms Z contain only the rows

$$A_1, A_2, \ldots, A_n$$

and are derived from θ by polar processes P.

We now consider the system of ground forms (F'), consisting of the N forms of the same order and the same number of variables

$$F_1, F_2, \ldots, F_N,$$

whose coefficients are taken respectively as the N rows

$$A_1, A_2, \ldots, A_N.$$

[119] F. Mertens: “Über eine Formel der Determinantentheorie.” (Formula 5), Sitzb. d. Ak. d. Wiss. Wien, vol. 91, II. Abt. (1885), and more fully (with applications) in: “Über ganze Funktionen von m Systemen von je n Unbestimmten”. Monatsh. f. Math. u. Phys. 4th year (1893). For some of Capelli’s proofs (since 1882) cf., for example, Math. Ann. 37, or the autographed “Lezioni sulla teoria delle forme algebriche” (1902). J. Deruyts: *Essai d’une théorie générale des formes algébriques*, Mém. d. 1. Soc. Roy. des Sciences de Liège (2) 17 (1892).

[120] E. Fischer: Über algebraische Modulsysteme und lineare homogene partielle Differentialgleichungen mit konstanten Koeffizienten, § 1–4, J. f. M. 140 (1911).

Let (J) denote the full system — consisting of finitely many invariants — of $F_1, \ldots, F_n$, in such a way that every invariant of $F_1, \ldots, F_n$ can be expressed integrally and rationally by the invariants (J). Hilbert's conjecture to be proved is then that, for the simultaneous invariants S of $F_1, \ldots, F_N$, a full system is given by the finitely many invariants (J, PJ), where PJ denotes all invariants derivable from J by polar processes P.

Indeed, every simultaneous invariant S with rational integer coefficients of (F') is a form homogeneous in each of the N rows $A_1, \ldots, A_N$ — with rational integer coefficients — and identity (1) can therefore be applied to it. Since application of the polar processes P to S is known to produce invariants again, the forms Z are also invariants, namely simultaneous invariants of $F_1, \ldots, F_n$, since they contain only the rows $A_1, \ldots, A_n$; hence, by assumption, they are integral rational functions of the invariants (J). Thus identity (1) becomes

$$S = \sum PY,$$

where the Y denote power products of the invariants (J). But, by the definition of P, the actual execution of the processes P on Y consists in the successive, finitely repeated execution of the simple polar operations P_{hk}, which, by the differentiation rule for products, leads to sums of products of invariants (J) and (PJ). The same holds for the application of P_{hk} to a product from (J) and (PJ); hence PY too yields only integral rational combinations of (J, PJ). *The integral rational representability of all invariants S by (J, PJ) is thereby proved.*

II.

Before proving the reduction theorem, we recall some facts about linear families of forms. By a *linear family of forms over K* we mean a system of forms of the same dimension, complete in the sense that together with θ_1 and θ_2 it always contains $c_1\theta_1 + c_2\theta_2$, where c_1, c_2 are quantities of a prescribed domain of rationality K, and where K contains the coefficient domain of the θ's. Among these forms there is always a finite number — say ρ — linearly independent over K, while any $\rho + 1$ forms satisfy a linear relation with coefficients from K; the family has *rank ρ* with respect to K.[121] We shall use the easily seen fact that if $\mathcal{L}$ and $\mathcal{T}$ are two linear families over K of the same rank ρ, and if $\mathcal{T}$ is a subfamily of $\mathcal{L}$, then $\mathcal{L}$ and $\mathcal{T}$ are identical (equivalent).

With this in hand, we pass to the reduction theorem. The identity from I,

$$\theta = \sum PZ,$$

passes into an analogous identity for $P\theta$ when one applies the same polar process P to both sides; thus the reduction theorem at the same time gives a representation of all polars of θ by sums of the special polars PZ. Since, conversely, these special polars are surely contained among all polars, the reduction theorem asserts the equivalence of these two linear families of forms; in particular, it also asserts the equivalence of those subfamilies whose forms have, in each individual row, the same dimension as θ. To prove this equivalence we insert an intermediate family determined by the forms Z. For simplicity we first give the proof for the case of three binary rows ($N = 3, n = 2$), and then briefly indicate the completely parallel general proof.

Thus let $\theta(x, y, z)$ have dimensions α, β, p respectively in the rows

$$x_1x_2; \quad y_1y_2; \quad z_1z_2;$$

and let the coefficients of θ first be indeterminates a (or rational numbers). We consider the following three linear families of forms.

1. The family $\mathcal{L}$, consisting of all forms $f(x, y, z)$ which arise from θ by polar processes P and which, in the individual rows x, y, z, have the same dimension as θ; $\mathcal{L}$ contains θ in particular. If $f(x, y, z)$ is regarded as a form in x, y, z and the indeterminates a, then the coefficient field is the field R of rational numbers; for K as well one must take R, since only for rational integral θ_1, θ_2 does $c_1P_1 + c_2P_2$ again become a process P; let $\mathcal{L}$ have rank ρ with respect to R.

2. The family $\mathcal{S}$, consisting of all forms $g(\lambda; x, y)$ defined by

$$g(\lambda; x, y) = f(x, y, \lambda_1 x + \lambda_2 y),$$

or, expressed by polar processes,

$$\begin{aligned} g(\lambda; x, y) &= \frac{1}{p!}\left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right]^p f(x, y, z) \\ &= g_0(xy)\lambda_1^p + g_1(x, y)\lambda_1^{p-1}\lambda_2 + \cdots + g_p(xy)\lambda_2^p, \end{aligned}$$

[121] More general linear families of forms — whose rank need not be finite — are obtained by dropping the restriction of equal dimension, or also by dropping the restriction that K contain the coefficient domain of the θ's.

where

$$i!(p-i)!\, g_i(xy) = \left(y\frac{\partial}{\partial z}\right)^i \left(x\frac{\partial}{\partial z}\right)^{p-i} f(xyz).$$

[122] Thus the $g_i(xy)$ give forms Z of identity (1). The g are rational-integer forms in x, y, λ and the indeterminates a; let $\mathcal{S}$ have rank σ with respect to R.

3. The family $\mathcal{T}$, obtained from $\mathcal{S}$ by the inverse polar process just as $\mathcal{S}$ is obtained from $\mathcal{L}$; the forms h of $\mathcal{T}$ are defined by

$$\begin{aligned} h(x,y,z) &= \frac{1}{p!}\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right]^p g(\lambda; xy) \\ &= h_0(xyz) + h_1(xyz) + \cdots + h_p(xyz), \end{aligned}$$

where, by 2.,

$$h_i(xyz) = \left(z\frac{\partial}{\partial x}\right)^{p-i} \left(z\frac{\partial}{\partial y}\right)^i g_i(xy).$$

The individual h_i, and consequently also h, are therefore forms PZ and sums of them; let $\mathcal{T}$ have rank τ with respect to R.

As the construction of the forms $h(x,y,z)$ of $\mathcal{T}$ shows, these forms arise from θ by polar processes P, and in the rows x, y, z have individually the same dimension as θ; hence the family $\mathcal{T}$ is a subfamily of $\mathcal{L}$. By the fact mentioned above, it remains only to prove equality of the ranks. In this proof no further use is made of the special structure of the $f(xyz)$ — namely, that they are polars of one and the same form θ.

a) For comparing the ranks of $\mathcal{L}$ and $\mathcal{S}$ we use Lemma a): from $f(x,y,\lambda_1 x+\lambda_2 y) = 0$ it follows necessarily that $f(x,y,z) = 0$. This follows immediately from the identical relation between three binary rows

$$(xy)z + (yz)x + (zx)y = 0, \qquad (xy) = (x_1y_2 - x_2y_1), \ldots,$$

which implies

$$f[x, y, (xy)x + (xz)y] = (xy)^p f(x,y,z).$$

Thus from $f(x,y,\lambda_1 x+\lambda_2 y) = 0$ (identically in λ) it follows, as is shown by the substitution $\lambda_1 = (zy)$, $\lambda_2 = (xz)$, and since $(xy) \neq 0$, that necessarily $f(x,y,z) = 0$.

By this lemma there is a one-to-one correspondence between the forms of $\mathcal{S}$ and those of $\mathcal{L}$. For from

$$f_1(x,y,\lambda_1 x+\lambda_2 y) = g(\lambda; xy), \qquad f_2(x,y,\lambda_1 x+\lambda_2 y) = g(\lambda; xy)$$

it follows necessarily that

$$f_1(xyz) = f_2(xyz).$$

Furthermore, every relation in $\mathcal{S}$ is preserved between the uniquely corresponding forms in $\mathcal{L}$ (and conversely). For from

$$c_1 g^{(1)}(\lambda; xy) + c_2 g^{(2)}(\lambda; xy) + \cdots + c_r g^{(r)}(\lambda; xy) = 0$$

it follows by the lemma that necessarily

$$c_1 f^{(1)}(x,y,z) + c_2 f^{(2)}(x,y,z) + \cdots + c_r f^{(r)}(x,y,z) = 0.$$

This proves equality of the ranks of $\mathcal{L}$ and $\mathcal{S}$; $\rho = \sigma$.

b) For comparing the ranks of $\mathcal{S}$ and $\mathcal{T}$ we use the corresponding Lemma b): from $h(x,y,z) = 0$ it follows necessarily that $g(\lambda; xy) = 0$. For the proof, observe that by 3. $h(x,y,z) = 0$ is equivalent to the vanishing of the individual power products

$$\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_1} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_1}\right)^{p_1} \left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_2} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_2}\right)^{p_2} g(\lambda; x, y), \qquad p_1 + p_2 = p.$$

[122] As in I,

$$\left(t\frac{\partial}{\partial x}\right) = t_1\frac{\partial}{\partial x_1} + t_2\frac{\partial}{\partial x_2},$$

and hence in particular

$$\left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right] = (\lambda_1 x_1 + \lambda_2 y_1)\frac{\partial}{\partial z_1} + (\lambda_1 x_2 + \lambda_2 y_2)\frac{\partial}{\partial z_2},$$

$$\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right] = z_1\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_1} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_1}\right) + z_2\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_2} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_2}\right).$$

Further differentiation then also yields the vanishing of the power products

$$\left(\frac{\partial}{\partial x_1}\right)^{\alpha_1}\left(\frac{\partial}{\partial x_2}\right)^{\alpha_2}\left(\frac{\partial}{\partial y_1}\right)^{\beta_1}\left(\frac{\partial}{\partial y_2}\right)^{\beta_2}$$
$$\cdot\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_1}+\frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_1}\right)^{p_1}\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_2}+\frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_2}\right)^{p_2} g(\lambda; xy).$$

Consequently every linear combination of these power products vanishes, and hence every form

$$f\left(\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x}+\frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right) g(\lambda, xy).$$

Choosing f in particular so that

$$f(x, y, \lambda_1 x+\lambda_2 y) = g(\lambda; xy),$$

one also obtains

$$g\left(\frac{\partial}{\partial x};\frac{\partial}{\partial x},\frac{\partial}{\partial y}\right) g(\lambda, x, y) = 0;$$

or, if

$$g(\lambda, xy) = \sum G_i \lambda_1^{\nu_1}\lambda_2^{\nu_2}x_1^{\alpha_1}x_2^{\alpha_2}y_1^{\beta_1}y_2^{\beta_2},$$

where the G_i are rational-integer linear forms in the indeterminates a, then

$$\sum \nu_1!\nu_2!\alpha_1!\alpha_2!\beta_1!\beta_2!\, G_i^2 = 0,$$

and hence $g(\lambda; xy) = 0$.

It follows from Lemma b), exactly as in a), that *the ranks of $\mathcal{S}$ and $\mathcal{T}$ are equal; $\sigma=\tau$.*

Thus by a) and b) one has $\rho=\tau$, and consequently — since $\mathcal{T}$ is a subfamily of $\mathcal{L}$ — the equivalence of these two linear families is proved. Hence every form of $\mathcal{L}$, in particular also θ, can be represented as a linear rational-integer combination of ρ linearly independent forms $h(x, y, z)$:

$$\theta = \sum c_i h^{(i)}(x, y, z) = \sum PZ.$$

This identity, derived for indeterminate coefficients a of θ, remains valid when the indeterminates a are replaced by quantities of any domain of rationality; hence *the reduction theorem is generally proved for the case of three binary rows.*

The general proof, for N rows of n variables each, is completely parallel. Let $\theta(A)$ have dimension α_i in the row A_i. We again consider the three linear families of forms:

1. the family $\mathcal{L}$, consisting of all forms $f(A)$ which arise from $\theta(A)$ by polar processes P and, in each individual row A_i, have the same dimension as $\theta(A)$;
2. the family $\mathcal{S}$, consisting of all forms $g(\lambda; A_1\cdots A_n)$ which arise from $f(A)$ by the substitutions

$$A_{n+1} = \lambda_{n+1}^{(1)}A_1+\lambda_{n+1}^{(2)}A_2+\cdots+\lambda_{n+1}^{(n)}A_n,$$
$$\vdots$$
$$A_N = \lambda_N^{(1)}A_1+\lambda_N^{(2)}A_2+\cdots+\lambda_N^{(n)}A_n;$$

in this process, the coefficients of the individual power products of the λ's in $g(\lambda; A_1\cdots A_n)$ give forms Z of identity (1);

3. the family $\mathcal{T}$, consisting of all forms $h(A)$ defined by

$$h(A) = \left[A_{n+1}\left(\frac{\partial}{\partial\lambda_{n+1}^{(1)}}\frac{\partial}{\partial A_1}+\cdots+\frac{\partial}{\partial\lambda_{n+1}^{(n)}}\frac{\partial}{\partial A_n}\right)\right]^{\alpha_{n+1}}$$
$$\cdots\left[A_N\left(\frac{\partial}{\partial\lambda_N^{(1)}}\frac{\partial}{\partial A_1}+\cdots+\frac{\partial}{\partial\lambda_N^{(n)}}\frac{\partial}{\partial A_n}\right)\right]^{\alpha_N} g(\lambda; A_1\cdots A_n).$$

Here $h(A)$ is a sum of forms PZ.

Because of the identical relation between any $n+1$ rows, Lemma a) remains valid; Lemma b) likewise remains valid, since the number of variables did not enter into its proof. Thus $\mathcal{L}$ and $\mathcal{T}$ have the same rank and — since $\mathcal{T}$ is a subfamily of $\mathcal{L}$ — are identical. Consequently the identity

$$\theta = \sum PZ$$

holds for indeterminate coefficients, and hence for coefficients that are quantities of any domain of rationality; *the reduction theorem is thereby proved in general.*

III.

We still indicate briefly how analogous considerations also yield the *general series expansion* (for $N \leq n$). Let Z be a separately homogeneous form in the n rows $A_1, \ldots, A_n$ of n quantities each; and let Δ be the determinant of these rows. We first prove the identity corresponding to (1):

$$Z = \sum PH \pmod{\Delta}, \tag{2}$$

where the forms H contain only the rows $A_1, \ldots, A_{n-1}$ and are derived from Z by processes P.

The proof consists in converting the relations derived in II into congruences modulo Δ. For simplicity take three ternary rows x, y, z; the coefficients are assumed to be indeterminates.

The three linear families $\mathcal{L}, \mathcal{S}, \mathcal{T}$ are defined as in II for the case of binary rows; let their ranks be ρ, σ, τ, while ρ^* and τ^* denote the ranks of $\mathcal{L}$ and $\mathcal{T}$ modulo Δ (that is, there are ρ^*, but not $\rho^* + 1$, forms of $\mathcal{L}$ that are linearly independent mod Δ). Then, as in II, one has

Lemma a): from $f(x, y, \lambda_1 x + \lambda_2 y) = 0$ *it follows necessarily that* $f(xyz) \equiv 0 \pmod{\Delta}$ *(and conversely).* For, by the relation between four ternary rows,

$$\Delta_1 x - \Delta_2 y + \Delta_3 z = \Delta t, \qquad \Delta_1 = (yzt), \ldots,$$

there is always a linear relation modulo Δ between three ternary rows:

$$\Delta_1 x - \Delta_2 y + \Delta_3 z \equiv 0 \pmod{\Delta},$$

and hence, as in II,

$$f(x, y, -\Delta_1 x + \Delta_2 y) = \Delta_3^p f(x, y, z) \pmod{\Delta}.$$

It follows from $f(x, y, \lambda_1 x + \lambda_2 y) = 0$ (identically in λ), since Δ is irreducible and Δ_3 is not divisible by Δ, that necessarily $f(x, y, z) \equiv 0 \pmod{\Delta}$. The converse is clear from $\Delta(x, y, \lambda_1 x + \lambda_2 y) = 0$. Hence, as in II, $\rho^* = \sigma$.

Lemma b) remains valid with its proof, and one has therefore $\sigma = \tau$.

For proving $\tau = \tau^*$ we use

Lemma c): from $h(x, y, z) \equiv 0 \pmod{\Delta}$ *it follows necessarily that* $h(x, y, z) = 0$. Namely, as a polar, $h(x, y, z)$ vanishes under the application of the known Ω-process; this is expressed by the differential equation

$$\Delta\Big(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\Big)\, h(x, y, z) = 0.$$

If now $h(x, y, z) = \Delta(x, y, z) \cdot k(x, y, z)$, then by further differentiation one also has the vanishing of

$$k\Big(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\Big)\, \Delta\Big(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\Big) = h\Big(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\Big)\, h(x, y, z),$$

and hence of $h(x, y, z)$. Thus $\rho^* = \sigma = \tau = \tau^*$; since $\mathcal{T}$ is a subfamily of $\mathcal{L}$, identity (2) is proved. The same argument gives the proof for n variables.[123]

Applying identity (2) to the multiplier of Δ, one obtains an expansion

$$Z = \varphi_0 + \Delta\varphi_1 + \Delta^2\varphi_2 + \cdots + \Delta^\mu \varphi_\mu,$$

where the φ_i each vanish under application of the Ω-process. The i-fold repetition of the process gives, since in general $\Omega(\Delta \cdot \psi) = c_1\psi + c_2\Delta \cdot \Omega(\psi)$,

$$c \cdot \Omega^i(Z) \equiv \varphi_i \pmod{\Delta};$$

on the other hand, by (2) one has

$$c \cdot \Omega^i(Z) \equiv \sum PH_i \pmod{\Delta},$$

[123] After 3. one has the corresponding relation, vanishing because of the determinant factor, for $\Omega(h)$; in operator notation it is the equation arising from products of

$$\Big(\lambda_1 \frac{\partial}{\partial \xi} + \lambda_2 \frac{\partial}{\partial \eta}\Big), \qquad \Big[z\Big(\frac{\partial}{\partial \lambda_1}\frac{\partial}{\partial \xi} + \frac{\partial}{\partial \lambda_2}\frac{\partial}{\partial \eta}\Big)\Big],$$

and it vanishes because of the vanishing of the determinant factor.

where the H_i are derived from the form $\Omega^i(Z)$ in the same way as H is derived from Z. By Lemma c) (extended to n variables) one therefore gets $\varphi_i = \sum PH_i$, and consequently

$$Z = \sum PH + \Delta \cdot \sum PH_1 + \Delta^2 \cdot \sum PH_2 + \cdots + \Delta^\mu \cdot \sum PH_\mu \tag{3}$$

as the most general series expansion of a form in n rows of n variables each, according to polars of forms in only $n-1$ rows.

The series expansion for forms in N rows of n variables ($N < n$) is then obtained, as is known, by a simple substitution. We put (for $N = 3$, $n > 3$)

$$u = \xi_1 x + \xi_2 y + \xi_3 z, \quad v = \eta_1 x + \eta_2 y + \eta_3 z, \quad w = \zeta_1 x + \zeta_2 y + \zeta_3 z,$$

and expand $H(u, v, w) = Z(\xi, \eta, \zeta)$ according to (3). Then, since ξ, η, ζ occur only in the combinations u, v, w, one has

$$\left(\eta \frac{\partial}{\partial \xi}\right) = \left(v \frac{\partial}{\partial u}\right) \cdots, \qquad \Omega = \sum_{ikl} \left(\frac{\partial}{\partial u} \frac{\partial}{\partial v} \frac{\partial}{\partial w}\right)_{ikl} (xyz)_{ikl} = \nabla_{uvw}(xyz).$$

Under the specialization $\xi_1 = \eta_2 = \zeta_3 = 1$; $\xi_2 = \xi_3 = \cdots = \zeta_2 = 0$, $H(u, v, w)$ goes over into $H(x, y, z)$, and $\nabla_{uvw}(xyz)$ into $\nabla_{xyz}(xyz) = \nabla_{xyz}$ up to a numerical factor, since the application of $(y\partial/\partial x)$ etc. to the determinants $(xyz)_{ikl}$ makes them vanish. Since the corresponding assertion holds for N rows, (3) gives the expansion

$$H = \sum P\Phi + \sum P\Phi_1 + \cdots + \sum P\Phi_\mu, \tag{4}$$

where the Φ_i arise from $\nabla^i_{A_1 \ldots A_N} H$ by processes P and contain only the rows $A_1, \ldots, A_{N-1}$, apart from the determinant combinations occurring in ∇^i, which, as mentioned, do not come into consideration under polarization.

If these determinants in ∇^i are then replaced by indeterminates p, the resulting forms can again be expanded according to (4); in this way one finally obtains an expansion

$$H = \left[\sum P\psi(p)\right]_{p=\text{determinants of } A_1 \cdots A_N}, \tag{5}$$

where the ψ arise from the original form by the processes P and $\nabla_{A_1 \ldots A_N}(p)$, $\nabla_{A_1 \ldots A_{N-1}}(p)$, and so on. With these expansions and their combinations — for example by applying (3), and then (4) and (5), to the forms Z occurring in (1) — all known expansions for forms in arbitrarily many rows of n cogredient variables each are exhausted.

Erlangen, 5 January 1915.

9. The Most General Domains from Integral Transcendental Numbers

Math. Ann. 77 (1916), pp. 103–128

By means of the well-ordering theorem, E. Zermelo constructed a domain of "integral transcendental numbers," that is, a numerical domain $\mathfrak{G}$ having the following abstract properties:

I. The sum, difference, and product of two numbers from $\mathfrak{G}$ is again a number from $\mathfrak{G}$.

II. Every real or complex number is the quotient of two numbers from $\mathfrak{G}$.

III. Every integral rational (respectively algebraic) number is an element of $\mathfrak{G}$.

IV. No non-integral rational (respectively algebraic) number belongs to $\mathfrak{G}$.[124]

The construction rests on the fact that the well-ordering theorem implies the existence of an *algebraic basis* of all numbers: namely, a *system H of numbers* η among which no algebraic relations exist, but by means of which all remaining numbers can be expressed algebraically. Because of their algebraic independence, these basis numbers η can be regarded as indeterminates, and the desired domain therefore becomes an integral domain of rational and algebraic functions of indeterminates η,[125] whose coefficients belong to the field K of all algebraic numbers. The special domain $\mathfrak{G}_\eta$ constructed by Zermelo is then characterized by the following basis properties:

$\mathfrak{G}_\eta$ contains:

1) all basis numbers η,

2) all polynomials in the η with integral[126] coefficients,

3) all integral algebraic functions of the η with integral coefficients.

$\mathfrak{G}_\eta$ excludes:

4) all rationally fractional functions of the η with integral coefficients,

5) all algebraically fractional functions of the η with integral coefficients.[127]

It is now easy to see (compare the examples in §§ 2 and 3) that there are domains $\mathfrak{G}$ essentially different from the Zermelo domain $\mathfrak{G}_\eta$: that is, domains $\mathfrak{G}$ for which, under no well-ordering, do all basis properties 1)–5) hold, and which consequently cannot be generated by Zermelo's construction under any well-ordering. In other words, there are domains $\mathfrak{G}$ that cannot be mapped one-to-one and isomorphically onto $\mathfrak{G}_\eta$. Thus the question arises of the most general domains from integral transcendental numbers; it will be answered completely in what follows.

For this purpose one must first determine which of the basis properties are consequences of the abstract defining properties and which are caused by the special construction. It is shown (§ 1) that for every prescribed domain $\mathfrak{G}$ there is a well-ordering such that basis properties 1) and 2) are satisfied; hence every domain $\mathfrak{G}$ can be constructed by means of an algebraic basis of all numbers. None of the further basis properties need be satisfied (§§ 2 and 3). In particular it follows that a further abstract property of the Zermelo domain does not belong to all domains $\mathfrak{G}$: algebraic integral closedness. It can be formulated as follows:

V. Every number that depends algebraically-integrally and integrally on finitely many numbers from $\mathfrak{G}$ belongs to $\mathfrak{G}$.

The special domains for which V holds in addition to I–IV will be called domains $\mathfrak{H}$ from algebraically-integral transcendental numbers. For the most general domains $\mathfrak{H}$ one obtains the simple results (§ 4):

[124] E. Zermelo, Über ganze transzendente Zahlen, Math. Ann. 75, p. 434 (1914).

[125] For this reason the considerations of the present paper partly run parallel to those of the author's earlier paper: Körper und Systeme rationaler Funktionen, Math. Ann. 76, p. 161 (1915).

[126] Throughout, "integral" is to mean that the coefficients belong to the integral domain $[K]$ of all algebraic integers. For the definition of $\mathfrak{G}_\eta$, it would also suffice in 1)–5) to consider only rational integer coefficients; for more general domains, however, the basis properties would then become less simple.

[127] Zermelo, loc. cit., § 3.

a) The intersection of all domains $\mathfrak{H}$ constructible by means of the same basis H is given by all integral algebraic functions of the basis, that is, by the Zermelo domain $\mathfrak{G}_\eta$, which is itself a domain $\mathfrak{H}$. (This also gives an abstract characterization of the Zermelo domain.)

b) Every domain $\mathfrak{H}$ arises by algebraically-integral adjunction[128] of an arbitrary system $\mathfrak{S}(\eta)$ to $\mathfrak{G}_\eta$, where $\mathfrak{S}(\eta)$ has to satisfy only the single condition that no algebraically fractional number enters the domain through the adjunction.

The results for the most general domains $\mathfrak{G}$, when an algebraic basis is taken as foundation, are less simple. Here the intersection of all domains $\mathfrak{G}$ is itself not a domain $\mathfrak{G}$, and the construction of the most general domains is consequently harder to survey (§ 5).

In order to obtain results analogous to those for the domains $\mathfrak{H}$, the algebraic basis is replaced by a *rational basis* Θ, that is, by a *system* Θ *of numbers* ϑ which permits all numbers to be expressed rationally, while under at least one well-ordering no basis number can be expressed rationally by finitely many preceding ones. This rational basis Θ must still be subject to the side condition that the integral domain $[\Theta]$ consisting of all integral polynomials in the ϑ contains no algebraically fractional number. (The construction of such a basis with the side condition is shown in § 7.) Then for the most general domains $\mathfrak{G}$ — which one could also call domains from rationally integral transcendental numbers — one again obtains the simple results (§ 6):

a) The intersection of all domains $\mathfrak{G}$ constructible by means of the same rational basis Θ is given by all integral rational functions of the basis, that is, by the domain $[\Theta]$, which is itself a domain $\mathfrak{G}$.

b) Every domain $\mathfrak{G}$ arises by rationally integral adjunction of an arbitrary system $\mathfrak{S}(\vartheta)$ to $[\Theta]$, where $\mathfrak{S}(\vartheta)$ has to satisfy only the single condition that no algebraically fractional number enters the domain through the adjunction.

To have complete analogy with the domains $\mathfrak{H}$, one would have to omit the algebraic numbers from the defining properties III and IV for the domains $\mathfrak{G}$. With this modification of III and IV, the construction of the integral elements by means of a rational basis can be transferred to an arbitrary field (§ 10), whereas Zermelo's construction presupposes the algebraic closedness of the field.

As H runs through every algebraic basis, respectively as Θ runs through every rational basis satisfying the side condition, the results a) and b) give the totality of all domains $\mathfrak{H}$ and $\mathfrak{G}$. If all isomorphic domains are collected into a class, then from this totality one can select a subset which represents the totality of all classes — at least countably infinitely many by § 9 (§ 8).

§ 1. Proof of basis properties 1) and 2) for all domains $\mathfrak{G}$.

Let $\mathfrak{G}$ be any prescribed domain from integral transcendental numbers, so that it has the abstract properties I–IV. Following the procedure applied by E. Zermelo to the field of all complex numbers, we select from $\mathfrak{G}$ an *algebraic basis* H *of all numbers from* $\mathfrak{G}$. Since by II every real or complex number can be represented as the quotient of two numbers from $\mathfrak{G}$, H is at the same time an algebraic basis of *all* numbers. Thus, for every prescribed domain $\mathfrak{G}$, the algebraic basis of all numbers can be chosen so that it belongs to $\mathfrak{G}$; hence basis property 1) is satisfied. Moreover, by III, $\mathfrak{G}$ contains the integral domain $[K]$ of all algebraic integers, and therefore, by I, also all polynomials in the basis elements η with coefficients from $[K]$, that is, all integral polynomials in the η, which form the integral domain $[H]$. Hence basis property 2) is also always satisfied; in other words, every domain $\mathfrak{G}$ can be constructed by means of an algebraic basis H of all numbers in such a way that $\mathfrak{G}$ contains the integral domain $[H]$ as a divisor.

Remark. Nevertheless, starting from a basis H, one can of course construct a domain $\mathfrak{G}$ for which basis properties 1) and 2) do not hold with respect to H itself, but do hold with respect to some other basis H'. For example, let $\mathfrak{G}'$ arise from the Zermelo domain $\mathfrak{G}_\eta$ by replacing, throughout the whole construction, some basis element η by a polynomial $f(\eta)$. Then $\mathfrak{G}'$ contains neither η nor $[H]$; but if in the basis H the basis element η is replaced by $\xi = f(\eta)$ — whereby H again goes over into an algebraic basis H' —, then basis properties 1) and 2) are satisfied for $\mathfrak{G}'$ with respect to H'.

§ 2. Exclusion of basis properties 4) and 5).

We now show that the domains $\mathfrak{G}$ need not satisfy any of the further basis properties. First 4) and 5) are excluded by replacing, in Zermelo's construction, the integral domain $[H]$ by a domain of integral rational

[128] Explanation of the expression in § 4.

“functionals.”

According to Weber,[129] a *rational functional* means every rational function with rational number coefficients in the following uniquely determined representation:

$$A(\eta_1 \cdots \eta_t) = a \cdot \frac{E_1(\eta_1 \cdots \eta_t)}{E_2(\eta_1 \cdots \eta_t)},$$

where E_1, E_2 are relatively prime primitive polynomials with rational integer coefficients, and a is a positive rational number (the absolute value of A). If, in particular, a is a *rational integer*, then A is called an *integral rational functional.* As special cases, the integral rational functionals include the rational integers and the polynomials with rational integer coefficients, while rationally fractional numbers and polynomials with rationally fractional coefficients are to be counted among the fractional rational functionals.

Let the domain $\mathfrak{G}$ now consist of the totality $\mathfrak{F}$ of all integral rational functionals of finitely many basis elements η at a time, and of all functions which depend algebraically-integrally on $\mathfrak{F}$ — that is, which satisfy some equation whose leading coefficient is unity and whose remaining coefficients are integral rational functionals.

The proof of properties I–IV for $\mathfrak{G}$ runs exactly parallel to the corresponding proof for Zermelo and will only be indicated briefly. First, II and III are clear, since $\mathfrak{G}$ contains the Zermelo domain $\mathfrak{G}_\eta$ as a divisor. By Gauss’s theorem on polynomials with rational integer coefficients, sums, differences, and products of integral rational functionals are again integral rational functionals; hence $\mathfrak{F}$ forms an integral domain, and by familiar arguments[130] $\mathfrak{G}$ also becomes an integral domain; thus I is fulfilled. Further, from the extended Gauss theorem[131] follow the two auxiliary propositions: “If z depends algebraically-integrally on $\mathfrak{F}$ by means of *some* equation, then it also does so by means of the *irreducible* equation which z satisfies with respect to the field of all rational functionals,”[132] and “the equation irreducible over the field of all rational numbers and satisfied by an algebraic number is also irreducible over the field of all rational functionals.” Thus $\mathfrak{G}$ can contain no fractional algebraic number; hence IV, and therefore all abstract properties I–IV, have been verified.

On the other hand, under no well-ordering can a basis be selected from $\mathfrak{G}$ in such a way that $\mathfrak{G}$ excludes all rationally and algebraically fractional functions of the basis elements. For let $z(\eta)$ be any function of the domain to be chosen as a basis element, hence defined by an equation

$$z^k + A_1(\eta) z^{k-1} + \cdots + A_k(\eta) = 0,$$

where

$$A_i(\eta) = a_i \cdot \frac{E_1(\eta)}{E_2(\eta)}$$

are integral rational functionals. Then the function $\frac{a_k}{z}$ belongs to $\mathfrak{G}$, and likewise $\frac{a_k}{\sqrt{z}}$, $\frac{a_k}{\sqrt[3]{z}}$, etc.[133] Hence basis properties 4) and 5) are excluded for the totality of the domains $\mathfrak{G}$.

Remark. An example in § 3 will show that $\mathfrak{G}$ may, under every well-ordering, contain rationally fractional functionals without containing a rationally or algebraically fractional number.

§ 3. Exclusion of basis property 3).

For excluding basis property 3), we use a generalized *module domain* in which the arguments of the polynomials of the domain are no longer the indeterminates, but all integral algebraic functions of the indeterminates, [134] while the indeterminates themselves take over the role of a module basis.

Let $\mathfrak{G}$ consist of *all elements*

$$z = \alpha + g_1(\eta)\eta_1 + g_2(\eta)\eta_2 + \cdots + g_t(\eta)\eta_t, \tag{1}$$

where α runs through all algebraic integers, $\eta_1 \cdots \eta_t$ through all basis elements, and for each fixed combination $\eta_1 \cdots \eta_t$, the functions $g_1(\eta) \cdots g_t(\eta)$ individually run through all integral algebraic functions of the η.[135]

[129] H. Weber, Lehrbuch der Algebra. Kleine Ausgabe §§ 96 and 97.

[130] Compare, for example, Weber, § 97, 8.

[131] Weber, § 20, 5.

[132] Compare Weber, § 97, 4.

[133] Quite analogously, every algebraic function of the η can be transformed into an element of the functional domain $\mathfrak{G}$ by multiplication by a rational integer. Thus this domain has the property that every complex number can be represented as the quotient of an element from $\mathfrak{G}$ and a rational integer, in analogy with the algebraic numbers.

[134] By “integral algebraic functions” are always meant functions with *integral* coefficients; that is, functions satisfying some equation whose leading coefficient is unity and whose remaining coefficients are polynomials in the η with algebraic-integer coefficients — polynomials from $[H]$. In particular, all polynomials from $[H]$ belong to the integral algebraic functions.

[135] The module property belongs to the elements $z - \alpha$.

It is easy to verify that properties I–IV hold for this module domain. First I holds, since the sum, difference, and product of two elements z again takes the form (1). The domain $\mathfrak{G}$ also contains all basis elements η, and besides all elements $\eta \cdot g(\eta)$, where $g(\eta)$ runs through all integral algebraic functions of the η. Hence all integral algebraic functions of the η, and therefore all algebraic functions of the η altogether, can be represented as quotients of two elements of $\mathfrak{G}$, proving II. In the representation (1), all algebraic integers are also included in particular — for $g_1 = 0 \cdots g_t = 0$ —; but z cannot be equal to a fractional algebraic number. For if z represents an algebraic number β, then from

$$\beta = \alpha + g_1(\eta)\eta_1 + \cdots + g_t(\eta)\eta_t$$

identically in the η, necessarily $\beta = \alpha$. This is shown by the specialization

$$\eta_1 = 0 \cdots \eta_t = 0,$$

under which the multipliers $g_1(\eta) \cdots g_t(\eta)$ remain finite as integral algebraic functions, since they go over into integral algebraic functions or numbers.[136] Thus conditions I–IV are satisfied for $\mathfrak{G}$.

On the other hand, under no well-ordering can a basis be selected from $\mathfrak{G}$ in such a way that all integral algebraic functions of the basis elements belong to $\mathfrak{G}$. For let $z(\eta)$ be any function of the domain to be chosen as a basis element — which therefore must actually contain the indeterminates η —. We shall show that, from a certain finite value of ν onward, depending on z, the functions

$$\xi = \sqrt[\nu]{z - \alpha}$$

can no longer belong to $\mathfrak{G}$.

Indeed, suppose that ξ belongs to $\mathfrak{G}$. Then, by (1), it is of the form

$$\xi = \gamma + h_1(\eta)\eta_{i_1} + h_2(\eta)\eta_{i_2} + \cdots + h_\tau(\eta)\eta_{i_\tau},$$

where again $h_1(\eta) \cdots h_\tau(\eta)$ are integral algebraic functions of the η, and $\eta_{i_1} \cdots \eta_{i_\tau}$ are arbitrary basis elements, which in particular may also coincide with $\eta_1 \cdots \eta_t$. First it must be shown that the algebraic integer γ is zero. Since $h_1(\eta) \cdots h_\tau(\eta)$ are integral algebraic functions, ξ becomes equal to γ for $\eta_{i_1} = 0 \cdots \eta_{i_\tau} = 0$ with arbitrary values of the remaining η; in particular, then, ξ is equal to γ for

$$\eta_{i_1} = 0 \cdots \eta_{i_\tau} = 0; \qquad \eta_1 = 0 \cdots \eta_t = 0.$$

But under this specialization the function $(z - \alpha)$ vanishes by (1), and therefore so does ξ; hence $\gamma = 0$, and one has

$$\xi = h_1(\eta)\eta_{i_1} + h_2(\eta)\eta_{i_2} + \cdots + h_\tau(\eta)\eta_{i_\tau}.$$

Raising this to the ν-th power gives

$$\xi^\nu = z - \alpha = F_\nu(\eta),$$

where $F_\nu(\eta)$ denotes a homogeneous, not identically vanishing, form of ν-th dimension in $\eta_{i_1} \cdots \eta_{i_\tau}$, whose coefficients are integral algebraic functions of the η. Now $z - \alpha$, which by (1) is an integral algebraic function of the η, satisfies an equation irreducible with respect to $[H]$:

$$(z - \alpha)^\chi + B(\eta)(z - \alpha)^{\chi - 1} + \cdots + C(\eta) = 0,$$

where the last coefficient $C(\eta)$ — the product of $(z - \alpha)$ with its conjugates — is a polynomial; let its degree be λ. On the other hand, from $z - \alpha = F_\nu(\eta)$, by multiplying $F_\nu(\eta)$ by the corresponding conjugates, one obtains

$$C(\eta) = G_{\nu\chi}(\eta),$$

where $G_{\nu\chi}(\eta)$ denotes a homogeneous form of $\nu\chi$-th dimension in $\eta_{i_1} \cdots \eta_{i_\tau}$, whose coefficients are polynomials in the η. Thus $G_{\nu\chi}$ is at least of degree $\nu\chi$ in η, or $\lambda \geq \nu\chi$; that is, the functions $\sqrt[\nu]{z - \alpha}$ for which $\nu > \lambda/\chi$ do not belong to $\mathfrak{G}$. Basis property 3) is therefore excluded for the totality of the domains $\mathfrak{G}$.

Remark. A slight generalization of the domain just constructed leads to a domain $\mathfrak{G}$ which, under every well-ordering, also contains fractional functionals. Let $\mathfrak{G}$ consist of all elements

$$z = \alpha + \gamma_1(\eta)\eta_1 + \gamma_2(\eta)\eta_2 + \cdots + \gamma_t(\eta)\eta_t,$$

where now $\gamma(\eta)$ runs through all algebraically integral, but not necessarily integral, functions of the η. The proof of properties I–IV remains the same as above. But since $\mathfrak{G}$, besides every function z, also contains $a \cdot (z - \alpha)$, where a denotes any rational or algebraically fractional number, fractional functionals also occur under every well-ordering.

[136]This more detailed proof of IV is given because of the expanded example at the end of the paragraph. Here the remark would suffice that, by construction, z is an integral algebraic function.

§ 4. The most general domains from algebraically-integral transcendental numbers.

For a more convenient formulation of what follows, introduce the expression: "an element z depends algebraically-integrally on $\mathfrak{T}$," meaning that z satisfies some equation whose leading coefficient is unity and whose remaining coefficients are integral polynomials in the elements of $\mathfrak{T}$, where $\mathfrak{T}$ denotes any prescribed domain.[137]

A domain $\mathfrak{T}$ is called *algebraically-integrally closed* if it satisfies the condition (compare the introduction):

V. Every element that depends algebraically-integrally on $\mathfrak{T}$ belongs to $\mathfrak{T}$.

We first show that the abstract property V belongs to the Zermelo domain $\mathfrak{G}_\eta$. Let an element z depend algebraically-integrally on $\mathfrak{G}_\eta$ by means of an equation $f(z) = 0$. Then multiplication of $f(z)$ by its conjugates with respect to $[H]$ shows that z also depends algebraically-integrally on $[H]$, and hence belongs to $\mathfrak{G}_\eta$. The algebraically-integral closedness of the functional domain considered in § 2 is shown in the same way.

That property V does not belong to every domain $\mathfrak{G}$ is shown by the module domain considered in § 3, which indeed does not possess basis property 3). More precisely, § 3 showed that the module domain is algebraically-integrally closed with respect to no transcendental element of the domain, whereas, by III and IV, this is of course the case with respect to every algebraic element of the domain.

This leads us first to select, from the totality of domains $\mathfrak{G}$, those which also possess the abstract property V; these are to be called "domains $\mathfrak{H}$ from algebraically-integral transcendental numbers."

Let $\mathfrak{H}$ be such a domain, and let H be an algebraic basis of all numbers from $\mathfrak{H}$; by II, H is at the same time an algebraic basis of all numbers. By III, I, and V, $\mathfrak{H}$ contains, besides H, all integral algebraic functions of the η, that is, the Zermelo domain $\mathfrak{G}_\eta$. Thus, in analogy with § 1: every domain $\mathfrak{H}$ from algebraically-integral transcendental numbers can be constructed by means of an algebraic basis of all numbers in such a way that $\mathfrak{H}$ contains the Zermelo domain $\mathfrak{G}_\eta$ as a divisor.

Now let H be any algebraic basis of all numbers, and let $\mathfrak{M}(H)$ denote the totality of all domains $\mathfrak{H}$ containing H. Every domain $\mathfrak{H}$ in $\mathfrak{M}(H)$ contains, as just proved, the Zermelo domain $\mathfrak{G}_\eta$ as a divisor; and since $\mathfrak{G}_\eta$ itself is a domain $\mathfrak{H}$, $\mathfrak{G}_\eta$ is the greatest common divisor, or intersection, of all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$. Thus $\mathfrak{G}_\eta$ has been defined abstractly. It also follows directly that $\mathfrak{M}(H)$ is closed under intersection: that is, the intersection of all domains $\mathfrak{H}$ of any subsystem of $\mathfrak{M}(H)$ again belongs to $\mathfrak{M}(H)$. For, as an intersection, I and V are satisfied; II and III follow because $\mathfrak{G}_\eta$ is a divisor; IV follows because the intersection is a divisor of $\mathfrak{H}$.

As the example of the functional domain in § 2 shows, the domains $\mathfrak{H}$ in general contain further functions besides $\mathfrak{G}_\eta$. Let $\psi(\eta)$ be such a rational or algebraic function of the η. Then, by I, $\mathfrak{H}$ also contains all integral rational combinations of ψ and finitely many functions from $\mathfrak{G}_\eta$ at a time. The integral domain so arising — consisting of all polynomials in ψ with coefficients from $\mathfrak{G}_\eta$ — will be denoted by $[\mathfrak{G}_\eta, \psi]$, and the process leading to it will be called "rationally integral adjunction of ψ to $\mathfrak{G}_\eta$." By V, $\mathfrak{H}$ further contains all functions which depend algebraically-integrally on $[\mathfrak{G}_\eta, \psi]$; the domain so arising will be denoted by $\{\mathfrak{G}_\eta, \psi\}$, and the process leading to it by "algebraically integral adjunction of ψ to $\mathfrak{G}_\eta$." The domain $\{\mathfrak{G}_\eta, \psi\}$ consists of all integral algebraic functions of ψ with coefficients from $\mathfrak{G}_\eta$, and is therefore an integral domain by familiar arguments.[138]

We show that $\{\mathfrak{G}_\eta, \psi\}$ is also algebraically-integrally closed, that is, satisfies condition V. Indeed, let z depend algebraically-integrally on $\{\mathfrak{G}_\eta, \psi\}$ by means of an equation

$$f(z; a \cdots c) = z^\nu + az^{\nu-1} + \cdots + c = 0,$$

where $a \cdots c$, as elements of $\{\mathfrak{G}_\eta, \psi\}$, satisfy equations with coefficients from $[\mathfrak{G}_\eta, \psi]$:

$$a^\lambda + A_1 a^{\lambda-1} + \cdots + A_\lambda = 0,$$
$$\vdots$$
$$c^\mu + C_1 c^{\mu-1} + \cdots + C_\mu = 0.$$

Let their roots be denoted by $a, a' \cdots a^{(\lambda-1)} \cdots c, c' \cdots c^{(\mu-1)}$. If one now forms the product over all combinations of the roots,

$$g(z) = f(z; a \cdots c) f(z; a' \cdots c') \cdots f(z; a^{(\lambda-1)} \cdots c^{(\mu-1)}),$$

[137] For an integral domain $\mathfrak{T}$, the leading coefficient is unity and the remaining coefficients are elements of $\mathfrak{T}$.

[138] Compare, for example, Weber, § 97, 6 and 7.

then $g(z)$ has unity as leading coefficient and elements of $[\mathfrak{G}_\eta, \psi]$ as its remaining coefficients. Consequently z belongs to $\{\mathfrak{G}_\eta, \psi\}$.[139]

Thus the domain $\{\mathfrak{G}_\eta, \psi\}$ has properties I and V; and, since $\mathfrak{G}_\eta$ is a divisor of the domain, it also has II and III. Whether IV is satisfied depends on the choice of ψ; for example, IV is not satisfied for $\psi = \frac{1}{n \cdot \eta}$, since then $\frac{1}{n}$ belongs to the domain. It is satisfied, however, by § 2 for $\psi = \frac{1}{\eta}$, or also for $\psi = \frac{\eta}{n}$. For in the latter case, if a function from $\{\mathfrak{G}_\eta, \psi\}$ represents an algebraic number, its value remains unchanged for $\eta = 0$; thereby it passes into a function from $\mathfrak{G}_\eta$, and consequently into an algebraic integer.

Thus $\{\mathfrak{G}_\eta, \psi\}$ becomes a domain $\mathfrak{H}$ as soon as one imposes on ψ the condition that no algebraically fractional number enters the domain through the algebraically integral adjunction.

Quite analogously one defines the rationally integral, respectively algebraically integral, adjunction of an arbitrary system $\mathfrak{S}$ of rational or algebraic functions of the η to $\mathfrak{G}_\eta$. The rationally integral adjunction leads to the integral domain $[\mathfrak{G}_\eta, \mathfrak{S}]$, consisting of all integral polynomials in finitely many elements from $\mathfrak{G}_\eta$ and $\mathfrak{S}$ at a time. The algebraically integral adjunction gives the domain $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ consisting of all elements that depend algebraically-integrally on $[\mathfrak{G}_\eta, \mathfrak{S}]$. Exactly as above, one shows that the integral domain $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ is algebraically-integrally closed and also satisfies II and III. Thus, as above:

$\{\mathfrak{G}_\eta, \mathfrak{S}\}$ becomes a domain $\mathfrak{H}$ as soon as one imposes on $\mathfrak{S}$ the condition that no algebraically fractional number enters the domain through the algebraically integral adjunction — that is, as soon as $\mathfrak{S}$ becomes an admissible system.[140]

It is now important that the converse also holds, so that the domains $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ actually exhaust all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$ as $\mathfrak{S}$ runs through all admissible systems. Indeed, let $\mathfrak{H}$ be any domain from $\mathfrak{M}(H)$ and denote by $\mathfrak{L} = \mathfrak{H} - \mathfrak{G}_\eta$ the system consisting of all functions from $\mathfrak{H}$ not belonging to $\mathfrak{G}_\eta$. By V, $\mathfrak{H}$ contains the domain $\{\mathfrak{G}_\eta, \mathfrak{L}\}$ as a divisor; but $\{\mathfrak{G}_\eta, \mathfrak{L}\}$ is also divisible by $\mathfrak{H}$, because it contains $\mathfrak{G}_\eta$ and $\mathfrak{L}$, and since these have no common element, it contains $\mathfrak{H}$ as well. Hence $\mathfrak{H} = \{\mathfrak{G}_\eta, \mathfrak{L}\}$; and since IV is satisfied for $\mathfrak{H}$, $\mathfrak{L}$ is of course an admissible system.[141]

As H runs through every possible algebraic basis, $\mathfrak{M}(H)$ runs, as proved at the beginning, through the totality of all domains $\mathfrak{H}$. Consequently the question of the most general domains $\mathfrak{H}$ from algebraically-integral transcendental numbers is completely answered by the results of this paragraph:

a) The intersection of all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$ is given by the Zermelo domain $\mathfrak{G}_\eta$, which itself is a domain $\mathfrak{H}$; $\mathfrak{M}(H)$ is closed with respect to forming intersections.

b) Every domain $\mathfrak{H}$ from $\mathfrak{M}(H)$ arises by algebraically integral adjunction of an admissible system $\mathfrak{S}$ to $\mathfrak{G}_\eta$. As H runs through every possible algebraic basis, $\mathfrak{M}(H)$ runs through the totality of all domains $\mathfrak{H}$.[142]

§ 5. The most general domains made from integral transcendental numbers, on the basis of an algebraic basis.

Let H be any algebraic basis of all numbers, and let $\mathfrak{M}(\mathfrak{G})$ denote the totality of all domains $\mathfrak{G}$ which contain H and therefore $[H]$. As H runs through every possible basis, $\mathfrak{M}(\mathfrak{G})$ runs through the totality of all domains $\mathfrak{G}$; hence, as in § 4, we may restrict ourselves to considering the domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$.

The domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ all contain $[H]$ as a divisor; but, since $[H]$ itself is not a domain $\mathfrak{G}$, one cannot conclude as in § 4 that $[H]$ is the greatest common divisor. That this is nevertheless the case is shown by a countable set of domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$, obtained by a slight generalization from the module domain of § 3, and in fact having no element in common besides $[H]$.

For every fixed ν, let the domains $\mathfrak{G}_\nu$ ($\nu = 1, 2, \ldots$ in inf.) consist of all quantities

$$z = f(\eta) + g_1(\eta)\eta_1^\nu + g_2(\eta)\eta_2^\nu + \cdots + g_t(\eta)\eta_t^\nu, \tag{1}$$

[139] Thus, by defining algebraic integrality by means of *some* equation, the concept of irreducibility with respect to a ground field becomes unnecessary for the proof of I and V. Compare also § 2, where, only for the proof of IV, the auxiliary proposition is needed which passes from the definition of algebraic integrality by an arbitrary equation to the irreducible equation. Since this auxiliary proposition no longer holds in general in the present case, IV has to be imposed on the function ψ as a special condition.

[140] More generally, one shows analogously: if $\mathfrak{T}$ is any domain and $\{\mathfrak{T}\}$ denotes the totality of elements depending algebraically-integrally on $\mathfrak{T}$, then $\{\mathfrak{T}\}$ is algebraically-integrally closed.

[141] The adjunction of a subsystem of $\mathfrak{L}$ can already suffice; for example, the functional domain of § 2 arises by algebraically integral adjunction of all integral rational functionals — with the exception of the polynomials — to $\mathfrak{G}_\eta$.

[142] The admissible systems $\mathfrak{S}$ can be found as follows. Put all algebraically fractional functions of the η under some well-ordering Ω, and take into $\mathfrak{S}$ all functions except those which, when algebraically-integrally adjoined to $\mathfrak{G}_\eta$ and the previously admitted functions, generate an algebraically fractional number. This procedure can be broken off at any arbitrary place. As Ω runs through every well-ordering, and as the process is broken off at every arbitrary place for each fixed Ω, all admissible systems $\mathfrak{S}$ are thereby exhausted.

where $f(\eta)$ runs through all integral polynomials in the η; $\eta_1,\ldots,\eta_t$ run through the basis quantities, and, for each fixed combination $\eta_1^\nu,\ldots,\eta_t^\nu$, the functions $g_1(\eta),\ldots,g_t(\eta)$ run independently through all algebraic integers functions of the η.

As in § 3, one sees that the domains $\mathfrak{G}_\nu$ have the abstract properties I–IV; $\mathfrak{G}_\nu$ contains only algebraic-integer functions and, for $\nu=1$, is identical with the domain of § 3. We now show that the domains $\mathfrak{G}_\nu$ have no further common element besides the polynomials $f(\eta)$ from $[H]$. Indeed, let z be any algebraic-integer — but not rational — function of the η, satisfying the equation, irreducible with respect to $[H]$,

$$z^\chi + a_1(\eta)z^{\chi-1} + a_2(\eta)z^{\chi-2} + \cdots + a_\chi(\eta) = 0, \tag{2}$$

where $\chi>1$; and let λ be the greatest of the degrees of the polynomials $a_1(\eta),\ldots,a_\chi(\eta)$. If z belongs to $\mathfrak{G}_\nu$, then the quantity $\xi = z - f(\eta)$ satisfies the likewise irreducible equation, again with respect to $[H]$,

$$\xi^\chi + b_1(\eta)\xi^{\chi-1} + b_2(\eta)\xi^{\chi-2} + \cdots + b_\chi(\eta) = 0, \tag{3}$$

where, by (1), $b_1(\eta), b_2(\eta),\ldots,b_\chi(\eta)$, as symmetric functions of the ξ, contain as terms of lowest dimension terms of at least the dimensions $\nu, 2\nu,\ldots,\chi\nu$. Comparing (2) and (3) gives

$$b_1(\eta) = a_1(\eta) + \chi f(\eta).$$

If, therefore, the quantity $\xi = z + \frac{a_1(\eta)}{\chi}$ satisfies

$$\xi^\chi + c_2(\eta)\xi^{\chi-2} + \cdots + c_\chi(\eta) = 0, \tag{4}$$

then

$$\xi - \zeta = \frac{b_1(\eta)}{\chi}; \qquad c_i(\eta) \equiv b_i(\eta) \pmod{\frac{b_1(\eta)}{\chi}}, \tag{5}$$

where $b_1(\eta)$ begins with terms of at least dimension ν. Now the $c_i(\eta)$, as the comparison of (2) with (4) shows, are of degree χ in the coefficients $a_i(\eta)$ of (2), hence of degree at most $\chi\lambda$ in the η; on the other hand, all the $b_i(\eta)$ begin with terms of at least dimension ν. Thus, if one chooses $\nu > \chi\lambda$, (5) gives

$$c_i(\eta) = 0; \qquad \xi^\chi = 0 \quad \text{or} \quad \left(z + \frac{a_1(\eta)}{\chi}\right)^\chi = 0.$$

Thus z, contrary to the assumption, becomes rational in the η. Hence:

If z is an algebraic-integer, non-rational function of the η, then z can belong to no domain $\mathfrak{G}_\nu$ for which $\nu > \chi\lambda$, or:

The intersection of all domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ is given by the integral domain $[H]$, which itself is not a domain $\mathfrak{G}$; consequently $\mathfrak{M}(\mathfrak{G})$ is not closed with respect to forming intersections.

Consequently the construction of the most general domains $\mathfrak{G}$ by means of an algebraic basis is also harder to survey than that of the domains $\mathfrak{H}$. The domain $[H]$ has properties I, III and IV; if one adjoins any system $\mathfrak{S}$ rationally-integrally, then for the resulting integral domain $[H,\mathfrak{S}]$ the properties I and III remain valid, whereas IV forms a special condition on $\mathfrak{S}$. In order that a domain $\mathfrak{G}$ arise, II must also be imposed on $\mathfrak{S}$ as a condition; or, expressed differently: for every field $\mathfrak{K}$, finite over (H)[143], there must be a field $\mathfrak{L}$ over $\mathfrak{K}$, finite over $\mathfrak{K}$, such that $[H,\mathfrak{S}]$ contains a primitive element from $\mathfrak{L}$.[144] Conversely, every domain $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ can also be represented in the form $[H,\mathfrak{S}]$; hence one obtains all domains $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ when $\mathfrak{S}$ runs through all systems restricted in this way. The structure of the systems $\mathfrak{S}$ will appear more simply in the next paragraph by means of the "rational basis"; at the same time this gives a complete analogy with the theorems of § 4.

§ 6. The most general domains made from integral transcendental numbers, on the basis of a rational basis.

In analogy with the algebraic basis, the rational basis is to be defined as follows: "A system Θ of numbers ϑ is called a rational basis of all numbers if Θ permits every number to be expressed rationally — with

[143] (H) means the field consisting of all rational functions of the η with algebraic-number coefficients.

[144] Cf. § 7.

coefficients in K[145] — while, for at least one well-ordering, no basis number can be expressed rationally — with coefficients in K — by finitely many preceding basis numbers."[146]

Let a domain $\mathfrak{G}$ made from integral transcendental numbers be subjected to a well-ordering Ω. Then it can happen that a quantity z from $\mathfrak{G}$ is rationally expressible by finitely many preceding quantities from $\mathfrak{G}$; that is, it satisfies an equation $z = \varphi(\xi)$, where $\varphi(\xi)$ is a rational function, with coefficients in K, of $\xi_1, \ldots, \xi_t$. In particular, every algebraic number α of the domain satisfies the equation $z = \alpha$. The remaining quantities ϑ from $\mathfrak{G}$ form a system Θ which is to be shown to be a rational basis of all quantities from $\mathfrak{G}$. For by assumption each quantity ϑ is rationally independent of the preceding quantities from Θ. The induction argument[147] further shows that every relation $z = \varphi(\xi)$ entails a relation $z = \psi(\vartheta)$; for otherwise there would have to be a first element $z_0 = \varphi_0(\xi_1, \ldots, \xi_t)$ which cannot be expressed rationally by the ϑ, while the preceding $\xi_1, \ldots, \xi_t$ are rational functions of the ϑ; this gives a contradiction. By II, Θ is at the same time a rational basis of all numbers; by III and I, besides Θ, $\mathfrak{G}$ contains as a divisor the integral domain $[\Theta]$ consisting of all integral polynomials in the ϑ, whereas by IV $[\Theta]$ contains no algebraically fractional number. Thus the rational basis Θ of all numbers satisfies the side condition that the integral domain $[\Theta]$ derived from Θ contains no algebraically fractional number;[148] Θ is a *rational basis with side condition* for all numbers. Thus, in analogy with § 2: *Every domain $\mathfrak{G}$ can be constructed by means of a rational basis with side condition, in such a way that $\mathfrak{G}$ contains the integral domain $[\Theta]$ as a divisor.*

Let Θ be any rational basis with side condition for all numbers — whose existence is therefore guaranteed by the existence of the domains $\mathfrak{G}$. Let $\mathfrak{N}(\mathfrak{G})$ denote the totality of all domains $\mathfrak{G}$ which contain Θ and therefore $[\Theta]$. As Θ runs through every possible basis with side condition, $\mathfrak{N}(\mathfrak{G})$ runs, by the above, through the totality of all domains $\mathfrak{G}$; hence we may again restrict ourselves to the domains $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$.

Now $[\Theta]$ itself is a domain $\mathfrak{G}$; for every number can be expressed rationally by Θ, and therefore as a quotient of two polynomials from $[\Theta]$; and by its construction $[\Theta]$ also satisfies the remaining conditions I, III, IV. *Thus* $[\Theta]$ *is the greatest common divisor, or intersection, of all domains $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$; moreover $\mathfrak{N}(\mathfrak{G})$ is closed with respect to intersection; that is, the intersection of all domains $\mathfrak{G}$ of any subsystem of $\mathfrak{N}(\mathfrak{G})$ also belongs to* $\mathfrak{N}(\mathfrak{G})$. For I holds for the intersection; II and III follow from the fact that $[\Theta]$ is a divisor; and IV from the fact that the intersection domain is contained as a divisor in a domain $\mathfrak{G}$.

The rationally integral adjunction of any system $\mathfrak{S}$ of rational functions of the ϑ to $[\Theta]$ — every algebraic function of the ϑ can, by the property of the rational basis, be expressed rationally through certain other ϑ — gives an integral domain $[\Theta, \mathfrak{S}]$ having properties I, II, III, and hence becomes a domain $\mathfrak{G}$ as soon as $\mathfrak{S}$ is an "admissible system," i.e. as soon as it satisfies the condition that no algebraically fractional number enters the domain through the rationally integral adjunction. Conversely, every domain $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$ permits the representation $[\Theta, \mathfrak{S}]$, provided only that $\mathfrak{S}$ denotes the residual system $\mathfrak{G} - [\Theta]$. Thus, for the most general domains $\mathfrak{G}$, in complete analogy with § 4, one has:

a) *The intersection of all domains $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$ is given by the integral domain* $[\Theta]$*, which itself is a domain $\mathfrak{G}$; $\mathfrak{N}(\mathfrak{G})$ is closed with respect to forming intersections.*

b) *Every domain $\mathfrak{G}$ from $\mathfrak{N}(\mathfrak{G})$ arises by rationally integral adjunction of an admissible system $\mathfrak{S}$ to* $[\Theta]$*. As Θ runs through every possible rational basis with side condition, $\mathfrak{N}(\mathfrak{G})$ runs through the totality of all domains $\mathfrak{G}$.*[149]

Remark. If one extracts from Θ an algebraic basis H of all quantities from Θ, then H is at the same time an algebraic basis of all numbers; conversely, every algebraic basis can be extended to a rational one by putting H first in the well-ordering. Accordingly, the construction in § 5 of the domains $\mathfrak{G}$ by means of an algebraic basis can be interpreted as follows. Decompose the system $\mathfrak{S}$ to be adjoined to $[H]$ into two subsystems $\mathfrak{S}_1$ and $\mathfrak{S}_2$; the adjunction of $\mathfrak{S}_1$ amounts to extending H to a rational basis with side condition, to which an admissible system $\mathfrak{S}_2$ is then still rationally-integrally adjoined.

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[145] By rational functions one should always understand such functions with algebraic-number coefficients; these algebraic-number coefficients are included in the definition because of III and IV. It would be more appropriate to require rational-number coefficients both in III and IV and in the definition of the rational basis. The arguments of §§ 6 and 7 remain literally unchanged if one merely replaces the field K of all algebraic numbers by the field of all rational numbers.

[146] In contrast to the linear and algebraic bases, the ordering of the elements plays a role here. For example, in the ordering η, $\sqrt[3]{\eta}$ both elements have to be admitted, whereas in the reverse ordering only $\sqrt[3]{\eta}$ has to be admitted.

[147] Cf. Zermelo, loc. cit., § 1.

[148] That this is indeed a condition is shown, for instance, by the fact that the two quantities η, $\frac{1}{2\sqrt{\eta}}$ are not admissible as basis elements, whereas η, $\frac{1}{\sqrt{\eta}}$ are.

[149] For the most general admissible systems $\mathfrak{S}$, what was said in § 4 applies, with "rational" replacing "algebraic."

§ 7. Construction of the most general rational basis with side condition.

The results of the preceding paragraph still need a supplement, since there the *existence* of a rational basis with side condition for all numbers was proved, but not a method for constructing it from a prescribed well-ordering. On account of the side condition, this construction takes a more complicated form when, instead of a domain $\mathfrak{G}$, the field of all complex numbers is present.[150]

Let the field of all complex numbers be subjected to a well-ordering Ω. Then quantities of three kinds can occur:

1. Quantities y whose rationally integral adjunction to the preceding basis quantities — defined recursively in 3. — forces an algebraically fractional number to enter the resulting integral domain; this domain consists of all integral polynomials in y if no basis quantity precedes it. In particular, all algebraically fractional numbers are quantities y, since the resulting integral domain contains the quantities $y = \beta$.

2. Quantities z which do not have property 1, but which can be expressed rationally by finitely many preceding numbers different from the y; thus they satisfy a relation $z = \varphi(\xi_1, \dots, \xi_t)$, where φ is a rational function with coefficients in K and none of the ξ are y. In particular, all algebraic integer numbers α belong here, since property 1 does not hold for them and they satisfy the relation $z = \alpha$.

3. Quantities ϑ for which neither 1 nor 2 holds and which are to be called basis quantities. In particular, the first transcendental number ϑ_0 in the well-ordering is such a basis quantity; for the integral polynomials in ϑ_0 cannot represent an algebraically fractional number, and ϑ_0 also cannot be rationally dependent on the preceding algebraic numbers.

The system Θ of all basis quantities ϑ is now to be shown to be a rational basis with side condition for all numbers.

Since no quantity y occurs among the ϑ, the integral domain $[\Theta]$ cannot contain any algebraically fractional number; hence the side condition is satisfied. Since, furthermore, no quantity z occurs among the ϑ, each ϑ is rationally independent of the preceding basis quantities. The induction argument shows, exactly as in § 6, that every relation $z = \varphi(\xi)$ — where no y occurs among the ξ — entails a relation $z = \psi(\vartheta)$, so that all quantities z are rationally expressible by the ϑ.

It remains only to prove the more difficult assertion that the y are also rationally expressible by the ϑ. First we show that the y depend algebraically on the ϑ. By assumption, for each y there exists at least one algebraically fractional number β admitting the representation

$$\beta = f_0(\vartheta) + f_1(\vartheta)y + \cdots + f_\chi(\vartheta)y^\chi, \tag{1}$$

where $f_0(\vartheta), \dots, f_\chi(\vartheta)$ are polynomials from $[\Theta]$ and hence cannot represent algebraically fractional numbers. If y were algebraically independent of the ϑ, then (1) would hold identically in y and one would have $\beta = f_0(\vartheta)$, contradicting the properties of $[\Theta]$.

In order to infer rational dependence from algebraic dependence, we extract from Θ an algebraic basis H of all quantities from Θ. Then every y, as an algebraic function of the ϑ, is also algebraically dependent on the η; through $y = y(\eta)$ an algebraic field $(H, y(\eta))$ over (H) is therefore determined. Since all η are at the same time quantities ϑ, the proof of rational dependence amounts to showing that for every such field there are primitive elements which are rational functions of the ϑ; more precisely, it will be shown that $y(\eta)$ loses property 1 after multiplication by a polynomial in the η, and consequently passes into such an element.

It is known that between (H) and (H, y) there are only finitely many intermediate fields. Let $\Omega_0 = (H), \Omega_1, \dots, \Omega_\sigma$ be those among them which possess a primitive element rationally expressible by the ϑ, so that every element of the field becomes a rational function of the ϑ; this condition is at least always satisfied for $\Omega_0 = (H)$. Let the primitive function belonging to Ω_i be given by $\frac{g_i(\vartheta)}{h_i(\vartheta)}$, where g_i, h_i are polynomials from $[\Theta]$. Then — with α_i denoting a suitably chosen integer — the polynomial from $[\Theta]$

$$\xi_i = g_i(\vartheta) + \alpha_i h_i(\vartheta)$$

[150] In § 6 b) it would suffice to let Θ run only through those special rational bases which consist of an algebraic basis and algebraically integral functions of the basis quantities; then the side condition is automatically satisfied. To see this, one need only, by § 6, extend an algebraic basis from $\mathfrak{G}$ to a rational one and, in the resulting basis, multiply each algebraically fractional function of the η by a suitably chosen polynomial in the η, whereby the basis property is preserved. This construction transfers unchanged to the field of all complex numbers. The question arising from § 6 concerning the most general rational basis with side condition, however, also has independent interest.

is a primitive function of Ω_i, or of an algebraic field over Ω_i,[151] and every function $\psi(\eta)$ from Ω_i admits the representation

$$\psi(\eta) = \frac{a_1(\eta)\xi_i^{\kappa-1} + a_2(\eta)\xi_i^{\kappa-2} + \cdots + a_\kappa(\eta)}{a(\eta)} = \frac{A(\eta, \xi_i)}{a(\eta)}, \tag{2}$$

where $a(\eta), a_1(\eta), \ldots, a_\kappa(\eta)$ are integral polynomials in the η, and hence $A(\eta, \xi_i)$ and $a(\eta)$ are quantities from $[\Theta]$.

The irreducible equation satisfied by y with respect to Ω_i is, by (2), of the form

$$f^{(i)}(\eta)y^{\lambda_i} + f_1^{(i)}(\eta, \xi_i)y^{\lambda_i - 1} + \cdots + f_{\lambda_i}^{(i)}(\eta, \xi_i) = 0,$$

and all coefficients of this equation belong to $[\Theta]$. Then the function $y_i = f^{(i)}(\eta) \cdot y$ satisfies the likewise irreducible equation, with respect to Ω_i,

$$y_i^{\lambda_i} + f_1^{(i)}(\eta, \xi_i)y_i^{\lambda_i - 1} + \cdots + \left(f^{(i)}(\eta)\right)^{\lambda_i - 1} f_{\lambda_i}^{(i)}(\eta, \xi_i) = 0,$$

by virtue of which it depends algebraically-integrally on $[H, \xi_i]$; the same holds for the product of y_i with any polynomial in the η. In particular, the function

$$Y = f^{(0)}(\eta) f^{(1)}(\eta) \cdots f^{(\sigma)}(\eta) \cdot y \tag{3}$$

depends algebraically-integrally on all domains $[H, \xi_i]$ by means of the equation, irreducible in each case with respect to Ω_i,

$$Y^{\lambda_i} + F_1^{(i)}(\eta, \xi_i)Y^{\lambda_i - 1} + \cdots + F_{\lambda_i}^{(i)}(\eta, \xi_i) = 0 \qquad (i = 0, 1, \ldots, \sigma). \tag{4}$$

We now show that Y is the desired primitive element rationally expressible by the ϑ. Suppose, for example, that the algebraic number γ is represented by Y and $[\Theta]$:

$$\gamma = k_0(\vartheta) + k_1(\vartheta)Y + \cdots + k_\nu(\vartheta)Y^\nu, \tag{5}$$

where $k_0(\vartheta), \ldots, k_\nu(\vartheta)$ are polynomials from $[\Theta]$. We now determine the irreducible equation satisfied by Y with respect to the field $\mathfrak{K} = (H; k_0(\vartheta), \ldots, k_\nu(\vartheta))$ arising from the coefficient domain of (5). It is known to be given by the irreducible equation of Y with respect to the intersection field of $\mathfrak{K}$ and (H, Y), lying between (H) and (H, Y). Now (H, Y) is identical with (H, y) by (3); and since all elements of $\mathfrak{K}$, hence also all elements of the intersection field, can be expressed rationally by finitely many ϑ, this latter field is necessarily one of the fields $\Omega_0, \Omega_1, \ldots, \Omega_\sigma$, say Ω_τ. The desired irreducible equation is therefore, by (4), of degree λ_τ and is

$$Y^{\lambda_\tau} + F_1^{(\tau)}(\eta, \xi_\tau)Y^{\lambda_\tau - 1} + \cdots + F_{\lambda_\tau}^{(\tau)}(\eta, \xi_\tau) = 0 \tag{6}$$

with polynomials from $[\Theta]$ as coefficients. By means of (6), (5) can be rewritten in the form

$$\gamma = K_0(\vartheta) + K_1(\vartheta)Y + \cdots + K_{\lambda_\tau - 1}(\vartheta)Y^{\lambda_\tau - 1}, \tag{7}$$

where $K_0(\vartheta), K_1(\vartheta), \ldots$ belong to $\mathfrak{K}$ and, on the other hand, are polynomials from $[\Theta]$. But since γ is an algebraic number, the relation (7) does not reach degree λ_τ in Y and is therefore satisfied *identically* in Y. Hence

$$\gamma = K_0(\vartheta),$$

i.e. γ belongs to $[\Theta]$ and is consequently an algebraic integer.

As γ runs through all algebraic numbers representable by Y and $[\Theta]$, the intersection field of (H, Y) and the corresponding field $\mathfrak{K}$ always runs only through the fields $\Omega_0, \Omega_1, \ldots, \Omega_\sigma$, and all the preceding arguments remain valid. That is, *by rationally integral adjunction of* Y *to* $[\Theta]$ *no algebraically fractional number can be represented.* Thus Y no longer has property 1; according to 2) or 3) it is a quantity z or ϑ, and therefore rationally representable by finitely many ϑ; by (3), the same holds for y: Θ *has been proved to be a rational basis of all numbers.* Conversely, since every rational basis with side condition can be constructed according to 1), 2), 3) by putting Θ first in the well-ordering, we have proved: *The construction given by 1), 2), 3) leads to the most general rational basis with side condition for all numbers.*

[151] Cf. the end of § 5.

§ 8. Division of the domains $\mathfrak{G}$ and $\mathfrak{H}$ into classes.

Alongside the question of the most general domains $\mathfrak{G}$ and $\mathfrak{H}$ there arises the further question of the *essentially different* domains, which — in a sense to be made more precise below — cannot be generated by the same construction principle. This leads to a division of these domains into classes.

Two domains will be called belonging to the same class, or isomorphic, if their elements can be put into one-to-one correspondence in such a way that the sum and product of any two elements of one domain are always assigned to the sum and product of the corresponding elements of the other. From this definition it follows that under every isomorphic relation zero and the unit correspond to themselves, and that all algebraic relations between finitely many elements are preserved for the corresponding elements, and conversely. In particular, besides the unit, all rational numbers correspond to themselves; whereas the totality of algebraic numbers — and likewise the totality of algebraic integers — goes over into itself, although an individual algebraic number can very well correspond to a conjugate.

First note that every domain from the totality $\mathfrak{M}(\mathfrak{G})$ of all domains $\mathfrak{G}$ containing a fixed algebraic basis H is isomorphic to a domain from the totality $\mathfrak{M}^*(\mathfrak{G})$ belonging to another algebraic basis H^*. For the field of all complex numbers arises from every algebraic basis by algebraic extension, and therefore has the same cardinality as the basis;[152] hence H and H^* have the same cardinality, and the desired isomorphic mapping is obtained by assigning the basis quantities η_i to η_i^*.[153] On the other hand, $\mathfrak{M}(\mathfrak{G})$ contains non-isomorphic, or essentially different, domains $\mathfrak{G}$, as follows from the examples in §§ 2, 3 and 5. For since all algebraic relations are preserved under the isomorphic mapping, an algebraic basis must again correspond to an algebraic basis; but §§ 2 and 3 showed that the functional and module domains possess no algebraic basis for which all the basis properties of the Zermelo domain would be fulfilled. Since the functional domain of § 2 is algebraically integrally closed, the subset $\mathfrak{M}(\mathfrak{H})$ also contains essentially different domains $\mathfrak{H}$.

We now show, analogously to the basis construction, how one can extract from $\mathfrak{M}(\mathfrak{G})$ a subset $\mathfrak{L}(\mathfrak{G})$ such that all domains $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$ are essentially different, while every domain from $\mathfrak{M}(\mathfrak{G})$ — and hence every domain $\mathfrak{G}$ at all — is isomorphic to a domain from $\mathfrak{L}(\mathfrak{G})$. Subject $\mathfrak{M}(\mathfrak{G})$ to a well-ordering Ω; then a domain $\mathfrak{G}$ from $\mathfrak{M}(\mathfrak{G})$ may or may not be isomorphic to some preceding one. The domains $\mathfrak{G}$ isomorphic to no preceding one form a set $\mathfrak{L}(\mathfrak{G})$, which by construction contains only essentially different domains $\mathfrak{G}$. The induction argument also shows that every domain from $\mathfrak{M}(\mathfrak{G})$ is isomorphic to a domain from $\mathfrak{L}(\mathfrak{G})$. For if $\mathfrak{G}_1$ were the first domain for which this failed, then $\mathfrak{G}_1$ either belongs to $\mathfrak{L}(\mathfrak{G})$, or is isomorphic to a preceding domain $\mathfrak{G}_0$, which by assumption is isomorphic to a domain from $\mathfrak{L}(\mathfrak{G})$; hence the same would hold for $\mathfrak{G}_1$. Since every domain $\mathfrak{G}$ belongs to some set $\mathfrak{M}^*(\mathfrak{G})$ and is therefore isomorphic to a domain from $\mathfrak{M}(\mathfrak{G})$, *the totality of the classes of domains $\mathfrak{G}$ is represented by $\mathfrak{L}(\mathfrak{G})$.*

If, in a domain $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$, one replaces the basis quantities η by those η^* of some other basis, then, as shown above, a domain isomorphic to $\mathfrak{G}$ arises. Conversely, let $\mathfrak{G}^*$ be any domain and suppose it is isomorphic to a domain $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$. If the basis quantities η correspond to the quantities η^* from $\mathfrak{G}^*$, then these again form an algebraic basis, and $\mathfrak{G}^*$ contains the same algebraic functions of η^* as $\mathfrak{G}$ contains of η — or their conjugates. *Thus every domain isomorphic to $\mathfrak{G}$ arises by letting the algebraic basis H run through every possible algebraic basis in $\mathfrak{G}$ and in its conjugates with respect to (H)*[154] *— if these differ from $\mathfrak{G}$.* From each fixed domain $\mathfrak{G}_i$ in $\mathfrak{L}(\mathfrak{G})$ one obtains in this way the class $\mathfrak{R}_i$, which contains all and only the domains $\mathfrak{G}$ isomorphic to $\mathfrak{G}_i$, although possibly each domain occurs several times or even infinitely often. From $\mathfrak{R}_i$ one can again, by the procedure above, extract a subset $\mathfrak{R}_i^*$ which contains every domain isomorphic to $\mathfrak{G}_i$ once and only once. As $\mathfrak{R}_i^*$ runs through all classes, one obtains the totality of all domains $\mathfrak{G}$, every domain once and only once. To characterize the class $\mathfrak{R}_i$, one can of course use, instead of $\mathfrak{G}_i$, any other domain of the class from which all domains of $\mathfrak{R}_i$ can be derived as above. In summary:

From $\mathfrak{M}(\mathfrak{G})$ one can extract a subset $\mathfrak{L}(\mathfrak{G})$ such that the domains $\mathfrak{G}$ from $\mathfrak{L}(\mathfrak{G})$ are in one-to-one correspondence with the totality of classes (without repetition). From $\mathfrak{G}_i$ the corresponding class $\mathfrak{R}_i$ arises by letting H run through every possible algebraic basis in $\mathfrak{G}_i$ and its conjugates with respect to (H). If $\mathfrak{R}_i^*$ denotes all mutually distinct domains from $\mathfrak{R}_i$, and $\mathfrak{R}_i^*$ runs through all classes, then the totality of all domains $\mathfrak{G}$ arises, every domain once and only once.[155]

[152] Steinitz, §§ 23 and 24.

[153] This conclusion of course does not hold for the sets $\mathfrak{N}(\mathfrak{G})$ and $\mathfrak{N}^*(\mathfrak{G})$ belonging to two rational bases Θ and Θ^*; nor can the isomorphism of $\mathfrak{N}(\mathfrak{G})$ and $\mathfrak{N}^*(\mathfrak{G})$ be inferred from the mutual rational expressibility of Θ and Θ^*. What is obtained, however, is an isomorphic relation between the fields (Θ) and (Θ^*).

[154] The elements of these domains conjugate to $\mathfrak{G}$ can certainly be expressed rationally by the elements of $\mathfrak{G}$ by II, but not necessarily integrally and rationally; the conjugates can therefore differ from $\mathfrak{G}$.

[155] Here $\mathfrak{M}(\mathfrak{G})$ is to be constructed according to §§ 6 and 7: by § 7 one extends an algebraic basis H to every rational basis Θ with side condition arising from it, and for each Θ forms, by § 6 b), the totality $\mathfrak{N}(\mathfrak{G})$ of all domains $\mathfrak{G}$ containing Θ.

The analogous statement holds for the domains $\mathfrak{H}$ made from algebraically integral transcendental numbers.

§ 9. An example of countably infinitely many classes of domains $\mathfrak{G}$.

The cardinality of the constructions given in § 4 b) and § 6 b) can be read off from the remarks thereto as equal to the cardinality of all well-orderings, hence as 2^c, if c denotes the cardinality of the continuum. But from this no conclusion can be drawn about the cardinality of the totality of all domains $\mathfrak{G}$, counted once and only once, and still less about the cardinality of all classes. That the number of classes is certainly not finite will now be shown by an example of countably infinitely many domains $\mathfrak{G}$ — analogous to those of § 5 — which will all turn out to be essentially different and hence supply countably infinitely many classes.[156]

Analogously to § 5, (1), let the domain $\mathfrak{G}_\sigma$ consist of the totality of quantities

$$z = a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta) \pmod{\mathfrak{M}_\sigma(\eta)}, \tag{1}$$

where $a_i(\eta)$ denotes homogeneous forms of i-th dimension in the η; and the module $\mathfrak{M}_\sigma$ contains as basis all power products of σ-th dimension in the η[157] and as multipliers all algebraic-integer functions of the η. We show that under an isomorphic relation between two domains $\mathfrak{G}_\sigma$ and $\mathfrak{G}_\tau$ $(\sigma \neq \tau)$ the modules $\mathfrak{M}_\sigma$ and $\mathfrak{M}_\tau$ must necessarily correspond isomorphically; the impossibility then follows easily.

Let the indeterminates in $\mathfrak{G}_\tau$ be denoted by ξ; under the isomorphic relation, let the quantities $f(\eta)$ from $\mathfrak{G}_\sigma$ correspond to these ξ. Since the isomorphic relation is preserved under forming quotients, the domain $\mathfrak{G}_\xi$ of all algebraic-integer functions of the ξ corresponds to an algebraically integrally closed domain $\mathfrak{T}$ consisting of all functions algebraically integral over the $f(\eta)$, and hence also algebraically integral in the η.

Suppose (1) corresponds to a quantity from $\mathfrak{M}_\tau$. Then, by the module property of $\mathfrak{M}_\tau$, the product of (1) with an arbitrary quantity $\psi(\eta)$ from $\mathfrak{T}$ must also belong to $\mathfrak{G}_\sigma$; in formulas:

$$(a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta))\psi(\eta) \equiv b_0 + b_1(\eta) + \cdots + b_{\sigma-1}(\eta) \pmod{\mathfrak{M}_\sigma}. \tag{2}$$

Setting all η equal to zero gives $a_0\psi(0) = b_0$, so (2) becomes

$$(a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta))(\psi(\eta) - \psi(0)) \equiv c_1(\eta) + \cdots + c_{\sigma-1}(\eta) \pmod{\mathfrak{M}_\sigma}. \tag{3}$$

Now, besides $\psi(\eta)$, the functions

$$\chi_\nu(\eta) = \sqrt[\nu]{\psi(\eta) - \psi(0)}$$

also belong to $\mathfrak{T}$ for every ν; and since $\chi_\nu(0) = 0$, the formulas analogous to (3) are:

$$(a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta))\chi_\nu(\eta) \equiv c_1^{(\nu)}(\eta) + \cdots + c_{\sigma-1}^{(\nu)}(\eta) \pmod{\mathfrak{M}_\sigma} \quad (\nu = 1, 2, \ldots). \tag{4}$$

Let $A(\eta)$, of degree λ, denote the product of $\chi_1 = \psi(\eta) - \psi(0)$ with its $\kappa - 1$ conjugates. Since $\chi_1(0) = 0$, $A(\eta)$ must begin with terms of at least first dimension. From (4) one obtains

$$a_0\chi_\nu(\eta) \equiv 0 \pmod{\mathfrak{M}_1}; \qquad a_0^\nu\chi_1(\eta) \equiv 0 \pmod{\mathfrak{M}_\nu}; \qquad a_0^{\nu\kappa}A(\eta) \equiv 0 \pmod{\mathfrak{M}_{\nu\kappa}}, \tag{5}$$

and therefore, for $a_0 \neq 0$, as in § 3, $\lambda > \nu\kappa$. But since $\nu < \frac{\lambda}{\kappa}$ contradicts the algebraically integral closedness of $\mathfrak{T}$, necessarily $a_0 = 0$. From (4) one now obtains

$$a_1(\eta)\chi_\nu(\eta) \equiv c_1^{(\nu)}(\eta) \pmod{\mathfrak{M}_2}; \qquad [a_1(\eta)]^{\nu\kappa}A(\eta) \equiv [c_1^{(\nu)}(\eta)]^{\nu\kappa} \pmod{\mathfrak{M}_{\nu\kappa+1}}, \tag{6}$$

and from this, since $A(\eta)$ begins with terms of at least first dimension, $c_1^{(\nu)}(\eta) = 0$. Thus, in complete analogy with (5), for every ν:

$$a_1(\eta)\chi_\nu(\eta) \equiv 0 \pmod{\mathfrak{M}_2}; \qquad [a_1(\eta)]^{\nu\kappa}A(\eta) \equiv 0 \pmod{\mathfrak{M}_{2\nu\kappa}},$$

and hence $a_1(\eta) = 0$. Continuing in this way, the alternating application of (5) and (6) gives

$$a_0 = 0, \qquad a_1(\eta) = 0, \quad \ldots, \quad a_{\sigma-1}(\eta) = 0.$$

[156] The domains $\mathfrak{G}$ of § 6 likewise give countably infinitely many classes, but the proof is somewhat more complicated.

[157] Naturally, in each individual z only finitely many power products of the η occur in the module.

Since the analogous consideration also holds with respect to $\mathfrak{G}_\tau$: *Under every isomorphic relation between* $\mathfrak{G}_\sigma$ *and* $\mathfrak{G}_\tau$ $(\sigma \neq \tau)$, *the modules* $\mathfrak{M}_\sigma$ *and* $\mathfrak{M}_\tau$ *correspond isomorphically.*

Suppose, for instance, that $\sigma > \tau$. To the indeterminates ξ there corresponded the quantities $f(\eta)$ from $\mathfrak{G}_\sigma$; and, since all quantities from $\mathfrak{G}_\tau$ can be represented by polynomials in the ξ mod. $\mathfrak{M}_\tau$, the correspondence of $\mathfrak{M}_\tau$ and $\mathfrak{M}_\sigma$ implies that all quantities from $\mathfrak{G}_\sigma$ are obtained as polynomials in the $f(\eta)$ mod. $\mathfrak{M}_\sigma$. Since $\sigma > \tau \geq 1$, the indeterminates η certainly do not belong to $\mathfrak{M}_\sigma$, and hence, being integrally and rationally expressible by the $f(\eta)$ mod. $\mathfrak{M}_\sigma$, there must be an $f(\eta)$ with non-vanishing linear terms. Thus let

$$\xi : f(\eta) \equiv a_0 + a_1(\eta) \pmod{\mathfrak{M}_2} \quad [a_1(\eta) \neq 0],$$

$$\begin{aligned} \xi^\tau : [f(\eta)]^\tau &\equiv a_0^\tau \pmod{\mathfrak{M}_1} \quad \text{for } a_0 \neq 0, \\ &\equiv [a_1(\eta)]^\tau \pmod{\mathfrak{M}_{\tau+1}} \quad \text{for } a_0 = 0. \end{aligned}$$

Now ξ^τ belongs to $\mathfrak{M}_\tau$, while $[f(\eta)]^\tau$ begins with non-vanishing terms of zeroth or τ-th dimension and, since $\tau < \sigma$, cannot belong to $\mathfrak{M}_\sigma$, contrary to the correspondence of $\mathfrak{M}_\sigma$ and $\mathfrak{M}_\tau$ derived above. The case $\tau > \sigma$ is dealt with at the same time by interchanging η and ξ. It is therefore shown that the domains $\mathfrak{G}_\sigma$ and $\mathfrak{G}_\tau$ are *not isomorphic, or essentially different.* Thus *the domains* $\mathfrak{G}_i$ $(i = 1, 2, \ldots$ *in inf.*) *give an example of countably infinitely many classes of domains* $\mathfrak{G}$.

Remark. Although any two domains $\mathfrak{G}_i$ are non-isomorphic, it can very well happen, for two domains, that each is isomorphic to a part of the other.[158] Let, for example, $\tau = 1$ and σ be arbitrary:

$$\mathfrak{G}_1 : a_0 + \mathfrak{M}_1(\xi); \qquad \mathfrak{G}_\sigma : a_0 + a_1(\eta) + \cdots + a_{\sigma-1}(\eta) + \mathfrak{M}_\sigma(\eta).$$

The assignment $\xi = \eta$ makes $\mathfrak{G}_\sigma$ a proper divisor of $\mathfrak{G}_1$; conversely, the mutually one-to-one assignment in $\mathfrak{G}_1$ and $\mathfrak{G}_\sigma$ given by $\xi = \eta^\sigma$, $\eta = \sqrt[\sigma]{\xi}$ makes $\mathfrak{G}_1$ a proper divisor of $\mathfrak{G}_\sigma$. Thus every $\mathfrak{G}_i$ is also isomorphic to a proper divisor of itself. In contrast with the notion of intersection, the notion of an intersection class — that is, a class such that each of its domains would be isomorphic to a proper divisor of every domain of every other class — does not exist, at least not as a uniquely determined notion.

§ 10. The integral quantities of an arbitrary field.

The developments of the preceding paragraphs have, for the domains $\mathfrak{G}$ — provided that the algebraic numbers are not included in the defining properties III and IV for the domains $\mathfrak{G}$ — relied only on the fact that the totality of all complex numbers forms a field;[159] for the domains $\mathfrak{H}$, they relied on the further fact that this field is algebraically closed. The results obtained therefore directly admit the *generalization to the most general field*, which, following Weber and E. Steinitz,[160] is abstractly defined as "a system of elements with two operations, addition and multiplication, which are subject to the associative and commutative laws, are connected by the distributive law, and admit unrestricted and unique inverse operations."[161]

Every such field contains a unit element, and consequently the "prime field" derived from it by the operations of addition and multiplication and their inverses. This prime field is either of the type of the rational numbers (characteristic 0), or of the type of the residue-class system of a prime number p (characteristic p).[162] The *integral quantities of the prime field* are then well-defined as the quantities arising from the unit element by addition and subtraction; for prime fields of characteristic p, the integral quantities already exhaust the prime field, and there are no fractional quantities of the prime field. Every algebraically closed field contains an "absolutely algebraic field," arising from the prime field by adjoining all elements algebraic over the prime field; for fields of characteristic 0 this is therefore of the type of the field of all algebraic numbers. The *algebraically integral quantities of the absolutely algebraic field* are then well-defined as all quantities algebraically integral over the integral quantities of the prime field (or, what is the same thing, over the unit); for fields of characteristic p there are therefore also no algebraically fractional quantities of the absolutely algebraic field. After these preliminary remarks, the definition of the "integral" and "algebraically integral" quantities can be transferred to arbitrary fields, respectively to algebraically closed fields: *The*

[158] This behavior can also occur for fields; cf. Steinitz, loc. cit., conclusion.

[159] Only in § 7 was the further assumption used that the "prime field" of the rational numbers contains infinitely many elements; in the case of a prime field with only finitely many elements, however, this paragraph simply disappears, as we shall see.

[160] loc. cit., introduction.

[161] Only division by zero is to be excluded.

[162] Steinitz, § 4.

integral quantities of a field $\mathfrak{K}$ are defined as an integral domain $\mathfrak{G}$ of quantities from $\mathfrak{K}$ whose quotient field[163] *exhausts $\mathfrak{K}$, and which contains the unit element but no fractional quantity of the prime field.*

The algebraically integral quantities of an algebraically closed field $\mathfrak{K}$ are defined as an algebraically integrally closed integral domain $\mathfrak{H}$ of quantities from $\mathfrak{K}$ whose quotient field exhausts $\mathfrak{K}$, and which contains the unit element but no algebraically fractional quantity of the absolutely algebraic field.

For fields of characteristic zero, the results obtained in the preceding paragraphs apply literally, once one merely replaces the "algebraic numbers" by the quantities of the prime field, respectively of the absolutely algebraic field. For fields of characteristic p, the results simplify still further because the prime field contains no fractional quantities, so that the side condition for the rational basis and for the systems to be adjoined simply drops away.[164] Thus the result is: *The abstract definition permits as integral quantities of a field of characteristic p: every integral domain derived from an arbitrary rational basis, the field itself, and every integral domain lying between these domains and the field. Correspondingly, as algebraically integral quantities of an algebraically closed field of prime characteristic one has: every algebraically integrally closed integral domain lying between the domain derived from an algebraic basis and the field itself — including the two boundary domains.* For the most general fields of prime characteristic, therefore, as with the corresponding prime fields, the distinction between integral and fractional is effaced.

Erlangen, 30 March 1915.

[163]That is, the field consisting of all quotients of pairs of quantities from $\mathfrak{G}$.
[164]Cf. the first note to this paragraph.

10. The Functional Equations of the Isomorphic Mapping

Math. Ann. 77 (1916), pp. 536–545

Following Dedekind,[165] by an *isomorphic mapping of the field* $\mathfrak{A}$ *onto a system* $\mathfrak{B}$ – which subsequently also proves to be a field – one understands *such a single-valued correspondence that to every element of* $\mathfrak{A}$ *there corresponds one and only one element of* $\mathfrak{B}$; *and that to the sum, the difference, the product, and the quotient of any two elements of* $\mathfrak{A}$ *there are again assigned the sum, difference, product, and quotient of the uniquely corresponding elements of* $\mathfrak{B}$.

Let such a mapping be mediated by a – by the preceding, single-valued – function $f(z)$. If z then runs through all elements $x, y, \dots$ of the field $\mathfrak{A}$, then $f(z)$ is characterized by the functional equations

$$(1)\quad f(x+y) = f(x) + f(y);\quad (2)\quad f(x-y) = f(x) - f(y);$$
$$(3)\quad f(x \cdot y) = f(x) \cdot f(y);\qquad (4)\quad f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}.$$

In what follows the most general single-valued solutions $f(z)$ *of these functional equations are to be given when* $x, y, \dots$ *run through all real and complex numbers; thus, the most general function* $f(z)$ *which effects an isomorphic mapping of the field of all complex numbers.*[166] The solution for arbitrary fields follows quite analogously.

G. Hamel[167] constructed the most general solutions of the functional equation (1) by means of a *linear* basis of all numbers. The construction given here for the solutions of (1) to (4) – hence of the mapping function relative to which the rational and algebraic operations are invariant – uses the *rational* and *algebraic* basis of all numbers. After the isomorphic mapping is discussed more closely in 1, necessary conditions for $f(z)$ are given in 2; in 3 these are recognized as sufficient and lead to the actual construction.[168] – The solutions of (1) to (4), with the known exceptions $f(z) = z$ or $\overline{z}$, are discontinuous; in 4 it is further shown that they are "extremely discontinuous" – a fact which G. Hamel proved for the real solutions of (1), but which need not be fulfilled there in the complex domain.[169]

1. We first draw some consequences from the functional equations (1) to (4), directly for general fields $\mathfrak{A}$. By (1), single-valuedness implies $f(0) = 0$. If $f(y)$ does not vanish, then the original y cannot vanish either, and the functional equations (1) to (4) therefore show that *the mapping system* $\mathfrak{B}$ *of all values* $f(z)$ *again forms a field.* Conversely, if the initial condition $f(0) = 0$ is prescribed, then (2) implies the single-valuedness of the function $f(z)$: *the requirement of single-valuedness and the initial condition* $f(0) = 0$ *are therefore equivalent.* From (4) it follows that $f(z)$ cannot vanish identically; from (3) further, that $f(z)$ vanishes only for $z = 0$. For from $f(y) = 0$ and $y \neq 0$ it would follow, since z/y also belongs to $\mathfrak{A}$, that

$$f(z) = f\left(\frac{z}{y} \cdot y\right) = f\left(\frac{z}{y}\right) \cdot f(y) = 0,$$

which contradicts (4) by making $f(z)$ vanish identically. From $f(x) = f(y)$ it follows therefore that $x = y$; that is, to each value of $f(z)$ there also corresponds one and only one value of z: $f(z)$ *is a reversibly single-valued function of* z. As the functional equations show, the mapping mediated by the inverse function is again isomorphic. Since every rational operation is composed of a finite number of additions, multiplications, and their inverses, one can also say: *By the isomorphic mapping a one-to-one relation is established between the elements of* $\mathfrak{A}$ *and of* $\mathfrak{B}$, *in such a way that all rational relations holding among finitely many elements of* $\mathfrak{A}$,

$$F(x, y, \dots, t) = 0,$$

[165] Cf. for example Dirichlet-Dedekind, *Lectures on Number Theory* (4th ed.), Supplement XI, § 161. The designation "isomorphic mapping" or "isomorphism", modeled on group theory, occurs only in the later literature; Dedekind speaks of the "permutations of a field".

[166] This question was occasionally put to me by Mr. Landau. As I noticed afterward, part of the solutions already occurs in H. Lebesgue, Sur les transformations ponctuelles, transformant les plans en plans, Atti Acad. Torino, 1906/07; and likewise implicitly in A. Ostrowski, Über einige Fragen der allgemeinen Körpertheorie, Crelle's Journal 143 (1913), § 2.

[167] G. Hamel, Eine Basis aller Zahlen und die unstetigen Lösungen der Funktionalgleichung: $f(x+y) = f(x) + f(y)$, Math. Ann. 60, p. 459 (1905).

[168] Cf. the parallel considerations in my paper: Die allgemeinsten Bereiche aus ganzen transzendenten Zahlen. Math. Ann. 77, p. 103 (1915), especially §§ 8 and 10.

[169] Hamel's term "totally discontinuous" is used in a different sense in the rest of the literature.

are preserved in $\mathfrak{B}$, and conversely.

It should also be noted that for a field $f(z)$ is already characterized by the functional equations (1) and (3), by the requirement that $f(z)$ not vanish identically, and by the initial condition $f(0) = 0$. Indeed, (2) follows from (1) by replacing x by the element $(x - y)$ belonging to $\mathfrak{A}$; then (2) and $f(0) = 0$ give the single-valuedness of $f(z)$. As was shown above, it follows further from (3), by replacing x by the element x/y belonging to $\mathfrak{A}$, that $f(y)$ cannot vanish for $y \neq 0$, and hence the validity of (4) and at the same time one-to-oneness. If instead of a field one takes an integral domain $\mathfrak{S}$ as basis, x/y need not belong to $\mathfrak{S}$, and from (3) one cannot infer one-to-oneness. Here, however, the following most familiar formulation suffices: an isomorphic mapping is a *one-to-one* correspondence between the elements of $\mathfrak{S}$ and $\mathfrak{S}'$ such that to the sum and the product there always correspond sum and product.[170]

2. From now on, for simplicity, instead of $\mathfrak{A}$ let the field $\mathfrak{C}$ of all real and complex numbers be taken as the basis. The first matter is to discuss the systems of values which $f(z)$ assumes. From (3) or (4) it follows that $f(1) = 1$, and thence, by the functional equations, for every *rational number* α, $f(\alpha) = \alpha$; *therefore every rational number corresponds to itself under the mapping.* Since an *algebraic number* β satisfies an irreducible equation with rational numerical coefficients $F(\beta, \alpha) = 0$, no. 1 and the preceding observation give, for $f(\beta)$, the equation $F(f(\beta), \alpha) = 0$; *every algebraic number therefore goes into itself or into one of its conjugates*, and by one-to-oneness the totality of algebraic numbers again corresponds to the field of all algebraic numbers. In order to recognize the behavior of the *transcendental* numbers, and at the same time to separate the conjugate values, we take as basis a *rational basis of all numbers*,[171] i.e. a system Θ of numbers ϑ which permits all numbers to be expressed rationally – with rational numerical coefficients[172] – while in at least one well-ordering no basis number is rationally expressible through finitely many preceding ones. The existence of this basis follows exactly analogously to linear and algebraic bases from the well-ordering theorem.

Let $\mathfrak{C}$ be subjected to a well-ordering Ω. Then every number is either expressible rationally through finitely many preceding numbers, or is rationally independent of the preceding ones. These latter numbers form the basis Θ, and induction shows that indeed every number is rationally expressible through finitely many ϑ's.

For each basis number ϑ, the "segment field $\mathfrak{K}(\vartheta)$" is defined: it consists of all rational combinations of those basis numbers which precede ϑ in the well-ordering. As the segment field of the first basis number one is to regard the field $\mathfrak{R}$ of all rational numbers. The number ϑ can display two sorts of behavior with respect to $\mathfrak{K}(\vartheta)$: it can be algebraically dependent on $\mathfrak{K}(\vartheta)$, or algebraically independent of it. Since, by no. 1, all relations valid in $\mathfrak{C}$ are also valid in the mapping field, and conversely, the totality of values $f(\vartheta)$ corresponding to the ϑ's forms a basis of the mapping range, well-ordered by means of the well-ordering of Θ itself. In particular, the segment field $\mathfrak{K}(\vartheta)$ corresponds to the segment field $\mathfrak{K}(f(\vartheta))$; and according as ϑ is algebraically dependent on, or independent of, $\mathfrak{K}(\vartheta)$, the same holds for $f(\vartheta)$ with respect to $\mathfrak{K}(f(\vartheta))$. In the dependent case the irreducible equations for ϑ and $f(\vartheta)$ also correspond. The totality of those values ϑ which are in each case algebraically independent of $\mathfrak{K}(\vartheta)$ form – as induction again shows – an *algebraic basis of all numbers*,[173] i.e. a system H of numbers η which permits all numbers to be expressed algebraically, while no algebraic relations hold among the η's themselves. To this algebraic basis H there corresponds in the mapping range again an algebraic basis Z, which by one-to-oneness has the same cardinality as H, namely the cardinality of the continuum, since algebraic extension is known not to increase cardinality.[174] From the knowledge of the values $f(\vartheta)$, however, $f(z)$ is immediately obtained for all systems of values, since from $z = \psi(\vartheta)$ there always follows $f(z) = \psi(f(\vartheta))$.

3. We now show that the necessary conditions found above actually also provide the construction of $f(z)$. The basis Z corresponding one-to-one to the algebraic basis H is constructed and assigned to H as follows. Put a second well-ordering Ω' in place, which supplies an algebraic basis Z' of all numbers. Let Z, with elements ξ, be a subset of Z' of the same cardinality as Z', and hence as H. To each element η_i of H assign one and only one element ξ_i, with the same numbering, by the rule: ξ_i is the *first* of all elements of Z in the well-ordering Ω' which has not been assigned to any element of H preceding η_i.

Now define $f(z)$ as follows:

(a) *for all rational numbers* α, *by* $f(0) = 0$, $f(\alpha) = \alpha$;

(b) *for all quantities of the algebraic basis* H, *by* $f(\eta_i) = \xi_i$;

[170] Cf. on no. 1: Dedekind, loc. cit., § 161.

[171] Cf. my cited paper on "whole transcendental numbers", § 6.

[172] By a "rational function" I shall always mean one whose coefficients are also rational numbers.

[173] Cf. E. Zermelo, Über ganze transzendente Zahlen, § 1, Math. Ann. 75 (1914).

[174] Cf. E. Steinitz, Algebraische Theorie der Körper, Crelle's Journal 137 (1910), § 24.

(c) *for all remaining quantities ϑ of the rational basis Θ – those algebraically dependent on the segment field $\mathfrak{K}(\vartheta)$, and hence satisfying an equation $F(\vartheta; \vartheta_{i_1}, \ldots, \vartheta_{i_\nu}) = 0$ irreducible in $\mathfrak{K}(\vartheta)$ – as a root of the equation $F(f(\vartheta); f(\vartheta_{i_1}), \ldots, f(\vartheta_{i_\nu})) = 0$;*

(d) *for all remaining values z, which by the definition of Θ are representable as $z = \psi(\vartheta_{k_1}, \ldots, \vartheta_{k_\nu})$, by $f(z) = \psi(f(\vartheta_{k_1}), \ldots, f(\vartheta_{k_\nu}))$.*[175]

To show that the so-defined $f(z)$ actually satisfies the functional equations (1) to (4), by no. 1 it suffices to prove this for (1) and (3), since $\mathfrak{C}$ is a field and $f(z)$, because of (a) and (b), cannot vanish identically and also satisfies the initial condition $f(0) = 0$. By means of the rational basis Θ, express $x, y, x + y$ as

$$x = \psi_1(\vartheta_1 \cdots \vartheta_t), \quad y = \psi_2(\vartheta_1 \cdots \vartheta_t), \quad (x + y) = \psi_3(\vartheta_1 \cdots \vartheta_t).$$

Thus the proof that $f(z)$ satisfies the functional equation (1),

$$f(x) + f(y) - f(x + y) = 0,$$

is, by (d), identical with showing that from

$$\psi_1(\vartheta) + \psi_2(\vartheta) - \psi_3(\vartheta) = \frac{H_1(\vartheta)}{H_2(\vartheta)} = 0$$

there always follows

$$\psi_1(f(\vartheta)) + \psi_2(f(\vartheta)) - \psi_3(f(\vartheta)) = \frac{H_1(f(\vartheta))}{H_2(f(\vartheta))} = 0,$$

where H_1, H_2 denote integral rational functions of their arguments. The functional equation (3) leads to precisely the same form of condition; hence the satisfaction of all the functional equations is identical with the following condition – necessary also by nos. 1 and 2: for *every* integral rational function $H(\vartheta)$, from $H(\vartheta) = 0$ there always follows $H(f(\vartheta)) = 0$, and from $H(\vartheta) \neq 0$ there follows $H(f(\vartheta)) \neq 0$, the latter because of the occurrence of the denominator.

That this is indeed the case is shown by induction. Suppose that in $H(\vartheta_1 \cdots \vartheta_t)$ the basis quantity ϑ_t is the last, in the well-ordering, among $\vartheta_1 \cdots \vartheta_t$.[176] Then $H(\vartheta_1 \cdots \vartheta_t) = 0$ may be regarded as a relation satisfied by ϑ_t with respect to the segment field $\mathfrak{K}(\vartheta_t)$, and likewise $H(f(\vartheta_1) \cdots f(\vartheta_t)) = 0$ as a relation satisfied by $f(\vartheta_t)$ with respect to $\mathfrak{K}(f(\vartheta_t))$. If not all relations $H(\vartheta) = 0$ were also fulfilled by $f(\vartheta)$, and conversely, then there would have to be a first basis quantity, say ϑ_0, for which at least one relation valid with respect to $\mathfrak{K}(\vartheta_0)$ failed to be preserved in the mapping field, or conversely, while the preceding segment fields $\mathfrak{K}(\vartheta_0)$ and $\mathfrak{K}(f(\vartheta_0))$ were isomorphic. In particular, by (a), the segment field of the first basis quantity of Θ, the field $\mathfrak{R}$ of all rational numbers, is isomorphic to its mapping field. Suppose now that $H(\vartheta_0)$ has the form

$$H(\vartheta_0) = a_0(\vartheta) \cdot \vartheta_0^\chi + a_1(\vartheta) \cdot \vartheta_0^{\chi-1} + \cdots + a_\chi(\vartheta),$$

where the $a_i(\vartheta)$ are quantities from $\mathfrak{K}(\vartheta_0)$. Then there are two possibilities.

1) ϑ_0 is algebraically independent of $\mathfrak{K}(\vartheta_0)$. Then $H(\vartheta_0) = 0$ is fulfilled identically in ϑ_0,[177] so that $a_i(\vartheta) = 0$ for all coefficients, and hence by the isomorphism of $\mathfrak{K}(\vartheta_0)$ and $\mathfrak{K}(f(\vartheta_0))$ also $a_i(f(\vartheta)) = 0$. From $H(\vartheta_0) = 0$ there follows, therefore, always $H(f(\vartheta_0)) = 0$. Conversely, suppose $H(f(\vartheta_0)) = 0$. Since ϑ_0, being algebraically independent of $\mathfrak{K}(\vartheta_0)$, belongs to the algebraic basis H, $f(\vartheta_0)$ belongs by (b) to the algebraic basis Z and is algebraically independent of $\mathfrak{K}(f(\vartheta_0))$. Thus all $a_i(f(\vartheta))$ vanish, and by the isomorphism all $a_i(\vartheta)$ vanish. From $H(f(\vartheta_0)) = 0$ follows $H(\vartheta_0) = 0$, or from $H(\vartheta_0) \neq 0$ also $H(f(\vartheta_0)) \neq 0$.

2) ϑ_0 is algebraically dependent on $\mathfrak{K}(\vartheta_0)$. Then $H(\vartheta_0)$ is divisible by the equation $F(\vartheta_0)$ irreducible with respect to $\mathfrak{K}(\vartheta_0)$; that is, identically in t,

$$H(t; a_i(\vartheta)) = F(t; \vartheta) \cdot G(t; \vartheta_i).$$

Since all the coefficients which occur belong to $\mathfrak{K}(\vartheta_0)$, the corresponding relation is fulfilled in the isomorphic field $\mathfrak{K}(f(\vartheta_0))$:

$$H(t; a_i(f(\vartheta))) = F(t; f(\vartheta)) \cdot G(t; f(\vartheta)).$$

[175] In (d), (a) is contained as a special case.

[176] The expressions $H(\vartheta)$ which are of degree zero in the ϑ's go over into themselves under the mapping and therefore need not be investigated further.

[177] Because $x, y, \ldots$ can be expressed by the ϑ's, this form of relation can very well occur.

But by (c), $f(\vartheta_0)$ was chosen to be a root of $F(t; f(\vartheta))$. Hence from $H(\vartheta_0) = 0$ there follows $H(f(\vartheta_0)) = 0$.

Conversely, if $H(f(\vartheta_0)) = 0$, then the divisibility of $H(t; a_i(f(\vartheta)))$ by the function $F(t; f(\vartheta))$, irreducible with respect to $\mathfrak{K}(f(\vartheta_0))$, implies for the isomorphic field $\mathfrak{K}(\vartheta_0)$ the divisibility of $H(t; a_i(\vartheta))$ by $F(t; \vartheta)$. And since $F(t; f(\vartheta))$ arose from the irreducible equation for ϑ_0 by the isomorphic relation, and this relation is one-to-one, $F(t; \vartheta)$ necessarily represents the irreducible function whose root is ϑ_0. Thus from $H(f(\vartheta_0)) = 0$ follows $H(\vartheta_0) = 0$, or from $H(\vartheta_0) \neq 0$ also $H(f(\vartheta_0)) \neq 0$.

From isomorphic segment fields $\mathfrak{K}(\vartheta_0)$ and $\mathfrak{K}(f(\vartheta_0))$, therefore, isomorphic fields also arise by adjunction of ϑ_0 and $f(\vartheta_0)$, respectively; the assumption that this fails for some ϑ_0 leads to a contradiction. It is thus proved that *the function $f(z)$ defined by (a) to (d) actually represents a solution of the functional equations (1) to (4), and, by 2, the most general one.*

All considerations used in constructing $f(z)$ have used only the fact that the totality $\mathfrak{C}$ of all numbers forms a field; hence they hold for every abstractly defined field $\mathfrak{A}$. Here one should note that the field $\mathfrak{R}$ of rational numbers is replaced by the "prime field" of $\mathfrak{A}$, i.e. by the field derived from the unit element of $\mathfrak{A}$ by the operations of addition and multiplication and their inverses. If, therefore, in (a) to (d) one replaces the rational numbers by the elements of the prime field, then *the function $f(z)$ so obtained effects the most general isomorphic mapping of an arbitrary abstractly defined field.*

4. It remains to show that the functions $f(z)$ defined for the field $\mathfrak{C}$ of all real and complex numbers – which, except for $f(z) = z$ or $\overline{z}$, are known to be discontinuous – are *extremely discontinuous*; that is, in every neighborhood of any real or complex number z_0 there are values z for which $f(z)$ comes arbitrarily close to any prescribed value Z_0. This extreme discontinuity belongs, as G. Hamel proved loc. cit., to the *real* solutions $\varphi(x)$, defined for all *real* numbers, of the functional equation (1). But, as the examples below show, it need not belong to the solutions defined on the complex numbers. Hamel argues as follows. If $\varphi(x)$ differs from $C \cdot x$, then there are at least two values of the linear basis, say x_1 and x_2, such that $\varphi(x_1) : x_1 \neq \varphi(x_2) : x_2$. Then the two equations

$$\alpha x_1 + \beta x_2 = a; \qquad \alpha\varphi(x_1) + \beta\varphi(x_2) = b$$

have a solution in real numbers α, β; consequently a and b can be approximated arbitrarily closely by rational numbers α, β. Because α, β are rational, $\alpha\varphi(x_1) + \beta\varphi(x_2)$ is then actually $\varphi(\alpha x_1 + \beta x_2)$. This argument breaks down for complex systems of values. In fact one can give, also in the complex domain, solutions of (1) which are discontinuous but not extremely discontinuous; for example,

$$\varphi(z) = \varphi(x + iy) = \varphi(x) + i(y + \varphi(x)),$$

where $\varphi(x)$ denotes a real solution which is extremely discontinuous in the real domain; or, still more simply, $\varphi(z) = \varphi(x)$. In both cases the real part of $\varphi(z)$ is extremely discontinuous, while the imaginary part is determined by the real part.

This behavior, and at the same time the behavior of the function $f(z)$, becomes completely clear if one introduces the concept of rank. We put $z = x + iy$, $\varphi(z) = X + iY$, and call $\varphi(z)$ of rank four, three, two, or one, respectively, according as there exist no, one, two, or three linearly independent relations

$$c_1 x + c_2 y + c_3 X + c_4 Y = 0,$$

where of course the c's are real numbers.

1) Let $\varphi(z)$ have *rank four.* Then there exist at least four values of the linear basis, say z_1, z_2, z_3, z_4, such that the determinant

$$\begin{vmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \\ X_1 & X_2 & X_3 & X_4 \\ Y_1 & Y_2 & Y_3 & Y_4 \end{vmatrix}$$

does not vanish. The equations

$$\begin{aligned} \alpha x_1 + \beta x_2 + \gamma x_3 + \delta x_4 &= a, \\ \alpha y_1 + \cdots + \delta y_4 &= b, \\ \alpha X_1 + \cdots + \delta X_4 &= A, \\ \alpha Y_1 + \cdots + \delta Y_4 &= B \end{aligned}$$

therefore have a solution in real numbers $\alpha, \ldots, \delta$, and a, b, A, B can be approximated arbitrarily closely by rational numbers $\alpha, \beta, \gamma, \delta$. Moreover one has

$$\alpha(X_1 + iY_1) + \beta(X_2 + iY_2) + \gamma(X_3 + iY_3) + \delta(X_4 + iY_4) = \varphi(\alpha z_1 + \beta z_2 + \gamma z_3 + \delta z_4).$$

"In general", therefore, the solutions $\varphi(z)$ are extremely discontinuous also in the complex domain; the *discontinuity values* of $\varphi(z)$ in the neighborhood of $z_0 = a + ib$ *form a two-dimensional linear manifold.*

2) Let $\varphi(z)$ have *rank three.* Then there are at least three systems of values of the linear basis, say z_1, z_2, z_3, such that the above determinant has rank three. The values a, b, A, B satisfying the relation

$$c_1 a + c_2 b + c_3 A + c_4 B = 0,$$

and only these, can be approximated arbitrarily closely; the *discontinuity values* form a *one-dimensional linear manifold* determined by $z_0 = a + ib$. As the examples given show, this case can actually occur.

3) Let $\varphi(z)$ have *rank two.* Since one can always choose two complex numbers z_1, z_2 such that

$$\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} \neq 0,$$

the relations have the form

$$X = \lambda_1 x + \lambda_2 y; \qquad Y = \mu_1 x + \mu_2 y.$$

The resulting expression

$$\varphi(z) = (\lambda_1 x + \lambda_2 y) + i(\mu_1 x + \mu_2 y)$$

is precisely the most general *continuous* solution of the functional equation (1) in the domain of complex numbers.

4) Let $\varphi(z)$ have *rank one.* This case cannot occur in the domain of complex numbers; in the domain of real numbers it means the *continuous* solution $\varphi(x) = C \cdot x$.

We now show that the *solutions $f(z)$ of the functional equations (1) to (4) defined for all real and complex numbers can have only rank two or rank four.* Here rank two gives only the two *continuous* functions $f(z) = z$ or $\overline{z}$, as the specialization $f(1) = 1$ and $f(i) = \pm i$, combined with 3), shows. Since rank one has already been excluded, it remains to exclude rank three. We therefore assume that at least one relation exists,

$$c_1 x + c_2 y + c_3 X + c_4 Y = 0,$$

so that $f(z)$ has rank three, or possibly lower rank, and show that under this assumption the latter case always occurs. As again the specialization $f(1) = 1$, $f(i) = \pm i$ shows, the relation becomes

$$c_3(X - x) + c_4(Y \mp y) = 0,$$

where c_3 and c_4 do not both vanish. First suppose, say, $c_4 = 0$. The relation $(X - x) = 0$ then means that to a purely imaginary z there also corresponds a purely imaginary $f(z)$. Thus for every real y one has $f(iy) = i \cdot \mathfrak{Y}$, where $\mathfrak{Y}$ is again real. From this, because

$$f(iy) = \pm i f(y),$$

one obtains also $f(y) = \pm\mathfrak{Y}$; hence to every real z there also corresponds a real $f(z)$, and because of $(X - x)$ all real values go into themselves. Thus one has only $f(z) = z$ or $\overline{z}$, so actually rank two. The conclusion is entirely analogous when $c_3 = 0$, $c_4 \neq 0$.

Now let c_3 and c_4 be different from zero, so that the relation can be written in the form

$$(Y \mp y) = c \cdot (X - x)$$

with real, nonzero, finite c. But this means that for $y = \pm cx$ one also has $Y = c \cdot X$. Hence, taking account of the functional equations, for every real x one has

$$f\{x \cdot (1 \pm ic)\} = \mathfrak{X}(1 + ic) = f(x) \cdot (1 + if(c)),$$

where $\mathfrak{X}$ is again real. But here, by no. 1, the factor of $f(x)$ cannot vanish identically, and division gives

$$f(x) = \mathfrak{X}(\sigma + i\tau)$$

with real σ and τ, independent of x and $\mathfrak{X}$. In particular, to $x = 1$ there also belongs a real X_0, and the condition $1 = X_0(\sigma + i\tau)$ shows that necessarily $\tau = 0$, so that $f(x) = \sigma \cdot \mathfrak{X}$. Thus to every real z there also corresponds a real $f(z)$; equivalently, from $y = 0$ there necessarily follows $Y = 0$. Hence the linear relation for real z becomes $X - x = 0$;[178] every real value corresponds to itself. One again has only $f(z) = z$ or $\overline{z}$, so actually rank two. Rank three is thereby excluded in all cases. Since, however – as the investigations of the first sections show – discontinuous solutions actually exist, and these must therefore necessarily have rank four, the following theorem holds by 1): *"All solutions $f(z)$ of the functional equations (1) to (4) – with the exception of the continuous solutions $f(z) = z$ or $\overline{z}$ – are extremely discontinuous in the real and in the complex domain."*[179]

Erlangen, 30 October 1915.

[178] Here one uses the fact that $c \neq 0$, so that neither c_3 nor c_4 vanishes, whereas above only $c_4 \neq 0$ was used; therefore the case where one of these two vanishes had to be dealt with in advance.

[179] Lebesgue shows loc. cit. that the real and imaginary parts of $f(z)$ are both "non-measurable" functions of x and y; as the example at the beginning of no. 4, $\varphi(z) = \varphi(x) + i(y + \varphi(x))$, shows, this does not yet imply extreme discontinuity in the complex domain.

11. Equations with Prescribed Group

Math. Ann. 78 (1918), pp. 221–229

The problem of constructing equations with prescribed group can be attacked in two directions, which one may briefly call the "irrational" direction, characterizing the roots, and the "rational" direction, characterizing the coefficients.

In the "irrational" direction, which works function-theoretically and arithmetically, lies Kronecker's theorem that all Abelian fields in the domain of the rational numbers are cyclotomic fields, and the corresponding theorems for relatively Abelian fields with respect to quadratic number fields. These theorems give, on the one hand, the realizability of all Abelian groups with respect to these special domains of rationality; on the other hand, they supply the totality of the equations and at the same time give a deeper insight into the structure of the number fields thereby defined. Each individual theorem, however, is restricted to a special domain of rationality; and every new domain of rationality requires an entirely new treatment.

The "rational", algebraic direction rests on Hilbert's irreducibility theorem, according to which in the coefficient representation one may restrict oneself to a parameter representation. To this direction belongs, for example, the existence of arbitrarily many equations with alternating group with respect to any prescribed domain of rationality. Here the insight, described above, into the structure of the fields defined by the equations is naturally absent; but the advantage of this "rational" direction lies in the fact that the realizability of the groups is proved for *all domains of rationality simultaneously*; nevertheless, the parameter representations known up to now do not in general yield the totality of the equations.[180]

What follows is a contribution to the "rational" direction; *sufficient conditions* are given for the *existence of parameter representations which, for any domain of rationality, supply the totality of the equations with prescribed group.* These conditions consist in this: the invariant field belonging to the group – Lagrange's genus domain – can be represented rationally by independent functions of the domain, that is, it possesses a "minimal basis"; or, expressed otherwise, it is isomorphic to the field of all rational functions of n indeterminates. As will still be shown in § 2, this is satisfied, for example, for all groups occurring in equations of the third and fourth degrees. The actual construction and more exact discussion of these equations of the third and fourth degrees is found in the dissertation of F. Seidelmann,[181] an extract from which immediately follows this communication.

The question of the minimal basis of Lagrange's genus domains was – as already mentioned in the paper "Körper und Systeme rationaler Funktionen" (Math. Ann. 76) – raised to me by E. Fischer independently of the connection just described, and gave the impetus for the present investigations.[182]

§ 1.

Sufficient Conditions for the Parameter Representation.

1. The possibility of reducing the construction of equations with prescribed group to parameter representation follows from Hilbert's irreducibility theorem, which says the following. Suppose that the equation

$$f(x) = x^n + F_1(\lambda_1 \cdots \lambda_r)x^{n-1} + \cdots + F_n(\lambda_1 \cdots \lambda_r) = 0 \tag{1}$$

– where $F_i(\lambda_1 \cdots \lambda_r)$ denotes rational functions of the independent parameters $\lambda_1 \cdots \lambda_r$, with coefficients from a number field[183] Ω – has, with respect to the field $\Omega(\lambda_1 \cdots \lambda_r)$ of all rational functions of $\lambda_1 \cdots \lambda_r$ with coefficients from Ω, the group Γ. Then one can replace the parameters λ by arbitrarily many rational numbers in such a way that the resulting numerical equation

$$g(x) = x^n + a_1x^{n-1} + \cdots + a_n = 0 \tag{2}$$

[180] For solvable equations, the parameter representation of the coefficients can be replaced by that of the roots. Recently F. Mertens has given, for certain groups of degree 8, most general root representations: "Gleichungen 8ten Grades mit Quaternionengruppen", Sitzb. d. Ak. d. Wiss., Wien, Abt. IIa, Bd. 125, p. 735. As I gather from that note, G. Bucht had already earlier given the most general root expressions for always constructible normal forms of equations of degree 3 and 4 and of equations of degree 8 with the groups mentioned: "Über einige algebraische Körper achten Grades", Arkiv for Math., Astron. och Physik, Bd. 6, Nr. 30. [Addition at correction.]

[181] Die Gesamtheit der kubischen und biquadratischen Gleichungen mit Affekt bei beliebigem Rationalitätsbereich. Erlangen 1916.

[182] Cf. Part 2 of the preliminary communication on "Rationale Funktionenkörper", Jahresb. d. d. Math. Vereinig., Bd. 22, 1913.

[183] Instead of the number field, a general domain of rationality could also occur.

has, with respect to Ω, precisely the group Γ. Such a $g(x)$ shall be denoted more precisely by g_Γ.

The question now is whether *such a parameter representation* (1) *can be given that the resulting numerical equation* (2) *runs through the totality of all* g_Γ *when* $\lambda_1 \cdots \lambda_r$ *run through all quantities from* Ω – with the exclusion of those quantities from Ω which cause a reduction of the group.[184] In other words: the nonlinear system of equations

$$F_1(\lambda_1 \cdots \lambda_r) = a_1; \quad \cdots; \quad F_n(\lambda_1 \cdots \lambda_r) = a_n \tag{3}$$

should possess, for every system of values $a_1 \cdots a_n$ corresponding to a g_Γ, a solution, and indeed in quantities $\lambda_1 \cdots \lambda_r$ from Ω. Since the system (3) is nonlinear, one must from the outset introduce a restriction when demanding that the totality of the g_Γ be obtained through a parameter representation. Namely, in such nonlinear systems there can always occur singular systems of values $a_1 \cdots a_n$, satisfying an algebraic relation $H(a_1 \cdots a_n) = 0$, for which *no* solution exists;[185] for them the parameter representation (1) fails and a supplementary representation is necessary. But in the present case these singular values of $a_1 \cdots a_n$, as will be shown, can be characterized simply.

2. To obtain such a parameter representation, we impose the stronger requirement that the λ_i be *natural irrationalities belonging to the group, identically in the roots regarded as indeterminates.* By this we mean the following. If in (1) one replaces the parameters $\lambda_1 \cdots \lambda_r$ by suitably chosen *rational* functions $\varphi_1(x) \cdots \varphi_r(x)$ of the indeterminates $x_1 \cdots x_n$ – with coefficients from Ω – then (1) is to pass into

$$h(x) = x^n - \sigma_1(x)x^{n-1} + \sigma_2(x)x^{n-2} \cdots \pm \sigma_n(x), \tag{4}$$

where $\sigma_1(x) \cdots \sigma_n(x)$ denote the elementary symmetric functions of $x_1 \cdots x_n$; and $h(x)$, with respect to the domain of rationality $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ thereby arising from $\Omega(\lambda_1 \cdots \lambda_r)$, is to have the group Γ. Because of the independence of the parameters, $\varphi_1(x) \cdots \varphi_r(x)$ are also to be algebraically independent functions.

To make this requirement precise, one must clarify the connection of $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ with the group Γ. For this we first determine the most general domain of rationality K with respect to which $h(x)$ becomes h_Γ. Let $\mathfrak{L}_\Gamma$ denote the field of all rational functions with rational integral coefficients of $x_1 \cdots x_n$ which admit Γ – also called the "Lagrangian genus domain" belonging to Γ, or the invariant field of Γ – and let $\Omega(\mathfrak{L}_\Gamma)$ denote the corresponding field arising by adjoining $\mathfrak{L}_\Gamma$ to Ω. Thus $\mathfrak{L}_\Gamma$ is a subfield of the field $\mathfrak{R}(x_1 \cdots x_n)$ of all rational functions with rational integral coefficients of $x_1 \cdots x_n$. According to the definition of the group, the most general domain K is then given by *any field which contains* $\mathfrak{L}_\Gamma$ *as a subfield and whose intersection with* $\mathfrak{R}(x_1 \cdots x_n)$ *is exactly* $\mathfrak{L}_\Gamma$. Correspondingly, for every domain K containing Ω, the intersection with $\Omega(x_1 \cdots x_n)$ is given by $\Omega(\mathfrak{L}_\Gamma)$. Therefore, in particular, since by our requirement $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ is to be a domain K, the intersection of $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ with $\Omega(x_1 \cdots x_n)$ is given by $\Omega(\mathfrak{L}_\Gamma)$; and since on the other hand $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ is a subfield of $\Omega(x_1 \cdots x_n)$, $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ is exactly equal to $\Omega(\mathfrak{L}_\Gamma)$. The requirement therefore says that *Lagrange's genus domain* $\Omega(\mathfrak{L}_\Gamma)$ *is to arise by adjoining the algebraically independent functions* $\varphi_1(x) \cdots \varphi_r(x)$ *to* Ω; here one must have $r = n$, since $\mathfrak{L}_\Gamma$ contains the n algebraically independent elementary functions, while on the other hand no more than n functions can be algebraically independent. The functions $\varphi_1(x) \cdots \varphi_n(x)$ shall be said, as we shall also put it, to form a *minimal basis* of $\Omega(\mathfrak{L}_\Gamma)$; or also, $\mathfrak{L}_\Gamma$ possesses a minimal basis with coefficients from Ω.

3. It is now essential that this whole consideration can be reversed, and thus actually leads to sufficient conditions[186] for the desired parameter representation. We therefore assume that $\varphi_1(x) \cdots \varphi_n(x)$ is a minimal basis of $\Omega(\mathfrak{L}_\Gamma)$; and let Ω be the smallest field containing the coefficients of $\varphi_1(x) \cdots \varphi_n(x)$. Then Ω is necessarily an algebraic number field, and can in particular be the field $\mathfrak{R}$ of all rational numbers. Since $\sigma_1(x), \ldots, \sigma_n(x)$ belong to $\Omega(\mathfrak{L}_\Gamma)$, there exist identities in $x_1 \cdots x_n$ – where the F_i denote rational functions of their arguments:

$$\sigma_1(x) = F_1(\varphi_1(x) \cdots \varphi_n(x)); \quad \cdots; \quad \sigma_n(x) = F_n(\varphi_1(x) \cdots \varphi_n(x)); \tag{5}$$

whereas conversely $\varphi_1(x) \cdots \varphi_n(x)$, and hence also $\varphi(x) = \mu_1\varphi_1(x) + \cdots + \mu_n\varphi_n(x)$, depend algebraically on $\sigma_1(x) \cdots \sigma_n(x)$ by means of the equation irreducible with respect to $\Omega(\sigma_1(x) \cdots \sigma_n(x))$:

$$G(\varphi(x); \sigma_1(x) \cdots \sigma_n(x)) = 0, \tag{6}$$

which is again an identity in the x.

[184] The occurrence of a double root of $g(x) = 0$ is also to be included here.

[185] This is to be carried out shortly in general in a paper on the "alternative in nonlinear systems of equations."

[186] The requirement in 2. was enlarged relative to 1., so that the conditions need not be necessary.

By virtue of (5), this identity (6) passes into

$$G(\varphi(x); F_1(\varphi_1 \cdots \varphi_n); \cdots ; F_n(\varphi_1 \cdots \varphi_n)) = H(\varphi_1(x) \cdots \varphi_n(x)) = 0; \tag{7}$$

and because of the algebraic independence of $\varphi_1(x) \cdots \varphi_n(x)$, (7) is fulfilled not only identically in x, but also in the φ_i; that is, identically in independent parameters λ one has

$$H(\lambda_1 \cdots \lambda_n) = G(\mu_1\lambda_1 + \cdots + \mu_n\lambda_n; F_1(\lambda_1 \cdots \lambda_n); \cdots ; F_n(\lambda_1 \cdots \lambda_n)) = 0. \tag{8}$$

From the identity (8) it follows that the equation

$$f(x) = x^n - F_1(\lambda_1 \cdots \lambda_n)x^{n-1} + F_2(\lambda_1 \cdots \lambda_n)x^{n-2} \cdots \pm F_n(\lambda_1 \cdots \lambda_n) = 0 \tag{9}$$

has the group Γ *with respect to* $\Omega(\lambda_1 \cdots \lambda_n)$. For this identity shows that the rationally known $\lambda = \mu_1\lambda_1 + \cdots + \mu_n\lambda_n$ is a function of the roots belonging to Γ; but no further function belonging to a subgroup of Γ can be rationally known, as is shown by the substitution $\lambda_i = \varphi_i(x)$; and under the same substitution λ is also different from all its conjugates. If further Ω^* is any number field containing Ω, then $f(x)$ *has the group* Γ *with respect to* $\Omega^*(\lambda_1 \cdots \lambda_n)$; for by 2. the group is not reduced under the substitution $\lambda_i = \varphi_i(x)$ even after adjoining Ω^*.

It remains to prove that, in the sense modified in 1., $f(x)$ actually runs through the totality of all g_Γ. For this, observe that by (5) the $F_i(\lambda)$ are algebraically independent functions of their arguments λ. The system of equations

$$F_1(\lambda_1 \cdots \lambda_n) = -a_1; \quad F_2(\lambda_1 \cdots \lambda_n) = a_2; \quad \cdots ; \quad F_n(\lambda_1 \cdots \lambda_n) = \pm a_n \tag{10}$$

therefore possesses solutions for every arbitrary system of values $a_1 \cdots a_n$ – with the exception of the singular ones still to be specified. For every system of values $a_1 \cdots a_n$ corresponding to a g_Γ with respect to Ω^*, the corresponding λ_i, as natural irrationalities belonging to the group,[187] become quantities from Ω^*. Thus as $\lambda_1 \cdots \lambda_n$ run through all systems of values, $g(x)$ runs through the totality of all equations (in the modified sense); and this totality contains the totality of the g_Γ to which values from Ω^* correspond. Since, conversely, by Hilbert's irreducibility theorem the values λ from Ω^* "in general" also correspond to equations g_Γ, one actually obtains, in the sense modified in 1., through the parameter representation (9) the totality of all g_Γ with respect to Ω^*.

It remains now to determine the singular systems of values for which the parameter representation fails. Split into numerator and denominator, let the functions of the minimal basis be given by

$$\varphi_i(x) = \frac{\psi_i(x)}{\chi_i(x)}.$$

Substituting this into the identities (5), suppose that these identities become, without cancellation,

$$\sigma_k(x) = \frac{G_k(x)}{H_k(x)} \quad \text{or} \quad H_k(x) \cdot \sigma_k(x) = G_k(x), \tag{11}$$

so that the product over all $H_k(x)$ is divisible by the product of all denominators $\chi_i(x)$. For all root systems $\alpha_1 \cdots \alpha_n$ for which the product

$$H_1(\alpha) \cdot H_2(\alpha) \cdots H_n(\alpha) \neq 0$$

and consequently no denominator $\chi_i(\alpha)$ vanishes either, one can divide (11) by $H_k(\alpha)$ and obtains

$$\sigma_k(\alpha) = \frac{G_k(\alpha)}{H_k(\alpha)} = F_k(\varphi_1(\alpha); \cdots ; \varphi_n(\alpha)), \tag{12}$$

where the $\varphi_i(\alpha)$, because of the nonvanishing of the denominators, assume finite values – and for equations g_Γ become quantities from Ω^*. Equation (12) represents the solution system of (10), as soon as $\alpha_1 \cdots \alpha_n$ are determined so that $a_k = (-1)^k\sigma_k(\alpha)$.[188]

[187] Cf. also below!

[188] The solution of the equation is thereby reduced to the solution of the system, consisting of independent functions,

$$\varphi_1(\alpha_1 \cdots \alpha_n) = \lambda_1; \quad \cdots ; \quad \varphi_n(\alpha_1 \cdots \alpha_n) = \lambda_n.$$

Thus only such coefficient systems $a_k = (-1)^k \sigma_k(\alpha)$ can become *singular* for which

$$H_1(\alpha) \cdot H_2(\alpha) \cdots H_n(\alpha) = 0;$$

for these, (11), instead of becoming a representation, passes into $0 = 0$. A special investigation is then needed as to whether among the singular $a_1 \cdots a_n$ so defined there are also such which correspond to equations g_Γ, and for which number fields.[189] In summary, we obtain the following.

Theorem. If the Lagrangian genus domain belonging to the group Γ possesses a minimal basis[190] *with coefficients from Ω, then for every number field Ω^* containing Ω, the totality of equations with prescribed group Γ with respect to Ω^* can be constructed rationally by the parameter representation (2) – possibly with the exception of those which are singular with respect to the parameter representation and can be given separately. The parameters are to run through all values from Ω^* which do not cause a reduction of the group. The construction is possible for every number field if the coefficient field Ω of the minimal basis is equal to the field $\mathfrak{R}$ of all rational numbers.*

Since, by Hilbert's irreducibility theorem, the parameter manifold is not lowered by excluding those values from Ω^* which cause a reduction of the group, it follows further: *from the existence of a minimal basis it follows that the manifold of equations with the group Γ with respect to Ω^* is equal to the manifold of the affect-free equations; the manifold is not lowered by the affect.*

§ 2.

Application to Special Equations.

We now prove the existence of the minimal basis for all groups occurring in equations of the third and fourth degrees; and for this we first show quite generally that, for equations of degree n, the problem can be reduced to one of $n-2$ indeterminates. The first reduction corresponds to the linear Tschirnhaus transformation for removing the second-highest term. We take as first basis function $\varphi_1(x) = \sigma_1(x)$; and we observe further that $\mathfrak{L}_\Gamma$ can be obtained rationally from all *homogeneous* forms in $x_1 \cdots x_n$ admitting Γ. Let $p(x_1 \cdots x_n)$ be such a form; then also

$$q(x) = p\left(x_1 - \frac{\sigma_1(x)}{n}; \cdots; x_n - \frac{\sigma_1(x)}{n}\right) = p\left(x_i - \frac{\sigma_1(x)}{n}\right) = p(y_i)$$

belongs to $\mathfrak{L}_\Gamma$, since it admits Γ, and moreover one has

$$p(x) \equiv p\left(x_i - \frac{\sigma_1(x)}{n}\right) \bmod \sigma_1(x)$$

or

$$p(x) = q(x) + \sigma_1(x) \cdot p_1(x)$$

with $p_1(x)$ belonging to $\mathfrak{L}_\Gamma$, and of smaller dimension than $p(x)$. Continuing the procedure thus shows that all homogeneous forms from $\mathfrak{L}_\Gamma$ – and consequently all rational functions from $\mathfrak{L}_\Gamma$ at all – can be rationally expressed by $\varphi_1(x) = \sigma_1(x)$ and by homogeneous forms

$$q(x) = p\left(x_i - \frac{\sigma_1(x)}{n}\right) = p(y_i).$$

Between these quantities y_i there is, however, the linear relation $\sum y_i = 0$; hence the $q(x)$ *depend only on* $(n-1)$ *arguments.*

The second reduction rests on the fact that the $q(x)$ are homogeneous forms of their arguments. We take as second basis function

$$\varphi_2(x) = \frac{\tau_3(x)}{\tau_2(x)}, \quad \text{where} \quad \tau_3(x) = \sigma_3\left(x_i - \frac{\sigma_1(x)}{n}\right) = \sigma_3(y_i), \quad \tau_2(x) = \sigma_2\left(x_i - \frac{\sigma_1(x)}{n}\right) = \sigma_2(y_i);$$

[189] For example, as F. Seidelmann has shown at the cited place, only for the alternating group of equations of the third and fourth degrees do such equations – corresponding to singular systems of values – occur, and indeed only for those number fields Ω^* which contain the field of third roots of unity. In each case a simple supplementary representation can be given.

[190] From the existence of one minimal basis there follows at once the existence of arbitrarily many, derivable from the original one by reversible rational transformations.

thus $\varphi_2(x)$ is a homogeneous fractional function of first degree of $(n-1)$ arguments, namely of the y_i with $\sum y_i = 0$. But by means of $\varphi_2(x)$ all $q(x)$ can be expressed rationally by the functions, likewise belonging to $\mathfrak{L}_\Gamma$,

$$r(x) = \frac{q(x)}{\varphi_2(x)^\lambda} = \frac{q(x)\cdot\tau_2(x)^\lambda}{\tau_3(x)^\lambda} = \frac{p(y_i)\cdot\sigma_2(y_i)^\lambda}{\sigma_3(y_i)^\lambda},$$

where λ denotes the degree of $q(x)$. Thus $r(x)$ is homogeneous of degree zero and therefore depends only on $(n-2)$ arguments, the ratios $y_1 : y_2 : \cdots : y_n$ with $\sum y_i = 0$. Hence $\mathfrak{L}_\Gamma$ can be rationally expressed by

$$\varphi_1(x) = \sigma_1(x); \quad \varphi_2(x) = \frac{\tau_3(x)}{\tau_2(x)}$$

and by the *field of all functions* $r(x)$ *from* $\mathfrak{L}_\Gamma$ *depending on these* $(n-2)$ *arguments*,[191] whereby the desired reduction is accomplished.

For equations of the third and fourth degrees, this gives in particular a reduction to function fields of one and two arguments respectively. But for these a minimal basis always exists by the theorems of Lüroth and Castelnuovo.[192] In fact, in the case of one indeterminate the minimal basis can be derived by rational operations, and therefore, specifically for $\mathfrak{L}_\Gamma$, contains only rational numerical coefficients. In the case of two indeterminates, according to Castelnuovo, one might still be dealing with a basis with coefficients from an algebraic number field Ω; the actual investigation (cf. F. Seidelmann, loc. cit.) shows that here too, for every $\mathfrak{L}_\Gamma$, one obtains a basis with rational integral coefficients. One sees this almost without computation already from the fact that for the four group such a rational-integral basis can be chosen, namely

$$\varphi_1(x) = \sigma_1(x); \quad \varphi_2(x) = \frac{vw}{u}; \quad \varphi_3(x) = \frac{u\cdot w}{v}; \quad \varphi_4(x) = \frac{u\cdot v}{w},$$

where

$$u = x_1 + x_2 - x_3 - x_4; \quad v = x_1 - x_2 + x_3 - x_4; \quad w = x_1 - x_2 - x_3 + x_4,$$

such that the alternating group passes into the cyclic group of the three basis functions $\varphi_2, \varphi_3, \varphi_4$, and each group of order eight into the symmetric group of two of these functions, whereby a reduction to the groups of equations of the third and second degrees is effected. For the cyclic group, however, a rational-integral basis is already known from Abel, although not formulated as such.[193] Thus it is proved: *The totality of equations of the third and fourth degrees with prescribed group – possibly with the exception of those singular with respect to the parameter representation – can be constructed rationally by parameter representation for any prescribed domain of rationality; their manifold is equal to the manifold of affect-free equations.*

The actual investigation shows (cf. F. Seidelmann, loc. cit.) that a supplementary representation is needed only for the alternating group and only with respect to domains of rationality containing the field of third roots of unity, whereas in all other cases the parameter representation supplies the totality without qualification.

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[191] For an equation of degree n without second term,

$$f(x) = x^n + a_2x^{n-2} + a_3x^{n-3} + \cdots + a_n = 0,$$

this second reduction corresponds to the substitution, always possible for $a_2 \neq 0$, $a_3 \neq 0$,

$$y = \frac{x\cdot a_2}{a_3},$$

by which $f(x)$, up to a factor, passes into

$$g(y) = y^n + \frac{a_2^3}{a_3^2}y^{n-2} + \frac{a_2^3}{a_3^2}y^{n-3} + \frac{a_4\cdot a_2^4}{a_3^4}y^{n-4} + \cdots + a_n\frac{a_2^n}{a_3^n} = 0;$$

that is, into an equation containing only $n-2$ parameters:

$$g(y) = y^n + c\cdot y^{n-2} + c\cdot y^{n-3} + c_4y^{n-4} + \cdots + c_n = 0.$$

[192] J. Lüroth: Beweis eines Satzes über rationale Kurven, Math. Ann. 9 (1875); G. Castelnuovo: Sulla razionalità delle involuzioni piane, Math. Ann. 44 (1893). That for fields of more than two indeterminates a minimal basis need not exist was shown by F. Enriques by a counterexample, Rend. Acc. Linc., Vol. 21, 21 Jan. 1912.

[193] Cf. Weber Algebra (small edition), § 93, Formula (12).

It should also be noted that the minimal basis is known for all *Abelian groups.*[194] Here, however, the basis has coefficients from the cyclotomic field derived from the group characters. Thus, in this case, no new results can be obtained for the construction of equations with prescribed group.

Göttingen, July 1916.

[194] E. Fischer: Die Isomorphie der Invariantenkörper der endlichen Abelschen Gruppen linearer Transformationen. Gött. Nachr. 1915, and Zur Theorie der endlichen Abelschen Gruppen. Math. Ann. 77 (1915).

12. Invariants of Arbitrary Differential Expressions

Nachr. v. d. Ges. d. Wiss. zu Göttingen 1918, pp. 37–44

Presented in the session of 25 January 1918 by F. Klein.

By a differential expression I mean a function

$$f(x, dx) = f(x_1, \dots, x_n;\, dx_1, \dots, dx_n)$$

of the n variables $x_1, \dots, x_n$ and of their differentials $dx_1, \dots, dx_n$, which is assumed to be analytic in the two systems of n arguments, but not necessarily real. I now take as underlying group the group of all analytic transformations of the variables, to which corresponds the group of all linear transformations of the differentials:

$$x_i = x_i(y_1, \dots, y_n); \qquad dx_i = \sum_k \frac{\partial x_i}{\partial y_k}\, dy_k; \qquad \delta x_i = \sum_k \frac{\partial x_i}{\partial y_k}\, \delta y_k, \tag{1}$$

under which $f(x, dx)$ is to pass over into $g(y, dy)$. By an invariant of $f(x, dx)$ I understand an invariant with respect to this group, and with respect to the group thereby induced for the higher differentials $d^2x, ddx, \dots$ and for the derivatives of $f(x, dx)$; thus an (analytic) function J such that, for arbitrary f, and by means of (1), the following identity holds in the variables and in all occurring differentials:

$$\begin{aligned} &J\left(f, \frac{\partial f}{\partial dx} \cdots \frac{\partial^{\rho+\sigma} f}{\partial x^\rho \partial dx^\sigma} \cdots dx, \delta x, d^2x, \dots\right) \\ &\quad = J\left(g, \frac{\partial g}{\partial dy} \cdots \frac{\partial^{\rho+\sigma} g}{\partial y^\rho \partial dy^\sigma} \cdots dy, \delta y, d^2y, \dots\right). \end{aligned}$$ [195]

In precisely the same way one defines the invariant of a simultaneous system of differential expressions. If, in particular, such an invariant contains only first differentials $dx, \delta x$, and only derivatives with respect to these differentials, not with respect to the variables, then the occurrence of the variables in the functions and the linear transformation of the differentials do not enter into consideration; these special invariants are invariants with respect to the group of all linear transformations of the differentials $dx, \delta x$ with undetermined coefficients, and shall be called projective invariants of the simultaneous system.

For quadratic homogeneous differential forms Christoffel (Crelle 70) and Ricci (Math. Annalen 54) have now set up such a system of invariant formations that all invariants become projective invariants of this simultaneous system, whereby the questions concerning the totality of invariants and concerning equivalence are reduced to questions of linear invariant theory. This is what I should like to call the "reduction theorem." The complete system consists of countably infinitely many forms; but for invariants of any finite order ρ (that is, those which contain no higher than ρ-th derivatives with respect to the variables), it consists only of finitely many forms.

In what follows I show the validity of the "reduction theorem" for arbitrary differential expressions; and here again the issue is a complete system of countably infinitely many invariant formations, but only finitely many of them for each finite order. Whereas, however, Christoffel and Ricci base the generation of the invariants and the proof of the reduction theorem on eliminations that are difficult to survey, I generate the invariants directly by invariant differentiation and variation processes, the latter of which can simply be regarded as differentiation processes with respect to other parameters. The algorithm of these invariant differentiation processes was developed, following Lagrange (*Mécanique analytique*, part 2, section IV), by Riemann (Works XXII) and Lipschitz (Crelle 70, 72) for quadratic differential forms, without, however, advancing as far as

[195] $\dfrac{\partial^{\rho+\sigma} f}{\partial x^\rho \partial dx^\sigma}$ will throughout be used as an abbreviation for

$$\frac{\partial^{\rho+\sigma} f}{\partial x_{i_1} \cdots \partial x_{i_\rho} \partial dx_{k_1} \cdots \partial dx_{k_\sigma}},$$

where the indices denote any of the numbers 1 to n.

the reduction theorem. The auxiliary means for proving the reduction theorem are provided by the "normal coordinates" introduced by Riemann for quadratic forms (Works XIII); these transform the extremals of the associated variational problem emanating from a point into straight lines, but they exist only for homogeneous differential expressions $f(x, \varkappa \cdot dx) = \Phi(\varkappa) \cdot f(x, dx)$[196]. The inhomogeneous case, however, can be reduced to this one.

I. Generation of invariants.

A sequence of projective invariants of $f(x, dx)$, in general not terminating, is furnished by the polars. Because of the linearity of the transformation of the differentials, from $f(x, dx) = g(y, dy)$ it also follows that $f(x, dx + \lambda\delta x) = g(y, dy + \lambda\delta y)$; and from this, by expansion in λ, there arise the relations characteristic of invariance:

$$\begin{gathered} f(dx) = g(dy), \\ f_\delta = \sum \frac{\partial f(dx)}{\partial dx}\,\delta x = \sum \frac{\partial g(dy)}{\partial dy}\,\delta y, \\ \vdots \\ f_{\delta^\sigma} = \sum \frac{\partial^\sigma f(dx)}{\partial dx^\sigma}\,\delta x^\sigma = \sum \frac{\partial^\sigma g(dy)}{\partial dy^\sigma}\,\delta y^\sigma, \\ \vdots \end{gathered} \tag{2}$$

Each polar, by application of a variation process $\bar{\delta}$, gives rise to a non-terminating sequence of general invariants:

$$\begin{gathered} \bar{\delta}^\rho f(x, dx) = \bar{\delta}^\rho g(y, dy), \\ \vdots \\ \bar{\delta}^\rho f_{\delta^\sigma} = \bar{\delta}^\rho \sum \frac{\partial^\sigma f(dx)}{\partial dx^\sigma}\,\delta x^\sigma = \bar{\delta}^\rho \sum \frac{\partial^\sigma g(dy)}{\partial dy^\sigma}\,\delta y^\sigma, \\ \vdots \\ (\rho = 0, 1, 2, \ldots). \end{gathered} \tag{3}$$

By means of (1), the relations (2) and (3) allow the transformation laws for the derivatives of $f(x, dx)$ to be read off; this, however, will not be needed in what follows. From the invariants (3) one can now obtain by linear combination the "normal form of the ρ-th variation," which is found for $\rho = 1$ in Lagrange and for $\rho = 2$ in Riemann; it is characterized by the fact that the mixed differentials of order $\rho + 1$, namely $d^\rho\delta, d^{\rho-1}\delta^2, \ldots$, are eliminated. One obtains for it:

$$\begin{gathered} \Omega_1 = \delta f(dx) - df_\delta(dx), \\ \Omega_2 = \delta^2 f - \delta d f_\delta + \frac{1}{2} d^2 f_{\delta^2}, \\ \vdots \\ \Omega_\rho = \delta^\rho f - \delta^{\rho-1} d f_\delta + \frac{1}{2}\delta^{\rho-2} d^2 f_{\delta^2} + \cdots + \frac{(-1)^\rho}{\rho!}\, d^\rho f_{\delta^\rho}, \\ \vdots \end{gathered} \tag{4}$$

From the invariants (4), by generalizing an ansatz of Riemann, one can now arrive at invariants that contain only first differentials; such invariants shall be called "fundamental functions." Namely, if in Ω_1 the differentials are replaced by the differential quotients, then $\Omega_1 = 0$ (identically in δx) gives exactly the Lagrange equations of the variational problem belonging to

$$f\left(\frac{dx}{dt}\right).$$

I now impose the invariant condition that the direction $(d + \lambda\delta)$ satisfy these Lagrange equations:

$$\Omega_1(d + \lambda\delta) = \bar{\delta} f(dx + \lambda\delta x) - (d + \lambda\delta) f_{\bar{\delta}}(dx + \lambda\delta x) = 0 \quad \text{(identically in } \delta x\text{)}, \tag{5}$$

[196]It is easy to see that $\Phi(\varkappa) = \varkappa^c$, where c denotes an arbitrary real or complex quantity.

from which, by the substitution $\bar{\delta} = d + \lambda\delta$, there still follows for homogeneous $f(dx)$:

$$(d + \lambda\delta)f(dx + \lambda\delta x) = 0, \tag{6}$$

except in the case of homogeneity of first order. In the latter case (6) is to be added as a new condition, since then the system of equations (5) represents only $n-1$ independent conditions. If in (5), respectively in (5) and (6), one sets the coefficients of the zeroth, first, and second powers of λ equal to zero, one thus obtains three invariant systems of equations $\varphi = 0$, $\psi = 0$, $\chi = 0$ (identically in δx), by means of which $d^2x, ddx, \delta^2 x$ can be expressed through first differentials. The further invariant systems of equations

$$\begin{gathered} d^\sigma\varphi = 0, \qquad d^\sigma\psi = 0, \qquad d^\sigma\chi = 0, \qquad d^{\sigma-1}\delta\chi = 0, \\ d^{\sigma-2}\delta^2\chi = 0, \ldots, \delta^\sigma\chi = 0 \qquad (\sigma = 0, 1, 2, \ldots) \end{gathered} \tag{7}$$

then suffice precisely to express all higher differentials by first ones.[197] The functions (4), combined with the invariant system of equations (7), therefore lead to fundamental functions $[\Omega_\rho]$ with only first differentials; moreover $[\Omega_1]$ vanishes identically.

Similarly, from the differential $\delta h(dx, \delta x)$ of a fundamental function h, one can by means of (7) again obtain a fundamental function, which is to be called the "covariant derivative" $[h^{(1)}]$ of h; in general let $[h^{(\nu)}]$ denote the covariant derivative of $[h^{(\nu-1)}]$. These two processes thus lead to a doubly infinite sequence of fundamental functions:

$$\begin{array}{llllll} [\Omega_2], & [\Omega_2^{(1)}], & [\Omega_2^{(2)}], & \ldots, & [\Omega_2^{(\nu)}], & \ldots \\ \vdots & & & & \vdots & \\ [\Omega_\rho], & [\Omega_\rho^{(1)}], & \ldots, & \ldots, & [\Omega_\rho^{(\nu)}], & \ldots \\ \vdots & \vdots & & & \vdots & \end{array} \tag{8}$$

It further turns out that the left-hand sides of the equations which express the higher differentials through first ones are cogredient to the first differentials, that is, they undergo the same linear transformations. If, therefore, instead of the higher differentials, one introduces these left-hand sides $p, q, r, \ldots$ as arguments of the invariant, then, apart from the derivatives, it depends only on quantities cogredient to one another. The formation of the functions (8) indicated above amounts simply to the introduction of these new quantities; every invariant thereby passes over into a sum of invariants, and from this sum one selects the one which contains precisely $dx, \delta x$, but not $p, q, r, \ldots$.

In what follows it is shown that under the homogeneity assumption

$$f(\varkappa \cdot dx) = \Phi(\varkappa) \cdot f(dx)$$

the fundamental functions (8) exhaust the complete system sought.

II. Normal coordinates, equivalence, and the reduction theorem.

I arrive at the normal coordinates by integrating the variational problem belonging to $f\left(\frac{dx}{dt}\right)$:

$$\begin{gathered} \Omega_1 = \delta f\left(\frac{dx}{dt}\right) - \frac{d}{dt} f_\delta\left(\frac{dx}{dt}\right) = 0 \quad \text{(identically in } \delta x), \\ \frac{d}{dt} f\left(\frac{dx}{dt}\right) = 0 \quad \text{(as a consequence or additional condition).} \end{gathered} \tag{9}$$

Because of the assumed homogeneity of $f(dx)$, this system of equations goes over into itself under the parameter substitution $t = \varkappa \cdot \tau$. Hence, if by means of (9) one expresses $\frac{d^\rho x_i}{dt^\rho}$ as a function $\varphi_\rho^{(i)}\left(\frac{dx}{dt}\right)$ of the first derivatives, then $\varphi_\rho^{(i)}$ is homogeneous of order ρ:

$$\varphi_\rho^{(i)}\left(\varkappa \cdot \frac{dx}{dt}\right) = \varkappa^\rho \varphi_\rho^{(i)}\left(\frac{dx}{dt}\right). \tag{10}$$

If, in integrating (9), one now prescribes for $t = t_0$ the initial values

$$x = (x)_0, \qquad \frac{dx}{dt} = \left(\frac{dx}{dt}\right)_0$$

[197] Only the linear form $\sum A_i\, dx_i$ requires special treatment, since here the system of equations (5) contains no second differentials.

and sets

$$u_i = (t - t_0)\left(\frac{dx_i}{dt}\right)_0, \tag{11}$$

where, by virtue of $f\left(\frac{dx}{dt}\right) = \text{const.}$, the parameter $(t - t_0)$ becomes proportional to the "length of the extremal," then the expansion in powers of $(t - t_0)$, by (10), gives

$$x_i - (x_i)_0 = u_i + \frac{1}{2}\varphi_2^{(i)}(u) + \cdots + \frac{1}{\rho!}\varphi_\rho^{(i)}(u) + \cdots. \tag{12}$$

This expansion can be regarded as a transformation of variables $x_i = x_i(u)$, where the quantities u are precisely the Riemann normal coordinates. Thus, by (11) and (12), these coordinates transform the extremals passing through $(x)_0$ into straight lines. Moreover, to an arbitrary transformation of variables $x = x(y)$ there corresponds, by (11), a linear transformation $u = u(v)$ of the associated normal coordinates, where the coefficients depend only on the arbitrary system of values $(x)_0$; hence the differentials $du, \delta u$ are subject to the same transformation as the u themselves.

By means of (12), let $f(x, dx)$ pass over into $F(u, du)$, or, expanded in powers of the u,

$$F(u, du) = F_0(u, du) + F_1(u, du) + \cdots + F_\rho(u, du) + \cdots, \tag{13}$$

where $F_\rho(u, du)$ denotes a homogeneous form of dimension ρ in u. Because of the linearity of the transformation $u = u(v)$, the identity $F(u, du) = G(v, dv)$ also carries the individual homogeneous components into the corresponding components:

$$F_\rho(u, du) = G_\rho(v, dv). \tag{14}$$

If one therefore takes any fixed constant system of values for $(x)_0$, then (13) and (14) show that the equivalence of $f(dx)$ with $g(dy)$ is equivalent to the equivalence of the associated systems of functions $F_\rho(u, du)$ with respect to linear transformation. Furthermore, the $F_\rho(u, du)$, or the corresponding $F_\rho(\delta u, du)$, represent invariants of $f(dx)$, taken at the point $x = (x)_0$; indeed, they form a complete system of fundamental functions for this point $x = (x)_0$. Namely,

$$F_\rho(\delta u, du) = \frac{1}{\rho!}\sum \frac{\partial F_\rho}{\partial \delta u^\rho}\,\delta u^\rho; \qquad \frac{\partial F_\rho}{\partial \delta u^\rho} = \left(\frac{\partial F(du)}{\partial u^\rho}\right)_0,$$

and this latter derivative can, by means of (12), be expressed in just the same way through derivatives of $f(dx)$, taken for $x = (x)_0$, as the correspondingly formed derivative of $G(dv)$ can be expressed through derivatives of $g(dy)$. Because

$$\left(\frac{\partial^{\rho+\sigma} F(du)}{\partial u^\rho \partial du^\sigma}\right)_0 = \frac{\partial^{\rho+\sigma} F_\rho(\delta u, du)}{\partial \delta u^\rho \partial du^\sigma},$$

every invariant of $f(dx) = F(du)$, taken at the point $x = (x)_0$, also becomes a projective invariant of the $F_\rho(\delta u, du)$, once the higher differentials have merely been replaced by the $p, q, r, \ldots$ cogredient to $dx, \delta x$. But now $(x)_0$ can be regarded as a parameter; to every invariant at the point $x = (x)_0$ there corresponds an invariant of $f(dx)$, if only $(x)_0$ is replaced again by x. Thus, if the $F_\rho(\delta u, du)$ become invariants $\Psi_\rho(\delta x, dx)$, taken for $x = (x)_0$ — for $x = (x)_0$, $du, \delta u$ pass over into $dx, \delta x$ —, then the Ψ_ρ themselves form the fundamental functions of a complete system. It remains only to express this system of fundamental functions by explicit invariants of $f(dx)$, namely by the functions (8). That this is possible is shown by their generation through differentiation processes. For the terms of highest order these processes pass into simple polarization processes, and a transformation analogous to the Clebsch–Gordan series expansion, combined with induction on account of the neglected terms, proves the assertion. If one is dealing with invariants of a simultaneous system, then the covariant derivatives of the remaining differential expressions must still be added to the fundamental functions (8), as is shown by introducing into these expressions the normal coordinates belonging to $f(dx)$.

The equivalence of $f(dx)$ with $g(dy)$, which had been reduced to (14), is therefore also equivalent to the equivalence of the Ψ_ρ, or equally of the functions (8), with respect to linear transformation of the differentials, without any special integrability conditions being required; this follows from the possibility of reduction to (14). Thus the reduction theorem is proved in all its parts. In particular, the identical vanishing of the sequence of functions $[\Omega_2], [\Omega_3], \ldots, [\Omega_\rho], \ldots$ is necessary and sufficient for $f(dx)$ to be transformable into an expression with constant coefficients.

Finally, if instead of the group of all analytic transformations one takes a subgroup as the basis, then the group of the corresponding linear transformations of the u also passes over into a subgroup of the projective group; the invariants of $f(dx)$ again become invariants of the functions (8) with respect to linear transformation, but now with respect to this subgroup. Thus, by homogenizing, the case of nonhomogeneous functions $f(dx)$ can be reduced to the affine group; the complete system can here be derived from the functions (8) formed for one additional variable.

From the reduction theorem one obtains the special case of quadratic forms, and more generally of forms of p-th dimension, from the fact that here the system of functions terminates with $[\Omega_2]$, more generally with $[\Omega_p]$ and its covariant derivatives, since all later fundamental functions become projective invariants of these. This explains the distinguished position of the Riemannian curvature form $[\Omega_2]$ for quadratic forms. The question of the equivalence of quadratic forms, respectively of forms of p-th dimension, comes down precisely to the equivalence of $[\Omega_2]$, respectively $[\Omega_2]\dots[\Omega_p]$, and of their covariant derivatives under linear transformation; and the identical vanishing of $[\Omega_2]$, respectively of $[\Omega_2]\dots[\Omega_p]$, is necessary and sufficient for transformation into forms with constant coefficients, which, for quadratic forms, states one of the best-known results.

A more detailed presentation is to appear shortly in the *Mathematische Annalen*.

13. Invariant Variational Problems

Nachr. v. d. Ges. d. Wiss. zu Göttingen 1918, pp. 235–257

Presented by F. Klein in the session of 26 July 1918.[198]

The subject is variational problems that admit a continuous group (in Lie's sense); the consequences which arise from this for the corresponding differential equations find their most general expression in the theorems formulated in §1 and proved in the following sections. For these differential equations arising from variational problems, much more precise assertions can be made than for arbitrary differential equations admitting a group, which form the subject of Lie's investigations. What follows therefore rests on a combination of the methods of the formal calculus of variations with those of Lie group theory. For special groups and variational problems this combination of methods is not new; I mention Hamel and Herglotz for special finite groups, Lorentz and his students (for example Fokker), Weyl and Klein for special infinite groups.[199] In particular, the second note of Klein and the present developments have mutually influenced one another; for this I may refer to the closing remarks of Klein's note.

§1. Preliminaries and formulation of the theorems

All functions occurring below are to be assumed analytic, or at least continuous and continuously differentiable a finite number of times, and single-valued in the domain under consideration.

By a "transformation group" one understands, as usual, a system of transformations such that for each transformation there exists an inverse transformation contained in the system, and such that the composition of any two transformations of the system again belongs to the system. The group is called a finite continuous $\mathfrak{G}_\rho$ when its transformations are contained in a most general transformation depending analytically on ρ essential parameters ε (that is, the ρ parameters are not to be representable as ρ functions of fewer parameters). Correspondingly, by an infinite continuous $\mathfrak{G}_{\infty\rho}$ one understands a group whose most general transformations depend on ρ essential arbitrary functions $p(x)$ and their derivatives, analytically or at least continuously and with a finite number of continuous derivatives. An intermediate case between the two is the group depending on infinitely many parameters but not on arbitrary functions. Finally, a mixed group denotes one that depends both on arbitrary functions and on parameters.[200]

[198] The final version of the manuscript was submitted only at the end of September.

[199] Hamel: *Math. Ann.* vol. 59 and *Zeitschrift f. Math. u. Phys.* vol. 50. Herglotz: *Ann. d. Phys.* (4) vol. 36, especially §9, p. 511. Fokker, Verslag d. Amsterdamer Akad., 27 Jan. 1917. For the further literature compare the second note of Klein: Göttinger Nachrichten, 19 July 1918. In a paper by Kneser that has just appeared (*Math. Zeitschrift* vol. 2), the issue is the construction of invariants by a similar method.

[200] Lie defines, in *Grundlagen für die Theorie der unendlichen kontinuierlichen Transformationsgruppen* (Reports of the Royal Saxon Society of Sciences, 1891), the infinite continuous group as a transformation group whose transformations are given by the most general solutions of a system of partial differential equations, once these solutions do not depend only on a finite number of parameters. This gives one of the types indicated above, different from the finite group; conversely, however, the limiting case of infinitely many parameters need not necessarily satisfy a system of differential equations.

Let $x_1, \ldots, x_n$ be independent variables, and let $u_1(x), \ldots, u_\mu(x)$ be functions depending on them. If one subjects the x's and u's to the transformations of a group, then, because the transformations are assumed invertible, among the transformed quantities there must again be exactly n independent ones, denoted $y_1, \ldots, y_n$; the remaining quantities depending on these are denoted $v_1(y), \ldots, v_\mu(y)$. The transformations may also contain the derivatives of the u's with respect to the x's, thus $\partial u/\partial x$, $\partial^2 u/\partial x^2, \ldots$.[201] A function is called an invariant of the group if a relation holds:

$$P\left(x, u, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \ldots\right) = P\left(y, v, \frac{\partial v}{\partial y}, \frac{\partial^2 v}{\partial y^2}, \ldots\right).$$

In particular, an integral I is therefore an invariant of the group if a relation holds:

$$\begin{aligned} I &= \int \cdots \int f\left(x, u, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \ldots\right) \mathrm{d}x \\ &= \int \cdots \int f\left(y, v, \frac{\partial v}{\partial y}, \frac{\partial^2 v}{\partial y^2}, \ldots\right) \mathrm{d}y, \end{aligned} \tag{1}$$

where the integration is over an arbitrary real x-domain and the corresponding y-domain.[202]

On the other hand, for an arbitrary integral I, not necessarily invariant, I form the first variation δI, and transform it by partial integration according to the rules of the calculus of variations. As is well known, once δu, together with all its derivatives occurring at the boundary, is taken to vanish at the boundary, but is otherwise arbitrary, one obtains:

$$\delta I = \int \cdots \int \delta f \, \mathrm{d}x = \int \cdots \int \left(\sum_i \psi_i\left(x, u, \frac{\partial u}{\partial x}, \ldots\right) \delta u_i\right) \mathrm{d}x, \tag{2}$$

where the ψ's denote the Lagrange expressions, that is, the left-hand sides of the Lagrange equations of the corresponding variational problem $\delta I = 0$. This integral relation corresponds to an identity free of integrals in δu and its derivatives, which arises when one writes the boundary terms as well. As partial integration shows, these boundary terms are integrals over divergences, that is, over expressions

$$\operatorname{Div} A = \frac{\partial A_1}{\partial x_1} + \cdots + \frac{\partial A_n}{\partial x_n},$$

where A is linear in δu and its derivatives. Thus one obtains:

$$\sum_i \psi_i \, \delta u_i = \delta f + \operatorname{Div} A. \tag{3}$$

In particular, if f contains only first derivatives of the u's, then, in the case of the simple integral, the identity (3) is identical with Herr's so-called "Lagrangian central equation":

$$\sum_i \psi_i \, \delta u_i = \delta f - \frac{\mathrm{d}}{\mathrm{d}x}\left(\sum_i \frac{\partial f}{\partial u_i'} \delta u_i\right), \qquad \left(u_i' = \frac{\mathrm{d}u_i}{\mathrm{d}x}\right), \tag{4}$$

whereas for the n-fold integral (3) becomes:

$$\sum_i \psi_i \, \delta u_i = \delta f - \frac{\partial}{\partial x_1}\left(\sum_i \frac{\partial f}{\partial\left(\frac{\partial u_i}{\partial x_1}\right)} \delta u_i\right) - \cdots - \frac{\partial}{\partial x_n}\left(\sum_i \frac{\partial f}{\partial\left(\frac{\partial u_i}{\partial x_n}\right)} \delta u_i\right). \tag{5}$$

For the simple integral and κ derivatives of the u's, (3) is given by:

$$\begin{aligned} \sum_i \psi_i \, \delta u_i = \delta f &- \frac{\mathrm{d}}{\mathrm{d}x}\left\{\sum_i \left[\binom{1}{1} \frac{\partial f}{\partial u_i^{(1)}} \delta u_i + \binom{2}{1} \frac{\partial f}{\partial u_i^{(2)}} \delta u_i^{(1)} + \cdots + \binom{\kappa}{1} \frac{\partial f}{\partial u_i^{(\kappa)}} \delta u_i^{(\kappa-1)}\right]\right\} \\ &+ \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left\{\sum_i \left[\binom{2}{2} \frac{\partial f}{\partial u_i^{(2)}} \delta u_i + \binom{3}{2} \frac{\partial f}{\partial u_i^{(3)}} \delta u_i^{(1)} + \cdots + \binom{\kappa}{2} \frac{\partial f}{\partial u_i^{(\kappa)}} \delta u_i^{(\kappa-2)}\right]\right\} \\ &+ \cdots + (-1)^\kappa \frac{\mathrm{d}^\kappa}{\mathrm{d}x^\kappa}\left\{\sum_i \binom{\kappa}{\kappa} \frac{\partial f}{\partial u_i^{(\kappa)}} \delta u_i\right\}. \end{aligned} \tag{6}$$

[201] I suppress the indices, as far as possible also in summations; thus, for example, $\partial^2 u/\partial x^2$ denotes $\partial^2 u_\nu/\partial x_i \partial x_k$, and so on.

[202] I write $\mathrm{d}x, \mathrm{d}y$, for short, for $\mathrm{d}x_1 \cdots \mathrm{d}x_n, \mathrm{d}y_1 \cdots \mathrm{d}y_n$. All arguments $x, u, \varepsilon, p(x)$ occurring in the transformations are to be taken as real, whereas the coefficients may be complex. But since the final results are identities in the x, u, the parameters, and the arbitrary functions, they also hold for complex values as soon as all occurring functions are assumed analytic. Moreover, a large part of the results can be founded without integrals, so that the restriction to the real is not necessary even for the justification. By contrast, the considerations at the end of §2 and the beginning of §5 do not seem to be feasible without integrals.

A corresponding identity holds for the n-fold integral; in particular A contains δu up to the $(\kappa-1)$-st derivative. That (4), (5), and (6) do in fact define the Lagrange expressions ψ_i follows from the fact that, in the combinations on the right-hand sides, all higher derivatives of δu are eliminated, whereas the relation (2) is fulfilled, to which partial integration leads uniquely.

In what follows the issue is the two theorems:

I. If the integral I is invariant with respect to a $\mathfrak{G}_\rho$, then ρ linearly independent combinations of the Lagrange expressions become divergences; conversely, from this follows the invariance of I with respect to a $\mathfrak{G}_\rho$. The theorem also holds in the limiting case of infinitely many parameters.

II. If the integral I is invariant with respect to a $\mathfrak{G}_{\infty\rho}$ in which the arbitrary functions occur up to the σ-th derivative, then there exist ρ identical relations between the Lagrange expressions and their derivatives up to order σ; here too the converse holds.[203]

For mixed groups, the assertions of both theorems apply; hence there occur both dependencies and divergence relations independent of them.

Passing from these identities to the corresponding variational problem, and thus setting $\psi = 0$,[204] Theorem I says in the one-dimensional case — where the divergence becomes a total differential — that ρ first integrals exist, although nonlinear dependencies may occur among them;[205] in the multidimensional case one obtains the divergence equations that recently have often been called "conservation laws". Theorem II says that ρ of the Lagrange equations are consequences of the remaining ones.

The simplest example of Theorem II — without the converse — is the Weierstrass parameter representation; here, under homogeneity of first order, the integral is known to be invariant if one replaces the independent variable x by an arbitrary function of x, while leaving u unchanged ($y = p(x);\ v_i(y) = u_i(x)$). Thus one arbitrary function occurs, but without derivatives; and to this corresponds the well-known linear relation between the Lagrange expressions themselves:

$$\sum_i \psi_i \frac{\mathrm{d}u_i}{\mathrm{d}x} = 0.$$

A further example is offered by the physicists' "general theory of relativity". Here one is dealing with the group of all transformations of the x's, $y_i = p_i(x)$, while the u's (denoted $g_{\mu\nu}$ and q) are subjected to the transformations thereby induced for the coefficients of a quadratic and a linear differential form, transformations that contain the first derivatives of the arbitrary functions $p(x)$. To this correspond the known n dependencies between the Lagrange expressions and their first derivatives.[206]

If, in particular, one specializes the group by admitting no derivatives of $u(x)$ in the transformations, and in addition by letting the transformed independent quantities depend only on the x's and not on the u's, then, as will be shown in §5, the invariance of I implies the relative invariance of $\sum_i \psi_i \delta u_i$,[207] and likewise of the divergences occurring in Theorem I, as soon as the parameters are subjected to suitable transformations. It follows further that the first integrals mentioned above also admit the group. For Theorem II one similarly obtains the relative invariance of the left-hand sides of the dependencies, combined by means of the arbitrary functions; and, as a consequence, also a function whose divergence vanishes identically and which admits the group — the function that in the physicists' theory of relativity mediates the connection between dependencies and energy theorem.[208] Finally, in group-theoretic form, Theorem II gives the proof of a related assertion of Hilbert concerning the failure of proper energy theorems in "general relativity". With these supplementary remarks Theorem I contains all the theorems on first integrals known in mechanics and so forth, whereas Theorem II can be described as the greatest possible group-theoretic generalization of the "general theory of relativity".

§2. Divergence relations and dependencies

Let $\mathfrak{G}$ be a continuous group, finite or infinite. It can always be arranged that the identical transformation corresponds to the value zero of the parameters ε, respectively of the arbitrary functions $p(x)$.[209] Thus the

[203] For certain trivial exceptional cases compare §2, second note.

[204] Somewhat more generally one may also set $\psi_i = T_i$; compare §3, first note.

[205] Compare the end of §3.

[206] Compare, for example, Klein's presentation.

[207] That is, $\sum_i \psi_i \delta u_i$ acquires a factor under transformations.

[208] Compare the second note of Klein.

[209] Compare, for example, Lie, *Grundlagen*, p. 331. If arbitrary functions are involved, then the special values a^σ of the parameters are to be replaced by fixed functions $p^\sigma,\ \partial p^\sigma/\partial x, \ldots$; and correspondingly the values $a^\sigma + \varepsilon$ by $p^\sigma + p(x),\ \partial p^\sigma/\partial x + \partial p/\partial x$, and so on.

most general transformation will be of the form

$$y_i = A_i\Big(x, u, \frac{\partial u}{\partial x}, \dots\Big) = x_i + \Delta x_i + \cdots, \qquad v_i(y) = B_i\Big(x, u, \frac{\partial u}{\partial x}, \dots\Big) = u_i + \Delta u_i + \cdots,$$

where $\Delta x_i, \Delta u_i$ denote the terms of lowest dimension in ε, respectively in $p(x)$ and its derivatives; they are to be assumed linear in these quantities. As will be shown later, this is no restriction of generality.

Now let the integral I be an invariant with respect to $\mathfrak{G}$, so that relation (1) is fulfilled. In particular, I is then also invariant with respect to the infinitesimal transformation contained in $\mathfrak{G}$:

$$y_i = x_i + \Delta x_i, \qquad v_i(y) = u_i + \Delta u_i.$$

For this, relation (1) becomes:

$$\begin{aligned} 0 = \Delta I = \int \cdots \int f\Big(y, v(y), \frac{\partial v}{\partial y}, \dots\Big)\, \mathrm{d}y \\ - \int \cdots \int f\Big(x, u(x), \frac{\partial u}{\partial x}, \dots\Big)\, \mathrm{d}x. \end{aligned} \tag{7}$$

The first integral is to be extended over the $x + \Delta x$-domain corresponding to the x-domain. But this integration can also be transformed into an integration over the x-domain by means of the transformation valid for infinitesimal Δx:

$$\begin{aligned} \int \cdots \int f\Big(y, v(y), \frac{\partial v}{\partial y}, \dots\Big)\, \mathrm{d}y = \int \cdots \int f\Big(x, v(x), \frac{\partial v}{\partial x}, \dots\Big)\, \mathrm{d}x \\ + \int \cdots \int \mathrm{Div}(f \cdot \Delta x)\, \mathrm{d}x. \end{aligned} \tag{8}$$

If one therefore introduces, in place of the infinitesimal transformation Δu, the variation

$$\bar{\delta} u_i = v_i(x) - u_i(x) = \Delta u_i - \sum_\lambda \frac{\partial u_i}{\partial x_\lambda} \Delta x_\lambda, \tag{9}$$

then (7) and (8) become:

$$0 = \int \cdots \int \{\bar{\delta} f + \mathrm{Div}(f \cdot \Delta x)\} \mathrm{d}x. \tag{10}$$

Since the relation (10) is fulfilled when integrating over every arbitrary domain, the integrand must vanish identically. Thus the Lie differential equations for the invariance of I pass over into the relation

$$\bar{\delta} f + \mathrm{Div}(f \cdot \Delta x) = 0. \tag{11}$$

If one expresses $\bar{\delta} f$ here, by (3), in terms of the Lagrange expressions, one obtains

$$\sum_i \psi_i \bar{\delta} u_i = \mathrm{Div}\, B, \qquad (B = A - f \cdot \Delta x), \tag{12}$$

and this relation therefore represents, for every invariant integral I, an identity in all occurring arguments; it is the desired form of the Lie differential equations for I.[210]

First assume $\mathfrak{G}$ to be a finite continuous group $\mathfrak{G}_\rho$. Since, by assumption, Δu and Δx are linear in the parameters $\varepsilon_1, \dots, \varepsilon_\rho$, by (9) the same holds for $\bar{\delta} u$ and its derivatives; hence A and B are linear in the ε. I therefore put

$$B = B^{(1)} \varepsilon_1 + \cdots + B^{(\rho)} \varepsilon_\rho, \qquad \bar{\delta} u = \bar{\delta} u^{(1)} \varepsilon_1 + \cdots + \bar{\delta} u^{(\rho)} \varepsilon_\rho,$$

where $\bar{\delta} u^{(1)}, \dots$ are functions of $x, u, \partial u/\partial x, \dots$. From (12) follow the desired divergence relations:

$$\sum_i \psi_i \bar{\delta} u_i^{(1)} = \mathrm{Div}\, B^{(1)}, \dots, \qquad \sum_i \psi_i \bar{\delta} u_i^{(\rho)} = \mathrm{Div}\, B^{(\rho)}. \tag{13}$$

Thus ρ linearly independent combinations of the Lagrange expressions become divergences. The linear independence follows from the fact that, by (9), $\bar{\delta} u = 0, \Delta x = 0$ would imply $\Delta u = 0, \Delta x = 0$, and hence

[210](12) becomes $0 = 0$ for the trivial case — which can occur only if $\Delta x, \Delta u$ also depend on derivatives of u — when $\mathrm{Div}(f \cdot \Delta x) = 0,\ \bar{\delta} u = 0$. These infinitesimal transformations must therefore always be split off from the groups, and in the formulation of the theorems only the number of the remaining parameters or arbitrary functions is to be counted. Whether the remaining infinitesimal transformations still always form a group must be left open.

a dependency between the infinitesimal transformations. But by hypothesis no such dependency holds for any parameter value; otherwise the $\mathfrak{G}_\rho$ arising again by integration from the infinitesimal transformations would depend on fewer than ρ essential parameters. The further possibility $\bar{\delta} u = 0,\ \mathrm{Div}(f\Delta x) = 0$ has been excluded. These conclusions remain valid in the limiting case of infinitely many parameters.

Now let $\mathfrak{G}$ be an infinite continuous group $\mathfrak{G}_{\infty\rho}$. Then again $\bar{\delta} u$ and its derivatives, hence also B, are linear in the arbitrary functions $p(x)$ and their derivatives.[211] Substitution of the values of $\bar{\delta} u$, still independently of (12), gives:

$$\sum_i \psi_i \bar{\delta} u_i = \sum_{\lambda,i} \psi_i \left\{ a_i^{(\lambda)}(x,u,\ldots) p^{(\lambda)}(x) + b_i^{(\lambda)}(x,u,\ldots) \frac{\partial p^{(\lambda)}}{\partial x} + \cdots + c_i^{(\lambda)}(x,u,\ldots) \frac{\partial^\sigma p^{(\lambda)}}{\partial x^\sigma} \right\}.$$

Now, analogously to the formula for partial integration, by means of the identity

$$\varphi(x,u,\ldots) \frac{\partial^\tau p(x)}{\partial x^\tau} = (-1)^\tau \frac{\partial^\tau \varphi}{\partial x^\tau} p(x) \quad \text{mod divergences},$$

the derivatives of p can be replaced by p itself and by divergences that are linear in p and its derivatives. Thus:

$$\sum_i \psi_i \bar{\delta} u_i = \sum_\lambda \left\{ a_i^{(\lambda)} \psi_i - \frac{\partial}{\partial x}(b_i^{(\lambda)} \psi_i) + \cdots + (-1)^\sigma \frac{\partial^\sigma}{\partial x^\sigma}(c_i^{(\lambda)} \psi_i) \right\} p^{(\lambda)} + \mathrm{Div}\,\Gamma, \tag{14}$$

with summation over i in the brace. In combination with (12) this gives:

$$\sum_\lambda \left\{ a_i^{(\lambda)} \psi_i - \frac{\partial}{\partial x}(b_i^{(\lambda)} \psi_i) + \cdots + (-1)^\sigma \frac{\partial^\sigma}{\partial x^\sigma}(c_i^{(\lambda)} \psi_i) \right\} p^{(\lambda)} = \mathrm{Div}(B - \Gamma). \tag{15}$$

I now form the n-fold integral over (15), extended over an arbitrary domain, and choose the $p(x)$'s so that they, together with all derivatives occurring in $B - \Gamma$, vanish at the boundary. Since the integral over a divergence reduces to a boundary integral, the integral over the left-hand side of (15) also vanishes for arbitrary $p(x)$'s that vanish at the boundary with sufficiently many derivatives. From the usual arguments it follows that the integrand vanishes for every $p(x)$, giving the ρ relations

$$\sum_i \left\{ a_i^{(\lambda)} \psi_i - \frac{\partial}{\partial x}(b_i^{(\lambda)} \psi_i) + \cdots + (-1)^\sigma \frac{\partial^\sigma}{\partial x^\sigma}(c_i^{(\lambda)} \psi_i) \right\} = 0, \qquad (\lambda = 1, 2, \ldots, \rho). \tag{16}$$

These are the desired dependencies between the Lagrange expressions and their derivatives when I is invariant with respect to $\mathfrak{G}_{\infty\rho}$. The linear independence follows as above, since the converse leads back to (12); and since one can again conclude from the infinitesimal transformations to the finite ones, as is carried out more fully in §4. Accordingly, for a $\mathfrak{G}_{\infty\rho}$, ρ arbitrary transformations already occur in the infinitesimal transformations. From (15) and (16) it follows further that $\mathrm{Div}(B - \Gamma) = 0$.

If, corresponding to a "mixed group", one takes Δx and Δu to be linear in the ε's and the $p(x)$'s, then by setting once the $p(x)$'s and once the ε's equal to zero, one sees that both the divergence relations (13) and the dependencies (16) exist.

§3. The converse in the case of the finite group

To prove the converse, the preceding considerations must first be run through, in essence, in reverse order. From the validity of (13), multiplying by the ε's and adding gives the validity of (12); and by means of the identity (3) one obtains a relation

$$\bar{\delta} f + \mathrm{Div}(A - B) = 0.$$

Thus, if one sets $\Delta x = \frac{1}{f}(A - B)$, one thereby arrives at (11); by integration this finally gives (7), $\Delta I = 0$, hence the invariance of I with respect to the infinitesimal transformation determined by $\Delta x, \Delta u$, where the Δu's are determined by (9) from Δx and $\bar{\delta} u$, and $\Delta x, \Delta u$ become linear in the parameters. But $\Delta I = 0$ implies, in the known way, the invariance of I with respect to the finite transformations arising by integration of the simultaneous system:

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = \Delta x_i, \qquad \frac{\mathrm{d}u_i}{\mathrm{d}t} = \Delta u_i, \qquad \left\{ \begin{array}{c} x_i = y_i, \\ u_i = v_i \end{array} \right\} \quad \text{for } t = 0. \tag{17}$$

[211] That it is no restriction to assume the p's independent of $u, \partial u/\partial x, \ldots$ is shown by the converse.

These finite transformations contain ρ parameters $a_1, \dots, a_\rho$, namely the combinations $t\varepsilon_1, \dots, t\varepsilon_\rho$. From the assumption that there are to be ρ, and only ρ, linearly independent divergence relations (13), it follows further that the finite transformations, as soon as they do not contain the derivatives $\partial u/\partial x$, always form a group. In the contrary case, at least one infinitesimal transformation arising by Lie's bracket process would not be a linear combination of the remaining ρ; and since I also admits this transformation, there would be more than ρ linearly independent divergence relations. Or else this infinitesimal transformation would be of the special form $\bar{\delta}u = 0,\ \mathrm{Div}(f\Delta x) = 0$. But then Δx or Δu would depend on derivatives, contrary to the assumption. Whether this case can occur when derivatives appear in Δx or Δu must be left open; then all functions Δx for which $\mathrm{Div}(f\Delta x) = 0$ must still be added to the Δx determined above in order to obtain the group property again, but by agreement the parameters thereby added are not to be counted. This proves the converse.

It also follows from this converse that Δx and Δu may indeed be assumed linear in the parameters. If Δu and Δx were forms of higher degree in ε, then, because of the linear independence of the power products of the ε's, entirely corresponding relations (13), only in larger number, would follow; by the converse, these imply invariance of I with respect to a group whose infinitesimal transformations contain the parameters linearly. If this group is to contain exactly ρ parameters, then there must be linear dependencies among the divergence relations originally obtained from the terms of higher degree in ε.

It should also be noted that, in the case where Δx and Δu also contain derivatives of the u's, the finite transformations may depend on infinitely many derivatives of the u's; for in this case the integration of (17), when determining

$$\frac{\mathrm{d}^2 x_i}{\mathrm{d}t^2}, \quad \frac{\mathrm{d}^2 u_i}{\mathrm{d}t^2},$$

leads to

$$\Delta\left(\frac{\partial u}{\partial x_\lambda}\right) = \frac{\partial\,\Delta u}{\partial x_\lambda} - \sum_\nu \frac{\partial u}{\partial x_\nu}\frac{\partial\,\Delta x_\nu}{\partial x_\lambda},$$

so that in general the number of derivatives of u grows at every step. An example is:

$$f = \frac{1}{2}u'^2, \qquad \psi = -u'', \qquad \psi \cdot x = \frac{\mathrm{d}}{\mathrm{d}x}\{(u - u'x)\}, \qquad \bar{\delta}u = x\varepsilon,$$

$$\Delta x = -\frac{2u}{u'}\,\varepsilon, \qquad \Delta u = \left(x - \frac{2u}{u'}\right)\varepsilon.$$

Since the Lagrange expressions of a divergence vanish identically, the converse finally shows the following: if I admits a $\mathfrak{G}_\rho$, then every integral that differs from I only by a boundary integral, that is, by an integral over a divergence, also admits a $\mathfrak{G}_\rho$ with the same $\bar{\delta}u$'s, while its infinitesimal transformations will in general contain derivatives of the u's. Thus, corresponding to the preceding example,

$$f^* = \frac{1}{2}\left\{u'^2 - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u^2}{x}\right)\right\}$$

admits the infinitesimal transformation $\Delta u = x\varepsilon,\ \Delta x = 0$; whereas derivatives of u occur in the infinitesimal transformations corresponding to f.

If one passes to the variational problem, that is, if one sets $\psi_i = 0$,[212] then (13) becomes the equations

$$\mathrm{Div}\,B^{(1)} = 0, \dots, \qquad \mathrm{Div}\,B^{(\rho)} = 0,$$

which are often called "conservation laws". In the one-dimensional case this gives:

$$B^{(1)} = \text{const.}, \dots, \qquad B^{(\rho)} = \text{const.}.$$

Here the B's contain at most $(2\kappa-1)$-st derivatives of u (by (6)), as soon as Δu and Δx contain no derivatives higher than the κ-th derivatives occurring in f. Since ψ contains, in general, 2κ-th derivatives,[213] one thus has the existence of ρ first integrals. That nonlinear dependencies may exist among them is shown again by the above f. To the linearly independent $\Delta u = \varepsilon_1,\ \Delta x = \varepsilon_2$ correspond the linearly independent relations

$$u'' = \frac{\mathrm{d}}{\mathrm{d}x}u', \qquad u''u' = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}(u')^2,$$

[212] $\psi_i = 0$, or somewhat more generally $\psi_i = T_i$, where the T_i's are newly adjoined functions, are called "field equations" in physics. In the case $\psi_i = T_i$, the identities (13) pass into the equations $\mathrm{Div}\,B^{(\lambda)} = \sum_i T_i \bar{\delta}u_i^{(\lambda)}$, which in physics are also still called conservation laws.

[213] As soon as f is nonlinear in the κ-th derivatives.

whereas among the first integrals $u' = \text{const.}$, $u'^2 = \text{const.}$ there is a nonlinear dependency. This is the elementary case in which $\Delta u, \Delta x$ contain no derivatives of u.[214]

§4. The converse in the case of the infinite group

First it is to be shown that the assumption of linearity of Δx and Δu is no restriction. Here this follows, even without the converse, from the fact that $\mathfrak{G}_{\infty\rho}$ depends formally on ρ, and only ρ, arbitrary functions. Namely, in the nonlinear case, when transformations are composed, the terms of lowest order add and the number of arbitrary functions would increase. Indeed, suppose for instance that

$$\begin{aligned} y &= A\Big(x, u, \frac{\partial u}{\partial x}, \dots ; p\Big) \\ &= x + \sum a(x,u,\dots)p^\nu + b(x,u,\dots)p^{\nu-1}\frac{\partial p}{\partial x} + c\,p^{\nu-2}\left(\frac{\partial p}{\partial x}\right)^2 + \cdots + d\left(\frac{\partial p}{\partial x}\right)^\nu + \cdots, \end{aligned}$$

where

$$p^\nu = (p^{(1)})^{\nu_1}\cdots(p^{(\rho)})^{\nu_\rho},$$

and correspondingly

$$v = B\Big(x, u, \frac{\partial u}{\partial x}, \dots ; p\Big).$$

Then, by composition with

$$z = A\Big(y, v, \frac{\partial v}{\partial y}, \dots ; q\Big),$$

the terms of lowest order give

$$z = x + \sum a(p^\nu + q^\nu) + b\left\{p^{\nu-1}\frac{\partial p}{\partial x} + q^{\nu-1}\frac{\partial q}{\partial x}\right\} + c\left\{p^{\nu-2}\left(\frac{\partial p}{\partial x}\right)^2 + q^{\nu-2}\left(\frac{\partial q}{\partial x}\right)^2\right\} + \cdots.$$

If here some coefficient other than a and b is different from zero, so that a term

$$p^{\nu-\sigma}\left(\frac{\partial p}{\partial x}\right)^\sigma + q^{\nu-\sigma}\left(\frac{\partial q}{\partial x}\right)^\sigma$$

actually occurs for $\sigma > 1$, then it cannot be written as the differential quotient of a single function, nor as a product of powers of such a quotient. Thus the number of arbitrary functions has increased against the hypothesis. If all coefficients other than a and b vanish, then, depending on the values of the exponents $\nu_1, \dots, \nu_\rho$, the second term is the differential quotient of the first (as, for example, always for a $\mathfrak{G}_{\infty 1}$), so that linearity actually occurs; or else the number of arbitrary functions increases here as well. Hence, because of linearity in $p(x)$, the infinitesimal transformations satisfy a system of linear partial differential equations; and, since the group property is fulfilled, they form an "infinite group of infinitesimal transformations" in Lie's sense (Grundlagen, §10).

The converse now results similarly to the case of the finite group. The existence of the dependencies (16), after multiplication by $p^{(\lambda)}(x)$ and addition, leads by means of the identical transformation (14) to

$$\sum_i \psi_i \bar{\delta} u_i = \operatorname{Div}\Gamma.$$

From this, as in §3, one obtains the determination of Δx and Δu, and the invariance of I with respect to these infinitesimal transformations, which in fact depend linearly on ρ arbitrary functions and their derivatives up to order σ. That these infinitesimal transformations, when they contain no derivatives $\partial u/\partial x, \dots$, certainly form a group follows as in §3, since otherwise more arbitrary functions would occur by composition, whereas by assumption there are to be only ρ dependencies (16). Thus they form an "infinite group of infinitesimal transformations". But such a group consists (Grundlagen, Theorem VII, p. 391) of the most general

[214] Otherwise one has additionally $u''u'^{\lambda-1} = \text{const.}$ for every λ, corresponding to

$$u''(u')^{\lambda-1} = \frac{1}{\lambda}\frac{\mathrm{d}}{\mathrm{d}x}(u')^\lambda.$$

infinitesimal transformations of a certain thereby defined "infinite group $\mathfrak{G}$ of finite transformations" in Lie's sense. Every finite transformation is thereby generated from infinitesimal ones (Grundlagen, §7),[215] and therefore arises by integration of the simultaneous system:

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = \Delta x_i, \qquad \frac{\mathrm{d}u_i}{\mathrm{d}t} = \Delta u_i, \qquad \left\{ \begin{array}{l} x_i = y_i, \\ u_i = v \end{array} \right\} \quad \text{for } t = 0.$$

It may be necessary, however, to let the arbitrary $p(x)$'s still depend on t. Thus $\mathfrak{G}$ in fact depends on ρ arbitrary functions. In particular, if it is sufficient to take $p(x)$ free of t, then this dependency is analytic in the arbitrary functions $q(x) = t\,p(x)$.[216] If derivatives $\partial u/\partial x, \ldots$ occur, it may be necessary to adjoin infinitesimal transformations $\bar{\delta} u = 0$, $\mathrm{Div}(f\Delta x) = 0$ before drawing the same conclusions.

Following an example of Lie (Grundlagen, §7), I now indicate a rather general case in which one can get as far as explicit formulae, which also show that the derivatives of the arbitrary functions occur up to order σ; in this case the converse is therefore complete. These are groups of infinitesimal transformations to which there corresponds the group of all transformations of the x's and the transformations of the u's thereby "induced"; that is, transformations of the u's in which Δu, and hence u, depends only on the arbitrary functions occurring in Δx. It is still assumed that the derivatives $\partial u/\partial x, \ldots$ do not occur in Δu. Thus:

$$\Delta x_i = p^{(i)}(x), \qquad \Delta u_i = \sum_{\lambda=1}^{n} \left\{ a_i^{(\lambda)}(x,u) p^{(\lambda)} + b_i^{(\lambda)} \frac{\partial p^{(\lambda)}}{\partial x} + \cdots + c_i^{(\lambda)} \frac{\partial^\sigma p^{(\lambda)}}{\partial x^\sigma} \right\}.$$

Since the infinitesimal transformation $\Delta x = p(x)$ generates every transformation $x = y + g(y)$ with arbitrary $g(y)$, one can in particular determine $p(x)$ as a function of t so that the one-parameter group is generated:

$$x_i = y_i + t\, g_i(y), \tag{18}$$

which for $t = 0$ becomes the identity and for $t = 1$ becomes the desired $x = y + g(y)$. Differentiation of (18) gives

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = g_i(y) = p^{(i)}(x,t), \tag{19}$$

where $p(x,t)$ is determined from $g(y)$ by inversion of (18); conversely, (18) arises from (19) by means of the side condition $x_i = y_i$ for $t = 0$, by which the integral is uniquely fixed. By means of (18), the x's in Δu can be replaced by the integration constants y and by t; the $g(y)$'s then occur exactly up to their σ-th derivatives, since in

$$\frac{\partial p}{\partial x} = \sum_k \frac{\partial g}{\partial y_k} \frac{\partial y_k}{\partial x}$$

the $\partial y/\partial x$ are expressed by $\partial x/\partial y$, and in general $\partial^\sigma p/\partial x^\sigma$ is replaced by its value in

$$\frac{\partial g}{\partial y}, \ldots, \frac{\partial^\sigma g}{\partial y^\sigma}, \ldots, \frac{\partial^\sigma x}{\partial y^\sigma}.$$

For the determination of the u's one therefore obtains the system of equations

$$\frac{\mathrm{d}u_i}{\mathrm{d}t} = F_i\left(g(y), \frac{\partial g}{\partial y}, \ldots, \frac{\partial^\sigma g}{\partial y^\sigma}, u, t \right), \qquad (u_i = v_i \text{ for } t = 0),$$

in which only t and u are variables, while $g(y), \ldots$ belong to the coefficient domain; hence integration gives

$$u_i = v_i + B_i\left(v, g(y), \frac{\partial g}{\partial y}, \ldots, \frac{\partial^\sigma g}{\partial y^\sigma}, t \right)_{t=1}.$$

Thus one obtains transformations that depend exactly on σ derivatives of the arbitrary functions. The identity is contained in this for $g(y) = 0$, by (18); and the group property follows from the fact that the procedure just described yields every transformation $x = y + g(y)$, whereby the induced transformation of the u's is uniquely determined, so that the group $\mathfrak{G}$ is exhausted.

[215] From this it follows in particular that the group $\mathfrak{G}$ generated from the infinitesimal transformations $\Delta x, \Delta u$ of a $\mathfrak{G}_{\infty\rho}$ leads back again to $\mathfrak{G}_{\infty\rho}$. For $\mathfrak{G}_{\infty\rho}$ contains no infinitesimal transformations depending on arbitrary functions and different from $\Delta x, \Delta u$, and cannot contain independent transformations depending on parameters either, since otherwise it would be a mixed group. But by the preceding argument the finite transformations are determined by the infinitesimal ones.

[216] The question whether this latter case perhaps always occurs was raised by Lie in another formulation (Grundlagen, §7 and §13, end).

It also follows incidentally from the converse that it is no restriction to assume that the arbitrary functions depend only on the x's, and not on $u, \partial u/\partial x, \ldots$. In the latter case, in the identical transformation (14), and hence also in (15), besides the $p^{(\lambda)}$'s there would occur the derivatives $\partial p^{(\lambda)}/\partial u,\ \partial p^{(\lambda)}/\partial(\partial u/\partial x), \ldots$. If one now successively takes the $p^{(\lambda)}$'s to be of zeroth, first, and higher degrees in $u, \partial u/\partial x, \ldots$, with arbitrary functions of x as coefficients, then dependencies (16) again arise, only in greater number. But by the preceding converse they can be brought back to the earlier case by being combined with arbitrary functions depending only on x. Similarly one shows that mixed groups correspond to the simultaneous occurrence of dependencies and independent divergence relations.[217]

§5. Invariance of the individual components of the relations

If the group $\mathfrak{G}$ is specialized to the simplest case, the one ordinarily considered, by admitting no derivatives of the u's in the transformations and by letting the transformed independent variables depend only on the x's and not on the u's, then one can infer the invariance of the individual components in the formulae. First, by familiar arguments, one obtains the invariance of

$$\int \cdots \int \left(\sum_i \psi_i \delta u_i\right) \mathrm{d}x,$$

hence the relative invariance of $\sum_i \psi_i \delta u_i$, where δ denotes any variation. Namely, on the one hand,

$$\delta I = \int \cdots \int \delta f\left(x, u, \frac{\partial u}{\partial x}, \ldots\right) \mathrm{d}x = \int \cdots \int \delta f\left(y, v, \frac{\partial v}{\partial y}, \ldots\right) \mathrm{d}y.$$

On the other hand, for $\delta u, \delta(\partial u/\partial x), \ldots$ vanishing at the boundary, to which, because of the linear homogeneous transformation of $\delta u, \delta(\partial u/\partial x), \ldots$, there corresponds $\delta v, \delta(\partial v/\partial y), \ldots$ also vanishing at the boundary, one has

$$\int \cdots \int \delta f\left(x, u, \frac{\partial u}{\partial x}, \ldots\right) \mathrm{d}x = \int \cdots \int \left(\sum_i \psi_i(u, \ldots) \delta u_i\right) \mathrm{d}x,$$

$$\int \cdots \int \delta f\left(y, v, \frac{\partial v}{\partial y}, \ldots\right) \mathrm{d}y = \int \cdots \int \left(\sum_i \psi_i(v, \ldots) \delta v_i\right) \mathrm{d}y.$$

Thus, for $\delta u, \delta(\partial u/\partial x), \ldots$ vanishing at the boundary,

$$\int \cdots \int \left(\sum_i \psi_i(u, \ldots) \delta u_i\right) \mathrm{d}x = \int \cdots \int \left(\sum_i \psi_i(v, \ldots) \delta v_i\right) \mathrm{d}y$$
$$= \int \cdots \int \left(\sum_i \psi_i(v, \ldots) \delta v_i\right) \left|\frac{\partial y_i}{\partial x_k}\right| \mathrm{d}x.$$

[217] As in §3, it follows here too from the converse that besides I, every integral I^* differing from it by an integral over a divergence also admits an infinite group with the same $\bar{\delta}u$'s; here, however, Δx and Δu will in general contain derivatives of the u's. Einstein introduced such an integral I^* into general relativity in order to obtain a simpler formulation of the energy theorems. I give the infinitesimal transformations admitted by this I^*, using exactly the notation of Klein's second note. The integral $I = \int \cdots \int K\,\mathrm{d}\omega = \int \cdots \int \mathfrak{R}\,\mathrm{d}S$ admits the group of all transformations of the w's and the group thereby induced for the $g_{\mu\nu}$'s. To this correspond the dependencies ((30) in Klein):

$$\sum \mathfrak{R}_{\mu\nu} g_{\mu\nu}^{\alpha\beta} + 2 \sum \frac{\partial g_{\mu\nu}^{\alpha\beta} \mathfrak{R}_{\mu\nu}}{\partial w^\sigma} = 0.$$

Now $I^* = \int \cdots \int \mathfrak{R}^*\,\mathrm{d}S$, where $\mathfrak{R}^* = \mathfrak{R} + \mathrm{Div}$, and consequently $\mathfrak{R}^*_{\mu\nu} = \mathfrak{R}_{\mu\nu}$, where $\mathfrak{R}^*_{\mu\nu}, \mathfrak{R}_{\mu\nu}$ denote the Lagrange expressions respectively. The given dependencies are therefore also dependencies for $\mathfrak{R}^*_{\mu\nu}$; and after multiplication by p^τ and addition, by reversing the transformations of product differentiation, one obtains

$$\sum \mathfrak{R}_{\mu\nu} p^{\mu\nu} + 2\,\mathrm{Div}\left(\sum g^{\mu\sigma} \mathfrak{R}_{\mu\tau} p^\tau\right) = 0,$$

$$\delta \mathfrak{R}^* + \mathrm{Div}\left(\sum 2 g^{\mu\sigma} \mathfrak{R}_{\mu\tau} p^\tau - \frac{\partial \mathfrak{R}^*}{\partial g_{\mu\nu}^\sigma} p^{\mu\nu}\right) = 0.$$

Comparison with the Lie differential equation $\delta \mathfrak{R}^* + \mathrm{Div}(\mathfrak{R}^* \Delta w) = 0$ gives

$$\Delta w^\sigma = \frac{1}{\mathfrak{R}^*}\left(\sum 2 g^{\mu\sigma} \mathfrak{R}_{\mu\tau} p^\tau - \frac{\partial \mathfrak{R}^*}{\partial g_{\mu\nu}^\sigma} p^{\mu\nu}\right), \qquad \Delta g^{\mu\nu} = p^{\mu\nu} + \sum g_\sigma^{\mu\nu} \Delta w^\sigma,$$

as infinitesimal transformations admitted by I^*. These infinitesimal transformations therefore depend on the first and second derivatives of the $g^{\mu\nu}$'s, and contain the arbitrary p's up to the first derivative.

If one expresses $y, v, \delta v$ in the third integral by $x, u, \delta u$, and sets it equal to the first, then one obtains a relation

$$\int \cdots \int \left(\sum_i \chi_i(u, \ldots) \delta u_i \right) \mathrm{d}x = 0$$

for δu vanishing at the boundary but otherwise arbitrary. From this it follows, as is known, that the integrand vanishes for arbitrary δu; hence the relation

$$\sum_i \psi_i(u, \ldots) \delta u_i = \left| \frac{\partial y_i}{\partial x_k} \right| \left(\sum_i \psi_i(v, \ldots) \delta v_i \right)$$

holds identically in δu. This expresses the relative invariance of $\sum_i \psi_i \delta u_i$, and consequently the invariance of

$$\int \cdots \int \left(\sum_i \psi_i \delta u_i \right) \mathrm{d}x.$$

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To apply this to the derived divergence relations and dependencies, one must first show that the $\bar{\delta} u$ derived from $\Delta u, \Delta x$ actually satisfies the transformation laws for the variation δu, provided only that the parameters, respectively arbitrary functions, in $\bar{\delta} v$ are determined as they correspond to the similar group of infinitesimal transformations in y, v. Let T_q denote the transformation that carries x, u into y, v; and let T_p be an infinitesimal transformation in x, u. Then the similar one in y, v is given by $T_r = T_q T_p T_q^{-1}$, where the parameters, respectively arbitrary functions, r are determined by p and q. In formulae this is expressed as follows:

$$T_p: \ \xi = x + \Delta x(x, p), \qquad u^* = u + \Delta u(x, u, p);$$

$$T_q: \ y = A(x, q), \qquad v = B(x, u, q);$$

$$T_q T_p: \ \eta = A(x + \Delta x(x, p), q), \quad v^* = B(x + \Delta x(p),\, u + \Delta u(p), q).$$

But from this arises $T_r = T_q T_p T_q^{-1}$, hence

$$\eta = y + \Delta y(r), \qquad v^* = v + \Delta v(r),$$

where, by means of the inverse of T_q, one regards the x's as functions of the y's and retains only the infinitesimal terms. Thus one has the identity

$$\eta = y + \Delta y(r) = y + \sum \frac{\partial A(x, q)}{\partial x} \Delta x(p), \tag{20a}$$

$$v^* = v + \Delta v(r) = v + \sum \frac{\partial B(x, u, q)}{\partial x} \Delta x(p) + \sum \frac{\partial B(x, u, q)}{\partial u} \Delta u(p). \tag{20b}$$

If, in this, one replaces $\xi = x + \Delta x$ by $\xi - \Delta \xi$, so that ξ goes back into x and hence Δx vanishes, then by the first formula (20) η also goes back into $y = \eta - \Delta \eta$. If under this substitution $\Delta u(p)$ goes over into $\bar{\delta} u(p)$, then $\Delta v(r)$ also goes over into $\bar{\delta} v(r)$, and the second formula (20) gives

$$v + \bar{\delta} v(y, v, \ldots, r) = v + \sum \frac{\partial B(x, u, q)}{\partial u} \bar{\delta} u(p),$$

$$\bar{\delta} v(y, v, \ldots, r) = \sum \frac{\partial B}{\partial u_\kappa} \bar{\delta} u_\kappa(x, u, p).$$

Thus the transformation formulae for variations are indeed satisfied, provided only that $\bar{\delta} v$ is taken to depend on the parameters, respectively arbitrary functions, r.[219]

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[218]These arguments fail if y also depends on the u's, because then $\delta f(y, v, \partial v/\partial y, \ldots)$ also contains terms $\sum (\partial f/\partial y) \delta y$, so that the divergence transformation does not lead to the Lagrange expressions. They likewise fail if derivatives of the u's are admitted; then the δv's become linear combinations of $\delta u, \delta(\partial u/\partial x), \ldots$, and hence only after a further divergence transformation lead to an identity $\int \cdots \int (\sum \chi_i(u, \ldots) \delta u_i) \mathrm{d}x = 0$, so that on the right the Lagrange expressions again do not occur. The question whether the invariance of $\int \cdots \int (\sum \psi_i \delta u_i) \mathrm{d}x$ already allows one to conclude the existence of divergence relations is, by the converse, equivalent to the question whether it implies invariance of I with respect to a group that need not lead to the same $\Delta u, \Delta x$, but does lead to the same $\bar{\delta} u$. In the special case of the simple integral and only first derivatives in f, for the finite group one can infer the existence of first integrals from the invariance of the Lagrange expressions (compare, for example, Engel, Gött. Nachr. 1916, p. 270).

[219]It is again seen that y must be assumed independent of u, and so on, if the conclusions are to hold. As an example, one may cite the $\delta g^{\mu\nu}$ and δq_σ indicated by Klein, which satisfy the transformations for variations as soon as the p's are subjected to a vector transformation.

It follows in particular that $\sum_i \psi_i \bar{\delta} u_i$ is relatively invariant; hence also, by (12), since the divergence relations are also fulfilled in y, v, the relative invariance of $\operatorname{Div} B$; and further, by (14) and (13), the relative invariance of $\operatorname{Div}\Gamma$ and of the left-hand sides of the dependencies combined with the $p^{(\lambda)}$'s, where in the transformed formulae the arbitrary $p(x)$'s (respectively the parameters) are everywhere to be replaced by the r's. From this follows also the relative invariance of $\operatorname{Div}(B-\Gamma)$, that is, of the divergence of a system of functions $B-\Gamma$ that does not vanish identically, but whose divergence vanishes identically.

From the relative invariance of $\operatorname{Div} B$, in the one-dimensional case and for a finite group, one can draw a further conclusion about the invariance of the first integrals. The transformation of the parameters corresponding to the infinitesimal transformation becomes, by (20), linear and homogeneous; and because all transformations are invertible, the ε's are also linear and homogeneous in the transformed parameters ε^*. This invertibility is certainly preserved when one sets $\psi = 0$, since no derivatives of u occur in (20). By equating the coefficients of the ε^* in

$$\operatorname{Div} B(x, u, \ldots, \varepsilon) = \frac{\mathrm{d}y}{\mathrm{d}x} \operatorname{Div} B(y, v, \ldots, \varepsilon^*)$$

the quantities $\frac{\mathrm{d}}{\mathrm{d}y} B^{(\lambda)}(y, v, \ldots)$ also become linear homogeneous functions of the $\frac{\mathrm{d}}{\mathrm{d}x} B^{(\lambda)}(x, u, \ldots)$. Thus from

$$\frac{\mathrm{d}}{\mathrm{d}x} B^{(\lambda)}(x, u, \ldots) = 0 \quad \text{or} \quad B^{(\lambda)}(x, u) = \text{const.}$$

it follows that

$$\frac{\mathrm{d}}{\mathrm{d}y} B^{(\lambda)}(y, v, \ldots) = 0 \quad \text{or} \quad B^{(\lambda)}(y, v) = \text{const.}$$

The ρ first integrals corresponding to a $\mathfrak{G}_\rho$ therefore each admit the group, so that the further integration is also simplified. The simplest example of this is that f is free of x or of a u, corresponding to the infinitesimal transformations $\Delta x = \varepsilon, \Delta u = 0$, respectively $\Delta x = 0, \Delta u = \varepsilon$. Then $\bar{\delta} u = -\varepsilon\, \mathrm{d}u/\mathrm{d}x$, respectively ε; and since B is derived from f and $\bar{\delta} u$ by differentiation and rational operations, it is also free of x, respectively u, and admits the corresponding groups.[220]

§6. A theorem of Hilbert

From the foregoing there finally results the proof of an assertion of Hilbert concerning the connection between the failure of proper energy theorems and "general relativity" (Klein's first note, Göttinger Nachr. 1917, Reply, first paragraph), in a generalized group-theoretic formulation.

Suppose the integral I admits a $\mathfrak{G}_{\infty\rho}$, and let $\mathfrak{G}_\sigma$ be any finite group arising by specialization of the arbitrary functions, thus a subgroup of $\mathfrak{G}_{\infty\rho}$. To the infinite group $\mathfrak{G}_{\infty\rho}$ there correspond dependencies (16), to the finite group $\mathfrak{G}_\sigma$ divergence relations (13); conversely, from the existence of any divergence relations follows the invariance of I with respect to a finite group, which is identical with $\mathfrak{G}_\sigma$ if and only if the $\bar{\delta} u$'s are linear combinations of those arising from $\mathfrak{G}_\sigma$. Hence the invariance with respect to $\mathfrak{G}_\sigma$ can lead to no divergence relations different from (13). But since from the existence of (16) follows the invariance of I with respect to the infinitesimal transformations $\Delta u, \Delta x$ of $\mathfrak{G}_{\infty\rho}$ for arbitrary $p(x)$, it follows in particular that I is already invariant with respect to the infinitesimal transformations of $\mathfrak{G}_\sigma$ arising from these by specialization, and hence with respect to $\mathfrak{G}_\sigma$. The divergence relations

$$\sum_i \psi_i \bar{\delta} u_i^{(\lambda)} = \operatorname{Div} B^{(\lambda)}$$

must therefore be consequences of the dependencies (16), which can also be written

$$\sum_i \psi_i a_i^{(\lambda)} = \operatorname{Div} \chi^{(\lambda)},$$

[220] In those cases where the existence of first integrals already follows from the invariance of $\int(\sum \psi_i \delta u_i)\mathrm{d}x$, these do not admit the complete group $\mathfrak{G}_\rho$. For example, $\int(u''\delta u)\mathrm{d}x$ admits the infinitesimal transformation $\Delta x = \varepsilon_2, \Delta u = \varepsilon_1 + x\varepsilon_3$; whereas the first integral $u - u'x = \text{const.}$, corresponding to $\Delta x = 0, \Delta u = x\varepsilon_3$, does not admit the two other infinitesimal transformations, since it contains both u and x explicitly. To this first integral there correspond infinitesimal transformations for f that contain derivatives. Thus one sees that the invariance of $\int \cdots \int(\sum \psi_i \delta u_i)\mathrm{d}x$ in any case accomplishes less than the invariance of I, a point relevant to the question raised in a preceding note.

where the $\chi^{(\lambda)}$'s are linear combinations of the Lagrange expressions and their derivatives. Since the ψ's enter linearly both in (13) and in (16), the divergence relations must in particular be linear combinations of the dependencies (16). Thus

$$\operatorname{Div} B^{(\lambda)} = \operatorname{Div}\left(\sum \alpha_\mu \chi^{(\mu)}\right),$$

and the $B^{(\lambda)}$'s themselves are thus composed linearly from the χ's, that is, from the Lagrange expressions and their derivatives, and from functions whose divergence vanishes identically, as for instance the $B-\Gamma$ occurring at the end of §2, for which $\operatorname{Div}(B-\Gamma)=0$, and where the divergence has at the same time the character of an invariant. I shall call divergence relations in which the $B^{(\lambda)}$'s can be composed in the indicated way from the Lagrange expressions and their derivatives "improper", and all the remaining ones "proper".

Conversely, if the divergence relations are linear combinations of the dependencies (16), hence "improper", then the invariance with respect to $\mathfrak{G}_\sigma$ follows from that with respect to $\mathfrak{G}_{\infty\rho}$; $\mathfrak{G}_\sigma$ becomes a subgroup of $\mathfrak{G}_{\infty\rho}$. Thus the divergence relations corresponding to a finite group $\mathfrak{G}_\sigma$ become improper if and only if $\mathfrak{G}_\sigma$ is a subgroup of an infinite group with respect to which I is invariant.

By specialization of the groups, Hilbert's original assertion follows from this. By the "translation group" one means the finite group

$$y_i = x_i + \varepsilon_i, \qquad v_i(y) = u_i(x);$$

hence

$$\Delta x_i = \varepsilon_i, \qquad \Delta u_i = 0, \qquad \bar{\delta} u_i = -\sum_\lambda \frac{\partial u_i}{\partial x_\lambda}\varepsilon_\lambda.$$

Invariance with respect to the translation group says, as is known, that in

$$I = \int \cdots \int f\left(x, u, \frac{\partial u}{\partial x}, \ldots\right) \mathrm{d}x$$

the x's do not occur explicitly in f. The associated n divergence relations

$$\sum_i \psi_i \frac{\partial u_i}{\partial x_\lambda} = \operatorname{Div} B^{(\lambda)} \qquad (\lambda = 1, 2, \ldots, n)$$

are to be called "energy relations", since the "conservation laws" $\operatorname{Div} B^{(\lambda)} = 0$ corresponding to the variational problem correspond to the "energy theorems", and the $B^{(\lambda)}$'s to the "energy components". Thus one obtains: if I admits the translation group, then the energy relations become improper if and only if I is invariant with respect to an infinite group containing the translation group as a subgroup.[221]

An example of such infinite groups is given by the group of all transformations of the x's and of the induced transformations of the $u(x)$'s in which only derivatives of the arbitrary functions $p(x)$ occur. The translation group arises by the specialization $p^{(i)}(x) = \varepsilon_i$. It must remain undecided, however, whether by this means — and by the groups arising through changing I by a boundary integral — the most general such groups have already been given. Induced transformations of the indicated kind arise, for example, by subjecting the u's to the coefficient transformations of a "total differential form", that is, of a form

$$\sum a\, \mathrm{d}^2 x_i + \sum b\, \mathrm{d}x_i \mathrm{d}x_k + \cdots,$$

which contains, besides the $\mathrm{d}x$'s, also higher differentials. More special induced transformations, in which the $p(x)$'s occur only in first derivatives, are given by the coefficient transformations of ordinary differential forms $\sum c\, \mathrm{d}x_{i_1} \cdots \mathrm{d}x_{i_\lambda}$; these are the ones that have usually been considered.

A further group of the indicated kind — which, because of the occurrence of the logarithmic term, cannot be a coefficient transformation — is, for example, the following:

$$y = x + p(x), \qquad v_i = u_i + \lg(1 + p'(x)) = u_i + \lg \frac{\mathrm{d}y}{\mathrm{d}x};$$

$$\Delta x = p(x), \qquad \Delta u_i = p'(x), \qquad \bar{\delta} u_i = p'(x) - u_i' p(x).$$

Here the dependencies (16) become

$$\sum_i \left(\psi_i u_i' + \frac{\mathrm{d}\psi_i}{\mathrm{d}x}\right) = 0,$$

[221] The energy theorems of classical mechanics, and likewise those of the old "theory of relativity" (where $\sum \mathrm{d}x^2$ goes over into itself), are "proper", since no infinite groups occur here.

and the improper energy relations become

$$\sum_i \left(\psi_i u_i' + \frac{\mathrm{d}(\psi_i + \mathrm{const.})}{\mathrm{d}x} \right) = 0.$$

A simplest invariant integral of the group is

$$I = \int \frac{e^{-2u_1}}{u_1' - u_2'}\, \mathrm{d}x.$$

The most general I is determined by integration of the Lie differential equation (11)

$$\bar{\delta} f + \frac{\mathrm{d}}{\mathrm{d}x}(f \cdot \Delta x) = 0,$$

which, after substituting the values for Δx and $\bar{\delta} u$, becomes, as soon as one assumes f to depend on only first derivatives of the u's,

$$\frac{\partial f}{\partial x} p(x) + \left\{ \sum_i \frac{\partial f}{\partial u_i} - \sum_i \frac{\partial f}{\partial u_i'} u_i' + f \right\} p'(x) + \left\{ \sum_i \frac{\partial f}{\partial u_i'} \right\} p''(x) = 0$$

(identically in $p(x), p'(x), p''(x)$). Already for two functions $u(x)$, this system of equations has solutions that actually contain the derivatives, namely

$$f = (u_1' - u_2')\, \Phi \left(u_1 - u_2, \frac{e^{-u_1}}{u_1' - u_2'} \right),$$

where Φ denotes an arbitrary function of the indicated arguments.

Hilbert states his assertion by saying that the failure of proper energy theorems is a characteristic feature of the "general theory of relativity". For this assertion to hold literally, the designation "general relativity" must therefore be understood more broadly than usual, and extended also to the preceding groups depending on n arbitrary functions.[222]

14. The Arithmetic Theory of Algebraic Functions of One Variable in Its Relation to the Other Theories and to Algebraic Number Theory

Jahresbericht der Deutschen Mathematiker-Vereinigung 38 (1919), pp. 182–203

The following report arose from the wish to have a connecting link between the individual theories of algebraic functions; at the same time it may be regarded as a supplement to the report by Brill–Noether,[223] in which the discussion of the arithmetic theory is essentially absent.[224]

For discussions of the individual theories, besides Brill–Noether, the following encyclopedia articles, with the literature indicated there, come into consideration:

For the theory of Riemann: Wirtinger (II B 2), nos. 1–19;

for the theory of Weierstrass: Wirtinger (II B 2), nos. 23, 32, 33, 34;

for the algebraic-geometric theory of Brill–Noether: Wirtinger (II B 2), nos. 21–31; Berzolari (III C 4), nos. 23–38;

for the arithmetic theory of Dedekind–Weber and Hensel–Landsberg: Hensel (II C 5), [cited as Hensel];[225]

more generally, for the arithmetic theory of algebraic quantities: Landsberg (I C 5), and for the arithmetic theory of algebraic functions of several variables: Jung (II C 6).

For algebraic number theory compare Hilbert's "Zahlbericht"[226] and the lectures on number theory by Dirichlet–Dedekind [cited as Dedekind–number theory].[227]

[222] This again confirms the correctness of a remark of Klein, that the term "relativity" customary in physics should be replaced by "invariance relative to a group". (Über die geometrischen Grundlagen der Lorentzgruppe. *Jhrber. d. d. Math. Vereinig.* vol. 19, p. 287, 1910; reprinted in the *Physikalische Zeitschrift.*)

[223] The development of the theory of algebraic functions in earlier and more recent times. *Jahresbericht der Deutschen Mathematiker-Vereinigung* vol. 3 (1894) [cited as Brill–Noether].

[224] With the exception of a discussion of Kronecker's investigation of the discriminant. (*J. f. M.* vol. 91.)

[225] The parts concerning Dedekind–Weber are due to A. Ostrowski.

[226] The theory of algebraic number fields. *Jahresbericht der Deutschen Mathematiker-Vereinigung* vol. 4 (1894/95).

[227] Braunschweig, Vieweg, 4th ed. 1894.

Contents

1. Foundations of the various theories

Riemann's theory is the only purely transcendental one, resting on existence theorems (Dirichlet's principle, Schwarz–Neumann methods). The integrals of algebraic functions are here primary, the algebraic functions themselves secondary. The functions on the Riemann surface are characterized by their zeros and poles, corresponding to the polygons (respectively divisors) of the arithmetic theory; whereas Weierstrass also speaks of the functions themselves together with their zeros and poles. The main problem, apart from the existence theorems, is the setting up of all functions with given poles, which is accomplished by integrals of the first and second kind; and, further, the question of the number of arbitrary constants that still enter when the points at infinity are prescribed. This latter question is answered by the Riemann–Roch theorem; corresponding to it in the arithmetic theory is the question of the dimension of a polygon class (respectively divisor class), and in the algebraic-geometric theory the question of the dimension of a complete linear system, whose answer is also provided here by the Riemann–Roch theorem. The actual construction of the functions is carried out in the arithmetic theory by means of the basis theorems; in the algebraic-geometric theory, by the residue theorem. In particular, the zeros of the integrands of the first kind correspond to the differential class in the arithmetic theory, and to the doctrine of special groups (that is, the pencil of point groups cut out by the adjoint curves of order $(n-3)$, the φ-curves) in the algebraic-geometric theory.

Just as Riemann's theory historically preceded the other theories, even though the rigorous proof of the existence theorems was obtained only later, so also later, with its transcendental aids, algebraic questions were settled which to this day are not yet accessible to a purely algebraic or arithmetic treatment; here above all Hurwitz's theory of singular correspondences is to be mentioned.[228]

Analogies with the transcendental questions in the theory of algebraic number fields will be discussed in the last section.

For the remaining theories the algebraic functions are primary, the integrals secondary. In this respect the Weierstrass theory rests on the transcendental foundations of power-series expansion and the residue theorem; but it then proceeds algebraically, by explicit specification of all functions under consideration.

The arithmetic theory in the form of Hensel–Landsberg also uses the transcendental foundations of series expansion. But by assigning divisors to the individual points on this foundation, it then proceeds purely arithmetically and can thus be viewed as a fusion of Weierstrass and Dedekind–Weber. In contrast with the latter, it simultaneously supplies methods for the actual construction of all basis functions in question. In detail it exhibits a number of simplifications compared with Dedekind–Weber, essentially by the introduction of "fractional" divisors. Recently Hensel[229] has given an arithmetic foundation for the theory of series expansions (apart from a majorant method for the convergence proof), whereby the theory has become almost purely arithmetic.

The algebraic-geometric theory of Brill–Noether presupposes only the concept of "successive points" of a branch, which is reduced to the series expansion at an ordinary point[230] and corresponds to the Weierstrass

[228] A. Hurwitz, Über algebraische Korrespondenzen und das erweiterte Korrespondenzprinzip. *Math. Ann.* vol. 28. (Compare Brill–Noether, section 10.)

[229] *Mathematische Zeitschrift* vol. 4; the foundation is also used as the basis of his report.

[230] For successive "singular" points compare Brill–Noether, section 6, especially nos. 10 and 11. Ordinary successive points are, for example, the points of intersection of the tangent with the curve or higher points of contact (branch points).

element concept; it then continues with algebraic-rational methods, essentially those of ternary elimination theory; it too reaches explicit representations.

The original, purely arithmetic theory of Dedekind–Weber[231] is akin to Riemann's in that its theorems again have the character of existence proofs; it preserves complete purity of method.[232] The arithmetic foundation of the theory, where the variables play only the role of indeterminates, runs parallel to the theory of algebraic numbers, as will be explained more closely in 2.[233] This foundation gives the means of introducing the point concept arithmetically (compare Hensel no. 10a) and thus of setting up the concept of the absolute Riemann surface without any consideration of continuity. The theory thereby acquires a formal character; for example, the concept of the differential is also introduced purely formally. On the basis obtained by the point concept, the close kinship with the methods and concepts of the algebraic-geometric theory can be demonstrated, since the latter too, apart from the element concept, is free of transcendental aids; this is to be done in the following sections.

2. Parallelism between algebraic numbers and algebraic functions. (Conjugate elements, basis theorems, module concept)

An algebraic number field is defined as a field of degree n over the field R of rational numbers; a field of algebraic functions of one variable as a field of degree n over the field $K(z)$ of all rational functions of one indeterminate z with arbitrary complex coefficients. Thus for both fields all theorems hold in common which abstract completely from the special nature of the ground field; these are the theorems on bases, norms, traces, and discriminants.[234] All these concepts are introduced by Dedekind–Weber without the use of conjugate elements. It should be noted, however, that the concept of the conjugate field can be defined in an entirely formal way, as Steinitz showed following Kronecker.[235] Namely, if K is an arbitrary field and $\varphi(x)$ a polynomial in x irreducible over K, then the system of residue classes modulo $\varphi(x)$ forms a field. If one assigns to the residue class of x an element ξ, then under the isomorphic assignment the element $f(\xi)$ corresponds to the residue class represented by the polynomial $f(x)$. Since all polynomials divisible by $\varphi(x)$ represent the zero class, $\varphi(\xi)$ is assigned to the zero class; the element ξ can thus symbolically be regarded as a zero of $\varphi(x) = 0$ and generates the algebraic field $K(\xi)$. Repetition of the procedure leads to the n conjugate fields $K(\xi), K(\xi'), \dots, K(\xi^{(n-1)})$. Thus one may in particular speak of conjugate elements also for algebraic functions, without leaving the purely arithmetic theory.

The two ground fields R and $K(z)$ have the property in common that in them the system of all integral quantities is defined, as rational integers and as integral rational functions of one indeterminate, respectively. These integral quantities have the characteristic property that any two a and b possess a greatest common divisor d that is representable in the form $ma + nb$, where d, m, n are also integral quantities from R or $K(z)$; and indeed d is the absolutely smallest number, respectively the polynomial of lowest degree, representable in this form. On this property alone, which entails the unique decomposition into prime factors for rational integers and for functions of one indeterminate, rest the theorems on the integral quantities of the algebraic or upper field. The argument proving the existence of the greatest common divisor also gives the existence of a fundamental system or a basis of the integral quantities of the algebraic field. If one considers only such bases of the field as consist of integral quantities, then their discriminant becomes a rational integer, respectively a rational function of one indeterminate. Every basis that corresponds to the absolutely smallest number, respectively to the function of lowest degree, forms a fundamental system.

A more precise insight into the basis questions, also for more general fields, is given by the introduction of the module concept, which shall be formulated so as to include the definitions that occur differently in the literature according to the nature of the elements. A module — with respect to a basic domain of integrity $\mathfrak{S}$ — is a system of elements such that the difference of two elements again belongs to the system, and likewise the product of an element with an arbitrary quantity from $\mathfrak{S}$. (If $\mathfrak{S}$ consists of the rational integers, then the second requirement is a consequence of the first.)[236]

[231] *J. f. M.* 92 (cited as D.–W.).

[232] For this reason it is also more suitable for comparison with the other theories than, for instance, the theory of Hensel–Landsberg, which will occur only occasionally in the following comparisons. — The theory of Abelian integrals and their inversion, insofar as it presupposes continuity, is however not treated in D.–W. nor in the further development of the algebraic theory (M. Noether, *Ann.* 37).

[233] A comparison of the theory of Hensel–Landsberg with number theory, especially the theory of p-adic numbers, is found in Hensel.

[234] D.–W., § 1 and 2; Hensel no. 4 (with note 5).

[235] Steinitz, Algebraische Theorie der Körper. (*J. f. M.* 137, § 6 and 8.)

[236] The module concept is due to Dedekind, who first introduced the number module ($\mathfrak{S}$ = rational integers) in the second edition of the number theory; the module of integral linear forms is also implicitly contained there. In Dedekind–Weber the "function modules" occur, whose elements consist of algebraic functions, whereas $\mathfrak{S}$ consists of all integral rational functions of one indeterminate. — The module of homogeneous forms in n indeterminates, more generally of polynomials, occurs first in Hilbert (*Annalen* 36); $\mathfrak{S}$ consists of all polynomials, and since the elements are also

Every module that has a basis of finitely many elements through which every element can be represented linearly with coefficients from $\mathfrak{S}$ is called finite (finitely generated); in particular, every one-term module is therefore of the form $c \cdot \alpha$, where α denotes the basis element and c runs through all quantities from $\mathfrak{S}$. *A module whose elements themselves belong to* $\mathfrak{S}$ *is called an ideal in* $\mathfrak{S}$*;* whence follows the concept of a finite ideal, and in particular of a one-term ideal (principal ideal).

Now all basis theorems for integral quantities and ideals rest on the following theorem:

If M is a module with respect to $\mathfrak{S}$, whose elements consist of linear forms in n indeterminates with coefficients from $\mathfrak{S}$, and if every ideal in $\mathfrak{S}$ is finite, then M is also a finite module. If in particular every ideal in $\mathfrak{S}$ is a principal ideal, then M possesses a basis of linearly independent forms.[237]

Indeed, if the elements of M are given by $l(x) = a_1x_1 + \cdots + a_nx_n$, then the totality of the coefficients a_1 runs through an ideal in $\mathfrak{S}$, which by hypothesis is of the form $b_1a^{(1)} + \cdots + b_\rho a^{(\rho)}$. The linear form belonging to M,

$$m(x) = l(x) - b_1l^{(1)}(x) - \cdots - b_\rho l^{(\rho)}(x),$$

therefore depends only on $(n-1)$ indeterminates, from which the assertion follows by finitely many repetitions. If in each case $\rho = 1$, then the basis consists of n forms in respectively $n, n-1, \ldots, 1$ indeterminates, which gives the linear independence of the non-vanishing forms.

The theorem on the fundamental system follows from this in the various cases as follows:

If the algebraic field is obtained by adjoining the integral algebraic element δ to the ground field, and if $\mathfrak{S}$ denotes the totality of all integral quantities of the ground field, then every integral quantity of the upper field admits the representation (compare, for instance, Zahlbericht § 3)

$$\omega = \frac{a_1\delta^{n-1} + a_2\delta^{n-2} + \cdots + a_n}{\Delta(\delta)},$$

where the a_i are quantities from $\mathfrak{S}$ and $\Delta(\delta)$ denotes the discriminant of δ. By the substitution

$$x_i = \frac{\delta^{n-i}}{\Delta(\delta)}$$

the module of all integral quantities is transformed into a module of linear forms; hence the existence of the fundamental system follows in all cases where every ideal in $\mathfrak{S}$ is finite. The same argument shows more generally that every ideal in the domain of integral quantities of the upper field is finite. Now in particular for rational integers and for functions of one indeterminate every ideal is a principal ideal; hence it follows that the fundamental systems in algebraic number fields and in fields of algebraic functions of one indeterminate consist of exactly n quantities, when n denotes the degree. And since, in these fields, every ideal is finite by the preceding argument, this at the same time yields the fundamental system for relative fields, which in general will consist of more quantities than the relative degree indicates.[238] Further, if $\mathfrak{S}$ denotes the domain of all polynomials in several indeterminates, every ideal in $\mathfrak{S}$ is also finite;[239] hence the existence of the fundamental system is obtained also for algebraic function fields of several indeterminates, and it will again consist of more quantities than the degree indicates.

3. Parallelism and differences in the ideal theory of algebraic numbers and functions

The proof of the fundamental theorem of ideal theory, that every ideal can be represented uniquely as a product of powers of prime ideals, can be carried out jointly for algebraic numbers and functions; solely on the basis of the property that, for rational integers and for functions of one indeterminate, every ideal is a principal ideal.[240] The main point of the proof lies in showing that, from the divisibility of an ideal $\mathfrak{c}$ by an

polynomials, this module falls under our ideal definition. Kronecker's "module systems" of polynomials start from the basis, not from the abstract definition.

[237] The theorem appears in the literature separately for the different coefficient domains $\mathfrak{S}$. For rational integers compare, for instance, Hilbert, Zahlbericht § 3 and 4; for polynomials in several indeterminates: Hilbert, Über die vollen Invariantensysteme: end of § 2 (*Math. Ann.* 42).

[238] According to Steinitz's investigations on modules of linear forms, where $\mathfrak{S}$ consists of the integers of a (finite) algebraic number field, $\nu + 1$ basis elements suffice when ν denotes the relative degree (*Math. Ann.* 71, § 4).

[239] By Hilbert's theorem on the module basis (*Math. Ann.* 36). The modules of polynomials here called ideals for the sake of consistency are usually called "form modules" or simply "modules". Hilbert's theorem that such a module N is finite can also be reduced to the theorem on linear forms. Namely, let f be a polynomial of degree n contained in N in the $r+1$ indeterminates $x_1, \ldots, x_r, x_{r+1}$, containing the term x_{r+1}^n (which can always be achieved by a linear transformation); then all polynomials from N are congruent mod. f to those representable in the form $a_1(x)x_{r+1}^{n-1} + \cdots + a_n(x)$; hence, by $x_{r+1}^{n-i} = X_i$, they form a module of linear forms for which every ideal in the domain of r indeterminates may be assumed finite, since this is the case for one indeterminate.

[240] Compare a remark in the introduction to Hurwitz: Über die Theorie der Ideale (*Gött. Nachr.* 1894). In Weber's textbook of algebra the proof, following Kronecker, is carried out in parallel for both fields by means of the introduction of functionals. [The connection between Kronecker's theory and ideal theory is mediated by the "extended Gauss theorem"; Hurwitz, loc. cit.]

ideal $\mathfrak{a}$ (that is, every element of $\mathfrak{c}$ is contained in $\mathfrak{a}$), the product representation $\mathfrak{c} = \mathfrak{a}\mathfrak{b}$ follows.[241] In Hurwitz this proof rests on the "extended Gauss theorem", which says that an integral quantity ω which divides every coefficient of the product of two polynomials φ and ψ also divides the product of all coefficients of φ and ψ; in Dedekind it rests on a module theorem equivalent to this theorem.[242]

Going further, the concept of the complementary module (Hensel no. 12), and from it the ramification ideal (ground ideal, different) $\mathfrak{D}$, can be defined jointly (its norm becomes the field discriminant D); and the fundamental theorem holds that the ramification ideal $\mathfrak{D}$ is the *greatest common divisor of all principal ideals* $F'(\delta)$; here δ runs through all integral quantities of the field and $F(\delta) = 0$ denotes in each case the corresponding irreducible equation. From this it follows that the *field discriminant contains all and only those rational primes, respectively linear factors in z, which are divisible by the square of a prime ideal.* For $F'(\delta)$ itself one obtains the representation

$$F'(\delta) = \mathfrak{f} \cdot \mathfrak{D},$$

and by taking norms

$$\Delta(\delta) = R^2 \cdot D;$$

here $\mathfrak{f}$ denotes the conductor of the ring derived from δ (order, domain of integrity), that is, $\mathfrak{f}$ consists of the totality of the (integral) quantities which, after multiplication by any integral quantity, are divisible by the module $[1, \delta, \dots, \delta^{n-1}]$. $R^2 = \operatorname{Norm} \mathfrak{f}$ represents the square of the determinant of substitution that carries the basis $1, \delta, \dots, \delta^{n-1}$ into a basis of the integral quantities.[243]

Differences in the further development are conditioned by special properties of the ground fields. Thus, in the case of algebraic numbers, the fact that the *elements of the ground field are at the same time exponents* leads to the possibility that the ramification ideal may be divisible by a higher power than the $(e-1)$-st power of a prime ideal $\mathfrak{p}$ which occurs in p to the e-th power (namely when e is divisible by p); and this leads to the further concepts of the inertia field and ramification field,[244] which have no analogue in the theory of algebraic functions. Above all, however, because of the *existence of infinitely many constants* in the ground field, the finiteness of the number of ideal classes disappears for algebraic functions, and with it all questions connected with determining the class number. Nevertheless, in a certain sense the domain of all (transcendental) prime forms[245] may be regarded as a transcendental analogue of the class field; for here every point is generated by a single form, a principal ideal.

On the other hand, the existence of infinitely many constants in the ground field $K(z)$ leads, for algebraic functions, to theorems and questions which have no analogue for algebraic numbers. First, a certain simplification in the arrangement of the proof follows from this; for example in the proof given by D.–W. of the unique factorization of ideals into prime ideals, where the "extended Gauss theorem" or a theorem equivalent to it can be avoided by using the fact that every linear form $(z - c)$ has infinitely many residue classes, namely all constants (whereas the number of residue classes modulo a prime number is finite).[246]

The further fact that every polynomial with integral rational functions of z as coefficients decomposes into linear factors mod. $(z - \alpha)$ (which is not the case for an integral polynomial mod. p) leads to genuinely new results; this fact also still holds if, instead of all constants, one takes only all algebraic numbers.[247] From this it follows, namely, that every prime ideal is an ideal of first degree;[248] and further, that the field discriminant D becomes the intersection of all discriminants $\Delta(\delta)$, where δ runs through all integral quantities;[249] both are in contrast to algebraic numbers. Subsequently it again follows from this that the ramification ideal is the intersection of all principal ideals $F'(\delta)$.

Further differences are conditioned by the fact that, for algebraic numbers, the ground field R of rational numbers is something absolutely distinguished, namely the prime field contained in the field (that is, the field derived from the unit); whereas the ground field $K(z)$ is something relative. Every nonconstant function ξ

[241] Emphasized especially by Dedekind (number theory); Supplement XI, theorem VII, § 177; compare the note on p. 554. The theorem uses the fact that algebraic fields are involved; it no longer holds for ideals (modules) of polynomials, where the least common multiple takes the place of the product.

[242] Number theory, Suppl. XI; theorem VI, § 173. Dedekind points out the equivalence of the two theorems (Über die Begründung der Idealtheorie, *Göttinger Nachr.* 1895).

[243] Dedekind, Über die Diskriminante endlicher Körper (*Abhandl. der Gött. Ges. d. Wissensch.* vol. 29, 1882). The definitions given there for algebraic numbers can be transferred directly to algebraic functions; the proofs too can be transferred almost exactly. The order of proof in D.–W. is different: see below. For the cited theorems compare also Zahlbericht, § 31 and 32. ($F'(\delta)$ is there called the "different" of the integer δ.)

[244] Compare Zahlbericht, § 39–45.

[245] Klein, Zur Theorie der Abelschen Funktionen (*Ann.* 36, 1890).

[246] D.–W. § 9. Compare the notes on pp. 212 and 213.

[247] D.–W. note in the introduction that in this case the whole theory remains intact.

[248] D.–W., § 9, no. 7.

[249] D.–W., p. 224.

may be chosen as independent variable; the field then becomes an algebraic field over $K(\xi)$. The theorem that every prime ideal $\mathfrak{p}$ is an ideal of first degree, hence every function is congruent modulo $\mathfrak{p}$ to a constant, combined with the possibility of taking an arbitrary function as independent variable, leads to the most important concept going beyond algebraic numbers, the point concept in purely arithmetic form. (Compare Hensel 10a.) "Every point is represented here by a field of constants isomorphic to the function field."[250] The totality of points so defined forms the "absolute Riemann surface". This definition of point depends only on the field, not on a special variable. The existence proof, however, is carried out by distinguishing some function as independent variable z; a point P is generated by a prime ideal $\mathfrak{p}$ by assigning to every function its constant residue modulo $\mathfrak{p}$; thus in particular the functions divisible by $\mathfrak{p}$ vanish at P, and conversely. As $\mathfrak{p}$ runs through all prime ideals in z, all points in which z is finite are thereby obtained; the remaining points arise by introducing the independent variable $\xi = 1/z$ and correspond to the prime ideals which divide the principal ideal ξ. Naturally, the concepts attached to the point concept also can have no analogue for algebraic numbers; such as the concept of a polygon, of a polygon class, of the dimension of a polygon class, and so on.

It should also be noted that the free choice of the independent variable leads to a further interpretation of the decomposition $F'(\vartheta) = \mathfrak{f} \cdot \mathfrak{D}$. Namely, if one considers ϑ as independent variable, then correspondingly one obtains

$$\left(\frac{\partial F}{\partial z}\right) = \mathfrak{f} \cdot \mathfrak{a},$$

since the ring derived from ϑ was defined symmetrically with respect to z and ϑ; and $\mathfrak{f}$ becomes the greatest common divisor of the two principal ideals $\frac{\partial F}{\partial z}$ and $\frac{\partial F}{\partial \vartheta}$; the conductor $\mathfrak{f}$ becomes at the same time the "double-point ideal" with respect to ϑ, z.[251]

4. Connection of the arithmetic foundation of series expansion with ideal theory

In Hensel–Landsberg the transcendental aid of series expansion and the point concept borrowed from function theory take the place of ideal theory. Hensel's recently given arithmetic foundation of series expansion for algebraic numbers and functions[252] can therefore be regarded as an equivalent of ideal theory. In fact this arithmetic foundation amounts to passing, for each prime ideal, to such a (transcendental) extension field in which this prime ideal becomes a principal ideal, so that the usual decomposition theorems hold; and it suffices to carry out this consideration for all prime ideals occurring in one prime number p, respectively one linear function $z - a$.

First one introduces a transcendental extension field formed by all series expansions in integral powers of $z - a$, respectively p (p-adic numbers), with in each case at most finitely many negative powers. In this field the irreducible equation defining the algebraic number field or function field decomposes into the product of ρ irreducible factors, corresponding to the decomposition of p, respectively $z - a$, into the product of the ρ "primary" ideals $\mathfrak{q}_1, \ldots, \mathfrak{q}_\rho$. The further representation of each primary ideal as a power of a prime ideal,

$$\mathfrak{q}_i = \mathfrak{p}_i^{\ell_i},$$

corresponds in Hensel to the introduction of a further field, algebraic with respect to the transcendental extension field. And in the case of algebraic functions this field can be defined by a pure equation; in the case of algebraic numbers, more generally, by an algebraically solvable equation. The simplification for algebraic functions is due to the fact that here no inertia and ramification fields (compare no. 3) occur.

The "terminating series" in a prime element π[253] that occurs in these investigations (and which does not yet require a transcendental convergence proof)

$$v = c_0 + c_1\pi + \cdots + c_{\sigma-1}\pi^{\sigma-1} - v_\sigma \pi^\sigma,$$

where $c_0, \ldots, c_{\sigma-1}$ are constants and v_σ denotes an integral element of the field, is found in exactly corresponding form in D.–W.[254] There it is derived from ideal theory and gives the basis for defining the order of vanishing (respectively becoming infinite) of a function at a point.

[250] Two fields are, as is well known, called "isomorphic" if they can be put in one-to-one correspondence in such a way that sum, difference, product, and quotient also correspond.

[251] D.–W. p. 263.

[252] Eine neue Theorie der algebraischen Zahlen (*Math. Zeitschr.* vol. 2, 1918) and Neue Begründung der arithmetischen Theorie der algebraischen Funktionen einer Variablen (*Math. Zeitschr.* vol. 4, 1919). For a more detailed exposition compare Hensel no. 44ff.

[253] Neue Begründung . . ., formula (11).

[254] D.–W., p. 231 and § 15.

5. Comparison of the arithmetic theory of algebraic functions with the geometric theory. Different concept of invariance

"Invariant" means any concept characterized by the field alone. In the arithmetic theory this invariance is in general already given in the definition, which is stated with respect to all elements of the field, not with respect to some basis or distinguished variable; the point concept given above is an example. By contrast, the remaining theories start from some special variable or basis and show afterwards the independence of the concept from this special choice; invariance here becomes invariance under birational transformation.

Namely, suppose that the field arises on the one hand by adjoining the algebraic element s to $K(z)$, and on the other by adjoining σ to $K(\xi)$, where $f(z,s)=0$, respectively $F(\xi,\sigma)=0$, denote the irreducible equations for s, respectively σ; or, still more generally, by adjoining a finite number of elements to $K(\eta)$, with the corresponding irreducible equations

$$g_1(\eta,\tau_1)=0, \qquad g_2(\eta,\tau_1,\tau_2)=0,\ldots.$$

Then, by definition, all elements can be expressed rationally by z and s, just as by ξ and σ, and also by $\eta,\tau_1,\tau_2,\ldots$. In particular z and s are therefore rationally expressible by ξ and σ, and conversely; likewise z and s rationally by $\eta,\tau_1,\tau_2,\ldots$, and conversely. The plane curve $f(z,s)=0$ thus goes by "birational transformation" into the plane curve $F(\xi,\sigma)=0$, or also into the curve in the space of variables $\eta,\tau_1,\tau_2,\ldots$ defined by $g_1(\eta,\tau_1)=0; g_2(\eta,\tau_1,\tau_2)=0,\ldots$. Invariance therefore appears in the fact that a definition stated for one special curve remains valid for all curves obtained from it by birational transformation (in a space of arbitrarily many dimensions); for all curves of the "class" in Riemann's sense. A "point" is here defined as a coherent system of values $z=a, s=b$, so that $f(a,b)=0$; under birational transformation this point generally goes again into a coherent system of values α,β of $F(\xi,\sigma)=0$, but in special cases, when a,b was a "singular" point, it can correspond to several "points" on $F(\xi,\sigma)=0$; and one can always specify curves on which a "singular" point of $f(z,s)=0$ corresponds to a finite number of simple points lying separately or successively (resolution of singularities).[255] Likewise a point group again goes into a point group; and the number of its points is invariant as soon as one counts a singular point as as many points as the simple points corresponding to it under a suitable transformation. Since every curve can be transformed birationally into one with "ordinary multiple" points,[256] the geometric theory develops its concepts only for such special curves and then proves the invariance under every birational transformation, including one which leads to the most general singular points.

In particular, as a distinguished curve of the "class" — excluding the hyperelliptic case — one can introduce the so-called "normal curve of the φ's" in the space of $(p-1)$ dimensions, which has no singular points; here the φ's are the p "adjoint curves of order $(n-3)$" for the homogenized $f(z,s)=0$. A birational transformation of $f(z,s)=0$ into $F(\xi,\sigma)=0$ corresponds to a linear transformation of the φ's, so that this is invariance in the projective sense. The representation of all functions of the field as rational functions of the φ's is called the "invariant representation" in the geometric theory.[257]

The linear system of the φ's, or also of the "special groups" cut out by them, therefore goes into itself under every birational transformation, and is consequently characterized by the field alone. It corresponds to the differential class in the arithmetic theory, where the invariance is again shown directly in the definition, which is given by properties with respect to every function of the field as independent variable.

6. Continuation of the comparison. Singular points

The different conception of the point concept is connected with the fact that the "singular points" of curves, which cause special difficulties in the algebraic-geometric theory, do not occur at all in the foundations of the arithmetic theory; and consequently the "adjoint functions" are not required for the construction of the arithmetic theory, as they are in the geometric theory, but belong to special higher parts of the theory. On the other hand, the branch points do not occur in the foundations of the geometric theory; in fact, they also drop out in the representation of the integrals in Aronhold's normal form (Hensel no. 27, formula 80).

The definition of singular point given in 5 can be formulated as follows: $f(z,s)=0$ (respectively a curve in a higher-dimensional space) has a singular point at $z=a, s=b$ (respectively $z=a; s_i=b_i$) if this system of values is assumed by more than one point of the absolute Riemann surface (point in the arithmetic sense), or

[255] Compare, for instance, Brill–Noether, section 6. More precise information on "singular" points appears in the next section.
[256] Ibid.
[257] Compare, for instance, Brill–Noether, section 8. For algebraic functions of several variables the question whether invariance can be reduced to such invariance under linear transformation is not treated.

if it is assumed at one or several points with higher order. In the latter case one has a point of the ramification polygon of z as well as of s; this corresponds to the "successive points" of the geometric theory; the point becomes a (higher) cusp point of $f(z, s) = 0$ (respectively of the curve in the higher-dimensional space).

Since by hypothesis every arbitrary function η of the field can be represented rationally by z and s, because of the isomorphism every point of the absolute Riemann surface at which $z = a, s = b$ is obtained by assigning to every function η, represented by z and s, its value for $z = a, s = b$. Thus, for the point to be singular, there must be at least one function η which assumes several values or one system of values several times for $z = a, s = b$. And because of rational representability by z and s, this can be the case only when in this representation numerator and denominator vanish simultaneously for $z = a, s = b$. It follows that the above definition is identical with the usual one, namely that

$$f, \quad \frac{\partial f}{\partial z}, \quad \frac{\partial f}{\partial s}$$

vanish for $z = a, s = b$. For in this case

$$\eta = \frac{\frac{\partial f}{\partial z}}{\frac{\partial f}{\partial s}}$$

is a function of the indicated kind which is multivalued for $z = a, s = b$, whereas in the opposite case every function, even if it becomes $0/0$, assumes only one value.

With the arithmetic point concept the singular point cannot occur, since two points are regarded as different as soon as only one function is assigned different systems of values. But even in the ideal theory serving to found the arithmetic point concept, the concept of a singular point does not occur, precisely because the totality of all integral quantities of the field is always considered at the same time. All singular points in the finite part of $F(z, \delta) = 0$, where δ is algebraically integral with respect to z, are generated (according to the second definition and the end of no. 3) by the "double-point ideal" $\mathfrak{f}$. Now, since by the fundamental theorem the ramification ideal $\mathfrak{D}$ is the greatest common divisor of all principal ideals $F'(\delta)$, for every prime ideal $\mathfrak{p}$ one can specify a δ such that the corresponding $\mathfrak{f}$ is not divisible by $\mathfrak{p}$, and consequently — since $\mathfrak{f}$ is the greatest common divisor of $F'(\delta)$ and $F'(z)$ — at least one of the two principal ideals $F'(\delta)$ and $F'(z)$ is not divisible by $\mathfrak{p}$. Thus there is *no system of values for which all $F'(\delta)$ and $F'(z)$ vanish simultaneously*, and consequently the system of values $z = a, \delta_i = b_i$ is assumed at only one point of the absolute Riemann surface; the *totality of integral algebraic functions of the field* (which may be interpreted as a curve in a space of infinitely many dimensions) *has no singular point in the finite part*; only finite values enter into consideration in ideal theory.

But the basis functions $\omega_1, \ldots, \omega_n$ of all integral quantities also have no singular points in the finite part. By the preceding, each point in the arithmetic sense is already uniquely determined by a system of constants isomorphically assigned to all integral functions. If, therefore, a "space curve" defined by any integral functions s_i has a singular point in the finite part for $s_i = \alpha_i$, then there must be at least one integral function η which assumes several systems of values for $s_i = \alpha_i$. If now $\omega_1, \ldots, \omega_n$ denotes a basis of all integral quantities, then, because $\delta = a_1\omega_1 + \cdots + a_n\omega_n$ with integral rational a_i, each system of values of the ω's corresponds to only one system of values of δ; hence the space curve defined by $\omega_1, \ldots, \omega_n$ can possess no singular point in the finite part; and *every point of the absolute Riemann surface at which z is finite is already uniquely defined by a system of constants isomorphically assigned to the ω's. The basis functions are precisely the functions η which become multivalued for $s_i = \alpha_i$.* Thus by introducing the basis of integral quantities, for example in place of a basis $1, \delta, \ldots, \delta^{n-1}$, the difficulty of singular points is avoided.

In the geometric theory the task of generating all points of the absolute Riemann surface uniquely — more generally, of forming all functions which become infinite at given points — is accomplished by means of adjoint forms and their quotients. A form ψ is called adjoint if $\psi = 0$ has at least an $(i-1)$-fold point at each i-fold point of the base curve $f(z, s) = 0$; a higher singular point must first be resolved into ordinary multiple points. It was shown above that, for the most general function $\eta = \frac{\psi}{\chi}$, numerator and denominator must vanish at all singular points of $f(z, s) = 0$; that they must be adjoint is shown by considering the everywhere finite differentials. That, conversely, the conditions of adjointness are also sufficient for forming the most general function is shown by the "residue theorem", which will be discussed more closely below. Thus the adjoint functions take the place of the basis functions of the arithmetic theory.

Arithmetically, an adjoint form corresponds to the numerator of a function of the field divisible by the "double-point ideal" $\mathfrak{f}$, as soon as one assumes the singular points — which can always be achieved by transformation — all to lie in the finite part. From this the representation of the basis quantities $\omega_1, \ldots, \omega_n$

as quotients of adjoint forms, and hence the connection among the various ways of representing them, is shown formally as follows:

Let $R(z)$ denote the determinant of substitution of a basis $1, \delta, \dots, \delta^{n-1}$ into a basis $\omega_1, \dots, \omega_n$. Conversely, every ω can then be expressed as a quotient of an integral function of z and δ by $R(z)$:

$$\omega_i = \frac{G_i(z,\delta)}{R(z)}.$$

Since $R(z)\omega_i$ belongs to the ring derived from δ, by the definition of the conductor $R(z)$ is divisible by $\mathfrak{f}$ [from $(R(z))^2 = \operatorname{Norm}\mathfrak{f}$ this divisibility follows only for $(R(z))^2$]. Since now ω_i, as an integral function, remains finite for all finite z, the numerator function $G_i(z,\delta)$ must also be divisible by $\mathfrak{f}$; numerator and denominator of all ω_i are therefore adjoint. From this representation of the ω's follows, however, the representation of all functions of the field by quotients of adjoints.

According to the preceding, the geometric theory can be characterized as the *theory of the ring derived from a quantity* δ (order), in contrast with the arithmetic theory, which considers *all integral quantities* simultaneously. Thus it is clear that the conductor of the ring — the singular points — plays a decisive role. Further analogies between the arithmetic theory of rings (orders) and the geometric theory can also be given. Thus the geometric theory does not count the intersection points that fall into the singular points in the intersection with adjoint curves, but considers only the point groups different from these. Corresponding to this is that Dedekind[258] considers only such ring ideals as are prime to the conductor ideal $\mathfrak{f}$, and for this domain of ideals sets up all divisibility laws, including unique decomposition into prime ideals.

Also the "fundamental theorem of algebraic functions"[259] underlying the geometric theory, or at least an immediate consequence of it, has in a certain sense an analogue in the arithmetic theory of rings. This theorem gives the conditions for a representability

$$\psi(z,s) = A(z,s)f(z,s) + B(z,s)\varphi(z,s),$$

where all quantities occurring are integral rational in s, z (more generally integral and homogeneous in x_1, x_2, x_3). From these conditions it follows,[260] by virtue of $f(z,s) = 0$, that the quotient $\frac{\psi}{\varphi}$ is always equal to an integral rational function $B(z,s)$ when it is adjoint to f. Corresponding to this is the theorem from no. 3, which underlies all these comparisons, that the conductor $\mathfrak{f}$ of a ring derived from δ is equal to the double-point ideal of $f(z,\delta) = 0$. For, by the definition of the conductor, every algebraic integral quantity divisible by $\mathfrak{f}$ belongs to the ring, and is therefore integral and rationally representable by z and δ; whereas the cited theorem says that such a quantity is an adjoint function.[261]

7. Comparison between the residue theorem and theorems of ideal theory

The just mentioned consequence of the "fundamental theorem of algebraic functions" leads in the geometric theory directly to the "residue theorem", and thus to the foundation on which the whole geometric theory is built. In a certain respect, however, the residue theorem again has a more elementary character than the remaining foundations, insofar as it can be stated arithmetically as a direct consequence of the unique factorization of ideals into prime ideals, or rather of the facts underlying this decomposition theorem, and therefore does not rest on the higher theory of rings. One is led to this arithmetic formulation as follows:

Assume the base curve $f(z,\delta) = 0$ — as can always be achieved by a linear fractional transformation of the variables — to be such that the singular points and the finitely many point groups under consideration lie in the finite part; and further that δ depends algebraically integrally on z, which can always be achieved by a subsequent linear transformation integral in the variables. Then to a prime ideal $\mathfrak{p}$ generating a point P of the absolute Riemann surface at which $z = a, \delta = b$, there corresponds the point $z = a, \delta = b$ of $f(z,\delta) = 0$; more generally, an ideal $\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_\rho^{e_\rho}$ generates the point group consisting of the points $z = a_i, \delta = b_i$ with the corresponding multiplicity; and all points of $f(z,\delta) = 0$ lying in the finite part are so generated. Since at a point P all functions $g_1, g_2, \dots$ divisible by $\mathfrak{p}$ vanish, and since conversely this totality of functions g generates the ideal $\mathfrak{p}$, the corresponding point on f is the intersection of

$$g_1(z,\delta) = 0, \quad g_2(z,\delta) = 0, \dots$$

[258] Über die Anzahl der Idealklassen in den verschiedenen Ordnungen eines endlichen Körpers. (Festschrift Braunschweig 1877); § 3–5. (The theorems stated in these paragraphs for algebraic numbers hold literally for algebraic functions of one variable.)

[259] M. Noether, Über einen Satz aus der Theorie der algebraischen Funktionen. *Math. Ann.* vol. 6 (1873).

[260] Brill–Noether, no. 55.

[261] Following Dedekind, the assumption is made here throughout that δ is an algebraic integer (which can always be achieved by transformation). The case more exactly corresponding to geometry, $f(z,s) = 0$, where s may also depend algebraically fractionally on z, is treated by Hensel–Landsberg; compare Hensel no. 26 and 29.

with $f = 0$; the same holds for the point group corresponding to an ideal $\mathfrak{a}$. Indeed, by the theorem that every ideal is the greatest common divisor of two principal ideals, two such curves always suffice to cut out a point group. In particular, the point group corresponding to a principal ideal is cut out by a curve $g(z, \delta) = 0$ and conversely; the principal ideal corresponds to the complete intersection.

Two point groups A and B are called *coresidual* in the geometric theory if they can be completed with the same point group R to a full intersection. If these point groups are generated respectively by the ideals $\mathfrak{a}, \mathfrak{b}, \mathfrak{r}$, then between these ideals one has the relation

$$\mathfrak{a} \cdot \mathfrak{r} = (\alpha), \qquad \mathfrak{b} \cdot \mathfrak{r} = (\beta),$$

which shows that the ideals $\mathfrak{a}$ and $\mathfrak{b}$ are equivalent. Conversely, from the equivalence of two ideals, $\mathfrak{a} = \eta\mathfrak{b}$, this relation always follows on the basis of the theorem that every ideal can be transformed into a principal ideal by multiplication by another ideal;[262] *equivalent ideals and coresidual point groups are therefore identical concepts.*

The residue theorem now says: *If A and B are coresidual, then they can be cut out by adjoint curves with the same residue (outside the singular points).* Since the zeros and poles of a function are always coresidual, this proves at the same time that the most general functions of the field are representable as quotients of adjoint forms. In the language of ideal theory the residue theorem simply amounts to saying that, besides $\mathfrak{a}$ and $\mathfrak{b}$, $\mathfrak{a}\mathfrak{f}$ and $\mathfrak{b}\mathfrak{f}$ are also equivalent, where $\mathfrak{f}$ denotes the double-point ideal. For then, by the above, there always exists an ideal $\mathfrak{m}$ such that

$$\mathfrak{a}\mathfrak{f}\mathfrak{m} = (\alpha), \qquad \mathfrak{b}\mathfrak{f}\mathfrak{m} = (\beta)$$

become principal ideals; $\alpha = 0$ and $\beta = 0$ are the adjoint curves which cut out the point groups A and B, respectively their subgroups A' and B' different from the singular points.[263] The proof of the residue theorem is therefore arithmetically contained in the possibility of assigning points to prime ideals, and in the proof that equivalent ideals can be transformed into principal ideals by multiplication by one and the same ideal.

Coresidual point groups need not consist of the same number of points; just as equivalent ideals need not be of the same degree. Thus, for example, all principal ideals are equivalent, and all point groups generated by complete intersections are coresidual. This difference as against equivalent polygons (divisors)[264] is explained by the fact that those points at which z becomes infinite are not generated by prime ideals in z. In the representation of a function as an ideal quotient $\eta = \frac{\mathfrak{a}}{\mathfrak{b}}$, which states the equivalence of $\mathfrak{a}$ and $\mathfrak{b}$, the zeros or poles of η at which z becomes infinite therefore do not appear. For instance, the principal ideal (α) becomes the product of the prime ideals corresponding to the zeros of the integral function α; the poles of α do not occur, since α becomes infinite only at points where z becomes infinite. In this respect ideal theory is the exact analogue of form theory in the geometric theory, where there are likewise only zeros and no poles.

The passage from forms to functions is made geometrically by forming quotients of forms of the same order; arithmetically, this corresponds to the passage from ideal classes to the invariantly defined polygon (divisor) classes. For, since no variable is distinguished here any longer, the representation of a function by polygon quotients gives all zeros and poles. The relation of ideal class (class of coresidual point groups) to polygon class is the following: if $\mathcal{A}$ and $\mathcal{B}$ are the polygons generated by the ideals $\mathfrak{a}$ and $\mathfrak{b}$, and $\mathcal{N}$ is the polygon of the points at which z becomes infinite (the denominator polygon of z and of the integral functions of z), then from the equivalence of $\mathfrak{a}$ and $\mathfrak{b}$ follows the equivalence of $\mathcal{A}$ and $\mathcal{B}$ if and only if $\mathfrak{a}$ and $\mathfrak{b}$ have the same degree; in general only the equivalence of $\mathcal{A}\mathcal{N}^{\nu}$ with $\mathcal{B}\mathcal{N}^{\nu'}$ $(\nu \neq \nu')$ follows. The division of ideals (respectively of coresidual point groups) into classes is thus a gathering together of infinitely many polygon classes into one class; it may also be viewed as a division of the polygons into classes modulo a power of $\mathcal{N}$.[265]

The residue theorem for the case of coresidual point groups consisting of different numbers of points occurs only in purely geometric questions. In the application to algebraic functions, in fact only the case corresponding to polygon classes occurs, that of coresidual point groups of the same number. The "complete linear system" generated by a point group, that is, the totality of point groups coresidual to this point group and of the same number of points, corresponds exactly to the polygon class generated by the corresponding

[262] Compare Dedekind, number theory § 181; namely, if $\mathfrak{r}$ is chosen so that $\mathfrak{a}\mathfrak{r}$ becomes a principal ideal (α), then $\mathfrak{b}\mathfrak{r} = \eta^{-1} \cdot (\alpha)$; and since $\mathfrak{b}\mathfrak{r}$ contains only integral quantities, the same holds for $\eta^{-1} \cdot (\alpha)$, which is therefore also a principal ideal (β).

[263] If $\mathfrak{a} = \mathfrak{f}\mathfrak{a}_1$, $\mathfrak{b} = \mathfrak{f}\mathfrak{b}_1$, $\mathfrak{r} = \mathfrak{f}\mathfrak{r}_1$, where $\mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{r}_1$ are relatively prime, then already $\mathfrak{a}_1\mathfrak{f}\mathfrak{n}$, $\mathfrak{b}_1\mathfrak{f}\mathfrak{n}$ become principal ideals, where $\mathfrak{n}$ is prime to $\mathfrak{f}$. For every ideal can be transformed into a principal ideal by multiplication with an ideal that is prime to a given one (D.–W. p. 214; or number theory, § 178, theorem XI).

[264] A polygon A consists of finitely many points of the absolute Riemann surface; two polygons A and B are called equivalent if they are the zeros and poles of a function η; η then admits the representation by polygon quotients $\eta = \frac{A}{B}$. (D.–W. § 17 and 18.)

[265] Hensel–Landsberg, 25th lecture, p. 438. — If one represents the two variables z and s by polygon quotients, then this can be understood as binary homogenizing of each of these variables; if one chooses in particular z and s so that they have the same denominator polygon, then the ternary homogenizing customary in geometry arises.

polygon; the concepts "sum of two complete systems" and "product of two polygon classes" also coincide. That such a complete system contains only finitely many linearly independent point groups is a direct consequence of the residue theorem, which reduces the dimension of the system (the number of linearly independent point groups) to the dimension of all adjoint curves of arbitrary but fixed order passing through the residual group. In order to arrive at the actual determination of this dimension, the Riemann–Roch theorem, the geometric theory first derives from the residue theorem the "reduction theorem" (theorem on the fixed points);[266] and from this the Riemann–Roch theorem.

Quite correspondingly, the arithmetic theory shows that every class of polygons is of finite dimension, and determines its dimension by a special basis of the integral quantities, a normal basis defined by means of the complementary module; thus it obtains the Riemann–Roch theorem as a consequence of the connection of the complementary module with the ramification polygon. In the presentation of D.–W. the reduction theorem also occurs,[267] though not as a foundation as in the geometric theory. In consequence of the correspondence mentioned in no. 6 between basis and adjoint functions, a certain correspondence of the auxiliary means in the two theories can also be established here. In summary, one may say that the formations of concepts in the two theories essentially coincide, even though, apart from formulation, they differ in their arrangement. The two theories arose independently of one another, the geometric one a decade earlier. It should also be noted that the Riemann–Roch theorem originally, in Riemann and Roch and also in Weierstrass, occurs as a rank theorem; what is at issue is the rank of the matrix $(\varphi_i(x_j))$, where the φ's run through the p adjoint curves of order $n-3$, and the x's through the points of the residual group.

8. Transcendental questions for algebraic functions and algebraic numbers

Hilbert in particular pointed out analogies between transcendental questions for algebraic functions and for algebraic numbers;[268] these are essentially existence questions. Thus Riemann's question concerning the existence of an algebraic function for a given Riemann surface (hence also for given branch points) corresponds to the question of the existence of algebraic number fields with prescribed group and discriminant.[269] Whereas in the case of algebraic functions the existence proof can be carried out in general, in the domain of algebraic number fields it has been achieved only for special ground fields (rational numbers, quadratic number fields) and only for Abelian groups.[270] This concerns Kronecker's theorem that every Abelian number field over the rational numbers can be assembled from fields of roots of unity, and the corresponding theorems of Fueter[271] and Hecke[272] for relatively Abelian number fields. Thus the desired number fields are actually supplied by a transcendental means, exponential functions, modular functions; in contrast with the pure existence proof for algebraic functions.

The existence proof of algebraic functions is preceded by the existence of everywhere finite integrals. An analogue of this is the existence of the "singular primary numbers" of an algebraic number field defined with respect to a prime number ℓ, whose ℓ-th roots are relatively unramified and supply the first building blocks for the class field.[273] This existence proof is carried out with essential help from the theorem that in the number field there always exist prime ideals with prescribed residue characters; this theorem may therefore be regarded as an analogue of the boundary-value problem on which the existence of the everywhere finite integrals rests.

The everywhere finite integrals provide a transcendental criterion for determining whether two polygons belong to the same class (two point groups of the same number are corresidual), namely by Abel's theorem, which says that the integrals of the first kind, extended over "corresidual paths", vanish; or by the corresponding differential theorem together with its converse.[274]

As the analogue of Abel's theorem for algebraic number fields one must take the "reciprocity law for ℓ-th

[266] Brill–Noether, no. 58.

[267] D.–W., § 30, no. 2.

[268] Mathematische Probleme, problem 12. (*Göttinger Nachrichten* 1900).

[269] The treatment of algebraic functions suggested thereby, starting from the group, was carried out by R. König as a special case of more general questions; compare especially: Riemannsche Funktionen- und Differentialsysteme in der Ebene (*J. f. M.* vol. 148) and *Math. Ann.* 78, 79. In place of the algebraic field, König has a special finite module with respect to $K(z)$: there is only "class property".

[270] Only the simple case of the groups occurring for equations of third and fourth degree, and some quite special groups for equations of degree 8, can be handled for arbitrary ground fields: G. Bucht, Über einige algebraische Körper 8. Grades (*Arkiv f. Math., Astron. och Physik*, vol. 6, no. 30.) Compare also E. Noether, Gleichungen mit vorgeschriebener Gruppe and the Seidelmann dissertation cited there. (*Math. Ann.* 78).

[271] *Math. Ann.* 75.

[272] *Math. Ann.* 76.

[273] Furtwängler, Reziprozitätsgesetze für Potenzreste mit Primzahlexponenten in algebraischen Zahlkörpern. *Math. Ann.* 67 and 72.

[274] Compare, for example, the formulation in M. Noether (*Ann.* 37). Since the residue theorem historically grew out of Abel's theorem, the geometric theory speaks of the "sum" of point groups and linear systems, following the integral sums; in contrast to the "product" of the classes in the arithmetic theory.

power residues", which gives, for the field itself or at least for a suitable upper field, a criterion that two ideals belong to the same class. Its content can also be formulated by saying that in this upper field the singular primary numbers have the same residue character with respect to all ideals of the same class. This latter fact also holds in the field itself, but there it does not coincide with the reciprocity law.[275]

Finally, Hensel's theory of p-adic numbers supplies the analogue of the power-series expansion of algebraic functions; for this, see Hensel's report.

[275] Furtwängler, loc. cit.

15. The Finiteness of the System of Integral Invariants of Binary Forms

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Presented by F. Klein in the session of 27 March 1919.

In what follows, as an extension of the known finiteness theorem, the following theorem is proved: *All polynomial integral invariants of a binary system of ground forms are polynomial rational functions, with integral coefficients, of a finite number among them.* Here, as usual, by an "invariant" one always means a polynomial rational invariant of a ground form or of a system of ground forms, namely an invariant with respect to the coefficient transformation induced by the group of all linear transformations.

The proof rests on an integrality sharpening of the Mertens-Hilbert proof of finiteness.[276] This proof of finiteness may be characterized as follows. One decomposes the ground forms into their linear factors - more precisely, one associates to each ground form a product of linear forms - whereby each invariant I is changed into an invariant of these linear forms, hence into a function of the homogenized roots. This function has to satisfy three conditions: 1) the condition of invariance, 2) weight conditions, and 3) symmetry conditions. Conditions 2) and 3) are necessary for I to become polynomial and rational in the coefficients of the ground forms. Condition 1) is satisfied by representing I as a polynomial rational function of the determinants of the linear forms; condition 2) by taking, instead of the individual determinants, certain products of powers of them as new arguments. Condition 3) then simply says that I becomes a polynomial rational symmetric function of rows of quantities, and that conversely every such symmetric function of rows of quantities becomes an invariant of the ground forms. The elementary symmetric functions of these rows of quantities therefore give the complete system.

If one considers only integral invariants, condition 1) can be satisfied just as above as soon as, for invariants of linear forms, the sharper theorem is proved that *all integral invariants of binary linear forms are polynomial integral functions of the determinants of these linear forms.* I show that this is in fact the case in § 3, in the form that the totality of all relations modulo any prime between these determinants coincides with the totality of all identical relations; that is, the module corresponding to these relations remains a prime module modulo every prime number. This verification is carried out by normalizing the residue classes with respect to the module of quadratic forms corresponding to the quadratic relations between the determinants; these quadratic forms thereby turn out to form an integral basis of the module, modulo every prime, whereas previously they were known as a module basis only when integrality was disregarded.

To satisfy condition 2) no alteration is needed, since the introduction of products of powers does not affect the integer coefficients. Condition 3) now says that $n_1! \cdots n_\mu! I$ (where $n_1, \ldots, n_\mu$ are the degrees of the individual ground forms) becomes an integral symmetric function of rows of quantities, and conversely every integral symmetric function of these rows becomes an integral invariant. But, as is easily shown, these integral symmetric functions of rows possess an integral basis, for which the elementary functions alone no longer suffice. The application of Hilbert's theorem on the integral module basis then also yields an integral basis for the I themselves.

Section 1 collects the special finiteness theorems corresponding to conditions 1), 2), and 3), from which the proof of finiteness follows in § 2. The special finiteness theorems corresponding to condition 3) also yield the finiteness of the integral invariants of finite groups of integral or algebraic-integral substitutions, as is briefly carried out in § 4.

§ 1. Special Finiteness Theorems

I first state the following special finiteness theorems, which will serve to satisfy the conditions of invariance, weight, and symmetry mentioned in the introduction.

I. Every integral invariant of a system of binary linear forms is a polynomial rational function, with integral coefficients, of the determinants of these linear forms. The proof will be given in § 3.

II. Every system $\mathfrak{S}$ of products of powers of n variables with nonnegative exponents, which together with any two products of powers f and g also contains their quotient whenever this quotient is polynomial in the

[276] F. Mertens, Wiener Sitzungsberichte Jan. 1889; D. Hilbert, *Math. Ann.* vol. 33 (1889) [dated March 1888]. The two proofs, which arose independently of one another following Mertens, Crelle vol. 100, agree in content.

variables, has a finite multiplicative basis $f_1, \dots, f_k$ consisting of functions from $\mathfrak{S}$, such that for every f in $\mathfrak{S}$ one has a representation

$$f = f_1^{\lambda_1} \cdots f_k^{\lambda_k}$$

with nonnegative exponents $\lambda_1, \dots, \lambda_k$.

Indeed, by Hilbert's theorem on the module basis there exists a finite number of functions from $\mathfrak{S}$, say $f_1, \dots, f_k$, each of which really contains the x's, such that every nonconstant f from $\mathfrak{S}$ belongs to the module generated by them:

$$f \equiv 0 \pmod{(f_1, \dots, f_k)}.$$

It follows that for every $f = x_1^{i_1} \cdots x_n^{i_n}$ there is at least one $f_\nu = x_1^{\nu_1} \cdots x_n^{\nu_n}$ such that $\nu_1 \le i_1, \dots, \nu_n \le i_n$. In the representation

$$f = x_1^{i_1 - \nu_1} \cdots x_n^{i_n - \nu_n} f_\nu = A f_\nu$$

the factor A belongs to $\mathfrak{S}$ by hypothesis, but it has lower degree in the variables than f. Hence by repeating the argument finitely many times the assertion follows. In order that the representation also hold for $f = 1$, one need only put all exponents λ equal to zero.[277]

III. All polynomial integral symmetric functions of n rows of quantities are polynomial integral functions of a finite number among them.

Let $\tau_1(x), \dots, \tau_n(x)$ denote the elementary symmetric functions of the indeterminates $x_1, \dots, x_n$. Then for every exponent k there is an identity

$$x_i^k = G_0^{(k)}(\tau) + x_i G_1^{(k)}(\tau) + \cdots + x_i^{n-1} G_{n-1}^{(k)}(\tau), \qquad (i = 1, 2, \dots, n),$$

where $G_0^{(k)}, \dots, G_{n-1}^{(k)}$ are polynomial integral functions of the $\tau(x)$. If $x_i, y_i, \dots, z_i$ are the elements of the i-th row, then by these identities and the corresponding ones for $y, \dots, z$, all special, one-rowed symmetric functions

$$\sum_{i=1}^{n} x_i^a y_i^b \cdots z_i^c$$

become polynomial integral functions of the finitely many

$$\sum_{i=1}^{n} x_i^a y_i^b \cdots z_i^c \qquad (0 \le a < n,\ 0 \le b < n,\ \dots,\ 0 \le c < n)$$

and of the elementary symmetric functions $\tau(x), \tau(y), \dots, \tau(z)$.[278] If one has the most general symmetric functions of the n rows - a case not occurring below - it is shown in the same way that the functions

$$\sum x_{i_1}^{a_1} y_{i_1}^{b_1} \cdots z_{i_1}^{c_1} x_{i_2}^{a_2} y_{i_2}^{b_2} \cdots z_{i_2}^{c_2} \cdots x_{i_n}^{a_n} y_{i_n}^{b_n} \cdots z_{i_n}^{c_n}$$

$$(0 \le a_\nu < n,\ 0 \le b_\nu < n,\ \dots,\ 0 \le c_\nu < n),$$

where $i_1, i_2, \dots, i_n$ runs through all permutations, together with $\tau(x), \tau(y), \dots, \tau(z)$, form an integral basis.

IV. Let $F_1(x), \dots, F_k(x)$ be polynomial integral functions of $x_1, \dots, x_n$, and let a be a fixed integer. Then all functions

$$\frac{G(F)}{a},$$

where G denotes a polynomial integral function, which become integral in the x's, are polynomial integral functions of a finite number among them.

Indeed, by Hilbert's theorem on the integral module basis, for each such $G(F)/a$ one has a representation

$$\frac{G(F)}{a} = A_1(F_1 \dots F_k) \frac{G_1(F)}{a} + \cdots + A_\rho(F_1 \dots F_k) \frac{G_\rho(F)}{a},$$

[277] This representation can also be proved directly, and conversely Hilbert's theorem follows from it: compare P. Gordan, Neuer Beweis des Hilbertschen Satzes über homogene Funktionen, Gött. Nachr. 1899. Another direct proof of this representation, from which Theorem II is also derived, is in A. Ostrowski, Über die Existenz einer endlichen Basis bei gewissen Funktionensystemen, *Math. Ann.* 78 (1916), § 2,1 and § 3,2. The special case of Theorem II used in § 2, where the exponents run through all nonnegative solutions of a Diophantine system of equations, is already found in Gordan-Kerschensteiner, *Vorlesungen über Invariantentheorie*, vol. I, p. 199.

[278] Hilbert, loc. cit., takes the power sums instead of the $\tau(x), \dots, \tau(z)$ into the basis; this makes the basis consist only of one-rowed functions, but the integrality of the representation is then lost.

where $A_1, \dots, A_\rho$ are polynomial integral functions of the F's, and

$$\frac{G_1(F)}{a}, \dots, \frac{G_\rho(F)}{a}$$

belong to the system. The functions

$$F_1 = \frac{aF_1}{a}, \dots, F_k = \frac{aF_k}{a}; \quad \frac{G_1(F)}{a}, \dots, \frac{G_\rho(F)}{a}$$

therefore form the required basis.

§ 2. Finiteness of the Integral Binary Invariants

Let x, y be binary variables subjected to a transformation with indeterminate coefficients

$$x = s_{11}x' + s_{12}y', \qquad y = s_{21}x' + s_{22}y', \tag{1}$$

and let f be a binary ground form whose coefficients are indeterminates. Then by (1),

$$f = a_0x^n + a_1x^{n-1}y + \cdots + a_ny^n = a_0'x'^n + a_1'x'^{n-1}y' + \cdots + a_n'y'^n. \tag{2}$$

By an integral invariant of f one understands, as is well known, a polynomial integral function $I(a)$ such that

$$I(a') = |s_{ik}|^\rho I(a), \qquad \text{identically in } a, s_{ik}. \tag{3}$$

The integral invariants of a system of ground forms are defined correspondingly; one may include homogeneity in the coefficients of the individual ground forms in the definition, since from such individually homogeneous invariants the remaining ones are composed in an integral-linear way.

In addition to (2), consider a special binary form g, namely the product of n linear forms whose coefficients are again indeterminates:[279]

$$\begin{aligned} g &= (\alpha_1x + \beta_1y)(\alpha_2x + \beta_2y)\cdots(\alpha_nx + \beta_ny) \\ &= \sigma_0(\alpha,\beta)x^n + \sigma_1(\alpha,\beta)x^{n-1}y + \cdots + \sigma_n(\alpha,\beta)y^n \\ &= (\alpha_1'x' + \beta_1'y')(\alpha_2'x' + \beta_2'y')\cdots(\alpha_n'x' + \beta_n'y') \\ &= \sigma_0'x'^n + \sigma_1'x'^{n-1}y' + \cdots + \sigma_n'y'^n. \end{aligned} \tag{4}$$

Thus $\sigma_0(\alpha,\beta), \sigma_1(\alpha,\beta), \dots, \sigma_n(\alpha,\beta)$ denote the homogenized elementary symmetric functions, and the transformed coefficients σ_i' become equal to $\sigma_i(\alpha',\beta')$.

Because of the algebraic independence of these elementary functions, to every relation

$$G(\sigma_0, \sigma_1, \dots, \sigma_n) = 0 \quad \text{identically in } \alpha, \beta \tag{5}$$

there corresponds a further relation

$$G(\sigma_0, \sigma_1, \dots, \sigma_n) = 0 \quad \text{identically in } \sigma_0, \dots, \sigma_n, \tag{6}$$

and hence, for indeterminates $a_0, \dots, a_n$,

$$G(a_0, a_1, \dots, a_n) = 0 \quad \text{identically in } a_0, \dots, a_n. \tag{7}$$

Now every integral invariant $I(a)$ of f, by the substitution $a_i = \sigma_i(\alpha,\beta)$, gives an integral invariant $I(\sigma) = H(\alpha,\beta)$ of g, where H becomes an integral function of α, β. For this invariant the relation (3) becomes

$$I(\sigma') = |s_{ik}|^\rho I(\sigma) \quad \text{identically in } \sigma, s_{ik}, \tag{8}$$

and since $\sigma' = \sigma(\alpha',\beta')$, (8) gives

$$H(\alpha',\beta') = |s_{ik}|^\rho H(\alpha,\beta) \quad \text{identically in } \alpha, \beta; s_{ik}. \tag{9}$$

[279] If, as is often done in invariant theory, one writes f with binomial coefficients, $f = b_0x^n + \binom{n}{1}b_1x^{n-1}y + \cdots + b_ny^n$, then the domain of all integral invariants $I(b)$ is enlarged relative to that of the $I(a)$, and the finiteness arguments used here fail; $I(b)$ is no longer an integral function of the roots.

Thus $I(\sigma) = H(\alpha, \beta)$ becomes an integral invariant of the n linear forms $(\alpha_i x + \beta_i y)$. Conversely, suppose $H(\alpha, \beta)$ is an integral invariant of these n linear forms, hence satisfies (9), and is at the same time equal to an integral function $I(\sigma)$ of the σ's. Since then

$$H(\alpha', \beta') = I(\sigma(\alpha', \beta')) = I(\sigma'),$$

equation (9) can also be written

$$I(\sigma') = |s_{ik}|^\rho I(\sigma) \quad \text{identically in } \alpha, \beta; s_{ik}. \tag{9a}$$

It follows from (5) and (6) that (8) holds, and from (7) that (3) holds. Therefore to every integral invariant $I(a)$ there corresponds an integral invariant $H(\alpha, \beta) = I(\sigma)$ of the n linear forms, and conversely. Moreover, if $I_1(a), \ldots, I_k(a)$ is an integral basis of all $I(a)$, then $H_1(\alpha, \beta), \ldots, H_k(\alpha, \beta)$ is an integral basis of all $H(\alpha, \beta)$ which are at the same time integral functions $I(\sigma)$; here too the converse holds by (5), (6), and (7). Thus, to prove the existence of an integral basis for all $I(a)$, it suffices to prove it for all integral invariants $H(\alpha, \beta)$ which are at the same time integral functions $I(\sigma)$; and the basis

$$H_1(\alpha, \beta) = I_1(\sigma), \ldots, H_k(\alpha, \beta) = I_k(\sigma)$$

then supplies the basis $I_1(a), \ldots, I_k(a)$ for the $I(a)$.

Let $H(\alpha, \beta) = I(\sigma)$ run through the totality of all integral invariants of (4). Since by (3) every invariant is homogeneous in the σ's, and by (4) each σ is homogeneous and linear in the coefficients of each linear form, $H(\alpha, \beta)$ is homogeneous in each row α_i, β_i and of the same dimension in each row; moreover it is of course a symmetric function of the n rows. Conversely, by the fundamental theorem on symmetric functions, every integral symmetric function $H(\alpha, \beta)$ of the n rows, homogeneous separately and of the same dimension in each row, is a polynomial integral function of the σ's. Thus, by the preceding argument, it is an invariant $I(\sigma)$ as soon as $H(\alpha, \beta)$ is invariant. The totality of the integral invariants of (4) is therefore characterized as the totality of integral functions $H(\alpha, \beta)$ with the following properties:

1) integral invariant of the n linear forms $\alpha_i x + \beta_i y$;
2) separately homogeneous and of the same dimension in each of the n rows α_i, β_i;
3) symmetric in the n rows α_i, β_i.

By the special finiteness theorem I, $H(\alpha, \beta)$, because of 1), becomes a polynomial integral function of the determinants

$$(ik) = \alpha_i \beta_k - \beta_i \alpha_k;$$

hence

$$H(\alpha, \beta) = G((ik)).$$

In this representation the symmetry can also be expressed formally by applying all permutations of the n rows to $G((ik))$:

$$n!\, H(\alpha, \beta) = G^{(1)}((ik)) + G^{(2)}((ik)) + \cdots + G^{(n!)}((ik)) = G^*((ik)). \tag{10}$$

Along with each product of powers P of the (ik), G^* therefore also contains the sum $Q = P^{(1)} + \cdots + P^{(n!)}$ over all permutations of P. In fact G^* becomes an integral linear combination of finitely many Q's, $G^* = \sum cQ$. Since the Q's satisfy conditions 1), 2), and 3), and hence themselves become invariants, it suffices to consider the Q's in place of $G^*((ik))$.

Now let P run through the system $\mathfrak{S}$ of all products of powers of the (ik) occurring in the Q's, i.e. satisfying condition 2). If the quotient of two P's is polynomial in the (ik), then it also satisfies condition 2), and hence belongs to $\mathfrak{S}$. Therefore, by Theorem II, there are finitely many such P's, say $P_1, \ldots, P_\rho$, such that every P is representable in the form

$$P = P_1^{\lambda_1} \cdots P_\rho^{\lambda_\rho}$$

with nonnegative exponents λ. Applying all permutations gives

$$Q = P_1^{(1)\lambda_1} \cdots P_\rho^{(1)\lambda_\rho} + \cdots + P_1^{(n!)\lambda_1} \cdots P_\rho^{(n!)\lambda_\rho}.$$

Thus Q becomes an integral symmetric function of the $n!$ rows of quantities, each with ρ elements:

$$P_1^{(1)} \ldots P_\rho^{(1)}; \quad P_1^{(2)} \ldots P_\rho^{(2)}; \quad \ldots; \quad P_1^{(n!)} \ldots P_\rho^{(n!)};$$

hence, by Theorem III, it is a polynomial integral function of finitely many symmetric functions of these rows. These basis functions are themselves quantities Q or elementary symmetric functions of the $n!$ elements $P_i^{(1)}, P_i^{(2)}, \ldots, P_i^{(n!)}$, and therefore satisfy conditions 1), 2), and 3) and become integral invariants

$$H_1(\alpha,\beta) = I_1(\sigma); \ldots; \quad H_l(\alpha,\beta) = I_l(\sigma). \tag{11}$$

By (10), since $G^* = \sum cQ$, every $H(\alpha,\beta) = I(\sigma)$ is therefore of the form

$$H(\alpha,\beta) = I(\sigma) = \frac{1}{n!}\Phi(I_1,\ldots,I_l),$$

where Φ denotes a polynomial integral function; conversely, $\frac{1}{n!}\Phi$ is an integral invariant as soon as it is polynomial in α,β. Hence by Theorem IV finitely many further invariants $I_{l+1}(\sigma),\ldots,I_k(\sigma)$ can be adjoined to (11) in such a way that

$$H_1(\alpha,\beta) = I_1(\sigma); \ldots; H_l(\alpha,\beta) = I_l(\sigma);$$
$$H_{l+1}(\alpha,\beta) = I_{l+1}(\sigma); \ldots; H_k(\alpha,\beta) = I_k(\sigma)$$

form an integral basis of all invariants $H(\alpha,\beta) = I(\sigma)$. This proves the theorem for the case of one ground form.

If one is dealing with a system of ground forms, then, as noted above, one may assume homogeneity of the invariants in the coefficients of each ground form. One again replaces the general ground forms by ones which are respectively products of linear forms. Their invariants then become: 1) invariants of these linear forms; 2) separately homogeneous in the rows formed from the coefficients of the linear forms, and of equal dimension in the rows belonging to each individual ground form; 3) symmetric in the rows belonging to each individual ground form. Conversely, each integral invariant of all the linear forms satisfying these conditions again corresponds to an integral invariant of the system of ground forms. The proof is absolutely parallel to that just given. To each product of powers $P = P_1^{\lambda_1}\cdots P_\rho^{\lambda_\rho}$ of the individual determinants satisfying the weight conditions (Theorems I and II) one must apply all

$$N = n_1!n_2!\cdots n_\mu!$$

permutations, where $n_1,\ldots,n_\mu$ denote the degrees of the ground forms. In this way $N \cdot I$ is converted into an integral symmetric function of N rows, each with ρ elements. From this one obtains the integral basis for all $N \cdot I$, and consequently, by Theorem IV, the integral basis for all I.

§ 3. The Basis of the Integral Invariants of Binary Linear Forms

It remains to supply the proof of the special finiteness theorem I. By a known theorem, first proved by Clebsch, all polynomial rational invariants of the n linear forms $\alpha_i x + \beta_i y$ are polynomial rational functions of the determinants (ik). In particular, therefore, all integral invariants of these linear forms become polynomial rational, though not necessarily integral, functions of these determinants.

It will be shown below that, in view of the identical relations existing between the (ik), this representation is always possible integrally.

In formulas, and in the language of module theory, this is expressed as follows. Let ξ_{ik}, $i<k$, be indeterminates. Then the totality of all integral polynomials $F(\xi_{ik})$ with the property that $F((ik))$ vanishes identically in α,β forms a module $\mathfrak{M}$; the sum of two such F's, and also the product of such an F with an arbitrary integral polynomial in the ξ_{ik}, again belongs to this totality. In formulas:

$$F((ik)) = 0 \quad \text{identically in } \alpha,\beta$$

implies

$$F(\xi_{ik}) \equiv 0 \pmod{\mathfrak{M}} \quad \text{identically in } \xi_{ik},$$

and conversely.

I now show that the following completely analogous statement holds. From

$$F((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha,\beta$$

$$\text{it follows that} \qquad F(\xi_{ik}) \equiv 0 \pmod{(\mathfrak{M},p)} \quad \text{identically in } \xi_{ik}, \tag{1}$$

and conversely, for every prime number p. Hence $\mathfrak{M}$ is also identical with the module $\mathfrak{M}_p$ corresponding to the totality of all relations modulo p among the (ik); this module consists of all integral polynomials $F(\xi_{ik})$ for which $F((ik))$ vanishes identically modulo p in α, β.

That (1) asserts the integral representability is seen as follows. It can also be written in the form: from $F((ik)) = p \cdot H(\alpha, \beta)$, identically in α, β, it follows that

$$F(\xi_{ik}) - p \cdot G(\xi_{ik}) \equiv 0 \pmod{\mathfrak{M}} \quad \text{identically in } \xi_{ik},$$

and hence, by the definition of $\mathfrak{M}$,

$$F((ik)) = p \cdot G((ik)) \quad \text{identically in } \alpha, \beta.^{280}$$

Thus $p \cdot H(\alpha, \beta) = p \cdot G((ik))$, or

$$H(\alpha, \beta) = G((ik)).$$

If, therefore, a representation is given

$$H(\alpha, \beta) = \frac{1}{p_1^{\lambda_1} \cdots p_\rho^{\lambda_\rho}} F((ik)),$$

where $p_1, \ldots, p_\rho$ are prime numbers, then it follows also that

$$H(\alpha, \beta) = \frac{1}{p_1^{\lambda_1 - 1} \cdots p_\rho^{\lambda_\rho}} G((ik)),$$

and by repeating this finitely many times one obtains the integrality of the representation.

For the proof of (1), denote by

$$f_{ik,rs} = \xi_{ik}\xi_{rs} - \xi_{ir}\xi_{ks} + \xi_{is}\xi_{kr}, \qquad i < k < r < s,$$

the forms in the indeterminates ξ_{ik} corresponding to the quadratic relations between the (ik),

$$(ik)(rs) - (ir)(ks) + (is)(kr) = 0,$$

and denote by $\mathfrak{N} = (f_{ik,rs})$ the module of integral polynomials generated from them. From

$$F(\xi_{ik}) \equiv 0 \pmod{\mathfrak{N}}$$

it follows that

$$F((ik)) = 0 \quad \text{identically in } \alpha, \beta,$$

and

$$\begin{array}{rcll} \text{from } F(\xi_{ik}) & \equiv & 0 & \pmod{(\mathfrak{N}, p)} \\ \text{it follows that } F((ik)) & \equiv & 0 & \pmod{p} \quad \text{identically in } \alpha, \beta. \end{array} \tag{2}$$

I shall now show, by normalizing the residue classes with respect to $\mathfrak{N}$, that the converse of (2) also holds. Then $\mathfrak{N} = \mathfrak{M} = \mathfrak{M}_p$, and (1) will be proved. The equality $\mathfrak{N} = \mathfrak{M}$ is obtained as the special case $\mathfrak{N} = \mathfrak{M}_p$ for $p = 0$; the case $p = 0$ is always included in the following considerations. In other words, the $f_{ik,rs}$ form an integral module basis of $\mathfrak{M}_p$, and in particular also of $\mathfrak{M}_0 = \mathfrak{M}$.

a) The case of two or three rows α_i, β_i

Since no four different indices occur, $\mathfrak{N}$ is the zero module. It must therefore be shown that from

$$F((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta \tag{3}$$

it follows that

$$F(\xi_{ik}) \equiv 0 \pmod{p} \quad \text{identically in } \xi_{ik}.$$

This proves (1) with $\mathfrak{M} = \mathfrak{M}_p = 0$.

Since the condition $F((ik)) = p \cdot H(\alpha, \beta)$ must be satisfied separately by the components homogeneous in the individual rows, it is enough throughout to assume $F((ik))$ homogeneous in each individual row, and

[280] This latter conclusion simultaneously proves the converse of (1).

hence $F(\xi_{ik})$ homogeneous in each index. In the case of two rows there is only the one determinant (12) and the corresponding indeterminate ξ_{12}. By the assumed homogeneity,

$$F = c\,\xi_{12}^{\nu}.$$

From

$$c\,(12)^{\nu} \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta$$

it follows, since $(12) \not\equiv 0 \pmod{p}$ identically in α, β, that $c \equiv 0 \pmod{p}$; hence

$$F(\xi_{ik}) \equiv 0 \pmod{p} \quad \text{identically in } \xi_{ik}.$$

This proves (3), and hence (1).

In the case of three rows there are the determinants $(12), (13), (23)$, corresponding to the indeterminates $\xi_{12}, \xi_{13}, \xi_{23}$. Thus

$$F = \sum c\,\xi_{12}^{\tau_{12}} \xi_{13}^{\tau_{13}} \xi_{23}^{\tau_{23}}.$$

Let F have respectively the degrees $\sigma_1, \sigma_2, \sigma_3$ in the indices $1, 2, 3$. Then

$$\sigma_1 = \tau_{12} + \tau_{13}, \qquad \sigma_2 = \tau_{12} + \tau_{23}, \qquad \sigma_3 = \tau_{13} + \tau_{23},$$

with the unique inverse

$$\tau_{12} = \frac{1}{2}(\sigma_1 + \sigma_2 - \sigma_3), \quad \tau_{13} = \frac{1}{2}(\sigma_1 - \sigma_2 + \sigma_3), \quad \tau_{23} = \frac{1}{2}(-\sigma_1 + \sigma_2 + \sigma_3).$$

Thus each separately homogeneous F contains at most one term,

$$F = c\,\xi_{12}^{\tau_{12}} \xi_{13}^{\tau_{13}} \xi_{23}^{\tau_{23}},$$

from which, as above, (3), and consequently (1), follows.

b) The case of four rows α_i, β_i

The six indeterminates $\xi_{12}, \xi_{13}, \xi_{14}, \xi_{23}, \xi_{24}, \xi_{34}$, and the corresponding determinants, occur; and $\mathfrak{N} = (f_{12,34})$. Since

$$\xi_{14}\xi_{23} \equiv \xi_{13}\xi_{24} - \xi_{12}\xi_{34} \pmod{\mathfrak{N}}, \tag{4}$$

every product of powers of the ξ_{ik} is congruent modulo $\mathfrak{N}$ to an integral linear combination of such products of powers in which the product $\xi_{14}\xi_{23}$ does not occur. Thus for every polynomial $F(\xi_{ik})$,

$$F(\xi_{ik}) \equiv F_0(\xi_{ik}) \pmod{\mathfrak{N}}, \tag{5}$$

where F_0 contains no product $\xi_{14}\xi_{23}$. The polynomial F_0 will be called the normalized polynomial of the residue class of F. Since (4) is separately homogeneous, a separately homogeneous F again corresponds to a separately homogeneous F_0.

Now put

$$F_0(\xi_{ik}) = G(\xi_{ik}) + \xi_{34} F_1(\xi_{ik}), \tag{6}$$

where G contains no product of powers divisible by ξ_{34}, and further put

$$[G]_{\xi_{14} = \xi_{13}} = K(\xi_{12}, \xi_{13}, \xi_{23}). \tag{7}$$

From

$$F((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta$$

it follows by (5), taking account of (2), that

$$F_0((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta. \tag{8}$$

By specializing $\alpha_4 = \alpha_3,\ \beta_4 = \beta_3$, (6) and (7) give

$$K((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta.$$

By the result proved under a), therefore,

$$K(\xi_{ik}) \equiv 0 \pmod p \quad \text{identically in } \xi_{ik}.$$

From this follows what remains to be shown below:

$$G(\xi_{ik}) \equiv 0 \pmod p \quad \text{identically in } \xi_{ik}.$$

Thus (6) gives

$$F_0(\xi_{ik}) \equiv \xi_{34} F_1(\xi_{ik}) \pmod p \quad \text{identically in } \xi_{ik}.$$

Since $(34) \not\equiv 0 \pmod p$ identically in α, β, it follows from (8) that

$$F_1((ik)) \equiv 0 \pmod p \quad \text{identically in } \alpha, \beta,$$

where $F_1(\xi_{ik})$ is again normalized and of lower degree in ξ_{34}. Repeating the procedure finitely many times yields

$$F_0(\xi_{ik}) \equiv \xi_{34}^{\lambda} F_\lambda(\xi_{ik}) \pmod p,$$

where F_λ is of degree zero in ξ_{34}, and hence vanishes modulo p by what has just been proved. Therefore

$$\begin{aligned} F_0(\xi_{ik}) &\equiv 0 \pmod p \quad \text{identically in } \xi_{ik}, \quad \text{or} \\ F(\xi_{ik}) &\equiv 0 \pmod{(\mathfrak{N}, p)} \quad \text{identically in } \xi_{ik}. \end{aligned} \tag{9}$$

This proves the converse of (2), and hence (1), at the same time in the sharper assertion for $F_0(\xi_{ik})$.[281]

It remains only to add that $G(\xi_{ik}) \equiv 0 \pmod p$ follows from $K(\xi_{ik}) \equiv 0 \pmod p$; equivalently, from $G(\xi_{ik}) \not\equiv 0 \pmod p$ it follows that $K(\xi_{ik}) \not\equiv 0 \pmod p$. The normalization is needed only for this part of the induction. Under the substitution $\xi_{14} = \xi_{13}$, the individual products of powers of G do not vanish. A vanishing of K could therefore occur only because distinct products of powers in G correspond to equal products in K; and that in turn can happen only when these products of powers differ merely by an interchange of the indices 3 and 4. Suppose that such a normalized product of powers is, for example,

$$\xi_{12}^{\tau_{12}} \xi_{13}^{\tau_{13}} \xi_{14}^{\tau_{14}} \xi_{24}^{\tau_{24}}.$$

Because of the assumed separate homogeneity in each index, a replacement of 3 by 4 in one ξ_{ik} must be accompanied by a replacement of 4 by 3 in another; a relevant term would therefore contain, instead of $\xi_{13}\xi_{24}$, the product $\xi_{14}\xi_{23}$, which is excluded by the normalization. In exactly the same way, against

$$\xi_{12}^{\tau_{12}} \xi_{13}^{\tau_{13}} \xi_{14}^{\tau_{14}} \xi_{24}^{\tau_{24}}$$

only a term containing the factor $\xi_{14}\xi_{23}$ could cancel, which is again excluded by the normalization. Hence $K(\xi_{ik}) \equiv 0 \pmod p$ implies $G(\xi_{ik}) \equiv 0 \pmod p$.

c) The case of n rows α_i, β_i

The case of n rows is now settled by complete induction, as an exact generalization of the just treated case of four rows.[282] First I again show that for every polynomial F there is a congruence

$$F(\xi_{ik}) \equiv F_0(\xi_{ik}) \pmod{\mathfrak{N}} \quad \text{identically in } \xi_{ik}, \tag{10}$$

where $\mathfrak{N} = (f_{ik,rs})$ and where $F_0(\xi_{ik})$ is normalized. The normalization is to consist in the following: if i, k, r, s are four different indices and $i < k < r < s$, then F_0 is not to contain the product $\xi_{is}\xi_{kr}$; that is, the indices of one ξ are not both to lie between those of the other. This normalization imposes no restriction in the case of two and three rows, where every polynomial is already normalized; in the case of four rows, as shown, there also exists a normalized polynomial for every residue class. The existence of a normalized polynomial for every residue class may therefore be assumed proved for $(n-1)$ rows.

Let an arbitrary product of powers in F be written in the form

$$\xi_{i_1 n} \cdots \xi_{i_\nu n} P,$$

[281] This sharper assertion means that each residue class contains only one normalized polynomial.

[282] The proof still applies for $n = 3$.

where P no longer contains the index n. By the induction hypothesis, P is congruent modulo $\mathfrak{N}$ to an integral normalized polynomial P_0:

$$\xi_{i_1 n} \cdots \xi_{i_\nu n} P \equiv \xi_{i_1 n} \cdots \xi_{i_\nu n} P_0 \pmod{\mathfrak{N}}.$$

If P_0 contains no ξ_{rs} such that for at least one index i_λ the two indices r, s lie between i_λ and n, i.e. $i_\lambda < r < s < n$, then the normalization has already been reached, since the hypotheses are fulfilled for the product of any two ξ's from P_0. Otherwise let i_λ be such an index, $i_\lambda < r < s < n$, and let P_0^* be the product of powers from P_0 containing ξ_{rs}. Since

$$\xi_{i_\lambda n} \xi_{rs} \equiv \xi_{i_\lambda s} \xi_{rn} - \xi_{i_\lambda r} \xi_{sn} \pmod{\mathfrak{N}},$$

one obtains

$$\begin{aligned}
&\xi_{i_1 n} \cdots \xi_{i_\lambda n} \cdots \xi_{i_\nu n} P_0^* \\
&\equiv \xi_{i_1 n} \cdots \xi_{rn} \cdots \xi_{i_\nu n} Q + \xi_{i_1 n} \cdots \xi_{sn} \cdots \xi_{i_\nu n} R \pmod{\mathfrak{N}} \\
&\equiv \xi_{i_1 n} \cdots \xi_{rn} \cdots \xi_{i_\nu n} Q_0 + \xi_{i_1 n} \cdots \xi_{sn} \cdots \xi_{i_\nu n} R_0 \pmod{\mathfrak{N}},
\end{aligned}$$

where Q_0 and R_0 again denote the normalized polynomials of the residue classes of Q and R. These exist by the induction hypothesis, since Q and R do not contain the index n, and they are certainly integral polynomials, since Q and R are simply products of powers. Since $i_\lambda < r < s < n$, one also has

$$((n - i_1) + \cdots + (n - r) + \cdots + (n - i_\nu)) < ((n - i_1) + \cdots + (n - i_\lambda) + \cdots + (n - i_\nu)),$$

$$((n - i_1) + \cdots + (n - s) + \cdots + (n - i_\nu)) < ((n - i_1) + \cdots + (n - i_\lambda) + \cdots + (n - i_\nu)).$$

Thus, in passing from P_0^* to Q_0, R_0, this sum of differences has decreased; since the sum is necessarily $\geq \sigma$, the procedure must terminate after finitely many steps for each product of powers P_0^*. Hence

$$F(\xi_{ik}) = \sum \xi_{i_1 n} \cdots \xi_{i_\nu n} P^{(1)} \equiv \sum \xi_{j_1 n} \cdots \xi_{j_\sigma n} S_0^{(j)} = F_0(\xi_{ik}) \pmod{\mathfrak{N}},$$

where the integral polynomial $F_0(\xi_{ik})$ is necessarily normalized, since otherwise the difference sum could be lowered further. This proves (10). Since, by the induction hypothesis for $(n-1)$ indices, every separately homogeneous F has a separately homogeneous normalized F_0, the procedure just given shows that this is also the case for n indices. Hence in what follows only separately homogeneous polynomials need be considered.

As in b), set

$$F_0(\xi_{ik}) = G(\xi_{ik}) + \xi_{n-1,n} F_1(\xi_{ik}),^{283} \tag{11}$$

where G contains no product of powers divisible by $\xi_{n-1,n}$, and put further

$$[G]_{\xi_{in} = \xi_{i,n-1}} = K(\xi_{ik}). \tag{12}$$

As the definition of normalization shows, K, as well as G, is normalized.

Again there is a one-to-one correspondence between G and K. Distinct products of powers from G can correspond to the same product in K only if they differ merely by an interchange of the indices n and $(n-1)$. Suppose, for instance, that G is of degree ρ in the index $(n-1)$ and of degree σ in the index n, and that

$$\chi_1 \leq \chi_2, \ldots, \leq \chi_\rho \leq \chi_{\rho+1}, \ldots, \leq \chi_{\rho+\sigma}$$

are the indices connected with $(n-1)$ and n in a product of powers of G. Their number must be $\rho + \sigma$, since by (11) the indeterminate $\xi_{n-1,n}$ does not occur in G. By the normalization, the product of powers is therefore of the form

$$\xi_{\chi_1, n-1} \cdots \xi_{\chi_\rho, n-1} \xi_{\chi_{\rho+1}, n} \cdots \xi_{\chi_{\rho+\sigma}, n} P(\xi_{ik}),$$

[283] The identity corresponding to (10) and (11),

$$F((ik)) = F_0((ik)) = G((ik)) + (n-1, n) F_1((ik)),$$

is an analogue of the Clebsch-Gordan series expansion, as soon as the two rows $(n-1)$ and (n) are replaced by rows of variables x, y, so that $(n-1, n)$ is replaced by (xy). But whereas Clebsch-Gordan involves an expansion by polars, the normalization of F_0 corresponds to "initial terms" of polars, and thereby forces the integrality which does not hold for Clebsch-Gordan. The known proof that, when integrality is disregarded, the $f_{ik,rs}$ form a basis of $\mathfrak{M}$ is carried out precisely by means of the Clebsch-Gordan series expansion (Gordan-Kerschensteiner, *Vorlesungen über Invariantentheorie*, p. 132).

where $P(\xi_{ik})$ no longer contains the indices n and $(n-1)$. Each interchange of n with $(n-1)$ would, for $i \neq j$, give the factor $\xi_{in}\xi_{j,n-1}$ with $i < j$; hence $j, n-1$ lies between i and n, and the term would not be normalized. Thus, for fixed $\chi_1, \chi_2, \ldots, \chi_{\rho+\sigma}$ and fixed P, G contains only one product of powers. This proves that different products of powers from G correspond to different products in K. Therefore

$$K(\xi_{ik}) \equiv 0 \pmod{p} \quad \Longrightarrow \quad G(\xi_{ik}) \equiv 0 \pmod{p}. \tag{13}$$

From

$$F((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta$$

it follows now from (10), taking account of (2), that

$$F_0((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta. \tag{14}$$

Specializing $\alpha_n = \alpha_{n-1},\ \beta_n = \beta_{n-1}$, (11) and (12) give

$$K((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta. \tag{15}$$

Since $K(\xi_{ik})$ is normalized and contains only $(n-1)$ indices, it follows by what was proved under a) and under b), formula (9), that

$$K(\xi_{ik}) \equiv 0 \pmod{p} \quad \text{identically in } \xi_{ik}.$$

Then by (13)

$$G(\xi_{ik}) \equiv 0 \pmod{p} \quad \text{identically in } \xi_{ik}.$$

Hence (11) gives

$$F_0(\xi_{ik}) \equiv \xi_{n-1,n} F_1(\xi_{ik}) \pmod{p} \quad \text{identically in } \xi_{ik}.$$

Since $(n, n-1) \not\equiv 0 \pmod{p}$ identically in α, β, it follows from (14) that

$$F_1((ik)) \equiv 0 \pmod{p} \quad \text{identically in } \alpha, \beta,$$

where $F_1(\xi_{ik})$ is again normalized and has lower degree in $\xi_{n-1,n}$ than F_0. Repeating the process finitely many times yields

$$F_0(\xi_{ik}) \equiv \xi_{n-1,n}^{\lambda} F_\lambda(\xi_{ik}) \pmod{p},$$

where F_λ is of degree zero in $\xi_{n-1,n}$, and hence vanishes modulo p by what has just been proved. Thus one obtains

$$\begin{array}{rcl} F_0(\xi_{ik}) & \equiv & 0 \pmod{p} \quad \text{identically in } \xi_{ik}, \quad \text{or} \\ F(\xi_{ik}) & \equiv & 0 \pmod{(\mathfrak{N}, p)} \quad \text{identically in } \xi_{ik}. \end{array} \tag{16}$$

This proves the converse of (2), and hence (1). At the same time, it proves in the case of n indices, and therefore in general, the sharper assertion used in the induction step for $F_0(\xi_{ik})$.

§ 4. Finiteness of the Integral Invariants of Finite Groups

The elementary proof of finiteness given in my note "The finiteness theorem for invariants of finite groups"[284] admits, by means of the special finiteness theorems III and IV of § 1, the sharpening to integrality, whereas the usual proof based on Hilbert's theorem on the module basis does not supply this integrality. Even the use of the integral module basis only gives a representation in which an arbitrarily high power of the integer h, the order of the group, appears in the denominator.

I use throughout the notation of the note just mentioned. Let the finite group $\mathfrak{H}$ consist of the h linear transformations $A_1, \ldots, A_h$ of nonzero determinant,[285] where A_k denotes the linear transformation

$$x_1^{(k)} = \sum_{\nu=1}^{n} a_{1\nu}^{(k)} x_\nu, \quad \ldots, \quad x_n^{(k)} = \sum_{\nu=1}^{n} a_{n\nu}^{(k)} x_\nu,$$

[284] *Math. Ann.* 77, p. 89 (1915).

[285] It would suffice to suppose that $\mathfrak{H}$ is simply isomorphic to an abstract finite group; that is, if a transformation determinant of the A's vanishes, then $\mathfrak{H}$ should be similar to a group $\begin{pmatrix} \mathfrak{S} & 0 \\ 0 & 0 \end{pmatrix}$, where the determinants of the substitutions of $\mathfrak{S}$ do not vanish. In this case the invariants of $\mathfrak{H}$ coincide with the invariants of $\mathfrak{S}$, so the condition of the nonvanishing of the determinants is in fact no restriction of generality.

or, briefly, $(x^{(k)}) = A_k(x)$. Thus the group $\mathfrak{H}$ sends the row (x) with elements $x_1, \dots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \dots, x_n^{(k)}$. The coefficients $a_{i\nu}^{(k)}$ are assumed to be algebraic integers; this is no restriction of generality, since by a theorem of J. Schur[286] every finite group of linear transformations is similar to one for which the $a_{i\nu}^{(k)}$ become algebraic integers.

By an integral absolute invariant of the group is meant a polynomial rational function of $x_1, \dots, x_n$, with algebraic integers of the number field $\mathfrak{K}$ generated by the $a_{i\nu}^{(k)}$ as coefficients, which remains identically unchanged under application of $A_1, \dots, A_h$. For such an invariant $f(x)$ one has

$$f(x) = f(x^{(1)}) = \cdots = f(x^{(h)}) = \frac{1}{h}\sum_{k=1}^{h} f(x^{(k)}). \tag{1}$$

Conversely, every integral symmetric function of the h rows of quantities $(x^{(k)})$ also becomes an integral invariant of the group.

Let $\omega_1, \omega_2, \dots, \omega_\rho$ denote a basis of all integers of $\mathfrak{K}$. Then every invariant can be represented as

$$f(x) = \omega_1 f_1(x) + \cdots + \omega_\rho f_\rho(x),$$

where $f_1(x), \dots, f_\rho(x)$ have rational integer coefficients. Thus by (1)

$$h\, f(x) = \omega_1 \sum_{k=1}^{h} f_1(x^{(k)}) + \cdots + \omega_\rho \sum_{k=1}^{h} f_\rho(x^{(k)}).$$

Here the individual sums are symmetric functions of the h rows of quantities $(x^{(k)})$, and indeed with rational integer coefficients. By Theorem III they can therefore be expressed polynomially and rationally, with rational integer coefficients, in terms of finitely many integral-rational symmetric functions of these rows. By the preceding argument, these basis functions $I_1(x), \dots, I_\sigma(x)$ become integral invariants of the group, in general with algebraic-integral coefficients, since $\mathfrak{H}$ need not contain together with A all of its algebraically conjugate transformations.

There is therefore a representation

$$h \cdot f(x) = \omega_1 G_1(I) + \cdots + \omega_\rho G_\rho(I), \tag{2}$$

where $G_1, \dots, G_\rho$ have rational integer coefficients.

Now let $w_1, \dots, w_\rho$ be indeterminates, and let

$$L = w_1 G_1(I) + \cdots + w_\rho G_\rho(I)$$

run through all linear forms in the w's which, by the substitution $w_i = \omega_i/h$, correspond to invariants $f(x)$ in the representation (2). Hilbert's theorem on the integral module basis gives

$$L = A_1(I)L_1 + \cdots + A_\tau(I)L_\tau, \tag{3}$$

where the A_i have rational integer coefficients, and where, since all L's are linear in the w's, the A_i no longer contain the w's. If the substitution $\omega_i = w_i/h$ sends $L_1, \dots, L_\tau$ to the invariants $\mathfrak{I}_1(x), \dots, \mathfrak{I}_\tau(x)$, then (3) becomes

$$f(x) = A_1(I)\mathfrak{I}_1(x) + \cdots + A_\tau(I)\mathfrak{I}_\tau(x). \tag{4}$$

This representation shows that

$$I_1(x), \dots, I_\sigma(x); \mathfrak{I}_1(x), \dots, \mathfrak{I}_\tau(x)$$

form a basis of all integral invariants of the group $\mathfrak{H}$, such that every invariant is representable as a polynomial rational function, with rational integer coefficients, of these finitely many invariants.

If, in particular, the group $\mathfrak{H}$ has the property that with each transformation A it also contains all $A', A'', \dots$ with algebraically conjugate number coefficients, then the integral-rational symmetric functions of the h rows $(x^{(k)})$ become functions of x with rational integer coefficients. This holds in particular also for the basis functions of the symmetric functions. Thus, if one restricts oneself to invariants $f(x)$ with rational integer coefficients, then the basis functions

$$I_1(x), \dots, I_\sigma(x); \mathfrak{I}_1(x), \dots, \mathfrak{I}_\tau(x)$$

also have rational integer coefficients. Under this special assumption, which is in general not satisfied by the groups $\mathfrak{H}$, integral finiteness therefore also holds for the subdomain of invariants with rational integer coefficients.

[286] Über Gruppen linearer Substitutionen mit Koeffizienten aus einem algebraischen Zahlkörper. Satz VII. *Math. Ann.* 71, p. 555 (1912).

16. On Series Expansion in Form Theory

Math. Ann. 81 (1920), pp. 25–30

By the "series expansion of form theory" one understands, as is well known, on the one hand the generalization of the Clebsch–Gordan series expansion – that is, the expansion of a form containing two binary rows according to powers of the determinant of these rows, where the coefficients are polars of forms in only one row of variables, or, what is identical with this, vanish under application of the Ω-process – to forms in arbitrarily many series of n variables each; and, on the other hand, as a generalization of the normal forms occurring for the "complex coordinates" t_{ik}, an expansion according to normal forms for forms which contain simultaneously the quantities of different levels $t_i, t_{ik}, t_{ikl}, \ldots$. This second kind of expansion can, however, be obtained from the first by invariant differentiation processes, as was carried out above all by Mertens and Deruyts.[287]

The various expansions of the first kind all arise – as is briefly sketched, for instance, in Math. Ann. 77, loc. cit. III – as consequences of the expansion of a form in n rows of n variables each according to powers of the determinant of these rows, where the coefficients are polars of forms in only $(n-1)$ rows, which themselves are derived from the initial form by polar formation and the Ω-process.[288] The original Clebsch–Gordan expansion ($n = 2$) had gone one step further: the special polar processes which occur there are given explicitly, even including the numerical coefficients.

What I now wish to add to the indicated note – where the series expansion is conceived more conceptually, as a problem of linear pencils of forms – is an explicit representation up to numerical coefficients, obtained by specializing E. Fischer's investigations concerning general module systems.[289]

This classification is led to by interpreting the normal form in the t_{ik}. Namely, if one replaces the "complex coordinates" t_{ik} by the "line coordinates"

$$p_{ik} = (x_i y_k - x_k y_i),$$

then, because of the identical relations existing among the p_{ik}, the representation of a form $F(p_{ik})$ is no longer unique. Uniqueness can be obtained by requiring that among the coefficients of F there hold the same linear relations as among the power products of the p_{ik} occurring in F.[290] Thus for this normal form Φ one has:

$$\begin{aligned} F(p_{ik}) &= \Phi(p_{ik}) \quad \text{[identically in } x, y], \quad \text{or} \\ F(t_{ik}) &\equiv \Phi(t_{ik}) \qquad \operatorname{mod} M, \end{aligned} \tag{1}$$

where M denotes the module of all identical relations among the p_{ik}, that is, consists of the totality of all forms $G(t_{ik})$ for which $G(p_{ik})$ vanishes identically in x, y. $\Phi(t_{ik})$ is the "normal form" in complex coordinates, while the basis functions of M become quadratic forms in the t_{ik}. By iterating the procedure with respect to the factors of these basis functions, the complete series expansion arises. Corresponding considerations hold when $t_i, t_{ik}, t_{ikl}, \ldots$ occur simultaneously.

By an analogous line of thought one can reduce the expansion according to polars. If, for example, in a form $F(x, y, z)$ one replaces the three ternary rows $x = (x_1, x_2, x_3), y, z$ by rows ξ, η, ζ which are linearly composed of only two of them (for instance $\xi = x, \eta = y; \zeta = \lambda_1 x + \lambda_2 y$), then the representation of $F(\xi, \eta, \zeta)$ is again not unique. One can again prescribe for the coefficients the same linear relations as for the power products of ξ, η, ζ occurring in F. Since here the module M of all identical relations has as basis function the determinant $\Delta = (xyz)$ of the three rows (loc. cit. III, auxiliary lemma a)), [1] is replaced by:

$$\begin{aligned} F(\xi, \eta, \zeta) &= \Phi(\xi, \eta, \zeta) \quad \text{[identically in } x, y], \quad \text{or} \\ F(x, y, z) &\equiv \Phi(x, y, z) \qquad \operatorname{mod} \Delta. \end{aligned} \tag{2}$$

It follows that Φ arises by polar formation (compare formula (2) loc. cit.); the complete series expansion is again obtained by iteration.

[287] F. Mertens, Über invariante Gebilde quaternärer Formen (Sitzber. d. Ak. d. Wiss. Wien 98, Abt. IIa (1889)) for the quaternary case. The general normal forms are found in J. Deruyts: Sur la réduction des fonctions invariantes dans le système des variables géométriques, Bulletins de l'Ac. R. de Belgique 24 (1892). Without knowing this work, and in close dependence on Mertens, I gave the general expansion according to normal forms in § 7 of the paper: Zur Invariantentheorie der Formen von n Variabeln, J. f. M. 139 (1910).

[288] Compare the bibliographical references loc. cit. Since then there has appeared, also based on the formal method: Schouten, Over reeksontwikkelingen van algebraische vormen met verschillende rijen van variabelen van verschillenden graad, Verslag Ak. Amsterdam 27 (1919), p. 1481.

[289] E. Fischer, Über die Differentiationsprozesse der Algebra, J. f. M. 148 (1917).

[290] Compare Clebsch, Über eine Fundamentalaufgabe der Invariantentheorie, § 4, Abh. d. K. G. d. Wissensch. zu Göttingen 17 (1872).

The formation of the normal forms Φ is therefore subordinated in both cases to the following problem: if in $F(z)$ the indeterminates z_i are replaced by polynomials $f_i(\xi)$ in parameters ξ, so that the representation $F(f_i(\xi))$ is therefore no longer unique, then the uniqueness is to be fixed by requiring that among the coefficients there hold the same linear relations as among the power products of the $f_i(\xi)$ occurring in F. Thus one has

$$\begin{aligned} F(f_i(\xi)) &= \Phi(f_i(\xi)) \quad \text{[identically in } \xi\text{]}, \quad \text{or} \\ F(z) &\equiv \Phi(z) \qquad \mod M, \end{aligned} \tag{3}$$

where M denotes the module of all identical relations among the $f_i(\xi)$.

For this normal form $\Phi(z)$, E. Fischer, in § 11, I of the cited paper, gave an explicit expansion up to numerical coefficients, namely by means of linear operations. At the same time, however, in Introduction III and in § 6 II, he also gave, for the difference $F - \Phi$ belonging to M – assuming knowledge of a module basis – an explicit expression in the same sense; and, by combining these developments in § 11 II, he thereby replaced formula [3] by an explicit representation. The question of the numerical coefficients, still open here, is according to § 12 of the same paper identical with the determination of the eigenvalues and eigenfunctions of the linear operations (formula 82).

I now specialize Fischer's formulas, in 1., to the case of the series expansion according to polars, and obtain from them, by iteration, the series expansion. The operation for determining Φ thereby passes into definite polar processes, which is considerably more precise than the previously known formulation given above. In the operation for determining $F - \Phi$, multiplication by the determinant Δ and the Ω-process occur in combination. To get to the usual, non-explicit form, one must also note that $\Omega\Delta$ can be expressed by polar processes.

At this point a correction should also be added. In the work cited above (p. 101, line 6 from the top) I inadvertently assumed as generally valid the identical transformation

$$\Omega(\Delta\psi) = c_1\psi + c_2\Delta \cdot \Omega(\psi),$$

which holds only in the binary domain, and thereby made the passage from the fundamental formula (2), which gives the expression for the normal form Φ in [2], to the series expansion (3). The considerations there therefore yield only formula (2), and as a special case of it the reduction theorem (1); but, since an explicit expression for the difference $F - \Phi$ is missing, they do not suffice to justify the series expansion.

In 2. I similarly obtain, by specialization, the normal form $\Phi(t_{ik})$ of [1]; more generally, the normal form $\Phi(t_i, t_{ik}, t_{ikl}, \ldots)$, which leads to known explicit formulas. But since setting up the complete series expansion requires knowing a module basis of M, the invariant-theoretic route (compare the cited literature) is preferable here; it does not require this knowledge, but on the contrary supplies a module basis by means of the series expansion.

1. According to Fischer, § 11 (formula (19) and the following lines), the normal form $\Phi(z)$ of [3] – there denoted $N(\zeta)$ – is given by:

$$\Phi(z) = (a_1 S + a_2 S^2 + \cdots + a_r S^r)F(z), \tag{4}$$

where the constants $a_1 \ldots a_r$ belong to the coefficient domain of the $f(\xi)$ and depend only on the degree of $F(z)$; the operator S is defined by

$$\begin{aligned} S = AB; \qquad BF(z) &= F(f(\xi)); \\ AF(f(\xi)) &= \left\{\sum z_i f_i\left(\frac{\partial}{\partial\xi}\right)\right\}^p F(f(\xi))^{291}. \end{aligned} \tag{5}$$

If in [2] one regards F as a form of p-th dimension in z alone, then $f_i(\xi)$ specializes to $\lambda_1 x_i + \lambda_2 y_i$; consequently one obtains:

$$\begin{aligned} BF &= F(x, y, \lambda_1 x + \lambda_2 y), \\ SF &= \left\{\sum z_i \left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x_i} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y_i}\right)\right\}^p F(x, y, \lambda_1 x + \lambda_2 y), \end{aligned}$$

or, written in polar processes (compare loc. cit. II, 2 and 3; also for the abbreviated notation):

$$SF(x, y, z) = \frac{1}{p!}\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right]^p \left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right]^p F(x, y, z). \tag{6}$$

[291] Compare formula (75). In the case of [3], the sum α runs over all power products of p-th dimension, and (75) therefore specializes to [5].

Equations [4] *and* [6] *give the desired explicit expression for* $\Phi(x,y,z)$; at the same time $\Phi(x,y,z)$ falls under the general form mentioned at the start ($\sum PH$ in (2), loc. cit.) – polars of expressions which depend on only 2 rows and are obtained from F by polar processes. The still undetermined coefficients a_i are rational numbers, since here the $f_i(\xi)$ have integral-rational numerical coefficients. If [2] is written in the form

$$F = \Phi + \Delta F_1, \tag{2a}$$

then specialization of the formula given by Fischer in Introduction III and in § 6 II (p. 41) gives

$$F_1 = (c_1\Omega + c_2\Omega\Delta\Omega + \cdots + c_i\Omega\Delta\Omega\cdots\Delta\Omega)F^{292}, \qquad \left[\Omega = \left(\frac{\partial}{\partial x}\frac{\partial}{\partial y}\frac{\partial}{\partial z}\right)\right], \tag{7}$$

where the c_i are again rational numbers depending only on the degree of F.

The iteration leads to:

$$F_1 = \Phi_1 + \Delta F_2; \quad F_2 = \Phi_2 + \Delta F_3; \quad \ldots; \quad F_i = \Phi_i + \Delta F_{i+1}; \quad \ldots,$$

where in each case the Φ_i are the normal forms corresponding to the F_i. Since the F_i have degree $(p-i)$ in z, [4] and [6] give:

$$\begin{gathered} \Phi_i = (a_{i1}S_i + a_{i2}S_i^2 + \cdots)F_i, \\ S_iF_i = \frac{1}{(p-i)!}\left[z\left(\frac{\partial}{\partial\lambda_1}\frac{\partial}{\partial x} + \frac{\partial}{\partial\lambda_2}\frac{\partial}{\partial y}\right)\right]^{p-i}\left[(\lambda_1 x + \lambda_2 y)\frac{\partial}{\partial z}\right]^{p-i} F_i. \end{gathered} \tag{8}$$

Formula [7] corresponds to

$$F_i = (\alpha_{i1}\Omega + \alpha_{i2}\Omega\Delta\Omega + \cdots)F_{i-1}, \quad \text{and by iteration}$$

$$F_i = (\beta_1\Omega^i + \beta_2\Omega^i\Delta\Omega + \cdots + \beta_\rho\Omega^{k_1}\Delta\Omega^{k_2}\cdots\Delta\Omega^{k_\lambda} + \cdots)F, \quad (k_1 + k_2 + \cdots + k_\lambda - \lambda + 1 = i). \tag{9}$$

By means of [8] *and* [9], *the expansion*

$$F = \Phi + \Delta\Phi_1 + \Delta^2\Phi_2 + \cdots + \Delta^i\Phi_i + \cdots + \Delta^p\Phi_p$$

has reached an explicit representation in the stated sense.

If one applies to [7] the identical transformation $\Omega\Delta F = PF$, where P denotes polar processes,[293] then, since these processes commute with the Ω-process, one obtains the known formula

$$F_1 = P\Omega F; \quad \ldots; \quad F_i = P\Omega^i F;$$

and from this, in conjunction with [8] or with the considerations loc. cit., follows the known formulation of the series expansion, as for instance in (3) loc. cit. If one is dealing with n rows of n variables, $x^{(1)}, x^{(2)}, \ldots, x^{(n-1)}, z$, then correspondingly

$$f_i(\xi) = \lambda_1 x_i^{(1)} + \lambda_2 x_i^{(2)} + \cdots + \lambda_{n-1}x_i^{(n-1)};$$

and likewise Δ and Ω are to be defined for n rows.

2. For the normal form of $F(t_i, t_{ik}, t_{ikl}, \ldots)$, the functions $f_i(\xi), f_{ik}(\xi), \ldots$ are given by the one-row, two-row, up to $(n-1)$-row determinants

$$\xi_i^{(1)}, \quad (\xi^{(1)}\xi^{(2)})_{ik}, \quad \ldots, \quad (\xi^{(1)}\xi^{(2)}\cdots\xi^{(n-1)})_{i_1 i_2\cdots i_{n-1}}.$$

Thus, as a specialization of [5], one obtains:

$$\begin{gathered} S = AB; \qquad BF = F(\xi_i^{(1)}, (\xi^{(1)}\xi^{(2)})_{ik}, \ldots), \\ ABF = \left(\sum t_i \frac{\partial}{\partial\xi_i^{(1)}}\right)^{\lambda_1}\left(\sum t_{ik}\left(\frac{\partial}{\partial\xi^{(1)}}\frac{\partial}{\partial\xi^{(2)}}\right)_{ik}\right)^{\lambda_2}\cdots F(\xi_i^{(1)}, (\xi^{(1)}\xi^{(2)})_{ik}, \ldots), \end{gathered}$$

where $\lambda_1, \lambda_2, \ldots$ denote the degrees in $t_i, t_{ik}, \ldots$.

[292] Kerim uses this specialized formula in his Erlangen dissertation, soon to appear, in the application to “inertia forms”.

[293] Compare, for example, F. Mertens, Über ganze Funktionen von m Systemen von je n Unbestimmten, Monatsh. f. Math. u. Phys. 4, formula (5).

From invariant-theoretic considerations (namely, all invariants of F which are linear in the coefficients can be linearly composed from Φ and from forms out of M, thus in particular also the SF not belonging to M)[294] the congruence follows here: $\Phi \equiv a_1 SF \mod M$; and because of the normal-form property one obtains from this, instead of [4], the simple relation

$$\Phi = a_1 SF = a_1 S\Phi. \tag{10}$$

The second equality holds because applying S makes every form from M vanish. Formula [10] shows that the normal form Φ is itself at the same time an eigenfunction; and indeed, by the invariant-theoretic consideration above, there are no further eigenfunctions which are at the same time invariants of F (compare, concerning [10], Deruyts, formulas (10) and (10')).

Corrections.

1. In the paper "Die allgemeinsten Bereiche aus ganzen transzendenten Zahlen" (Math. Ann. 77), on p. 121, line 8 from the top, line 9 from the top, and line 11 from the bottom, the field (H, Y) is to be replaced by the "field $(H, Y, Y', \ldots, Y^{(\lambda_0-1)})$, where $Y', \ldots, Y^{(\lambda_0-1)}$ denote the conjugates of Y with respect to H." Correspondingly, on p. 120, line 5 from the top, and p. 121, line 9 from the top, (H, y) is to be replaced by $(H, y, y', \ldots, y^{(\lambda_0-1)})$.

2. In the paper "Gleichungen mit vorgeschriebener Gruppe" (Math. Ann. 78), in the formulation of Hilbert's irreducibility theorem – p. 222, line 3 from the bottom – instead of "number field Ω" it must more precisely say "finite algebraic number field Ω", as Mr. Ostrowski has pointed out to me. In fact, with respect to the field of all algebraic numbers, for instance, the group of every equation with algebraic numerical coefficients is the identity group. – Correspondingly, in the introduction, "arbitrary domain of rationality" is to be understood as a finite algebraic number field to which finitely many parameters can still be adjoined, or as one of the intermediate fields which arise in this way.

(Accepted September 1919.)

[294] This follows, for instance, from § 6 and § 7, 2. of my cited paper "Zur Invariantentheorie der Formen von n Variabeln".

17. Together with W. Schmeidler: Modules in Noncommutative Domains, Especially from Difference Expressions

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Contents.

Introduction.

The formal theory of differential expressions in one variable leads to a decomposition into irreducible factors which is unique in a certain sense.[1] But whereas the corresponding theorem in algebra can be carried over to polynomials in several variables, this is no longer possible for partial differential expressions.[2] For differential expressions in one variable, however, alongside the product representation there is also a representation as least common multiple, with similar laws holding; for, in two different decompositions, the individual constituents can be assigned to one another pairwise: they are "of the same kind".[3]

This concept of the least common multiple can now be interpreted as a module concept and so transferred to n variables; in the present paper this gives the natural generalization of the theorems just mentioned. Beyond this, however, we also note that for differential expressions only the following fact is used: they can be interpreted as polynomials with noncommutative multiplication, for which the degree of a product is equal to the sum of the degrees of the factors. These conditions are fulfilled, for example, by difference expressions as well, and hence the theorems also hold for these, where until now they were likewise known only for one variable.[4]

We therefore develop, in general, a theory of modules whose elements are polynomials with noncommutative multiplication; in doing so we reduce the study of the modules to that of residue classes.[5] In contrast with the commutative special case, one can indeed speak here of the sum of residue classes, but not of their product; one can speak only of the product of a residue class with a polynomial. The concept of a "residue group" is therefore to be modified relative to the commutative case. In the special case of polynomials in one variable, the residue group is moreover "finite", that is, it possesses only a finite number of linearly independent residue classes. In general there is no such finite "order"; the groups are "infinite". For our methods, however, this difference is immaterial.

Just as in the commutative case, to the representation of a module as the least common multiple of relatively prime modules there corresponds the simpler process of additive decomposition of the residue group. In this connection the isomorphism of residue groups proves essential; upon specialization to differential expressions in one variable this isomorphism becomes the concept of the same kind for the associated modules, while upon specialization to commutative modules it becomes identity. Incidentally, the concept of the same

[1] Cf. E. Landau, Ein Satz über die Zerlegung homogener linearer Differentialausdrücke in irreduzible Faktoren, Journal für die reine und angewandte Mathematik 124 (1902), pp. 115–120, and subsequently A. Loewy, Über reduzible lineare homogene Differentialgleichungen, Mathematische Annalen 56 (1903), pp. 549–584.

[2] Cf. a counterexample of Landau published in H. Blumberg, Über algebraische Eigenschaften von linearen homogenen Differentialausdrücken, dissertation, Göttingen 1912, 52 pp., pp. 51–52.

[3] Cf., besides the papers just cited, A. Loewy, Über vollständig reduzible lineare homogene Differentialgleichungen, Mathematische Annalen 62 (1906), pp. 89–117; cited as Loewy.

[4] Cf. G. Wallenberg–A. Guldberg, Theorie der linearen Differenzengleichungen, Leipzig and Berlin 1911.

[5] For the case of commutative domains, cf. W. Schmeidler, Über Moduln und Gruppen hyperkomplexer Größen, Mathematische Zeitschrift 3 (1919), pp. 29–42.

kind can also be transferred to n variables, and its agreement with the isomorphism concept can be proved. The fact that in the additive decomposition of the residue group only finitely many summands ever occur follows by transferring Hilbert's theorem on the module basis, using Gordan's method of proof; Hilbert's original method of proof would require restrictions on the multiplication laws (in formula (6), a on the right instead of a_i, which is fulfilled for differential expressions but not for difference expressions).

In order to arrive at a uniqueness theorem, we must specialize the modules – as Loewy also does – to "completely reducible" ones, that is, to those representable as least common multiples of prime modules. Given two different decompositions, the constituents are then either pairwise identical, or at least to every constituent of one decomposition there corresponds an isomorphic constituent of the other decomposition, and conversely; at the same time, each decomposition then contains at least two isomorphic constituents. Furthermore, from the occurrence of two isomorphic constituents in one and the same decomposition there follows the existence of infinitely many decompositions; this even holds without the restriction to prime modules, which until now was not known even in the special case of differential expressions in one variable.

Finally, for modules from differential expressions, we discuss the connection with integrals and show that, for finite residue groups, the order is equal to the largest number of linearly independent integrals. Thus arbitrary functions occur in the general integral precisely when the residue group is not finite. The concept that two modules are of the same kind means, here as in the one-variable case, a "birational" relation between the integrals. Some examples conclude the paper.

The first impulse for the present paper was given several years ago by a question of E. Landau to E. Noether concerning the generalization of the product theorem. At that time E. Noether only observed that the question was one of module theory in the noncommutative polynomial domain, but could not yet be attacked. The Lasker module theorems then known were not directly transferable to this domain, because Lasker's method of proof rests on the theorem that a polynomial is uniquely decomposable into irreducible factors.[6] Only Schmeidler's methods offered the possibility of a transfer, which we then carried out jointly. In detail, let it be mentioned that the generalization of the residue group, the transfer and application of Hilbert's finiteness theorem, and the existence of infinitely many decompositions are due to E. Noether; the generalization of Loewy's concept of complete reducibility, the concept of isomorphism, and its connection with the concept of the same kind are due to W. Schmeidler.

§ 1. Formal Matters Concerning Differential and Difference Expressions.

1. For differential expressions in one variable, a symbolic product formation has been introduced. Let

$$a_0(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_1(x)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_n(x)y = A(y),$$

$$b_0(x)\frac{\mathrm{d}^m y}{\mathrm{d}x^m} + b_1(x)\frac{\mathrm{d}^{m-1} y}{\mathrm{d}x^{m-1}} + \cdots + b_m(x)y = B(y).$$

Then one defines the symbolic product $AB(y)$ as the differential expression

$$a_0(x)\frac{\mathrm{d}^n B(y)}{\mathrm{d}x^n} + a_1(x)\frac{\mathrm{d}^{n-1} B(y)}{\mathrm{d}x^{n-1}} + \cdots + a_n(x)B(y).$$

This product formation is in reality a multiplication of two operators; it is most simply written by replacing, as usual, $\frac{\mathrm{d}^n}{\mathrm{d}x^n}$ by $\left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^n$. Then AB becomes

$$\left(a_0\left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^n + \cdots + a_n\right)\left(b_0\left(\frac{\mathrm{d}}{\mathrm{d}x}\right)^m + \cdots + b_m\right)y,$$

where the ordinary rules for differentiating sums and products apply in carrying out the product. Every such operator can now be regarded as a polynomial in the single variable $\frac{\mathrm{d}}{\mathrm{d}x} = \xi$, by not writing explicitly the function to which the operator is applied and by fixing for the variable ξ the multiplication rule corresponding to product differentiation:

$$\frac{\mathrm{d}}{\mathrm{d}x}(a(x)y) = a(x)\frac{\mathrm{d}}{\mathrm{d}x}(y) + a'(x)y,$$

or, in our notation $(n = 1, m = 0)$,

$$\xi \cdot a = a \cdot \xi + a'. \tag{1}$$

[6]Recently E. Noether observed that the Lasker theorems too can be established without using this theorem, by methods related to those used here.

One proceeds in exactly the same way for linear partial differential expressions

$$\left(\sum a_{\alpha_1\ldots\alpha_n}(x_1,\ldots,x_n)\frac{\partial^{\alpha_1+\cdots+\alpha_n}}{\partial x_1^{\alpha_1}\cdots\partial x_n^{\alpha_n}}\right)y$$

of a function y. For the n operators $\frac{\partial}{\partial x_1},\ldots,\frac{\partial}{\partial x_n}$ we introduce variables $\xi_1,\ldots,\xi_n$ and thereby obtain the polynomial

$$\sum a_{\alpha_1\ldots\alpha_n}(x_1,\ldots,x_n)\xi_1^{\alpha_1}\cdots\xi_n^{\alpha_n}.$$

The product of two polynomials is then computed according to the laws

$$\xi_i a = a\xi_i + a_i \qquad (i=1,\ldots,n), \tag{2}$$

where a_i denotes the partial derivative of a with respect to x_i; the products of the ξ's among themselves are commutative. Apart from the commutative law of multiplication, all laws of addition and multiplication hold for the polynomials so defined just as in ordinary algebra. In particular, for the product of two such polynomials the total degree, and the degree in each individual variable, is equal to the sum of the corresponding degrees of the two factors.

2. For difference expressions (cf. Wallenberg, loc. cit.) of the form

$$a_x^{(0)}y_{x+n} + a_x^{(1)}y_{x+n-1} + \cdots + a_x^{(n)}y_x = A_x,$$

$$b_x^{(0)}y_{x+m} + b_x^{(1)}y_{x+m-1} + \cdots + b_x^{(m)}y_x = B_x$$

one has a corresponding product definition

$$(AB)_x = a_x^{(0)}B_{x+n} + a_x^{(1)}B_{x+n-1} + \cdots + a_x^{(n)}B_x,$$

which can be carried out with the aid of the multiplication rule

$$(ay)_{x+1} = a_{x+1}y_{x+1}.$$

If we introduce the operator ξ for the transition from x to $x+1$, then we obtain $\xi\{ay\} = a^*\xi\{y\}$, where $a^* = a_{x+1}$. Again one can omit the function y to which the operator is applied and obtains the law

$$\xi a = a^*\xi, \tag{3}$$

where, along with a, also a^* does not vanish identically. In exactly the same way, for n variables,

$$\xi_i a = a_i\xi_i \qquad (i=1,\ldots,n), \tag{4}$$

where ξ_i stands for the transition from x_i to x_i+1, and a_i denotes the function obtained from a by replacing x_i by x_i+1; hence it does not vanish identically if a does not. With these conventions, all rules of addition and multiplication, including the rule concerning the degree of a product, again hold, except for the commutative law.

Both cases will be subordinated, in the following section, to a quite generally defined polynomial domain with arbitrary coefficients.

§ 2. The General Polynomial Domain.

We start from an arbitrary abstractly defined field P (whose elements we denote by small Latin letters), for which all rules of algebra hold, in particular also the commutative law of multiplication with unique inversion. To this field we adjoin n indeterminates $\xi_1,\ldots,\xi_n$; that is, we consider the integral domain J consisting of the products $a\xi_1^{\nu_1}\xi_2^{\nu_2}\cdots\xi_n^{\nu_n}$ and their finite sums, where all laws of addition and multiplication are to hold with the exception of the commutative law of multiplication. In particular, we call finite sums of the form

$$F(\xi_1,\ldots,\xi_n) = \sum a_{\nu_1\ldots\nu_n}\xi_1^{\nu_1}\cdots\xi_n^{\nu_n}$$

"polynomials" (which throughout are to be denoted by capital Latin letters). In place of the commutative law, we now subject the elements of J to product laws such that *the product of two polynomials is again a* *polynomial*; thus the domain of all polynomials from J already exhausts J. For this it is plainly necessary and sufficient that, for the elements $\xi_1, \ldots, \xi_n$ and an arbitrary element a of P, product rules of the form

$$\xi_i a = F_i(\xi_1, \ldots, \xi_n) \qquad (i = 1, \ldots, n) \tag{5}$$

hold, where $F_i(\xi_1, \ldots, \xi_n)$ denotes some polynomial depending naturally on ξ_i and a. (In particular, the rules (2) and (4) for differential and difference expressions, respectively, are of this form.) For since $\xi_i a$ is not in itself a polynomial, while by hypothesis it belongs to the domain, the equation is necessary. Conversely, by finitely repeated application of formula (5), every element of the domain J can be transformed into a polynomial.

For the unit of the field, which we denote by 1, we shall further require $\xi_i \cdot 1 = 1 \cdot \xi_i = \xi_i$. We shall also stipulate that the multiplication of $\xi_1, \ldots, \xi_n$ among themselves is commutative.[7] Every polynomial in J then writes itself just like an ordinary polynomial in $\xi_1, \ldots, \xi_n$ with coefficients from P, and two polynomials agree only when all their coefficients agree.

More specifically, in what follows[8] we require that, in place of (5), the rule

$$\xi_i a = a_i \xi_i + b_i \qquad (a_i \neq 0) \quad (i = 1, \ldots, n) \tag{6}$$

shall hold. Here too, in particular, the laws (2) and (4) are included. Then the degree of a product is equal to the sum of the degrees of the factors; this holds both for total degree and for the degree in each individual variable, so that the product of two polynomials which do not vanish identically is likewise different from 0.

§ 3. One-Sided Modules and Their Residue Groups.

For our domain J, because multiplication is noncommutative, the following kinds of modules[9] can be distinguished:[10]

1. The *right-sided module*, which, along with M, contains also FM for an arbitrary polynomial F, and along with M_1 and M_2 contains also $M_1 + M_2$.

2. The *left-sided module*, which, along with M, contains also MF, and along with M_1 and M_2 contains also $M_1 + M_2$.

We refer to these two kinds by the common name *one-sided modules*; modules which are both right-sided and left-sided will be called *two-sided*. In what follows we restrict ourselves to right-sided modules and note that the theory of the left-sided ones can be built up in exactly the same way.

We call two polynomials F and G congruent modulo $\mathfrak{M}$, written $F \equiv G(\mathfrak{M})$, if their difference is divisible by $\mathfrak{M}$, that is, if it belongs to $\mathfrak{M}$. The totality of all polynomials congruent to a given polynomial forms a *residue class* modulo $\mathfrak{M}$. (Residue classes are to be denoted by small German letters.) Since from

$$F_1 \equiv F_2(\mathfrak{M}) \qquad \text{and} \qquad G_1 \equiv G_2(\mathfrak{M})$$

it follows that $F_1 + G_1 \equiv F_2 + G_2(\mathfrak{M})$, the sum of two residue classes $\mathfrak{x}$ and $\mathfrak{y}$ is defined and is again a residue class $\mathfrak{x} + \mathfrak{y}$. By contrast, because the module is one-sided, one cannot speak of the product of two residue classes. Indeed, from

$$F_1 = F_2 + M_1, \qquad G_1 = G_2 + M_2$$

there follows only

$$F_1 G_1 = F_2 G_2 + F_2 M_2 + M_1 G_2 + M_1 M_2;$$

but since $M_1 G_2$ need not belong to $\mathfrak{M}$ when M_1 belongs to $\mathfrak{M}$, $F_1 G_1$ and $F_2 G_2$ need not be congruent in general. On the other hand, if $M_1 = 0$, hence $F_1 = F_2 = F$, one sees that $G_1 \equiv G_2$ implies $FG_1 \equiv FG_2$. Thus an arbitrary residue class $\mathfrak{y}$ can be multiplied from the front by a polynomial F, and this gives a well-defined residue class $F\mathfrak{y} = \mathfrak{z}$.

For computation with residue classes, the commutative and associative laws of addition hold, as does the law of unique reversal of addition; moreover, for multiplication of residue classes by polynomials and addition, the distributive law $F(\mathfrak{y}_1 + \mathfrak{y}_2) = F\mathfrak{y}_1 + F\mathfrak{y}_2$ holds, as follows directly from computing with polynomials modulo $\mathfrak{M}$.

[7] This is needed only from § 6 onward for the finiteness theorem and its consequences.

[8] This too is used only from § 6 onward.

[9] We denote modules by capital German letters.

[10] One-sided ideals were already considered by A. Hurwitz; cf. Vorlesungen über die Zahlentheorie der Quaternionen (Berlin, Springer 1919), Lecture 6. There, however, what is essentially involved are principal ideals; the same holds for the works of Du Pasquier cited there in note 1.

The residue class to which the unit belongs will be called the unit class, or unit, and denoted $\mathfrak{e}$. Then $F\mathfrak{e} = \mathfrak{x}$ is the residue class to which the polynomial F belongs. In contrast to the general residue classes, however, here the product $\mathfrak{x}\mathfrak{e}$ is also well-defined and equal to $\mathfrak{x}$; for from

$$F_1 = F_2 + M_1, \qquad G_1 = 1 + M_2$$

and $M_1 \cdot 1 = M_1$ it follows that

$$F_1 G_1 = F_2 + M.$$

The totality of the residue classes modulo $\mathfrak{M}$ is called the *residue group of* $\mathfrak{M}$. A residue group therefore has the following properties: 1. Along with every residue class $\mathfrak{x}$ and an arbitrary polynomial F, it contains the residue class $F\mathfrak{x}$; and along with the residue classes $\mathfrak{x}_1$ and $\mathfrak{x}_2$ it contains the residue class $\mathfrak{x}_1 + \mathfrak{x}_2$.[11] 2. It possesses a unit $\mathfrak{e}$, so that $F\mathfrak{e}$ represents, for every polynomial F, the residue class $\mathfrak{x}$ of F. *Thus, as F runs through all polynomials, $F\mathfrak{e}$ runs through the whole residue group.*

§ 4. Reducibility of the Residue Group into Two Subgroups and Its Relation to the Module.

For one-sided modules, the concepts *divisibility*, *greatest common divisor*, and *least common multiple* remain as in the two-sided case, as the following definitions show.

A module $\mathfrak{M}$ is said to be divisible by another module $\mathfrak{N}$ if from $M \equiv 0(\mathfrak{M})$ it follows also that $M \equiv 0(\mathfrak{N})$.

For two modules $\mathfrak{M}$ and $\mathfrak{N}$, the *greatest common divisor* $(\mathfrak{M}, \mathfrak{N})$ is defined as the totality of all polynomials representable in the form $M + N$, where $M \equiv 0(\mathfrak{M})$ and $N \equiv 0(\mathfrak{N})$; it is again a module. The same holds for several modules.

The totality of all polynomials divisible both by $\mathfrak{M}$ and by $\mathfrak{N}$ forms a module $[\mathfrak{M}, \mathfrak{N}]$, the *least common multiple* of $\mathfrak{M}$ and $\mathfrak{N}$. This definition too applies correspondingly to several modules, indeed even to a set of modules of arbitrary cardinality.

Two modules are called relatively prime when their greatest common divisor is the unit module, that is, the totality of all polynomials.

We now start from a module $\mathfrak{M}$ which is representable as the least common multiple of two relatively prime modules $\mathfrak{N}$ and $\mathfrak{L}$, both assumed to be proper divisors:

$$\mathfrak{M} = [\mathfrak{N}, \mathfrak{L}]. \tag{7}$$

We investigate the residue group $\mathfrak{G}$ of $\mathfrak{M}$. By hypothesis,

$$1 \equiv N + L(\mathfrak{M}) \tag{8}$$

for two polynomials $N \equiv 0(\mathfrak{N})$ and $L \equiv 0(\mathfrak{L})$. From this we show

$$\mathfrak{L} = (\mathfrak{M}, L), \qquad \mathfrak{N} = (\mathfrak{M}, N), \tag{9}$$

where, for example, $(\mathfrak{M}, L)$ means the module consisting of all polynomials of the form $M + FL$. In fact, first $(\mathfrak{M}, L) \equiv 0(\mathfrak{L})$. Conversely, from (8) it follows, for an arbitrary polynomial F, that

$$F \equiv FN(\mathfrak{L}). \tag{10}$$

Now if $F \equiv 0(\mathfrak{L})$, then $FN \equiv 0(\mathfrak{L})$, hence $\equiv 0(\mathfrak{M})$, and consequently $F \equiv FL(\mathfrak{M})$. In the same way one proves $\mathfrak{N} = (\mathfrak{M}, N)$. It follows incidentally, since $\mathfrak{N}$ and $\mathfrak{L}$ are proper divisors of $\mathfrak{M}$, that neither $L \equiv 0$ nor $N \equiv 0(\mathfrak{M})$ can hold.

Let $\mathfrak{a}$ be the residue class of N modulo $\mathfrak{M}$, and $\mathfrak{b}$ the residue class of L modulo $\mathfrak{M}$. Then by (8)

$$\mathfrak{e} = \mathfrak{a} + \mathfrak{b} \qquad (\mathfrak{a} \neq 0,\ \mathfrak{b} \neq 0).$$

Since every residue class is of the form $F\mathfrak{e}$, the distributive law gives, for every residue class,

$$F\mathfrak{e} = F\mathfrak{a} + F\mathfrak{b} \qquad (\mathfrak{a} \neq 0,\ \mathfrak{b} \neq 0). \tag{11}$$

This representation of an arbitrary residue class as a sum of two residue classes of the form $G\mathfrak{a}$ and $H\mathfrak{b}$ is, however, *unique.* For from a relation $0 = G\mathfrak{a} + H\mathfrak{b}$, if K is a polynomial in the residue class $G\mathfrak{a} = -H\mathfrak{b}$, then

[11] This property could, in a more general sense, be called the "module property" of the residue group.

plainly $K \equiv GN(\mathfrak{M})$, $K \equiv -HL(\mathfrak{M})$, hence $K \equiv 0(\mathfrak{N})$, $K \equiv 0(\mathfrak{L})$, and therefore $K \equiv 0(\mathfrak{M})$; that is, $G\mathfrak{a} = 0$, $H\mathfrak{b} = 0$.

Conversely, suppose now that relation (11) is fulfilled for every polynomial F, and indeed uniquely in the above sense. Let N then be a polynomial from $\mathfrak{a}$ and L a polynomial from $\mathfrak{b}$. We form the two modules (9) and assert (7) and (8). The latter follows from (11) for $F = 1$. Furthermore, by the definition of $\mathfrak{N}$ and $\mathfrak{L}$, $\mathfrak{M}$ is a common multiple of these two modules, and indeed the least one. For from $F \equiv 0(\mathfrak{L})$ and $F \equiv 0(\mathfrak{N})$ there follows $F \equiv HL \equiv GN(\mathfrak{M})$; hence $G\mathfrak{a} = H\mathfrak{b}$, $G\mathfrak{a} - H\mathfrak{b} = 0$, and therefore, by uniqueness of the representation, $G\mathfrak{a} = 0$, $H\mathfrak{b} = 0$, $F \equiv 0(\mathfrak{M})$. It has therefore been shown that the *representation of a module $\mathfrak{M}$ as the least common multiple of two relatively prime modules is equivalent to the unique additive decomposition of the residue group $\mathfrak{G}$ of $\mathfrak{M}$, as given by formula* (11).

We now investigate the system $\mathfrak{A}$ of residue classes of the form $F\mathfrak{a}$, which we wish to relate to the residue group $\overline{\mathfrak{A}}$ of $\mathfrak{L}$. By (11), every polynomial F in $\mathfrak{L}$ belongs to the residue class $F\mathfrak{a}$ when $\overline{\mathfrak{a}}$ is the residue class of N modulo $\mathfrak{L}$, and hence the unit class of $\overline{\mathfrak{A}}$. If we now associate the residue classes $F\mathfrak{a}$ in $\mathfrak{A}$ with the residue classes $F\overline{\mathfrak{a}}$ in $\overline{\mathfrak{A}}$, then to every residue class of $\mathfrak{A}$ there is assigned a residue class of $\overline{\mathfrak{A}}$, in such a way that the sum of two residue classes of $\mathfrak{A}$ corresponds to the sum of the assigned residue classes of $\overline{\mathfrak{A}}$, and the product of a residue class of $\mathfrak{A}$ with a polynomial corresponds to the product of the assigned residue class of $\overline{\mathfrak{A}}$ with the same polynomial. This assignment is moreover one-to-one: from $F\mathfrak{a} = 0$ follows $F \equiv 0(\mathfrak{L})$ by (11), hence $F\overline{\mathfrak{a}} = 0$; conversely, $F\overline{\mathfrak{a}} = 0$, hence $F \equiv 0(\mathfrak{L})$, also says by (10) that $FN \equiv 0(\mathfrak{L})$, hence $FN \equiv 0(\mathfrak{M})$, that is, $F\mathfrak{a} = 0$.

This leads us to a fundamental concept, which we define as follows:

A one-to-one correspondence between two systems $\mathfrak{A}$ and $\overline{\mathfrak{A}}$ of residue classes is called isomorphic if the sum of two residue classes of $\mathfrak{A}$ is assigned the sum of the corresponding residue classes of $\overline{\mathfrak{A}}$, and the product of an arbitrary residue class of $\mathfrak{A}$ with a polynomial is assigned the product of the corresponding residue class of $\overline{\mathfrak{A}}$ with the same polynomial.

A subsystem $\mathfrak{A}$ of residue classes in $\mathfrak{G}$ which is isomorphic to the residue group of a module is called a *subgroup* of $\mathfrak{G}$. As the proof above shows, the assignment here is specifically such that every residue class of $\mathfrak{A}$ arises from the corresponding residue class of $\overline{\mathfrak{A}}$ by adjoining certain polynomials, namely all polynomials belonging to $\mathfrak{L}$ but not to $\mathfrak{M}$.

Similarly one shows that the totality of all residue classes of the form $F\mathfrak{b}$ forms a subgroup $\mathfrak{B}$ of $\mathfrak{G}$ which is isomorphic to the residue group $\overline{\mathfrak{B}}$ of $\mathfrak{N}$.

A residue group $\mathfrak{G}$ whose residue classes admit a unique additive decomposition (11) will be called *reducible* into the two subgroups $\mathfrak{A}$ and $\mathfrak{B}$, written $\mathfrak{G} = \mathfrak{A} + \mathfrak{B}$; a group which is not reducible is called *irreducible.* We also occasionally call the associated modules reducible or irreducible. Thus the following holds.

Theorem 1.[12] *A module $\mathfrak{M}$ is representable as the least common multiple of two relatively prime proper divisors $\mathfrak{L}$ and $\mathfrak{N}$ if and only if its residue group $\mathfrak{G}$ is reducible into two subgroups $\mathfrak{A}$ and $\mathfrak{B}$. In this case $\mathfrak{A}$ and $\mathfrak{B}$ are respectively isomorphic to the residue groups of $\mathfrak{L}$ and $\mathfrak{N}$.*

[12] Cf. Schmeidler, loc. cit., p. 39.

17. Together with W. Schmeidler: Modules in Noncommutative Domains, Especially from Differential and Difference Expressions

Math. Zs. 8 (1920), pp. 1–35

§ 5. Computation with Units and Reducibility into k Subgroups.

The residue classes $\mathfrak{a}$ and $\mathfrak{b}$ of $\mathfrak{G}$ share with the unit $\mathfrak{e}$ the property that the product of $\mathfrak{a}$ and $\mathfrak{b}$ with an arbitrary residue class $\mathfrak{x}$ is defined; we therefore likewise call $\mathfrak{a}$ and $\mathfrak{b}$ units of $\mathfrak{G}$. Indeed, from (11), for $F = M$, there follows

$$0 = M\mathfrak{e} = M\mathfrak{a} + M\mathfrak{b},$$

and hence, by uniqueness, $M\mathfrak{a} = 0$, $M\mathfrak{b} = 0$, and therefore

$$(F + M)\mathfrak{a} = F\mathfrak{a} = \mathfrak{x}\mathfrak{a},$$

where $\mathfrak{x}$ denotes the residue class of F. In place of relation (11), we may therefore write the equivalent class relation

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{a} + \mathfrak{x}\mathfrak{b}.$$

In particular, putting $\mathfrak{x} = \mathfrak{a}$ gives

$$\mathfrak{a} = \mathfrak{a}^2 + \mathfrak{a}\mathfrak{b},$$

and hence, by uniqueness, $\mathfrak{a}^2 = \mathfrak{a}$, $\mathfrak{a}\mathfrak{b} = 0$. Similarly one obtains $\mathfrak{b}^2 = \mathfrak{b}$, $\mathfrak{b}\mathfrak{a} = 0$.

Corresponding to the representation of a residue group as a sum of two constituents, we may now also define such a representation as a sum of k constituents. Namely, suppose a unique representation is given

$$F\mathfrak{e} = F\mathfrak{a}_1 + F\mathfrak{a}_2 + \cdots + F\mathfrak{a}_k \qquad (\mathfrak{a}_1 \neq 0, \ldots, \mathfrak{a}_k \neq 0),$$

where from $0 = G_1\mathfrak{a}_1 + \cdots + G_k\mathfrak{a}_k$ it also follows that $G_i\mathfrak{a}_i = 0$. Then, in particular for $F = M$, one obtains the relation $M\mathfrak{a}_i = 0$, so that the representation may also be written in the form[1]

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{a}_k \qquad (\mathfrak{a}_1 \neq 0, \ldots, \mathfrak{a}_k \neq 0). \tag{12}$$

This representation may also be regarded as having arisen from a successive decomposition into two constituents. If this can be continued without limit, with all $\mathfrak{a}_i \neq 0$, then we speak of a sum of infinitely many constituents.

Putting $\mathfrak{x} = \mathfrak{a}_i$ in (12) gives

$$\mathfrak{a}_i^2 = \mathfrak{a}_i, \qquad \mathfrak{a}_i\mathfrak{a}_j = 0 \quad (i \neq j), \qquad i, j = 1, \ldots, k, \tag{13}$$

with $\mathfrak{a}_i \neq 0$. Conversely, suppose a system of residue classes $\mathfrak{a}_1, \ldots, \mathfrak{a}_k$ in $\mathfrak{G}$ is present for which the orthogonality conditions (13) are fulfilled and for which

$$\mathfrak{e} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_k \tag{14}$$

holds. We shall show that this gives a representation of the group as a sum of k constituents. First, from the hypothesis, for every polynomial A_i from $\mathfrak{a}_i$, since $A_i\mathfrak{a}_i = (A_i + M)\mathfrak{a}_i$, it follows that $M\mathfrak{a}_i = 0$; hence $\mathfrak{a}_i$ can be multiplied by every class. Thus (14) implies representation (12), and this representation is unique. For if $0 = \mathfrak{x}_1\mathfrak{a}_1 + \cdots + \mathfrak{x}_k\mathfrak{a}_k$, then right multiplication by $\mathfrak{a}_i$ gives, by (13), $0 = \mathfrak{x}_i\mathfrak{a}_i$, proving uniqueness and hence the assertion. The residue classes $\mathfrak{a}_1, \ldots, \mathfrak{a}_k$ occurring in the decomposition of the unit class will be called the units belonging to the decomposition; equivalently they are characterized by the orthogonality relations (13) and condition (14). The later developments will essentially amount to computation with units.

It remains to establish the connection between the additive decomposition of the group into k constituents and the corresponding decomposition of the module. As in the case $k = 2$, a representation of the module as the least common multiple of k relatively prime modules will result, by induction.

Let the decomposition (12) be given, which we write in the form

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{b} + \mathfrak{x}\mathfrak{a}_k, \tag{15}$$

[1] For the additive decomposition of residue groups, the commutative and associative laws hold; in contrast to the commutative case, however, $\mathfrak{U}_1 + \mathfrak{U}_2 = \mathfrak{U}_1 + \mathfrak{U}_3$ does not imply $\mathfrak{U}_2 = \mathfrak{U}_3$; cf. example 1 in § 12.

$$\mathfrak{x}\mathfrak{b} = \mathfrak{x}\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{a}_{k-1}. \tag{16}$$

From (15) it follows by Theorem I that the residue classes $\mathfrak{x}\bar{\mathfrak{b}}$ are isomorphic to the residue group of the module

$$\mathfrak{N} = (\mathfrak{M}, A_k). \tag{17}$$

For their residue classes $\mathfrak{x}\bar{\mathfrak{b}}$, the representation corresponding to (16) is therefore also valid:

$$\mathfrak{x}\bar{\mathfrak{b}} = \mathfrak{x}\bar{\mathfrak{a}}_1 + \cdots + \mathfrak{x}\bar{\mathfrak{a}}_{k-1}. \tag{18}$$

Suppose now it has already been proved that

$$\mathfrak{N} = [\mathfrak{M}_1, \ldots, \mathfrak{M}_{k-1}], \tag{19}$$

where

$$\begin{cases} \mathfrak{M}_1 = (\mathfrak{N}, A_2 + \cdots + A_{k-1}), \\ \mathfrak{M}_2 = (\mathfrak{N}, A_1 + A_3 + \cdots + A_{k-1}), \\ \qquad \vdots \\ \mathfrak{M}_{k-1} = (\mathfrak{N}, A_1 + \cdots + A_{k-2}). \end{cases} \tag{20}$$

For two summands this assertion agrees with our earlier result. Moreover, the polynomials A_i may here be chosen specifically as polynomials from $\bar{\mathfrak{a}}_i$ which at the same time are contained in $\mathfrak{a}_i$, since by § 4 the polynomials of the latter residue class are contained in the former. Then, by (15),

$$\mathfrak{M} = [\mathfrak{N}, \mathfrak{M}_k], \tag{21}$$

where

$$\mathfrak{M}_k = (\mathfrak{M}, A_1 + \cdots + A_{k-1}).$$

From (19) and (21) one obtains

$$\mathfrak{M} = [[\mathfrak{M}_1, \ldots, \mathfrak{M}_{k-1}], \mathfrak{M}_k] = [\mathfrak{M}_1, \ldots, \mathfrak{M}_{k-1}, \mathfrak{M}_k]$$

by the definition of least common multiple; and from (17) and (20) one gets

$$\mathfrak{M}_1 = (\mathfrak{M}, A_k, A_2 + \cdots + A_{k-1}).$$

It follows that

$$\mathfrak{M}_1 = (\mathfrak{M}, A_2 + \cdots + A_k),$$

for the latter module contains, by (13), also the polynomial

$$A_k = A_k(A_2 + \cdots + A_k) \pmod{\mathfrak{M}}.$$

Corresponding expressions are obtained for $\mathfrak{M}_2, \ldots, \mathfrak{M}_{k-1}$. Furthermore, the modules $\mathfrak{M}_1, \ldots, \mathfrak{M}_k$ are relatively prime, since by (14)

$$\frac{1}{k-1}\Big((A_2 + \cdots + A_k) + (A_1 + A_3 + \cdots + A_k) + \cdots + (A_1 + \cdots + A_{k-1})\Big) \equiv 1 \pmod{\mathfrak{M}}.$$

They moreover form a totally relatively prime system, that is, the least common multiple of any of them is relatively prime to the least common multiple of the remaining ones.[2] This follows directly by grouping the summands in formula (12) into two suitable groups.

It has thus been shown that to a decomposition of the residue group into k summands there corresponds a representation of the module as the least common multiple of k modules forming a totally relatively prime system. We now prove the converse theorem, likewise by induction.

Suppose therefore that

$$\mathfrak{M} = [\mathfrak{M}_1, \ldots, \mathfrak{M}_k] = [[\mathfrak{M}_1, \ldots, \mathfrak{M}_{k-1}], \mathfrak{M}_k] = [\mathfrak{N}, \mathfrak{M}_k],$$

[2]This is a generalization of a concept introduced by Loewy.

where $\mathfrak{M}_1, \ldots, \mathfrak{M}_k$ form a totally relatively prime system. In combination with Theorem I this gives representation (15) for the residue group. The modules $\mathfrak{M}_1, \ldots, \mathfrak{M}_{k-1}$ also form a totally relatively prime system. For if, under some division of the modules of this system into two groups, the least common multiple of one group and the least common multiple of the other had a divisor in common, this divisor would remain if the module $\mathfrak{M}_k$ were added to one of the two groups and the least common multiple were then formed.[3] For $k = 2$, "totally relatively prime" means the same as "relatively prime"; hence our assertion may be assumed proved up to $k-1$. Thus for the residue class $\mathfrak{r}\overline{\mathfrak{b}}$ the unique representation (18) holds, hence, by the isomorphism, also (16), and so the assertion follows.

Theorem II. *A module $\mathfrak{M}$ is representable as the least common multiple of k modules $\mathfrak{M}_1, \ldots, \mathfrak{M}_k$ which form a totally relatively prime system if and only if its residue group $\mathfrak{G}$ is reducible into k subgroups $\mathfrak{U}_1, \ldots, \mathfrak{U}_k$. With suitable notation, $\mathfrak{U}_i$ is then isomorphic to the residue group of $\mathfrak{M}_i$.*

§ 6. Finiteness Theorems.

In this paragraph it will be shown that every residue group can be represented as a sum of finitely many irreducible constituents. To do this we first transfer Hilbert's theorem on the module basis.

From now on we use the more special hypotheses formulated in § 2: the multiplication of $\xi_1, \ldots, \xi_n$ among themselves is commutative, and the multiplication law of the polynomials has the form

$$\xi_i a = a_i \xi_i + b_i, \qquad (a_i \neq 0) \quad (i = 1, \ldots, n). \tag{6}$$

It suffices to prove the assertion for modules only. For if $\mathfrak{S}$ is any system of polynomials, we form the module $\mathfrak{M}$ derived from it by adjoining to $\mathfrak{S}$ all polynomials which can be composed from finitely many polynomials $G_1, \ldots, G_k$ of $\mathfrak{S}$ in the form $A_1 G_1 + \cdots + A_k G_k$. If $F_1, \ldots, F_l$ is then a module basis of $\mathfrak{M}$, and if

$$F_i = \sum_j A_{ij} G_j \qquad (i = 1, \ldots, l),$$

then the finitely many polynomials G_j occurring in these representations form a basis of $\mathfrak{S}$.

We prove the module-basis theorem following Gordan (Göttingen Nachrichten 1899). The proof splits into two parts. The first concerns a system $\mathfrak{P}$ of infinitely many power products and remains essentially unchanged because of the commutativity of $\xi_1, \ldots, \xi_n$. The power products

$$P = \xi_1^{a_1} \cdots \xi_n^{a_n},$$

each of which is entered into the system only once, are to be ordered by increasing degree and, within the same degree, in some fixed way. We now consider the subsystem $\mathfrak{Q}$ of $\mathfrak{P}$ consisting of all power products $P = Q$ not divisible by any preceding P. Then no Q is divisible by a later Q either, since later ones are either of higher degree or, in the same degree, different from the earlier ones. On the other hand, every P of $\mathfrak{P}$ is divisible by at least one Q. For suppose, to the contrary, that P_m is the first P divisible by no Q. Then P_m cannot itself be a Q, and so it is divisible by at least one preceding P; since by hypothesis this preceding P is divisible by some Q, P_m itself is divisible by a Q, a contradiction.

We now assert that the number of the Q is finite. Indeed, let

$$Q_0 = \xi_1^{h_1} \cdots \xi_n^{h_n}$$

be the first Q. Since an arbitrary $Q_\lambda = \xi_1^{h_{1\lambda}} \cdots \xi_n^{h_{n\lambda}}$ is not divisible by Q_0, for every Q at least one $h_{\rho\lambda} < h_\rho$ must hold; let λ denote the smallest index for which this is the case. We group all Q for which $h_{\rho\lambda}$ has the same fixed value c_λ into one class $K(c_\lambda)$. Since $c_\lambda < h_\lambda$, the number of these classes is finite. But each class contains only finitely many elements. For if we put a Q in the class $K(c_\lambda)$ equal to $\xi_\lambda^{c_\lambda} Q'$, then the Q' also form a system of power products none of which is divisible by another, and they contain only $n-1$ variables. For $n = 2$ the Q' contain only one variable; their number is then 1, hence finite, so that the number may also be assumed finite for $n-1$ variables. This proves the assertion.

Second, let $\mathfrak{M}$ be a module of polynomials whose terms are lexicographically ordered. Let $\mathfrak{P}$ be the system of the power products which occur in the highest term, each written only once. Then to the system $\mathfrak{Q} = (Q_1, \ldots, Q_\rho)$ there corresponds a system of finitely many polynomials $F_1, \ldots, F_\rho$, obtained by assigning

[3] Continuing this argument shows that among $\mathfrak{M}_1, \ldots, \mathfrak{M}_k$ every two are relatively prime; the converse, that this would imply $\mathfrak{M}_1, \ldots, \mathfrak{M}_k$ to form a totally relatively prime system, is generally valid only in the commutative case; cf. the counterexample 1 in § 12.

to each Q_i some polynomial from $\mathfrak{M}$ whose highest term is equal to b_iQ_i, so that $F_i = b_iQ_i + \cdots$ with lower terms.

Now let $F = bP + \cdots$ be an arbitrary polynomial from $\mathfrak{M}$, where $P = \xi_1^{\beta_1}\cdots\xi_n^{\beta_n}Q_i$. Form the product

$$\xi_1^{\beta_1}\cdots\xi_n^{\beta_n}F_i = \xi_1^{\beta_1}\cdots\xi_n^{\beta_n}(b_iQ_i+\cdots) = c\,\xi_1^{\beta_1}\cdots\xi_n^{\beta_n}Q_i + \cdots = cP + \cdots .$$

Here $c \neq 0$ by (6), and all following terms are lower terms in the lexicographic order, since multiplication is commutative as regards the exponents, up to lower terms.[4] The difference

$$F - \frac{b}{c}\,\xi_1^{\beta_1}\cdots\xi_n^{\beta_n}F_i$$

then contains only lower terms and again belongs to $\mathfrak{M}$. The procedure therefore terminates after finitely many steps, so that $F_1, \ldots, F_\rho$ are recognized as a module basis.

Theorem III. *Every system $\mathfrak{S}$ of infinitely many polynomials has a finite module basis $G_1, \ldots, G_k$, so that every polynomial G from $\mathfrak{S}$ admits a representation $G = A_1G_1 + \cdots + A_kG_k$.*

The following is immediate.

Corollary. *Every system $\mathfrak{S}$ of infinitely many residue classes mod $\mathfrak{M}$ has a basis $\mathfrak{x}_1, \ldots, \mathfrak{x}_k$, so that every $\mathfrak{x}$ from $\mathfrak{S}$ is representable in the form $\mathfrak{x} = A_1\mathfrak{x}_1 + \cdots + A_k\mathfrak{x}_k$.*

Namely, assign to each residue class some representative and consider the system of these representatives together with all polynomials from $\mathfrak{M}$.

We now turn to the proof of the following theorem.

Theorem IV. *Every residue group can be represented as a sum of finitely many irreducible constituents; equivalently, by Theorem II, every module can be represented as the least common multiple of finitely many irreducible modules forming a totally relatively prime system.*

Suppose, to the contrary, that $\mathfrak{G}$ is a sum of infinitely many constituents $\mathfrak{U}_1, \mathfrak{U}_2, \ldots$ (cf. § 5), and let $\mathfrak{a}_1, \mathfrak{a}_2, \ldots$ be the corresponding units, all of which are nonzero by definition. We form the system $\mathfrak{S}$ of all these units; by the corollary to Theorem III it has a finite basis $\mathfrak{a}_{i_1}, \ldots, \mathfrak{a}_{i_v}$ with $i_1 < \cdots < i_v$. Let $\mu > i_v$. Then

$$\mathfrak{a}_\mu = \mathfrak{x}_1\mathfrak{a}_{i_1} + \cdots + \mathfrak{x}_v\mathfrak{a}_{i_v}.$$

This is a linear relation between units, which is excluded by the definition. Hence in every additive decomposition of a residue group an irreducible constituent occurs after finitely many steps; because of commutativity of the decomposition, we can place it at the beginning. Thus, without loss of generality, we may assume that the $\mathfrak{U}_i$ themselves are irreducible. But by the preceding argument only finitely many constituents $\mathfrak{U}_i$ can occur. The assertion is proved.

§ 7. A Case of Unique Decomposition.

Let

$$\mathfrak{G} = \mathfrak{U}_1 + \cdots + \mathfrak{U}_k = \mathfrak{B}_1 + \cdots + \mathfrak{B}_l$$

be a residue group whose constituents $\mathfrak{U}_1, \ldots, \mathfrak{U}_k$, $\mathfrak{B}_1, \ldots, \mathfrak{B}_l$ are irreducible. Then

$$\mathfrak{x}\mathfrak{e} = \mathfrak{x}\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{a}_k = \mathfrak{x}\mathfrak{b}_1 + \cdots + \mathfrak{x}\mathfrak{b}_l, \tag{22}$$

where $\mathfrak{a}_\chi$ is the residue class of $\mathfrak{G}$ isomorphically assigned to the unit class of $\mathfrak{U}_\chi$, and $\mathfrak{b}_\lambda$ is the one assigned to the unit class of $\mathfrak{B}_\lambda$. Replacing $\mathfrak{x}$ by $\mathfrak{x}\mathfrak{b}_\lambda$, with λ a fixed index in the range $1, \ldots, l$, gives

$$\mathfrak{x}\mathfrak{b}_\lambda = \mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_k. \tag{23}$$

This representation of the group $\mathfrak{B}_\lambda$ is unique, but it is not an additive decomposition of $\mathfrak{B}_\lambda$, since the individual residue classes on the right need not belong to $\mathfrak{B}_\lambda$. Since $\mathfrak{b}_\lambda$ does not vanish, not every product $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ can be zero. Without loss of generality, let

$$\mathfrak{b}_\lambda\mathfrak{a}_1 \neq 0, \ldots, \mathfrak{b}_\lambda\mathfrak{a}_\rho \neq 0, \qquad \mathfrak{b}_\lambda\mathfrak{a}_{\rho+1} = 0, \ldots, \mathfrak{b}_\lambda\mathfrak{a}_k = 0.$$

[4] This shows that in place of law (6) one could also use the somewhat less restrictive law

$$\xi_i a = a_i\xi_i + a_{i,i+1}\xi_{i+1} + \cdots + a_{in}\xi_n + b_i \qquad (a_i \neq 0)$$

for some fixed numbering of the ξ.

Then

$$\mathfrak{x}\mathfrak{b}_\lambda = \mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_1 + \cdots + \mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\rho.$$

For a fixed χ in the range $1, \ldots, \rho$, consider the totality of those residue classes $\mathfrak{x}\mathfrak{b}_\lambda$ which are nonzero and for which also $\mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$. The two systems $\mathfrak{x}\mathfrak{b}_\lambda$ and $\mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\chi$ are isomorphic. First, the assignment is one-to-one: from $\mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$ follows, by hypothesis, $\mathfrak{x}\mathfrak{b}_\lambda = 0$, and from the latter, by uniqueness of representation (23), $\mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$. Furthermore, sums correspond to sums, and products with arbitrary polynomials to the corresponding products. Keeping χ fixed, the same argument can be made for every λ with $\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$. For each $\mathfrak{a}_\chi$ there must be at least one such $\mathfrak{b}_\lambda$, since $\mathfrak{a}_\chi = (\mathfrak{b}_1 + \cdots + \mathfrak{b}_l)\mathfrak{a}_\chi \neq 0$.

We first assume that for each χ there is only one λ, and for each λ only one χ, such that $\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$.

This correspondence implies $k = l$; moreover the notation may be chosen so that $\mathfrak{b}_\chi\mathfrak{a}_\chi \neq 0$ and $\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$ for $\lambda \neq \chi$. We then say that the products $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ form a diagonal scheme. In this case, by (23), the unit is $\mathfrak{b}_\chi = \mathfrak{b}_\chi\mathfrak{a}_\chi$, and since

$$\mathfrak{a}_\chi = \mathfrak{b}_1\mathfrak{a}_\chi + \cdots + \mathfrak{b}_l\mathfrak{a}_\chi = \mathfrak{b}_\chi\mathfrak{a}_\chi = \mathfrak{b}_\chi, \tag{24}$$

one has, in general, $\mathfrak{U}_\chi = \mathfrak{B}_\chi$. Thus there is uniqueness of decomposition, and the products $\mathfrak{a}_\chi\mathfrak{b}_\lambda$ form a diagonal scheme.

If, for example, the commutative law holds specifically for the residue classes $\mathfrak{a}_\chi\mathfrak{b}_\lambda$, then the preceding hypothesis is always fulfilled. For in (23) each constituent $\mathfrak{x}\mathfrak{b}_\lambda\mathfrak{a}_\chi = \mathfrak{x}\mathfrak{a}_\chi\mathfrak{b}_\lambda$ is then contained in $\mathfrak{B}_\lambda$, so that (23) gives an additive decomposition of the group $\mathfrak{B}_\lambda$. Since $\mathfrak{B}_\lambda$ is irreducible, all but one of the $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ are zero. By (24), correspondingly, for fixed χ only one $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ is different from zero, since otherwise (24), by interchangeability of $\mathfrak{a}_\chi$ and $\mathfrak{b}_\lambda$, would yield an additive decomposition of $\mathfrak{U}_\chi$, which is supposed to be irreducible.[5]

The same conclusions hold in the case where the conditions $\mathfrak{b}_\lambda\mathfrak{a}_\lambda \neq 0$, $\mathfrak{b}_\lambda\mathfrak{a}_\chi = 0$ for $\lambda \neq \chi$, $\chi = 1, \ldots, k$, and likewise, for each $\mu = 1, \ldots, l$, $\mathfrak{b}_\mu\mathfrak{a}_\chi = 0$, $\mathfrak{b}_\mu\mathfrak{a}_\mu \neq 0$, $\chi = 1, \ldots, \sigma$, are fulfilled not for all, but only for part of the $\mathfrak{b}_\lambda$ $(\lambda = 1, \ldots, \sigma)$. One finds $\mathfrak{U}_\chi = \mathfrak{B}_\chi$ for $\chi = 1, \ldots, \sigma$; and if one puts

$$\mathfrak{U}_{\sigma+1} + \cdots + \mathfrak{U}_k = \mathfrak{U}, \qquad \mathfrak{B}_{\sigma+1} + \cdots + \mathfrak{B}_l = \mathfrak{B},$$

then, also for the units $\mathfrak{a}$ and $\mathfrak{b}$ of $\mathfrak{U}$ and $\mathfrak{B}$, the conditions $\mathfrak{b}\mathfrak{a}_\chi = 0$, $\mathfrak{b}_\chi\mathfrak{a} = 0$ $(\chi = 1, \ldots, \sigma)$, and $\mathfrak{b}\mathfrak{a} \neq 0$ are fulfilled, whence $\mathfrak{U} = \mathfrak{B}$.

§ 8. Additive Decomposition of Completely Reducible Groups into Prime Groups. The Theorem on Isomorphism.

We now consider another case, in which uniqueness does not hold, but isomorphism of the different decompositions does. This is the generalization of the case considered by Loewy.

Definition. *A module is called a prime module if it has no divisor other than itself and the unit module containing all polynomials. The residue group of a prime module is called a prime group.*

Every prime group is, a fortiori, irreducible. There are also infinite prime groups (cf. the definition in § 11 and example § 12, 2).

Let the given residue group $\mathfrak{G}$ now be representable in two ways as a sum of finitely many prime groups $\mathfrak{U}_1, \ldots, \mathfrak{U}_k$ and $\mathfrak{B}_1, \ldots, \mathfrak{B}_l$. A residue group which admits at least one such representation as a sum of prime groups, and the corresponding module $\mathfrak{M}$, will be called completely reducible, following Loewy's terminology.

We consider the products $\mathfrak{a}_\chi\mathfrak{b}_\lambda$ $(\chi = 1, \ldots, k,\ \lambda = 1, \ldots, l)$, and show that either $\mathfrak{a}_\chi\mathfrak{b}_\lambda = 0$, or else $\mathfrak{x}\mathfrak{a}_\chi\mathfrak{b}_\lambda$ runs through the whole group $\mathfrak{B}_\lambda$ as $\mathfrak{x}$ runs through all residue classes of $\mathfrak{G}$. Indeed, the module belonging to $\mathfrak{B}_\lambda$ is

$$(\mathfrak{M}, \mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l) = \mathfrak{M}_{\mathfrak{b}_\lambda};$$

it arises from $\mathfrak{M}$ by adjoining all polynomials of the residue class $\mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l$. This module has the divisor

$$(\mathfrak{M}, \mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l, \mathfrak{a}_\chi\mathfrak{b}_\lambda).$$

But since $\mathfrak{M}_{\mathfrak{b}_\lambda}$ is, by hypothesis, a prime module, the divisor is either equal to $\mathfrak{M}_{\mathfrak{b}_\lambda}$ or to the unit module. In the first case $\mathfrak{a}_\chi\mathfrak{b}_\lambda$ belongs to $\mathfrak{M}_{\mathfrak{b}_\lambda}$; hence there is a relation

$$\mathfrak{a}_\chi\mathfrak{b}_\lambda = \mathfrak{x}(\mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l),$$

[5] The same applies in the noncommutative two-sided case treated by Schmeidler, loc. cit., where the product of every residue class of one subgroup with every residue class of the other subgroup vanishes. Then, since $\mathfrak{b}_\lambda\mathfrak{a}_\chi = \mathfrak{b}_\lambda\mathfrak{a}_\chi\mathfrak{b}_1 + \cdots + \mathfrak{b}_\lambda\mathfrak{a}_\chi\mathfrak{b}_l$ and $\mathfrak{b}_\lambda(\mathfrak{a}_\chi\mathfrak{b}_\mu) = 0$ for $\lambda \neq \mu$, one has $\mathfrak{b}_\lambda\mathfrak{a}_\chi = \mathfrak{b}_\lambda\mathfrak{a}_\chi\mathfrak{b}_\lambda$, hence this class is contained in $\mathfrak{B}_\lambda$; by irreducibility of $\mathfrak{B}_\lambda$, it follows that for only one, and hence for exactly one, χ the product $\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$. In the same way one shows that for every $\mathfrak{a}_\chi$ there is one and only one $\mathfrak{b}_\lambda$ with $\mathfrak{a}_\chi\mathfrak{b}_\lambda \neq 0$, whence $k = l$. Thus the $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ form a diagonal scheme, proving the uniqueness established there.

which gives $\mathfrak{a}_\chi\mathfrak{b}_\lambda = 0$. In the second case there are two classes $\mathfrak{x}_0$ and $\mathfrak{x}_1$ such that

$$\mathfrak{x}_0\mathfrak{a}_\chi\mathfrak{b}_\lambda + \mathfrak{x}_1(\mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l) = \mathfrak{e} = \mathfrak{b}_1 + \cdots + \mathfrak{b}_{\lambda-1} + \mathfrak{b}_{\lambda+1} + \cdots + \mathfrak{b}_l + \mathfrak{b}_\lambda,$$

and therefore, by uniqueness, $\mathfrak{x}_0\mathfrak{a}_\chi\mathfrak{b}_\lambda = \mathfrak{b}_\lambda$. Thus $F\mathfrak{x}_0\mathfrak{a}_\chi\mathfrak{b}_\lambda$, and hence also $\mathfrak{x}\mathfrak{a}_\chi\mathfrak{b}_\lambda$, runs through the whole group $\mathfrak{B}_\lambda$ as F or $\mathfrak{x}$ runs through all polynomials or all residue classes of $\mathfrak{G}$. This proves the assertion.

If $\mathfrak{a}_\chi\mathfrak{b}_\lambda \neq 0$, then $\mathfrak{U}_\chi$ is also isomorphic to $\mathfrak{B}_\lambda$. For by § 7 the totality of all classes $\mathfrak{x}\mathfrak{a}_\chi \neq 0$ for which $\mathfrak{x}\mathfrak{a}_\chi\mathfrak{b}_\lambda \neq 0$ is isomorphic to the system of classes $\mathfrak{x}\mathfrak{a}_\chi\mathfrak{b}_\lambda$; and these exhaust the subgroup $\mathfrak{B}_\lambda$. It remains only to show that the indicated classes $\mathfrak{x}\mathfrak{a}_\chi$ exhaust $\mathfrak{U}_\chi$. To this end, consider all residue classes $\mathfrak{x}\mathfrak{a}_\chi$ for which $\mathfrak{x}\mathfrak{a}_\chi\mathfrak{b}_\lambda = 0$. This system has the property that the sum of two of its classes and the product with an arbitrary polynomial again belong to the system. If all these residue classes are adjoined to the module $\mathfrak{M}_{\mathfrak{a}_\chi}$, one again obtains a module which is a divisor of $\mathfrak{M}_{\mathfrak{a}_\chi}$; thus it is either $\mathfrak{M}_{\mathfrak{a}_\chi}$ or the unit module. In the latter case, however, $\mathfrak{a}_\chi$ itself would be contained in it, so that $\mathfrak{a}_\chi\mathfrak{b}_\lambda = 0$, contrary to the hypothesis. Hence there are no classes $\mathfrak{x}\mathfrak{a}_\chi \neq 0$ for which $\mathfrak{x}\mathfrak{a}_\chi\mathfrak{b}_\lambda = 0$, and the isomorphism of $\mathfrak{U}_\chi$ with $\mathfrak{B}_\lambda$ is proved.

Suppose now, for a fixed χ, that $\mathfrak{a}_\chi\mathfrak{b}_{\lambda_1} \neq 0, \ldots, \mathfrak{a}_\chi\mathfrak{b}_{\lambda_i} \neq 0$ with $i > 1$. Then $\mathfrak{U}_\chi$ is isomorphic to the groups $\mathfrak{B}_{\lambda_1}, \ldots, \mathfrak{B}_{\lambda_i}$, so these groups are mutually isomorphic.

We now show that at least two of the $\mathfrak{U}$ must then also be isomorphic. For if in the products $\mathfrak{b}_\lambda\mathfrak{a}_\chi$ there belonged to each $\mathfrak{b}_\lambda$ only a single $\mathfrak{a}_\chi$ with $\mathfrak{b}_\lambda\mathfrak{a}_\chi \neq 0$, so that $\mathfrak{b}_\lambda = \mathfrak{b}_\lambda\mathfrak{a}_\chi$, then $\mathfrak{B}_\lambda = \mathfrak{U}_\chi$, since $\mathfrak{U}_\chi$, as a prime group, cannot contain a proper subgroup. Thus $k = l$ and, with a correct numbering, $\mathfrak{a}_\chi = \mathfrak{b}_\chi$; but then the scheme $\mathfrak{a}_\chi\mathfrak{b}_\lambda$ would be a diagonal scheme, which is not the case. The following theorem is therefore proved.

Theorem V. *If a completely reducible group $\mathfrak{G}$ is represented in two ways as a sum of prime groups,*

$$\mathfrak{G} = \mathfrak{U}_1 + \cdots + \mathfrak{U}_k = \mathfrak{B}_1 + \cdots + \mathfrak{B}_l,$$

then every group $\mathfrak{U}$ is isomorphic to at least one group $\mathfrak{B}$, and conversely. If every $\mathfrak{U}$ is isomorphic to exactly one $\mathfrak{B}$, then the decompositions are identical. In the other case at least two groups $\mathfrak{U}$ are mutually isomorphic, and likewise at least two groups $\mathfrak{B}$ are mutually isomorphic.

§ 9. Connection Between the Concepts "Isomorphic" and "of the Same Kind".

In this paragraph we show that the concept "of the same kind", known for differential expressions in one variable, when generalized in the natural way, leads to the isomorphism of the associated residue groups.

Definition.[6] *Two modules $\mathfrak{M}$ and $\mathfrak{N}$ are called of the same kind if there are two polynomials P and Q, with P relatively prime to $\mathfrak{M}$ and Q relatively prime to $\mathfrak{N}$, such that for every $M \equiv 0(\mathfrak{M})$ and $N \equiv 0(\mathfrak{N})$*

$$MQ \equiv 0(\mathfrak{N}), \tag{25}$$
$$NP \equiv 0(\mathfrak{M}) \tag{26}$$

hold. Relative primeness here means the existence of two polynomials X and Y such that

$$YQ \equiv 1(\mathfrak{N}), \tag{27}$$
$$XP \equiv 1(\mathfrak{M}) \tag{28}$$

hold.

We now have the following theorem.

Theorem VI. *Two modules are of the same kind if and only if their residue groups are isomorphic.*

1. Let $\mathfrak{A}$ be the residue group of $\mathfrak{M}$, $\mathfrak{B}$ the residue group of $\mathfrak{N}$, and suppose $\mathfrak{A}$ is isomorphic to $\mathfrak{B}$. Let $\mathfrak{a}$ and $\mathfrak{b}$ be the unit classes of $\mathfrak{A}$ and $\mathfrak{B}$. Further, let $Q\mathfrak{b}$ be the class in $\mathfrak{B}$ corresponding to the unit $\mathfrak{a}$ in $\mathfrak{A}$, and let $P\mathfrak{a}$ be the class in $\mathfrak{A}$ corresponding to the unit $\mathfrak{b}$ of $\mathfrak{B}$. We write this assignment as

$$\mathfrak{a} \sim Q\mathfrak{b}, \tag{29}$$
$$P\mathfrak{a} \sim \mathfrak{b}. \tag{30}$$

It follows that

$$0 = M\mathfrak{a} \sim MQ\mathfrak{b}, \qquad \text{that is,} \qquad MQ \equiv 0(\mathfrak{N}),$$

which proves (25). Likewise,

$$NP\mathfrak{a} \sim N\mathfrak{b} = 0, \qquad \text{that is,} \qquad NP \equiv 0(\mathfrak{M}),$$

[6] For differential expressions in one variable, cf. Blumberg's formal definition, loc. cit., p. 25.

which proves (26). Moreover, by (29), $P\mathfrak{a} \sim PQ\mathfrak{b}$; since by (30) $P\mathfrak{a} \sim \mathfrak{b}$ was assumed, uniqueness of the assignment gives $PQ\mathfrak{b} = \mathfrak{b}$, hence $PQ \equiv 1(\mathfrak{N})$, and similarly $QP \equiv 1(\mathfrak{M})$. Thus (27) and (28) are proved, with $X = Q$, $Y = P$.

2. Conversely, suppose $\mathfrak{M}$ and $\mathfrak{N}$ are of the same kind, so that relations (25) through (28) hold. We assign to an arbitrary class $F\mathfrak{a}$ of $\mathfrak{A}$ the class $FQ\mathfrak{b}$ of $\mathfrak{B}$. This assigns to each class in $\mathfrak{A}$ only one class in $\mathfrak{B}$: for from $F\mathfrak{a} = 0$, hence $F = M$, it follows by (25) that $FQ\mathfrak{b} = 0$. Moreover, the whole group $\mathfrak{B}$ is exhausted by this assignment, since the special class $Y\mathfrak{a}$ corresponds to $YQ\mathfrak{b}$, which equals $\mathfrak{b}$ by (27). Thus we have an unambiguous map Φ from $\mathfrak{A}$ onto $\mathfrak{B}$; it has not yet been shown to be one-to-one. Similarly, equations (26) and (28) give an unambiguous map Ψ from $\mathfrak{B}$ onto $\mathfrak{A}$, which again has not yet been shown to be one-to-one.

If either of these two maps is one-to-one, then the isomorphism is proved, since the requirements concerning sums and products are fulfilled. But if both maps Φ and Ψ were not one-to-one, then there would be classes $G\mathfrak{a}$ in $\mathfrak{A}$ corresponding to the class $GQ\mathfrak{b} = 0$ in $\mathfrak{B}$. The classes $FQ\mathfrak{b}$ corresponding to all the other residue classes $F\mathfrak{a}$ would then already exhaust the group $\mathfrak{B}$, and consequently the map Ψ would return the whole group $\mathfrak{A}$ in the classes $FQP\mathfrak{a}$. As we know, there are also nonzero classes in $\mathfrak{B}$ which, under Ψ, correspond to the zero class in $\mathfrak{A}$; we denote by $G_1\mathfrak{a}$ all those classes for which $G_1\mathfrak{a} \neq 0$ but $G_1QP\mathfrak{a} = 0$. All remaining polynomials F have the property that $FQP\mathfrak{a}$ exhausts the group $\mathfrak{A}$. Those polynomials F for which $FQP\mathfrak{a} = G_1\mathfrak{a}$ will be denoted by G_2; such polynomials exist whenever polynomials G_1 exist. For each G_2 one has $G_2QP\mathfrak{a} \neq 0$, but $G_2(QP)^2\mathfrak{a} = 0$. Correspondingly, for every integer ν there are classes $G_\nu\mathfrak{a}$ such that $G_\nu(QP)^{\nu-1}\mathfrak{a} \neq 0$ but $G_\nu(QP)^\nu\mathfrak{a} = 0$.

Now consider the system $\mathfrak{S}$ consisting of all polynomials $G_1, G_2, \ldots$. It has a finite module basis $G_{\lambda_1}, \ldots, G_{\lambda_e}$. Choose ν greater than the largest of the indices $\lambda_1, \ldots, \lambda_e$, which we denote by λ. Then

$$G_\nu = A_1G_{\lambda_1} + \cdots + A_eG_{\lambda_e}.$$

Multiplying by $(QP)^\lambda$ gives

$$G_\nu(QP)^\lambda \equiv 0(\mathfrak{M}),$$

a contradiction to our hypothesis. Hence the isomorphism follows, and therefore, by part 1, the sharper relations (27), (28) hold with $X = Q$, $Y = P$.

By Theorem VI, Theorem V may also be expressed as follows.

Theorem VII.[7] *If a completely reducible module $\mathfrak{M}$ is representable in two ways as the least common multiple of prime modules $\mathfrak{P}_1, \ldots, \mathfrak{P}_k$ and $\mathfrak{D}_1, \ldots, \mathfrak{D}_l$, each forming a totally relatively prime system, then every $\mathfrak{P}$ is of the same kind as at least one $\mathfrak{D}$, and conversely. If every $\mathfrak{P}$ is of the same kind as exactly one $\mathfrak{D}$, then the decompositions are identical. In the other case at least two of the modules $\mathfrak{P}$ are of the same kind as one another, and likewise at least two of the modules $\mathfrak{D}$ are of the same kind as one another.*

[7] Cf. Loewy, pp. 100–101.

§ 10. Existence of Infinitely Many Decompositions of a Group.

If the representation of a completely reducible group as a sum is not unique, then by Theorem V every representation contains at least two mutually isomorphic subgroups. We now consider the subgroup formed from such a system of isomorphic subgroups,

$$\mathfrak{H} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_\alpha,$$

where $\mathfrak{A}_1, \mathfrak{A}_2, \ldots, \mathfrak{A}_\alpha$ are all isomorphic, and assert that $\mathfrak{H}$ then admits infinitely many distinct decompositions of this kind, provided only that the rationality domain P contains infinitely many "constants", that is, quantities a for which $\xi_i a = a\xi_i$. This theorem is valid even when no further hypothesis is imposed on the groups $\mathfrak{A}_i$, not even irreducibility. Thus we have:

Theorem VIII. *If $\mathfrak{G}$ can be represented as a sum of finitely many mutually isomorphic subgroups, then this representation is possible in infinitely many distinct ways, provided only that the rationality domain P contains infinitely many constants. Here constants mean those quantities a of the domain for which $\xi_i a = a\xi_i$.*

Proof. Let

$$\mathfrak{G} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_\alpha,$$

and let $\mathfrak{A}_i$ be isomorphic to $\mathfrak{A}_\varkappa$ for $i, \varkappa = 1, \ldots, \alpha$ $(\alpha \geq 2)$. In order to specify a definite isomorphic assignment, we first single out one subgroup, say $\mathfrak{A}_1$. Let the isomorphism between $\mathfrak{A}_1$ and $\mathfrak{A}_i$ be fixed by

$$a_1 \sim t_{1i}a_i, \qquad a_i \sim t_{i1}a_1, \qquad t_{11}a_1 = a_1 \quad (i, \varkappa = 1, \ldots, \alpha), \tag{31}$$

so that, in the group $\mathfrak{A}_1$, each residue class is assigned to itself. Since $Ma_1 = 0$ and $Ma_i = 0$, it follows that $Mt_{1i}a_i = 0$ and $Mt_{i1}a_1 = 0$; hence the classes $t_{1i}a_i$ and $t_{i1}a_1$, analogously to units, admit multiplication by classes, not merely by polynomials. It follows from the definition of isomorphism that, for two assigned classes $\mathfrak{y}$ and $\bar{\mathfrak{y}}$ admitting this class multiplication, the classes $\mathfrak{r}\mathfrak{y}$ and $\mathfrak{r}\bar{\mathfrak{y}}$ are again assigned to each other. Therefore the relations $a_r a_s = 0$ $(r \neq s)$ $(r = 1, \ldots, \alpha;\ s = 1, \ldots, \alpha)$ give, for $s = 1$,

$$\left\{\begin{array}{lll} & a_r t_{1\varkappa} a_\varkappa = 0 & (\varkappa = 1, \ldots, \alpha,\ r \neq 1), \\ \text{and for } s > 1 & a_r t_{s1} a_1 = 0 & (r \neq s), \\ \quad\text{hence also} & t_{1i}a_i = a_1 t_{1i} a_i, \quad t_{i1}a_1 = a_i t_{i1} a_1. & \end{array}\right. \tag{32}$$

Furthermore, from (31), generalizing (27) and (28), the uniqueness of the assignment gives

$$t_{1i}t_{i1}a_1 = a_1, \qquad t_{\varkappa 1}t_{1\varkappa}a_\varkappa = a_\varkappa. \tag{33}$$

Let the isomorphism between $\mathfrak{A}_i$ and $\mathfrak{A}_\varkappa$ now be the relation obtained by composing (31); such a stipulation is necessary, since the individual groups may admit isomorphisms into themselves. From (31), (32), and (33) we therefore get

$$a_i \sim t_{i1}t_{1\varkappa}a_\varkappa = t_{i\varkappa}a_\varkappa, \qquad t_{i\varkappa}a_i = a_i, \qquad a_r t_{s\varkappa} a_\varkappa = 0 \quad (r \neq s), \tag{34}$$

where $t_{i1}t_{1\varkappa} = t_{i\varkappa}$ has been put. From (33) and (34) one obtains further

$$t_{i\varkappa}t_{\varkappa l}a_l = t_{i1}t_{1\varkappa}t_{\varkappa 1}t_{1l}a_l = t_{i1}t_{1l}a_l = t_{il}a_l;$$

thus the singling out of the group $\mathfrak{A}_1$ is again removed. The isomorphic relation between $\mathfrak{A}_i$ and $\mathfrak{A}_\varkappa$ remains the same, whichever group $\mathfrak{A}_\varkappa$ is used in place of $\mathfrak{A}_1$ for the definition. Hence the $t_{i\varkappa}$ are classes which, in summary, satisfy the relations

$$Mt_{i\varkappa}a_\varkappa = 0; \quad a_r t_{i\varkappa}a_\varkappa = 0 \ (r \neq i); \quad t_{i\varkappa}t_{\varkappa l}a_l = t_{il}a_l; \quad t_{\varkappa i}a_i = t_{\varkappa i}t_{i\varkappa}a_\varkappa = a_\varkappa. \tag{35}$$

These relations say precisely that the modules $\mathfrak{M}_i$ and $\mathfrak{M}_\varkappa$ belonging to $\mathfrak{A}_i$ and $\mathfrak{A}_\varkappa$ are of the same kind (cf. the definition in § 9); in place of the P and Q there we have here the classes $t_{i\varkappa}$ and $t_{\varkappa i}$, and the first two relations (35), taken together, correspond to formula (25) or (26). Conversely, these relations imply the isomorphism by Theorem VI.[20]

From the classes $t_{i\varkappa}a_\varkappa$ we now compose classes b_ρ which will again turn out to be units of groups $\mathfrak{B}_\rho$, whence a representation

$$\mathfrak{G} = \mathfrak{B}_1 + \cdots + \mathfrak{B}_\alpha$$

[20] And in fact without applying the finiteness theorem, since the formulas already display respectively a map $\varphi_{i\varkappa}$ and its inverse $\varphi_{\varkappa i}$. For from $ra_i = ra_i t_{i\varkappa}a_\varkappa t_{\varkappa i}a_i$ it follows that $ra_i \neq 0$ also implies $ra_i t_{i\varkappa}a_\varkappa \neq 0$.

will follow. Let $\lambda_{i\varkappa}$ be constants whose determinant $|\lambda_{i\varkappa}|$ is nonzero, and let $\lambda^{\varkappa r}$ be the corresponding cofactors divided by $|\lambda_{i\varkappa}|$, so that the relations

$$\begin{cases} \displaystyle\sum_{r=1}^{\alpha} \lambda_{ri}\lambda^{r\varkappa} = \delta_{i\varkappa} = \begin{cases} 0, & i \neq \varkappa, \\ 1, & i = \varkappa, \end{cases} \\ \displaystyle\sum_{r=1}^{\alpha} \lambda_{ir}\lambda^{\varkappa r} = \delta_{i\varkappa} \end{cases} \tag{36}$$

hold. We then define

$$b_\rho = \sum_{i,\varkappa} \lambda_{\rho i}\lambda^{\rho\varkappa} t_{i\varkappa} a_\varkappa \qquad (\rho = 1, \ldots, \alpha). \tag{37}$$

From (35), (36), and (37) one obtains $Mb_\rho = 0$ and

$$\sum_{\rho=1}^{\alpha} b_\rho = \sum_{i,\varkappa,\rho} \lambda_{\rho i}\lambda^{\rho\varkappa} t_{i\varkappa} a_\varkappa = \sum_{i,\varkappa} \delta_{i\varkappa} t_{i\varkappa} a_\varkappa = \sum_i t_{ii} a_i = \sum_i a_i = e, \tag{38}$$

and also

$$\begin{aligned} b_\rho b_\sigma &= \sum_{\substack{i,\varkappa \\ \mu,\nu}} \lambda_{\rho i}\lambda^{\rho\varkappa}\lambda_{\sigma\mu}\lambda^{\sigma\nu} t_{i\varkappa} a_\varkappa t_{\mu\nu} a_\nu \\ &= \sum_{i,\mu,\nu} \lambda_{\rho i}\lambda^{\rho\mu}\lambda_{\sigma\mu}\lambda^{\sigma\nu} t_{i\nu} a_\nu = \delta_{\rho\sigma} \sum_{i,\nu} \lambda_{\rho i}\lambda^{\sigma\nu} t_{i\nu} a_\nu = \begin{cases} 0 & (\rho \neq \sigma), \\ b_\rho & (\rho = \sigma), \end{cases} \end{aligned}$$

$(\rho, \sigma = 1, \ldots, \alpha)$. Thus the representation $\mathfrak{r}e = \mathfrak{r}b_1 + \cdots + \mathfrak{r}b_\alpha$ is an additive decomposition of the group $\mathfrak{G}$. The groups $\mathfrak{B}_\rho$ are distinct from the groups $\mathfrak{A}_\sigma$ as soon as the $\lambda_{i\varkappa}$ do not merely form a diagonal matrix. For if we form $a_\sigma b_\rho$, then by (35) we get

$$a_\sigma b_\rho = \sum_{i,\varkappa} \lambda_{\rho i}\lambda^{\rho\varkappa} a_\sigma t_{i\varkappa} a_\varkappa = \lambda_{\rho\sigma} \sum_\varkappa \lambda^{\rho\varkappa} t_{\sigma\varkappa} a_\varkappa \neq 0 \quad \text{for } \lambda_{\rho\sigma} \neq 0,$$

since otherwise each individual summand in the last sum would have to vanish, which is not the case. Thus if, for fixed σ, there are at least two ρ with $\lambda_{\rho\sigma} \neq 0$, then $\mathfrak{A}_\sigma$ cannot be identical with any $\mathfrak{B}_\rho$.

But the groups $\mathfrak{B}_\rho$ are also mutually isomorphic and isomorphic to the groups $\mathfrak{A}_\sigma$. We first show the existence of classes $r_{\rho\sigma}a_\sigma$ and $r^{\sigma\rho}b_\rho$ which mediate the isomorphism of $\mathfrak{A}_\sigma$ and $\mathfrak{B}_\rho$ $(\rho, \sigma = 1, \ldots, \alpha)$. Indeed, we show that for the classes

$$r_{\rho\sigma}a_\sigma = \sum_i \lambda_{\rho i} t_{i\sigma} a_\sigma, \qquad r^{\sigma\rho} = \sum_\varkappa \lambda^{\rho\varkappa} t_{\sigma\varkappa} a_\varkappa = r^{\sigma\rho} b_\rho$$

the relations "of the same kind" are fulfilled between the modules

$$(\mathfrak{M}, b_1 + \cdots + b_{\rho-1} + b_{\rho+1} + \cdots + b_\alpha)$$

and

$$(\mathfrak{M}, a_1 + \cdots + a_{\sigma-1} + a_{\sigma+1} + \cdots + a_\alpha):$$

$$\begin{cases} b_\varkappa r_{\rho\sigma} a_\sigma = 0 \quad (\varkappa \neq \rho), \quad r^{\sigma\rho} r_{\rho\sigma} a_\sigma = a_\sigma, \\ a_\varkappa r^{\sigma\rho} b_\rho = 0 \quad (\varkappa \neq \sigma), \quad r_{\rho\sigma} r^{\sigma\rho} b_\rho = b_\rho, \end{cases} \tag{39}$$

from which, as in (35), the isomorphism follows through the assignment

$$b_\rho \sim r_{\rho\sigma} a_\sigma, \qquad a_\sigma \sim r^{\sigma\rho} b_\rho. \tag{40}$$

Indeed,

$$\begin{aligned} b_\varkappa r_{\rho\sigma} a_\sigma &= \sum_{\mu,\nu,i} \lambda_{\varkappa\mu}\lambda^{\varkappa\nu} t_{\mu\nu} a_\nu \lambda_{\rho i} t_{i\sigma} a_\sigma \\ &= \sum_{\mu,i} \lambda_{\varkappa\mu}\lambda^{\varkappa i}\lambda_{\rho i} t_{\mu\sigma} a_\sigma = \delta_{\varkappa\rho} \sum_\mu \lambda_{\varkappa\mu} t_{\mu\sigma} a_\sigma = \delta_{\varkappa\rho} r_{\rho\sigma} a_\sigma = 0 \quad (\varkappa \neq \rho), \end{aligned}$$

and further

$$r^{\sigma\rho} r_{\rho\sigma} a_\sigma = \sum_{\nu,i} \lambda^{\rho\nu} t_{\sigma\nu} a_\nu \lambda_{\rho i} t_{i\sigma} a_\sigma = \sum_i \lambda^{\rho i}\lambda_{\rho i} a_\sigma = a_\sigma.$$

Conversely, in the same way one finds

$$a_\varkappa r^{\sigma\rho} b_\rho = a_\varkappa \sum_\mu \lambda^{\rho\mu} t_{\sigma\mu} a_\mu = 0 \quad (\varkappa \neq \sigma),$$

and

$$r_{\rho\sigma} r^{\sigma\rho} b_\rho = \sum_{i,\mu} \lambda_{\rho i} t_{i\sigma} a_\sigma \lambda^{\rho\mu} t_{\sigma\mu} a_\mu = \sum_{i,\mu} \lambda_{\rho i} \lambda^{\rho\mu} t_{i\mu} a_\mu = b_\rho.$$

From the relations of the same kind just proved, the isomorphism of $\mathfrak{A}_\sigma$ and $\mathfrak{B}_\rho$ follows (again without the finiteness theorem). The isomorphism between $\mathfrak{B}_\rho$ and $\mathfrak{B}_\tau$ is obtained from this by composition; we record the formulas. Put

$$\ell_{\rho\tau} b_\tau = r_{\rho\sigma} r^{\sigma\tau} b_\tau = \sum_{i,j} \lambda_{\rho i} t_{i\sigma} a_\sigma \lambda^{\tau j} t_{\sigma j} a_j = \sum_{i,j} \lambda_{\rho i} \lambda^{\tau j} t_{ij} a_j;$$

then the assignment is $b_\rho \sim \ell_{\rho\tau} b_\tau$. Here too the relations of the same kind can again be verified formally; if the $\ell_{\rho\tau}$ are put in place of the $t_{i\varkappa}$, they satisfy (35). Indeed,

$$b_\sigma \ell_{\rho\tau} b_\tau = b_\sigma r_{\rho\sigma} r^{\sigma\tau} b_\tau = 0 \quad (\sigma \neq \rho),$$

$$\ell_{\rho\sigma} \ell_{\sigma\tau} b_\tau = r_{\rho\varkappa} r^{\varkappa\sigma} r_{\sigma\lambda} r^{\lambda\tau} b_\tau = r_{\rho\varkappa} a_\varkappa r^{\varkappa\tau} b_\tau = \ell_{\rho\tau} b_\tau.$$

This proves Theorem VIII, since one need only choose the $\lambda_{i\varkappa}$ in infinitely many ways so that they form no diagonal matrix and also are not transformed into one another by diagonal matrices.

A consequence of Theorem VIII, in connection with Theorem V or VII, is:

Theorem IX.[21] *Let the module $\mathfrak{M}$ be completely reducible and the least common multiple of prime modules $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ which form a totally relatively prime system, and suppose the rationality domain P contains infinitely many constants. Then the necessary and sufficient condition that $\mathfrak{M}$ be representable in infinitely many distinct ways as the least common multiple of prime modules is that at least two of the modules $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ be of the same kind.*

We shall now derive general criteria for a module $\mathfrak{M}$ to be divisible by infinitely many prime modules, and first prove the following:

Auxiliary theorem.[22] *If a completely reducible module $\mathfrak{M} = [\mathfrak{P}_1, \dots, \mathfrak{P}_k]$ is divisible by a prime module $\mathfrak{P}'$ different from all the $\mathfrak{P}_i$, then at least two of the $\mathfrak{P}_i$ are of the same kind.*

We first select from $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ a totally relatively prime system by successively deleting every one of these prime modules that is contained in the least common multiple of those not yet deleted. Hence assume now that $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ themselves form a totally relatively prime system. Let a' be a class of $\mathfrak{M}$ corresponding to the unit class modulo $\mathfrak{P}'$. Then

$$Fa' = Fa'a_1 + \dots + Fa'a_k$$

for every polynomial F. At least two of the products $a'a_i$ are nonzero, say $a'a_1$ and $a'a_2$; for if, for instance, only $a'a_1 \neq 0$, then $Fa' = Fa'a_1$, and the residue classes Fa' would form a subgroup of $\mathfrak{A}_1$, hence, since $\mathfrak{A}_1$ is a prime group, would be identical with $\mathfrak{A}_1$, so that the modules $\mathfrak{P}'$ and $\mathfrak{P}_1$ would also coincide. By the reasoning of § 8, it follows that $\mathfrak{A}'$ is isomorphic to $\mathfrak{A}_1$ and $\mathfrak{A}_2$.

Let $\mathfrak{M}$ now be an arbitrary module. We consider the least common multiple $\mathfrak{N}$ of all prime modules by which $\mathfrak{M}$ is divisible.[23] Two cases are possible. Either $\mathfrak{N}$ is not representable as the least common multiple of finitely many prime modules; in this case $\mathfrak{N}$, and hence also $\mathfrak{M}$, has infinitely many prime modules as divisors. Or $\mathfrak{N}$ is representable as the least common multiple of finitely many prime modules; then $\mathfrak{N}$ is called the greatest completely reducible module of $\mathfrak{M}$ (following a corresponding construction of Loewy). If among these finitely many prime modules two are of the same kind, then by Theorem IX the module $\mathfrak{N}$, and therefore $\mathfrak{M}$, is divisible by infinitely many prime modules. Conversely, if the latter is the case, then there is at least one prime divisor of $\mathfrak{N}$ different from all the finitely many modules whose least common multiple represents $\mathfrak{N}$. By the auxiliary theorem, $\mathfrak{N}$, hence $\mathfrak{M}$, has two prime divisors of the same kind. Thus we have proved:

Theorem X. *If the rationality domain P contains infinitely many constants, then the necessary and sufficient condition that a module be divisible by infinitely many distinct prime modules is that either no greatest completely reducible module exists, or, if such a module exists, that it be divisible by at least two distinct prime modules of the same kind.*

[21] Cf. Loewy, pp. 106–107.

[22] Cf. Loewy, p. 96.

[23] If there are infinitely many of these, then by Theorem IV no additive decomposition of the residue group can correspond to this representation. Cf. also the example § 12, 4.

§ 11. Relations to Systems of Differential Equations and Their Integrals.

By virtue of the finiteness theorem, to every module of differential expressions there can be assigned as a basis a system of finitely many differential expressions, so that the totality of simultaneous integrals of all differential equations obtained by setting a differential expression of the module equal to zero coincides with the totality of integrals of the finite system of differential equations obtained by setting the basis differential expressions equal to zero. This gives the connection with the theory of systems of linear partial differential equations; we pursue this connection a little further in the case where the residue group of the module is "finite".

Definition. *A residue group is called finite if there is a number μ, the order of the group, such that between any $\mu + 1$ residue classes there is a linear relation with coefficients from P, while there is at least one system of μ residue classes between which no relation exists, and which we shall call a linear basis of the residue group. In the other case the residue group is called infinite.*

The sum of two finite residue groups has finite order, namely the sum of their orders. The sum of two infinite groups, or of a finite and an infinite group, is an infinite group. Examples of finite groups are furnished by all residue groups of modules in one variable; but for several variables the order can also be finite, for instance for the module with basis ξ_1^2, ξ_2^3, where all residue classes can be linearly composed from those of $1, \xi_1, \xi_2, \xi_1\xi_2$. For examples of infinite groups see § 12.

We now have the following:

Theorem XI. *If a module has a finite residue group, then every system of differential equations obtained by setting a basis of the module equal to zero admits exactly as many linearly independent integrals as the order of its residue group indicates. Conversely, for a given system of finitely many linear partial differential equations that possess only a finite number of linearly independent integrals, this number is equal to the order of the residue group of the module generated by the differential expressions.*

Here the rationality domain P consists of analytic functions of n variables $x_1, \dots, x_n$, with the occurring singularities, inside a certain n-dimensional domain, forming at most manifolds of lower dimensions, so that there are arbitrarily many points with regular n-dimensional neighborhoods.

Let $\mathfrak{M}$ be a module with finite residue group. We show that there is a linear basis of power products with the property that *when an arbitrary residue class is expressed through this basis, only powers of finitely many different quantities from P can occur in the denominator.*

For this purpose we order all power products $\xi_1^{a_1} \cdots \xi_n^{a_n}$ lexicographically and take as representatives $Q_1, \dots, Q_m$ of the residue classes all those that are linearly independent modulo $\mathfrak{M}$ of the preceding ones. By hypothesis there are only finitely many such products. All the others can be expressed linearly through these.

Let $\mathfrak{S}$ be the system of all power products not admitted into the linear basis. Then, by the first part of the proof of Theorem III, $\mathfrak{S}$ has a finite subsystem $\mathfrak{T}$ of power products such that every power product from $\mathfrak{S}$ is divisible by at least one from $\mathfrak{T}$. Let $P_1, \dots, P_\rho$ be the power products of $\mathfrak{T}$, and let $M_1 = 0, \dots, M_\rho = 0$ $(\mathfrak{M})$ be the relations by means of which $P_1, \dots, P_\rho$ are each expressed modulo $\mathfrak{M}$ by a linear combination of lower power products; for example,

$$M_i = b_i P_i + \text{lower terms}.$$

If P is an arbitrary power product from $\mathfrak{S}$, hence of the form

$$\xi_1^{h_1} \cdots \xi_n^{h_n} P_i,$$

then

$$\begin{aligned} \xi_1^{h_1} \cdots \xi_n^{h_n} M_i &= b_i \xi_1^{h_1} \cdots \xi_n^{h_n} P_i + \text{lower terms} \\ &= b_i P + \text{lower terms}. \end{aligned}$$

Assume now that it has already been proved for all power products lower than P that only powers of $b_1, \dots, b_\rho$ occur in denominators, while the numerators are formed from the coefficients of the M_i by addition, multiplication, and differentiation. Then the same holds for P itself. This assumption is permitted, since it is fulfilled for P_1, the first element from $\mathfrak{S}$ and $\mathfrak{T}$.

By Theorem III, moreover, $M_1, \dots, M_\rho$ is a module basis of $\mathfrak{M}$. Now consider a system of values $\beta_1, \dots, \beta_n$ of $x_1, \dots, x_n$ for which $b_1 \neq 0, \dots, b_\rho \neq 0$ and the remaining coefficients of the M_i are regular, that is, a regular point of the system of differential equations $M_1 = 0, \dots, M_\rho = 0$. The power products $Q_i = \xi_1^{h_{i1}} \cdots \xi_n^{h_{in}}$ $(i = 1, \dots, m)$ correspond to the partial derivatives

$$\frac{\partial^{h_{i1} + \cdots + h_{in}}}{\partial x_1^{h_{i1}} \cdots \partial x_n^{h_{in}}} y,$$

for which we prescribe arbitrary constant values $(\gamma_1, \dots, \gamma_m)$ at the point $(\beta_1, \dots, \beta_n)$. Then the differential equations $M_i = 0$ determine uniquely, as finite values at this point, the values of all higher derivatives of y, since only the b_i occur in denominators; and it is known that this implies the existence of m linearly independent analytic integrals.

On the other hand, under our hypotheses there cannot exist more than m linearly independent integrals, since otherwise at a regular point one could prescribe arbitrarily the values of $m+1$ suitably chosen derivatives, contrary to the fact that linear dependencies modulo $\mathfrak{M}$ must exist among any $m+1$ power products.

Conversely, if a module has an infinite residue group, the same method shows that there are infinitely many linearly independent integrals; therefore, in the case of only m linearly independent integrals, the residue group is finite, and its order is, by the above, equal to m.

To close this paragraph we show that, for two modules of differential expressions, the concept of isomorphism, or equivalently of being of the same kind, has for the integrals of the associated systems of differential equations the same meaning as Poincaré's concept of kind for one variable (cf. Blumberg, loc. cit., Appendix I).

Theorem XII. *If two modules $\mathfrak{M}$ and $\mathfrak{N}$ of differential expressions are of the same kind, then there are two differential expressions P and Q such that $P(y)$ and $Q(z)$ run through all integrals of $\mathfrak{N}$ and $\mathfrak{M}$, respectively, when y runs through all integrals of $\mathfrak{M}$ and z through all integrals of $\mathfrak{N}$.*

By hypothesis, namely (by § 9), for two suitably chosen polynomials P and Q the relations

$$MQ = 0(\mathfrak{N}), \qquad PQ = 1(\mathfrak{N}),$$
$$NP = 0(\mathfrak{M}), \qquad QP = 1(\mathfrak{M})$$

hold. Thus, since $NP(y) = 0$, the function $P(y)$ is an integral of $\mathfrak{N}$. Likewise $Q(z)$ is an integral of $\mathfrak{M}$. Since $PQ(z) = z$ and $QP(y) = y$, $P(y)$ runs through all integrals of $\mathfrak{N}$, and $Q(z)$ all integrals of $\mathfrak{M}$. In this way, to each nonzero y there corresponds a nonzero z, and conversely.

§ 12. Examples.

1. As a first example we consider an ordinary linear differential equation of second order whose coefficients are represented as symmetric functions of the integrals. The rationality domain P consists here of all rational functions of the integrals and their derivatives; we restrict ourselves to a sufficiently small neighborhood of a regular point of the differential equation. We define the module $\mathfrak{M}$ as the totality of all differential expressions that have y_1 and y_2 as integrals, and for which we again write polynomials in one indeterminate ξ. The module basis is then one-membered, and is given by the polynomial

$$M = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \xi^2 - \begin{vmatrix} y_1 & y_2 \\ y_1'' & y_2'' \end{vmatrix} \xi + \begin{vmatrix} y_1' & y_2' \\ y_1'' & y_2'' \end{vmatrix},$$

which is uniquely determined up to a factor from P. Now

$$\mathfrak{M} = [\mathfrak{M}_1, \mathfrak{M}_2], \tag{41}$$

where $\mathfrak{M}_i$ consists of all differential expressions having the integral y_i, and whose basis is determined up to a factor from P by

$$M_i = y_i\xi - y_i' \qquad (i = 1, 2).$$

Indeed, $\mathfrak{M}$ is divisible by $\mathfrak{M}_1$ and $\mathfrak{M}_2$, since it vanishes for the integrals y_1 and y_2, and hence is divisible by $[\mathfrak{M}_1, \mathfrak{M}_2]$; and since the order of the basis polynomial of $[\mathfrak{M}_1, \mathfrak{M}_2]$ must be at least 2, one has $\mathfrak{M} = [\mathfrak{M}_1, \mathfrak{M}_2]$. Further, $\mathfrak{M}_1$ and $\mathfrak{M}_2$ are relatively prime, because

$$\frac{y_2}{D}(y_1\xi - y_1') - \frac{y_1}{D}(y_2\xi - y_2') = 1, \qquad D = D(y) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}. \tag{42}$$

They are also prime modules, since their residue group has order 1.

But besides y_1 and y_2, $\mathfrak{M}$ also has all integrals

$$z_1 = \alpha_{11}y_1 + \alpha_{12}y_2, \qquad z_2 = \alpha_{21}y_1 + \alpha_{22}y_2, \qquad A = |\alpha_{ik}| \neq 0,$$

and hence is identical with the least common multiple of the two relatively prime modules $\mathfrak{N}_1$ and $\mathfrak{N}_2$, which have respectively the integrals z_1 and z_2, with basis polynomials

$$N_i = z_i\xi - z_i' = (\alpha_{i1}y_1 + \alpha_{i2}y_2)\xi - (\alpha_{i1}y_1' + \alpha_{i2}y_2').$$

Thus $\mathfrak{M}$ admits infinitely many distinct representations as the least common multiple of relatively prime prime modules; by Theorem VII the two modules $\mathfrak{N}_1$ and $\mathfrak{N}_2$ are therefore of the same kind as one another and as all $\mathfrak{M}_1$ and $\mathfrak{M}_2$. Indeed, the residue groups of all these modules have order 1, and so are isomorphic.

We shall now write the additive decomposition of the residue group $\mathfrak{G} = \mathfrak{A}_1 + \mathfrak{A}_2$ corresponding to the representation (41). To do this we start, as in § 4, from relation (42), which expresses the relative primeness of $\mathfrak{M}_1$ and $\mathfrak{M}_2$. For the units a_1 and a_2 we obtain the residue classes generated by the polynomials

$$-\frac{y_1}{D}(y_2\xi - y_2') \quad \text{and} \quad \frac{y_2}{D}(y_1\xi - y_1').$$

Thus

$$\mathfrak{M}_1 = \left(\mathfrak{M}, \frac{y_2}{D}(y_1\xi - y_1')\right), \qquad \mathfrak{M}_2 = \left(\mathfrak{M}, -\frac{y_1}{D}(y_2\xi - y_2')\right).$$

If we now form in the same way the units b_1 and b_2 corresponding to the decomposition $\mathfrak{M} = [\mathfrak{N}_1, \mathfrak{N}_2]$, replacing y_i by z_i, we get

$$-\frac{z_1}{D(z)}(z_2\xi - z_2') = -\frac{\alpha_{11}y_1 + \alpha_{12}y_2}{AD(y)}((\alpha_{21}y_1 + \alpha_{22}y_2)\xi - (\alpha_{21}y_1' + \alpha_{22}y_2')),$$

$$\frac{z_2}{D(z)}(z_1\xi - z_1') = \frac{\alpha_{21}y_1 + \alpha_{22}y_2}{AD(y)}((\alpha_{11}y_1 + \alpha_{12}y_2)\xi - (\alpha_{11}y_1' + \alpha_{12}y_2')),$$

and therefore

$$\begin{cases} b_1 = \dfrac{\alpha_{11}y_1 + \alpha_{12}y_2}{y_1}\left(\dfrac{\alpha_{22}}{A}\right)a_1 + \dfrac{\alpha_{11}y_1 + \alpha_{12}y_2}{y_2}\left(-\dfrac{\alpha_{21}}{A}\right)a_2 = \dfrac{z_1}{y_1}\alpha^{11}a_1 + \dfrac{z_1}{y_2}\alpha^{12}a_2, \\ b_2 = \dfrac{\alpha_{21}y_1 + \alpha_{22}y_2}{y_1}\left(-\dfrac{\alpha_{12}}{A}\right)a_1 + \dfrac{\alpha_{21}y_1 + \alpha_{22}y_2}{y_2}\left(\dfrac{\alpha_{11}}{A}\right)a_2 = \dfrac{z_2}{y_1}\alpha^{21}a_1 + \dfrac{z_2}{y_2}\alpha^{22}a_2. \end{cases} \tag{43}$$

Here (α^{ik}) is the inverse matrix of (α_{ik}).

Conversely, exchanging z_i with y_i gives

$$\begin{cases} a_1 = \dfrac{y_1}{z_1}\alpha_{11}b_1 + \dfrac{y_1}{z_2}\alpha_{12}b_2, \\ a_2 = \dfrac{y_2}{z_1}\alpha_{21}b_1 + \dfrac{y_2}{z_2}\alpha_{22}b_2. \end{cases} \tag{44}$$

These formulas display the isomorphism of the groups $\mathfrak{A}$ and $\mathfrak{B}$. Namely, by our general considerations, if $a_1 b_\sigma \neq 0$ the corresponding assignment is $a_1 \sim a_1 b_\sigma$, and hence by (44)

$$a_1 \sim \frac{y_1}{z_\sigma}\alpha_{1\sigma}b_\sigma, \qquad \text{provided } \alpha_{1\sigma} \neq 0,$$

and by (43) likewise

$$b_\sigma \sim \frac{z_\sigma}{y_2}\alpha^{\sigma 2}a_2, \qquad \text{provided } \alpha^{\sigma 2} \neq 0,$$

so that

$$\frac{y_1}{z_\sigma}\alpha_{1\sigma}b_\sigma \sim \frac{y_1}{z_\sigma}\alpha_{1\sigma}\frac{z_\sigma}{y_2}\alpha^{\sigma 2}a_2 = \frac{y_1}{y_2}\alpha_{1\sigma}\alpha^{\sigma 2}a_2.$$

Hence

$$a_1 \sim \alpha_{1\sigma}\alpha^{\sigma 2}\frac{y_1}{y_2}a_2 \quad \text{for } \alpha_{1\sigma}\alpha^{\sigma 2} \neq 0, \qquad \frac{1}{\alpha_{1\sigma}\alpha^{\sigma 2}}\frac{y_2}{y_1}a_1 \sim a_2,$$

which gives the inverse. If (α_{ik}) is not a diagonal matrix, this condition $\alpha_{1\sigma}\alpha^{\sigma 2} \neq 0$ is always fulfilled for some σ. The case of a diagonal matrix must be excluded, since then the modules are identical. If, for example, only $\alpha_{12} = 0$, then $\mathfrak{A}_1 = \mathfrak{B}_1$, but not $\mathfrak{A}_2 = \mathfrak{B}_2$. Thus from $\mathfrak{A}_1 + \mathfrak{A}_2 = \mathfrak{B}_1 + \mathfrak{B}_2$ and $\mathfrak{A}_1 = \mathfrak{B}_1$ it does not follow that $\mathfrak{A}_2 = \mathfrak{B}_2$ (cf. note 12). The three modules $\mathfrak{M}_1$, $\mathfrak{M}_2$, and $\mathfrak{N}_2$ are then pairwise relatively prime, but do not form a totally relatively prime system, since $[\mathfrak{M}_1, \mathfrak{M}_2]$ is divisible by $\mathfrak{N}_2$ (cf. note 15).

Putting further

$$t_{12} = \alpha_{1\sigma}\alpha^{\sigma 2}\frac{y_1}{y_2}a_2, \qquad t_{21} = \frac{1}{\alpha_{1\sigma}\alpha^{\sigma 2}}\frac{y_2}{y_1}a_1,$$

one verifies by direct calculation that the relations (35) are fulfilled.

2. Every module in more than one variable whose basis consists of a single polynomial gives an example of an infinite group. The following somewhat more complicated example is meant to show that infinite groups can also occur for multi-membered modules, that is, for modules whose basis contains more than one polynomial.[24]

$$\mathfrak{M} = [(\xi + x\eta), (\xi + 1)] = [(P), (Q)] \qquad \left(\xi = \frac{\partial}{\partial x},\ \eta = \frac{\partial}{\partial y}\right).$$

The multiplication rule is that for differential expressions. The residue group is here infinite because the residue groups of $(\xi + x\eta)$ and $(\xi + 1)$ are both infinite, whereas the modules $(\xi + x\eta)$ and $(\xi + 1)$ are relatively prime, since

$$1 = (-x\xi - x^2\eta + 1)Q + (x\xi + x - 1)P.$$

It is now very easy to show that $\mathfrak{M}$ possesses no polynomial of the second degree: one sets

$$(u_1\xi + u_2\eta + u_3)(\xi + x\eta) = (v_1\xi + v_2\eta + v_3)(\xi + 1),$$

which gives $u_1 = u_2 = u_3 = v_1 = v_2 = v_3 = 0$. If one next looks for polynomials of the third degree in $\mathfrak{M}$, one finds two linearly independent ones, for example

$$(\xi^2 + x\xi\eta + \xi + (2 + x)\eta)Q = (\xi^2 + 2\xi + 1)P$$

and

$$(\xi^2 + 2x\xi\eta + x^2\eta^2 + \eta)Q = (\xi^2 + x\xi\eta + \xi + (x - 1)\eta)P.$$

If $\mathfrak{M}$ were a one-membered module, then both would have to be divisible by the basis polynomial, and, since $\mathfrak{M}$ contains no polynomial of the second degree, by a polynomial of the third degree; they would therefore have to be proportional, which they are not. The multi-memberedness of $\mathfrak{M}$ is the internal reason for the failure of Landau's product theorems, which are valid in the case of one variable.

3. A further example shows that infinite groups can also occur for prime modules.[25] Let P be the domain of all rational functions of two variables x, y. The module $\mathfrak{M}$ with basis

$$\xi - \frac{y}{x} = \xi - a$$

has an infinite group, since this group contains the residue class of every power of y; on the other hand, $\mathfrak{M}$ is a prime module. Indeed, we show that every divisor

$$\mathfrak{M}_1 = (\xi - a, F(\xi, \eta))$$

is equal to the unit module. Every polynomial $F(\xi, \eta)$ can be represented modulo $(\xi - a)$ by a polynomial $G(\eta)$:

$$F(\xi, \eta) = c_0\eta^\nu + c_1\eta^{\nu-1} + \cdots + c_\nu \quad (\mathfrak{M}).$$

Without loss of generality let $c_0 = 1$. By the multiplication law, $\xi\eta^\nu - \eta^\nu\xi = 0$; hence, since

$$\xi \equiv a(\mathfrak{M}_1), \qquad \eta^\nu \equiv F_1(\mathfrak{M}_1),$$

where $F_1 = -(c_1\eta^{\nu-1} + \cdots + c_\nu)$, one has

$$\xi F_1 - \eta^\nu a \equiv 0(\mathfrak{M}_1).$$

Expanding

$$\eta^\nu a = a\eta^\nu + \nu\frac{\partial a}{\partial y}\eta^{\nu-1} + \cdots,$$

and again replacing η^ν by F_1, one obtains

$$\xi(c_1\eta^{\nu-1} + \cdots + c_\nu) - a(c_1\eta^{\nu-1} + \cdots + c_\nu) + \nu\frac{\partial a}{\partial y}\eta^{\nu-1} + \cdots \equiv 0(\mathfrak{M}_1),$$

[24] Cf. the example of Landau communicated by Blumberg (loc. cit., pp. 51–52). Here and below we denote the independent variables by x and y.

[25] Our definition of "prime module" differs from what is called a prime module in the commutative case, since there divisors of lower dimension always exist, except when the dimension is 0, that is, when the residue group is finite.

or

$$c_1\eta^{\nu-1}a + \frac{\partial c_1}{\partial x}\eta^{\nu-1} + \cdots - ac_1\eta^{\nu-1} + \cdots + u\frac{\partial a}{\partial y}\eta^{\nu-1} + \cdots \equiv 0(\mathfrak{M}_1),$$

or finally

$$\left(\frac{\partial c_1}{\partial x} + \nu\frac{\partial a}{\partial y}\right)\eta^{\nu-1} + \cdots \equiv 0(\mathfrak{M}_1).$$

On the left now stands a polynomial in η of exact degree $\nu - 1$ which does not vanish identically, for the highest coefficient

$$\frac{\partial c_1}{\partial x} + \nu\frac{\partial a}{\partial y} = \frac{\partial c_1}{\partial x} + \nu\frac{1}{x} \quad \text{is } \neq 0,$$

since otherwise $c_1 = -\nu \log x + \varphi(y)$ would not be rational. Thus the procedure can be continued and leads to a non-identically-vanishing polynomial of degree zero, proving that the unit is also contained in $\mathfrak{M}_1$.[26]

4. On the other hand, if one adjoins the function $\log x$ to the rationality domain, then $\mathfrak{M}$ has the divisors

$$\left(\xi - \frac{y}{x},\ \eta - \log x + \varphi(y)\right),$$

where $\varphi(y)$ runs through all rational functions of y. These divisors are all distinct, the integrability condition is fulfilled for each, and the corresponding integral is

$$y \log x - \int \varphi(y)\, dy + c.$$

Therefore the unit cannot be contained in the module, since otherwise the zero function would be the only possible integral. On the other hand, ξ and η are congruent, with respect to the module, to a quantity of the domain; hence the residue group is finite and of order 1. Further, any two of the infinitely many divisors are relatively prime, and the given module $\mathfrak{M}$ is equal to the least common multiple $\mathfrak{N}$ of all these divisors. Indeed, it is divisible by all divisors, hence also by $\mathfrak{N}$. If $\mathfrak{N}$ were a proper divisor of $\mathfrak{M}$, then $\mathfrak{N}$ would still have to contain at least one polynomial F, say of degree ρ, depending only on η alone. Letting $\varphi(y)$ run through the functions $y, \ldots, y^{\rho+1}$, there is no linear relation with coefficients independent of y between the corresponding integrals

$$y \log x - \frac{y^{i+1}}{i+1} + c_i \qquad (i = 1, \ldots, \rho + 1).$$

But the differential equation of order ρ, $F = 0$, cannot possess $\rho + 1$ linearly independent integrals.

The module $\mathfrak{M} = \mathfrak{N}$ is, however, not completely reducible.[27] For otherwise it would be the least common multiple of finitely many of these prime modules, and its residue group would therefore be of finite order; yet the residue group of $\xi - y/x$ contains the residue classes of all powers of η, and so is infinite.

Göttingen, August 1, 1919.

(Received August 4, 1919.)

[26] This procedure amounts to proving that the integrability conditions necessary for a system of differential equations are not fulfilled.

[27] Thus we have an example of the first case of Theorem X.

18. On a Work of K. Hentzelt, Fallen in the War, on Elimination Theory

Jahresbericht der DMV 30 (1921), p. 101

11th meeting, Friday, September 23, 1921, 11 a.m.
(Chair: Loewy.)

1. E. Noether: On a work of K. Hentzelt, fallen in the war, on elimination theory.
Kurt Hentzelt has been missing before Dixmuiden since October 1914 and must be counted among the dead. I intend soon to publish, in a free redaction in the *Mathematische Annalen*, the essential parts of his Erlangen dissertation, which he left behind built up without gaps but almost incomprehensibly presented.

The matter concerns the general problem of **elimination theory**: to determine the common zeros of all polynomials of an ideal $\mathfrak{M}$ consisting of polynomials. If the variables are first subjected to a linear transformation with indeterminate coefficients, then the main result can be stated as follows:

To the ideal $\mathfrak{M}$ there can be assigned uniquely a resultant form

$$R_{\mathfrak{M}} = R^{(1)}(x_1, \ldots, x_n) \cdots R^{(i)}(x_i, \ldots, x_n) \cdots R^{(m)}(x_n) = 0(\mathfrak{M}),$$

such that $R_{\mathfrak{M}}$ vanishes for all zeros of $\mathfrak{M}$ and only for these. If $\mathfrak{N}$ is a divisor of $\mathfrak{M}$ and $R_{\mathfrak{N}}$ and $R_{\mathfrak{M}}$ agree, then $\mathfrak{N}$ and $\mathfrak{M}$ themselves also agree.

The second part of the theorem shows that the resultant form supplies the zeros in characteristic multiplicity. It rests on the fact that $\mathfrak{M}$ can be regarded as a module of linear forms in the power products of $x_1, \ldots, x_{i-1}$; the individual factors $R^{(i)}(x_i, \ldots, x_n)$ of $R_{\mathfrak{M}}$ then become the norms of the respective "basic module" with respect to this module, when $x_{i+1}, \ldots, x_n$ is adjoined to the coefficient domain. If one invokes abstract ideal theory, this yields the following interpretation: if

$$\mathfrak{M} = [\mathfrak{Q}, \mathfrak{Q}_1, \ldots, \mathfrak{Q}_r] = [\mathfrak{Q}, \mathfrak{L}]$$

is a decomposition of $\mathfrak{M}$ into primary ideals $\mathfrak{Q}_i$, and $\mathfrak{L}$ is the complement of $\mathfrak{Q}$, then to the ideal $\mathfrak{Q}$ there corresponds a primary factor $Q^{(i)}(x_i, \ldots, x_n)$ of $R^{(i)}(x_i, \ldots, x_n)$, which, in the sense specified above, represents the norm of $\mathfrak{L}$ with respect to $\mathfrak{Q}$.

19. Ideal Theory in Ring Domains

Math. Ann. 83 (1921), pp. 24–66

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Introduction.

The content of the present paper is the *transfer of the decomposition theorems for the rational integers, respectively of the ideals in algebraic number fields, to ideals in arbitrary integral domains, and more generally ring domains.* To understand this transfer, let us first state the decomposition theorems for the rational integers in a form somewhat different from the usual formulation.

If in

$$a = p_1^{\varrho_1} p_2^{\varrho_2} \cdots p_\sigma^{\varrho_\sigma} = q_1 q_2 \cdots q_\sigma$$

one regards the prime-power factors q_i as the components of the decomposition, then these components have the following characteristic properties:

1. They are *pairwise coprime*; but no q_i can be represented as a product of pairwise coprime numbers, so that in this sense irreducibility holds. From the pairwise coprimeness it further follows that the product $q_1 \cdots q_\sigma$ is equal to the least common multiple $[q_1 \cdots q_\sigma]$.

2. Any two of the components, q_i and q_k, are *relatively prime*; that is, if bq_i is divisible by q_k, then b is divisible by q_k. In this sense too irreducibility holds.

3. Every q is *primary*; that is, if a product $b \cdot c$ is divisible by q, but b is not divisible, then a power[1] of c is divisible. The representation is moreover one by *greatest primary components*, since the product of two distinct q is no longer primary. Also with respect to decomposition into greatest primary components the q are irreducible.

4. Every q is *irreducible* in the sense that it cannot be represented as the least common multiple of two proper divisors.

The connection of these primary numbers q with the prime numbers p consists in this: to each q there exists one, and apart from sign only one, p which is a divisor of q and a power of which is divisible by q: the associated prime number. If p^ϱ is the lowest power of this kind – ϱ the exponent of q – then here, in particular, p^ϱ is equal to q. The uniqueness theorem may now be stated as follows:

For two different decompositions of a rational integer into the irreducible, greatest primary components q, the number of components, the associated prime numbers (up to sign), and the exponents coincide. Since $p^\varrho = q$, it follows from this also that the q themselves coincide (up to sign).

The indeterminacy arising from the sign is, as is well known, removed when one considers, instead of the numbers, the ideals derived from them (all numbers divisible by a); then the formulation holds in exactly the same way for the unique decomposition of the ideals of finite algebraic number fields into powers of prime ideals.

[1] If this power is always the first, then one is, as is well known, dealing with prime numbers.

In what follows (§ 1), the basis will be a general ring domain satisfying only the finiteness condition that each ideal of the domain possess a finite ideal basis. Without such a finiteness condition, irreducible and prime ideals need not exist at all, as is shown by the domain of all algebraic integers, in which there is no decomposition into prime ideals.

It turns out that – corresponding to the four characteristic properties of the components q – in general four separate decompositions exist, each arising from the preceding one by subdivision. The decomposition into coprime-irreducible ideals is a product representation; the other three decompositions are reduced (§ 2) representations as least common multiples. The connection between a primary ideal – the irreducible ideals too are primary – and the associated prime ideal also persists: to every primary ideal $\mathfrak{Q}$ there is uniquely determined an associated prime ideal $\mathfrak{P}$ which is a divisor of $\mathfrak{Q}$ and a power of which is divisible by $\mathfrak{Q}$. If $\mathfrak{P}^{\varrho}$ is the lowest such power – ϱ the exponent of $\mathfrak{Q}$ – then here $\mathfrak{P}^{\varrho}$ need not coincide with $\mathfrak{Q}$. The uniqueness theorem is expressed as follows:

Decompositions 1 and 2 are unique; in two different decompositions 3 or 4 the number of components and the associated prime ideals coincide;[2] the isolated ideals occurring among the components (§ 7) are uniquely determined.

For the proof of the decomposition theorems, the finiteness condition first yields the "theorem on the finite chain", first stated by Dedekind for finite number modules; from it the representation 4 of each ideal as the least common multiple of finitely many irreducible ideals is derived. By transforming the concept of reducibility of a component, one obtains from this the fundamental uniqueness theorem for decomposition 4 into irreducible ideals. By combining finitely many components at a time one rises to the remaining decompositions, whose uniqueness theorems then follow from the uniqueness theorem 4.

Finally it is shown (§ 9) that the representation by finitely many irreducible components still holds under weaker hypotheses; commutativity of the ring domain is not required, and it is enough to consider, in place of an ideal, a module with respect to the domain. In this more general case the equality of the number of components in two different decompositions still holds, whereas the notions prime and primary are tied to commutativity and to the ideal concept; by contrast, the notion coprime remains valid for ideals in noncommutative domains.

The simplest ring domain for which the four separate decompositions actually occur is the domain of all polynomials in n variables with arbitrary complex coefficients. Here the individual decompositions can be interpreted, irrationally, by the behavior of algebraic configurations, and the uniqueness theorem for the associated prime ideals corresponds to the fundamental theorem of elimination theory on the unique decomposability of algebraic configurations into irreducible ones. Further examples are given by all finite integral domains made of polynomials (§ 10). But even the simple domain of all even numbers, more generally of all numbers divisible by a fixed number, already gives an example of partially separated decompositions (§ 11). An example of ideal theory in noncommutative domains is provided by elementary-divisor theory (§ 12), where unique decomposition into irreducible ideals, respectively classes, holds. These irreducible classes completely characterize the irreducible constituents of the elementary divisors, and may perhaps be viewed as their equivalent for domains in which the usual elementary-divisor theory fails.

On the existing literature the following is to be said. The decomposition into greatest primary ideals for the polynomial domain with arbitrary complex, respectively integral, coefficients was given by Lasker, and in individual points further developed by Macaulay.[3] Both rely on elimination theory, and hence use the fact that a polynomial can be represented uniquely as a product of irreducible polynomials. In fact the decomposition theorems for ideals are independent of this prerequisite, as ideal theory in algebraic number fields suggests and as the present paper shows. The primary ideal is also defined by Lasker and Macaulay on the basis of concepts from elimination theory.

The decomposition into irreducible ideals and the decomposition into relatively-prime irreducible ideals do not seem to have been noticed in the literature even for the polynomial domain; only in Macaulay is there a remark on the uniqueness of isolated primary ideals.

The decomposition into coprime-irreducible ideals was given for the polynomial domain by Schmeidler,[4] using elimination theory for the proof of finiteness. There, however, the uniqueness theorem is stated only for classes of ideals, not for the ideals themselves. This latter uniqueness theorem appears in a joint paper,[5]

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[2]Presumably, beyond this, the exponents also coincide, and still more generally corresponding components are isomorphic.

[3]E. Lasker, Zur Theorie der Moduln und Ideale. Math. Ann. 60 (1905), p. 20, Theorems VII and XIII. – F. S. Macaulay, On the Resolution of a given Modular System into Primary Systems including some Properties of Hilbert Numbers. Math. Ann. 74 (1913), p. 66.

[4]W. Schmeidler, Über Moduln und Gruppen hyperkomplexer Größen. Math. Zeitschr. 3 (1919), p. 29.

[5]E. Noether–W. Schmeidler, Moduln in nichtkommutativen Bereichen, insbesondere aus Differential- und Differenzenausdrücken. Math. Zeitschr. 8 (1920), p. 1.

where ideals in noncommutative polynomial domains are involved. Only the finite ideal basis is used there; hence theorems and methods remain valid for general ring domains, and are sharpened in the present paper with respect to equality of number (§ 11). The present investigations are a strong generalization and further development of the conceptual formations underlying those two papers. The essential feature of the two papers is the passage from representation as a least common multiple to an additive decomposition of the system of residue classes. Here, for simplicity of presentation, we remain with the least common multiple; the additive decomposition is then matched by the transformation of the notion of reducibility into a property of the complement (§ 3). Yet by considerations essentially corresponding to those of the joint paper, all the theorems stated here can also be understood as additive decomposition theorems for the system of residue classes and certain partial systems. This system of residue classes forms a ring of the same generality as the ring originally taken as basis; indeed every ring may be regarded as the system of residue classes of the ideal corresponding to the totality of identical relations among the ring elements; or also of a partial system of these relations, if the remaining relations are assumed to be satisfied already in the domain.

This remark also gives the place of Fraenkel's papers.[6] Fraenkel considers additive decompositions of rings that are subjected to such restrictive conditions (existence of regular elements, division by these, decomposability condition) that, for the corresponding ideal, the four decompositions coincide. Because of this coincidence, his finiteness condition, that the ideal should have only finitely many proper divisors – from which he in part departs – also means no stronger restriction than ours. Fraenkel's starting point is conditioned by the different, essentially algebraic, aims of his papers; by algebraic extension he then arrives at more general rings with less restrictive conditions.

§ 1. Ring domain, ideal, finiteness condition.

1. The domain Σ taken as basis is to be a (commutative) ring in the abstract definition;[7] that is, Σ consists of a system of elements $a, b, c, \dots, f, g, h, \dots$ in which a relation satisfying the usual conditions is defined as equality; and in which by two operations (modes of composition), addition and multiplication, from any two ring elements a and b a third is always uniquely obtained, respectively as sum $a+b$ and as product $a \cdot b$. The ring and the otherwise wholly arbitrary operations must satisfy the following laws:

1. The associative law of addition: $(a+b)+c = a+(b+c)$.
2. The commutative law of addition: $a+b = b+a$.
3. The associative law of multiplication: $(a \cdot b)c = a(b \cdot c)$.
4. The commutative law of multiplication: $a \cdot b = b \cdot a$.
5. The distributive law: $a(b+c) = ab+ac$.
6. The law of unrestricted and unique subtraction. There is in Σ a single element x satisfying the equation $a+x=b$. (One denotes $x = b-a$.)

From these properties follows the existence of zero; but a ring need not possess a unit; and the product of two elements can vanish without one of the factors vanishing. Rings for which, from the vanishing of a product, the vanishing of a factor always follows, and which in addition possess a unit, will be called proper integral domains. For the finite sum $a+a+\cdots+a$ we introduce the usual abbreviated notation na, where the integers n are to be regarded merely as abbreviating signs, not as ring elements, and are defined recursively by $a = 1a$, $na + a = (n+1)a$.

2. By an ideal $\mathfrak{M}$[8] in Σ is meant a system of elements from Σ satisfying the two conditions:

1. $\mathfrak{M}$ contains, together with f, also $a \cdot f$, where a is an arbitrary element of Σ.
2. $\mathfrak{M}$ contains, together with f and g, also the difference $f-g$; hence with f also nf for every integer n.

[6] A. Fraenkel, Über die Teiler der Null und die Zerlegung von Ringen. J. f. M. 145 (1914), p. 139. Über gewisse Teilbereiche und Erweiterungen von Ringen. Habilitationsschrift, Leipzig, Teubner, 1916. Über einfache Erweiterungen zerlegbarer Ringe. J. f. M. 151 (1920), p. 121.

[7] The definition is taken from Fraenkel's habilitation thesis, omitting his restrictive conditions 6, I and II; for this the commutative law of addition had to be included. Thus these are the laws defining a field with the invertibility of multiplication omitted.

[8] Ideals are denoted by capital German letters. $\mathfrak{M}$ is meant to recall the example of the ideal in polynomials usually called a "module" or form module. Incidentally, §§ 1–3 use only the module property and not the ideal property; compare § 9.

If f is an element of $\mathfrak{M}$, we express this as usual by $f \equiv 0(\mathfrak{M})$ and say that f is divisible by $\mathfrak{M}$. If every element of $\mathfrak{N}$ is at the same time an element of $\mathfrak{M}$, and hence divisible by $\mathfrak{M}$, we say: $\mathfrak{N}$ is divisible by $\mathfrak{M}$; in symbols, $\mathfrak{N} \equiv 0(\mathfrak{M})$. $\mathfrak{M}$ is called a proper divisor of $\mathfrak{N}$ when it contains elements different from those of $\mathfrak{N}$, and hence is not conversely divisible by $\mathfrak{N}$. From $\mathfrak{N} \equiv 0(\mathfrak{M})$ and $\mathfrak{M} \equiv 0(\mathfrak{N})$ follows $\mathfrak{N} = \mathfrak{M}$.

The other familiar notions are also retained word for word. By the greatest common divisor of two ideals $\mathfrak{A}$ and $\mathfrak{B}$ – $\mathfrak{D} = (\mathfrak{A}, \mathfrak{B})$ – we mean the totality of elements that can be represented in the form $a + b$, where a runs through all elements of $\mathfrak{A}$ and b through all elements of $\mathfrak{B}$; $\mathfrak{D}$ is again an ideal. Likewise, the greatest common divisor of infinitely many ideals – $\mathfrak{D} = (\mathfrak{A}_1, \mathfrak{A}_2, \dots, \mathfrak{A}_\nu, \dots)$ – is defined as the totality of elements d representable as sums of elements from finitely many of the ideals at a time: $d = a_{i_1} + a_{i_2} + \cdots + a_{i_n}$; here too $\mathfrak{D}$ is again an ideal.

If, in particular, the ideal $\mathfrak{M}$ contains a finite number of elements $f_1, f_2, \dots, f_\varrho$ such that

$$\mathfrak{M} = (f_1 \dots f_\varrho), \qquad f = a_1 f_1 + \cdots + a_\varrho f_\varrho + n_1 f_1 + \cdots + n_\varrho f_\varrho$$

for every $f \equiv 0(\mathfrak{M})$, where the a_i are quantities of the ring domain and the n_i are integers, then $\mathfrak{M}$ is called a finite ideal; $f_1, \dots, f_\varrho$ an ideal basis.

In what follows we take as basis only rings Σ satisfying the finiteness condition: *Every ideal in Σ is finite, hence possesses an ideal basis.*

3. From the finiteness condition there follows directly the result on which all subsequent considerations rest.

Theorem I (theorem on the finite chain).[9] *Let $\mathfrak{M}, \mathfrak{M}_1, \mathfrak{M}_2, \dots, \mathfrak{M}_\nu, \dots$ be a countably infinite system of ideals in Σ, each of which is divisible by the following one. Then from some finite index n onward all ideals are identical, $\mathfrak{M}_n = \mathfrak{M}_{n+1} = \cdots$. In other words: if $\mathfrak{M}, \mathfrak{M}_1, \mathfrak{M}_2, \dots, \mathfrak{M}_s, \dots$ form a simply ordered chain of ideals such that every ideal is a proper divisor of the one immediately preceding it, then the chain breaks off after finitely many steps.*

Indeed, let $\mathfrak{D} = (\mathfrak{M}, \mathfrak{M}_1, \mathfrak{M}_2, \dots, \mathfrak{M}_\nu, \dots)$ be the greatest common divisor of the system, and let $f_1, \dots, f_\varrho$ be a basis of $\mathfrak{D}$, which always exists by the finiteness condition. From the divisibility assumption it follows that every element of $\mathfrak{D}$ is at the same time an element of one ideal of the chain; for from

$$f = g + h, \qquad g \equiv 0(\mathfrak{M}_r), \qquad h \equiv 0(\mathfrak{M}_s), \qquad (r < s)$$

one gets $g \equiv 0(\mathfrak{M}_s)$ and hence $f \equiv 0(\mathfrak{M}_s)$. The same holds when f is a sum of several constituents. Hence there is a finite index n such that

$$f_1 \equiv 0(\mathfrak{M}_n), \quad f_2 \equiv 0(\mathfrak{M}_n), \quad \dots, \quad f_\varrho \equiv 0(\mathfrak{M}_n).$$

Since conversely $\mathfrak{M}_n \equiv 0(\mathfrak{D})$, we have $\mathfrak{M}_n = \mathfrak{D}$; and since further $\mathfrak{M}_n \equiv 0(\mathfrak{D})$ and $\mathfrak{D} = \mathfrak{M}_n \equiv 0(\mathfrak{M}_r)$, we also have $\mathfrak{M}_r = \mathfrak{D} = \mathfrak{M}_n$ for every $r > n$, proving the theorem.

We note that conversely the existence of an ideal basis again follows from this theorem, so that the finiteness condition could also have been stated in this basis-free form.

§ 2. Representation of an ideal as the least common multiple of finitely many irreducible ideals.

The least common multiple $[\mathfrak{B}_1, \mathfrak{B}_2, \dots, \mathfrak{B}_k]$ of the ideals $\mathfrak{B}_1, \mathfrak{B}_2, \dots, \mathfrak{B}_k$ is defined as usual as the totality of the elements divisible by $\mathfrak{B}_1$, by $\mathfrak{B}_2$, …, and by $\mathfrak{B}_k$; in symbols,

$$\text{from } f \equiv 0(\mathfrak{B}_i), \quad (i = 1, 2, \dots, k), \quad \text{follows} \quad f \equiv 0([\mathfrak{B}_1, \mathfrak{B}_2, \dots, \mathfrak{B}_k]),$$

and conversely. The least common multiple is again an ideal; the ideals $\mathfrak{B}_i$ are also called the components of the decomposition.

Definition I. A representation $\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k]$ is called a reduced representation if no $\mathfrak{B}_i$ is contained in the least common multiple $\mathfrak{U}_i$ of the remaining ideals, and if no $\mathfrak{B}_i$ can be replaced by a proper divisor.[10] If the conditions are satisfied only for the ideal $\mathfrak{B}_i$, the representation is called reduced with respect to $\mathfrak{B}_i$. The

[9] First stated for number modules by Dedekind: Zahlentheorie, Suppl. XI, § 172, Theorem VIII (4th ed.); our proof and the designation "chain" are taken from there. For ideals in polynomials, see Lasker, loc. cit., p. 56 (lemma). But in both cases the theorem finds only isolated application. Our applications rest throughout on the principle of choice.

[10] An example of a non-reduced representation is $(x, xy) = [(x), (x^2, xy, y^\lambda)]$ for every exponent $\lambda gtr 2$; the representation $[(x), (x^2, y)]$ corresponding to $\lambda = 1$ is an associated reduced one.

least common multiple $\mathfrak{U}_i = [\mathfrak{B}_1, \ldots, \mathfrak{B}_{i-1}, \mathfrak{B}_{i+1}, \ldots, \mathfrak{B}_k]$ is called the complement of $\mathfrak{B}_i$. Representations in which only the first condition is satisfied are called shortest representations.

For representing an ideal as a least common multiple it therefore suffices to restrict to reduced representations, by the following result.

Lemma I. *Every representation of an ideal as the least common multiple of finitely many ideals can be replaced, in at least one way, by a reduced representation; such a representation can in particular be attained by successive decomposition.*

Indeed, let $\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k]$ be an arbitrary representation of $\mathfrak{M}$. We omit, in order, those $\mathfrak{B}_i$ which are contained in the least common multiple of the ideals retained. Since the remaining ideals still yield $\mathfrak{M}$, in the resulting representation

$$\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k] = [\mathfrak{U}_i, \mathfrak{B}_i]$$

the first condition is satisfied, hence it is a shortest representation; and this condition remains satisfied if any $\mathfrak{B}_i$ is replaced by a proper divisor. The second condition, however, is always attainable by the theorem on the finite chain (Theorem I). For if

$$\mathfrak{B}_i \supset \mathfrak{B}_i' \supset \mathfrak{B}_i'' \supset \cdots \supset \mathfrak{B}_i^{(\nu)} \supset \cdots, \qquad \mathfrak{M} = [\mathfrak{U}_i, \mathfrak{B}_i] = [\mathfrak{U}_i, \mathfrak{B}_i'] = \cdots = [\mathfrak{U}_i, \mathfrak{B}_i^{(\nu)}] = \cdots,$$

where every $\mathfrak{B}_i^{(\nu)}$ is a proper divisor of the one immediately preceding, then by that theorem the chain $\mathfrak{B}_i, \mathfrak{B}_i', \ldots, \mathfrak{B}_i^{(\nu)}, \ldots$ must terminate after finitely many steps; hence in the representation $\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{B}_i^{(\nu)}]$ the ideal $\mathfrak{B}_i^{(\nu)}$ can no longer be replaced by a proper divisor; and this holds a fortiori if $\mathfrak{U}_i$ is replaced by a proper divisor. Applying the procedure successively to each $\mathfrak{B}_i$, always forming the complement with the already reduced $\mathfrak{B}$, gives a reduced representation.[11]

In order to obtain such a representation successively, it remains to show that from the individual reduced representations

$$\mathfrak{M} = [\mathfrak{B}_1, \mathfrak{C}_1], \qquad \mathfrak{C}_1 = [\mathfrak{B}_2, \mathfrak{C}_2], \quad \ldots, \quad \mathfrak{C}_{k-1} = [\mathfrak{B}_k, \mathfrak{C}_k]$$

it follows that the resulting representation $\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k, \mathfrak{C}_k]$ is reduced. For this it suffices to show that from reduced representations

$$\mathfrak{M} = [\mathfrak{B}, \mathfrak{C}], \qquad \mathfrak{C} = [\mathfrak{C}_1, \mathfrak{C}_2]$$

one obtains a reduced representation $\mathfrak{M} = [\mathfrak{B}, \mathfrak{C}_1, \mathfrak{C}_2]$. In fact, by hypothesis no $\mathfrak{B}$ lies in its complement; if this happened for a $\mathfrak{C}_i$, then, contrary to the hypothesis in the first representation, $\mathfrak{C}$ would be replaceable by a proper divisor, since by the second reduced representation $\mathfrak{C}_1$ and $\mathfrak{C}_2$ are proper divisors of $\mathfrak{C}$; so the representation is shortest. Moreover, by hypothesis no $\mathfrak{B}$ can be replaced by a proper divisor; if this were possible for a $\mathfrak{C}_i$, it would, contrary to the hypothesis, correspond to replacing $\mathfrak{C}$ by a proper divisor, since the representation for $\mathfrak{C}$ is reduced. Thus the lemma is proved.

Definition II. An ideal $\mathfrak{M}$ is called reducible if it can be represented as the least common multiple of two proper divisors; in the contrary case $\mathfrak{M}$ is called irreducible.

We now prove, by means of Theorem I on the finite chain and using reduced representations:

Theorem II. *Every ideal is representable as the least common multiple of finitely many irreducible ideals.*[12]

For an arbitrary ideal $\mathfrak{M}$ is either irreducible; then $\mathfrak{M} = [\mathfrak{M}]$ is a representation of the kind required by Theorem II; or else $\mathfrak{M} = [\mathfrak{B}_1, \mathfrak{C}_1]$, where $\mathfrak{B}_1$ and $\mathfrak{C}_1$ are proper divisors of $\mathfrak{M}$, and the representation may, by Lemma I, be assumed reduced. For $\mathfrak{C}_1$ the same alternative holds: either it is irreducible, or there is a reduced representation $\mathfrak{C}_1 = [\mathfrak{B}_2, \mathfrak{C}_2]$.

Continuing in this way, one obtains the series of reduced representations

$$\mathfrak{M} = [\mathfrak{B}_1, \mathfrak{C}_1] = [\mathfrak{B}_1, \mathfrak{B}_2, \mathfrak{C}_2] = \cdots = [\mathfrak{B}_1, \ldots, \mathfrak{B}_n, \mathfrak{C}_n] = \cdots. \tag{1}$$

In the chain $\mathfrak{C}_1, \mathfrak{C}_2, \ldots, \mathfrak{C}_n, \ldots$ each $\mathfrak{C}_n$ is a proper divisor of the one immediately preceding, and therefore the chain terminates after finitely many steps; there is an index n such that $\mathfrak{C}_n$ is irreducible. By Lemma I, moreover, the representation $\mathfrak{M} = [\mathfrak{U}_n, \mathfrak{C}_n]$ is reduced; $\mathfrak{C}_n$ therefore cannot lie in its complement $\mathfrak{U}_n$, and in the representation $\mathfrak{M} = [\mathfrak{U}_n, \mathfrak{C}_n]$ the ideal $\mathfrak{U}_n$ cannot be replaced by a proper divisor. Replacing, if necessary,

[11] That such a representation is not uniquely defined by the given representation is shown by the previous example. For $(x^2, xy) = [(x), (x^2, xy, y^\lambda)]$, $\lambda \geq 2$, besides $[(x), (x^2, y)]$ also $[(x), (x^2, ux + y)]$, for arbitrary u, is a reduced representation.

[12] That such a representation is not unique in general is shown by the previous example: $(x^2, xy) = [(x), (x^2, ux + y)]$ for arbitrary u. The two components are irreducible for arbitrary u. Indeed, all divisors of (x) are of the form $(x, g(y))$, where $g(y)$ denotes a polynomial in y; the least common multiple of any two therefore also has this form, hence is a proper divisor of (x). The ideal $(x^2, ux + y)$ possesses only the single divisor (x, y), and therefore is necessarily irreducible as well.

$\mathfrak{U}_n$ by a proper divisor,[13] so that the representation becomes reduced, shows that every reducible ideal admits a reduced representation as the least common multiple of an irreducible ideal and a complementary ideal. In the series (1), therefore, all $\mathfrak{B}_i$ may be assumed irreducible without loss of generality; repeating the preceding argument gives the existence of an irreducible $\mathfrak{C}_n$, proving Theorem II.

§ 3. Equality of the number of components in two different decompositions into irreducible ideals.

For the proof of equality of number, reducibility or irreducibility of an ideal must first be expressed by properties of its complement.

Theorem III.[14] *Let the shortest representation $\mathfrak{M} = [\mathfrak{U}, \mathfrak{C}]$ be reduced with respect to $\mathfrak{C}$. Then the necessary and sufficient condition that $\mathfrak{C}$ be reducible is the existence of two ideals $\mathfrak{N}_1$ and $\mathfrak{N}_2$, proper divisors of $\mathfrak{M}$, such that*

$$\mathfrak{N}_1\mathfrak{N}_2 \equiv 0(\mathfrak{M}), \qquad \mathfrak{N}_i \equiv 0(\mathfrak{U}), \qquad [\mathfrak{N}_1, \mathfrak{N}_2] = \mathfrak{M}. \tag{2}$$

It also follows: if conditions (2) are satisfied and $\mathfrak{C}$ is irreducible, then at least one $\mathfrak{N}_i$ is not a proper divisor of $\mathfrak{M}$; $\mathfrak{N}_i = \mathfrak{M}$.

Let $\mathfrak{C} = [\mathfrak{C}_1, \mathfrak{C}_2]$, where $\mathfrak{C}_1, \mathfrak{C}_2$ are proper divisors of $\mathfrak{C}$. Then

$$\mathfrak{M} = [\mathfrak{U}, \mathfrak{C}] = [\mathfrak{U}, \mathfrak{C}_1, \mathfrak{C}_2] = [[\mathfrak{U}, \mathfrak{C}_1], [\mathfrak{U}, \mathfrak{C}_2]].$$

Here the ideals $[\mathfrak{U}, \mathfrak{C}_i]$ are proper divisors of $\mathfrak{M}$, since otherwise $[\mathfrak{U}, \mathfrak{C}]$ would not be reduced with respect to $\mathfrak{C}$. Since divisibility by $\mathfrak{U}$ is also fulfilled, condition (2) is proved necessary. (The representation (2) is not reduced, since one $[\mathfrak{U}, \mathfrak{C}_i]$ can be replaced by $\mathfrak{C}_i$.)

Conversely, suppose (2) is satisfied. We form the ideals

$$\mathfrak{C}_1 = (\mathfrak{C}, \mathfrak{N}_1), \qquad \mathfrak{C}_2 = (\mathfrak{C}, \mathfrak{N}_2), \qquad \mathfrak{C}^* = [\mathfrak{C}_1, \mathfrak{C}_2].$$

Then $\mathfrak{C}$ is divisible by both $\mathfrak{C}_1$ and $\mathfrak{C}_2$, hence also by their least common multiple $\mathfrak{C}^*$. To show divisibility of $\mathfrak{C}^*$ by $\mathfrak{C}$, let

$$f \equiv 0(\mathfrak{C}^*); \quad f \equiv 0(\mathfrak{C}_1); \quad f \equiv 0(\mathfrak{C}_2), \quad \text{or also} \quad f = c + n_1 = t + n_2,$$

where c, t, n_1, n_2 are quantities in $\mathfrak{C}, \mathfrak{C}, \mathfrak{N}_1, \mathfrak{N}_2$, and in particular n_1 is divisible by $\mathfrak{N}_1$. Hence the difference

$$g = c - t = n_2 - n_1$$

is divisible both by $\mathfrak{C}$ and by $\mathfrak{U}$, and hence by $\mathfrak{M}$. Since $n_1 = n_2 + m$, furthermore, n_1 (and likewise n_2) is divisible by both $\mathfrak{N}_1$ and $\mathfrak{N}_2$, hence by $\mathfrak{M}$. Thus

$$f = c + m, \qquad f \equiv 0(\mathfrak{C}), \qquad \mathfrak{C}^* = \mathfrak{C}.$$

The ideals $\mathfrak{C}_1$ and $\mathfrak{C}_2$ are proper divisors of $\mathfrak{C}$; for if $\mathfrak{C}_1 = (\mathfrak{C}, \mathfrak{N}_1) = \mathfrak{C}$, then $\mathfrak{N}_1$ would be divisible by $\mathfrak{C}$, and hence, because of the divisibility by $\mathfrak{U}$, equal to $\mathfrak{M}$, contrary to the hypothesis. Thus $\mathfrak{C} = [\mathfrak{C}_1, \mathfrak{C}_2]$ has been recognized as reducible; Theorem III is proved.

Almost the same argument also shows the following.

Lemma II. *If in a shortest representation $\mathfrak{M} = [\mathfrak{U}, \mathfrak{C}]$ the ideal $\mathfrak{C}$ can be replaced by a proper divisor, then $\mathfrak{C}$ is reducible.*

Thus suppose

$$\mathfrak{M} = [\mathfrak{U}, \mathfrak{C}] = [\mathfrak{U}, \mathfrak{C}_1],$$

and set

$$\mathfrak{C}^* = [\mathfrak{C}_1, (\mathfrak{U}, \mathfrak{C})].$$

Again $\mathfrak{C}$ is divisible by $\mathfrak{C}^*$. From $f \equiv 0(\mathfrak{C}^*)$ follows

$$f = c_1 = a + c.$$

[13] In fact the representation is also reduced with respect to $\mathfrak{U}_n$, as will be shown in § 3 (Lemma IV) as the converse of Lemma I.

[14] Theorem III corresponds to the passage from modules to residue groups in the papers of Schmeidler and Noether–Schmeidler (compare the Introduction). The residue group corresponds to $\mathfrak{C}$, and $\mathfrak{N}_1$ and $\mathfrak{N}_2$ to the subgroups into which the residue group is decomposed.

The difference $a = c_1 - c$ is therefore divisible both by $\mathfrak{U}$ and by $\mathfrak{C}_1$, hence by $\mathfrak{M}$; thus $c_1 = c + m$, $f \equiv 0(\mathfrak{C})$, $\mathfrak{C}^* = \mathfrak{C}$. Since both $\mathfrak{C}_1$ and $(\mathfrak{U}, \mathfrak{C})$ are, by hypothesis, proper divisors of $\mathfrak{C}$, the ideal $\mathfrak{C} = \mathfrak{C}^*$ is recognized as reducible.

An irreducible $\mathfrak{C}$ therefore cannot be replaced by a proper divisor.

Now let two different shortest representations of $\mathfrak{M}$ as the least common multiple of finitely many irreducible ideals be given:

$$\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k] = [\mathfrak{D}_1 \cdots \mathfrak{D}_l].$$

By the remark after Lemma II, these representations are at the same time reduced. We first prove:

Lemma III. *For every complement* $\mathfrak{U}_i = [\mathfrak{B}_1 \cdots \mathfrak{B}_{i-1}\mathfrak{B}_{i+1} \cdots \mathfrak{B}_k]$ *there exists an ideal* $\mathfrak{D}_j$ *such that* $\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{D}_j]$.

Indeed, putting $\mathfrak{M} = [\mathfrak{B}_i, \mathfrak{C}_1]$, $\mathfrak{C}_1 = [\mathfrak{D}_1, \mathfrak{C}_2]$, and so on, we get

$$\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{B}_i] = [\mathfrak{U}_i, [\mathfrak{D}_1, \mathfrak{C}_2]] = [\mathfrak{B}_i, [\mathfrak{U}_i, \mathfrak{D}_1], \mathfrak{C}_2].$$

Here, for $\mathfrak{B}_i = [\mathfrak{U}_i, \mathfrak{D}_1]$ and $\mathfrak{C}_i = [\mathfrak{U}_i, \mathfrak{C}_2]$, the conditions (2) of Theorem III are satisfied, since $\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{B}_i]$ is reduced with respect to $\mathfrak{B}_i$ and the representation is shortest. Since $\mathfrak{B}_i$ was assumed irreducible, one $\mathfrak{N}_i$ must necessarily become equal to $\mathfrak{M}$.

If $[\mathfrak{U}_i, \mathfrak{D}_1] = \mathfrak{M}$, the lemma is proved; if $[\mathfrak{U}_i, \mathfrak{C}_2] = \mathfrak{M}$, then correspondingly $\mathfrak{M} = [[\mathfrak{U}_i, \mathfrak{D}_1], [\mathfrak{U}_i, \mathfrak{C}_2]]$, where by the same argument again one component must become equal to $\mathfrak{M}$. Continuing in this way, either $\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{D}_j]$ with $j < l$, or $\mathfrak{M} = [\mathfrak{U}_i, \mathfrak{D}_1, \ldots, \mathfrak{D}_{l-1}]$; since $\mathfrak{D}_1 \cdots \mathfrak{D}_{l-1} = \mathfrak{D}_l$, the lemma is proved.

It follows that:

Theorem IV. *In two different shortest representations of an ideal as the least common multiple of irreducible ideals, the number of components is the same.*

From the lemma, for $i = 1$,

$$\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k] = [\mathfrak{U}_1, \mathfrak{D}_{j_1}] = [\mathfrak{D}_{j_1}, \mathfrak{B}_2, \ldots, \mathfrak{B}_k].$$

Considering now the two decompositions

$$\mathfrak{M} = [\mathfrak{D}_{j_1}, \mathfrak{B}_2, \ldots, \mathfrak{B}_k] = [\mathfrak{D}_1, \mathfrak{D}_2, \ldots, \mathfrak{D}_l]$$

and repeating the previous argument with respect to the complement $\mathfrak{U}_2 = [\mathfrak{D}_{j_1}, \mathfrak{B}_3, \ldots, \mathfrak{B}_k]$ of $\mathfrak{B}_2$, one obtains

$$\mathfrak{M} = [\mathfrak{U}_2, \mathfrak{B}_2] = [\mathfrak{U}_2, \mathfrak{D}_{j_2}] = [\mathfrak{D}_{j_1}, \mathfrak{D}_{j_2}, \mathfrak{B}_3, \ldots, \mathfrak{B}_k],$$

and, continuing the procedure,

$$\mathfrak{M} = [\mathfrak{D}_{j_1}, \mathfrak{D}_{j_2}, \ldots, \mathfrak{D}_{j_k}].$$

Since by hypothesis the representation $\mathfrak{M} = [\mathfrak{D}_1 \cdots \mathfrak{D}_l]$ is shortest, so that no $\mathfrak{D}$ can be omitted, the distinct ideals among the $\mathfrak{D}_{j_i}$ must exhaust all $\mathfrak{D}$; hence $k \geq l$. If in the lemma and the following arguments the $\mathfrak{B}$ and $\mathfrak{D}$ are interchanged throughout, one obtains correspondingly $l \geq k$, and therefore $k = l$, proving equality of number. It also follows that the ideals $\mathfrak{D}_{j_i}$ are all mutually distinct, since otherwise in a shortest representation by the $\mathfrak{D}_{j_i}$ fewer than k components would occur; thus the notation can be chosen so that $\mathfrak{D}_{j_i} = \mathfrak{D}_i$. For the same reason, all intermediate representations $\mathfrak{M} = [\mathfrak{D}_{j_1} \cdots \mathfrak{D}_{j_r}, \mathfrak{B}_{r+1}, \ldots, \mathfrak{B}_k]$ are shortest and hence, by the remark after Lemma II, reduced.

Equality of number leads to a converse of Lemma I:

Lemma IV. *If in a reduced representation one combines the components into groups and forms their least common multiples, then the resulting representation is reduced. In other words, from a reduced representation*

$$\mathfrak{M} = [\mathfrak{C}_1 \cdots \mathfrak{C}_k]$$

it follows that $\mathfrak{M} = [\mathfrak{N}_1, \mathfrak{N}_2]$ *is also reduced, if* $\mathfrak{N}_1 = [\mathfrak{C}_1, \ldots, \mathfrak{C}_h]$ *and* $\mathfrak{N}_2 = [\mathfrak{C}_{h+1}, \ldots, \mathfrak{C}_k]$.

First, $\mathfrak{N}_1$ cannot lie in its complement $\mathfrak{N}_2$, since this is true of none of its divisors $\mathfrak{C}_i$; the representation is therefore shortest. To show that $\mathfrak{N}_1$ cannot be replaced by a proper divisor, decompose the $\mathfrak{C}$ into irreducible ideals $\mathfrak{B}$,[15] giving the reduced representations, by Lemma I,

$$\mathfrak{N}_1 = [\mathfrak{B}_1 \cdots \mathfrak{B}_\lambda], \qquad \mathfrak{N}_2 = [\mathfrak{B}_{\lambda+1} \cdots \mathfrak{B}_\kappa].$$

[15] This always means a shortest, hence reduced, representation by the $\mathfrak{B}$.

Let $\mathfrak{M} = [\mathfrak{N}_1, \mathfrak{N}_2]$ be reduced with respect to $\mathfrak{N}_2$, and let $\mathfrak{N}_1^*$ be a proper divisor of $\mathfrak{N}_1$. By Lemma II,

$$\mathfrak{N}_1 = [\mathfrak{N}_1^*, (\mathfrak{N}_1, \mathfrak{N}_2)],$$

and this representation is necessarily reduced with respect to $\mathfrak{N}_1^*$, since otherwise $\mathfrak{N}_1$ could also be replaced in $\mathfrak{M}$ by a proper divisor. If necessary, replace $(\mathfrak{N}_1, \mathfrak{N}_2)$ by a proper divisor as well so that a reduced representation for $\mathfrak{N}_1$ is obtained. Decomposing the two components of $\mathfrak{N}_1$ into irreducible ideals, the number λ of irreducible ideals of $\mathfrak{N}_1$ is then additively composed from the numbers of the components; the number of irreducible ideals corresponding to the proper divisor $\mathfrak{N}_1^*$ is therefore necessarily less than λ. But then decomposing $\mathfrak{M} = [\mathfrak{N}_1^*, \mathfrak{N}_2]$ into irreducible ideals would lead to fewer than κ ideals, contradicting equality of number. As the special case $\varrho = 2$, it follows also that the representation $\mathfrak{M} = [\mathfrak{N}_1, \mathfrak{N}_2]$ is reduced with respect to the complement $\mathfrak{N}_2$ as well.

§ 4. Primary ideals. Uniqueness of the associated prime ideals in two different decompositions into irreducible ideals.

In what follows the matter at issue is the connection between primary and irreducible ideals.

Definition III. An ideal $\mathfrak{Q}$ is called primary if from $a \cdot b \equiv 0(\mathfrak{Q})$ and $a \not\equiv 0(\mathfrak{Q})$ it necessarily follows that $b^{\varkappa} \equiv 0(\mathfrak{Q})$, where the exponent $\varkappa$ is a finite number.

The definition may also be stated as follows: if a product $a \cdot b$ is divisible by $\mathfrak{Q}$, then either one factor is divisible or a power of each factor is divisible. If in particular $\varkappa$ is always equal to 1, the ideal is called a prime ideal.

From the definition of the primary (respectively prime) ideal, by virtue of the existence of bases, there follows the formulation involving only products of ideals:[16]

Definition IIIa. An ideal $\mathfrak{Q}$ is called primary if from $\mathfrak{A}\mathfrak{B} \equiv 0(\mathfrak{Q})$ and $\mathfrak{A} \not\equiv 0(\mathfrak{Q})$ it necessarily follows that $\mathfrak{B}^{\varkappa} \equiv 0(\mathfrak{Q})$. If $\varkappa$ is always equal to 1, the ideal is called a prime ideal. For a prime ideal $\mathfrak{P}$, therefore, from $\mathfrak{A}\mathfrak{B} \equiv 0(\mathfrak{P})$ and $\mathfrak{A} \not\equiv 0(\mathfrak{P})$ it always follows that $\mathfrak{B} \equiv 0(\mathfrak{P})$.

Since IIIa contains Definition III as the special case $\mathfrak{A} = (a)$, $\mathfrak{B} = (b)$, every ideal primary according to IIIa is also primary according to III. Conversely, let $\mathfrak{Q}$ be primary according to III, and suppose the hypothesis of IIIa is fulfilled: $\mathfrak{A}\mathfrak{B} \equiv 0(\mathfrak{Q})$. Then either $\mathfrak{A} \equiv 0(\mathfrak{Q})$ follows, or else there exists at least one quantity $a \not\equiv 0(\mathfrak{Q})$ such that $a\mathfrak{B} \equiv 0(\mathfrak{Q})$ and $a \not\equiv 0(\mathfrak{Q})$. If $b_1, \ldots, b_s$ is an ideal basis of $\mathfrak{B}$, then by Definition III, since $ab_i \equiv 0(\mathfrak{Q})$,

$$b_i^{\varkappa_i} \equiv 0(\mathfrak{Q}) \qquad (i = 1, \ldots, s).$$

Consequently, because

$$b = n_1 b_1 + \cdots + n_s b_s + a_1 b_1 + \cdots + a_s b_s,$$

for $\varkappa = \varkappa_1 + \cdots + \varkappa_s$ every product of $\varkappa$ quantities b is divisible by $\mathfrak{Q}$, proving that ideals primary according to III satisfy Definition IIIa. In the special case of prime ideals $\mathfrak{P}$, from $a\mathfrak{B} \equiv 0(\mathfrak{P})$, hence also $ab \equiv 0(\mathfrak{P})$ for every $b \equiv 0(\mathfrak{B})$, and $a \not\equiv 0(\mathfrak{P})$, it follows that $b \equiv 0(\mathfrak{P})$ and therefore $\mathfrak{B} \equiv 0(\mathfrak{P})$. Thus the equivalence of the two definitions is proved.

The connection between primary and prime ideals is made by the observation that the totality $\mathfrak{P}$ of all elements p with the property that some power of p is divisible by $\mathfrak{Q}$ forms a prime ideal. First it is clear that $\mathfrak{P}$ is an ideal, since together with p_1 and p_2 the elements ap_1 and $p_1 - p_2$ also have the stated property. By the same basis argument as was applied in the definition of IIIa, there further follows the existence of a number λ such that $\mathfrak{P}^{\lambda} \equiv 0(\mathfrak{Q})$.

Now let

$$ab \equiv 0(\mathfrak{P}), \qquad a \not\equiv 0(\mathfrak{P}).$$

Then, by the definition of $\mathfrak{P}$,

$$a^{\lambda} b^{\lambda} \equiv 0(\mathfrak{Q}), \qquad a^{\lambda} \not\equiv 0(\mathfrak{Q}),$$

and hence, by the definition of $\mathfrak{Q}$,

$$b^{\lambda\varkappa} \equiv 0(\mathfrak{Q}), \qquad \text{and consequently} \qquad b \equiv 0(\mathfrak{P}),$$

so that $\mathfrak{P}$ is proved to be a prime ideal. $\mathfrak{P}$ is also defined as the greatest common divisor of all ideals $\mathfrak{B}$ having the property that some power of $\mathfrak{B}$ is divisible by $\mathfrak{Q}$. For each of these $\mathfrak{B}$ is divisible by $\mathfrak{P}$, hence so

[16] If $\mathfrak{A} = (a_1, \ldots, a_r)$ and $\mathfrak{B} = (b_1, \ldots, b_s)$, then $\mathfrak{A}\mathfrak{B}$ is understood as the ideal having the products $a_i b_k$ as basis; that these and only these elements form the product ideal follows from the basis representation, by commutativity.

is their greatest common divisor $\mathfrak{D}$. Conversely $\mathfrak{P}$ itself is an ideal $\mathfrak{B}$, and hence is divisible by $\mathfrak{D}$, which proves $\mathfrak{P} = \mathfrak{D}$. Thus $\mathfrak{P}$ is a prime ideal which is a divisor of $\mathfrak{Q}$ and a power of which is divisible by $\mathfrak{Q}$; this property defines it uniquely. For from

$$\mathfrak{Q} \equiv 0(\mathfrak{P}), \qquad \mathfrak{P}^\varrho \equiv 0(\mathfrak{Q}), \qquad \mathfrak{Q} \equiv 0(\mathfrak{B}), \qquad \mathfrak{B}^\sigma \equiv 0(\mathfrak{Q})$$

follows

$$\mathfrak{P}^\varrho \equiv 0(\mathfrak{B}), \qquad \mathfrak{B}^\sigma \equiv 0(\mathfrak{P}),$$

hence, by the property of prime ideals,

$$\mathfrak{P} \equiv 0(\mathfrak{B}), \qquad \mathfrak{B} \equiv 0(\mathfrak{P}), \qquad \mathfrak{B} = \mathfrak{P}.$$

In summary:

Theorem V. *To every primary ideal $\mathfrak{Q}$ there exists one and only one prime ideal $\mathfrak{P}$ which is a divisor of $\mathfrak{Q}$ and a power of which is divisible by $\mathfrak{Q}$; $\mathfrak{P}$ will be called the "associated prime ideal".*[17] *$\mathfrak{P}$ is defined as the greatest common divisor of all ideals $\mathfrak{B}$ with the property that a power of $\mathfrak{B}$ is divisible by $\mathfrak{Q}$. If ϱ is the smallest number such that $\mathfrak{P}^\varrho \equiv 0(\mathfrak{Q})$, then ϱ is called the exponent of $\mathfrak{Q}$.*[18]

We now prove the connection between primary and irreducible:

Theorem VI. *Every non-primary ideal is reducible; in other words, every irreducible ideal is primary.*[19]

Let $\mathfrak{K}$ be a non-primary ideal, so that by Definition III there exists at least one pair of quantities a, b such that

$$ab \equiv 0(\mathfrak{K}), \qquad a \not\equiv 0(\mathfrak{K}), \qquad b^\varkappa \not\equiv 0(\mathfrak{K}) \quad \text{for every } \varkappa. \tag{3}$$

We now form the two ideals

$$\mathfrak{K}_1 = (\mathfrak{K}, a), \qquad \mathfrak{N}_1 = (\mathfrak{K}, b),$$

which by (3) are proper divisors of $\mathfrak{K}$ and for which, by (3),

$$\mathfrak{K}_1\mathfrak{N}_1 \equiv 0(\mathfrak{K}). \tag{4}$$

For the elements f of the least common multiple $\mathfrak{K}_2 = [\mathfrak{K}_1, \mathfrak{N}_1]$, the following alternative holds.

Either from

$$f \equiv 0(\mathfrak{K}_1), \qquad f \equiv 0(\mathfrak{N}_1), \qquad \text{that is} \qquad f \equiv a_1 b \ (\mathfrak{K}),$$

there always follows a representation

$$f \equiv l_1 b \ (\mathfrak{K}), \qquad l_1 \equiv 0(\mathfrak{K}_1),$$

and hence, by (4), $f \equiv 0(\mathfrak{K})$, so that $\mathfrak{K}_2 \equiv 0(\mathfrak{K})$; since also $\mathfrak{K} \equiv 0(\mathfrak{K}_2)$, one has $\mathfrak{K} = \mathfrak{K}_2$, and $\mathfrak{K}$ is recognized as reducible.

Or there is at least one $f \equiv 0(\mathfrak{K}_2)$ for which no such l_1 exists. With the corresponding a_1 we then form

$$\mathfrak{K}_2 = (\mathfrak{K}, a, a_1), \qquad \mathfrak{N}_2 = (\mathfrak{K}, b^2).$$

Then because $a_1 b \equiv 0(\mathfrak{K}_1)$, (4) gives again

$$\mathfrak{K}_2\mathfrak{N}_2 \equiv 0(\mathfrak{K}), \tag{4'}$$

and $\mathfrak{K}_2$ is a proper divisor of $\mathfrak{K}_1$.

The same alternative holds for the elements f of $\mathfrak{K}_3 = [\mathfrak{K}_2, \mathfrak{N}_2]$. Either from

$$f \equiv 0(\mathfrak{K}_2), \qquad f \equiv 0(\mathfrak{N}_2), \qquad \text{that is} \qquad f \equiv a_2 b^2 \ (\mathfrak{K}),$$

there always follows

$$f \equiv l_2 b^2 \ (\mathfrak{K}), \qquad l_2 \equiv 0(\mathfrak{K}_2),$$

and hence, by (4'), $f \equiv 0(\mathfrak{K})$.

[17] That the converse does not hold is shown by the example $\mathfrak{M} = (x^2, xy)$. The prime ideal (x) satisfies all conditions, but $\mathfrak{M}$ is not primary.

[18] That in general $\mathfrak{P}^\varrho = \mathfrak{Q}$ need not hold, as it does in the domain of rational, respectively algebraic, integers, is shown by the example $\mathfrak{Q} = (x^2, y)$; $\mathfrak{P} = (x, y)$; $\mathfrak{P}^2 = (x^2, xy, y^2) \equiv 0(\mathfrak{Q})$, but $\mathfrak{Q} \not\equiv 0(\mathfrak{P}^2)$; thus $\mathfrak{Q}$ is different from $\mathfrak{P}^2$.

[19] That the converse does not hold here is shown, for example, by $\mathfrak{Q} = (x^2, xy, y^\lambda) = [(x^2, y), (x, y^\lambda)]$, where $\lambda \geq 2$. Here $\mathfrak{Q}$ is primary but reducible. That $\mathfrak{Q}$ is primary follows from the fact that it contains all power products of dimension λ in x, y; thus for every polynomial without constant term some power is divisible by $\mathfrak{Q}$. But if, in $ab \equiv 0(\mathfrak{Q})$, the polynomial b contains a constant term – hence $b^\varkappa \not\equiv 0(\mathfrak{Q})$ for every $\varkappa$ – then, because the basis polynomials of $\mathfrak{Q}$ are homogeneous and every homogeneous component of ab is divisible by $\mathfrak{Q}$, a must be divisible by $\mathfrak{Q}$.

Or for at least one f there is no such l_2, which leads to the formation of $\mathfrak{K}_3 = (\mathfrak{K}, a, a_1, a_2)$ and $\mathfrak{N}_3 = (\mathfrak{K}, b^3)$, with $\mathfrak{K}_3\mathfrak{N}_3 \equiv 0(\mathfrak{K})$, where $\mathfrak{K}_3$ is a proper divisor of $\mathfrak{K}_2$. Continuing in this way, we define in general

$$\mathfrak{K}_n = (\mathfrak{K}, a, a_1, \ldots, a_{n-1}), \qquad \mathfrak{N}_n = (\mathfrak{K}, b^n),$$

$$\mathfrak{K}_n\mathfrak{N}_n \equiv 0(\mathfrak{K}),$$

where the a_n are defined by the fact that there is an f such that

$$f \equiv 0(\mathfrak{K}_n), \quad f \equiv 0(\mathfrak{N}_n), \quad f \equiv a_n b^n\ (\mathfrak{K}), \quad \text{but} \quad f \not\equiv l_n b^n\ (\mathfrak{K}), \quad l_n \equiv 0(\mathfrak{K}_n).$$

Thus, in general, $\mathfrak{K}_n\mathfrak{N}_n \equiv 0(\mathfrak{K})$, $\mathfrak{N}_n$ is a proper divisor of $\mathfrak{K}$ by (3), and $\mathfrak{K}_n$ is a proper divisor of $\mathfrak{K}_{n-1}$. By Theorem I on the finite chain, the chain of the $\mathfrak{K}_n$ must therefore terminate after finitely many steps, say with $\mathfrak{K}_n$. For every $f \equiv 0(\mathfrak{K}_n)$ and $f \equiv 0(\mathfrak{N}_n)$, then, one has $f \equiv l_n b^n\ (\mathfrak{K})$ with $l_n \equiv 0(\mathfrak{K}_n)$, and consequently, by the preceding argument, $\mathfrak{K} = [\mathfrak{K}_n, \mathfrak{N}_n]$, so that $\mathfrak{K}$ is recognized as reducible.

The *uniqueness of the associated prime ideals* follows from what has just been proved.

Let

$$\mathfrak{M} = [\mathfrak{B}_1 \cdots \mathfrak{B}_k] = [\mathfrak{D}_1 \cdots \mathfrak{D}_k]$$

be two shortest, hence reduced, representations of $\mathfrak{M}$ as the least common multiple of irreducible ideals, whose number agrees by Theorem IV. Then by that theorem also the intermediate representations occurring there (where, as noted there, the index $j_i = i$ may be put)

$$\mathfrak{M} = [\mathfrak{D}_1 \cdots \mathfrak{D}_{i-1}\mathfrak{B}_i\mathfrak{B}_{i+1} \cdots \mathfrak{B}_k] = [\mathfrak{D}_1 \cdots \mathfrak{D}_{i-1}\mathfrak{D}_i\mathfrak{B}_{i+1} \cdots \mathfrak{B}_k] = [\mathfrak{U}_i, \mathfrak{B}_i] = [\mathfrak{U}_i, \mathfrak{D}_i]$$

are shortest representations. Thus

$$\mathfrak{U}_i\mathfrak{B}_i \equiv 0(\mathfrak{D}_i), \qquad \mathfrak{U}_i \not\equiv 0(\mathfrak{D}_i), \qquad \mathfrak{U}_i\mathfrak{D}_i \equiv 0(\mathfrak{B}_i), \qquad \mathfrak{U}_i \not\equiv 0(\mathfrak{B}_i).$$

Since by Theorem VI the irreducible ideals $\mathfrak{B}_i$ and $\mathfrak{D}_i$ are primary, there follows the existence of two numbers λ_i and μ_i such that

$$\mathfrak{B}_i^{\lambda_i} \equiv 0(\mathfrak{D}_i), \qquad \mathfrak{D}_i^{\mu_i} \equiv 0(\mathfrak{B}_i). \tag{5}$$

Let $\mathfrak{P}_i$ and $\bar{\mathfrak{P}}_i$ denote the associated prime ideals of $\mathfrak{B}_i$ and $\mathfrak{D}_i$, respectively; thus $\mathfrak{P}_i^{\varrho_i} \equiv 0(\mathfrak{B}_i)$ and $\bar{\mathfrak{P}}_i^{\sigma_i} \equiv 0(\mathfrak{D}_i)$. From (5) then follows

$$\mathfrak{P}_i^{\lambda_i\varrho_i} \equiv 0(\bar{\mathfrak{P}}_i), \qquad \bar{\mathfrak{P}}_i^{\mu_i\sigma_i} \equiv 0(\mathfrak{P}_i),$$

and by the property of prime ideals

$$\mathfrak{P}_i \equiv 0(\bar{\mathfrak{P}}_i), \qquad \bar{\mathfrak{P}}_i \equiv 0(\mathfrak{P}_i), \qquad \mathfrak{P}_i = \bar{\mathfrak{P}}_i.$$

Thus:

Theorem VII. *In two different shortest representations of an ideal as the least common multiple of irreducible ideals, the associated prime ideals coincide; among them equal ones can also occur,*[20] *and indeed occur equally often in every decomposition. Consequently the ideals themselves can be paired in at least one way such that in each case a power of one ideal $\mathfrak{B}_i$ is divisible by the assigned ideal $\mathfrak{D}_i$, and conversely. Their number coincides by Theorem IV.*[21]

§ 5. Representation of an ideal as the least common multiple of greatest primary ideals. Uniqueness of the associated prime ideals.

Definition IV. *A shortest representation $\mathfrak{M} = [\mathfrak{Q}_1 \cdots \mathfrak{Q}_\alpha]$ will be called a least common multiple of greatest primary ideals if all $\mathfrak{Q}$ are primary, but the least common multiple of two $\mathfrak{Q}$ is no longer primary.*

That at least one such representation always exists follows from the representation of $\mathfrak{M}$ as a least common multiple of irreducible ideals. These ideals are primary; either there is already a representation by greatest primary ideals, or the least common multiple of some two ideals is again primary. Since the number of

[20] This is shown, for example, by the example in footnote 19: $(x^2, xy, y^\lambda) = [(x^2, y), (x, y^\lambda)]$, where $\lambda \geq 2$. The two associated prime ideals here are (x, y).

[21] For the uniqueness of the "isolated" ideals contained among the irreducible ideals, compare § 7.

ideals has thereby decreased by one, repetition of the procedure leads after finitely many steps to the desired representation.

This representation is reduced by Lemma IV. Conversely, every reduced representation by greatest primary ideals arises in this way, as the decomposition of the $\mathfrak{Q}$ into irreducible ideals shows.

In order to derive here from Theorem VII a corresponding uniqueness theorem, the connection with the associated prime ideals has to be investigated.

Theorem VIII. *If the primary ideals $\mathfrak{N}_1, \mathfrak{N}_2, \ldots, \mathfrak{N}_\lambda$ all possess the same associated prime ideal $\mathfrak{P}$, then their least common multiple $\mathfrak{Q} = [\mathfrak{N}_1, \mathfrak{N}_2, \ldots, \mathfrak{N}_\lambda]$ is also primary and has $\mathfrak{P}$ as associated prime ideal. Conversely, if $\mathfrak{Q} = [\mathfrak{N}_1 \cdots \mathfrak{N}_\lambda]$ is a reduced representation of the primary ideal $\mathfrak{Q}$, then all $\mathfrak{N}_i$ are primary and have, as associated prime ideal, the associated prime ideal $\mathfrak{P}$ of $\mathfrak{Q}$.*

For the proof of the first part, first observe that from $\mathfrak{P}^{\varrho_i} \equiv 0(\mathfrak{N}_i)$ for every i it also follows that $\mathfrak{P}^{\tau} \equiv 0(\mathfrak{Q})$, where τ denotes the largest of the indices ϱ_i. Since $\mathfrak{P}$ is also a divisor of $\mathfrak{Q}$, $\mathfrak{P}$ is necessarily the associated prime ideal if $\mathfrak{Q}$ is primary. From

$$\mathfrak{A}\mathfrak{B} \equiv 0(\mathfrak{Q}), \qquad \mathfrak{B}^k \not\equiv 0(\mathfrak{Q}) \quad \text{(for every } k)$$

it follows that

$$\mathfrak{B} \not\equiv 0(\mathfrak{P}), \quad \text{hence} \quad \mathfrak{B}^k \not\equiv 0(\mathfrak{N}_i) \quad \text{(for every } k),$$

and consequently $\mathfrak{A} \equiv 0(\mathfrak{N}_i)$; hence $\mathfrak{A} \equiv 0(\mathfrak{Q})$. Thus $\mathfrak{Q}$ is proved primary, and $\mathfrak{P}$ the associated prime ideal.

Conversely, let first

$$\mathfrak{Q} = [\mathfrak{N}_1 \cdots \mathfrak{N}_\lambda] = [\mathfrak{N}_i, \mathfrak{N}'_i]$$

be a shortest representation of $\mathfrak{Q}$ by primary ideals $\mathfrak{N}_i$, and let $\mathfrak{P}_i$ be the associated prime ideals. From

$$\mathfrak{N}_i\mathfrak{N}'_i \equiv 0(\mathfrak{Q}), \qquad \mathfrak{N}'_i \not\equiv 0(\mathfrak{N}_i)$$

(because the representation is shortest) follows

$$\mathfrak{N}'_i \not\equiv 0(\mathfrak{P}_i), \qquad \text{but} \qquad \mathfrak{N}'_i \equiv 0(\mathfrak{P}).$$

Since $\mathfrak{P}_i$ is at the same time a divisor of $\mathfrak{Q}$, $\mathfrak{P}_i$ thus agrees, for every i, with the associated prime ideal $\mathfrak{P}$ of $\mathfrak{Q}$.

It remains to show that in every reduced representation $\mathfrak{Q} = [\mathfrak{N}_1 \cdots \mathfrak{N}_\lambda]$ the $\mathfrak{N}_i$ are primary.[22] To this end, decompose the $\mathfrak{N}_i$ into their irreducible ideals $\mathfrak{B}$; in the resulting shortest representation $\mathfrak{Q} = [\mathfrak{B}_1 \cdots \mathfrak{B}_s]$, every ideal is primary and, by what has just been proved, has $\mathfrak{P}$ as associated prime ideal. Then by the first part of the theorem proved above the same holds for each $\mathfrak{N}_i$, completing the proof of Theorem VIII.

Addendum. It also follows that a prime ideal is necessarily irreducible. For from the reduced representation $\mathfrak{P} = [\mathfrak{N}, \mathfrak{K}]$ one obtains by Theorem VIII $\mathfrak{N} \equiv 0(\mathfrak{P})$; hence, since $\mathfrak{P} \equiv 0(\mathfrak{N})$, also $\mathfrak{P} = \mathfrak{N}$, and similarly $\mathfrak{P} = \mathfrak{K}$. The irreducibility of $\mathfrak{P}$ also follows directly: for from $\mathfrak{P} = [\mathfrak{N}, \mathfrak{K}]$ follows $\mathfrak{N}\mathfrak{K} \equiv 0(\mathfrak{P})$, $\mathfrak{N} \not\equiv 0(\mathfrak{P})$, $\mathfrak{K} \not\equiv 0(\mathfrak{P})$, contrary to the defining property of a prime ideal.

Now let

$$\mathfrak{M} = [\mathfrak{Q}_1 \cdots \mathfrak{Q}_\alpha] = [\mathfrak{Q}'_1 \cdots \mathfrak{Q}'_\beta]$$

be two reduced representations of $\mathfrak{M}$ as the least common multiple of greatest primary ideals. Decomposing the $\mathfrak{Q}$ into irreducible ideals $\mathfrak{B}$, and the $\mathfrak{Q}'$ correspondingly into irreducible ideals $\mathfrak{D}$, gives two reduced representations of $\mathfrak{M}$ as the least common multiple of irreducible ideals; by Theorem VII, both the number of components and the associated prime ideals coincide. By Theorem VIII, all irreducible ideals $\mathfrak{B}$ occurring for a fixed $\mathfrak{Q}_i$ have the same associated prime ideal $\mathfrak{P}_i$; whereas the $\mathfrak{P}_j$ associated to $\mathfrak{Q}_j$ is necessarily different from this, since otherwise, by Theorem VIII, there would be no representation by greatest primary ideals. Hence the number α of the $\mathfrak{Q}$ is equal to the number of different associated prime ideals $\mathfrak{P}$ of the $\mathfrak{B}$; these different $\mathfrak{P}$ form the associated prime ideals of the $\mathfrak{Q}$. The same holds for the $\mathfrak{Q}'$ with respect to their decomposition into the $\mathfrak{D}$. Theorem VII therefore gives equality of number for the $\mathfrak{Q}$ and $\mathfrak{Q}'$, and agreement of their associated prime ideals. At the same time it is seen that the grouping of the irreducible ideals into greatest primary ideals described at the beginning of the paragraph consists in combining all, and only those, with the same associated prime ideal. Theorem VIII further shows the irreducibility property of the greatest primary ideals: they admit no reduced representation as the least common multiple of greatest primary ideals.

[22] That the reduced representation is essential here is shown by the example of the non-reduced representation $\mathfrak{Q} = [x^2, xy, y^\lambda] = [(x^2, xy, y^\lambda, yz), (x, y^\lambda)]$, where $\lambda \geq 2$. Here $(x^2, xy, y^\lambda, yz) = [(x^2, y), (x, y^\lambda, z)]$ is not primary by what has just been proved; for the latter representation is a shortest one by primary ideals, but the associated prime ideals (x, y) and (x, y, z) are different. ($\mathfrak{Q}$ is primary by footnote 19.)

In summary:

Theorem IX. *In two reduced representations of an ideal as the least common multiple of greatest primary ideals, the number of components and the associated prime ideals, all of which are distinct from each other, coincide. In other words, to each $\mathfrak{Q}$ there can be assigned one and only one $\mathfrak{Q}'$ such that a power of $\mathfrak{Q}$ is divisible by $\mathfrak{Q}'$, and conversely.*[23] *The $\mathfrak{Q}$ and $\mathfrak{Q}'$ have the irreducibility property with respect to decomposition into greatest primary ideals.*

Addendum. It should be noted that Theorem IX remains essentially valid if one assumes only a shortest representation instead of a reduced one. If, say,

$$\mathfrak{M} = [\mathfrak{Q}_1 \cdots \mathfrak{Q}_i^* \cdots \mathfrak{Q}_\alpha]$$

is reduced with respect to $\mathfrak{Q}_i^*$, and $\mathfrak{Q}_i^*$ is a proper divisor of $\mathfrak{Q}_i$, while $\mathfrak{U}_i$ is the complement of $\mathfrak{Q}_i$, then by Lemma IV

$$\mathfrak{Q}_i = [\mathfrak{Q}_i^*, (\mathfrak{U}_i, \mathfrak{Q}_i)],$$

and this representation is reduced with respect to $\mathfrak{Q}_i^*$. By Theorem VIII – replacing $(\mathfrak{U}_i, \mathfrak{Q}_i)$ by a proper divisor if necessary when applying the theorem – $\mathfrak{Q}_i^*$ is therefore primary and has the same associated prime ideal $\mathfrak{P}_i$ as $\mathfrak{Q}_i$. Continuing the procedure shows that to every such representation there can be assigned a reduced representation by greatest primary ideals in such a way that the number of components and the associated prime ideals coincide. Thus even for shortest representations it holds that in two different representations the number of components and the associated prime ideals coincide.

The associated, mutually distinct prime ideals thereby uniquely defined will be called simply "the associated prime ideals of $\mathfrak{M}$."

§ 6. Unique representation of an ideal as the least common multiple of relatively-prime irreducible ideals.

Definition V. *An ideal $\mathfrak{R}$ is called relatively prime to $\mathfrak{S}$ if from*

$$\mathfrak{T}\mathfrak{R} \equiv 0(\mathfrak{S})$$

it necessarily follows that $\mathfrak{T} \equiv 0(\mathfrak{S})$. If both $\mathfrak{R}$ is relatively prime to $\mathfrak{S}$ and $\mathfrak{S}$ is relatively prime to $\mathfrak{R}$, then $\mathfrak{R}$ and $\mathfrak{S}$ are called mutually relatively prime.[24]

An ideal is called relatively-prime irreducible if it cannot be represented as the least common multiple of mutually relatively prime proper divisors.

In particular, if instead of $\mathfrak{T}$ one takes the greatest common divisor $\mathfrak{T}_0$ of all $\mathfrak{T}$ for which $\mathfrak{T}\mathfrak{R} \equiv 0(\mathfrak{S})$, then also $\mathfrak{T}_0\mathfrak{R} \equiv 0(\mathfrak{S})$ and $\mathfrak{S} \equiv 0(\mathfrak{T}_0)$. Thus $\mathfrak{T}_0 = \mathfrak{S}$ when $\mathfrak{R}$ is relatively prime to $\mathfrak{S}$; and $\mathfrak{T}_0$ is a proper divisor of $\mathfrak{S}$ when $\mathfrak{R}$ is not relatively prime to $\mathfrak{S}$.[25]

The uniqueness proof rests on:

Theorem X. *1. If $\mathfrak{R}$ is relatively prime to the ideals $\mathfrak{S}_1, \ldots, \mathfrak{S}_\lambda$, then $\mathfrak{R}$ is also relatively prime to their least common multiple $\mathfrak{S}$.*

2. If the ideals $\mathfrak{S}_1, \ldots, \mathfrak{S}_\lambda$ are relatively prime to $\mathfrak{R}$, then their least common multiple $\mathfrak{S}$ is also relatively prime to $\mathfrak{R}$.

3. If $\mathfrak{R}$ is relatively prime to $\mathfrak{S}$ and $\mathfrak{S} = [\mathfrak{S}_1 \cdots \mathfrak{S}_\lambda]$ is a reduced representation for $\mathfrak{S}$, then $\mathfrak{R}$ is also relatively prime to each $\mathfrak{S}_i$.

4. If $\mathfrak{S}$ is relatively prime to $\mathfrak{R}$, then every divisor $\mathfrak{S}_i$ of $\mathfrak{S}$ is also relatively prime to $\mathfrak{R}$.

1. Since $\mathfrak{T}\mathfrak{R} \equiv 0(\mathfrak{S})$ necessarily gives $\mathfrak{T}\mathfrak{R} \equiv 0(\mathfrak{S}_i)$, the hypothesis gives $\mathfrak{T} \equiv 0(\mathfrak{S}_i)$, and hence $\mathfrak{T} \equiv 0(\mathfrak{S})$.

2. Put

$$\widehat{\mathfrak{S}}_1 = [\mathfrak{S}_2 \cdots \mathfrak{S}_\lambda], \quad \widehat{\mathfrak{S}}_{12} = [\mathfrak{S}_3 \cdots \mathfrak{S}_\lambda], \quad \ldots, \quad \widehat{\mathfrak{S}}_{12\ldots\lambda-1} = \mathfrak{S}_\lambda.$$

From $\mathfrak{T}\mathfrak{S} \equiv 0(\mathfrak{R})$ it follows that $\mathfrak{T}\mathfrak{S}_1\widehat{\mathfrak{S}}_1 \equiv 0(\mathfrak{R})$; hence, by the hypothesis, $\mathfrak{T}\widehat{\mathfrak{S}}_1 \equiv 0(\mathfrak{R})$. Applying the same argument successively gives $\mathfrak{T}\widehat{\mathfrak{S}}_{12} \equiv 0(\mathfrak{R})$ and finally $\mathfrak{T}\widehat{\mathfrak{S}}_{12\ldots\lambda-1} = \mathfrak{T}\mathfrak{S}_\lambda \equiv 0(\mathfrak{R})$; hence $\mathfrak{T} \equiv 0(\mathfrak{R})$.

[23]An example of different representations is the one given in footnote 12 for Theorem II: $(x^2, xy) = [(x), (x^2, \mu x + y)]$ for arbitrary μ. Since the associated prime ideals $\mathfrak{P}_1 = (x)$ and $\mathfrak{P}_2 = (x, y)$ are distinct, these are greatest primary ideals. – For the uniqueness of the "isolated" greatest primary ideals, compare § 7.

[24]The condition of being relatively prime is not symmetric. For example, $\mathfrak{R} = (x^2, y)$ is relatively prime to $\mathfrak{S} = (x)$; but $\mathfrak{S}$ is not relatively prime to $\mathfrak{R}$, since $\mathfrak{S}^2 \equiv 0(\mathfrak{R})$, whereas $\mathfrak{S} \not\equiv 0(\mathfrak{R})$.

[25]The $\mathfrak{T}_0$ so defined agrees with Lasker's "residual module"; and, in its extension to number modules in place of ideals, with Dedekind's "quotient" of two modules. Lasker, loc. cit., p. 49; Dedekind (Zahlentheorie), p. 504.

3. Let $\widehat{\mathfrak{S}}_i$ denote the complement of $\mathfrak{S}_i$. From $\mathfrak{T}_0\mathfrak{R} \equiv 0(\mathfrak{S}_i)$, together with $\widehat{\mathfrak{S}}_i\mathfrak{R} \equiv 0(\mathfrak{S}_i)$, it follows that $[\mathfrak{T}_0, \widehat{\mathfrak{S}}_i]\mathfrak{R} \equiv 0(\mathfrak{S})$. If $\mathfrak{T}_0$ were a proper divisor of $\mathfrak{S}_i$, then, since $\mathfrak{S} = [\mathfrak{S}_i, \widehat{\mathfrak{S}}_i]$ is reduced, $[\mathfrak{T}_0, \widehat{\mathfrak{S}}_i]$ would be a proper divisor of $\mathfrak{S}$, contrary to the hypothesis.

4. From $\mathfrak{T}\mathfrak{S}_i \equiv 0(\mathfrak{R})$, where $\mathfrak{T}$ is a proper divisor of $\mathfrak{R}$, it follows also that $\mathfrak{T}\mathfrak{S} \equiv 0(\mathfrak{R})$; hence $\mathfrak{T}$ would be a proper divisor of $\mathfrak{R}$, contrary to the hypothesis.

Definition V shows in particular that every ideal is relatively prime to the unit ideal $\mathfrak{O}$ consisting of all elements of Σ;[26] the following theorems, however, apply only to ideals distinct from $\mathfrak{O}$.

From Theorem X one first obtains:

Lemma V. *Every representation of an ideal as the least common multiple of mutually relatively prime ideals distinct from $\mathfrak{O}$ is reduced.*

Let

$$\mathfrak{M} = [\mathfrak{R}_1\mathfrak{R}_2 \cdots \mathfrak{R}_\sigma] = [\mathfrak{R}_i, \mathfrak{L}_i]$$

be such a representation. By Theorem X, 2, $\mathfrak{L}_i$ is also relatively prime to $\mathfrak{R}_i$; hence $\mathfrak{R}_i$ cannot be contained in $\mathfrak{L}_i$, and the representation is shortest. For from $\mathfrak{L}_i \equiv 0(\mathfrak{R}_i)$, with $\mathfrak{L}_i$ relatively prime to $\mathfrak{R}_i$, it would follow from $\mathfrak{O}\mathfrak{L}_i \equiv 0(\mathfrak{R}_i)$ that $\mathfrak{O} \equiv 0(\mathfrak{R}_i)$, hence $\mathfrak{R}_i = \mathfrak{O}$, contrary to the hypothesis.

If now $\mathfrak{R}_i$ could be replaced by a proper divisor $\mathfrak{R}_i^*$, then by Lemma II $\mathfrak{R}_i$ would be reducible and

$$\mathfrak{R}_i = [\mathfrak{R}_i^*, (\mathfrak{R}_i, \mathfrak{L}_i)].$$

If necessary, replace $(\mathfrak{R}_i, \mathfrak{L}_i)$ by a proper divisor $(\mathfrak{R}_i, \mathfrak{L}_i)^*$ and likewise replace $\mathfrak{R}_i^*$ by a proper divisor so that a reduced representation of $\mathfrak{R}_i$ is obtained. Then Theorem X, 3 shows that $\mathfrak{L}_i$ is also relatively prime to $(\mathfrak{R}_i, \mathfrak{L}_i)^*$, and this ideal is distinct from $\mathfrak{O}$ because the original representation was shortest. But $(\mathfrak{R}_i, \mathfrak{L}_i)^*$ is contained in $(\mathfrak{R}_i, \mathfrak{L}_i)$ and $(\mathfrak{R}_i, \mathfrak{L}_i)$ is contained in $\mathfrak{L}_i$, giving the contradiction.

Theorem X further gives the following theorem, which mediates the connection with the associated prime ideals.

Theorem XI. *If $\mathfrak{R}$ is relatively prime to $\mathfrak{S}$ and $\mathfrak{S}$ is distinct from $\mathfrak{O}$, then no associated prime ideal of $\mathfrak{R}$[27] is divisible by an associated prime ideal of $\mathfrak{S}$. Conversely, if no such divisibility holds, then $\mathfrak{R}$ is relatively prime to $\mathfrak{S}$, and of course $\mathfrak{S}$ is distinct from $\mathfrak{O}$.*

Let

$$\mathfrak{R} = [\mathfrak{Q}_1 \cdots \mathfrak{Q}_\alpha], \qquad \mathfrak{S} = [\mathfrak{Q}_1^* \cdots \mathfrak{Q}_\beta^*]$$

be reduced representations of $\mathfrak{R}$ and $\mathfrak{S}$ by greatest primary ideals, and let $\mathfrak{P}_1, \ldots, \mathfrak{P}_\alpha$ and $\mathfrak{P}_1^*, \ldots, \mathfrak{P}_\beta^*$ be their associated prime ideals. We prove the assertion in the form: if some $\mathfrak{P}$ is divisible by some $\mathfrak{P}^*$, then $\mathfrak{R}$ cannot be relatively prime to $\mathfrak{S}$, and conversely.

Suppose then that $\mathfrak{P}_\mu \equiv 0(\mathfrak{P}_\nu^*)$. Then also $\mathfrak{Q}_\mu^\varrho \equiv 0(\mathfrak{P}_\nu^*)$, and by the definition of $\mathfrak{P}_\nu^*$ this gives $\mathfrak{Q}_\mu^\varrho \equiv 0(\mathfrak{Q}_\nu^*)$, hence $\mathfrak{R}^\varrho \equiv 0(\mathfrak{Q}_\nu^*)$. Let $\mathfrak{R}^\tau$ be the lowest power of $\mathfrak{R}$ divisible by $\mathfrak{Q}_\nu^*$. If $\tau = 1$, then $\mathfrak{R}$ is divisible by $\mathfrak{Q}_\nu^*$; but since $\mathfrak{S} \neq \mathfrak{O}$, $\mathfrak{R}$ is not relatively prime to $\mathfrak{Q}_\nu^*$. If $\tau \geq 2$, then $\mathfrak{R}^{\tau-1}\mathfrak{R} \equiv 0(\mathfrak{Q}_\nu^*)$ while $\mathfrak{R}^{\tau-1} \not\equiv 0(\mathfrak{Q}_\nu^*)$, so again $\mathfrak{R}$ is not relatively prime to $\mathfrak{Q}_\nu^*$;[28] in both cases Theorem X, 3 then implies that $\mathfrak{R}$ is not relatively prime to $\mathfrak{S}$.

Conversely, if $\mathfrak{R}$ is not relatively prime to $\mathfrak{S}$, then by Theorem X, 1 it is not relatively prime to at least one $\mathfrak{Q}_\nu^*$. Thus

$$\mathfrak{T}_0\mathfrak{R} \equiv 0(\mathfrak{Q}_\nu^*), \qquad \mathfrak{T}_0 \not\equiv 0(\mathfrak{Q}_\nu^*),$$

and since $\mathfrak{Q}_\nu^*$ is primary,

$$\mathfrak{R}^\varrho \equiv 0(\mathfrak{Q}_\nu^*), \qquad \mathfrak{Q}_1^\varrho \cdots \mathfrak{Q}_\alpha^\varrho \equiv 0(\mathfrak{Q}_\nu^*).$$

For the associated prime ideals this implies

$$\mathfrak{P}_1^{\varrho_1} \cdots \mathfrak{P}_\alpha^{\varrho_\alpha} \equiv 0(\mathfrak{P}_\nu^*),$$

and therefore, by the property of prime ideals, $\mathfrak{P}_\nu^*$ is contained in at least one of the $\mathfrak{P}$. This proves Theorem XI.

Theorems X and XI now give the existence[29] and uniqueness of decomposition into relatively-prime irreducible ideals as follows.

[26] $\mathfrak{O}$ plays the role of the unit only with respect to divisibility and least common multiple, not with respect to product formation. For instance, $\mathfrak{O} = (x)$ for the domain of all integral polynomials in x without constant term; and $\mathfrak{O} = (2)$ for the domain of all even numbers.

[27] The associated prime ideals of $\mathfrak{R}$ were defined at the end of § 5.

[28] $\mathfrak{R}^0$ is not defined, because Σ need not contain a unit; hence the case $\tau = 1$ had to be dealt with first. The case $\tau = 0$ is also excluded, if Σ has a unit, by the hypothesis on $\mathfrak{S}$.

[29] The existence of the decomposition can also be proved directly, in exact analogy with the proof in § 2 of the existence of a decomposition into finitely many irreducible ideals.

Let $\mathfrak{M} = [\mathfrak{Q}_1 \cdots \mathfrak{Q}_\alpha]$ be a reduced (or at least shortest) representation of $\mathfrak{M}$ by greatest primary ideals, and let $\mathfrak{P}_1, \dots, \mathfrak{P}_\alpha$ be the associated prime ideals. We collect the $\mathfrak{P}$ into groups so that no ideal of one group is divisible by an ideal of a different group, while a single group cannot be split into two subgroups both having this property.

To construct such a grouping, note that by definition a single group G must contain, together with each ideal $\mathfrak{P}$, all divisors and multiples of $\mathfrak{P}$ occurring among $\mathfrak{P}_1, \dots, \mathfrak{P}_\alpha$. Let, for example, $\mathfrak{P}_j^{(i)}$ be all multiples of $\mathfrak{P}$, $\mathfrak{P}_{j_1}^{(i_1)}$ all divisors of $\mathfrak{P}_j^{(i)}$, $\mathfrak{P}_{j_1}^{(i_1 i_2)}$ all multiples of $\mathfrak{P}_{j_1}^{(i_1)}$, and so on. Since altogether only finitely many ideals $\mathfrak{P}$ occur, the procedure must stop after finitely many steps. The set of ideals so obtained is then a group G with the required properties: it contains all divisors and all multiples prescribed by construction, and no ideal outside it can be a divisor or a multiple of an ideal inside it.

The group G also satisfies the irreducibility condition. If it were divided into two subgroups $G^{(1)}$ and $G^{(2)}$, and if $G^{(1)}$ contained one of the ideals occurring at some stage of the alternating construction, then it would contain all preceding ones, since these are alternately multiples and divisors (or divisors and multiples), hence also $\mathfrak{P}$ and therefore the whole group G. Repeating the construction with the ideals not contained in G yields a decomposition into groups $G_1, \dots, G_\sigma$ of all the $\mathfrak{P}$, and the same argument shows that this grouping is unique.

Let $\mathfrak{Q}_{i\mu}$, where μ runs from 1 to λ_i, denote the ideals $\mathfrak{Q}$ whose associated prime ideals are collected in a group G_i. Since the $\mathfrak{P}$ are all distinct, the primary ideals $\mathfrak{Q}_{i\mu}$ belonging to a shortest representation are uniquely determined. Put

$$\mathfrak{R}_i = [\mathfrak{Q}_{i1} \cdots \mathfrak{Q}_{i\lambda_i}], \qquad \mathfrak{M} = [\mathfrak{R}_1 \cdots \mathfrak{R}_\sigma].$$

Theorem XI remains applicable even if $\mathfrak{R}_i$ is only a shortest representation, since by the addendum to Theorem IX its associated prime ideals are already well-defined. Because no associated prime ideal of $\mathfrak{R}_i$ is divisible by one of $\mathfrak{R}_j$, and conversely, Theorem XI shows that $\mathfrak{R}_i$ and $\mathfrak{R}_j$ are mutually relatively prime; and each $\mathfrak{R}_i$ is relatively-prime irreducible by the irreducibility property of the group G_i. By Lemma V the representation is reduced. Conversely, every decomposition into relatively-prime irreducible ideals leads, by Theorem XI, to the grouping just described.

Now let

$$\mathfrak{M} = [\bar{\mathfrak{R}}_1 \cdots \bar{\mathfrak{R}}_\tau]$$

be a second representation of $\mathfrak{M}$ by relatively-prime irreducible ideals. It is reduced by Lemma V. Decomposing the $\mathfrak{R}$ into greatest primary ideals shows that the associated prime ideals agree, and hence the groupings of these prime ideals agree. Thus $\tau = \sigma$, and the notation may be chosen so that $\mathfrak{R}_i$ and $\bar{\mathfrak{R}}_i$ belong to the same group. If

$$\mathfrak{M} = [\mathfrak{R}_i, \mathfrak{L}_i] = [\bar{\mathfrak{R}}_i, \bar{\mathfrak{L}}_i]$$

is the representation by ideal and complement, then, since $\bar{\mathfrak{R}}_i$ is assigned to the same group as $\mathfrak{R}_i$, Theorem XI gives that $\mathfrak{L}_i$ is relatively prime to $\bar{\mathfrak{R}}_i$, and $\bar{\mathfrak{L}}_i$ is relatively prime to $\mathfrak{R}_i$. From

$$\mathfrak{R}_i \mathfrak{L}_i \equiv 0(\bar{\mathfrak{R}}_i), \qquad \bar{\mathfrak{R}}_i \bar{\mathfrak{L}}_i \equiv 0(\mathfrak{R}_i)$$

therefore follows

$$\mathfrak{R}_i \equiv 0(\bar{\mathfrak{R}}_i), \qquad \bar{\mathfrak{R}}_i \equiv 0(\mathfrak{R}_i), \qquad \mathfrak{R}_i = \bar{\mathfrak{R}}_i.$$

Thus:

Theorem XII. *Every ideal can be represented uniquely as the least common multiple of finitely many mutually relatively prime and relatively-prime irreducible ideals.*

§ 7. Uniqueness of the isolated ideals.

Definition VI. *If the shortest representation* $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}]$ *is reduced with respect to* $\mathfrak{L}$*, then* $\mathfrak{R}$ *is called an isolated ideal if no associated prime ideal of* $\mathfrak{R}$ *is contained in an associated prime ideal of* $\mathfrak{L}$*; equivalently, if* $\mathfrak{L}$ *is relatively prime to* $\mathfrak{R}$*.*

Then the representation $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}]$ satisfies the conditions of the representation $\mathfrak{M} = [\mathfrak{R}_i, \mathfrak{L}_i]$ in Lemma V, and Lemma V proves:

Lemma VI. *If* $\mathfrak{R}$ *is an isolated ideal of the shortest representation* $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}]$*, reduced with respect to* $\mathfrak{L}$*, then the representation is also reduced with respect to* $\mathfrak{R}$*.*

Thus isolated ideals occur only in reduced representations. The associated prime ideals arising in the decompositions of $\mathfrak{R}$ and $\mathfrak{L}$ into irreducible ideals complement one another to the uniquely determined associated prime ideals arising in the corresponding decomposition of $\mathfrak{M}$. Hence no associated prime ideal belonging to the decomposition of $\mathfrak{R}$ into irreducible ideals is contained in the remaining associated prime ideals of $\mathfrak{M}$. Conversely, if this condition is satisfied and $\mathfrak{R}$ occurs in at least one representation $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}]$ reduced with respect to $\mathfrak{R}$, and hence also in a reduced representation $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}^*]$, then $\mathfrak{R}$ is isolated by Definition VI. This gives the following definition, independent of the particular complement $\mathfrak{L}$.

Definition VIa. *$\mathfrak{R}$ is called an isolated ideal if the prime ideals belonging to its decomposition into irreducible ideals are not contained in the other associated prime ideals belonging to the corresponding decomposition of $\mathfrak{M}$, and if $\mathfrak{R}$ occurs in at least one representation $\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}]$ reduced with respect to $\mathfrak{R}$.*[30]

Suppose

$$\mathfrak{M} = [\mathfrak{R}, \mathfrak{L}] = [\bar{\mathfrak{R}}, \bar{\mathfrak{L}}]$$

are two representations of $\mathfrak{M}$ by isolated ideals $\mathfrak{R}$ and $\bar{\mathfrak{R}}$, with complements $\mathfrak{L}$ and $\bar{\mathfrak{L}}$, such that the associated prime ideals of $\mathfrak{R}$ and $\bar{\mathfrak{R}}$ agree. Replace $\mathfrak{L}$ and $\bar{\mathfrak{L}}$ by divisors $\mathfrak{L}^*$ and $\bar{\mathfrak{L}}^*$ so that the representations become reduced. Then the associated prime ideals of $\mathfrak{L}^*$ and $\bar{\mathfrak{L}}^*$ also agree; by Theorem XI, $\mathfrak{L}^*$ is relatively prime to $\bar{\mathfrak{R}}$ and $\bar{\mathfrak{L}}^*$ is relatively prime to $\mathfrak{R}$. Since

$$\mathfrak{R}\mathfrak{L}^* \equiv 0(\bar{\mathfrak{R}}), \qquad \bar{\mathfrak{R}}\bar{\mathfrak{L}}^* \equiv 0(\mathfrak{R}),$$

we obtain

$$\mathfrak{R} \equiv 0(\bar{\mathfrak{R}}), \qquad \bar{\mathfrak{R}} \equiv 0(\mathfrak{R}), \qquad \mathfrak{R} = \bar{\mathfrak{R}}.$$

Thus isolated ideals are uniquely determined by their associated prime ideals. This sharpens Theorems VII and IX on decompositions into irreducible, respectively greatest primary, ideals; by the addendum to Theorem IX it is enough to assume a shortest representation. In summary:

Theorem XIII. *In every shortest representation of an ideal as the least common multiple of irreducible, respectively greatest primary, ideals, the isolated irreducible, respectively greatest primary, ideals are uniquely determined; the ambiguity concerns only the non-isolated irreducible, respectively greatest primary, ideals.*[31] *In general, isolated ideals are uniquely determined by their associated prime ideals.*

If in such a shortest representation by irreducible, respectively greatest primary, ideals the ideals $\mathfrak{B}_i$, respectively $\mathfrak{Q}_j$, are non-isolated, then by definition their complements $\mathfrak{U}_i$, respectively $\mathfrak{L}_j$, are divisible by $\mathfrak{P}_i$, respectively $\mathfrak{P}_j$. Hence

$$\mathfrak{U}_i^{\varrho_i} \equiv 0(\mathfrak{B}_i) \qquad \text{respectively} \qquad \mathfrak{L}_j^{\sigma_j} \equiv 0(\mathfrak{Q}_j).$$

Conversely, if these relations hold, then $\mathfrak{P}_i$, respectively $\mathfrak{P}_j$, is contained in at least one associated prime ideal of the complement, and hence, by the addendum to Theorem IX, in an associated prime ideal of the divisor $\mathfrak{L}_j^*$ of $\mathfrak{L}_j$ that supplies a representation reduced with respect to $\mathfrak{L}_j^*$. Thus $\mathfrak{B}_i$, respectively $\mathfrak{Q}_j$, is non-isolated. In particular, irreducible ideals $\mathfrak{B}_i$ whose associated prime ideal occurs more than once in the decomposition of $\mathfrak{M}$ are always non-isolated. *Non-isolated primary ideals are therefore also characterized by the fact that a power of every complement is divisible by them; isolated primary ideals by the fact that this cannot occur.*

§ 8. Unique representation of an ideal as a product of coprime-irreducible ideals.

If the underlying ring domain Σ has a unit, that is, an element ε such that $\varepsilon a = a$ for every element of Σ,[32] then coprime ideals can be defined as follows.

Definition VIII. *Two ideals $\mathfrak{R}$ and $\mathfrak{S}$ are called coprime if their greatest common divisor is the unit ideal $\mathfrak{D} = (\varepsilon)$ consisting of all elements of Σ. An ideal is called coprime-irreducible if it cannot be represented as the least common multiple of pairwise coprime ideals.*

Note that two coprime ideals are always mutually relatively prime. By definition there are elements $r \equiv 0(\mathfrak{R})$ and $s \equiv 0(\mathfrak{S})$ with $\varepsilon = r + s$. From $\mathfrak{T}\mathfrak{R} \equiv 0(\mathfrak{S})$ it follows that $\mathfrak{T}r \equiv 0(\mathfrak{S})$, hence $\mathfrak{T}\varepsilon = \mathfrak{T} \equiv 0(\mathfrak{S})$; correspondingly, from $\mathfrak{T}\mathfrak{S} \equiv 0(\mathfrak{R})$ it follows that $\mathfrak{T} \equiv 0(\mathfrak{R})$.[33] Thus Lemma V implies that every representation by pairwise coprime ideals is also reduced.

The uniqueness proof rests on the analogue of Theorem X:

[30] If one introduced ordinary associated prime ideals (end of § 5), then one would have to add the special condition that those of $\mathfrak{L}$ are all distinct from those of $\mathfrak{R}$. With Definition VIa, the representation no longer has to be assumed reduced with respect to the complement.

[31] For ideals in polynomial domains, this theorem was already stated without proof by Macaulay in the case of decomposition into greatest primary ideals; his definition of isolated and non-isolated (imbedded) primary ideals can be regarded as an irrational form of the one given below.

[32] Because multiplication is commutative, Σ can of course have at most one unit: for any two units ε_1 and ε_2, one has $\varepsilon_1\varepsilon_2 = \varepsilon_2 = \varepsilon_1$.

[33] The converse is false: for example, $\mathfrak{R} = (x)$ and $\mathfrak{S} = (y)$ are mutually relatively prime, but not coprime.

Theorem XIV. *If $\mathfrak{R}$ is coprime to each of the ideals $\mathfrak{S}_1, \ldots, \mathfrak{S}_\lambda$, then $\mathfrak{R}$ is also coprime to $\mathfrak{S} = [\mathfrak{S}_1 \cdots \mathfrak{S}_\lambda]$. Conversely, if $\mathfrak{R}$ and $\mathfrak{S}$ are coprime, then $\mathfrak{R}$ is coprime to each $\mathfrak{S}_j$. If $\mathfrak{R} = [\mathfrak{R}_1 \cdots \mathfrak{R}_\mu]$ and every $\mathfrak{R}_i$ is coprime to every $\mathfrak{S}_j$, then $\mathfrak{R}$ and $\mathfrak{S}$ are coprime; here too the converse holds.*

Indeed, if $\mathfrak{R}$ is coprime to every $\mathfrak{S}_j$, then there are elements s_j such that

$$s_j \equiv 0(\mathfrak{S}_j), \qquad s_j \equiv \varepsilon(\mathfrak{R}).$$

Consequently

$$s_1 s_2 \cdots s_\lambda \equiv 0(\mathfrak{S}), \qquad s_1 s_2 \cdots s_\lambda \equiv \varepsilon(\mathfrak{R}), \qquad (\mathfrak{R}, \mathfrak{S}) = (\varepsilon).$$

Since $(\mathfrak{R}, \mathfrak{S})$ is divisible by each $(\mathfrak{R}, \mathfrak{S}_j)$, the converse follows. Repeated application of the same argument gives the second part. For if $\mathfrak{R}_i$ is coprime to $\mathfrak{S}_1, \ldots, \mathfrak{S}_\lambda$ for a fixed i, then $\mathfrak{R}_i$ is coprime to $\mathfrak{S}$; if this holds for every i, then, since coprimeness is mutual, $\mathfrak{S}$ is coprime to $\mathfrak{R}$. Conversely, coprimeness of $\mathfrak{R}$ and $\mathfrak{S}$ gives coprimeness of $\mathfrak{S}$ with $\mathfrak{R}_i$, and hence of $\mathfrak{R}_i$ with $\mathfrak{S}$.

The proof of existence and uniqueness[34] of the decomposition into coprime-irreducible ideals is, as in the corresponding proof for relatively-prime ideals, a matter of a unique grouping. Since, however, the relatively-prime irreducible ideals $\mathfrak{R}_1, \ldots, \mathfrak{R}_\sigma$ of $\mathfrak{M}$ are uniquely defined by Theorem XII, it is unnecessary here to go back to the associated prime ideals.

We collect the uniquely defined relatively-prime irreducible ideals $\mathfrak{R}_1, \ldots, \mathfrak{R}_\sigma$ of $\mathfrak{M}$ into groups so that *every ideal of one group is coprime to every ideal of any different group, while a single group cannot be split into two subgroups such that every ideal of one subgroup is coprime to every ideal of the other subgroup.* Such a grouping is obtained as follows. By definition a group G must contain, together with every ideal $\mathfrak{R}$, all ideals not coprime to $\mathfrak{R}$. Let these be $\mathfrak{R}_{i_1}$; let $\mathfrak{R}_{i_1 i_2}$ be the ideals not coprime to these; in general let $\mathfrak{R}_{i_1 \ldots i_\lambda}$ be not coprime to $\mathfrak{R}_{i_1 \ldots i_{\lambda-1}}$. Since only finitely many ideals occur, this process terminates after finitely many steps. The set of ideals so obtained forms a group G with the desired properties: all ideals outside G are, by construction, coprime to all ideals in G. If G were split into two subgroups $G^{(1)}$ and $G^{(2)}$, and if $\mathfrak{R}_{i_1 \ldots i_\lambda}$ belonged to $G^{(1)}$, then, because coprimeness is mutual, $G^{(1)}$ would also have to contain $\mathfrak{R}_{i_1 \ldots i_{\lambda-1}}, \ldots, \mathfrak{R}_{i_1}$, hence also $\mathfrak{R}$ and therefore the whole group G. This proves irreducibility. Repeating the construction with the remaining ideals gives a grouping $G_1, \ldots, G_\tau$ of all $\mathfrak{R}$.

The grouping is unique: if $G_1', \ldots, G_\tau'$ is a second grouping and $\mathfrak{R}_{i_1 \ldots i_\lambda}$ is an element of G_i', then G_i' contains $\mathfrak{R}$ and hence G_1, and, by the irreducibility condition, cannot contain elements different from G_1; therefore $G_i' = G_1$.

Let $\mathfrak{T}_i$ be the least common multiple of the ideals $\mathfrak{R}$ united in a group G_i. We show that

$$\mathfrak{M} = [\mathfrak{T}_1 \cdots \mathfrak{T}_\tau]$$

is a representation of $\mathfrak{M}$ by coprime-irreducible ideals. Decomposing the $\mathfrak{T}$ into the $\mathfrak{R}$ first shows that $[\mathfrak{T}_1 \cdots \mathfrak{T}_\tau]$ really represents $\mathfrak{M}$. By Theorem XIV the $\mathfrak{T}$ are pairwise coprime, and each $\mathfrak{T}$ is coprime-irreducible. This proves the existence of such a representation.

For uniqueness, let $\mathfrak{M} = [\bar{\mathfrak{T}}_1 \cdots \bar{\mathfrak{T}}_\tau]$ be a second representation of this type. Decomposing the $\bar{\mathfrak{T}}$ into relatively-prime irreducible ideals $\mathfrak{R}$, Theorem XIV shows that ideals $\mathfrak{R}$ occurring in different $\bar{\mathfrak{T}}$ are coprime to one another, hence also mutually relatively prime; therefore they agree with the uniquely defined relatively-prime irreducible ideals $\mathfrak{R}$ of $\mathfrak{M}$. The ideals $\bar{\mathfrak{T}}_i$ also induce, by Theorem XIV, a grouping G_i' of the $\mathfrak{R}$ with the stated properties. Since this grouping is unique and each $\bar{\mathfrak{T}}_i$ is uniquely determined by the group $G_i' = G_i$, one has $\bar{\mathfrak{T}}_i = \mathfrak{T}_i$; uniqueness is proved.

For pairwise coprime ideals, the least common multiple is moreover equal to the product. For by Theorem XIV, if $\mathfrak{M} = [\mathfrak{T}_1 \cdots \mathfrak{T}_\tau]$, the complement $\mathfrak{L}_i$ is also coprime to $\mathfrak{T}_i$; hence there are elements

$$t_i \equiv 0(\mathfrak{T}_i), \qquad l_i \equiv 0(\mathfrak{L}_i), \qquad \varepsilon = t_i + l_i.$$

From $f \equiv 0(\mathfrak{T}_i)$ and $f \equiv 0(\mathfrak{L}_i)$ it follows that

$$f = f\varepsilon = ft_i + fl_i, \qquad f \equiv 0(\mathfrak{L}_i \mathfrak{T}_i).$$

Conversely $\mathfrak{L}_i \mathfrak{T}_i$ is divisible by $[\mathfrak{L}_i, \mathfrak{T}_i]$, whence $[\mathfrak{L}_i, \mathfrak{T}_i] = \mathfrak{L}_i \mathfrak{T}_i$; continuing the procedure on $\mathfrak{L}_i$ gives

$$\mathfrak{M} = [\mathfrak{T}_1 \cdots \mathfrak{T}_\tau] = \mathfrak{T}_1 \mathfrak{T}_2 \cdots \mathfrak{T}_\tau.$$

[34] The existence of the decomposition can again be proved directly in analogy with § 2; the uniqueness proof too can be carried out directly (compare what was said in the introduction about Schmeidler and Noether–Schmeidler). The proof given here also gives insight into the structure of coprime-irreducible ideals.

Therefore:

Theorem XV. *Every ideal can be represented uniquely as a product of finitely many pairwise coprime and coprime-irreducible ideals.*

§ 9. Extension of the investigation to modules. Equality of the number of components in decompositions into irreducible modules.

We now show that the content of the first three paragraphs, which concerns irreducible ideals and not primary and prime ideals, remains valid under weaker hypotheses. Those paragraphs do not use the commutative law of multiplication and refer only to the property of ideals of being modules. They therefore remain valid for modules with respect to non-commutative domains, which are now to be defined. The definition of modules is to be based on a double domain (Σ, T) with the following properties.

Σ is an abstractly defined non-commutative ring, that is, Σ is a system of elements $a, b, c, \ldots$ for which two modes of combination are defined, ring addition $(+)$ and ring multiplication $(\times)$, satisfying the laws stated in § 1, with the exception of the commutative law 4 for ring multiplication.

T is a system of elements $\alpha, \beta, \gamma, \ldots$, for which, in connection with Σ, two combinations are likewise defined: addition, which from any two elements α, β produces a unique third element $\alpha + \beta$; and multiplication of an element α of T by an element c of Σ, which uniquely produces an element $c\alpha$ of T.[35]

For these combinations the following laws hold:

1. The associative law of addition: $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$.
2. The commutative law of addition: $\alpha + \beta = \beta + \alpha$.
3. The law of unrestricted and unique subtraction: there is in T one and only one element ξ satisfying the equation $\alpha + \xi = \beta$ (one writes $\xi = \beta - \alpha$).
4. The associative law of multiplication: $a(b\gamma) = (a \times b)\gamma$.
5. The distributive law:
$$(a + b)\gamma = a\gamma + b\gamma, \qquad c(\alpha + \beta) = c\alpha + c\beta.$$

From these conditions there follow, as is well known, the existence of the zero element and the validity of the distributive law also for the combination of subtraction and multiplication:

$$(a - b)\gamma = a\gamma - b\gamma, \qquad c(\alpha - \beta) = c\alpha - c\beta,$$

where $(-)$ denotes subtraction in Σ. If Σ contains a unit ε, then $\varepsilon\alpha = \alpha$ is to hold for every element α of T.

By a module M in (Σ, T) we mean a system of elements of T satisfying the two conditions:

1. along with α, M contains $c\alpha$, where c is any element of Σ;
2. along with α and β, M contains the difference $\alpha - \beta$, hence also $n\alpha$ for every integer n.[36]

By this definition, T itself forms a module in (Σ, T). If in particular the domain T and the combinations established there coincide with the domain Σ and the combinations valid there, then the module M becomes a right ideal M in Σ. If Σ is also assumed commutative, the ordinary concept of ideal results, and is thus a special case of the concept of module.[37]

For modules all definitions of § 1 remain valid. Thus $\alpha \equiv 0\,(M)$, respectively $N \equiv 0\,(M)$, means that α, respectively every element of N, is an element of M; in other words, α, respectively N, is divisible by M. M is a proper divisor of N if M contains elements different from those of N; from $N \equiv 0\,(M)$ and $M \equiv 0\,(N)$ follows $M = N$. The definitions of greatest common divisor and least common multiple likewise remain literally the same. If the module M contains, in particular, a finite number of elements $\alpha_1, \ldots, \alpha_\rho$ such that

$$M = (\alpha_1, \ldots, \alpha_\rho),$$

[35]Thus here one is dealing with a "right-sided" multiplication, a "right-sided" domain T, and consequently with "right-sided" modules and ideals. If one were to base T instead on a left-sided multiplication αc, a corresponding theory of left-sided modules and ideals would result; M would contain αc along with α. The associative law would then have the form $(\gamma b)a = \gamma(b \times a)$.

[36]The integers are again to be understood as abbreviating symbols, not as ring elements.

[37]The simplest example of a module is the module of integral linear forms: here Σ consists of all rational integers and T of all integral linear forms. A somewhat more general module arises when, in Σ and T, one takes algebraic integers instead of rational integers, or for instance all even integers. If, instead of the linear forms, one considers each complex of coefficients as an element, then the combinations in Σ and T are in fact different. Ideals in non-commutative polynomial ring domains are the subject of the joint paper of Noether–Schmeidler. Ideals in other special non-commutative domains are treated in Hurwitz's lectures on the number theory of quaternions (Berlin, Springer 1919) and in the works of Du Pasquier cited there.

that is,

$$\alpha = c_1\alpha_1 + \dots + c_\rho\alpha_\rho + n_1\alpha_1 + \dots + n_\rho\alpha_\rho$$

for every $\alpha \equiv 0\,(M)$, where the c_i are elements of Σ and the n_i are integers, then M is called a finite module and $\alpha_1, \dots, \alpha_\rho$ a module basis.

In what follows we assume, just as in § 1, only those domains (Σ, T) that satisfy the finiteness condition: every module in (Σ, T) is finite, and hence has a module basis.

For such a domain (Σ, T), Theorem I on finite chains holds also for modules, as the proof given there shows, and therefore all hypotheses for §§ 2 and 3 are satisfied. Definition I and Lemma I on shortest and reduced representations carry over directly; likewise Theorem II on the representability of every module as the least common multiple of finitely many irreducible modules, where Lemma II shows that every shortest representation of this kind is at the same time reduced. Theorem III also remains valid, expressing reducibility of a module through properties of its complement; from it follows Lemma III, and finally Theorem IV, which asserts equality of the number of components in two different shortest representations of a module as the least common multiple of irreducible modules. Theorem IV also gives Lemma IV as the converse of Lemma I on reduced representations.

Addendum. The same argument shows that all these theorems and definitions remain valid if, for two-sided domains T, that is, domains which are both right- and left-sided, one everywhere understands by a module a two-sided module: one which, along with α, also contains $c\alpha$ and αc, and along with α and β also contains $\alpha - \beta$.

While all these theorems rest only on the concept of divisibility and of least common multiple, the further uniqueness theorems rest essentially on the product concept and therefore do not admit a direct transfer. Thus the definition of a primary ideal and of a prime ideal cannot be transferred to modules, since the product of two elements of T is not defined. The formally possible transfer to non-commutative rings loses its meaning, since here the existence of the associated prime ideal cannot be proved[38] and the proof that an irreducible ideal is primary also fails. These two facts form the basis of the following uniqueness theorems. By contrast, if the non-commutative ring has a unit, coprime and coprime-irreducible ideals can be defined, and the arguments used in the proof of Theorem II show that every ideal can be represented as the least common multiple of finitely many pairwise coprime and coprime-irreducible ideals.[39]

Finally, we mention a sufficient criterion for the finiteness condition to hold in (Σ, T): if Σ contains a unit and satisfies the finiteness condition, and if T itself is a finite module in (Σ, T), then every module in (Σ, T) is finite.

Indeed, from the existence of the unit in Σ, for

$$\mathfrak{M} = (f_1, \dots, f_e), \qquad f = \bar{b}_1 f_1 + \dots + \bar{b}_e f_e + n_1 f_1 + \dots + n_e f_e,$$

where the $\bar{b}_i$ are elements of Σ and the n_i are integers, there follows also a representation

$$f = b_1 f_1 + \dots + b_e f_e$$

with the b_i in Σ. For since $f_i = \varepsilon f_i$, one has $n_i f_i = n_i \varepsilon f_i$, and since $n_i\varepsilon = (\varepsilon + \dots + \varepsilon)$ belongs to Σ, $b_i = \bar{b}_i + n_i\varepsilon$ is an element of Σ. The hypothesis on T says that every element α of T admits a representation

$$\alpha = \bar{a}_1\tau_1 + \dots + \bar{a}_k\tau_k + n_1\tau_1 + \dots + n_k\tau_k,$$

and hence

$$\alpha = a_1\tau_1 + \dots + a_k\tau_k,$$

the second representation following from the first just as above because $\varepsilon\tau_i = \tau_i$.

As α runs through the elements of a module M in (Σ, T), the coefficients a_k of τ_k run through an ideal $\mathfrak{M}_k$ of Σ; by the preceding, for every $a_k \equiv 0\,(\mathfrak{M}_k)$ one has

$$a_k = b_1 a_k^{(1)} + \dots + b_e a_k^{(e)}.$$

Let $\alpha^{(i)}$ be an element of M for which the coefficient of τ_k is $a_k^{(i)}$. Then $\alpha - b_1\alpha^{(1)} - \dots - b_e\alpha^{(e)}$ is an element belonging to M which depends only on $\tau_1, \dots, \tau_{k-1}$. The procedure can be repeated for the totality of these elements, which no longer contain τ_k and which form a module M', and the assertion is proved by finitely many repetitions.

[38] For from $p_1^\rho \equiv 0\,(\mathfrak{Q})$ and $p_2^\sigma \equiv 0\,(\mathfrak{Q})$ it does not follow here that $(p_1 - p_2)^{\rho+\sigma} \equiv 0\,(\mathfrak{Q})$; likewise, from $(ab)^\rho \equiv 0\,(\mathfrak{Q})$ it does not follow that $a^\rho b^\rho \equiv 0\,(\mathfrak{Q})$. Thus $\mathfrak{P}$ can be proved neither to be an ideal nor to have the property of a prime ideal. For two-sided ideals $\mathfrak{P}$ can indeed be proved to be an ideal, but not a prime ideal.

[39] For special, "completely reducible" ideals, uniqueness theorems can be set up here too; compare the joint paper Noether–Schmeidler.

§ 10. Special case of the polynomial domain.

1. Let the underlying ring domain Σ consist of all polynomials in $x_1, \ldots, x_n$ with arbitrary complex coefficients. By Hilbert's basis theorem (Ann. 36) the finiteness condition holds for it. The issue is the relation of our theorems to the known theorems of elimination theory and module theory.

This relation is established by the following special case of a known theorem of Hilbert:[40]

If f vanishes for every finite system of values of $x_1, \ldots, x_n$ that is a zero of all polynomials of a prime ideal $\mathfrak{P}$ (a zero of $\mathfrak{P}$), then f is divisible by $\mathfrak{P}$. In other words, a prime ideal $\mathfrak{P}$ consists of the totality of polynomials which vanish at its zeros.[41]

Thus, if a product fg vanishes for all zeros of $\mathfrak{P}$, at least one factor vanishes; the zeros form an irreducible algebraic variety. Conversely, if this definition of irreducible variety is adopted, then the totality of polynomials vanishing on an irreducible variety forms a prime ideal. Prime ideal and irreducible variety therefore correspond one-to-one. Since $\mathfrak{P}^{\rho} \equiv 0\,(\mathfrak{Q})$ and $\mathfrak{Q} \equiv 0\,(\mathfrak{P})$, the zeros of a primary ideal coincide with those of its associated prime ideal.[42] Therefore the representation of an ideal as the least common multiple of greatest primary ideals gives a resolution of all zeros of the ideal into irreducible varieties; and, as Lasker showed, the converse also holds. The proof of uniqueness of the associated prime ideals thus corresponds here to the fundamental theorem of elimination theory on the unique decomposability of an algebraic variety into irreducible varieties, and for special polynomial domains, where no unique product representation of the polynomials by irreducible polynomials of the domain and consequently no elimination theory exists, it can serve as an equivalent of that theorem of elimination theory.

The irreducible varieties corresponding to the isolated primary ideals are exactly those which occur in the "minimal resolvent";[43] for the zeros of every non-isolated greatest primary ideal are at the same time zeros of at least one isolated one, namely one whose associated prime ideal is divisible by that of the non-isolated ideal. The uniqueness of isolated primary ideals therefore gives, in the exponents, new invariant multiplicity numbers. The uniqueness theorems for the resolution of primary ideals into irreducible ones can likewise be regarded as supplements to elimination theory in the sense of multiplicity.

After these remarks the different decompositions can be interpreted in terms of their behavior with respect to algebraic varieties. Pairwise coprime ideals correspond to varieties having no common zero; for mutually relatively prime ideals, no irreducible variety of one ideal is at the same time a common zero; the greatest primary ideals vanish only on irreducible varieties which are all distinct from one another; in the decomposition into irreducible ideals the same irreducible varieties may occur with multiplicity.

It should also be noted that, instead of the general polynomial domain, the domain of all homogeneous forms may also be taken as the basis; for one readily verifies that the general theorems remain valid for the combinations there in force, where addition is defined only for forms of the same dimension.[44]

A simplest example of the four different decompositions - for which the formulas follow below - is given, according to the preceding, by a line and two intersecting lines skew to it, one of the latter containing, with higher multiplicity, a point different from the point of intersection. The decomposition into coprime-irreducible ideals corresponds to the decomposition into the line and the variety skew to it; in the decomposition into relatively-prime irreducible ideals this variety splits into the two lines; the decomposition into greatest primary ideals corresponds to separating off the point of higher multiplicity, while the decomposition into irreducible ideals forces a resolution of that point.

Taking this point as the origin, the running line as the y-axis, the line intersecting it parallel to the x-axis, and the line skew to it parallel to the z-axis, such a configuration is represented, for example, by the following irreducible ideals:[45]

$$\mathfrak{B}_1 = (x-1, y), \quad \mathfrak{B}_2 = (y-1, z), \quad \mathfrak{B}_3 = (x, z), \quad \mathfrak{B}_4 = (x^3, y, z), \quad \mathfrak{B}_5 = (x^2, y^2, z).$$

[40] Über die vollen Invariantensysteme. Math. Ann. 42 (1893), § 3, p. 313.

[41] As Lasker showed [Math. Ann. 60 (1905), p. 607], this special case can also be proved directly for homogeneous forms; conversely, Hilbert's theorem follows again from it in the homogeneous and inhomogeneous cases. We can express that theorem as follows: if an ideal $\mathfrak{R}$ vanishes at all zeros of M, then a power of $\mathfrak{R}$ is divisible by M. This theorem, and also the special case, holds only if the range of values of the x is algebraically closed; hence it cannot follow from our theorems alone, but must use the existence of roots, for example at the point where one uses that an ideal whose zeros consist only of $x_1 = 0, \ldots, x_n = 0$ contains all products of powers of the x from some dimension on. The remaining proof can be somewhat simplified in comparison with Lasker by using our theorems. For Lasker must also appeal to Hilbert's theorem to prove the decomposition of an ideal into greatest primary ideals.

[42] Macaulay (compare the introduction) takes this property of a primary ideal, namely to possess an irreducible variety, as the definition; Lasker includes only the concept of the manifold of a variety in the definition and otherwise defines abstractly. The primary ideals vanishing only for $x_1 = 0, \ldots, x_n = 0$ occupy a special position for Lasker.

[43] Compare, for example, J. König, Einleitung in die allgemeine Theorie der algebraischen Größen (Leipzig, Teubner, 1903), p. 235.

[44] That in the ambiguous case inhomogeneous decompositions can exist beside homogeneous ones is shown by the example $(x^3, xy, y^3) = [(x^3, y), (y^3, x)] = [(xy, x^3, y^3, x + y^3), (xy, x^3, y^3, y + x^2)]$.

[45] Of these, the first three are irreducible as prime ideals; $\mathfrak{B}_4$ because it has only the divisors (x^2, y, z) and (x, y, z); $\mathfrak{B}_5$ because every divisor contains the polynomial xy.

The associated prime ideals are

$$\mathfrak{P}_1 = \mathfrak{B}_1, \quad \mathfrak{P}_2 = \mathfrak{B}_2, \quad \mathfrak{P}_3 = \mathfrak{B}_3, \quad \mathfrak{P}_4 = \mathfrak{P}_5 = (x, y, z).$$

The greatest primary ideals are

$$\mathfrak{Q}_1 = \mathfrak{B}_1, \quad \mathfrak{Q}_2 = \mathfrak{B}_2, \quad \mathfrak{Q}_3 = \mathfrak{B}_3, \quad \mathfrak{Q}_4 = [\mathfrak{B}_4, \mathfrak{B}_5] = (x^3, y^2, x^2y, z).$$

The relatively-prime irreducible ideals are

$$\mathfrak{R}_1 = \mathfrak{Q}_1, \quad \mathfrak{R}_2 = \mathfrak{Q}_2, \quad \mathfrak{R}_3 = [\mathfrak{Q}_3, \mathfrak{Q}_4] = (x^3, x^2y, xy^2, z).$$

The coprime-irreducible ideals are

$$\mathfrak{S}_1 = \mathfrak{R}_1, \quad \mathfrak{S}_2 = [\mathfrak{R}_2, \mathfrak{R}_3] = ((y-1)x^3, (y-1)x^2y, (y-1)xy^2, z).$$

Thus the total ideal is

$$\begin{aligned} \mathfrak{M} &= [\mathfrak{S}_1, \mathfrak{S}_2] = \mathfrak{S}_1\mathfrak{S}_2 \\ &= ((x-1)(y-1)x^3, (y-1)x^2y, (y-1)xy^2, (x-1)z, y(y-1)x^3, yz), \end{aligned}$$

where

$$1 = -(y-1)x^3 + (y-1)(x^3-1) + y,$$
$$(y-1)(x^3-1) + y \equiv 0\,(\mathfrak{S}_1), \qquad -(y-1)x^3 \equiv 0\,(\mathfrak{S}_2).$$

Here the ideals $\mathfrak{B}_1$, $\mathfrak{B}_2$, $\mathfrak{B}_3$ are isolated, and hence are uniquely determined also in the decompositions into irreducible and greatest primary ideals. The ideals $\mathfrak{B}_4$ and $\mathfrak{B}_5$, respectively $\mathfrak{Q}_4$, are non-isolated; they are not uniquely determined, but can, for instance, be replaced by

$$\mathfrak{D}_4 = (x^3, y + \lambda x^2, z), \qquad \mathfrak{D}_5 = (x^2 + \mu xy, y^2, z),$$

and $\mathfrak{Q}_4$ can likewise be replaced by

$$\overline{\mathfrak{Q}}_4 = (x^3, x^2y, y^2 + \lambda xy, z).$$

2. Just as the general, and the integral, polynomial domains satisfy the finiteness condition, so does every finite integral domain of polynomials, by Hilbert's theorem on module bases;[46] the coefficients may be assigned to an arbitrary field. We give one more example, for the domain of all even polynomials, the simplest domain in which, because $x^2y^2 = (xy)^2$, no unique product representation of the polynomials by irreducible polynomials of the domain exists. It is the same configuration as in the preceding example, now given by the irreducible ideals

$$\mathfrak{B}_1 = (x^2-1, xy, y^2, yz), \quad \mathfrak{B}_2 = (y^2-1, xz, yz, z^2),$$
$$\mathfrak{B}_3 = (x^2, xy, xz, yz, z^2), \quad \mathfrak{B}_4 = (x^4, x^3y, y^2, xz, yz, z^2),$$
$$\mathfrak{B}_5 = (x^2, y^2, xz, yz, z^2).$$

The associated prime ideals are

$$\mathfrak{P}_1 = \mathfrak{B}_1, \quad \mathfrak{P}_2 = \mathfrak{B}_2, \quad \mathfrak{P}_3 = \mathfrak{B}_3, \quad \mathfrak{P}_4 = \mathfrak{P}_5 = (x^2, xy, y^2, xz, yz, z^2).$$

That $\mathfrak{P}_1$ is a prime ideal follows from the fact that every polynomial of the domain has the form

$$f = \varphi(z^2) + xz\,\psi(z^2) \pmod{\mathfrak{P}_1}.$$

Thus if

$$f_1 = \varphi_1(z^2) + xz\,\psi_1(z^2), \qquad f_2 = \varphi_2(z^2) + xz\,\psi_2(z^2),$$

then from $f_1f_2 \equiv 0\,(\mathfrak{P}_1)$ one obtains the equations

$$\varphi_1\varphi_2 + z^2\psi_1\psi_2 = 0, \qquad \varphi_2\psi_1 + \varphi_1\psi_2 = 0,$$

[46] Conversely, if a polynomial domain satisfies the finiteness condition and every polynomial has at least one representation in which the multipliers are of lower degree in x, then the domain is a finite integral domain.

and hence $f_1 \equiv 0\,(\mathfrak{P}_1)$ or $f_2 \equiv 0\,(\mathfrak{P}_1)$. In exactly the same way one sees that $\mathfrak{P}_2$ is a prime ideal; $\mathfrak{P}_3$ is a prime ideal because every polynomial of the domain modulo $\mathfrak{P}_3$ is congruent to a polynomial in y^2; $\mathfrak{P}_4$ consists of all polynomials of the domain. Thus $\mathfrak{B}_1$, $\mathfrak{B}_2$, and $\mathfrak{B}_3$ are irreducible as prime ideals, while $\mathfrak{B}_4$ and $\mathfrak{B}_5$ each have only the single proper divisor $\mathfrak{P}_4$, and hence are necessarily irreducible.

From the irreducible ideals one obtains the greatest primary ideals

$$\mathfrak{Q}_1 = \mathfrak{B}_1, \quad \mathfrak{Q}_2 = \mathfrak{B}_2, \quad \mathfrak{Q}_3 = \mathfrak{B}_3, \quad \mathfrak{Q}_4 = [\mathfrak{B}_4, \mathfrak{B}_5] = (x^4, x^3y, y^2, xz, yz, z^2),$$

the relatively-prime irreducible ideals

$$\mathfrak{R}_1 = \mathfrak{Q}_1, \quad \mathfrak{R}_2 = \mathfrak{Q}_2, \quad \mathfrak{R}_3 = [\mathfrak{Q}_3, \mathfrak{Q}_4] = (x^4, x^3y, x^2y^2, xy^3, xz, yz, z^2),$$

and the coprime-irreducible ideals

$$\mathfrak{S}_1 = \mathfrak{R}_1, \quad \mathfrak{S}_2 = [\mathfrak{R}_2, \mathfrak{R}_3] = ((y^2-1)x^4, (y^2-1)x^3y, (y^2-1)x^2y^2, (y^2-1)xy^3, xz, yz, z^2).$$

As in the first example,

$$[\mathfrak{B}_4, \mathfrak{B}_5] = [\mathfrak{D}_4, \mathfrak{D}_5], \qquad [\mathfrak{Q}_3, \mathfrak{Q}_4] = [\mathfrak{Q}_3, \overline{\mathfrak{Q}}_4],$$

where

$$\mathfrak{D}_4 = (x^4, xy + \lambda x^2, \ldots), \qquad \mathfrak{D}_5 = (x^2 + \mu xy, \ldots), \qquad \lambda\mu \neq 1,$$

and

$$\overline{\mathfrak{Q}}_4 = (x^4, x^3y, y^2 + \lambda xy, \ldots).$$

The remaining irreducible, respectively greatest primary, ideals $\mathfrak{B}_1$, $\mathfrak{B}_2$, $\mathfrak{B}_3$ are uniquely determined as isolated ideals.

§ 11. Examples from number theory and from the theory of differential expressions.

1. Let the underlying domain Σ consist of all even rational integers. Thus Σ can be put in one-to-one correspondence with the domain of all rational integers, by assigning to each number $2a$ in Σ the number a. It follows at once that every ideal in Σ is a principal ideal $(2a)$, where in the basis representation $2c = n \cdot 2a$ for every element $2c$ of the ideal, the odd integers n are only abbreviations for finite sums.

The prime ideals of the domain are $\mathfrak{P}_0 = \mathfrak{D} = (2)$ and $\mathfrak{P} = (2p)$, where p denotes an odd prime number; every prime ideal is therefore divisible by $\mathfrak{P}_0$, but by no further prime ideal. The primary ideals are

$$\mathfrak{Q}_0 = (2 \cdot 2^{e_0}) \qquad \text{and} \qquad \mathfrak{Q}_p = (2p^e).$$

They are at the same time irreducible ideals, and by what was said about the prime ideals, two ideals belonging to different odd prime numbers are mutually relatively prime, but no $\mathfrak{Q}$ is relatively prime to any $\mathfrak{Q}_0$.[47] Thus the $\mathfrak{Q}_0$ are the only non-isolated primary ideals. The unique decomposition of a into prime powers corresponds to the unique representation of the ideal $(2a)$ by greatest primary, at the same time irreducible, ideals:

$$(2a) = [(2 \cdot 2^{e_0}), (2p_1^{e_1}), \ldots, (2p_\mu^{e_\mu})].$$

Thus here, in contrast to the examples from the polynomial domain, even the non-isolated greatest primary ideals are uniquely determined. If $e_0 = 0$ this is at the same time a representation by mutually relatively prime ideals, while if $e_0 > 0$ the ideal is relatively-prime-irreducible.

Thus, while in the domain of all integers the four different decompositions coincide, here this happens only for the two decompositions into greatest primary and into irreducible ideals on the one hand, and, for $e_0 > 0$, for coprime-irreducible (every ideal is coprime-irreducible, since the domain has no unit) and relatively-prime-irreducible ideals on the other; for $e_0 = 0$, however, the coprime-irreducible and relatively-prime-irreducible decompositions become different. At the same time one already obtains here an example in which one prime ideal can be divisible by another without being identical with it; more generally, one sees that product representation does not follow from divisibility. The latter fact - a consequence of the absence of a unit in the domain - is also why no unique product representation of the numbers of the domain by irreducible numbers of the domain exists, even though every ideal is a principal ideal; the introduction of least common multiples

[47] Indeed, from $2b \cdot 2p_1^{e_1} \equiv 0\,(2p_2^{e_2})$ for odd $p_1 \neq p_2$ one always obtains $2b \equiv 0\,(2p_2^{e_2})$; but from $2b \cdot 2p^e \equiv 0\,(2 \cdot 2^{e_0})$ one obtains only $2b = 2 \cdot 2^{e_0 - 1}$ (mod $2 \cdot 2^{e_0}$).

therefore proves necessary. It should also be noted that the situation remains exactly the same if, instead of all even integers, one takes all integers divisible by a fixed prime number or power of a prime.

Irreducible and primary ideals also become different, however, if Σ consists of all numbers divisible by a composite number $g = p_1^{\nu_1} \cdots p_s^{\nu_s}$. Again every ideal is a principal ideal $(g \cdot a)$, and the prime ideals are again $\mathfrak{P}_0 = \mathfrak{O} = (g)$ and $\mathfrak{P} = (g \cdot p)$, where p is a prime different from the primes occurring in g. But the irreducible ideals are $\mathfrak{B}_{\lambda_i} = (g \cdot p_i^{\lambda_i})$ and $\mathfrak{B}_e = (g \cdot p^e)$; the primary ideals are $\mathfrak{Q}_e = \mathfrak{B}_e$ and the ideals, different from the irreducible ones,

$$\mathfrak{Q}_{\lambda_1 \dots \lambda_s} = (g \cdot p_1^{\lambda_1} \cdots p_s^{\lambda_s}),$$

where the $\mathfrak{B}_{\lambda_i}$ and $\mathfrak{Q}_{\lambda_1 \dots \lambda_s}$ all have the same associated prime ideal $\mathfrak{P}_0 = (g)$. Here too the decomposition into irreducible ideals is unique, and consequently so is the decomposition into greatest primary ideals; thus here again the non-isolated ideals are uniquely determined.

2. An example of a non-commutative ring domain is supplied by the ideal theory in non-commutative polynomial domains treated in the paper of Noether–Schmeidler. In particular one has "completely reducible" ideals, that is, ideals for which the components of the decomposition are pairwise coprime and have no proper divisors; the components are therefore a fortiori irreducible. Thus, in addition to the isomorphism proved there according to § 9, one obtains equality of the number of components in two different decompositions. For the decomposition of systems of partial or ordinary linear differential expressions, which arises as a special case of that paper, this gives a result which does not seem to have been noticed even in the known case of a single ordinary linear differential expression.

At the same time the system T of all residue classes of a fixed ideal M, together with the non-commutative polynomial domain Σ, supplies a double domain (Σ, T), where T has the module property with respect to Σ. For the difference of two residue classes is again a residue class, and likewise the product of a residue class by an arbitrary polynomial; whereas the product of two residue classes does not exist (loc. cit., § 3). Thus the systems of residue classes called "subgroups" there furnish examples of modules in double domains (Σ, T), where the underlying ring domain Σ is non-commutative.

§ 12. Example from elementary divisor theory.

This is a conception of elementary divisor theory dictated by the general developments, but the theory itself is assumed as known.

Let Σ be the domain of all integral matrices with n^2 elements, with addition and multiplication defined in the usual sense for matrices. Then Σ is a non-commutative ring domain; the ideals are therefore in general one-sided, with two-sided ideals only as a special case.[48]

We first show that every ideal is principal. For this purpose, in the case of right ideals, to each matrix $A = (a_{ik})$ we assign the module

$$A = (a_{11}\xi_1 + \cdots + a_{1n}\xi_n, \dots, a_{n1}\xi_1 + \cdots + a_{nn}\xi_n)$$

of integral linear forms. Conversely, to this module there corresponds every matrix that supplies a basis of A, hence, beside A, also UA, where U is unimodular. More generally, to the product PA there corresponds a module B which is a multiple of A. A single linear form from A is given by such matrices P as contain only one non-zero row.

Let now $A_1, A_2, \dots, A_\nu, \dots$ be all elements of an ideal $\mathfrak{M}$; let $A_1, A_2, \dots, A_\nu, \dots$ be the modules assigned to them; let A be their greatest common divisor and UA the most general matrix assigned to this module A. To each single linear form in A there then corresponds, by the definition of greatest common divisor, a matrix $P_1 A_{i_1} + \cdots + P_\sigma A_{i_\sigma}$, where, as above, the P have only one non-zero row. It follows that the matrix A corresponding to a basis of A also admits such a representation, now with general P, and hence becomes an element of $\mathfrak{M}$. Since, furthermore, every module A_i is divisible by A, every matrix A_i is divisible by A; A, and in general UA, forms a basis of $\mathfrak{M}$. If one is dealing with left ideals, then correspondingly the columns of each matrix are to be regarded as the basis of a module; every ideal is a principal ideal, for which, beside A, also AV is a basis, with V an arbitrary unimodular matrix.

For the connection with elementary divisor theory, we now base the following on two-sided ideals; hence in particular PAQ belongs to the ideal whenever A does. By the preceding, the most general basis of such an ideal is UAV, where U and V are unimodular. The basis elements therefore exhaust a class of equivalent

[48]The ideal theory of these domains is the subject of the works of Du Pasquier: Zahlentheorie der Tettarionen, dissertation, Zürich, Vierteljahrsschr. d. Naturf. Ges. Zürich, 51 (1906); Zur Theorie der Tettarionenideale, ibid., 52 (1907). The content of the second paper is the proof that every ideal is a principal ideal.

matrices,[49] and there is a one-to-one relation between ideal and class, hence also between ideal and the elementary divisor system $(a_1 \mid a_2 \mid \cdots \mid a_n)$ of the class, where the a_i are, as is well known, non-negative integers, each dividing the following one. Thus the matrix of the class that occurs in the normal form determined by the elementary divisors may be regarded as a special basis of the ideal; divisibility of ideals, respectively of classes, implies divisibility of the elementary divisors, and conversely.

But Du Pasquier has shown, loc. cit. § 11, that for every two-sided ideal the rank is n and all elementary divisors agree. To be able to consider the case of general elementary divisors, we must therefore start not from the ideals but directly from the two-sided classes (which are likewise to be denoted by capital German letters). A class $\mathfrak{A} = UAV$ is divisible by another class $\mathfrak{B}$ if $A = PBQ$.

In general: *the least common multiple (the greatest common divisor) of two classes is obtained by forming the least common multiple (the greatest common divisor) of the corresponding elementary divisor systems.*[50] For let

$$(a_1 \mid a_2 \mid \cdots \mid a_n), \quad (b_1 \mid b_2 \mid \cdots \mid b_n), \quad (c_1 \mid c_2 \mid \cdots \mid c_n),$$

where $c_i = [a_i, b_i]$, be respectively the elementary divisor systems of $\mathfrak{A}$, $\mathfrak{B}$, and $\mathfrak{C}^*$, and let $\mathfrak{C} = [\mathfrak{A}, \mathfrak{B}]$. Then $\mathfrak{C}^*$ is divisible by $\mathfrak{A}$ and $\mathfrak{B}$, hence by $\mathfrak{C}$; conversely, the elementary divisors of $\mathfrak{C}$ are divisible by those of $\mathfrak{C}^*$, so $\mathfrak{C}$ is divisible by $\mathfrak{C}^*$ and therefore $\mathfrak{C} = \mathfrak{C}^*$. The proof for the greatest common divisor is obtained correspondingly.

The unique representation of the elementary divisors a_i as least common multiples of prime powers therefore corresponds to a representation of $\mathfrak{A}$ as the least common multiple of classes $\mathfrak{Q}$ whose elementary divisors are given by powers of one prime number, in symbols

$$\mathfrak{Q} \sim (p^{r_1} \mid p^{r_2} \mid \cdots \mid p^{r_\rho} \mid 0 \mid \cdots \mid 0), \qquad r_1 \leq r_2 \leq \cdots \leq r_\rho.$$

If in particular the rank ρ is equal to n, then one has here a decomposition into coprime and coprime-irreducible classes which is unique despite the non-commutative domain.

The classes $\mathfrak{Q}$ can be further decomposed into classes corresponding to the elementary divisor systems

$$\mathfrak{B}_1 \sim (p^{r_1} \mid \cdots \mid p^{r_1}), \quad \mathfrak{B}_2 \sim (1 \mid p^{r_2} \mid \cdots \mid p^{r_2}), \quad \ldots,$$
$$\mathfrak{B}_\nu \sim (1 \mid \cdots \mid 1 \mid p^{r_\nu} \mid \cdots \mid p^{r_\nu}), \quad \mathfrak{B}_{\rho+1} \sim (1 \mid \cdots \mid 1 \mid 0 \mid \cdots \mid 0),$$

where in $\mathfrak{B}_\nu$ the number 1 occurs $(\nu - 1)$ times. If, for example, $r_1 = r_2 = \cdots = r_\mu = 0$, then $\mathfrak{B}_1, \ldots, \mathfrak{B}_\mu$ are equal to the unit class and are to be omitted from the decomposition; the same holds for $\mathfrak{B}_{\rho+1}$ if $\rho = n$. If, furthermore, say $r_\nu = r_{\nu+1} = \cdots = r_{\nu+\lambda}$, then $\mathfrak{B}_{\nu+1}, \ldots, \mathfrak{B}_{\nu+\lambda}$ are proper divisors of $\mathfrak{B}_\nu$, and are likewise to be discarded. Denote by $\mathfrak{B}_{i_1}, \ldots, \mathfrak{B}_{i_k}$ those that remain and now give a shortest representation. Then

$$\mathfrak{Q} = [\mathfrak{B}_{i_1}, \ldots, \mathfrak{B}_{i_k}]$$

is the unique decomposition of $\mathfrak{Q}$ into irreducible classes. Indeed, suppose

$$\mathfrak{B}_\nu \sim (1 \mid \cdots \mid 1 \mid p^{r_\nu} \mid \cdots \mid p^{r_\nu})$$

can be represented as the least common multiple of

$$\mathfrak{C} \sim (1 \mid \cdots \mid 1 \mid p^{s_\nu} \mid \cdots \mid p^{s_i}) \quad \text{and} \quad \mathfrak{D} \sim (1 \mid \cdots \mid 1 \mid p^{t_\nu} \mid \cdots \mid p^{t_i}).$$

Then in the elementary divisor systems of $\mathfrak{C}$ and $\mathfrak{D}$ the number 1 must stand in the first $(\nu - 1)$ places; in the ν-th place one of the exponents s_ν or t_ν, say s_ν, must become equal to r_ν. But since

$$r_\nu = s_\nu \leq s_{\nu+1} \leq \cdots \leq s_i \leq r_\nu,$$

one obtains $\mathfrak{C} = \mathfrak{B}_\nu$, and $\mathfrak{B}_\nu$ is therefore irreducible.[51] The same holds for $\mathfrak{B}_{\rho+1}$, where p is to be replaced by 0. But, as the formation of the least common multiple shows, each of these irreducible classes gives

[49] The basis elements of one-sided ideals correspond to right or left classes.

[50] In the paper Zur Theorie der Moduln, Math. Ann. 52 (1899), p. 1, E. Steinitz defines the least common multiple (the greatest common divisor) of classes by the least common multiples and greatest common divisors of the elementary systems. Independently, the least common multiple of classes occurs as "congruence composition" in H. Brandt, Komposition der binären quadratischen Formen relativ einer Grundform, J. f. M. 150 (1919), p. 1.

[51] The $\mathfrak{B}$ are still reducible into one-sided classes; here there is no longer a unique relation between elementary divisors and class, and hence no unique decomposition into irreducible one-sided classes. The following example, supplied to me by H. Brandt, shows this for decomposition into right-sided classes (where the classes are represented by a basis matrix, or by the corresponding module):

$$\mathfrak{B} = [\mathfrak{C}_1, \mathfrak{C}_2] = [\mathfrak{D}_1, \mathfrak{D}_2], \quad \mathfrak{B} \sim \begin{pmatrix} p & 0 \\ 0 & p \end{pmatrix}, \quad \mathfrak{C}_1 \sim \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}, \quad \mathfrak{C}_2 \sim \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix},$$
$$\mathfrak{D}_1 \sim \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}, \quad \mathfrak{D}_2 \sim \begin{pmatrix} p & 0 \\ (p-1) & 1 \end{pmatrix}.$$

Indeed, the modules $(\xi, p\eta)$, $(p\xi, \eta)$, and $(p\xi, (p-1)\xi + \eta)$ each have as their only proper divisor the module (ξ, η), and are therefore irreducible and mutually distinct.

a definite exponent and the place where this exponent first occurs in the elementary divisor system of $\mathfrak{Q}$, respectively $\mathfrak{A}$, while $\mathfrak{B}_{\rho+1}$ gives the rank. Since these numbers are uniquely determined by the elementary divisor system of $\mathfrak{Q}$, respectively $\mathfrak{A}$, and since the relation between elementary divisors and class is one-to-one, the decomposition of $\mathfrak{Q}$, and likewise of any class, into irreducible classes is unique.

In summary: *Every two-sided class $\mathfrak{A}$ of integral matrices with bounded number of entries can be represented uniquely as the least common multiple of finitely many irreducible two-sided classes. Each irreducible class represents a fixed prime divisor of the elementary divisor system of $\mathfrak{A}$, an associated exponent, and the place where that exponent first occurs. The irreducible class corresponding to the divisor* 0 *gives the rank of $\mathfrak{A}$.*

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20. An Algebraic Criterion for Absolute Irreducibility

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A polynomial – with coefficients from an arbitrary, abstractly defined field – is called *absolutely irreducible* if it remains irreducible in the algebraically closed field to which the coefficient domain can be enlarged.[1] I show in what follows that absolute irreducibility, in contrast with irreducibility relative to a prescribed field, can be grasped algebraically. More precisely, the theorem is:

To every polynomial in $n \geq 2$ variables with indeterminate coefficients there can be assigned an integral rational function, with integral coefficients, of these coefficients and of further indeterminates – the reducibility form – in such a way that, for every special polynomial of the same degree, the necessary and sufficient condition for absolute irreducibility is the non-vanishing of the reducibility form. In other words: for every special system of values of the coefficients for which the reducibility form vanishes identically in the indeterminates and which does not cause a lowering of degree, the special polynomial is reducible, and conversely.

If, instead of polynomials, one considers homogeneous forms in one more variable, the degree condition is replaced by the simpler condition that the coefficient system not vanish identically.

As a direct consequence of the theorem one also obtains the following, first stated by A. Ostrowski:[2]

Every absolutely irreducible polynomial with algebraic numbers as coefficients remains irreducible modulo every prime ideal of a finite algebraic field Ω containing the coefficient domain, with the exception of at most finitely many prime ideals of Ω. In particular, Ω may also be the field of rational numbers.

I intend to discuss further applications to the theory of relatively integral functions[3] elsewhere.

1. We first show that *every polynomial in $n \geq 2$ variables with indeterminate coefficients is absolutely irreducible.* Set

$$\begin{aligned} F(x) &= \sum A_{i_1 \dots i_n} x_1^{i_1} \cdots x_n^{i_n} && (i_1 + \cdots + i_n \leq l), \\ G(x) &= \sum B_{j_1 \dots j_n} x_1^{j_1} \cdots x_n^{j_n} && (j_1 + \cdots + j_n \leq m), \\ H(x) = F(x)G(x) &= \sum C_{k_1 \dots k_n}(A, B) x_1^{k_1} \cdots x_n^{k_n} && (k_1 + \cdots + k_n \leq l + m), \end{aligned} \tag{1}$$

where the A and B denote indeterminates, and the $C(A, B)$ become integral bilinear combinations of the A and B. The quotients

$$D = C(A, B)/C_{0 \dots 0}(A, B)$$

are therefore rational functions of the quotients $A/A_{0 \dots 0}$ and $B/B_{0 \dots 0}$; but the number of these functions is greater than the number of their arguments, since these numbers are respectively

$$\binom{l+m+n}{n} - 1, \qquad \binom{l+n}{n} - 1, \qquad \binom{m+n}{n} - 1,$$

[1] That every field can be enlarged, essentially uniquely, to an algebraically closed one, i.e. to one in which every polynomial in one variable splits into linear factors, was shown by E. Steinitz in his *Algebraische Theorie der Körper* (J. f. M. 137 (1910), p. 167); he thereby gave the rational equivalent of the fundamental theorem of algebra.

[2] Zur arithmetischen Theorie der algebraischen Größen. Gött. Nachr. 1919, p. 279 (lemma p. 296). For the case of fields consisting of ordinary numbers, Ostrowski, as is known to me, also arrived at the main theorem above; he will publish his investigations on this soon. That, in my arrangement of the proof, the main theorem holds for arbitrary, abstractly defined fields rests on the fact that the reducibility form constructed for indeterminates has integral coefficients independent of the particular field taken as ground field.

[3] D. Hilbert, Mathematische Probleme. Gött. Nachr. 1900, p. 253, Problem 14.

and

$$\binom{l+m+n}{n}+1>\binom{l+n}{n}+\binom{m+n}{n} \quad \text{for } n \geq 2,\ l>0,\ m>0.$$

Thus there is at least *one* algebraic relation among the $D(A,B)$ and consequently also among the $C(A,B)$:

$$\Phi(Z) \neq 0 \quad [\text{identically in } Z_{k_1\dots k_n}], \qquad \Phi(C(A,B))=0 \quad [\text{identically in } A,B], \tag{2}$$

where Φ denotes an integral polynomial.

A polynomial with indeterminate Z as coefficients,

$$E(x)=\sum Z_{k_1\dots k_n} x_1^{k_1}\cdots x_n^{k_n} \qquad (k_1+\cdots+k_n \leq l+m), \tag{3}$$

is therefore necessarily absolutely irreducible for $n \geq 2$.[4]

2. The totality of these polynomials $\Phi(Z)$ which vanish identically in A,B under the substitution $Z=C(A,B)$ forms an *ideal of polynomials*,[5] more precisely a prime ideal $\mathfrak{P}$; for if, under the substitution, a product $\Phi_1\Phi_2$ vanishes, then at least one factor vanishes. Since (2) is satisfied identically in A,B, it remains valid for every special system of values A,B, where A,B and consequently also $\Gamma=C(A,B)$ denote quantities of some field. *The coefficients Γ of every polynomial reducible in an extension field therefore satisfy the conditions $\Phi(\Gamma)=0$; they are zeros of $\mathfrak{P}$,* or also of the finitely many basis polynomials of $\mathfrak{P}$. It remains to prove the converse.

For this it should be noted that all relations among A,B,C are preserved when, by introducing a new variable y, the polynomials (1) are transformed into homogeneous forms $F(x,y)$, $G(x,y)$, $H(x,y)$ of dimensions $l, m, l+m$. Thus coefficients Γ also become zeros of $\mathfrak{P}$ when, for instance, $F(x,y)$[6] becomes a power of y; for them the passage to inhomogeneous polynomials causes a lowering of degree of $\bar{H}(x)$ and not a factorization. It will be shown that, with the exception of this lowering of degree, the converse always holds; in particular, that for *homogeneous forms the converse holds without exception, provided not all Γ vanish; in other words, every zero of $\mathfrak{P}$ is supplied by the parameter representation $\Gamma=C(A,B)$ with quantities A,B from an extension field.*[7]

I prove this by the direct construction of finitely many functions $\Phi(Z)$: by means of Kronecker's substitution

$$x_i=\xi^{d^{n-i}}$$

I assign to the polynomials (1) polynomials in *one* variable; I show that gaps occur here in the exponents, and by forming the norm of the corresponding coefficients I arrive at the functions $\Phi(Z)$ and at the desired criterion.

3. Put:[8]

$$d=l+m+1, \qquad i=i_1d^{n-1}+i_2d^{n-2}+\cdots+i_n, \quad j=j_1d^{n-1}+\cdots+j_n, \quad k=k_1d^{n-1}+\cdots+k_n. \tag{4}$$

Then, by Kronecker's substitution, the polynomials F,G,H in (1) correspond respectively to the polynomials

$$\begin{aligned} f(\xi) &= \sum A_{i_1\dots i_n}\xi^i && (i_1+\cdots+i_n \leq l),\\ g(\xi) &= \sum B_{j_1\dots j_n}\xi^j && (j_1+\cdots+j_n \leq m),\\ h(\xi) &= \sum C_{k_1\dots k_n}(A,B)\xi^k && (k_1+\cdots+k_n \leq l+m). \end{aligned} \tag{5}$$

This correspondence is known to be one-to-one, since for all induced exponents occurring in (5), $k=0$ implies $k_1=0,\dots,k_n=0$; and therefore $i+j=i'+j'$ also implies $i_1+j_1=i'_1+j'_1;\dots;i_n+j_n=i'_n+j'_n$. Hence distinct products $\xi^i\xi^j$ can produce the same exponent ξ^k only when the corresponding products $x_1^{i_1}\cdots x_n^{i_n}x_1^{j_1}\cdots x_n^{j_n}$ also lead to the same exponent. Thus one obtains

$$h(\xi)=f(\xi)g(\xi), \tag{6}$$

[4] For, in the terminology of 2, the indeterminates Z are not zeros of $\mathfrak{P}$.

[5] Such totalities are usually called modules; but in fact their characteristic property – that with Φ_1 and Φ_2 they contain also $\Phi_1-\Phi_2$, and with Φ also $\Psi\Phi$, where Ψ is an arbitrary integral polynomial in Z – is the ideal property.

[6] Polynomials and forms which arise by replacing the indeterminates in $F,G,\dots,f,g,\dots$ by special systems of values will throughout be denoted by overlining: $\bar{F},\bar{G},\dots,\bar{f},\bar{g},\dots$.

[7] That "in general" the zeros of $\mathfrak{P}$ are supplied by the parameter representation follows from the property of the prime ideal $\mathfrak{P}$ of defining an irreducible algebraic variety. But, as is well known, this does not imply – something I had originally overlooked and to which A. Ostrowski called my attention some time ago – that the parameter representation cannot fail for any special system of values. The proof given here rests on more elementary notions.

[8] Festschrift, p. 11.

and from the validity of a relation (6), in such a way that in $f(\xi)$ and $g(\xi)$ no exponents other than the induced ones occur, one obtains back again

$$H(x) = F(x)G(x). \tag{7}$$

Here *gaps actually occur in the induced exponents in* $f(\xi)$ *and* $g(\xi)$. The highest exponent of $f(\xi)$ is ld^{n-1}, while the second highest is $(l-1)d^{n-1} + d^{n-2}$; their difference is $d^{n-2}(d-1) > 1$, since $d = l + m + 1 \geq 3$ and $n \geq 2$; and the same holds for $g(\xi)$.

The relations (6) and (7), valid for the indeterminates A, B, remain valid for every special system of values A, B. To preserve the degrees, I homogenize (6) and (7) by means of the variables η and y. *A homogeneous form* $H(x, y)$ *therefore splits in an algebraic extension field if and only if, in that field, the corresponding binary form* $h(\xi, \eta)$ *can be split into two factors in such a way that no exponents other than the induced ones occur in the factors.*

4. In order to carry out the formation of the norm mentioned at the end of 2, let

$$a_1, b_1; \ldots; a_p, b_p; \quad a_{p+1}, b_{p+1}; \ldots; a_{p+q}, b_{p+q}$$

be new indeterminates. Set

$$\begin{aligned}
\varphi(\xi, \eta) &= (a_1\xi + b_1\eta) \cdots (a_p\xi + b_p\eta) = r_p\xi^p + r_{p-1}\xi^{p-1}\eta + \cdots + r_0\eta^p, \\
\psi(\xi, \eta) &= (a_{p+1}\xi + b_{p+1}\eta) \cdots (a_{p+q}\xi + b_{p+q}\eta) = s_q\xi^q + s_{q-1}\xi^{q-1}\eta + \cdots + s_0\eta^q, \\
\chi(\xi, \eta) &= \varphi(\xi, \eta)\psi(\xi, \eta) = t_{p+q}\xi^{p+q} + t_{p+q-1}\xi^{p+q-1}\eta + \cdots + t_0\eta^{p+q},
\end{aligned} \tag{8}$$

so that r, s, t denote the homogenized elementary symmetric functions; that is, the symmetric functions of the rows a, b that are linear in each individual row and from which all integral symmetric functions that are homogeneous and of the same dimension in each individual row are built up integrally and with integral coefficients – and in only one way.

Now form, with further indeterminates $u_{\mu\nu}$, the linear form

$$\zeta(u, a, b) = \sum u_{\mu\nu} r_\mu s_\nu, \tag{9}$$

where the summation is extended over prescribed indices μ and ν. The product of $\zeta(u)$ with all its conjugates arising from permutation of the row a, b which are different from $\zeta(u)$ and from one another – the norm $N(\zeta(u))$ of $\zeta(u)$ with respect to the field of the $t(a, b)$, when $\zeta(u)$ is regarded as a function of the field of the $r_\mu(a, b)$ and $s_\nu(a, b)$ – is then an integral symmetric function of all rows a, b, homogeneous and of the same dimension in each row. Hence, by what was said above, it is a uniquely determined integral function, with integral coefficients, of the $t(a, b)$ and $u_{\mu\nu}$:

$$N(\zeta(u, a, b)) = T(t(a, b), u) \qquad \text{[identically in } a, b, u]. \tag{10}$$

The same reasoning shows that also

$$N(z - \zeta(u)) = z^\delta + T_{\delta-1}(t, u)z^{\delta-1} + \cdots + T(t, u) \qquad \text{[identically in } a, b, u, z]$$

where all T are integral functions, with integral coefficients, of t and u. Both sides of this identity vanish for $z = \zeta(u)$; and indeed, because of the algebraic independence of the elementary symmetric functions $r(a, b)$ and $s(a, b)$, they vanish identically in r, s when $t(a, b)$ is set, according to (8), equal to an integral bilinear combination $w(r, s)$ of the r and s, and $\zeta(u)$ is regarded as a function of r and s. Thus

$$\zeta(u)^\delta + T_{\delta-1}(w(r, s), u)\zeta(u)^{\delta-1} + \cdots + T(w(r, s), u) = 0 \quad \text{[identically in } r, s, u]. \tag{11}$$

[9]

The identities (10) and (11) remain valid for all special systems of values α, β of a, b, respectively ϱ, σ of r, s. If ϱ, σ are chosen in particular so that all products $\varrho_\mu\sigma_\nu$ vanish – where μ, ν denote the indices occurring in (9) – so that $\zeta(u)$ vanishes under the substitution, then, putting $w(\varrho, \sigma) = \tau$, (11) gives

$$T(\tau, u) = 0 \qquad \text{[identically in } u]. \tag{12}$$

[9] If the summation in (9) is extended over all indices, (11) represents Kronecker's extension of Gauss's theorem, essentially in Kronecker's arrangement of the proof (Zur Theorie der Formen höherer Stufen, Berliner Ber. 1883, Werke, 2, p. 417).

Conversely, let $\tau_0, \dots, \tau_{p+q}$ be systems of values, not all zero, for which (12) is satisfied. Since in an algebraic extension field there is a decomposition

$$\bar{\chi}(\xi,\eta) = \tau_{p+q}\xi^{p+q} + \cdots + \tau_0\eta^{p+q} = (\alpha_1\xi + \beta_1\eta)\cdots(\alpha_{p+q}\xi + \beta_{p+q}\eta), \tag{12'}$$

in which α and β do not vanish simultaneously in any factor,[10] although some α or β may vanish, we have $\tau = t(\alpha,\beta)$. Substituting these values into (10), using (12), shows that $\zeta(u,\alpha,\beta)$ or one of its conjugate factors vanishes identically in u. Thus, after a suitable numbering of α,β, setting $r(\alpha,\beta) = \varrho$ and $s(\alpha,\beta) = \sigma$ says precisely that $\bar{\chi}(\xi,\eta)$ admits at least one decomposition

$$\bar{\chi}(\xi,\eta) = \bar{\varphi}(\xi,\eta)\bar{\psi}(\xi,\eta) = \sum \varrho_\mu\xi^\mu\eta^{p-\mu} \cdot \sum \sigma_\nu\xi^\nu\eta^{q-\nu}, \tag{13}$$

in such a way that all products $\varrho_\mu\sigma_\nu$ occurring in $\zeta(u)$ vanish, but not all ϱ or all σ vanish.

5. The degrees p,q in (8) are now taken to be ld^{n-1} and md^{n-1}, i.e. the degrees of $f(\xi)$ and $g(\xi)$ in (5). The indices μ,ν are split into $\mu^{(1)},\nu^{(1)},\mu^{(2)},\nu^{(2)}$: here $\mu^{(1)}$ runs through all exponents missing from $f(\xi)$, $\nu^{(1)}$ through all exponents of ξ in $\psi(\xi,\eta)$; correspondingly, $\nu^{(2)}$ runs through all exponents missing from $g(\xi)$, and $\mu^{(2)}$ through all exponents of ξ in $\varphi(\xi,\eta)$. Further, let

$$e(\xi) = \sum Z_{k_1\dots k_n}\xi^k \qquad (k_1 + \cdots + k_n \leq l+m)$$

be the polynomial in one variable corresponding to the polynomial (3), and let

$$S(Z,u)$$

be the integral polynomial in Z and u obtained from $T(t,u)$ – the uniquely determined right hand side of (10), where the summation in (9) is extended over the indicated indices $\mu^{(1)},\nu^{(1)},\mu^{(2)},\nu^{(2)}$ – by replacing the t by the coefficients Z and the corresponding zeros of $e(\xi)$.

If now r,s are replaced by the coefficients A,B and the corresponding zeros of $f(\xi)$ and $g(\xi)$, then by this substitution $\zeta(u)$ vanishes for the indicated summation. Moreover $t = w(r,s)$ passes over into the coefficients $C(A,B)$ and the corresponding zeros of $h(\xi)$; hence $T(t,u)$ passes over into $S(C(A,B),u)$. By the identity (11) we therefore get

$$S(C(A,B),u) = 0 \qquad [\text{identically in } A,B,u]. \tag{14}$$

Since (14) is preserved for every special system of values, the coefficients $\Gamma = C(A,B)$ of such special $H(x)$, or more generally $H(x,y)$, that split in an extension field into two factors satisfy the condition

$$S(\Gamma,u) = 0 \qquad [\text{identically in } u]. \tag{15}$$

Conversely, suppose that for the coefficients Γ, not all zero, of a polynomial $\bar{H}(x)$, respectively form $\bar{H}(x,y)$, the condition (15) is satisfied. By the above this is equivalent to $T(\tau,u) = 0$, where τ denotes all coefficients of $\bar{h}(\xi,\eta)$. Hence by (13) there exists in an extension field of the Γ a decomposition

$$\bar{h}(\xi,\eta) = \bar{f}(\xi,\eta) \cdot \bar{g}(\xi,\eta),$$

where, because of the prescribed summation, no non-induced exponents occur in $\bar{f}$ and $\bar{g}$; consequently there is a relation

$$\bar{H}(x,y) = \bar{F}(x,y) \cdot \bar{G}(x,y).$$

If, for $y = 1$, no factor becomes of degree zero – in particular if the Γ do not cause a lowering of the degree of $H(x)$ – it follows further that

$$\bar{H}(x) = \bar{F}(x) \cdot \bar{G}(x).$$

Condition (15) is therefore necessary and sufficient for $\bar{H}(x,y)$, and, when degree lowering is excluded, also for $\bar{H}(x)$, to split into two factors in an extension field.

It follows moreover that $S(Z,u)$ cannot vanish identically in Z and u, since in 1 it was shown that for $n \geq 2$ polynomials with indeterminates as coefficients, hence also the corresponding homogeneous forms in one more variable, are absolutely irreducible. Thus, with U denoting power products of the u,

$$S(Z,u) = \sum \Phi_\lambda(Z)U_\lambda,$$

[10]This rational "fundamental theorem of algebra" is the existence theorem on which the exceptionless validity of the converse stated at the end of 2 rests.

where the $\Phi_\lambda(Z)$ are, by (14), polynomials belonging to $\mathfrak{P}$. It is thereby shown that $\mathfrak{P}$ has no further zeros beyond those given by the parameter representation $\Gamma = C(A, B)$, where A and B belong to an algebraic extension field of the Γ. The converse stated at the end of 2 is therefore proved.

6. *The condition of absolute irreducibility* can now be formulated directly. One need only decompose the degree of $E(x)$ in (3), in all possible ways, into two positive summands $l + m$, and form the form $S(Z, u)$ for each such decomposition. The product over these forms,

$$R(Z, u) = \prod S(Z, u), \tag{16}$$

then forms the reducibility form of $E(x)$; for by the above, every factorization of $E(x)$, respectively $E(x, y)$, corresponds to the vanishing of one factor of (16), and conversely. Thus the main theorem stated in the introduction is proved, and at the same time it is shown how the reducibility form can actually be constructed in finitely many steps.

7. From the existence of the reducibility form (16) the theorem of Ostrowski mentioned in the introduction follows immediately. Let

$$\bar{E}(x) = \sum \Gamma_{k_1 \ldots k_n} x_1^{k_1} \cdots x_n^{k_n} \qquad (k_1 + \cdots + k_n \leq v)$$

be an absolutely irreducible polynomial with algebraic integers Γ as coefficients; let Ω denote a finite algebraic number field containing all Γ, and let $\mathfrak{p}$ be a prime ideal of Ω.

Then the reducibility form $R(Z, u)$ of exact degree for $\bar{E}(x)$ remains different from zero under the substitution $Z = \Gamma$; that is,

$$R(\Gamma, u) = \sum P_\lambda U_\lambda \neq 0 \qquad \text{[identically in } u\text{]}.$$

The ideal $\mathfrak{r} = (P_1, P_2, \ldots)$ is therefore divisible by only finitely many prime ideals of Ω. If $\mathfrak{p}$ is different from these finitely many, then $R(\Gamma, u)$ does not vanish identically modulo $\mathfrak{p}$; and since the residue classes modulo $\mathfrak{p}$ again form a field, it follows that $\bar{E}(x)$ modulo $\mathfrak{p}$ is irreducible, more precisely absolutely irreducible. The same holds for $\bar{E}(x, y)$, so that no lowering of degree modulo $\mathfrak{p}$ occurs.

If the coefficients of $\bar{E}(x)$ are algebraically fractional numbers, then $\mathfrak{p}$ must also be taken different from the finitely many prime ideals dividing the common denominator Λ; modulo all remaining prime ideals, $E(x)$ may be replaced by $\Lambda\bar{E}(x)$, so that the preceding hypotheses are satisfied. This proves the theorem.

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21. Formal Variational Calculus and Differential Invariants

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28. Formal Variational Calculus and Differential Invariants.[149]

The generation of all differential invariants and the derivation of the main theorems can be carried out most uniformly by the formal methods of the calculus of variations. This principle goes back to Riemann[72)] and received its chief form-theoretic development from Lipschitz[73)].

Start from the variational problem belonging to a homogeneous form $f(dx)$ – where

$$\left|\frac{\partial^2 f}{\partial dx_i\, \partial dx_k}\right| \neq 0$$

is assumed – namely

$$\delta \int_{t_0}^{t_1} f(x')\, dt = 0, \qquad \left(x_i' = \frac{dx_i}{dt}\right). \tag{140}$$

By partial integration one is led to the invariant system of equations

$$2\psi_i = \frac{d}{dt}\frac{\partial f}{\partial x_i'} - \frac{\partial f}{\partial x_i} = 0,$$

where the Lagrange expressions ψ_i are contragredient to the dx. Formally this is seen most simply if the ψ_i are defined by the integral-free formal identity corresponding to partial integration (Lagrange central equation[150])

$$\delta f - df_\delta = -2\sum \psi_i(d,d)\, \delta x_i, \qquad \left(f_\delta = \sum \frac{\partial f}{\partial dx_i}\, \delta x_i\right),$$

where both f and df_δ are invariants, and hence the right hand side is one as well. In particular, for a quadratic form one obtains

$$\delta \sum g_{ik} dx_i dx_k - 2d \sum g_{ik} dx_i \delta x_k = -2 \sum \psi_\mu(d,d)\, \delta x_\mu.$$

Replacing d here by the cogredient $(d+\lambda\delta)$ and comparing powers of λ, one obtains the corresponding system of identities:

$$\begin{cases} D\sum g_{ik}dx_i dx_k - 2d\sum g_{ik}dx_i Dx_k = -2\sum \psi_\mu(d,d)\, Dx_\mu, \\ D\sum g_{ik}dx_i\delta x_k - \delta\sum g_{ik}dx_i Dx_k - d\sum g_{ik}\delta x_i Dx_k = -2\sum \psi_\mu(d,\delta)\, Dx_\mu, \\ D\sum g_{ik}\delta x_i \delta x_k - 2\delta\sum g_{ik}\delta x_i Dx_k = -2\sum \psi_\mu(\delta,\delta)\, Dx_\mu. \end{cases} \tag{141}$$

It follows that $\psi_\mu(d,\delta)$, and in particular also $\psi_\mu(d,d)$ and $\psi_\mu(\delta,\delta)$, are contragredient vectors. From (141) one computes $\psi_\mu(d,\delta)$ as

$$\psi_\mu(d,\delta) = \sum g_{i\mu}\, d\delta x_i + \sum \begin{bmatrix} ik \\ \mu \end{bmatrix} dx_i \delta x_k,$$

and from this one obtains the cogredient vector

$$p^\sigma(d,\delta) = \sum g^{\sigma\mu}\psi_\mu = d\delta x_\sigma + \sum \left\{ {ik \atop \sigma} \right\} dx_i \delta x_k. \tag{142}$$

Putting $p^\nu(d,d) = 0$ evidently gives the equations of the geodesic lines.

Knowledge of the cogredient vector $p^\sigma(d,\delta)$ leads directly to the covariant derivative (compare No. 19). If, for instance, $h(d,\delta)$ is a form of the two rows dx and δx, then the expression

$$h^{(1)}(dx,\delta x) = \delta h - \sum \frac{\partial h}{\partial dx_\sigma} p^\sigma(d,\delta) - \sum \frac{\partial h}{\partial \delta x_\sigma} p^\sigma(\delta,\delta) \tag{143}$$

is itself an invariant as a difference of invariants, and is formed so that the second differentials cancel. Thus $h^{(1)}(dx,dx)$ is a form involving only first differentials: the covariant derivative of h. The corresponding statement holds when h contains more rows $d^{(1)}x, \ldots, d^{(\lambda)}x$.[151]

[149] This number is due to E. Noether.

[150] For special f the designation is due to Heun; see his article IV, 1 11, p. 447.

[151] Lipschitz, J. f. Math. 72 (1870), p. 17.

The formation of the curvature form by Riemann and Lipschitz rests on the same principle. One passes from $\delta^2 f$ to the "normal form of the second variation"

$$\Omega = \delta^2 \sum g_{ik} dx_i dx_k - 2d\delta \sum g_{ik} dx_i \delta x_k + d^2 \sum g_{ik} \delta x_i \delta x_k, \tag{144}$$

which, as an aggregate of invariants, is itself again an invariant, and in which, now generalizing the Lagrange central equation, the third differentials are eliminated. The elimination of the second differentials is carried out, in analogy with covariant differentiation, by forming the difference

$$K = \Omega - 2 \sum \{p^\sigma(d,\delta)\psi_\sigma(d,\delta) - p^\sigma(\delta,\delta)\psi_\sigma(d,d)\}, \tag{145}$$

which may also be written as[152]

$$\begin{aligned} \frac{1}{2}K = d \sum \psi_\sigma(d,\delta)\delta x_\sigma - \delta \sum \psi_\sigma(d,d)\delta x_\sigma \\ - \sum \{p^\sigma(d,\delta)\psi_\sigma(d,\delta) - p^\sigma(\delta,\delta)\psi_\sigma(d,d)\}. \end{aligned} \tag{146}$$

The law of formation of K may also be stated by saying that in Ω the second differentials are eliminated by setting $p(d,\delta)$ and $p(\delta,\delta)$ – respectively the right hand sides of (141) – equal to zero. This is Riemann's definition of the curvature form (compare No. 21), but it must be explicitly noted – as (146) makes especially clear – that this setting to zero may take place only after the differentiation has been performed; Lipschitz's formal mode of formation avoids this difficulty. Correspondingly, the covariant derivative $h^{(1)}$ (143) could also be defined by eliminating in δh the second differentials by setting $p(d,\delta), \ldots$ equal to zero.

From the curvature form one obtains, by successive formation of covariant derivatives, a sequence of forms [$\Phi_4, \Phi_5, \ldots$ in Christoffel], whose projective invariants, after adjoining f, exhaust according to Christoffel and Ricci the totality of the differential invariants of the quadratic form, and whose equivalence under linear transformations, without any further integrability condition, expresses the equivalence of two quadratic differential forms (compare No. 22). The inner reason for this reduction theorem rests on the existence of the Riemannian normal coordinates arising from the variational problem (140) (compare No. 20): these transform the geodesic lines through a point into straight lines, and consequently transform linearly under arbitrary transformations of the x. The formation of all invariants and the equivalence problem are thereby reduced to the corresponding question for the individual homogeneous components in the expansion of f in normal coordinates, and one shows that these components are linearly composed from $f, \Phi_4, \Phi_5, \ldots$. For quadratic forms this is carried out in detail by Vermeil[92)], following a note by E. Noether[93)]. According to that note, the theorems and methods remain valid when an arbitrary differential expression is taken as the basis (compare Nos. 22 and 23); the scope of the methods of formal variational calculus as against those of pure elimination theory is thereby made apparent.[153]

Levi-Civita[134)] interpreted the setting $p(d,\delta) = 0$ geometrically as "parallel displacement of a vector" (compare also Hessenberg[133)]). This concept, cast axiomatically by Weyl, is essentially restricted to quadratic forms, but it admits a generalization of another kind. It remains valid when the group is enlarged (multiplication of the g_{ik} by an arbitrary function), as soon as, besides the quadratic form, a linear form is also made basic. Then $p(d,\delta)$ is uniquely determined, invariant constructions corresponding to those above can be carried out, and geodesic coordinates can be defined; but $p(d,d) = 0$ now no longer arises from a variational problem.[154] Questions about the reduction theorem and equivalence have not yet been attacked here in general; only for invariants of lowest order has R. Weitzenböck given a reduction theorem.[155]

Finally, attention should be drawn to the general "invariant variational problems", i.e. to those in which the simple or multiple integral J is invariant with respect to some group in Lie's sense. Special physically interesting cases are treated in works by Hamel[156], Herglotz[157], Lorentz[158], Fokker[159], Weyl[160] and Klein[161]. The principled formulation by E. Noether[162] shows that to the invariance of J under a G_ρ (a finite group

[152] Lipschitz, J. f. Math. 82 (1876), § 1.

[153] Compare these remarks, and for further geometric interpretations the seminar lectures of F. Klein listed under "Literature."

[154] H. Weyl, Reine Infinitesimalgeometrie, Math. Ztschr. 2 (1918), p. 384–411, and Raum, Zeit, Materie (3rd and 4th ed.).

[155] Wien. Ber. 129 (1920), p. 683 and 697.

[156] Math. Ann. 59 (1904), p. 416–434; Ztschr. Math. Phys. 50 (1904), p. 1–54.

[157] Ann. d. Phys. (4) 36 (1911), p. 511, especially § 9.

[158] Verslag der Amsterd. Akad., April, Sept. 1916, Oct. 1916, May 1917.

[159] Ibid., Jan. 1917.

[160] Ann. d. Phys. (4) 54 (1917), p. 117, § 2.

[161] Gött. Nachr. 1917, p. 469–482; ibid. 1918, p. 171–189.

[162] Gött. Nachr. 1918, p. 235–257.

with ρ essential parameters) there correspond ρ linearly independent divergences; to the invariance under an infinite group containing ρ arbitrary functions up to the σ-th derivative there correspond ρ dependencies between the Lagrange expressions and their derivatives up to order σ. In both cases the converse holds. Since the Lagrange expressions become relative invariants of the group, one has at the same time a process generating invariants.

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